AYURVEDA • POSTGRADUATE STUDIES

MD / MS AYURVEDA

Rasashastra & Bhaishajya Kalpana | Final Year | Paper IV

TEXTBOOK OF

Pharmacology and Pharmacy Management

PAPER IV

Point-wise Study Notes with University Questions

Prepared strictly as per the Paper IV syllabus

Part A • Part B • 18 Chapters • Every Syllabus Point

Pharmacology • Pharmacy Management • Drug Laws

University Questions 2018–2026 • Model Answers • 120 MCQs

AUTHOR

Dr Milind Pareshkumar Chatrabhuji

BAMS, MD (Ayurveda), PhD (Scholar)

Associate Professor, PG Department of Rasashastra and Bhaishajya Kalpana

Parul Institute of Ayurved and Research, Parul University

FREE TO DOWNLOAD • FREE TO PRINT • NOT FOR SALE

Parul Institute of Ayurved and Research

Parul University, Vadodara, Gujarat, Bharat

TEXTBOOK OF

Pharmacology and Pharmacy Management

For Postgraduate Scholars of Ayurveda

MD / MS (AYURVEDA) • FINAL YEAR • RASASHASTRA AND BHAISHAJYA KALPANA

TEXTBOOK OF

Pharmacology and Pharmacy Management

Point-wise Study Notes with University Questions

Prepared strictly as per the Paper IV syllabus

PAPER IV

AUTHOR

Dr Milind Pareshkumar Chatrabhuji

BAMS, MD (Ayurveda), PhD (Scholar)
Associate Professor
PG Department of Rasashastra and Bhaishajya Kalpana
Parul Institute of Ayurved and Research, Parul University

Parul Institute of Ayurved and Research

Parul University, Vadodara, Gujarat, Bharat

Textbook of Pharmacology and Pharmacy Management

Point-wise Study Notes for MD / MS (Ayurveda), Rasashastra and Bhaishajya Kalpana, Final Year, Paper IV — Pharmacology and Pharmacy Management (PG final-year syllabus, old scheme)

EDITION

First Edition, 2026

AUTHOR

Dr Milind Pareshkumar Chatrabhuji
BAMS, MD (Ayurveda — Rasashastra and Bhaishajya Kalpana), PhD (Scholar)
Associate Professor, PG Department of Rasashastra and Bhaishajya Kalpana
Parul Institute of Ayurved and Research, Parul University, Vadodara

PUBLISHED BY

Parul Institute of Ayurved and Research
Parul University, Vadodara, Gujarat, Bharat

COPYRIGHT

© 2026 Dr Milind Pareshkumar Chatrabhuji.
The copyright in this work is retained by the author. It is published by Parul Institute of Ayurved and Research, Parul University, with the author's permission.

ISBN

ISBN 978-81-181614-2-4

LICENCE — THIS BOOK MAY BE FREELY COPIED AND PRINTED

This work is licensed under a **Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International Licence (CC BY-NC-ND 4.0)**. You are free to download, copy, print and redistribute this book in any medium or format, for any non-commercial purpose, provided that you give appropriate credit to the author, do not sell it, and do not distribute an altered version. Full licence text: creativecommons.org/licenses/by-nc-nd/4.0/

NOTE ON PREPARATION

This book was prepared by the author with the assistance of Claude Opus, a large language model developed by Anthropic. The choice of subject matter, the mapping to the Paper IV syllabus, the selection and verification of references, and the review of all content were directed and carried out by the author, who accepts sole and full responsibility for its contents.

DISCLAIMER

This book is intended solely for the academic use of postgraduate scholars and teachers of Ayurveda. It discusses drugs, toxic metals and minerals, and procedures that are hazardous outside a controlled setting. No procedure described in this book should be attempted except in a licensed pharmacy, a recognised laboratory or a teaching Rasashala, under the direct supervision of a qualified teacher, and in accordance with the prescribed safety measures and statutory requirements. Doses and drug examples are given to explain principles; they are not prescriptions. Acts, rules, notifications, pharmacopoeial limits and official guidelines are revised from time to time; readers must verify the current position from the primary source before relying on it. Nothing in this book constitutes clinical or legal advice or a warranty of any kind. The author and the publisher accept no liability for any loss or injury arising from its use.

TYPESET IN

Liberation Serif and Liberation Sans. Set on A4 with mirrored margins for double-sided printing and binding.

Dedicated to

all the Postgraduate Scholars of

Rasashastra and Bhaishajya Kalpana

PREFACE

Every year, final-year scholars of Rasashastra and Bhaishajya Kalpana tell me the same thing about Paper IV. They know their Bhasma and their Kupipakva Rasayana. They have prepared Asava and Avaleha with their own hands. But when the paper asks them to explain the phases of a clinical trial, the chelation of lead, the duties of a pharmacovigilance centre or the labelling rules for an Ayurvedic drug, they do not know where to begin. The material exists, but it lives in five different worlds: the pharmacology textbook, the pharmacy-management textbook, the toxicology journal, the Gazette of India and the websites of regulators. No one book brings it together for a scholar of our speciality. This book tries to.

It was written backwards, from the examination hall to the syllabus. I first collected every question set in seven university papers of this subject, from July 2018 to March 2026: seventy questions in all. I then placed each question under the syllabus point it belongs to, and wrote each chapter so that those questions, and the ones likely to follow them, can be answered from it. Every unit therefore opens with the questions already asked, and closes with an exam corner that shows how to write the answer and how to divide the marks.

Three features of the book need a word of explanation.

The law is stated as it stands today. Paper IV is the only paper of the final year in which a part of the syllabus has gone out of date by itself. The Prevention of Food Adulteration Act named in the syllabus was repealed by the Food Safety and Standards Act. The Consumer Protection Act of 1986 was replaced in 2019. The Factories Act has been absorbed into a labour code. The rule that required prior approval of advertisements of Ayurvedic drugs has been omitted. The Drugs Rules for Ayurvedic, Siddha and Unani drugs were substantially amended in 2026. A scholar who writes the old law in 2027 loses marks, and a practitioner who follows it can break the new one. So every syllabus heading is kept exactly as printed, the present law is taught under it, and every change is marked in a box headed *Law update*, with the date on which the position was checked.

The book looks at pharmacology through the eyes of Rasashastra. General pharmacology is the same for every student of medicine. But a scholar of our subject has particular reasons to understand dose and response, the toxicology of mercury, lead and arsenic, the evidence on herbo-mineral formulations, and the reporting of adverse reactions. Wherever the syllabus allows, the examples are taken from our own drugs: Vatsanabha for the therapeutic index, Haridra with Maricha for bioavailability, Rasa Sindura and Swarna Bhasma for toxicity studies.

The boundary between evidence and claim is marked plainly. Our subject is defended too often by overstatement and attacked too often by ignorance. Neither helps a scholar. Where published data support a classical claim, this book says so and cites the data. Where they do not yet, the book says that as well. A reader will find the words *Established* and *Not established* throughout. They are not a verdict on the tradition. They are an honest map of what has been measured and what still waits to be.

The language is kept simple on purpose. Most of our scholars studied in Gujarati, Hindi, Marathi or another Indian language before they studied in English, and a pharmacology text written for native speakers makes them work twice: once on the language and once on the subject. Here the sentences are short, technical words are explained where they first appear, and Sanskrit terms are written in ordinary Roman letters. Figures and tables are used wherever a picture says more than a paragraph, because a well-drawn diagram earns marks and is remembered long after the paragraph is forgotten.

This book does not replace the primary sources. The text of an Act, the current Gazette notification and the Ayurvedic Pharmacopoeia of India remain the final authority, and laws will continue to change after this edition is printed. I have named the source of every regulatory statement so that a reader can check it.

The book is free. It may be downloaded, printed and shared by any scholar, teacher or institution, and I hope it will be. It will certainly contain errors, and I shall be grateful to every reader who points one out.

Dr Milind Pareshkumar Chatrabhuji
Vadodara, Gujarat • 2026

ACKNOWLEDGEMENTS

My first and deepest thanks are to my wife, **Dr Smita Chatrabhuji**. This book was written in hours that belonged to her and to our family, and it was written only because she gave those hours without once being asked to. Whatever is of use in these pages exists because of her patience.

I thank **my family**, who have borne, over many years and many academic assignments, a husband, a father and a son who was often present in the house but absent from it. Their forbearance is not acknowledged often enough, and I acknowledge it here.

I record my gratitude to **Dr Rohit Gokarn**, Professor and Principal, Shree Swaminarayan Ayurvedic College, Kalol, under whose guidance I am pursuing my doctoral work. What I understand of research in this subject, and of the discipline of saying only what the evidence permits, I learned substantially from him.

I thank **Parul Institute of Ayurved and Research, Parul University**, and my colleagues in the PG Department of Rasashastra and Bhaishajya Kalpana, for the academic environment in which this work was possible.

Finally, I thank **my students**. Their questions in the classroom and in the Rasashala decided what needed explaining and what did not. This book is theirs before it is mine.

Dr Milind Pareshkumar Chatrabhuji

CONTENTS

Page numbers refer to the number printed at the foot of each page.

How to Use This Book 1

Syllabus at a Glance 4

Scheme of Examination 6

PART A — PHARMACOLOGY 9

Chapter A1 — General Pharmacology 11

Unit A1(a) — Principles of Pharmacology, Pharmacodynamics and Pharmacokinetics 12

A1a.1 Principles of pharmacology 13

A1a.2 Pharmacokinetics 16

A1a.3 Pharmacodynamics and mechanism of action 29

A1a.4 Dose determination and dose–response 34

A1a.5 Structure–activity relationship 40

A1a.6 Pharmacodynamics and pharmacokinetics illustrated with Ayurvedic drugs 44

Unit A1(b) — Routes of Drug Administration 51

A1b.1 What is a route of administration? 51

A1b.2 Classification of routes 52

A1b.3 Oral route 53

A1b.4 Sublingual and buccal routes 54

A1b.5 Rectal route 54

A1b.6 Injections (parenteral routes in the narrow sense) 55

A1b.7 Inhalation 57

A1b.8 Nasal route 58

A1b.9 Transdermal route 59

A1b.10 Topical application to the eye, ear, mouth and vagina 60

A1b.11 Newer approaches: targeted drug delivery 60

A1b.12 Comparison of the main routes 61

A1b.13 Classical Ayurvedic routes 61

Unit A1(c) — Factors Modifying Drug Effect, Bioavailability, Interactions, ADR and Toxicity 65

A1c.1 Factors modifying drug effect 65

A1c.2 Bioavailability and bioequivalence 70

A1c.3 Drug interactions 73

A1c.4 Adverse drug reactions 78

A1c.5 Drug toxicity 82

Unit A1(d) — Preclinical Evaluation and Animal Ethics 88
A1d.1 Where preclinical evaluation fits 88
A1d.2 Components of preclinical evaluation 89
A1d.3 Experimental pharmacology 90
A1d.4 Toxicity studies 97
A1d.5 Preclinical evaluation of an Ayurvedic preparation: general approach 99
A1d.6 Worked design: the in vivo immunomodulatory effect of Swarna Bhasma 99
A1d.7 Animal ethics 102
Viva pointers — Chapter A1 108

Chapter A2 — Clinical Pharmacology 111
A2.1 Clinical pharmacology 112
A2.2 Evaluation of a new chemical entity: the pathway in India 113
A2.3 Phases of clinical research 114
A2.4 Methods of clinical research 117
A2.5 Clinical research on a new Ayurvedic formulation 121
A2.6 Ethics involved in human research 124
Viva pointers — Chapter A2 133

Chapter A3 — Elemental Constituents of the Human Body 135
A3.1 Elemental composition of the human body 136
A3.2 Bio-elementology: the principle of essentiality 138
A3.3 Macro-minerals 139
A3.4 Trace elements 140
A3.5 Absorption and interactions of minerals 142
A3.6 Deficiency and excess: general principles 143
A3.7 Elements in Rasashastra drugs 144
Viva pointers — Chapter A3 148

Chapter A4 — Toxicity of Heavy Metals and Chelation Therapy 149
A4.1 What are heavy metals? 150
A4.2 General mechanisms of metal toxicity 150
A4.3 Factors that change toxicity 151
A4.4 Lead 152
A4.5 Mercury 154
A4.6 Arsenic 157
A4.7 Cadmium and other metals 160
A4.8 Heavy metals in Ayurvedic medicines: the published evidence 160
A4.9 Chelation therapy 162

A4.10 Heavy metal toxicity: Ayurvedic and modern aspects 166
Viva pointers — Chapter A4 172

Chapter A5 — Toxicity and Pharmacological Activities of Herbo-mineral Compounds 173
A5.1 What are herbo-mineral compounds? 174
A5.2 Why herbo-mineral medicines? The classical rationale 175
A5.3 What are they, chemically? 176
A5.4 How might herbo-mineral compounds act? 177
A5.5 Pharmacological activities: what the evidence shows 178
A5.6 Toxicity of herbo-mineral compounds 180
A5.7 Safe use of herbo-mineral medicines 183
A5.8 Classical safeguards: the Ayurvedic view of toxicity of Rasaushadhi 184
Viva pointers — Chapter A5 189

Chapter A6 — Pharmacovigilance 191
A6.1 Definition and scope 192
A6.2 Why pharmacovigilance is needed 193
A6.3 Milestones 195
A6.4 Methods of pharmacovigilance 196
A6.5 The pharmacovigilance process 197
A6.6 The international scenario 202
A6.7 The national scenario: Pharmacovigilance Programme of India 205
A6.8 Pharmacovigilance of Ayurvedic drugs 207
Viva pointers — Chapter A6 215

Question Bank — Part A 217
A-QB.1 Section A questions, paper by paper 217
A-QB.2 Which topics are asked most often 219
A-QB.3 Practice questions, chapter by chapter 220
A-QB.4 Model question papers: Section A 224
A-QB.5 Multiple-choice questions for self-assessment 225

PART B — PHARMACY MANAGEMENT 231

Chapter B1 — Scope and Evolution of Pharmacy; Information Resources 233
B1.1 What is pharmacy? 234
B1.2 Evolution of pharmacy 235
B1.3 Scope of pharmacy 236
B1.4 Information resources in pharmacy and pharmaceutical science 238
Viva pointers — Chapter B1 243

Chapter B2 — Pharmaceutical Dosage Form Design (Pre-formulation) 245

B2.1 Dosage forms 246

B2.2 Dosage form design 248

B2.3 Pre-formulation 249

B2.4 Dosage form design in Bhaishajya Kalpana 256

B2.5 Classical concepts and modern pre-formulation parameters 257

B2.6 Drug design 259

Viva pointers — Chapter B2 263

Chapter B3 — Packaging Materials and Labelling 265

B3.1 Packaging 266

B3.2 Packaging materials 267

B3.3 Selecting packaging for Ayurvedic dosage forms 269

B3.4 Shelf life, stability and storage 270

B3.5 Labelling 274

Viva pointers — Chapter B3 279

Chapter B4 — Pharmacy Management and Good Manufacturing Practices 281

B4.1 Management: principles and functions 282

B4.2 Management of an Ayurvedic pharmacy 283

B4.3 Store management 287

B4.4 Inventory management 289

B4.5 Personnel management 292

B4.6 Good Manufacturing Practices for the Ayurvedic drug industry 293

Viva pointers — Chapter B4 301

Chapter B5 — Pharmaceutical Marketing, Product Release and Withdrawals 303

B5.1 Pharmaceutical marketing 303

B5.2 Product release 306

B5.3 Product withdrawal and recall 307

Viva pointers — Chapter B5 312

Chapter B6 — Hospital, Dispensing and Community Pharmacy 313

B6.1 Hospital pharmacy 314

B6.2 Dispensing 316

B6.3 Community pharmacy 318

Viva pointers — Chapter B6 323

Chapter B7 — Patenting and Intellectual Property Rights 325

B7.1 Intellectual property 326

B7.2 The Indian patent system 328
B7.3 The international framework 330
B7.4 The Biological Diversity Act, 2002 330
B7.5 Traditional knowledge and biopiracy 331
B7.6 IPR and the Ayurvedic scholar 332
Viva pointers — Chapter B7 336

Chapter B8 — Laws Governing Ayurvedic Drugs 337

Unit B8(i) — Drugs and Cosmetics Act, 1940 and Rules, 1945 340
B8(i).1 The Act and its history 341
B8(i).2 Definitions 341
B8(i).3 Chapter IV-A of the Act 342
B8(i).4 Licensing of manufacture (Part XVI of the Rules) 343
B8(i).5 Good Manufacturing Practices: Schedule T 344
B8(i).6 Standards, testing and inspection 344
B8(i).7 Labelling, packing and shelf life (Part XVII) 344
B8(i).8 Schedule E(1) and advertisement 345
B8(i).9 The 2026 amendments in brief 346
Unit B8(ii) — Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 349
B8(ii).1 Objective 349
B8(ii).2 Definitions (Section 2) 349
B8(ii).3 What is prohibited 350
B8(ii).4 Exceptions (Section 14) 350
B8(ii).5 Penalties and enforcement 351
B8(ii).6 Relevance to Ayurvedic drugs 351
Unit B8(iii) — Prevention of Food Adulteration Act 353
B8(iii).1 Objectives 353
B8(iii).2 Main provisions 353
B8(iii).3 Why it was replaced 354
B8(iii).4 The PFA Act and herbal supplements 354
Unit B8(iv) — Food Safety and Standards Act, 2006 356
B8(iv).1 Objectives 356
B8(iv).2 Structure of the food safety system 357
B8(iv).3 Key definitions 358
B8(iv).4 Special foods: Section 22 358
B8(iv).5 Ayurveda Aahara 358
B8(iv).6 Licensing, enforcement and penalties 359

Unit B8(v) — Laws Pertaining to Narcotics 360
B8(v).1 The international framework 360
B8(v).2 The Narcotic Drugs and Psychotropic Substances Act, 1985 360
B8(v).3 Relevance to Ayurvedic drugs 361
B8(v).4 Practical points for an Ayurvedic pharmacy 362
Unit B8(vi) — Factories and Pharmacy Acts 364
B8(vi).1 The Factories Act, 1948 and the OSH Code, 2020 364
B8(vi).2 The Pharmacy Act, 1948 366
Unit B8(vii) — Consumer Protection Act 368
B8(vii).1 The Consumer Protection Act, 1986 368
B8(vii).2 The Consumer Protection Act, 2019 368
Viva pointers — Chapter B8 372
Chapter B9 — Regulatory Affairs in International Trade of Ayurvedic Drugs 375
B9.1 India’s trade in Ayurvedic products 376
B9.2 Export documentation from India 377
B9.3 Regulation of Ayurvedic products in major markets 378
B9.4 Barriers to international trade and how to overcome them 379
B9.5 India’s international initiatives for Ayurveda 379
B9.6 International practice of Ayurveda 379
Viva pointers — Chapter B9 382
Chapter B10 — Ayurvedic Pharmacopoeia and Formulary of India 383
B10.1 Pharmacopoeia and formulary 384
B10.2 Why Ayurveda needs a pharmacopoeia and formulary 384
B10.3 History and institutions 385
B10.4 Publications 386
B10.5 Structure of an API monograph 387
B10.6 Legal status 389
Viva pointers — Chapter B10 391
Chapter B11 — Indian, British and United States Pharmacopoeias and the Pharmaceutical Codex 393
B11.1 Pharmacopoeias and their purpose 393
B11.2 The Indian Pharmacopoeia 394
B11.3 The British Pharmacopoeia 395
B11.4 The United States Pharmacopeia–National Formulary 395
B11.5 The European Pharmacopoeia 395
B11.6 The Pharmaceutical Codex 396

B11.7 Other pharmacopoeias 396
Viva pointers — Chapter B11 398

Chapter B12 — Traditional Knowledge Digital Library 399

B12.1 What is the TKDL? 399
B12.2 Contents and structure 400
B12.3 Access 401
B12.4 How the TKDL works against wrong patents 401
B12.5 Significance and limitations 402
Viva pointers — Chapter B12 404

Question Bank — Part B 405

B-QB.1 Section B questions, paper by paper 405
B-QB.2 Which topics are asked most often 407
B-QB.3 Practice questions, chapter by chapter 408
B-QB.4 Model question papers: Section B 410
B-QB.5 Multiple-choice questions for self-assessment 412

Practical Work and Practical Examination 417

PR.1 The practical requirements of the syllabus 417
PR.2 The practical examination 417
PR.3 How to write up a practical 418
PR.4 Worked examples of practical write-ups 419
PR.5 Drug testing and analysis 422
PR.6 Identification of specimens 423
PR.7 Reports of visits to GMP pharmacies 424
PR.8 Posting in an R&D or production unit 425
PR.9 Clinical cases studied for efficacy and ADRs 425
PR.10 Presentations, conferences, publication and documentation 427
PR.11 Viva voce 427

Appendix 1 — Specimen Documents for an Ayurvedic Pharmacy 429

AP1.1 How to use these specimens 429
AP1.2 The documents that travel with one batch 429
AP1.3 Good documentation practice 431
AP1.4 Goods receipt note and status labels 431
AP1.5 Bin card and the Rasa Dravya register 433
AP1.6 Master formula record 435
AP1.7 Batch manufacturing record 438

AP1.8 The standard operating procedure 440
AP1.9 SOP for handling mercury in the Rasashala 442
AP1.10 SOP for Puta in an electric muffle furnace 444
AP1.11 Certificate of analysis and batch release 445
AP1.12 Deviation report 447
AP1.13 Market complaint record 448
AP1.14 Recall notice and recall record 449

Appendix 2 — Numerical Exercises in Pharmacology and Pharmacy Management 451
AP2.1 How to solve a numerical problem 451
AP2.2 Formulae at a glance 451
AP2.3 Elimination and half-life 452
AP2.4 Volume of distribution and clearance 453
AP2.5 Area under the curve and bioavailability 454
AP2.6 Dosage regimens 455
AP2.7 Dose conversion between animals and humans 455
AP2.8 Doses for children 456
AP2.9 Classical measures and batch calculations 457
AP2.10 Pharmacy management calculations 458
AP2.11 Exercises for practice 459

Appendix 3 — Model Answers to Frequently Asked Long Questions 461
AP3.1 Principles of pharmacology, pharmacodynamics and pharmacokinetics 461
AP3.2 Phases and methods of clinical research, and ethics in human research 463
AP3.3 Heavy metal toxicity and chelation therapy 466
AP3.4 Pharmacovigilance of Ayurvedic drugs and its national programme 468
AP3.5 Preclinical evaluation of drugs and its importance 470
AP3.6 Good Manufacturing Practices related to the Ayurvedic drug industry 472
AP3.7 Laws governing Ayurvedic drugs and their necessity 475
AP3.8 Pharmaceutical dosage form design with special reference to pre-formulation 477
AP3.9 Ayurvedic Pharmacopoeia and Formulary of India 479

Appendix 4 — Chronology of Laws, Standards and Milestones 483
AP4.1 The story in one picture 484
AP4.2 Chronology 484

Principal Sources and References 489
Glossary of Ayurvedic Terms 495
About the Author 503
Index 505

ABBREVIATIONS

Regulatory, statutory and institutional

AFI	Ayurvedic Formulary of India
AIIA	All India Institute of Ayurveda, New Delhi
API	Ayurvedic Pharmacopoeia of India (in Chapter B2 and elsewhere it may also mean Active Pharmaceutical Ingredient; the sense is stated where it is used)
ASU	Ayurveda, Siddha and Unani
ASU&H	Ayurveda, Siddha, Unani and Homoeopathy
AYUSHEXCIL	Ayush Export Promotion Council
CBD	Convention on Biological Diversity
CCPA	Central Consumer Protection Authority
CCRAS	Central Council for Research in Ayurvedic Sciences
CCSEA	Committee for Control and Supervision of Experiments on Animals (formerly CPCSEA)
CDC	Centers for Disease Control and Prevention (United States)
CDSCO	Central Drugs Standard Control Organisation
COPP	Certificate of Pharmaceutical Product
CPA	Consumer Protection Act
CSIR	Council of Scientific and Industrial Research
CTRI	Clinical Trials Registry — India
DMR	Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954
EMA	European Medicines Agency
FDA	Food and Drug Administration (United States)
FSSAI	Food Safety and Standards Authority of India
GCP	Good Clinical Practice
GLP	Good Laboratory Practice
GMP	Good Manufacturing Practices
IAEC	Institutional Animal Ethics Committee
ICH	International Council for Harmonisation of Technical Requirements for Pharmaceuticals for Human Use
ICMR	Indian Council of Medical Research
IPC	Indian Pharmacopoeia Commission
IPR	Intellectual Property Rights
MHRA	Medicines and Healthcare products Regulatory Agency (United Kingdom)
NBA	National Biodiversity Authority
NDPS	Narcotic Drugs and Psychotropic Substances
OECD	Organisation for Economic Co-operation and Development
OSH Code	Occupational Safety, Health and Working Conditions Code, 2020
PCIM&H	Pharmacopoeia Commission for Indian Medicine and Homoeopathy

PCT	Patent Cooperation Treaty
PFA	Prevention of Food Adulteration
PLIM	Pharmacopoeial Laboratory for Indian Medicine
PvPI	Pharmacovigilance Programme of India
QCI	Quality Council of India
TKDL	Traditional Knowledge Digital Library
TKRC	Traditional Knowledge Resource Classification
TRIPS	Agreement on Trade-Related Aspects of Intellectual Property Rights
UMC	Uppsala Monitoring Centre
USP–NF	United States Pharmacopeia–National Formulary
WHO	World Health Organization
WIPO	World Intellectual Property Organization

Pharmacy management and quality

A.R.	Analytical report (number given by QC to each sample tested)
BCS	Biopharmaceutics Classification System
BMR	Batch manufacturing record
CAPA	Corrective and preventive action
CoA	Certificate of analysis
EOQ	Economic order quantity
FEFO	First expiry, first out
GRN	Goods receipt note
GTIN	Global Trade Item Number
HDPE	High-density polyethylene
HPTLC	High-performance thin-layer chromatography
LOD	Loss on drying
MFR	Master formula record
QA	Quality assurance
QC	Quality control
SOP	Standard operating procedure
TLC	Thin-layer chromatography
VED	Vital, Essential, Desirable (inventory classification)
XRD	X-ray diffraction

Pharmacological

ACh	Acetylcholine
ADME	Absorption, Distribution, Metabolism and Excretion
ADR	Adverse Drug Reaction
ATP	Adenosine triphosphate
AUC	Area Under the plasma concentration–time Curve

BAL	British anti-Lewisite (dimercaprol)
BBB	Blood–Brain Barrier
BSA	Body Surface Area
cAMP	Cyclic adenosine monophosphate
CL	Clearance
Cmax	Peak plasma concentration
COMT	Catechol-O-methyltransferase
COX	Cyclo-oxygenase
Css	Steady-state plasma concentration
CYP	Cytochrome P450 enzyme
DMSA	Dimercaptosuccinic acid (succimer)
EC50	Concentration producing 50 per cent of the maximal effect
ED50	Dose effective in 50 per cent of subjects
EDTA	Ethylenediaminetetraacetic acid (edetate)
F	Bioavailability (fraction of the dose reaching the systemic circulation)
GPCR	G-protein-coupled receptor
HED	Human Equivalent Dose
IA	Intrinsic activity
LD50	Dose lethal to 50 per cent of animals
MAO	Monoamine oxidase
MEC	Minimum Effective Concentration
MIC	Minimum inhibitory concentration
MRSD	Maximum Recommended Starting Dose
MTC	Minimum Toxic Concentration
NCE	New Chemical Entity
NOAEL	No Observed Adverse Effect Level
PD	Pharmacodynamics
P-gp	P-glycoprotein (an efflux transporter)
PK	Pharmacokinetics
pKa	Negative logarithm of the acid dissociation constant
PRR	Proportional reporting ratio
QSAR	Quantitative Structure–Activity Relationship
RCT	Randomised controlled trial
SAR	Structure–Activity Relationship
t½	Elimination half-life
TD50	Dose toxic to 50 per cent of subjects
TI	Therapeutic Index
Tmax	Time to peak plasma concentration
Vd	Apparent Volume of Distribution

VMAT	Vesicular Monoamine Transporter

Examination

Q.3a	Question 3, option a, of a section of the question paper

HOW TO USE THIS BOOK

These notes are written for the postgraduate scholar of **Rasashastra and Bhaishajya Kalpana** in the final year. They follow **Paper IV — Pharmacology and Pharmacy Management** of the PG final-year syllabus exactly as it is printed. No topic has been added. No topic has been left out.

How the book is arranged

The syllabus has two parts. **Part A, Pharmacology,** has six points. **Part B, Pharmacy Management,** has twelve. The question paper follows the same division: Section A of the paper is set from Part A and Section B from Part B. This book therefore has two Parts and eighteen chapters, one chapter for each syllabus point, numbered A1 to A6 and B1 to B12. Where a point has sub-points — A1 has four, (a) to (d), and B8 has seven, (i) to (vii) — each sub-point is a unit of its own. So you can keep the syllabus in one hand and these notes in the other, and never lose your place.

Each chapter opens with the syllabus point and its record in university examinations. Each unit is then arranged like this.

Part of the unit	What it gives you
Syllabus	The wording of the syllabus
Learning outcomes	What you should be able to do after reading. These are set by the author, because the old syllabus has no table of objectives.
Main notes	The full explanation, point by point, with figures and tables
Ayurveda correlation	The classical concept beside the modern one, with the boundary between what is established and what is proposed
Law update	Wherever the law has changed since the syllabus was written
Key points to remember	A short revision list
Exam corner	The questions actually set, likely questions, how to attempt them, and an answer frame
References used	The reference-book numbers from the syllabus list, with full citations and any further verified source

Each chapter ends with **viva pointers**. Each Part ends with a **question bank**. After Part B comes the **practical work** that the syllabus requires and that touches Paper IV — the clinical cases recorded for ADRs, the visits to GMP pharmacies and the R&D posting — written up with safety sections.

Four **appendices** follow: specimen GMP documents for an Ayurvedic pharmacy, numerical exercises with worked solutions, model answers to the long questions asked most often, and a chronology of the laws and standards. They add no new syllabus topic. They help you apply, write and revise what the chapters teach.

How to read

1. First read the syllabus and the questions already asked. Know what you have to learn.
2. Then read the main notes once, slowly, without trying to memorise.

3. Read them a second time and study the figures and tables. Most examiners give marks for a well-drawn table or diagram.
4. Revise from “Key points to remember”.
5. Practise writing answers from “Exam corner”. Write at least one full long answer for every chapter before the examination.

About the past questions

Seven Paper IV question papers were examined: those set by Parul University for MD/MS (Ayurved) in July 2018, September 2018, September 2020, August 2021, August 2022, January 2024 and March 2026. Every question in them, seventy in all, is printed in this book, word for word as it was set, under the topic it belongs to. A question is shown with its paper, for example `Sep 2018`. A question that draws on two topics is shown under both. The papers of other universities may differ in pattern, but the syllabus is the same.

A word about the language

The sentences here are kept short on purpose. Long sentences hide the meaning. If a sentence looks simple, that is not because the topic is simple. It is because you should be able to read it once and understand it.

Ayurvedic terms are written in plain Roman letters. No diacritical marks are used. Shodhana is written as Shodhana, not with special symbols. This is easier to read and easier to write in the examination.

About the references

The syllabus gives a list of 53 reference books. Where a unit uses one of them, it is cited by its number in that list, for example **[R29]**, with the full citation of the edition actually consulted. Some entries in the syllabus list need care: reference 29 is printed as “Clinical Pharmacology, KD Tripathi”, but KD Tripathi’s book is *Essentials of Medical Pharmacology*; reference 30 prints the authors’ names as “Lawrence Benette” for Laurence and Bennett; references 16 and 20 are the same book (Siddha Bhaishajya Manimala); and reference 49 is a description of the kind of sources to consult, not a citation. Where a topic needs a source that is not in the list — a journal paper, a gazette notification, an official guideline — it has been verified and is cited in full. Classical texts are cited by Sthana and chapter, and by verse where the verse number could be verified. No reference in this book is invented. The whole syllabus list, with a note against every entry that needs one, is printed in Principal Sources and References at the end of the book.

Corrections made to the syllabus text

The syllabus is reproduced as printed, except for these obvious errors, which are corrected throughout the book.

As printed	As corrected	Reason
Pharmcodynamics	Pharmacodynamics	spelling (Part A, 1a)
Drug and Cosmetics Act - 1940	Drugs and Cosmetics Act, 1940	official title of the Act (Part B, 8i)
Food Standards and Safety Act - 2006	Food Safety and Standards Act, 2006	official title of the Act (Part B, 8iv)
Factory and Pharmacy Acts	Factories and Pharmacy Acts	the Act is the Factories Act, 1948 (Part B, 8vi)
Pharmacoepia	Pharmacopoeia	spelling (Part B, 11)
Pharamcoepial Codex	Pharmaceutical Codex	spelling, and the book's title is The Pharmaceutical Codex (Part B, 11)
Minmum	Minimum	spelling (Practicals)

One request. These notes will help you pass. They will not replace the standard texts. Keep a pharmacology textbook, the Drugs and Cosmetics Act and Rules, and the Ayurvedic Pharmacopoeia of India beside you. Read them. This book only shows you the road.

SYLLABUS AT A GLANCE

Paper IV, **Pharmacology and Pharmacy Management**, as printed in the PG final-year syllabus of Rasa Shastra and Bhaishajya Kalpana, with the corrections listed in How to Use This Book. The last column shows how many questions were set on each point in the seven university papers examined.

Point	Syllabus	Asked
A1(a)	Principles of Pharmacology, Pharmacodynamics & Pharmacokinetics: Absorption, distribution, Metabolism & excretion, mechanism of action, dose determination and dose response, structure activity relationship.	4
A1(b)	Routes of drug administration	4
A1(c)	Factors modifying drug effect, Bioavailability and Bioequivalence, drug interactions, adverse drug reaction and drug toxicity	4
A1(d)	Preclinical evaluation: experimental pharmacology [bioassay, in vitro, in vivo, cell line studies] animal ethics.	3
A2	Clinical pharmacology: Evaluation of New Chemical Entity – phases and methods of clinical research. Ethics involved in human research.	5
A3	Elemental constituents of human body and its physiological importance. Deficiencies and excess of various elements (micro-nutrients).	5
A4	Toxicity of heavy metals and chelation therapy.	5
A5	Knowledge of toxicity and pharmacological activities of herbo-mineral compounds.	1
A6	Detailed Knowledge of Pharmacovigilance – National and International Scenario. Pharmacovigilance of Ayurvedic Drugs	7
B1	Scope and evolution of pharmacy. Information resources in pharmacy and pharmaceutical Science.	1
B2	Pharmaceutical dosage form design (Pre-formulation)	5
B3	Packaging materials and Labeling	4
B4	Management of pharmacy, store and inventory management, personnel management, Good Manufacturing Practices related to Ayurvedic drug industry.	7
B5	Pharmaceutical Marketing, product release and withdrawals.	1
B6	Hospital, Dispensing and Community pharmacy.	1
B7	Patenting and Intellectual Property Rights.	4
B8(i)	Relevant regulatory provisions of Ayurvedic drugs in Drugs and Cosmetics Act, 1940 and Rules - 1945	4
B8(ii)	Laws pertaining to Drugs and Magic remedies (Objectionable Advertisement) Act - 1954.	3
B8(iii)	Prevention of Food Adulteration (PFA) act.	3
B8(iv)	Food Safety and Standards Act, 2006	2
B8(v)	Laws pertaining to Narcotics	2

Point	Syllabus	Asked
B8(vi)	Factories and Pharmacy Acts	2
B8(vii)	Consumer Protection Act -1986	2
B9	Regulatory Affairs related to International Trade and Practices of Ayurvedic Drugs	1
B10	Introduction to Ayurvedic Pharmacopoeia of India, Ayurvedic Formulary of India.	5
B11	Introduction to Indian Pharmacopoeia, British and United States Pharmacopoeia, Pharmaceutical Codex	—
B12	Introduction to Traditional Knowledge Digital Library	3

Most asked: Pharmacovigilance (A6) was set in all seven papers; Good Manufacturing Practices and pharmacy management (B4) in six; drug laws (B8) in six. Never asked so far: the Indian, British and United States pharmacopoeias (B11).

SCHEME OF EXAMINATION

Theory: Paper IV

All seven papers examined follow the same pattern: 100 marks, three hours, two sections written on separate answer sheets.

Section	Questions	Marks
A (Part A)	Q.1 long question; Q.2 long question; Q.3 any two of three short notes	20 + 20 + (5 × 2) = 50
B (Part B)	Q.1 long question; Q.2 long question; Q.3 any two of three short notes	20 + 20 + (5 × 2) = 50
Total		**100**

This is the Parul University pattern seen in the papers of 2018 to 2026. Confirm the pattern of your own university.

Practicals, as printed in the syllabus

The practical examination is common to the four papers of the final year. It is reproduced here because Paper IV supplies part of it: drug testing and analysis, the documentation of clinical cases for ADRs, and the reports of the GMP pharmacy visits.

- Minimum 150 Practicals are to be performed covering all the dosage forms.
- Educational Visits to minimum Five (5) GMP approved Ayurvedic Pharmacies.
- Ten days posting in R & D / Production unit in GMP certified Ayurvedic Pharmacy.
- Minimum attendance in three National Conferences within three academic years.
- Minimum 10 clinical cases using different dosage forms are to be studied and recorded for efficacy and ADRs (if any).
- Minimum one research paper on the core contributory work of dissertation published or accepted in peer reviewed, indexed journal.
- Examinee has to submit supporting documents in relation to above mentioned academic activities at the time of Practical examination.

Pattern of Practical Examination (Total Marks 100; duration 2 days)

No.	Item	Marks
1	Identification of the specimen	10
2	Drug Processing — Major Practicals: 2 (1 Rasashastra and 1 Bhaishajya Kalpana); Minor Practicals: 2 (1 Rasashastra and 1 Bhaishajya Kalpana)	30 (10+10+5+5)
3	Drug testing and Analysis — Rasaushadhi 1, Kashthaushadhi 1	10 (5+5)
4	Power Point Presentation on dissertation work: 15 minutes	10
6	A Demo Lecture on any subject topic: 10 minutes	10
7	Documentation — Journals (Practical Records) of minimum 75 practicals (25 each from Rasa Shastra, Bhaishajya Kalpana and Analytical part); Reports of the Visits; Case Records	15
8	Viva Voce examination	15
	Total	**100**

The printed scheme numbers its items 1, 2, 3, 4, 6, 7 and 8; there is no item 5. The marks add up to 100 as printed.

PART A

PHARMACOLOGY

Paper IV · Section A of the university question paper (50 marks)

Chapter	Syllabus point
A1	General Pharmacology
A2	Clinical Pharmacology
A3	Elemental Constituents of the Human Body
A4	Toxicity of Heavy Metals and Chelation Therapy
A5	Toxicity and Pharmacological Activities of Herbo-mineral Compounds
A6	Pharmacovigilance

PART A · PHARMACOLOGY · CHAPTER A1

A1 · GENERAL PHARMACOLOGY

SYLLABUS POINT (PART A, 1)

General Pharmacology:

- **A1(a)** Principles of Pharmacology, Pharmacodynamics & Pharmacokinetics: Absorption, distribution, Metabolism & excretion, mechanism of action, dose determination and dose response, structure activity relationship.
- **A1(b)** Routes of drug administration
- **A1(c)** Factors modifying drug effect, Bioavailability and Bioequivalence, drug interactions, adverse drug reaction and drug toxicity
- **A1(d)** Preclinical evaluation: experimental pharmacology [bioassay, in vitro, in vivo, cell line studies] animal ethics.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A1(a)	Principles of Pharmacology, Pharmacodynamics and Pharmacokinetics	3	1	Sep 2018, Sep 2020, Aug 2021, Mar 2026
A1(b)	Routes of Drug Administration	2	2	Jul 2018, Sep 2018, Aug 2021, Aug 2022
A1(c)	Factors Modifying Drug Effect, Bioavailability, Interactions, ADR and Toxicity	1	3	Jul 2018, Jan 2024, Mar 2026
A1(d)	Preclinical Evaluation and Animal Ethics	3	0	Sep 2018, Jan 2024, Mar 2026

Why this chapter matters to you. Every other chapter of Paper IV rests on this one. Clinical trials (A2), heavy-metal toxicity (A4), herbo-mineral toxicity (A5) and pharmacovigilance (A6) all use the words defined here: absorption, half-life, dose–response, therapeutic index, adverse reaction. In the seven papers examined, General Pharmacology supplied **eight long questions and six short notes** across its four units (one long question, September 2018, drew on two units at once). It is the most heavily examined chapter of Part A.

UNIT A1(a)

PRINCIPLES OF PHARMACOLOGY, PHARMACODYNAMICS AND PHARMACOKINETICS

SYLLABUS

Principles of Pharmacology, Pharmacodynamics & Pharmacokinetics: Absorption, distribution, Metabolism & excretion, mechanism of action, dose determination and dose response, structure activity relationship.

LEARNING OUTCOMES

After studying this unit you should be able to:

1. Define pharmacology, pharmacokinetics and pharmacodynamics, and state the general principles that govern how every drug acts.
2. Describe absorption, distribution, metabolism and excretion, and the factors that control each.
3. Calculate volume of distribution, half-life, clearance, loading dose and maintenance dose from given values.
4. Explain mechanisms of drug action, with the four families of receptors and their signalling.
5. Draw and interpret graded and quantal dose–response curves; distinguish potency from efficacy; define therapeutic index.
6. Explain how a dose is determined, from animal data to the first human dose and to the individual patient.
7. Explain structure–activity relationship with examples, including one from Rasashastra.
8. Set these modern concepts beside Dravyaguna and Rasashastra, and say which parallels are established and which are only proposed.

These outcomes are set by the author. The old syllabus does not prescribe session objectives.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	A, Q.1	20	Explain the general principles of Pharmacology. Write Pharmacodynamics and Pharmacokinetics in detail. Explain any 2 routes of administration of Drug.
Sep 2020	A, Q.1	20	Write down principles of Pharmacology, Pharmacodynamics & Pharmacokinetics
Aug 2021	A, Q.1	20	Explain in detail regarding Principles of Pharmacodynamics and Pharmacokinetics by illustrating with any Ayurvedic Formulation
Mar 2026	A, Q.3a	5	Define “Dose Response Relationship” and “Structure-Activity Relationship”.

This unit has been set three times as a 20-mark question. Every time the examiner asked for the **principles first, then pharmacodynamics and pharmacokinetics in detail**. Once (August 2021) the question added

"by illustrating with any Ayurvedic formulation". Section A1a.6 of this unit is written for exactly that demand.

A1a.1 Principles of pharmacology

A1a.1.1 Starting point: what is a drug?

Take a tablet of paracetamol. Before it is swallowed it is only a chemical. After it is swallowed, two things happen at the same time. The body acts on the tablet: it dissolves it, absorbs it, carries it, changes it and throws it out. And the drug acts on the body: it reduces fever and pain. Pharmacology is the study of both.

> **Pharmacology** is the science of drugs: their sources, physical and chemical properties, actions on living systems, fate in the body, and therapeutic and toxic uses. The word comes from the Greek *pharmakon* (drug, and also poison) and *logos* (study).

> **Drug.** The World Health Organization defines a drug as any substance or product used, or intended to be used, to modify or explore physiological systems or pathological states for the benefit of the recipient [R29].

Note three things in this definition.

- A substance becomes a drug by its **use**, not by its nature. Water given in dehydration, or a mineral given as a Bhasma, is a drug in that use.
- A drug may be used to **explore** as well as to treat. A dye used in a diagnostic test is a drug.
- The use must be **for the benefit of the recipient**. The same substance used to harm is a poison.

The Drugs and Cosmetics Act, 1940 has its own legal definitions of a drug and of an Ayurvedic drug. They are explained in Chapter B8(i).

A1a.1.2 Branches of pharmacology

Pharmacology has grown into many branches. The two at its centre are pharmacokinetics and pharmacodynamics. Figure A1a.1 shows how they relate. Figure A1a.2 shows the main branches.

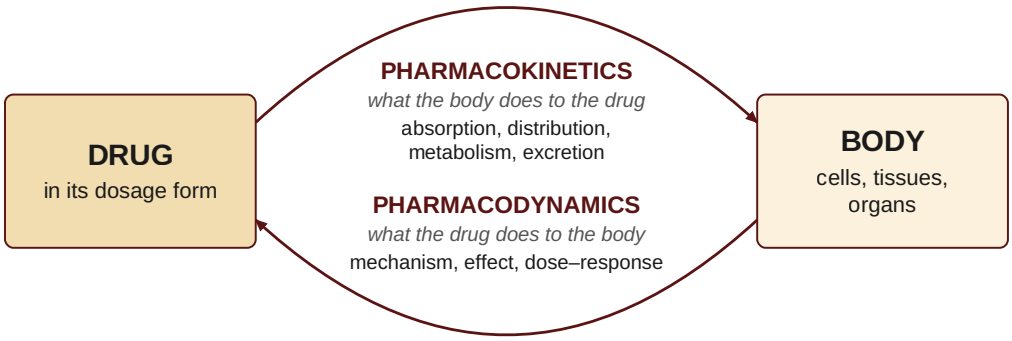
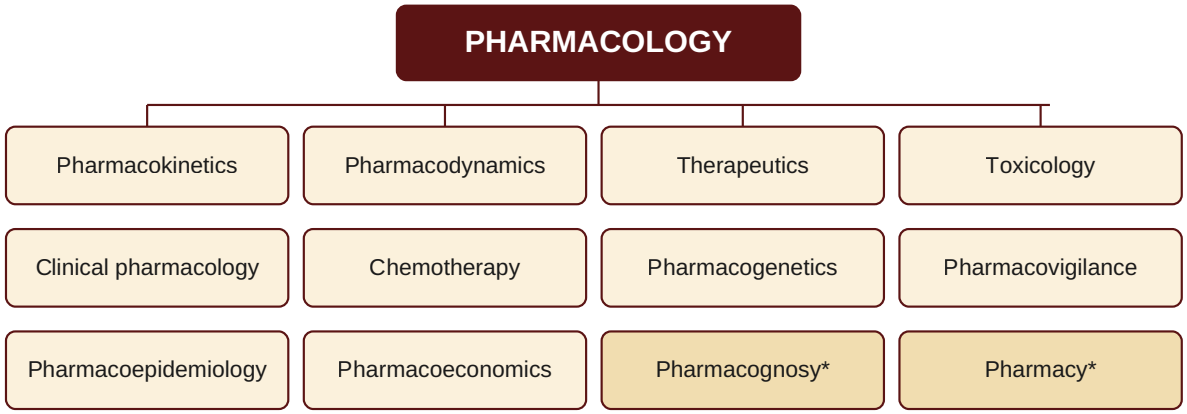


Figure A1a.1. Pharmacokinetics and pharmacodynamics: two directions of one relationship between drug and body.



*Allied sciences: they serve pharmacology but are subjects of their own.

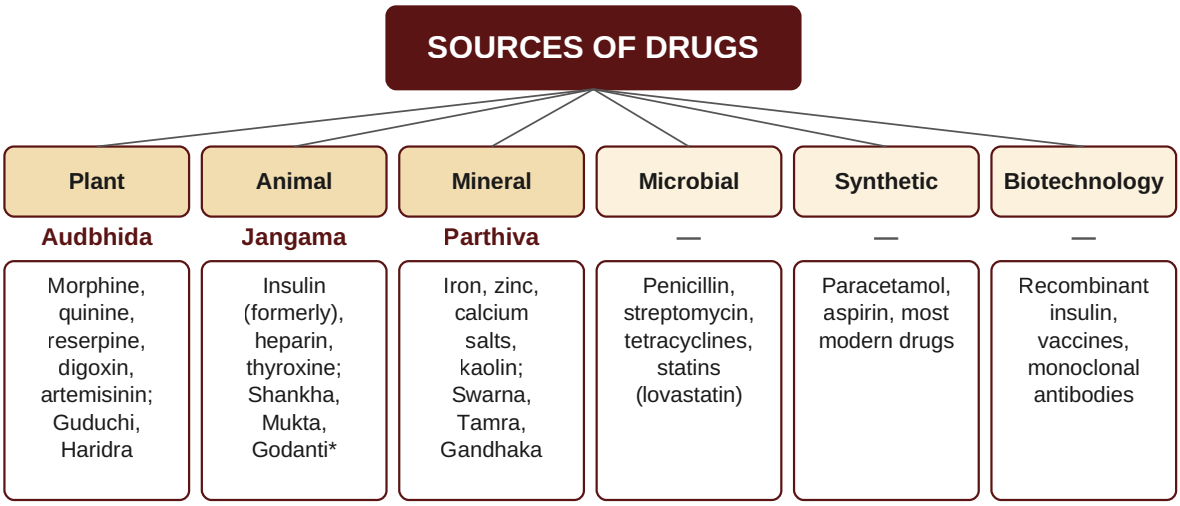
Figure A1a.2. Main branches of pharmacology. Pharmacognosy and pharmacy are allied sciences.

Branch	What it studies	Example of a question it answers
Pharmacokinetics (PK)	What the body does to the drug: absorption, distribution, metabolism, excretion	How long does one dose stay in the blood?
Pharmacodynamics (PD)	What the drug does to the body: mechanism and effect	On which receptor does salbutamol act?
Therapeutics	Use of drugs to prevent and treat disease	Which drug for which patient, at what dose?
Toxicology	Harmful effects of chemicals, including drugs	What dose of arsenic causes chronic poisoning?
Clinical pharmacology	Drugs in human beings, healthy and ill	Is the new drug better than the old one?
Chemotherapy	Drugs that kill or stop the growth of microbes, parasites or cancer cells	Which antibiotic kills this organism?
Pharmacogenetics and pharmacogenomics	Genetic causes of variation in drug response	Why does codeine fail in some patients?
Pharmacovigilance	Detection and prevention of adverse effects after marketing	Is this new rash caused by the drug?
Pharmacoepidemiology	Use and effects of drugs in large populations	How often does this drug cause liver injury?
Pharmacoeconomics	Costs and outcomes of drug therapy	Which treatment gives more benefit per rupee?
Pharmacognosy (allied)	Drugs of natural origin, their identification and quality	Is this root genuine Ashwagandha?
Pharmacy (allied)	Preparation, dispensing and proper use of medicines	How should this Taila be stored?

A1a.1.3 Sources of drugs

Drugs come from six broad sources (Figure A1a.3). Charaka classified Dravya into three by source: **Jangama** (from animals), **Audbhida** (from plants) and **Parthiva** (from the earth, that is, minerals and

metals) (Charaka Samhita, Sutra Sthana 1) [R26]. The first three modern sources match this classical scheme closely. Microbial, synthetic and biotechnological sources are modern additions.



Gold boxes: the three classical sources of Dravya (Charaka Samhita, Sutra Sthana 1).
**Godanti is gypsum, a mineral, though traditionally grouped with the Sudha Varga.*

Figure A1a.3. Sources of drugs, with the classical three-fold classification of Dravya.

For a scholar of Rasashastra, the mineral (Parthiva) source is the most important. Mercury, sulphur, mica, gold, copper, iron, zinc, lead and tin are all mineral sources. Their pharmacology and toxicology are the subject of Chapters A3, A4 and A5.

A1a.1.4 Names of a drug

One drug may have three kinds of name.

Kind of name	What it is	Example
Chemical name	Describes the chemical structure; long and exact	N-(4-hydroxyphenyl)acetamide
Non-proprietary (generic) name	Official short name, the same for all makers; in India usually as in the Indian Pharmacopoeia	Paracetamol (acetaminophen in the United States)
Proprietary (brand) name	The maker’s trade name; one generic drug may have many brands	Any registered brand of paracetamol

Ayurvedic medicines show the same pattern. A classical medicine carries the name given in the authoritative book, such as *Chyavanaprasha Avaleha*. A proprietary Ayurvedic medicine carries the maker’s own name. The legal difference between the two is explained in Chapter B8(i).

A1a.1.5 General principles of pharmacology

These are the principles on which the rest of pharmacology is built. A 20-mark answer on “principles of pharmacology” should state them before going into pharmacokinetics and pharmacodynamics.

No.	Principle	What it means in practice
1	A drug does not create a new function. It only increases or decreases a function the body already has.	Salbutamol relaxes bronchial muscle that can already relax. No drug can make a cell do something foreign to it.

No.	Principle	What it means in practice
2	Most drugs act by binding to a specific target molecule.	Receptors, enzymes, ion channels and transporters. Paul Ehrlich summed this up as “a substance does not act unless it is bound”.
3	The effect depends on the concentration at the site of action.	That concentration depends on the dose and on pharmacokinetics. Same dose, different absorption, different effect.
4	The effect increases with dose, up to a limit.	This is the dose–response relationship (A1a.4). Above the ceiling, more drug brings more toxicity, not more benefit.
5	No drug has only one effect.	Every drug has some adverse effects. Selectivity is relative and is lost at high doses.
6	People respond differently to the same dose.	Age, weight, genes, disease, other drugs and diet change the response (A1(c)). Doses must be individualised.
7	Every effect has a time course.	Onset, peak and duration follow the plasma concentration (A1a.2.8).
8	Drugs interact with other drugs and with food.	One drug may raise or lower the level or effect of another (A1(c)).
9	Benefit must outweigh risk.	A toxic drug is acceptable for a fatal disease; it is not acceptable for a minor one.

AYURVEDA CORRELATION

Dravyaguna is the classical pharmacology of Ayurveda. It studies the **Dravya** (the substance), its **Guna** (properties) and its **Karma** (actions), and it has its own principles.

- Charaka holds that there is no substance in the world which cannot be used as a medicine, given the right purpose and the right method (Charaka Samhita, Sutra Sthana 26/12) [R26]. This matches the modern view that a substance is a drug because of its use, not its nature.
- Charaka names four qualities of a good drug: it is available in plenty, it is fit for the purpose, it can be made into many dosage forms, and it is of good quality (Sutra Sthana 9/7) [R26]. Availability, efficacy, formulability and quality are still the first questions asked of any new drug.
- The classical teaching that a poison becomes a medicine by right use, and a medicine becomes a poison by wrong use (Sutra Sthana 1) [R26], is the dose–response principle in another language. It is discussed in A1a.4.

Established. Dravyaguna is a systematic and internally consistent classical framework for describing drugs and their actions.

Not established. That each classical term has an exact modern equivalent. Section A1a.6 sets out the parallels and marks how far each one is supported.

A1a.2 Pharmacokinetics

> **Pharmacokinetics** is the study of the movement of a drug into, through and out of the body, and of the time course of its concentration. In short, it is what the body does to the drug.

The movement has four stages, remembered as **ADME**: Absorption, Distribution, Metabolism and Excretion. Some books add **L** for Liberation, the release of the drug from its dosage form, and write **LADME**. For a

solid dosage form, liberation comes first and often decides everything that follows. Figure A1a.4 follows one tablet through the body.

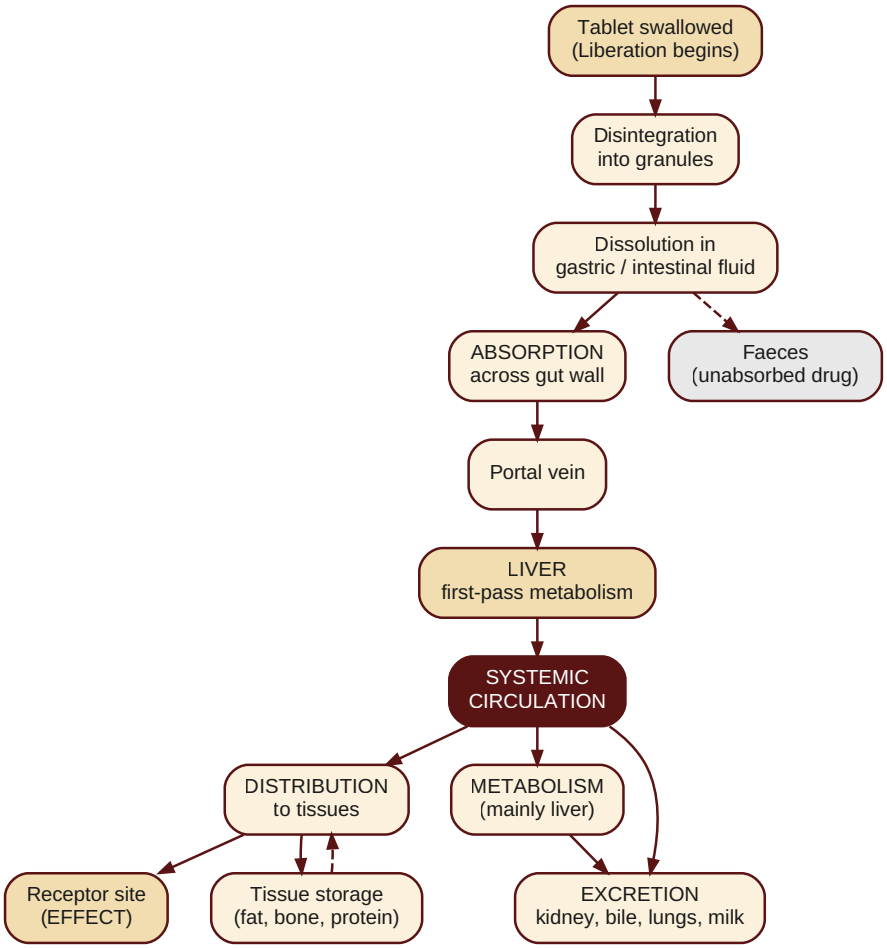


Figure A1a.4. The journey of an oral tablet: liberation, absorption, first-pass metabolism, distribution, metabolism and excretion.

A1a.2.1 How drugs cross membranes

Every stage of ADME needs the drug to cross cell membranes. The gut wall, the capillary wall, the blood–brain barrier and the renal tubule are all made of membranes. A cell membrane is a double layer of lipid with proteins set in it. So the rule is simple: **lipid-soluble drugs cross easily; water-soluble and charged drugs need a pore or a carrier.** There are five ways to cross (Figure A1a.5).

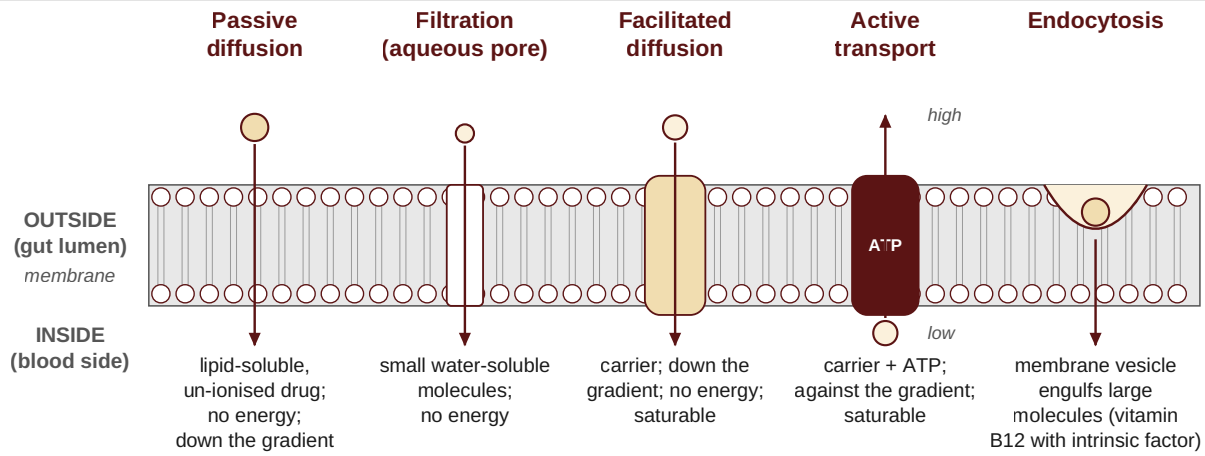


Figure A1a.5. Five ways a drug crosses a cell membrane.

Mechanism	Energy needed?	Carrier?	Saturable?	Typical drugs
Passive diffusion through lipid	No	No	No	Most drugs: small, lipid-soluble, un-ionised
Filtration through aqueous pores	No	No	No	Small water-soluble molecules; urea, ethanol
Facilitated diffusion	No	Yes	Yes	Glucose into cells; some vitamins
Active transport	Yes (ATP)	Yes	Yes	Levodopa and methyldopa across the gut (amino-acid carrier); iron; efflux by P-glycoprotein
Endocytosis (pinocytosis)	Yes	No (vesicle)	—	Large molecules; vitamin B12 with intrinsic factor

Passive diffusion is the most important. Its rate follows Fick's law: it is greater when the concentration difference is greater, the area is larger, the membrane is thinner and the drug is more lipid-soluble.

Carrier transport is selective and saturable. When all carriers are busy, raising the dose does not raise the rate. Two drugs that use the same carrier compete. The efflux carrier **P-glycoprotein (P-gp)** pushes some drugs back out of gut cells and brain capillaries. It is one reason why some drugs are poorly absorbed and do not reach the brain.

A1a.2.2 Ionisation and pH: why the pKa matters

Most drugs are weak acids or weak bases. In solution, part of each drug is ionised (charged) and part is un-ionised. **Only the un-ionised part is lipid-soluble and crosses membranes by diffusion.** The share that is ionised depends on two numbers: the pKa of the drug and the pH of the fluid. The link is the Henderson–Hasselbalch equation.

$$\text{For a weak acid: } pH - pKa = \log\left(\text{ionised} \div \text{un-ionised}\right)$$

$$\text{For a weak base: } pKa - pH = \log\left(\text{ionised} \div \text{un-ionised}\right)$$

The pKa is the pH at which a drug is exactly half ionised. Two practical rules follow (Figure A1a.6).

- **Weak acids** are mostly un-ionised in acid fluid (stomach) and mostly ionised in alkaline fluid.

• **Weak bases** are mostly un-ionised in alkaline fluid and mostly ionised in acid fluid.

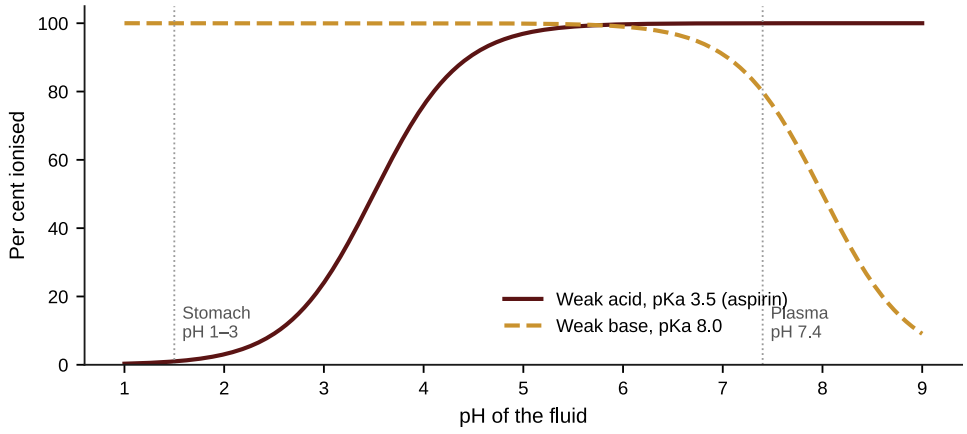


Figure A1a.6. Per cent ionised against pH for a weak acid (pKa 3.5) and a weak base (pKa 8.0), drawn from the Henderson–Hasselbalch equation.

WORKED EXAMPLE: ASPIRIN IN STOMACH AND PLASMA

Aspirin is a weak acid with a pKa of about 3.5.

- In the stomach at pH 1.5: pH − pKa = 1.5 − 3.5 = −2. So ionised ÷ un-ionised = 10^{-2} = 1 ÷ 100. About 99 per cent is **un-ionised**, and so can be absorbed.
- In plasma at pH 7.4: pH − pKa = 7.4 − 3.5 = 3.9. So ionised ÷ un-ionised = $10^{3.9}$, about 8000 ÷ 1. Almost all of it is **ionised**, and so it cannot diffuse back.

This is called **ion trapping**: a drug moves from the side where it is un-ionised to the side where it is ionised and is held there. In practice most aspirin is still absorbed from the small intestine, because its surface area is so much larger than that of the stomach. pH explains the direction; area decides the amount.

Ion trapping is used in treatment. In poisoning with a weak acid such as salicylate or phenobarbitone, making the urine alkaline ionises the drug in the tubule, so less is reabsorbed and more is excreted (A1a.2.6).

A1a.2.3 Absorption

> **Absorption** is the movement of a drug from its site of administration into the circulation.

Absorption is complete and instant after an intravenous injection. For every other route it depends on the factors in the table below. Routes are dealt with in detail in Unit A1(b).

Factor	How it changes absorption
A. Factors of the drug	
Lipid solubility	More lipid-soluble, faster absorbed.
Degree of ionisation	Un-ionised fraction is absorbed (A1a.2.2).
Molecular size	Small molecules are absorbed better; very large ones poorly.
Particle size	Smaller particles dissolve faster. Micronised griseofulvin is better absorbed than the ordinary powder.

Factor	How it changes absorption
Physical form and dosage form	Solution faster than suspension, faster than capsule, faster than tablet. Coating and sustained-release forms slow it on purpose.
Formulation (excipients)	Binders, disintegrants and lubricants change disintegration and dissolution. Two brands of one drug can differ (bioequivalence, A1(c)).
B. Factors of the body	
Area of absorbing surface	Small intestine has the largest area, and is the main site for most oral drugs.
Blood flow at the site	More blood flow, faster absorption. Shock slows absorption from intramuscular sites.
Gastric emptying and gut motility	Fast emptying speeds absorption of most drugs. Diarrhoea shortens the contact time.
Food	Usually delays absorption. A fatty meal increases absorption of some lipid-soluble drugs.
Disease of the gut or liver	Malabsorption reduces it; liver disease reduces first-pass loss.
First-pass metabolism	Drug destroyed in gut wall and liver before reaching the circulation.
Other drugs	Antacids, iron and calcium can bind some drugs in the gut (A1(c)).

First-pass metabolism. After oral absorption, blood from the gut goes by the portal vein to the liver before it reaches the general circulation. Part of the drug is metabolised in the gut wall and the liver on the way (Figure A1a.7). For some drugs the loss is so large that the oral dose must be much higher than the injected dose, or the oral route is useless. Glyceryl trinitrate is the classic example; it is given under the tongue, where the veins drain directly into the systemic circulation. Propranolol, lignocaine and morphine also undergo marked first-pass metabolism.

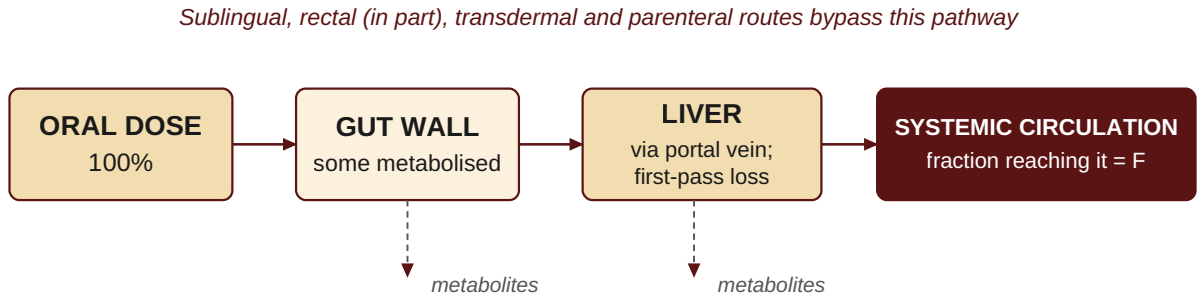


Figure A1a.7. First-pass metabolism. Only the fraction F reaches the systemic circulation.

The fraction of an administered dose that reaches the systemic circulation unchanged is called **bioavailability (F)**. It is 100 per cent for an intravenous dose by definition. Bioavailability and its measurement belong to Unit A1(c).

A1a.2.4 Distribution

> **Distribution** is the movement of a drug from the blood into the tissues and back.

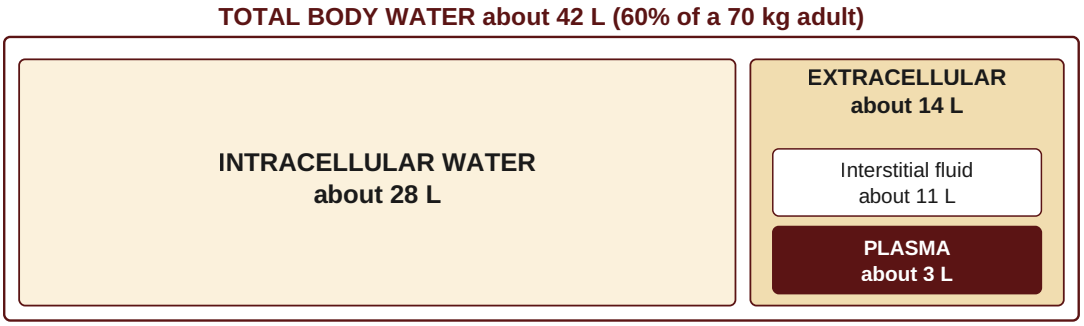
Once in the blood, a drug is carried everywhere. How much enters each tissue depends on blood flow to the tissue, the drug's ability to cross capillary and cell membranes, binding to plasma proteins and to tissues, and special barriers.

Apparent volume of distribution (Vd)

The **apparent volume of distribution** is the volume of fluid that would be needed to hold the whole amount of drug in the body at the same concentration as in plasma.

$$Vd = \text{Amount of drug in the body} \div \text{Plasma concentration}$$

It is an "apparent" volume because it is not a real space in the body. It is a number that tells where the drug has gone. To read it, compare it with the real fluid volumes of a 70 kg adult (Figure A1a.8).



Vd near 3 L: drug held in plasma (often heavily protein-bound).
Vd far above 42 L: drug concentrated in the tissues.

Figure A1a.8. Body water compartments of a 70 kg adult, the yardstick for reading Vd.

Vd (70 kg adult)	What it tells you	Typical reason
About 3 L (plasma volume)	Drug stays in the blood	Large molecule or heavy binding to plasma proteins (heparin; warfarin is mostly bound)
About 14 L (extracellular water)	Drug leaves blood but does not enter cells	Water-soluble, ionised drugs (aminoglycosides)
About 42 L (total body water)	Drug enters cells freely	Small lipid-soluble drugs (ethanol)
Far above 42 L	Drug is concentrated in tissues	Binding in tissues: digoxin in muscle, chloroquine in tissues, lipophilic drugs in fat

WORKED EXAMPLE: VOLUME OF DISTRIBUTION

A patient is given 500 mg of a drug intravenously. The plasma concentration, extrapolated to time zero, is 10 mg per litre.

Vd = 500 mg ÷ 10 mg/L = **50 L**.

50 L is more than the total body water of a 70 kg adult (about 42 L). So the drug is concentrated in the tissues. A large Vd also means that haemodialysis will remove little of the drug in an overdose, because most of it is not in the blood.

Plasma protein binding

Many drugs bind to plasma proteins. **Albumin** binds mainly acidic drugs (warfarin, phenytoin, most NSAIDs, sulphonamides). **Alpha-1-acid glycoprotein** binds mainly basic drugs (lignocaine, propranolol).

- Only the **free (unbound)** drug can cross membranes, act on receptors, be metabolised and be filtered by the kidney. The bound drug is a store.
- Binding is reversible. As free drug is removed, bound drug is released.
- Two drugs that bind to the same site compete. One may displace the other and raise its free level for a time. This is a **displacement interaction** (A1(c)).
- In low albumin states (malnutrition, liver disease, nephrotic syndrome) the free fraction of a highly bound drug rises.

Tissue storage and redistribution

Some drugs collect in particular tissues: lipid-soluble drugs and anaesthetics in **fat**; tetracyclines, fluoride and lead in **bone and teeth**; heavy metals and chloroquine in **liver** and other tissues; iodine in the **thyroid**. Storage prolongs the action and can cause late toxicity. Chronic heavy-metal toxicity (Chapter A4) is largely a story of tissue storage.

Redistribution. Thiopentone given intravenously reaches the brain within seconds, because the brain has a rich blood flow, and the patient sleeps. Within minutes it moves from brain to muscle and fat, and the patient wakes, though most of the drug is still in the body. The short action is due to redistribution, not to elimination.

Barriers

Barrier	Structure	Consequence
Blood–brain barrier (BBB)	Brain capillaries have tight junctions, few pores, and P-glycoprotein pumps	Only lipid-soluble drugs enter the brain easily. Inflammation (meningitis) makes it leakier, so penicillin can then reach the cerebrospinal fluid.
Placental barrier	Layers of trophoblast	Not a real barrier for lipid-soluble drugs. Most drugs reach the foetus. Hence caution in pregnancy.
Blood–testis barrier	Sertoli cell junctions	Limits entry of many drugs into the testis.

A1a.2.5 Metabolism (biotransformation)

> **Metabolism** (biotransformation) is the chemical change of a drug inside the body.

The kidney excretes water-soluble substances well, but it reabsorbs lipid-soluble ones (A1a.2.6). So the main purpose of metabolism is to turn a **lipid-soluble drug into a water-soluble, excretable product**. The **liver** is the main site. The gut wall, kidney, lungs, plasma and skin also metabolise drugs.

Metabolism usually inactivates a drug, but not always:

Result	Example
Active drug → inactive metabolite	Most drugs; phenobarbitone, propranolol
Active drug → active metabolite	Diazepam → desmethyldiazepam; codeine → morphine
Inactive drug (prodrug) → active drug	Levodopa → dopamine; enalapril → enalaprilat; prednisone → prednisolone
Drug → toxic metabolite	Paracetamol → NAPQI, which damages the liver in overdose

There are two phases (Figure A1a.9).

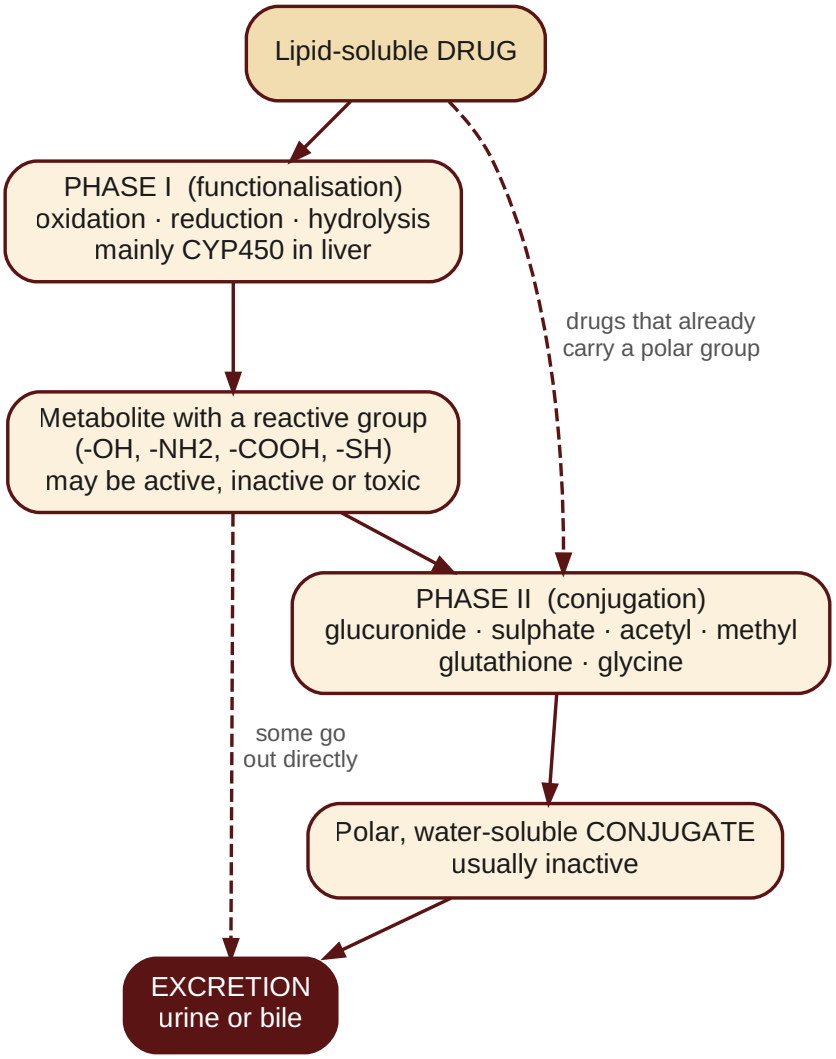


Figure A1a.9. Phase I and Phase II metabolism.

Phase I: functionalisation

Phase I reactions add or uncover a reactive group such as –OH, –NH_2, –COOH or –SH.

Reaction	Main enzymes	Examples
Oxidation	Cytochrome P450 (CYP) mono-oxygenases in liver microsomes; also monoamine oxidase, alcohol dehydrogenase, xanthine oxidase	Phenytoin, warfarin, most drugs; ethanol; catecholamines (MAO)
Reduction	Reductases	Chloramphenicol (nitro-reduction)
Hydrolysis	Esterases, amidases	Procaine, aspirin, succinylcholine (by plasma pseudocholinesterase)

The **cytochrome P450** system is the most important. It is a family of enzymes named by number and letter, such as CYP3A4, CYP2D6, CYP2C9 and CYP1A2. CYP3A4 handles the largest share of drugs in clinical use [R32]. Many drug interactions arise because one drug speeds up or slows down a CYP enzyme that metabolises another drug.

Phase II: conjugation

Phase II reactions join the drug, or its Phase I metabolite, to a large polar molecule supplied by the body. The product, the conjugate, is water-soluble and usually inactive.

Conjugation	Example of a drug conjugated
Glucuronidation (the commonest)	Morphine, paracetamol, chloramphenicol. Newborns have low capacity, hence the “grey baby syndrome” with chloramphenicol.
Acetylation	Isoniazid, hydralazine, sulphonamides. Rate is genetically fast or slow (“fast and slow acetylators”).
Sulphation	Paracetamol (important in children), steroids
Methylation	Catecholamines (by catechol-O-methyltransferase, COMT), histamine
Glutathione conjugation	NAPQI from paracetamol (detoxification)
Glycine conjugation	Salicylic acid → salicyluric acid

Enzyme induction and inhibition

	Enzyme induction	Enzyme inhibition
What happens	More enzyme is made; metabolism speeds up	Enzyme activity is blocked; metabolism slows down
Onset	Slow (days to weeks), needs new protein	Fast (within a dose or two)
Effect on other drugs	Their levels fall; therapy may fail	Their levels rise; toxicity may follow
Examples of the culprit drug	Rifampicin, phenytoin, carbamazepine, phenobarbitone, chronic alcohol use, smoking (CYP1A2), St John’s wort	Ketoconazole, erythromycin, clarithromycin, cimetidine, ritonavir, grapefruit juice (gut CYP3A4)
Classic consequence	Oral contraceptive failure with rifampicin	Bleeding with warfarin plus a CYP2C9 inhibitor

Herbal medicines can induce or inhibit drug-metabolising enzymes as well. Piperine, the pungent principle of Pippali and Maricha, inhibits drug metabolism and raises the blood levels of some co-administered drugs (A1a.6). This is useful when it is intended and dangerous when it is not.

A1a.2.6 Excretion

> **Excretion** is the removal of a drug or its metabolites from the body.

Route	Notes
Kidney (urine)	The main route for most drugs. Three processes: filtration, secretion, reabsorption.
Bile and faeces	Large polar molecules and conjugates. Unabsorbed oral drug also leaves in faeces.
Lungs	Gases and volatile liquids: general anaesthetics, alcohol (the basis of the breath test).
Saliva	Small amounts; some drugs (phenytoin) can be monitored in saliva.
Milk	Lipid-soluble and basic drugs pass into milk and reach the suckling infant.
Sweat, skin, hair, nails	Small amounts; heavy metals such as arsenic are laid down in hair and nails, which is useful in diagnosis (Chapter A4).

Renal excretion

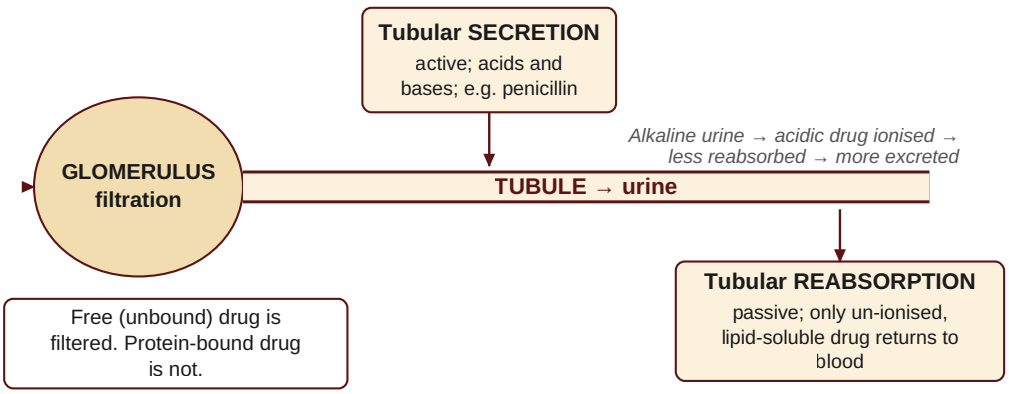


Figure A1a.10. Renal handling of drugs: filtration, active secretion and passive reabsorption.

1. **Glomerular filtration.** Free drug in plasma is filtered at the glomerulus. Protein-bound drug is not filtered. The rate depends on the glomerular filtration rate and on the free fraction.
2. **Active tubular secretion.** Carriers in the proximal tubule secrete acids (penicillin, probenecid) and bases. Secretion is fast and saturable, and drugs compete for it. Probenecid slows the secretion of penicillin and so prolongs its action.
3. **Passive tubular reabsorption.** As water is reabsorbed, drug in the tubule becomes concentrated. Lipid-soluble, un-ionised drug diffuses back into blood. Ionised drug stays in the urine and is excreted. This is where urine pH matters (Figure A1a.10).

Enterohepatic circulation. Some drugs are excreted in bile as conjugates, broken down by gut bacteria, and reabsorbed. This cycle prolongs their action. Oral contraceptive steroids are a known example; antibiotics that kill gut bacteria can reduce their reabsorption.

A1a.2.7 Kinetics of elimination

Elimination means metabolism plus excretion. Two patterns are seen (Figure A1a.11).

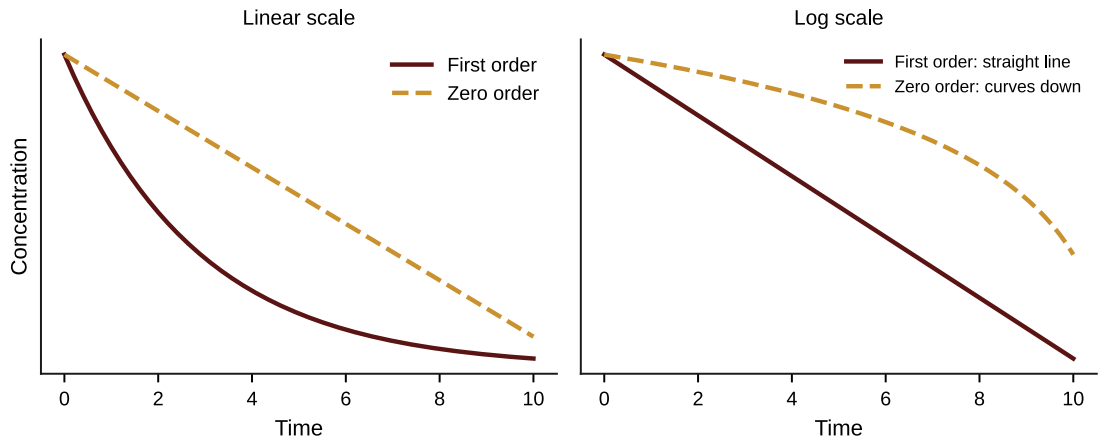


Figure A1a.11. First-order and zero-order elimination, on a linear scale and on a log scale (schematic).

	First-order kinetics	Zero-order kinetics
What is constant	A constant **fraction** is removed per unit time	A constant **amount** is removed per unit time
Cause	Enzymes and carriers are far from saturated	Enzymes or carriers are saturated
Half-life	Constant, whatever the dose	Not constant; rises with the dose
Plot of log concentration against time	Straight line	Curved
Effect of raising the dose	Level rises in proportion	Level rises out of proportion; toxicity comes suddenly
Examples	Most drugs at usual doses	Ethanol; phenytoin, aspirin and theophylline at high doses (also called saturation or Michaelis–Menten kinetics)

The four key parameters

Clearance (CL) = Rate of elimination ÷ Plasma concentration (litres per hour)

Elimination half-life (t½) = 0.693 × Vd ÷ CL

Elimination rate constant (k) = CL ÷ Vd = 0.693 ÷ t½

- **Clearance** is the volume of plasma cleared of the drug per unit time. Total clearance is the sum of renal, hepatic and other clearances.
- **Half-life** is the time taken for the plasma concentration to fall to half. It depends on both Vd and CL. A large Vd or a small CL gives a long half-life.

In first-order kinetics, each half-life removes half of what is left.

Half-lives passed	1	2	3	4	5
Drug eliminated (%)	50	75	87.5	93.75	96.9
Drug remaining (%)	50	25	12.5	6.25	3.1

So a drug is taken to be practically eliminated after **four to five half-lives**. The same rule tells how long it takes to reach steady state.

A1a.2.8 The plasma concentration–time curve

After a single oral dose the plasma concentration first rises, while absorption is faster than elimination, and then falls (Figure A1a.12). Every pharmacokinetic parameter can be read from this curve, and it is worth drawing in every long answer.

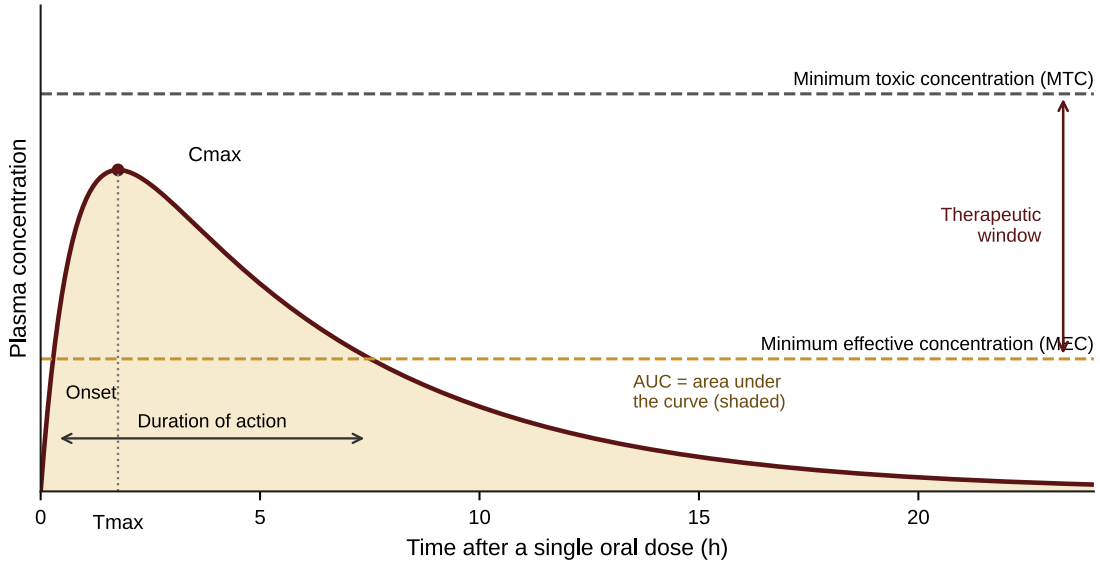


Figure A1a.12. Plasma concentration–time curve after a single oral dose (schematic).

Term	Meaning	What it tells you
Cmax	Peak plasma concentration	Intensity of effect; risk of toxicity
Tmax	Time to reach Cmax	Speed of absorption
AUC (area under the curve)	Total area under the curve	Total amount of drug that reached the circulation; used to calculate bioavailability
Minimum effective concentration (MEC)	Level below which there is no effect	Onset and end of action
Minimum toxic concentration (MTC)	Level above which toxicity appears	Upper limit of dosing
Therapeutic window	Range between MEC and MTC	Narrow window means careful dosing and monitoring
Onset, duration of action	Time when the curve crosses MEC upwards and downwards	When the effect begins and how long it lasts

A1a.2.9 Repeated dosing and steady state

Most drugs are given repeatedly. If each dose is given before the last one is fully eliminated, the drug accumulates. It keeps accumulating until the amount eliminated between doses equals the amount given. This balance is called **steady state** (Figure A1a.13).

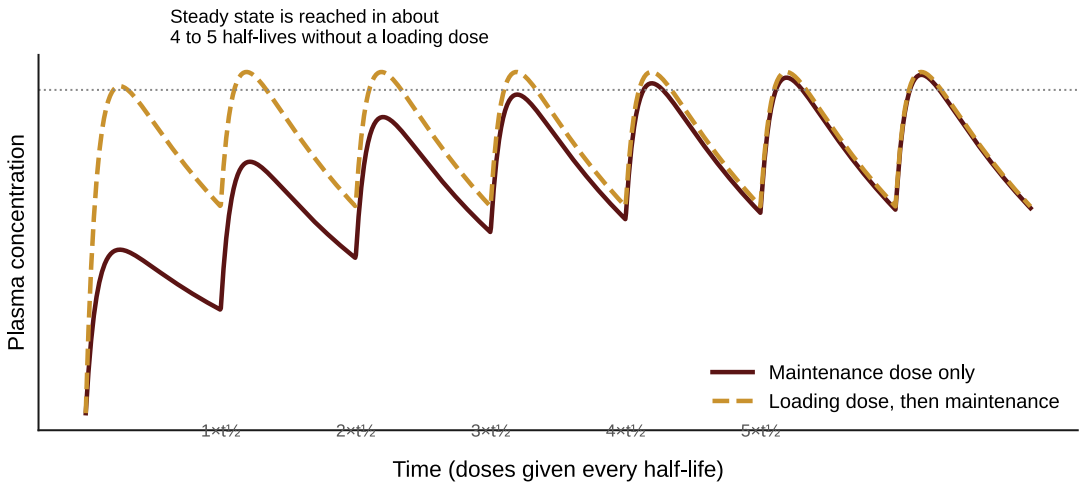


Figure A1a.13. Repeated dosing at intervals of one half-life, with and without a loading dose (schematic).

- Steady state is reached in about **four to five half-lives**, whatever the dose.
- A drug with a long half-life takes a long time to reach steady state. To get an effect quickly, a larger first dose, the **loading dose**, is given.
- Doubling the dose doubles the steady-state level (in first-order kinetics) but does not change the time to reach it.

Loading dose = Target plasma concentration × Vd ÷ F

Maintenance dose rate = Target plasma concentration × CL ÷ F

WORKED EXAMPLE: LOADING AND MAINTENANCE DOSE

A drug has Vd = 50 L, CL = 5 L per hour and oral bioavailability F = 1. The target steady-state concentration is 2 mg per litre.

- Half-life = 0.693 × 50 ÷ 5 = **6.9 hours**. Without a loading dose, steady state comes in about 28 to 35 hours.
- Loading dose = 2 × 50 ÷ 1 = **100 mg**.
- Maintenance dose rate = 2 × 5 ÷ 1 = 10 mg per hour, that is **80 mg every 8 hours**.

If the same drug had F = 0.5 by mouth, both oral doses would double.

Compartment models

To describe these curves mathematically, the body is imagined as one or more boxes, called compartments.

- In a **one-compartment model**, the drug spreads instantly and evenly through one space. The log concentration falls in one straight line.
- In a **two-compartment model**, the drug first fills a central compartment (blood and well-perfused organs) and then moves more slowly into a peripheral one (less perfused tissue). The log concentration falls in two phases: a fast distribution phase (alpha) and a slower elimination phase (beta).

These models are mathematical conveniences. They do not correspond to real anatomical spaces.

A1a.3 Pharmacodynamics and mechanism of action

> **Pharmacodynamics** is the study of the biochemical and physiological effects of drugs and of their mechanisms of action. In short, it is what the drug does to the body.

Pharmacodynamics answers three questions: **what** effect the drug produces, **how** it produces it (mechanism), and **how much** effect a given dose produces (dose–response, A1a.4).

A1a.3.1 Types of drug action

A drug can change the function of a cell in five basic ways.

Type of action	Meaning	Example
Stimulation	Increases the activity of a specialised cell	Adrenaline on the heart; pilocarpine on salivary glands
Depression	Decreases the activity of a specialised cell	Barbiturates on the brain; quinidine on the heart
Irritation	Non-selective, often harmful action on less specialised cells; mild irritation can stimulate function	Counter-irritants on the skin; bitters increase salivation and gastric juice
Replacement	Supplies a substance the body lacks	Insulin in diabetes; iron in iron-deficiency anaemia; levothyroxine
Cytotoxic action	Selective killing of microbes, parasites or cancer cells	Penicillin; chloroquine; anticancer drugs

A1a.3.2 Mechanisms of drug action

The syllabus asks for “mechanism of action”. The answer is that drugs act in a small number of ways (Figure A1a.14). Most act on one of four kinds of protein target: **receptors, enzymes, ion channels and transporters**. A few act by simple physical or chemical means.

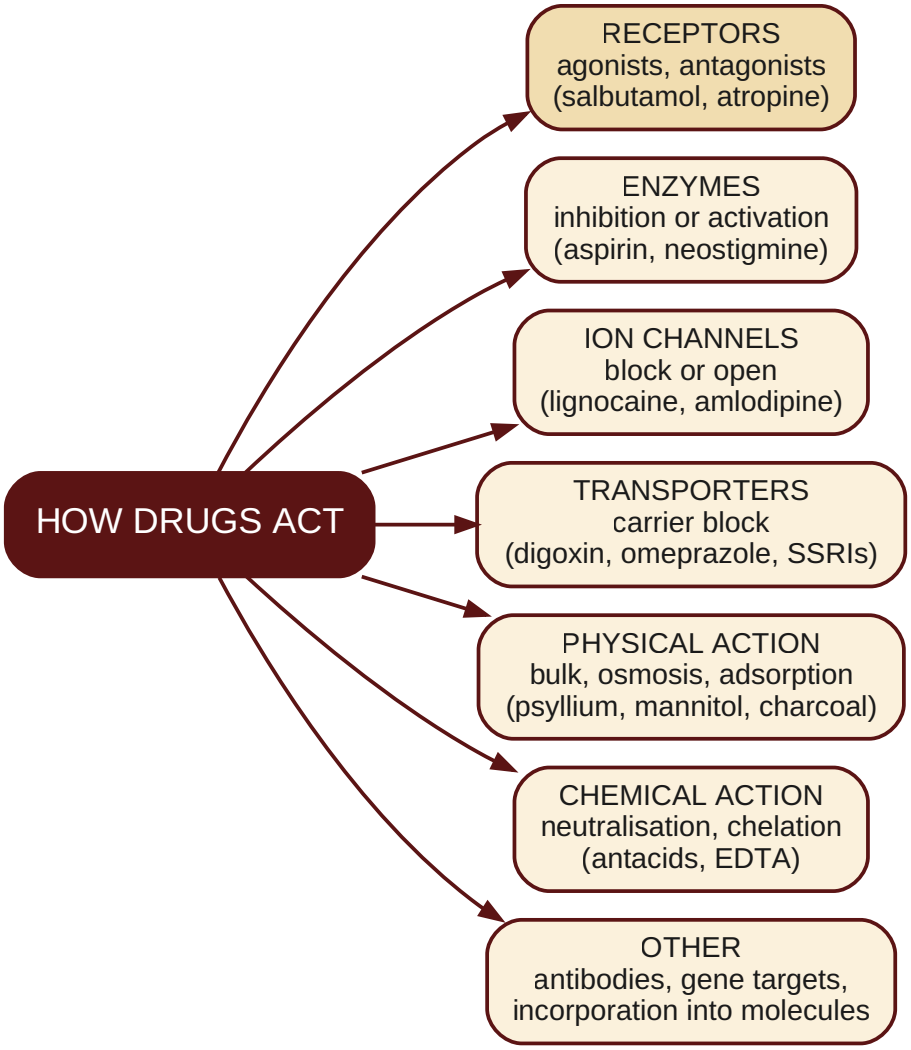


Figure A1a.14. How drugs act: the main mechanisms, with examples.

Mechanism	How it works	Examples
Physical action	Bulk, osmotic pressure, adsorption, demulcent or protective coating	Isabgol (psyllium) as a bulk laxative; mannitol as an osmotic diuretic; activated charcoal adsorbing poisons
Chemical action	Simple chemical reaction	Antacids neutralise gastric acid; chelating agents (EDTA, dimercaprol) bind heavy metals (Chapter A4)
Enzymes	Inhibit or activate an enzyme	Aspirin inhibits cyclo-oxygenase (COX); neostigmine inhibits cholinesterase; allopurinol inhibits xanthine oxidase
Ion channels	Block or open a channel	Lignocaine blocks sodium channels; amlodipine blocks calcium channels
Transporters	Block a carrier or pump	Digoxin inhibits the Na^+/K^+-ATPase pump; omeprazole inhibits the gastric proton pump; reserpine blocks the vesicular monoamine transporter (VMAT); SSRIs block serotonin reuptake

Mechanism	How it works	Examples
Receptors	Bind to a specific receptor and either activate it (agonist) or block it (antagonist)	Salbutamol on β_2 receptors; atropine on muscarinic receptors; morphine on opioid receptors
Other	Antibodies, incorporation into macromolecules, action on genes	Monoclonal antibodies; anticancer antimetabolites built into DNA

Enzyme inhibition in more detail

Type	What happens	Effect of raising substrate	Example
Competitive (reversible)	Drug and substrate compete for the same active site	Overcomes the inhibition	Physostigmine and neostigmine on cholinesterase
Non-competitive	Drug binds elsewhere and changes the enzyme	Does not overcome it	Acetazolamide on carbonic anhydrase
Irreversible	Drug binds covalently; new enzyme must be made	Does not overcome it	Aspirin acetylates COX in platelets; organophosphates on cholinesterase

A1a.3.3 Receptors

> A **receptor** is a specific macromolecule, usually a protein, to which a drug binds and through which it brings about its effect. The body's own messengers (hormones, neurotransmitters) act on the same receptors.

Two properties of a drug decide what happens at a receptor:

- **Affinity** — how strongly the drug binds to the receptor.
- **Intrinsic activity (IA)**, also called efficacy — how well the bound drug activates the receptor.

From these two properties come four kinds of ligand (Figure A1a.15).

Ligand	Affinity	Intrinsic activity	What it does	Examples
Full agonist	Yes	1 (maximum)	Activates the receptor fully	Adrenaline, morphine, salbutamol
Partial agonist	Yes	Between 0 and 1	Activates partly; in the presence of a full agonist it can reduce the full agonist's effect	Buprenorphine and pentazocine (opioid); pindolol (β)
Antagonist	Yes	0	Occupies the receptor without activating it; blocks the agonist	Atropine, propranolol, naloxone
Inverse agonist	Yes	Below 0	Reduces the basal (constitutive) activity of a receptor	Beta-carbolines at the benzodiazepine site

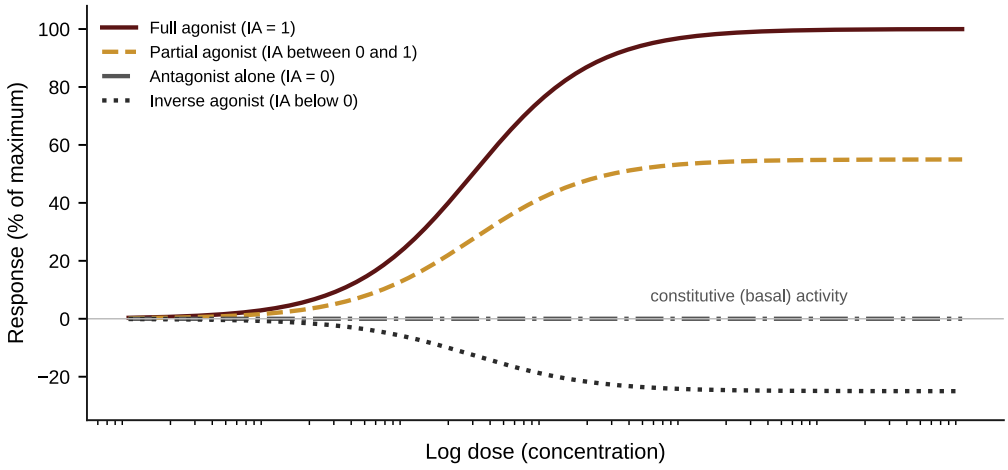


Figure A1a.15. Log dose–response curves of a full agonist, partial agonist, antagonist and inverse agonist (schematic).

Theories of drug–receptor interaction

Theory	Main idea
Occupation theory (Clark)	Response is proportional to the number of receptors occupied. Maximum response when all are occupied.
Modified occupation theory (Ariens, Stephenson)	Adds intrinsic activity (efficacy). Explains partial agonists. Introduces **spare receptors**: a full response may need only a fraction of receptors to be occupied.
Rate theory (Paton)	Response depends on the rate of association and dissociation, not on occupancy.
Two-state (conformational) model	A receptor exists in active and inactive forms in equilibrium. Agonists favour the active form, inverse agonists the inactive form, and antagonists bind both equally. This model best explains all four kinds of ligand.
Induced-fit model	Binding of the drug changes the shape of the receptor, which triggers the response.

The four families of receptors

All receptors known belong to four families (Figure A1a.16). They differ in structure, in how they pass on the signal, and in speed.

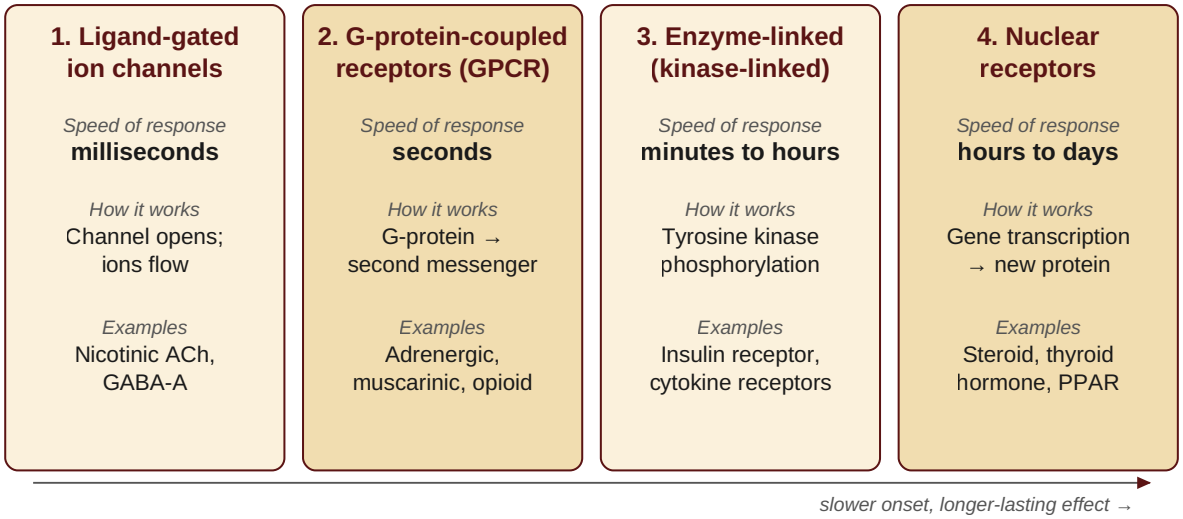


Figure A1a.16. The four receptor families, from the fastest (milliseconds) to the slowest (hours to days).

Family	Structure and signalling	Time scale	Examples
1. Ligand-gated ion channels (ionotropic)	Receptor is itself a channel; binding opens it and ions flow	Milliseconds	Nicotinic ACh receptor; GABA-A receptor (benzodiazepines act here); 5-HT_3 receptor
2. G-protein-coupled receptors (GPCR, metabotropic)	Seven membrane-spanning helices; activate a G-protein, which changes a second messenger	Seconds	Adrenergic (α, β), muscarinic, opioid, histamine and dopamine receptors
3. Enzyme-linked (kinase-linked) receptors	One membrane-spanning helix; an enzyme part inside the cell, usually a tyrosine kinase	Minutes to hours	Insulin receptor; growth factor receptors; cytokine receptors (JAK–STAT)
4. Nuclear (intracellular) receptors	In cytoplasm or nucleus; drug–receptor complex binds DNA and changes gene transcription	Hours to days	Corticosteroids, thyroid hormone, sex steroids, vitamin D, PPAR (pioglitazone)

G-proteins and second messengers

GPCRs are the largest family and the target of very many drugs. The signal passes through a G-protein to a second messenger inside the cell (Figure A1a.17).

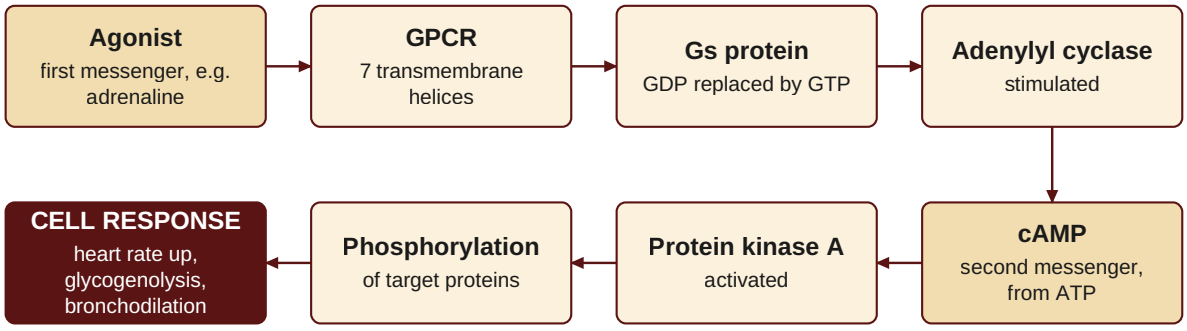


Figure A1a.17. The Gs–cAMP signalling cascade, as triggered by adrenaline at a β receptor.

G-protein	Effector	Second messenger	Typical receptors
Gs	Stimulates adenylyl cyclase	cAMP rises	β_1, β_2, H_2, D_1
Gi	Inhibits adenylyl cyclase; opens K^+ channels	cAMP falls	α_2, M_2, opioid, D_2
Gq	Stimulates phospholipase C	IP_3 (releases Ca^{2+}) and DAG (activates protein kinase C)	α_1, M_1, M_3, H_1

Regulation of receptors

Receptors are not fixed in number or sensitivity.

Change	Cause	Clinical consequence
Down-regulation and desensitisation	Continued exposure to an agonist	Tolerance: repeated β_2-agonist use in asthma gives less effect over time
Up-regulation (supersensitivity)	Continued exposure to an antagonist, or loss of the natural agonist	Rebound on sudden withdrawal: propranolol stopped abruptly can worsen angina
Tachyphylaxis	Rapid loss of response with repeated doses, within minutes or hours	Ephedrine and tyramine (they deplete the noradrenaline store they act through); nitrates

A1a.3.4 Selectivity and specificity

A drug is **selective** if it acts on one target much more than on others at the usual dose. No drug is perfectly selective. Salbutamol is selective for β_2 receptors at inhaled doses, but at high doses it also stimulates β_1 receptors in the heart and causes palpitations. Selectivity is therefore a matter of dose. This is why adverse effects increase as the dose rises.

A1a.4 Dose determination and dose–response

A1a.4.1 Words used for dose

Term	Meaning
Dose	The amount of drug given at one time
Dosage	The full scheme: dose, frequency and duration (for example 500 mg three times a day for five days)
Loading dose	A larger first dose to reach the target level quickly (A1a.2.9)
Maintenance dose	The dose that keeps the level steady
Minimal effective dose	The smallest dose that produces a detectable effect
Maximal tolerated dose	The largest dose that can be given without unacceptable toxicity
Therapeutic dose	The range of doses that gives the desired effect in most patients
ED50 (median effective dose)	Dose that produces the desired effect in 50 per cent of subjects
TD50 (median toxic dose)	Dose that produces a defined toxic effect in 50 per cent of subjects
LD50 (median lethal dose)	Dose that kills 50 per cent of test animals
NOAEL	No Observed Adverse Effect Level: the highest dose in an animal study that causes no significant adverse effect

A1a.4.2 The dose–response relationship

> The **dose–response relationship** describes how the size of a drug's effect changes with its dose or concentration.

There are two kinds of dose–response relationship. In a **graded** relationship, the response in one subject or tissue grows steadily with the dose (blood pressure, heart rate, contraction of a gut strip). In a **quantal**

relationship, the response is all-or-none (sleep or no sleep, death or survival), and what grows with the dose is the *percentage of subjects* who respond.

Graded dose–response curve

If the response is plotted against the dose on an ordinary scale, the curve is a **rectangular hyperbola**: steep at first, then flattening to a maximum. If the dose is plotted on a **log scale**, the same data give an **S-shaped (sigmoid) curve** (Figure A1a.18). The log scale is preferred because:

- a wide range of doses fits on one graph;
- the middle part of the curve (from about 20 to 80 per cent of the maximum) is nearly straight, which makes comparison easy;
- curves of different drugs can be compared by their position and height.

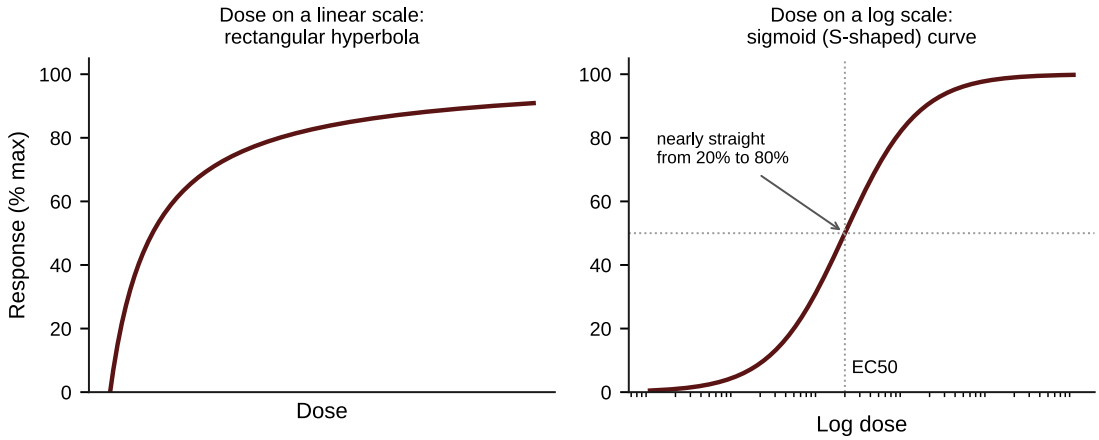


Figure A1a.18. Graded dose–response curve on a linear and on a log dose scale (schematic).

The shape is described by the **Hill equation**:

$$E = E_{max} \times C^n \div (EC50^n + C^n)$$

where E is the effect, Emax the maximum effect, C the concentration, EC50 the concentration that gives half the maximum effect, and n the Hill coefficient, which sets the steepness of the curve.

Potency and efficacy

These two words are often confused in examinations. They are different properties (Figure A1a.19).

	Potency	Efficacy
Meaning	The amount of drug needed to produce a given effect	The maximum effect a drug can produce, however large the dose
Read from the curve	Position along the dose axis (EC50). Further left = more potent	Height of the plateau (Emax)
Clinical importance	Usually small. A less potent drug is simply given in a larger dose	Large. A drug of low efficacy cannot do the work of a drug of high efficacy
Example	Bumetanide is more potent than furosemide (smaller dose), with similar maximum diuresis	Furosemide (a high-ceiling diuretic) has greater efficacy than a thiazide; morphine has greater analgesic efficacy than aspirin

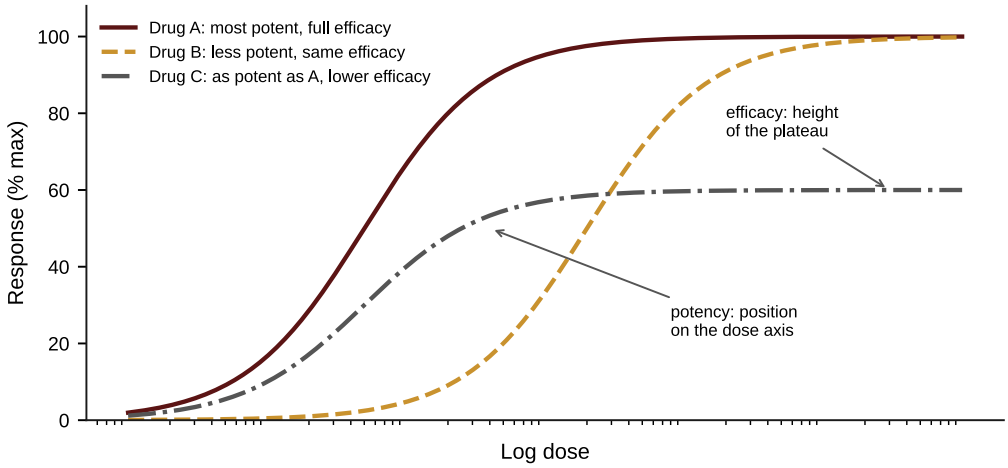


Figure A1a.19. Potency and efficacy compared (schematic).

Antagonism and the dose–response curve

An antagonist changes the agonist's curve in a way that reveals how it acts (Figure A1a.20).

	Competitive antagonist	Non-competitive antagonist
Binding	Same site as the agonist, reversible	Different site, or irreversible binding
Curve of the agonist	Shifted to the right, parallel	Flattened: the maximum falls
Can a high agonist dose overcome it?	Yes (surmountable)	No (not surmountable)
Example	Atropine against acetylcholine; naloxone against morphine	Phenoxybenzamine against noradrenaline at α receptors

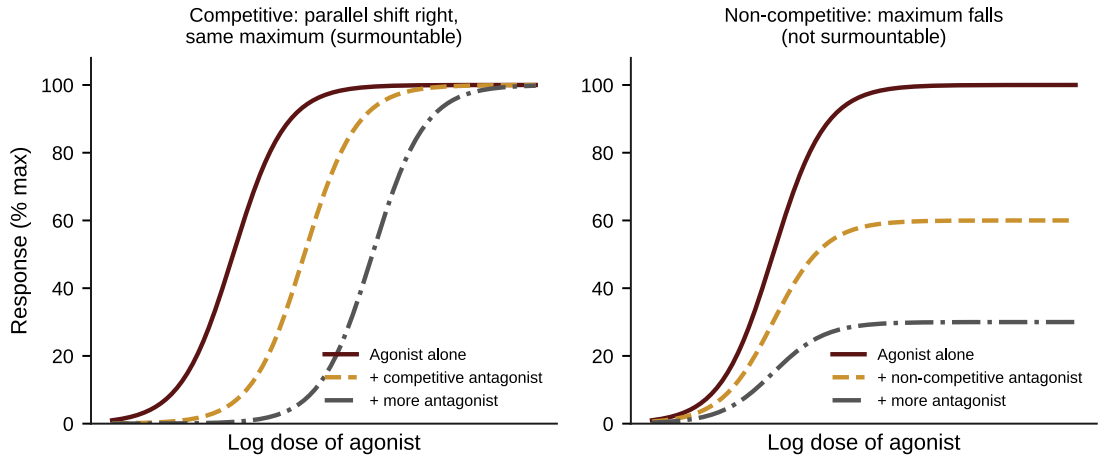


Figure A1a.20. Competitive and non-competitive antagonism, as seen on the agonist's log dose–response curve (schematic).

Quantal dose–response and the therapeutic index

When the percentage of subjects who show an all-or-none response is plotted against log dose, the result is again a sigmoid curve. Separate curves can be drawn for the therapeutic effect, for a toxic effect and, in animals, for death (Figure A1a.21). The distance between these curves is a measure of safety.

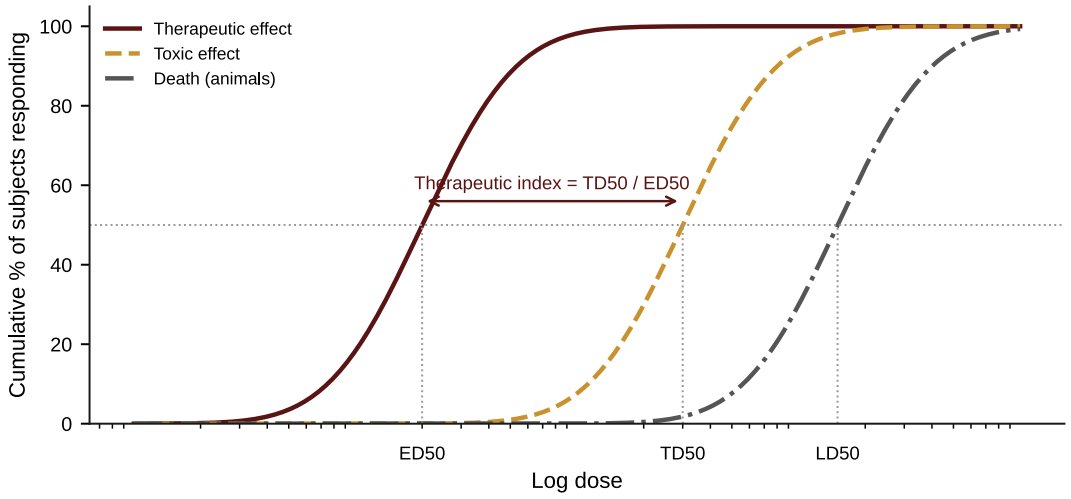


Figure A1a.21. Quantal dose–response curves for therapeutic effect, toxic effect and death, with the therapeutic index (schematic).

Therapeutic index (TI) = TD50 ÷ ED50 (in animals, often LD50 ÷ ED50)

- A **high** therapeutic index means a wide margin between the effective and the toxic dose: the drug is relatively safe.
- A **low** (narrow) therapeutic index means that a small rise in dose or level can cause toxicity. Such drugs need careful dosing and often monitoring of blood levels. Examples: digoxin, lithium, warfarin, phenytoin, aminoglycosides, theophylline.
- The TI compares *median* doses. It does not show whether the two curves overlap at their ends. A more cautious measure, the **certain safety factor**, compares the dose toxic to 1 per cent of subjects with the dose effective in 99 per cent (TD1 ÷ ED99).

The **therapeutic window** (A1a.2.8) is the same idea expressed in plasma concentrations: the range between the minimum effective and the minimum toxic concentration.

A1a.4.3 How a dose is determined

A dose is not guessed. It is worked out in stages, from the laboratory to the individual patient.

Stage 1: from animal data to the first human dose

Before a new drug is given to a human being for the first time, the starting dose is calculated from animal toxicity data. The United States Food and Drug Administration guidance of July 2005 sets out the most widely used method (Figure A1a.22) [FDA 2005].

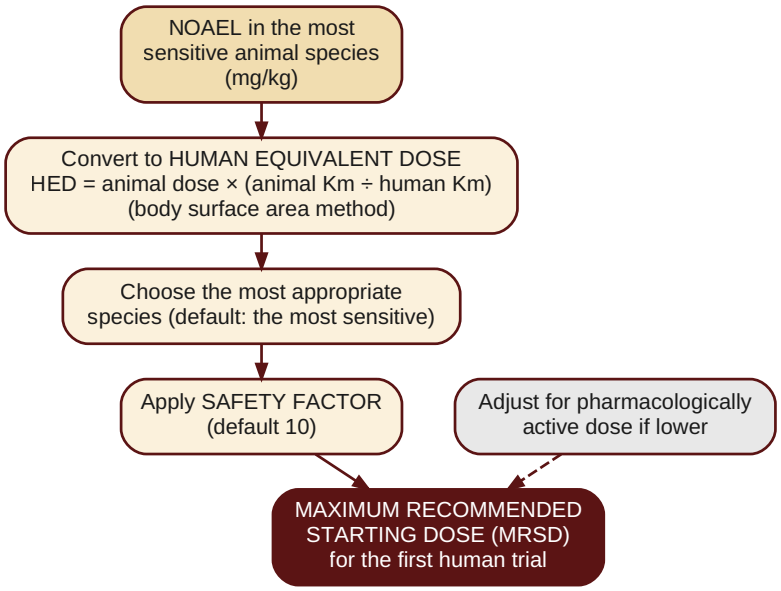


Figure A1a.22. Estimating the maximum recommended starting dose for a first-in-human trial, following the FDA guidance of 2005.

1. Find the **NOAEL** in each animal species studied.
2. Convert each NOAEL to a **human equivalent dose (HED)**. A dose in mg/kg does not scale directly between species, because smaller animals have a faster metabolism per kilogram. The guidance scales by **body surface area** using a factor called Km (body weight ÷ body surface area).
3. Choose the most appropriate species. By default this is the most sensitive one, the species with the lowest HED.
4. Divide by a **safety factor**. The default is 10.
5. The result is the **maximum recommended starting dose (MRSD)**. It may be lowered further if the dose that is merely pharmacologically active is lower.

$$\text{HED (mg/kg)} = \text{Animal dose (mg/kg)} \times (\text{Animal Km} \div \text{Human Km})$$

Species	Reference body weight (kg)	Km	Multiply animal dose (mg/kg) by
Human (adult)	60	37	—
Mouse	0.02	3	0.081
Rat	0.15	6	0.162
Guinea pig	0.4	8	0.216
Rabbit	1.8	12	0.324
Monkey	3	12	0.324
Dog	10	20	0.541

Selected rows from Table 3 of the FDA guidance [FDA 2005]. The guidance lists more species.

WORKED EXAMPLE: RAT TO HUMAN

In a 90-day study in rats, the NOAEL of a new extract is 300 mg/kg per day.

- HED = 300 × 0.162 = **48.6 mg/kg**. For a 60 kg adult: 48.6 × 60 ≈ 2.9 g per day.
- MRSD with the default safety factor of 10 = 4.86 mg/kg, or about **290 mg per day** for a 60 kg adult.

The reverse calculation is common in Ayurvedic research. A human dose is converted to an animal dose for an experiment by dividing by the same factor, that is, multiplying the human mg/kg dose by Human Km ÷ Animal Km (37 ÷ 6, about 6.2, for the rat) [Reagan-Shaw 2008]. A careful study states which conversion it used.

Stage 2: dose finding in clinical trials

The first human dose is only a start. Phase I trials raise the dose step by step to find the maximum tolerated dose. Phase II trials test several doses to find the dose that gives the best balance of benefit and harm. These are described in Chapter A2.

Stage 3: fitting the dose to the patient

Factor	Adjustment
Body weight	Many doses are given in mg per kg of body weight
Age: children	By weight or body surface area (preferred). Older formulae are still asked in examinations: **Young's formula**: child's dose = age (years) ÷ (age + 12) × adult dose. **Clark's formula**: child's dose = weight (lb) ÷ 150 × adult dose. **Dilling's formula**: child's dose = age (years) ÷ 20 × adult dose. **Body surface area**: child's dose = BSA (m^2) ÷ 1.73 × adult dose
Age: elderly	Lower doses; kidney and liver function decline; more sensitive to many drugs
Kidney function	Dose of renally excreted drugs reduced in proportion to creatinine clearance
Liver function	Dose of drugs with high hepatic metabolism reduced
Genetics	Slow acetylators, poor CYP2D6 metabolisers and others need different doses (A1(c))
Therapeutic drug monitoring	For narrow-TI drugs, the plasma level is measured and the dose adjusted

AYURVEDA CORRELATION: MATRA (DOSE)

Ayurveda places great weight on the right dose. Charaka states that the dose (Matra) must be fixed according to time, digestive power, age, strength, constitution, the Dosha involved and the place (Charaka Samhita, Kalpa Sthana 12/86) [R26]. These factors correspond closely to the modern list in the table above: age, body size and strength, the state of digestion and metabolism, the disease, and the environment.

Charaka also arranges the five basic forms of Kashaya in order of strength: Svarasa (fresh juice) is the strongest, then Kalka (paste), Shrita (decoction), Shita (cold infusion) and Phanta (hot infusion) (Sutra Sthana 4/7) [R26]. Today we would say that the dose of active matter per unit volume falls from the first to the last.

Classical doses are given in units such as Gunja (Ratti), Masha and Tola. The equivalents used by the Ayurvedic Formulary of India, and in the companion volume of this series, are: **1 Gunja = 125 mg; 8 Gunja = 1 Masha = 1 g; 12 Masha = 1 Tola = 12 g** [R37].

Established. The classical texts recognise that dose must be individualised and that the strength of a preparation depends on its form. This is a sound principle and agrees with modern pharmacology.

Not established. That a classical dose, expressed in Gunja or Masha, is the optimal dose in the modern sense. Very few classical doses have been tested by formal dose-ranging studies. Where a dose-finding trial exists for an Ayurvedic preparation, it should be cited; where none exists, the classical dose is a tradition, not a measured optimum.

"The dose makes the poison"

The sixteenth-century physician Paracelsus taught that all things are poison, and only the dose makes a thing not a poison. Charaka said the same earlier and from the other side: a sharp poison becomes an excellent medicine by right use, and a medicine wrongly used becomes a sharp poison (Charaka Samhita, Sutra Sthana 1) [R26]. Both statements are the dose–response principle. Rasashastra lives by it. Mercury, arsenic sulphides and Vatsanabha (aconite) are toxic substances, which the tradition processes and gives in small doses for a defined purpose. Whether a given processed product is in fact safe at its classical dose is a question for measurement, not for tradition alone (Chapters A4 and A5).

A1a.5 Structure–activity relationship

Structure–activity relationship (SAR) is the relationship between the chemical structure of a molecule and its biological activity. It asks: which part of the molecule produces the effect, and what happens to the effect when that part is changed?

A drug fits its target the way a key fits a lock. Change the shape of the key, and it may turn the lock better, worse or not at all. SAR is the systematic study of these changes. It is the main tool of medicinal chemistry for making a lead molecule more potent, more selective, longer-acting, better absorbed or less toxic.

A1a.5.1 Basic ideas

Idea	Meaning
Pharmacophore	The minimum set of structural features (shape, charged groups, hydrogen-bond donors and acceptors) a molecule needs to act on a target
Functional groups	–OH, $–NH_2$, –COOH, ester, ether and others; each changes binding, solubility and metabolism
Lipophilicity	How well the molecule dissolves in fat. Measured as the partition coefficient (log P). Governs absorption, entry into the brain and storage in fat
Isosteres and bioisosteres	Atoms or groups of similar size and electronic character that can replace each other with similar activity, for example –S– for –O–
Stereochemistry	Mirror-image forms (enantiomers) of a molecule can differ greatly in activity, because the target itself is three-dimensional
QSAR	Quantitative SAR: a mathematical equation linking activity to measured properties of the molecule

A1a.5.2 Classic examples

Catecholamines: one change at the nitrogen

Adrenaline, isoprenaline and salbutamol share the same backbone (Figure A1a.23). Two changes decide how they act.

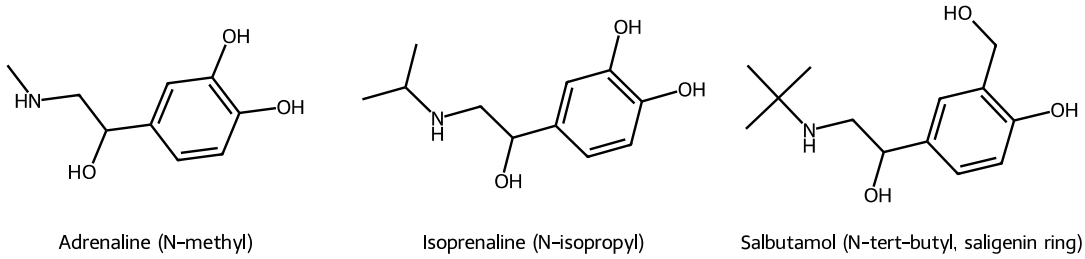


Figure A1a.23. Adrenaline, isoprenaline and salbutamol: how the group on the nitrogen and the ring decide receptor selectivity.

Change	Effect	Result
Larger group on the nitrogen: H (noradrenaline) → methyl (adrenaline) → isopropyl (isoprenaline)	α activity falls, β activity rises	Isoprenaline is an almost pure β agonist
tert-Butyl on the nitrogen and the ring 3-OH replaced by $–CH_2OH$ (salbutamol)	Selective for β_2 receptors; no longer a substrate for COMT	Bronchodilator with fewer cardiac effects, longer action, active by mouth

Salicylic acid and aspirin: one acetyl group

Acetylation of the phenolic –OH of salicylic acid gives aspirin, acetylsalicylic acid (Figure A1a.24). Aspirin was introduced as a better-tolerated form of salicylate. The acetyl group also gives aspirin a property salicylic acid lacks: it transfers the acetyl group to cyclo-oxygenase and inhibits it **irreversibly**. In platelets, which cannot make new enzyme, the effect lasts for the life of the platelet. This is why a small daily dose of aspirin prevents clotting.

Morphine and codeine: one methyl group

Codeine is morphine with a methyl group on the phenolic –OH at position 3 (Figure A1a.24). This single change lowers its affinity for the opioid receptor many times. Much of codeine's analgesic effect comes from its conversion back to morphine by the liver enzyme CYP2D6. Patients who lack this enzyme get little pain relief from codeine, while ultra-rapid metabolisers can form dangerous amounts of morphine [R32]. Adding acetyl groups at both positions 3 and 6 gives diamorphine (heroin), which is more lipid-soluble and enters the brain faster.

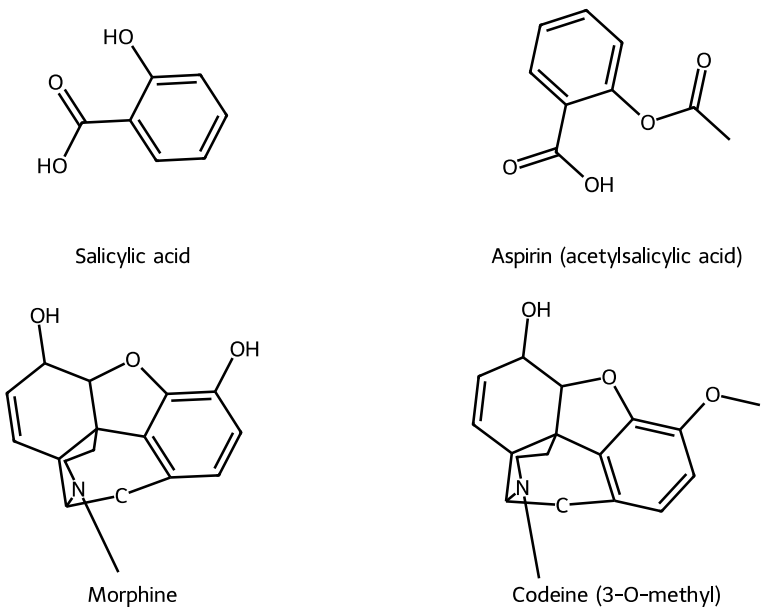


Figure A1a.24. Structure and activity: salicylic acid and aspirin; morphine and codeine.

Barbiturates: changes on one ring

Change	Effect
Phenyl group at carbon 5 (phenobarbitone)	Anticonvulsant activity
Sulphur in place of oxygen at carbon 2 (thiopentone)	Much more lipid-soluble; enters the brain within seconds; ultra-short action (A1a.2.4)

Stereoisomers

Drug	Observation
Warfarin	The S-form is several times more potent than the R-form
Salbutamol	The R-form (levosalbutamol) carries the bronchodilator activity
Omeprazole	The S-form is marketed separately as esomeprazole
Thalidomide	One form was linked to its teratogenic effect, but the two forms interconvert in the body, so giving a single form would not have prevented the harm

A1a.5.3 Rules for orally active drugs

In 1997 Lipinski and colleagues examined compounds that had reached clinical trials and proposed a simple screen, now called the **rule of five** [Lipinski 2001]. Poor absorption or permeation is more likely when a compound has:

- a molecular weight above 500;
- a calculated log P above 5;
- more than 5 hydrogen-bond donors;
- more than 10 hydrogen-bond acceptors.

(All the numbers are multiples of five, hence the name.) The rule is a screen, not a law. Many natural products and some important drugs break it and are still used, often because a carrier transports them.

A1a.5.4 Quantitative SAR and computer-aided design

QSAR expresses activity as an equation. The best known is the Hansch approach, which relates biological activity to a molecule's lipophilicity (log P), its electronic character and the size of its substituents. Once such an equation is fitted to a series of compounds, the activity of a new, unmade compound can be predicted.

Today SAR is supported by **computer-aided drug design**: the three-dimensional structure of the target is known, and candidate molecules are "docked" into it on a computer to predict how well they bind. Docking studies of plant compounds are now common in Ayurvedic research. A docking score is a prediction about binding to an isolated protein. It is not evidence that the compound reaches the target in the body, at that concentration, or that the patient benefits.

A1a.5.5 Natural products as starting points

Many drugs began as natural molecules that were then improved by SAR.

Natural lead	Source	Improved derivatives and why
Artemisinin	*Artemisia annua*	Artesunate (water-soluble, can be injected) and artemether (oil-soluble). Tu Youyou received the Nobel Prize in Physiology or Medicine in 2015 for its discovery
Morphine	Opium poppy	Codeine, many synthetic opioids
Salicin	Willow bark	Salicylic acid, then aspirin
Reserpine	*Rauvolfia serpentina* (Sarpagandha)	Isolated in 1952; it lowers blood pressure by depleting stored monoamines (A1a.6)

A CAUTION: CURCUMIN

Curcumin, the yellow pigment of Haridra, is one of the most studied natural compounds. In 2017, a review in the *Journal of Medicinal Chemistry* examined its chemistry and concluded that curcumin is an unstable, reactive and poorly bioavailable molecule. It is flagged as a frequent false positive in laboratory screens, and despite well over a hundred clinical trials, no double-blind, placebo-controlled trial of curcumin had succeeded at the time of the review [Nelson 2017]. This does not show that Haridra is useless. It shows that the activity of a whole drug cannot be assumed to be the activity of one isolated molecule, and that laboratory screens can mislead. Other authors have disputed the review's conclusions [Bahadori 2017], so the question remains open.

AYURVEDA CORRELATION: STRUCTURE AND ACTIVITY IN RASASHASTRA

Shodhana changes the chemistry of a toxic drug. Vatsanabha (aconite root) contains aconitine, a highly toxic alkaloid. Its toxicity depends on two ester groups in the molecule. When these are hydrolysed, aconitine is converted into benzoylaconine and then aconine, which are far less toxic. A study of Vatsanabha before and after traditional Shodhana in cow urine found, by thin-layer chromatography, that aconitine and pseudoaconitine were converted to less toxic derivatives, and toxicity in animals fell. A chemical

purification method used for comparison was less effective [Deore 2013]. Here a classical process works through a change in structure that modern chemistry can explain.

Rasa and chemical class. It has been proposed that taste (Rasa) predicts chemical class and action: bitter (Tikta) drugs are often rich in alkaloids and glycosides, and astringent (Kashaya) drugs in tannins. A statistical study of 183 Ayurvedic plants found significant associations between a plant's predominant taste and its recorded actions [Gilca 2015].

Established. For Vatsanabha, a laboratory study has shown that Shodhana converts the toxic diester alkaloids into less toxic forms. The chemical basis, ester hydrolysis, is well understood.

Not yet established. That every Shodhana procedure works through a known structural change; this must be shown drug by drug. The taste–action association is drawn from Ayurvedic texts on both sides of the comparison, so it shows internal consistency of the tradition. It has not yet been tested against independent pharmacological measurements.

A1a.6 Pharmacodynamics and pharmacokinetics illustrated with Ayurvedic drugs

The August 2021 paper asked for the principles of pharmacodynamics and pharmacokinetics *"by illustrating with any Ayurvedic Formulation"*. A good answer does two things. It sets the classical concepts beside the modern ones, honestly. And it gives at least one worked illustration in which an Ayurvedic drug's action or fate in the body is actually known.

A1a.6.1 Classical concepts beside modern ones

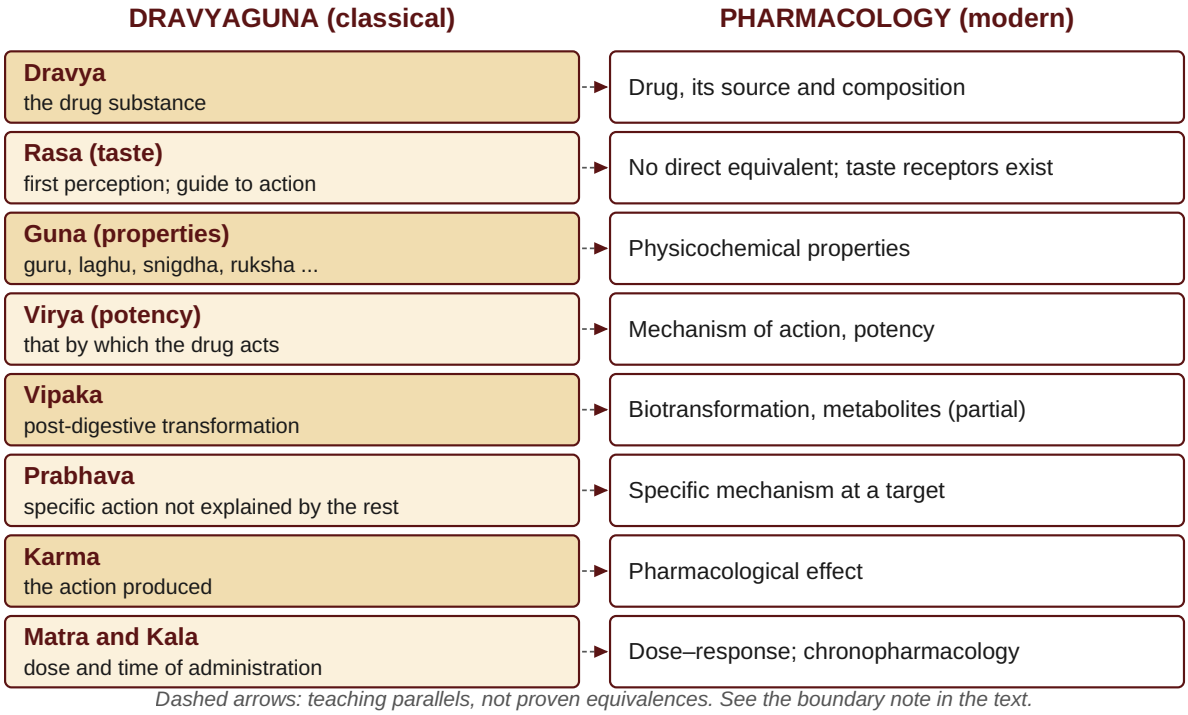


Figure A1a.25. Dravyaguna and pharmacology set side by side. The arrows are teaching parallels, not proven equivalences.

Classical concept	Closest modern idea	How far the parallel holds
Rasa (taste)	Taste perception; a possible clue to chemical class	Proposed. A statistical association with recorded actions has been shown from texts [Gilca 2015]; not tested against laboratory pharmacology
Guna (properties)	Physical and chemical properties	Partial. Guru, Snigdha, Ruksha and others describe physical qualities, but in a different frame
Virya (potency)	Mechanism of action; potency	Teaching parallel only. Virya is classed as Ushna or Shita, which has no single modern counterpart
Vipaka (post-digestive effect)	Biotransformation and metabolites	Teaching parallel only. Vipaka describes the final effect on the Doshas, not the chemical fate of a molecule
Prabhava (specific action)	A specific mechanism at a defined target	Teaching parallel. Charaka defines Prabhava as an action not explained by Rasa, Virya and Vipaka (Sutra Sthana 26/67) [R26]; modern pharmacology would look for the target
Agni (Jatharagni, Bhutagni, Dhatvagni)	Digestion, absorption, tissue metabolism	Teaching parallel
Srotas (channels)	Membranes, transport and distribution	Teaching parallel
Anupana (vehicle)	Effect of vehicle and food on absorption	The principle is established in modern pharmacy; most specific classical Anupana claims are untested
Yogavahi (carrier, enhancer)	Bioenhancer	Established for piperine with some drugs (A1a. 6.3)
Matra and Kala (dose and time)	Dose–response; chronopharmacology	Principle established; most classical doses and timings untested (A1a.4)

Write a table like this in the examination, but write the third column too. An answer that says “Vipaka is metabolism” without qualification states an equivalence that has not been shown, and a careful examiner will notice.

A1a.6.2 Illustration 1 — pharmacodynamics: Sarpagandha

Sarpagandha is the root of *Rauvolfia serpentina*. In Ayurvedic practice it is used for insomnia, anxiety and high blood pressure, for example as Sarpagandha Ghana Vati. Its best-known alkaloid, reserpine, has one of the clearest mechanisms in pharmacology [R32, R29].

Pharmacodynamic step	What happens
Target	Vesicular monoamine transporter (VMAT) in the storage vesicles of nerve endings
Mechanism	Reserpine binds to VMAT and blocks it. Noradrenaline, dopamine and serotonin can no longer be taken up into vesicles; left in the cytoplasm, they are destroyed by monoamine oxidase. The stores are depleted.
Peripheral effect	Less noradrenaline is released from sympathetic nerves: blood pressure and heart rate fall
Central effect	Monoamines are depleted in the brain: sedation, and at higher doses depression
Time course	The block is long-lasting; new transporter has to be made. The effect builds over days and outlasts the presence of the drug in plasma
Dose–response and adverse effects	Sedation, depression, nasal stuffiness, diarrhoea and increased gastric acid; worse at higher doses

Three lessons follow. First, a herbal drug can act by an exact, well-defined mechanism. Second, the effect need not follow the plasma level: an irreversible action outlasts the drug. Third, a whole-root preparation contains a variable amount of reserpine and other alkaloids. Its effect is predictable only if the alkaloid content is standardised.

Established: the mechanism and effects of reserpine as a pure compound.

Not established by this alone: that a given whole-root preparation, at a given dose, delivers a consistent dose of reserpine. That needs analysis of the product.

A1a.6.3 Illustration 2 — pharmacokinetics: Haridra with Maricha

Curcumin from Haridra is a good illustration of every barrier in A1a.2. It dissolves poorly in water (liberation and absorption). It is rapidly conjugated with glucuronic acid in the gut wall and liver (first-pass metabolism, Phase II). It is quickly eliminated. So very little reaches the blood after an oral dose.

Piperine, the pungent alkaloid of Maricha and Pippali, inhibits glucuronidation. In a study published in 1998, curcumin given alone to healthy volunteers produced undetectable or very low serum levels. Given with piperine, its bioavailability rose by about 2000 per cent; in rats the rise was about 154 per cent [Shoba 1998].

This is the modern explanation of one classical idea. Trikatu (Shunthi, Maricha and Pippali) is described as **Yogavahi**: it carries and enhances the action of the drugs it is combined with. For piperine, part of that enhancement is a measured pharmacokinetic effect.

Step of PK	What curcumin does alone	What piperine changes
Absorption	Poor (low solubility)	—
First-pass metabolism	Extensive glucuronidation	Glucuronidation inhibited
Blood level	Very low	Much higher (about 20-fold in the 1998 human study)

Established: piperine raises the blood level of curcumin in human volunteers, and it can raise the levels of other drugs that are cleared by the same enzymes.

Not established: that the higher blood level by itself produces a better clinical result, or that every Yogavahi acts by this mechanism. The same effect that helps curcumin can cause an **interaction** with a conventional drug taken at the same time (A1(c)).

A1a.6.4 Illustration 3 — dose–response and therapeutic index: Vatsanabha

Vatsanabha (*Aconitum* species) is a drug of very narrow therapeutic index. Aconitine binds to voltage-gated sodium channels and holds them open. Nerves and heart muscle are kept depolarised. The results are tingling and numbness, vomiting, and dangerous disturbances of heart rhythm.

Classical practice guards against its toxicity in three ways. It is used only after Shodhana, which reduces toxicity by hydrolysing the toxic alkaloids (A1a.5) [Deore 2013]. It is given in a very small dose. And it is given for defined indications only. The law adds a fourth safeguard: Vatsanabha is one of the poisonous drugs listed in Schedule E(1) of the Drugs Rules, 1945, and a medicine containing it must carry a caution on its label (Chapter B8(i)). Each of these safeguards widens the effective margin between the useful dose and the harmful one.

A1a.6.5 Illustration 4 — vehicle: Sneha Kalpana

Medicated Ghrita and Taila carry the fat-soluble constituents of the herbs cooked into them. Modern pharmaceutics recognises that a lipid vehicle can improve the absorption of lipophilic drugs, partly by keeping them in solution and partly by uptake through the intestinal lymph, which bypasses the liver. Sneha Kalpana is therefore a classical lipid-based dosage form.

Established: the general principle that lipid vehicles improve the absorption of many lipophilic drugs.

Not established for most products: the size of the effect for a particular Ghrita or Taila. That needs a bioavailability study of that product.

KEY POINTS TO REMEMBER

- Pharmacokinetics is what the body does to the drug (ADME); pharmacodynamics is what the drug does to the body.
- Only the un-ionised, lipid-soluble form crosses membranes by diffusion. Weak acids are un-ionised in acid, weak bases in alkali (Henderson–Hasselbalch).
- First-pass metabolism in gut wall and liver reduces oral bioavailability; sublingual and parenteral routes bypass it.
- Vd = amount in body ÷ plasma concentration. Compare it with 3 L, 14 L and 42 L to see where the drug has gone.
- Only free (unbound) drug acts, is metabolised and is filtered. Albumin binds acidic drugs.
- Phase I adds a reactive group (CYP450 oxidation, reduction, hydrolysis); Phase II conjugates it (glucuronidation is commonest). Inducers lower and inhibitors raise the levels of other drugs.
- The kidney filters free drug, secretes acids and bases actively, and reabsorbs lipid-soluble un-ionised drug. Alkaline urine speeds excretion of weak acids.
- First order: constant fraction per unit time, constant t½. Zero order: constant amount, t½ rises with dose (ethanol, phenytoin at high dose).
- t½ = 0.693 × Vd ÷ CL. Steady state and near-complete elimination each take 4 to 5 half-lives.
- Loading dose = Css × Vd ÷ F; maintenance dose rate = Css × CL ÷ F.
- Drugs act on receptors, enzymes, ion channels and transporters, or by physical and chemical means.

- Affinity decides binding; intrinsic activity decides response. Agonist IA 1, partial agonist 0–1, antagonist 0, inverse agonist below 0.
- Four receptor families: ion channel (ms), GPCR (s), kinase-linked (min–h), nuclear (h–days).
- Potency is position on the dose axis; efficacy is the height of the plateau. Efficacy matters more clinically.
- TI = TD50 ÷ ED50. Narrow-TI drugs (digoxin, lithium, warfarin, phenytoin) need careful dosing.
- First human dose: NOAEL → HED by body surface area (Km) → ÷ safety factor 10 → MRSD (FDA 2005).
- SAR: small structural changes alter activity, selectivity and toxicity (catecholamines, aspirin, codeine). Shodhana of Vatsanabha works partly by such a change.
- Classical parallels (Vipaka, Prabhava, Yogavahi) are useful for teaching; only some are supported by data. Say which.

EXAM CORNER — UNIT A1(a)

This unit has been asked as a **20-mark long question** three times and as a **5-mark short note** once (see the table at the start of the unit).

Likely long questions (20 marks)

1. Explain the general principles of pharmacology. Describe pharmacokinetics and pharmacodynamics in detail.
2. Explain the principles of pharmacodynamics and pharmacokinetics, illustrating them with an Ayurvedic formulation.
3. Describe the absorption, distribution, metabolism and excretion of drugs, with the factors that affect each.
4. Describe drug metabolism: Phase I and Phase II reactions, the cytochrome P450 system, and enzyme induction and inhibition, with their clinical importance.
5. Define a receptor. Describe the types of receptors, the theories of drug–receptor interaction and signal transduction, with examples of agonists and antagonists.
6. Describe the dose–response relationship. Explain potency, efficacy and therapeutic index, and explain how the dose of a new drug is determined.
7. Define structure–activity relationship. Explain it with examples, and discuss its use in developing drugs from natural products.

Likely short notes (5 marks)

1. Dose–response relationship and structure–activity relationship *(set in March 2026)*.
2. Henderson–Hasselbalch equation and ion trapping.
3. First-pass metabolism.
4. Apparent volume of distribution.
5. Plasma protein binding and its clinical importance.
6. Blood–brain barrier.
7. Phase II reactions.
8. Enzyme induction.
9. Enterohepatic circulation.
10. First-order and zero-order kinetics.
11. Half-life and steady state; loading dose.

12. The plasma concentration–time curve.
13. Partial agonist and inverse agonist.
14. G-protein-coupled receptors and second messengers.
15. Potency and efficacy.
16. Competitive and non-competitive antagonism.
17. Therapeutic index and therapeutic window.
18. Human equivalent dose.
19. Calculation of dose for children.
20. Bioenhancers (Yogavahi), with piperine as an example.

How to attempt

- **A 20-mark long question** has about 35 minutes in a three-hour paper. Use headings that follow the question's own words. Draw at least two figures: the ADME flow chart (Figure A1a.4) and either the concentration–time curve (Figure A1a.12) or the log dose–response curve (Figure A1a.18). Put at least two tables. Examiners give marks for structure.
- **A 5-mark short note** has about 9 minutes. Write a definition, four to six points or a small table, and one sketch if the topic allows. About one page.
- End with one or two lines linking the topic to Rasashastra and Bhaishajya Kalpana, and **say whether the link is established or proposed**. This is a paper of your speciality, not a general pharmacology paper, and an honest boundary earns more than an enthusiastic claim.

Answer frame: "Explain the general principles of pharmacology. Write pharmacodynamics and pharmacokinetics in detail." (20 marks)

Part	What to write	Suggested marks
1. Definitions	Pharmacology, drug (WHO), pharmacokinetics, pharmacodynamics; sketch Figure A1a.1	2
2. General principles	The principles in the table of A1a.1.5	3
3. Pharmacokinetics	Membrane transport (table); absorption and first pass; distribution with Vd and protein binding; Phase I and II metabolism, induction and inhibition; renal excretion; first- and zero-order kinetics; concentration–time curve with Cmax, Tmax, AUC and therapeutic window	7
4. Pharmacodynamics	Types of action; mechanisms (table); receptor families (table); agonist, partial agonist, antagonist; log dose–response curve with potency, efficacy and therapeutic index	6
5. Ayurveda correlation	Dravyaguna parallels with the established/proposed boundary; one illustration (A1a.6)	1
6. Conclusion	Pharmacokinetics decides how much drug reaches the target and for how long; pharmacodynamics decides what it does there; together they decide the dose	1

The mark split is the author's suggestion, to guide the length of each part. Examiners mark by their own judgement.

Answer frame: "Define dose–response relationship and structure–activity relationship." (5 marks)

Part	What to write	Suggested marks
Dose–response	Definition; graded and quantal; sketch of the log dose–response sigmoid with EC50 and Emax	2
Potency, efficacy, TI	One line each	0.5
SAR	Definition; pharmacophore; one example with the change and its result (catecholamines or aspirin)	2
Rasashastra link	Shodhana of Vatsanabha as a structural change that lowers toxicity [Deore 2013]	0.5

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **26, 29, 32, 37**.

- **[R26]** Charaka Samhita (syllabus reference 26, "Relevant parts from Charaka Samhita …"). Verse numbers verified in: Charak Samhita New Edition, Charak Samhita Research, Training and Skill Development Centre, Jamnagar; carakasamhitaonline.com (accessed 16 October 2026). Cited: Sutra Sthana 1, 4/7, 9/7, 26/12 and 26/67; Kalpa Sthana 12/86.
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024. (The syllabus lists this author's book as "Clinical Pharmacology, KD Tripathi".)
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.
- **[R37]** Government of India. *The Ayurvedic Formulary of India*, Part I: appendix on weights and measures.

Additional verified sources (not in the syllabus list):

- **[FDA 2005]** US Food and Drug Administration, Center for Drug Evaluation and Research. *Guidance for Industry: Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers*. July 2005. fda.gov/media/72309/download (accessed 16 October 2026).
- **[Reagan-Shaw 2008]** Reagan-Shaw S, Nihal M, Ahmad N. Dose translation from animal to human studies revisited. *FASEB J*. 2008;22(3):659–661.
- **[Lipinski 2001]** Lipinski CA, Lombardo F, Dominy BW, Feeney PJ. Experimental and computational approaches to estimate solubility and permeability in drug discovery and development settings. *Adv Drug Deliv Rev*. 2001;46(1–3):3–26. (First published in 1997.)
- **[Shoba 1998]** Shoba G, Joy D, Joseph T, Majeed M, Rajendran R, Srinivas PS. Influence of piperine on the pharmacokinetics of curcumin in animals and human volunteers. *Planta Med*. 1998;64(4):353–356.
- **[Nelson 2017]** Nelson KM, Dahlin JL, Bisson J, Graham J, Pauli GF, Walters MA. The essential medicinal chemistry of curcumin. *J Med Chem*. 2017;60(5):1620–1637.
- **[Bahadori 2017]** Bahadori F, Demiray M. A realistic view on "The Essential Medicinal Chemistry of Curcumin". *ACS Med Chem Lett*. 2017;8(9):893–896.
- **[Deore 2013]** Deore SL, Moon KV, Khadabadi SS, Deokate UA, Baviskar BA. Evaluation of toxicity of 'Vatsanabha' (*Aconitum ferox*, Ranunculaceae) before and after Shodhana. *J Young Pharm*. 2013;5(1):3–6.
- **[Gilca 2015]** Gilca M, Barbulescu A. Taste of medicinal plants: a potential tool in predicting ethnopharmacological activities? *J Ethnopharmacol*. 2015;174:464–473.

UNIT A1(b)

ROUTES OF DRUG ADMINISTRATION

SYLLABUS

Routes of drug administration

LEARNING OUTCOMES

After studying this unit you should be able to:

1. Classify the routes of drug administration into local and systemic, enteral and parenteral.
2. State the factors that decide the choice of route for a given drug and patient.
3. Describe each important route with its advantages, disadvantages and examples.
4. Compare routes for speed of onset, bioavailability and first-pass metabolism.
5. Explain the newer routes and delivery systems: transdermal, inhalational, nasal and targeted.
6. Set the classical Ayurvedic routes (Nasya, Basti, Anjana, Lepa, Dhumapana and others) beside their modern counterparts, and say what is known about absorption by each.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	A, Q.1	20	Routes of Drug administration.
Sep 2018	A, Q.1	20	Explain the general principles of Pharmacology. Write Pharmacodynamics and Pharmacokinetics in detail. Explain any 2 routes of administration of Drug.
Aug 2021	A, Q.3C	5	Brief about Routes of Drug Administration
Aug 2022	A, Q.3b	5	Explain routes of drug administration with their importance.

Routes have been set twice as a 20-mark question, once alone (July 2018) and once as part of a larger question (September 2018, “any 2 routes”). They have twice been a short note. The August 2022 wording, *“with their importance”*, is a reminder to give the clinical reason for each route, not only its description.

A1b.1 What is a route of administration?

> A **route of administration** is the path by which a drug is brought into contact with the body.

The route decides how quickly a drug acts, how much of it reaches the circulation, how long it acts and what adverse effects it can cause. The same drug can behave like a different drug when the route changes. Magnesium sulphate taken by mouth is a purgative. Given intravenously, it controls the convulsions of eclampsia. Applied as a paste, it reduces swelling. The drug is the same; the route changes the effect.

A1b.1.1 Factors that decide the route

Factor	Question to ask	Example
Nature of the drug	Is it destroyed by gastric acid or gut enzymes? Is it irritant? Is it absorbed from the gut at all?	Insulin and benzylpenicillin are destroyed in the gut; streptomycin is not absorbed orally
Site of the desired action	Is the effect wanted locally or throughout the body?	Antifungal cream on skin; salbutamol inhaled into the airways
Speed of action needed	Is it an emergency?	Intravenous adrenaline in anaphylaxis
Duration needed	Is a long action wanted?	Depot injection; transdermal patch
Condition of the patient	Conscious? Vomiting? Able to swallow? In shock?	Rectal diazepam in a convulsing child; intravenous fluids in shock
Accuracy of dose	Must the level be controlled exactly?	Intravenous infusion of heparin or dopamine
First-pass metabolism	Is the oral drug destroyed by the liver before it reaches the circulation?	Glyceryl trinitrate given sublingually
Convenience, cost and compliance	Can the patient take it at home for months?	Oral route for chronic disease

A1b.2 Classification of routes

Routes are first divided by **where the action is wanted**, local or systemic. Systemic routes are then divided by **whether the drug passes through the gut** (enteral) or not (parenteral) (Figure A1b.1).

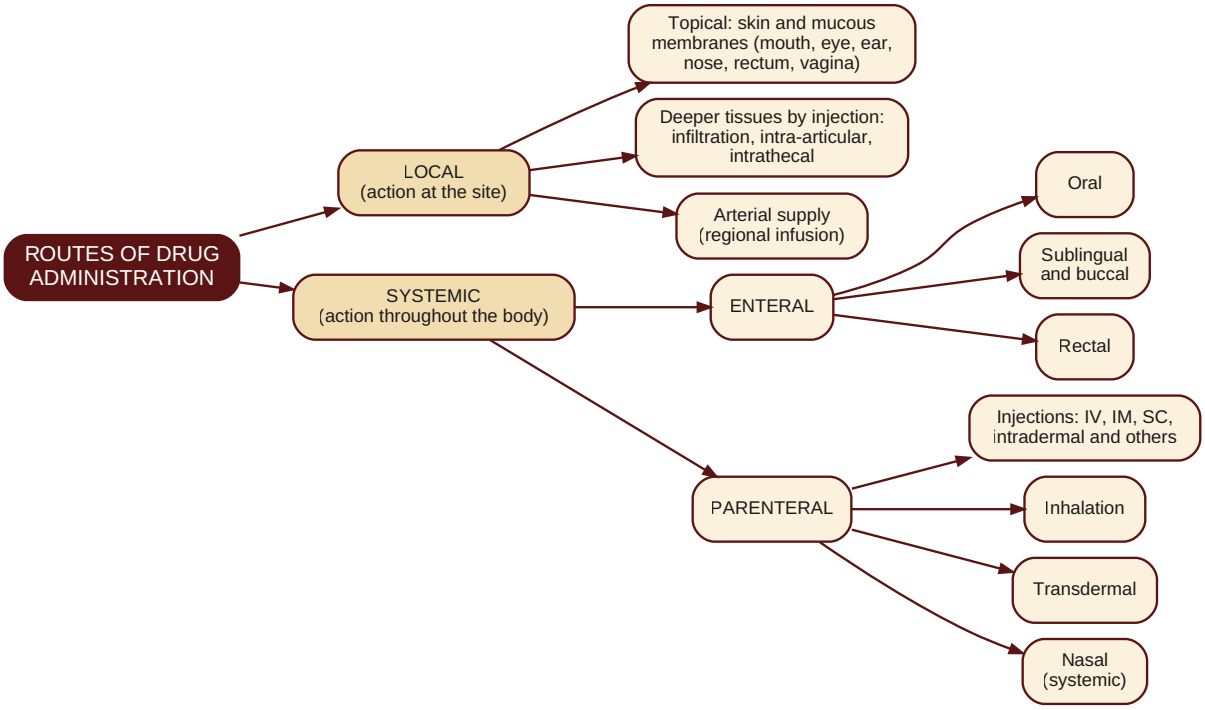


Figure A1b.1. Classification of the routes of drug administration.

Group	Routes	Main idea
Local	Topical on skin and mucous membranes (mouth, eye, ear, nose, rectum, vagina); deeper tissues by injection (infiltration, intra-articular, spinal); arterial supply	Drug acts where it is applied; little reaches the blood
Systemic, enteral	Oral; sublingual and buccal; rectal	Drug enters the body through the gastrointestinal tract
Systemic, parenteral	Injections (intravenous, intramuscular, subcutaneous, intradermal and others); inhalation; transdermal; nasal for systemic effect	Drug enters without passing through the gut (Greek *para*, beside, and *enteron*, intestine)

Note that the word “parenteral” is often used to mean “by injection”. Strictly, it means any route other than the gut, and so includes inhalation and transdermal absorption.

A1b.3 Oral route

The oral route is the commonest, the safest and the most economical. It is also the slowest and the least predictable of the systemic routes, because the drug has to survive the stomach, cross the gut wall and pass the liver before it reaches the circulation (Unit A1(a), A1a.2.3).

Advantages	Disadvantages
Safe: a dose already taken can, in part, be removed by emesis or gastric lavage	Slow onset; not for emergencies
Convenient; self-administration; good compliance	Cannot be used in an unconscious or vomiting patient
Painless; no need for sterile preparation	Irritant or unpleasant drugs cause nausea and vomiting
Economical	Some drugs are destroyed by acid or enzymes (insulin, benzylpenicillin)
Wide choice of dosage forms	Some are not absorbed (aminoglycosides)
	First-pass metabolism lowers bioavailability
	Absorption is affected by food, gut motility and other drugs

Dosage forms by mouth include solid forms (tablets, capsules, powders, granules), liquid forms (solutions, syrups, elixirs, suspensions, emulsions) and modified-release forms (enteric-coated, sustained-release, controlled-release). Ayurvedic oral forms are richer still: Churna, Vati and Gutika, Avaleha, Kashaya and Kwatha, Asava and Arishta, Ghrita and Taila for internal use, Bhasma and Pishti with a vehicle.

Enteric coating resists gastric acid and dissolves in the intestine. It protects acid-labile drugs and protects the stomach from irritant drugs. **Sustained-release forms** release the drug slowly over many hours; they must never be crushed or chewed, because the whole dose is then released at once.

A1b.4 Sublingual and buccal routes

The tablet is kept under the tongue (sublingual) or between the cheek and the gum (buccal) and allowed to dissolve. The mucosa is thin and rich in blood vessels. The veins drain into the superior vena cava, so the drug **bypasses the liver**.

Advantages	Limitations
Rapid onset, within a few minutes	Only for potent, lipid-soluble drugs given in small doses
Escapes first-pass metabolism	Unpleasant-tasting drugs are not suitable
Action can be stopped by spitting out the tablet	Irritation of the mucosa
Not affected by gastric acid	Patient must not swallow saliva too soon

Examples: glyceryl trinitrate and isosorbide dinitrate for angina, buprenorphine, nifedipine (no longer recommended this way), and fentanyl buccal preparations.

A1b.5 Rectal route

Drugs are given as suppositories or as enemas, either for a local effect (for example in piles or ulcerative colitis) or for a systemic effect. The rectum is drained by three veins. The upper part drains into the portal system and so to the liver; the middle and lower parts drain into the inferior vena cava, which bypasses the liver (Figure A1b.2). So **roughly half of a rectal dose escapes first-pass metabolism**, depending on how high in the rectum it is absorbed.

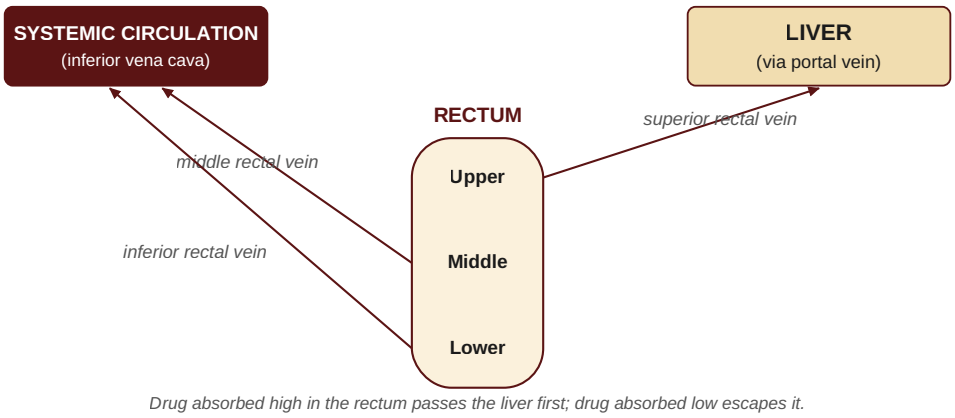


Figure A1b.2. Venous drainage of the rectum: why the rectal route only partly bypasses the liver.

Advantages	Disadvantages
Useful when the patient is vomiting, unconscious or cannot swallow	Absorption is irregular and incomplete
Useful in children who refuse oral medicine	Irritation of the rectal mucosa
Partial escape from first-pass metabolism	Less acceptable to many patients
Irritant drugs can be given	

Examples: diazepam rectal solution in a convulsing child; paracetamol suppositories in children; bisacodyl and glycerin suppositories as laxatives; mesalazine enema in ulcerative colitis.

AYURVEDA CORRELATION: BASTI

Basti, the administration of medicated decoction (Niruha or Asthapana) or oil (Anuvasana) through the rectum, is the principal procedure of Panchakarma. It is a rectal route in the modern sense, and Sharangadhara Samhita describes its method in detail (Uttara Khanda) [R11]. Uttara Basti, given through the urethra or into the uterus, is a local route to the urinary bladder and the female genital tract.

Established. The rectal mucosa absorbs many drugs, including lipid-soluble ones; a part of the absorbed drug bypasses the liver. Large-volume enemas reach well into the colon.

Not established. That the components of a classical Niruha or Anuvasana Basti are absorbed in quantities large enough to explain the systemic effects attributed to Basti. Very few pharmacokinetic studies of Basti preparations exist. The effects of Basti may also arise from local actions on the colon, its contents and its nerves. This is a question for measurement.

A1b.6 Injections (parenteral routes in the narrow sense)

Injections deliver the drug directly into the tissues or the blood. They are used when speed or accuracy is needed, when the patient cannot take the drug by mouth, or when the drug is destroyed or not absorbed in the gut.

General advantages: rapid and predictable action; usable in unconscious and vomiting patients; no first-pass metabolism; suitable for drugs not absorbed orally. **General disadvantages**: the preparation must be sterile and pyrogen-free; injection is painful and needs skill; there is a risk of infection, local injury and nerve damage; once injected the dose cannot be recalled; costlier.

The depth and angle of the needle decide which tissue receives the drug (Figure A1b.3).

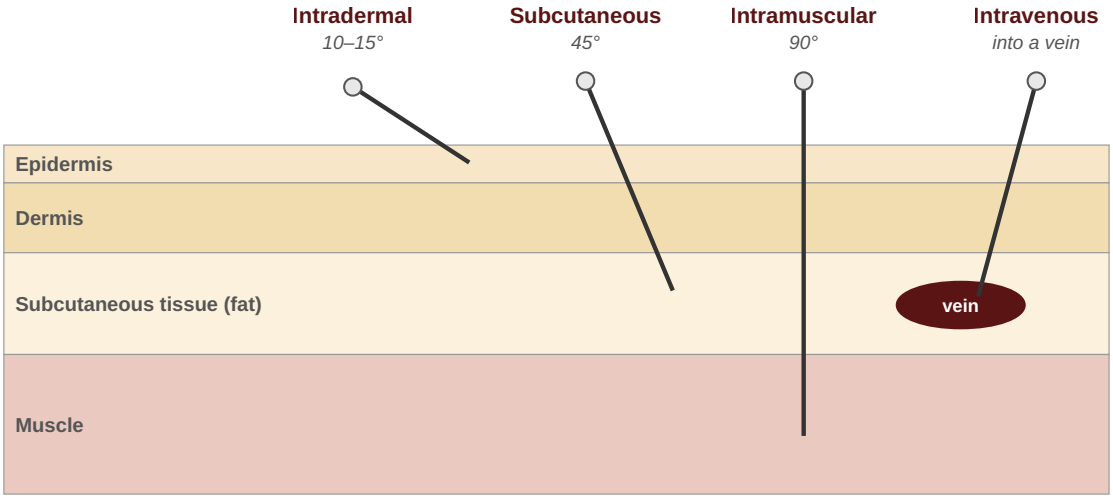


Figure A1b.3. Injection routes by depth: intradermal, subcutaneous, intramuscular and intravenous.

Route	Site and method	Advantages	Disadvantages and cautions	Examples
Intravenous (IV)	Into a vein; as a bolus (all at once) or an infusion (drip)	Fastest onset (seconds); 100 per cent bioavailability; dose can be titrated; large volumes and irritant drugs can be given (diluted)	Most dangerous route: an overdose or a reaction cannot be undone; thrombophlebitis; air embolism; infection; only solutions (not oily or particulate preparations)	Adrenaline in cardiac arrest, antibiotics in sepsis, fluids
Intramuscular (IM)	Into a large muscle: deltoid, gluteal (upper outer quadrant), lateral thigh	Fairly rapid absorption from a solution; oily and depot preparations give a slow, long action; moderate irritants can be given	Painful; limited volume; risk to the sciatic nerve in the gluteal region; not in patients on anticoagulants	Vaccines, vitamin B12, depot penicillin, depot antipsychotics
Subcutaneous (SC)	Into the loose tissue under the skin	Slow, steady absorption; self-injection possible; depot implants	Only small volumes of non-irritant drugs; poor absorption in shock	Insulin, low-molecular-weight heparin, some vaccines, hormone implants
Intradermal	Into the dermis, raising a small bleb	Very small volume; used for testing and some vaccines	Painful; only tiny volumes	BCG vaccine, tuberculin (Mantoux) test, drug sensitivity tests
Intra-arterial	Into an artery	Concentrates the drug in one organ or limb	Risk of spasm and gangrene; used only by specialists	Contrast for angiography; some anticancer drugs
Intrathecal	Into the subarachnoid space of the spine	Bypasses the blood–brain barrier	Strict asepsis; only preservative-free solutions	Spinal anaesthesia, some antibiotics and anticancer drugs
Intra-articular	Into a joint	High local concentration	Infection of the joint if asepsis fails	Corticosteroids in arthritis
Others	Intraperitoneal, intraosseous, intracardiac	Special situations		Intraosseous fluids in a child in shock when no vein can be found

The time course of action differs sharply between injection routes and the oral route (Figure A1b.4).

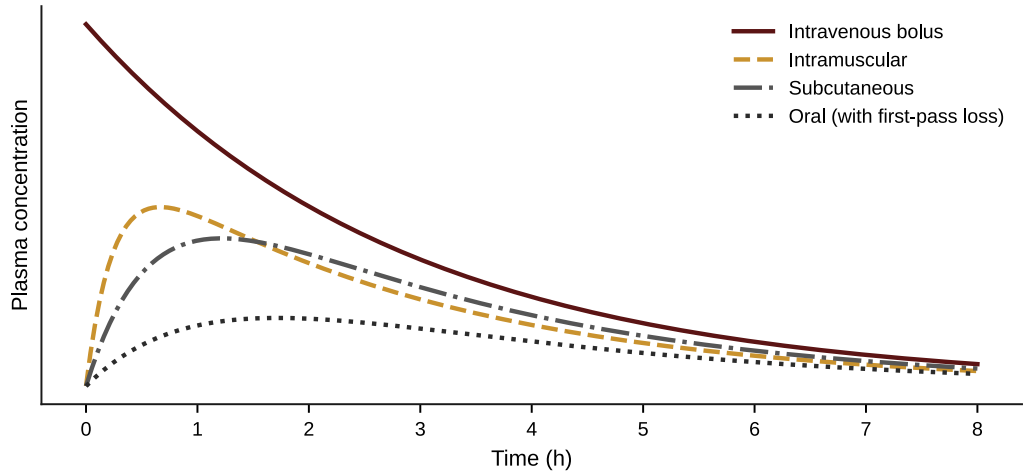


Figure A1b.4. Typical plasma concentration–time curves after the same dose by different routes (schematic).

Depot preparations release a drug slowly from the injection site for days or weeks. Methods include an oily vehicle (fluphenazine decanoate), a poorly soluble salt (benzathine penicillin), an implanted pellet, or a polymer microsphere. Their advantage is compliance; their danger is that a reaction cannot be stopped until the depot is exhausted.

A1b.7 Inhalation

Gases, volatile liquids and aerosols are inhaled into the lungs. The lungs offer a very large surface (tens of square metres) with a thin barrier and a rich blood supply, so absorption is rapid. There are two uses.

1. **Systemic effect**: volatile general anaesthetics (sevoflurane), oxygen and nitrous oxide act after absorption into the blood.
2. **Local effect on the airways**: bronchodilators (salbutamol), inhaled corticosteroids (budesonide) and others act on the bronchi. A small inhaled dose gives a high local concentration with few systemic effects.

Aerosol devices include the metered-dose inhaler (MDI, often used with a spacer), the dry-powder inhaler (DPI) and the nebuliser. **Particle size decides where an aerosol is deposited** (Figure A1b.5).

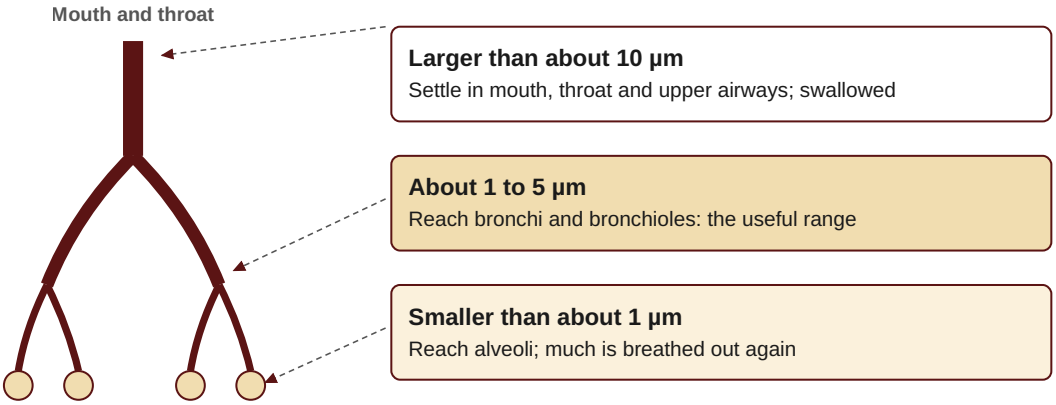


Figure A1b.5. Where inhaled particles are deposited, by particle size (approximate ranges).

Particle size (aerodynamic diameter)	Where it settles
Larger than about 10 µm	Mouth, throat and upper airways; swallowed
About 1 to 5 µm	Bronchi and bronchioles; the useful range for therapy
Smaller than about 1 µm	Reach the alveoli; much is exhaled again

Advantages	Disadvantages
Rapid onset	Needs correct technique, especially for an MDI
High local concentration in the airways with low systemic dose	Irritant drugs cause cough and bronchospasm
Avoids first-pass metabolism	Dose actually reaching the lungs varies
Anaesthetic depth can be controlled breath by breath	Oral thrush with inhaled steroids unless the mouth is rinsed

AYURVEDA CORRELATION: DHUMAPANA

Dhumapana, the inhalation of medicated smoke from a Dhumavarti, is a classical inhalational route for disorders of the head, neck and chest (Sharangadhara Samhita, Uttara Khanda) [R11].

Established. The respiratory tract absorbs volatile and smoke-borne substances rapidly.

Not established. Which constituents of a particular Dhumavarti reach the airways, in what amount, and with what local or systemic effect. Burning also produces combustion products that irritate the airways. A modern scholar should treat Dhumapana as a route that needs pharmacokinetic and safety study, not as one whose benefits are already measured.

A1b.8 Nasal route

Drugs are given as drops or sprays into the nose for a local effect (decongestants, steroid sprays in allergic rhinitis) or for a systemic effect. The nasal mucosa is thin and vascular, absorption is rapid, and the liver is bypassed.

Examples for systemic effect: desmopressin in diabetes insipidus; sumatriptan in migraine; naloxone in opioid overdose; midazolam for prolonged convulsions; several vaccines.

The nose-to-brain pathway. Some molecules can travel from the olfactory region of the nose along the olfactory and trigeminal nerves to the brain, partly bypassing the blood–brain barrier. This has been shown in animals for a number of molecules and is under study in humans. The amount that reaches the brain this way is usually a small fraction of the dose.

AYURVEDA CORRELATION: NASYA

Ayurveda holds that the nose is the gateway to the head, and Nasya is the principal treatment for disorders above the clavicle (Ashtanga Hridaya, Sutra Sthana 20) [R26]. Medicated oils, powders, juices and smoke are all given by the nose.

Established. The nasal mucosa absorbs drugs rapidly and bypasses the liver; local effects on the nasal and sinus mucosa are expected.

Proposed, partly supported. That Nasya delivers active constituents to the brain. A nose-to-brain pathway exists and has been demonstrated for some molecules in animals. Whether the constituents of a particular Nasya oil reach the brain in an effective amount in humans has not been shown for most classical preparations.

A1b.9 Transdermal route

A drug is applied to the skin in a patch that releases it at a controlled rate for absorption into the systemic circulation (Figure A1b.6). The skin's outer horny layer, the stratum corneum, is a strong barrier. So only drugs with suitable properties can be given this way.

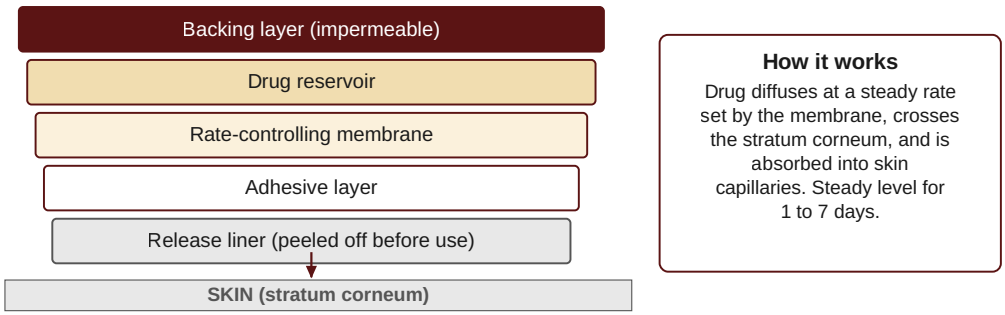


Figure A1b.6. Layers of a reservoir-type transdermal patch.

Requirement of a transdermal drug	Reason
Potent: small daily dose (a few milligrams)	Only a little crosses the skin each day
Lipid-soluble, with some water solubility	To enter and then leave the stratum corneum
Small molecule (roughly below 500 daltons)	Larger molecules cross poorly
Not irritant to skin	Patch stays on for days

Examples: glyceryl trinitrate (angina), fentanyl (chronic pain), nicotine (smoking cessation), hyoscine (motion sickness), oestradiol (hormone therapy), rivastigmine (dementia).

Advantages: steady blood level for one to seven days; avoids first-pass metabolism; good compliance; can be removed if adverse effects occur. **Disadvantages**: only suitable drugs; skin irritation; slow onset; cost. Absorption can be enhanced by chemical enhancers, by **iontophoresis** (an electric current) or by **microneedles**.

AYURVEDA CORRELATION: LEPA, ABHYANGA AND RELATED PROCEDURES

Lepa (medicated paste), Abhyanga (oil massage), Udvartana (powder massage), Parisheka (pouring of liquids) and Shirodhara are external routes used both for local and for systemic effects.

Established. Lipid-soluble constituents in an oil vehicle can penetrate the skin, and massage increases local blood flow. Local effects on the skin and the underlying tissues are plausible.

Not established. That the constituents of a given Taila are absorbed into the systemic circulation in amounts large enough to explain systemic effects. Skin permeation studies (for example in a Franz diffusion cell) and blood-level studies are the tools to settle this for each preparation.

A1b.10 Topical application to the eye, ear, mouth and vagina

Site	Forms	Notes
Eye	Drops, ointments, inserts	Drops drain through the nasolacrimal duct into the nose and are absorbed into the blood, bypassing the liver. Timolol eye drops can slow the heart and cause bronchospasm. Pressing on the inner corner of the eye for a minute after instilling reduces this (Figure A1b.7). Ophthalmic preparations must be sterile.
Ear	Drops	Local action; avoid ototoxic drugs if the eardrum is perforated
Mouth and throat	Gargles, mouthwashes, lozenges, paints	Local action on the oral mucosa
Vagina	Pessaries, creams, tablets	Local antifungal and antimicrobial action; some systemic absorption

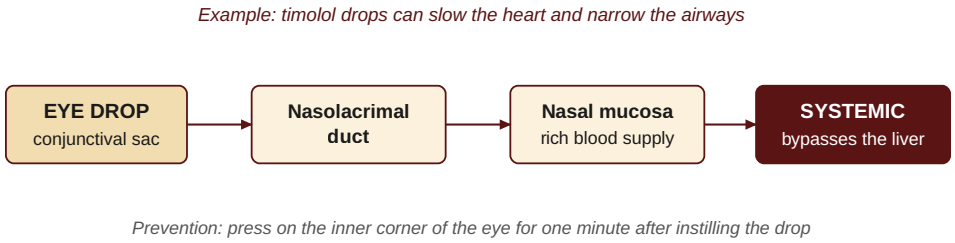


Figure A1b.7. Why eye drops can act on the whole body: drainage through the nasolacrimal duct.

A1b.11 Newer approaches: targeted drug delivery

Modern pharmaceutics tries to deliver a drug to its target and nowhere else. The main approaches are:

- **Liposomes**: phospholipid vesicles carrying a drug (liposomal amphotericin B, liposomal doxorubicin);
- **Nanoparticles and polymeric micelles**;
- **Antibody–drug conjugates**, which carry a cytotoxic drug to cells bearing a specific antigen;
- **Implants and pumps**: insulin pumps, intrauterine hormone systems;
- **Prodrugs** activated at the site of action.

These are covered more fully in courses on pharmaceutics. Here it is enough to know that the route and the carrier are now designed together.

A1b.12 Comparison of the main routes

Route	Onset (typical)	Bioavailability	First-pass metabolism	Typical use
Intravenous	Seconds	100% by definition	None	Emergencies, precise dosing
Inhalation	Seconds to minutes	Variable	None	Asthma, anaesthesia
Sublingual	1–5 minutes	Often high	Avoided	Angina
Intramuscular	10–30 minutes	High (for solutions)	None	Vaccines, depots
Subcutaneous	15–30 minutes	High	None	Insulin, heparin
Intranasal	Minutes	Variable	Avoided	Migraine, convulsions, opioid overdose
Rectal	5–30 minutes	Irregular	Partly avoided	Vomiting, children
Oral	30–90 minutes	Variable; often low	Yes	Most chronic treatment
Transdermal	Hours	Variable	Avoided	Steady long-term delivery

Onset times are typical ranges for rapidly absorbed preparations; they vary with the drug and the formulation.

A1b.13 Classical Ayurvedic routes

Ayurveda developed a very wide range of routes, many of them centuries before their modern counterparts. Sharangadhara Samhita gives a whole section, the Uttara Khanda, to these procedures [R11].

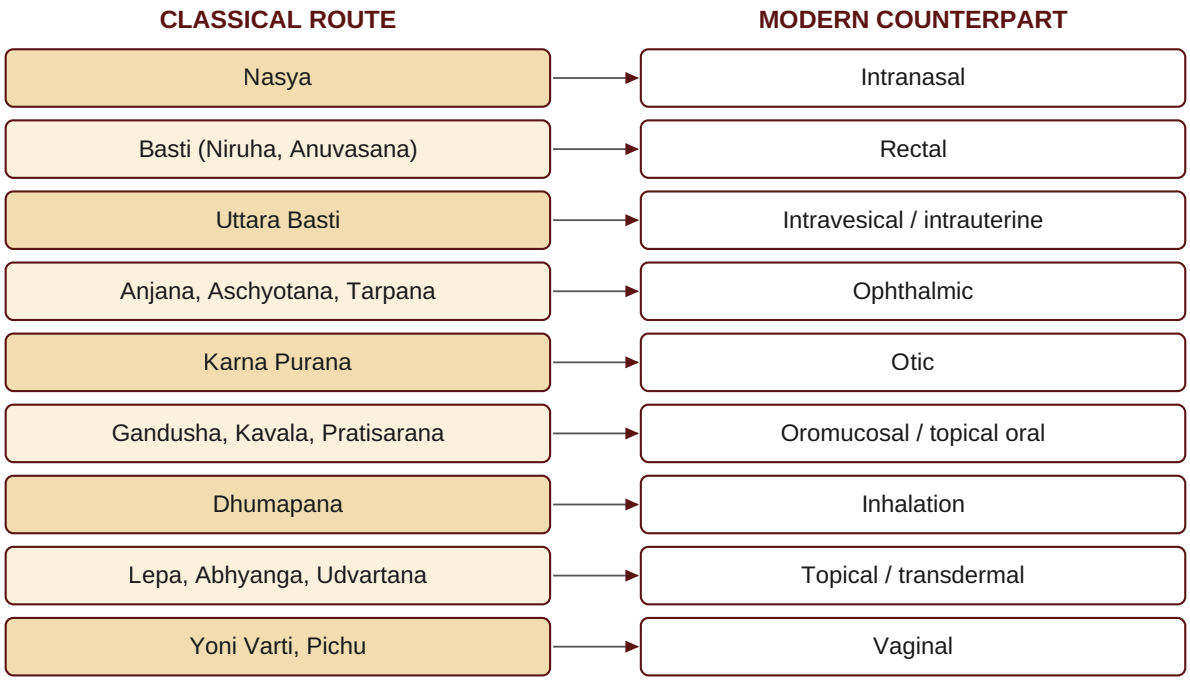


Figure A1b.8. Classical Ayurvedic routes and their closest modern counterparts.

Classical route	What is done	Closest modern route	What is known about absorption
Mukha (oral)	All internal medicines	Oral	As for any oral drug; Anupana may change absorption
Nasya	Oil, juice, powder or smoke into the nose	Intranasal	Mucosal absorption established; nose-to-brain delivery shown for some molecules in animals
Basti (Niruha, Anuvasana)	Decoction or oil per rectum	Rectal	Rectal absorption established; amounts from classical Basti little studied
Uttara Basti	Into urethra, bladder or uterus	Intravesical, intrauterine	Mainly local
Anjana, Aschyotana, Tarpana	Collyrium, drops, retention of Ghrita on the eye	Ophthalmic	Local; systemic absorption possible through the nasolacrimal duct
Karna Purana	Oil filled into the ear	Otic	Local
Gandusha, Kavala	Holding or swishing liquid in the mouth	Oromucosal	Mainly local; some buccal absorption possible
Pratisarana	Rubbing a medicine on the gums or tongue	Topical oral	Local
Dhumapana	Inhaled medicated smoke	Inhalation	Rapid absorption of smoke constituents; content and safety little studied
Lepa, Abhyanga, Udvartana	Paste, oil massage, powder massage	Topical, transdermal	Skin penetration of lipophilic constituents possible; systemic levels little studied
Avagaha, Parisheka	Immersion bath, pouring	Topical	Mainly local
Yoni Varti, Pichu	Vaginal pessary, oil-soaked swab	Vaginal	Mainly local

The pattern is clear. The classical routes are well matched to modern ones in **anatomy and intent**. What is mostly missing is **pharmacokinetic measurement**: how much of which constituent reaches the blood or the target tissue by each route. That gap is an opportunity for research in Bhaishajya Kalpana.

KEY POINTS TO REMEMBER

- The route can change the effect of a drug: magnesium sulphate is a purgative orally and an anticonvulsant intravenously.
- Local routes act at the site; systemic routes are enteral (oral, sublingual, rectal) or parenteral (injection, inhalation, transdermal, nasal).
- Oral: safest, cheapest, most convenient; slow, variable, and subject to first-pass metabolism.
- Sublingual and buccal: rapid; bypass the liver; only for potent lipid-soluble drugs (glyceryl trinitrate).
- Rectal: useful in vomiting and in children; about half the absorbed dose bypasses the liver.

- Intravenous: fastest, 100 per cent bioavailable, most dangerous. Intramuscular allows depots. Subcutaneous gives slow, steady absorption (insulin).
- Inhaled aerosols of about 1 to 5 µm reach the bronchi; inhalation gives high airway levels with low systemic dose.
- Transdermal patches need potent, small, lipid-soluble drugs; they give steady levels for days.
- Eye drops can act systemically via the nasolacrimal duct (timolol).
- Classical routes (Nasya, Basti, Anjana, Lepa, Dhumapana) match modern routes in anatomy; their pharmacokinetics are largely unmeasured.

EXAM CORNER — UNIT A1(b)

Likely long questions (20 marks)

1. Classify the routes of drug administration. Describe each with its advantages, disadvantages and examples.
2. Describe the parenteral routes of drug administration. Compare them with the oral route.
3. Describe the routes of drug administration with their importance. Add a note on the classical Ayurvedic routes and their modern counterparts.
4. Discuss the factors that govern the choice of route of administration, with examples.

Likely short notes (5 marks)

1. Sublingual route.
2. Advantages and disadvantages of the oral route.
3. Rectal route and first-pass metabolism.
4. Intravenous route.
5. Depot preparations.
6. Inhalational route and particle size.
7. Transdermal drug delivery systems.
8. Nasal route and the nose-to-brain pathway.
9. Systemic effects of eye drops.
10. Nasya, Basti or Dhumapana as routes of drug administration.

How to attempt

- For the 20-mark question, open with a definition and the classification diagram (Figure A1b.1). Then take each route under the same five sub-headings: method, advantages, disadvantages, examples, importance. A table for each route earns more than paragraphs. Close with the comparison table (A1b.12) and the classical routes table (A1b.13).
- When the question says “any two routes” (September 2018), choose two that let you show contrast: the oral and the intravenous routes are the safest choice.
- For a short note, give the definition, four to six points, two examples and one sketch.

Answer frame: “Routes of drug administration.” (20 marks)

Part	What to write	Suggested marks
Definition and factors	Definition; factors deciding the route (table)	2
Classification	Local and systemic; enteral and parenteral (diagram)	2
Enteral routes	Oral, sublingual and buccal, rectal: method, merits, demerits, examples	5
Parenteral routes	IV, IM, SC, intradermal and special injections (table); depot	5
Other routes	Inhalation (particle size), nasal, transdermal, topical, eye	3
Comparison	Onset, bioavailability, first pass (table)	1
Ayurveda	Classical routes and modern counterparts, with the evidence boundary	2

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **11, 26, 29, 32**.

- **[R11]** Sharangadhara. *Sharangadhara Samhita*, with the Dipika commentary of Adhamalla and the Gudartha Dipika of Kashirama. Shastri P, editor. 6th ed. Varanasi: Chaukhamba Orientalia; 2005. Uttara Khanda (procedures of Basti, Nasya, Dhumapana, Gandusha, Lepa and the eye).
- **[R26]** Ashtanga Hridaya, Sutra Sthana 20 (Nasya), one of the “relevant parts” of the classical texts named in syllabus reference 26.
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman’s The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

UNIT A1(c)

FACTORS MODIFYING DRUG EFFECT, BIOAVAILABILITY, INTERACTIONS, ADR AND TOXICITY

SYLLABUS

Factors modifying drug effect, Bioavailability and Bioequivalence, drug interactions, adverse drug reaction and drug toxicity

LEARNING OUTCOMES

After studying this unit you should be able to:

1. List and explain the factors that modify the effect of a drug, with examples, and relate them to the classical factors of Dashavidha Pariksha and Bheshaja Kala.
2. Define bioavailability and bioequivalence, calculate absolute and relative bioavailability, and describe the design and acceptance criteria of a bioequivalence study.
3. Classify drug interactions, explain their mechanisms, and give examples of herb–drug interactions relevant to Ayurvedic practice.
4. Define and classify adverse drug reactions, including allergic reactions and teratogenicity.
5. Explain drug toxicity and the principles of managing poisoning, with the main antidotes.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	A, Q.3c	5	Drug interaction and adverse drug reaction.
Jan 2024	A, Q.3a	5	Mention the factors modifying drug effect.
Mar 2026	A, Q.2	20	Discuss Adverse Drug Reactions (ADR) and drug toxicity. Explain the role of Pharmacovigilance in the safety monitoring of Ayurvedic drugs.
Mar 2026	A, Q.3b	5	What are drug-drug interactions? Give examples relevant to Ayurvedic practice.

This unit is asked mostly as short notes, but the March 2026 paper made ADR and toxicity a 20-mark question, joined to pharmacovigilance. The same paper asked for drug–drug interactions *"relevant to Ayurvedic practice"*. Section A1c.3.6 is written for that.

A1c.1 Factors modifying drug effect

The same dose of the same drug does not produce the same effect in every patient, nor in the same patient at every time. A drug's effect depends on the drug, on the patient and on the circumstances (Figure A1c.1). Knowing these factors is the basis of individualising a dose.

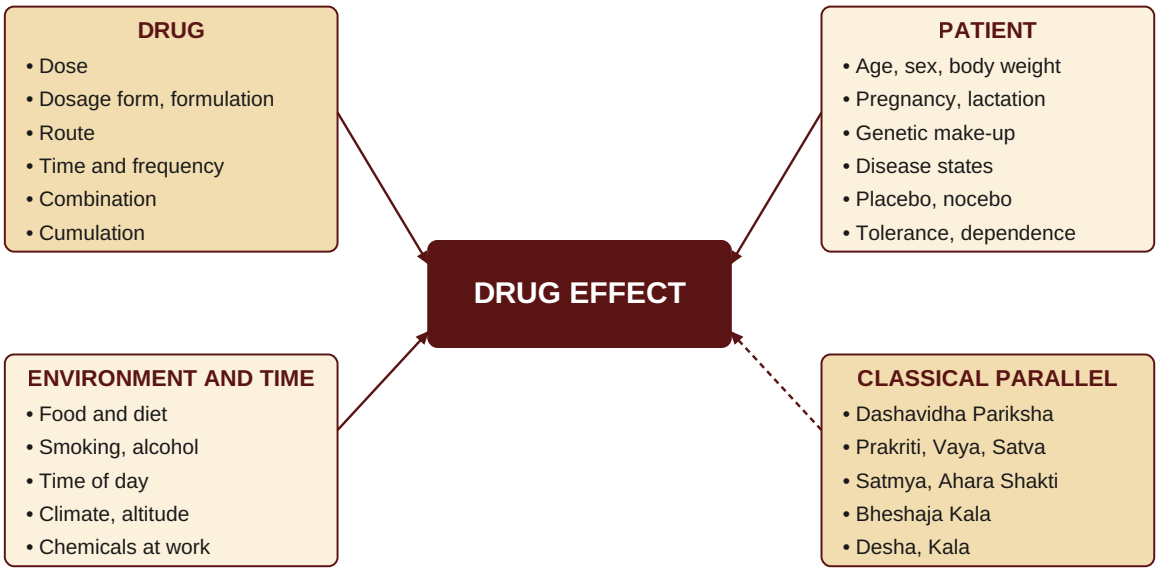


Figure A1c.1. Factors that modify the effect of a drug.

A1c.1.1 Factors related to the drug

Factor	Effect	Example
Dose	Effect rises with dose up to a ceiling; toxicity rises beyond it	Aspirin: analgesic in low doses, anti-inflammatory in high doses, toxic in overdose
Dosage form and formulation	Changes the rate and extent of absorption	Sustained-release tablet; two brands of phenytoin with different bioavailability
Route	Changes onset, intensity and even the kind of effect	Magnesium sulphate (A1(b))
Time and frequency of administration	Changes the effect and the adverse effects	Corticosteroids given in the morning suppress the adrenal axis less
Combination with other drugs	Interaction (A1c.3)	Rifampicin lowers the effect of oral contraceptives
Cumulation	A drug eliminated slowly accumulates with repeated doses	Digoxin, heavy metals

A1c.1.2 Factors related to the patient

Age

Children are not small adults, and the elderly are not merely old adults.

Age group	What is different	Consequence and example
Newborn (especially premature)	Liver enzymes immature (low glucuronidation); kidney function low; more body water; blood–brain barrier immature; less plasma protein	Chloramphenicol causes the grey baby syndrome; doses must be much lower; sulphonamides can displace bilirubin and cause kernicterus
Child	Faster metabolism of some drugs per kilogram; growing bones and teeth	Doses by weight or surface area (A1a. 4.3); tetracyclines stain teeth; fluoroquinolones avoided
Elderly	Kidney function falls with age; liver blood flow falls; less body water and more fat; more sensitive brain; many diseases and many drugs (polypharmacy)	Lower doses of renally excreted drugs (digoxin, aminoglycosides); more confusion with sedatives; more falls; more interactions

Sex

Women on average weigh less and have more body fat. Pregnancy, lactation and menstruation create special situations. Some drugs carry sex-specific risks: androgens cause virilism in women; some drugs prolong the QT interval more in women.

Body weight and body composition

The dose of many drugs is given per kilogram. In obesity, lipid-soluble drugs have a larger volume of distribution, while water-soluble drugs are better dosed by lean body weight.

Pregnancy

Change in pregnancy	Effect on drugs
Blood volume and body water rise	Volume of distribution of many drugs rises
Kidney blood flow and filtration rise	Renally excreted drugs are cleared faster
Plasma albumin falls	Free fraction of highly bound drugs rises
Gastric emptying slows	Absorption is delayed
Placenta transfers most drugs	The foetus is exposed; risk of teratogenicity (A1c.4.6)

Genetic factors (pharmacogenetics)

Inherited differences in enzymes, transporters and receptors explain much of the variation between people.

Genetic variation	Drugs affected	Consequence
Slow and fast acetylators (NAT2)	Isoniazid, hydralazine, procainamide, dapsone	Slow acetylators: more peripheral neuropathy with isoniazid; drug-induced lupus
CYP2D6 poor and ultra-rapid metabolisers	Codeine, tramadol, many antidepressants	Poor: little pain relief from codeine. Ultra-rapid: morphine toxicity
CYP2C19 poor metabolisers	Clopidogrel (a prodrug)	Less activation; less antiplatelet effect
CYP2C9 and VKORC1 variants	Warfarin	Some patients need much lower doses

Genetic variation	Drugs affected	Consequence
Glucose-6-phosphate dehydrogenase (G6PD) deficiency	Primaquine, dapsone, sulphonamides, nitrofurantoin	Haemolysis
Pseudocholinesterase deficiency	Succinylcholine	Prolonged paralysis and apnoea
HLA-B*15:02 allele (common in parts of Asia)	Carbamazepine	Stevens–Johnson syndrome; testing is advised before starting
Malignant hyperthermia susceptibility	Halothane, succinylcholine	Dangerous rise in body temperature in anaesthesia
Thiopurine methyltransferase (TPMT) deficiency	Azathioprine, mercaptopurine	Severe bone-marrow suppression

Disease states

Disease	Effect	Example
Liver disease	Less metabolism; less albumin; sometimes portal shunting so first-pass loss falls	Morphine, lignocaine and diazepam need lower doses
Kidney disease	Less excretion of renally cleared drugs and active metabolites	Aminoglycosides, digoxin, lithium; dose by creatinine clearance
Heart failure	Less blood flow to liver and kidney; gut oedema reduces absorption	Lignocaine toxicity
Thyroid disease	Hyperthyroidism speeds, hypothyroidism slows the metabolism of many drugs	Digoxin needs change
Gastrointestinal disease	Diarrhoea, malabsorption, achlorhydria change absorption	
Fever, shock	Altered distribution and absorption	Poor absorption from intramuscular sites in shock

Psychological factors: placebo and nocebo

A **placebo** is an inert substance or sham procedure given in the form of a medicine. The improvement that follows, the **placebo effect**, arises from the patient's expectation, the doctor–patient relationship and the natural course of the illness. It is largest in subjective symptoms such as pain, anxiety and nausea. A **nocebo** effect is the reverse: adverse symptoms that arise from negative expectation. These effects are the reason why new drugs are compared with placebo in **double-blind** trials (Chapter A2).

Tolerance and dependence

- **Tolerance** is a reduced response to a drug after repeated use, so a larger dose is needed for the same effect. It may be **pharmacokinetic** (the drug induces its own metabolism, as with barbiturates and carbamazepine) or **pharmacodynamic** (receptors adapt, as with opioids and nitrates). **Cross-tolerance** occurs between related drugs (alcohol and benzodiazepines). **Tachyphylaxis** is rapid tolerance (A1a. 3.3).

- **Drug dependence** is a state in which a person needs the drug to function normally or to avoid withdrawal. Psychological dependence is compulsive desire; physical dependence shows as a withdrawal syndrome when the drug is stopped.

A1c.1.3 Factors related to the environment and the time

Factor	Example
Time of day (chronopharmacology)	Cortisol peaks in the morning; corticosteroids given in the morning cause less adrenal suppression. Asthma and angina have daily rhythms.
Food	Food delays most drugs, increases absorption of some lipid-soluble drugs, and reduces others (tetracyclines with milk)
Smoking	Induces CYP1A2: theophylline is cleared faster in smokers
Alcohol	Acute use inhibits, chronic use induces drug metabolism; additive sedation
Altitude, temperature	Change in the body's handling of drugs and in their effects
Occupational and environmental chemicals	Pesticides and heavy metals can induce or inhibit enzymes and add to toxicity

AYURVEDA CORRELATION: DASHAVIDHA PARIKSHA AND BHESHAJA KALA

Ayurveda requires the physician to examine ten aspects of the patient before deciding the medicine and its dose (Dashavidha Pariksha, Charaka Samhita, Vimana Sthana 8) [R26]. The match with the modern list of patient factors is close.

Classical factor	Modern factor
Prakriti (constitution)	Genetic make-up; pharmacogenetics
Vikriti (morbidity)	Disease state
Sara, Samhanana, Pramana (tissue quality, build, measurements)	Body weight and body composition
Satmya (habituation)	Diet, habits, tolerance
Satva (mental strength)	Psychological factors; placebo and nocebo
Ahara Shakti (power of intake and digestion)	Absorption and metabolic capacity
Vyayama Shakti (exercise capacity)	General physiological reserve
Vaya (age)	Age

Ayurveda also prescribes the time of administration, Bheshaja Kala: on an empty stomach, before, during or after meals, between morsels, at night and so on, according to the disease and the Dosha (Charaka Samhita, Chikitsa Sthana 30/298) [R26]. This is chronopharmacology in principle.

Established. The classical texts recognise, in their own language, almost every modern factor that modifies drug response. This is a mature clinical insight.

Under study. Whether Prakriti types correspond to measurable genetic or metabolic differences is an active field of research, sometimes called Ayurgenomics. Some studies have reported such differences; the findings are not yet strong enough to guide drug dosing.

Not established. That a specific Bheshaja Kala changes the absorption or effect of a specific Ayurvedic preparation. This has rarely been measured.

A1c.2 Bioavailability and bioequivalence

A1c.2.1 Bioavailability

Bioavailability is the fraction of an administered dose of unchanged drug that reaches the systemic circulation, and the rate at which it does so. It is written as **F** and expressed as a fraction or a percentage.

Bioavailability has two parts: the **extent** (how much reaches the blood), measured by the area under the plasma concentration–time curve (AUC), and the **rate** (how fast), shown by Cmax and Tmax. An intravenous dose has a bioavailability of 100 per cent by definition, so it is the reference (Figure A1c.2).

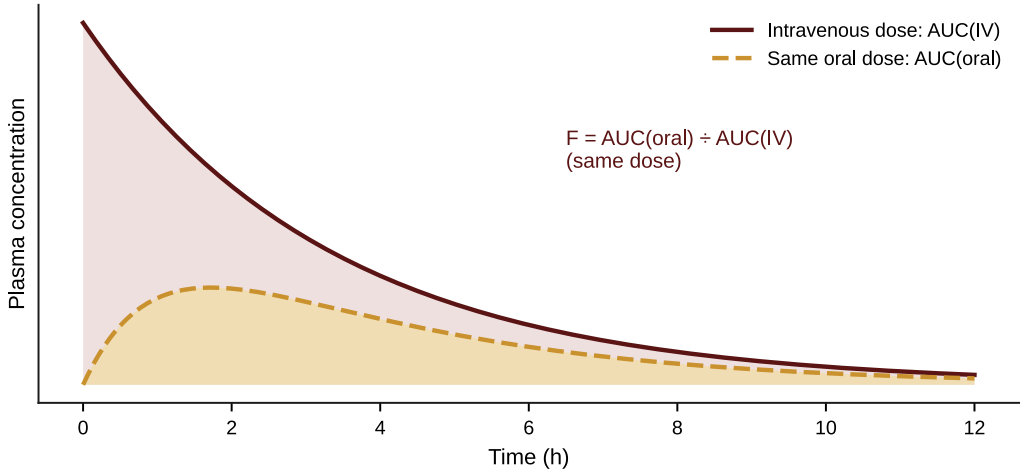


Figure A1c.2. Bioavailability: the AUC after an oral dose compared with the AUC after the same intravenous dose (schematic).

$$\textbf{Absolute bioavailability } F = (\,AUC_{oral} \times Dose_{IV}\,) \div (\,AUC_{IV} \times Dose_{oral}\,) \times 100$$

$$\textbf{Relative bioavailability} = (\,AUC_{test} \times Dose_{reference}\,) \div (\,AUC_{reference} \times Dose_{test}\,) \times 100$$

- **Absolute** bioavailability compares a non-intravenous route with the intravenous route.
- **Relative** bioavailability compares two formulations given by the same route, for example a new tablet against a standard tablet or a solution.

WORKED EXAMPLE: ABSOLUTE BIOAVAILABILITY

100 mg of a drug given intravenously gives an AUC of 40 mg·h/L. 200 mg given by mouth gives an AUC of 32 mg·h/L.

F = (32 × 100) ÷ (40 × 200) × 100 = 3200 ÷ 8000 × 100 = **40 per cent**.

So the oral dose must be about two and a half times the intravenous dose to give the same exposure.

Factors that affect bioavailability

Group	Factors
Drug	Solubility, particle size, crystal form (polymorphism), salt form, lipid solubility, pKa, stability in the gut
Formulation	Disintegration and dissolution rate; excipients; coating; manufacturing method (compression force, granulation)
Patient	Gastric emptying, gut motility, food, gut pH, gut wall enzymes and P-glycoprotein, first-pass metabolism, disease, age
Other drugs and food	Chelation, adsorption, enzyme inhibition or induction

How bioavailability is measured

1. **Plasma level method** (the usual one): measure the drug in plasma at intervals after the dose; calculate AUC, Cmax and Tmax.
2. **Urinary excretion method**: for a drug excreted unchanged in urine, the total amount in urine reflects the amount absorbed.
3. **Pharmacodynamic method**: when the drug cannot be measured, an effect (for example the fall in blood glucose) is measured.
4. **Clinical end-point**: used only when nothing else is possible.

A1c.2.2 Bioequivalence

> Two products are **bioequivalent** when they are pharmaceutically equivalent (or alternatives) and their rate and extent of absorption, when given in the same molar dose under similar conditions, do not differ significantly. In practice, bioequivalent products can be expected to give the same clinical effect and safety.

Term	Meaning
Pharmaceutical equivalents	Same active ingredient, same amount, same dosage form, same route
Pharmaceutical alternatives	Same active moiety but a different salt, ester or dosage form
Bioequivalence	Pharmaceutical equivalents (or alternatives) with no significant difference in rate and extent of absorption
Therapeutic equivalence	Bioequivalent products expected to have the same clinical effect and safety; they can be substituted

Why bioequivalence matters. When the patent of an innovator drug expires, other companies make the same drug as a “generic”. The generic need not repeat the full clinical trials. It must instead show that it is bioequivalent to the innovator product. This is why generic medicines are cheaper.

Design of a bioequivalence study

The standard design is a **randomised, two-period, two-sequence, crossover study** in healthy adult volunteers (Figure A1c.3).

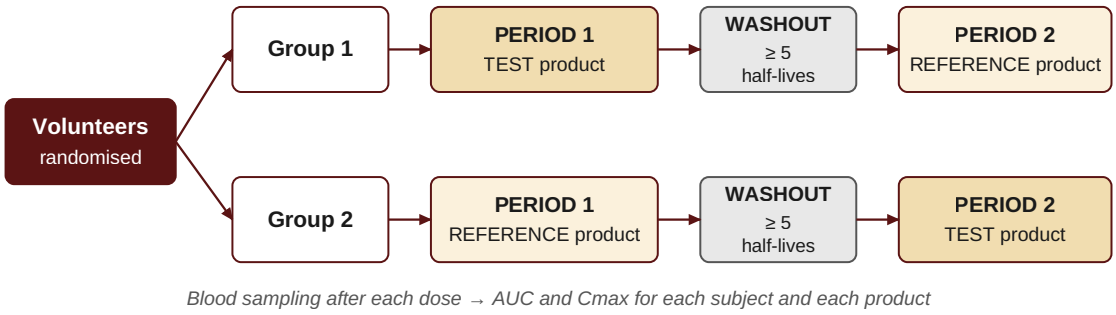


Figure A1c.3. Two-period, two-sequence crossover design of a bioequivalence study.

1. Volunteers are divided at random into two groups.
2. In period 1, one group takes the test product and the other the reference product.
3. A **washout** period follows, long enough for the drug to be eliminated (usually at least five half-lives).
4. In period 2, the groups cross over.
5. Blood is sampled at fixed times after each dose; AUC and Cmax are calculated for each subject and each product.
6. Because each subject receives both products, each acts as his own control, and fewer subjects are needed.

Acceptance criteria

The ratio of test to reference is calculated for AUC and Cmax on log-transformed values. **The 90 per cent confidence interval of the ratio of geometric means must lie within 80 to 125 per cent** for both parameters (Figure A1c.4). This is the criterion used by the United States FDA, the European Medicines Agency and Indian regulators. Narrower limits are applied to drugs with a narrow therapeutic index.

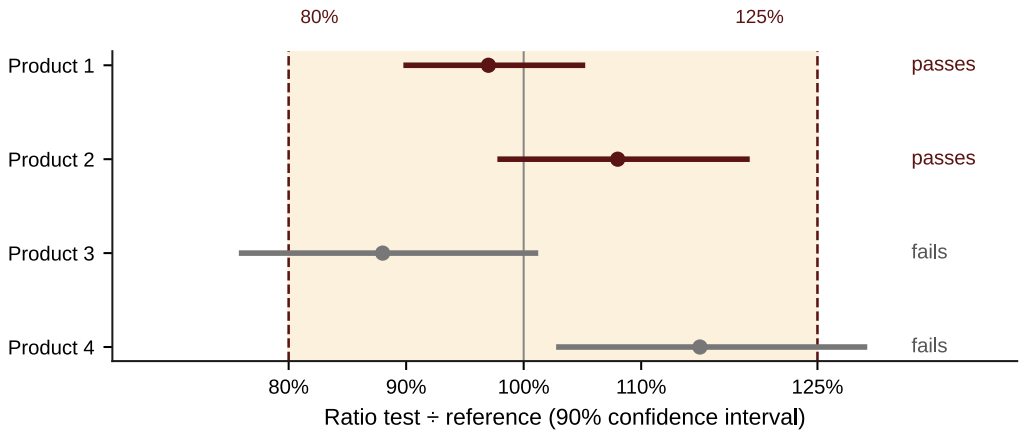


Figure A1c.4. Bioequivalence acceptance: the 90 per cent confidence interval of the test/reference ratio must fall within 80 to 125 per cent (schematic examples).

Note what the limits mean. They do not allow a generic to contain 20 per cent less drug. Both products must contain the labelled amount. The limits concern the statistical confidence about the *ratio of exposures*.

Biowaiver. For some drugs that are highly soluble and highly permeable (Class I of the Biopharmaceutics Classification System), regulators may accept comparative **dissolution tests** in place of a human study.

AYURVEDA CORRELATION: BIOAVAILABILITY AND BIOEQUIVALENCE OF AYURVEDIC MEDICINES

Classical texts are well aware that the same drug in different forms has different strength. The order of Svarasa, Kalka, Kwatha, Hima and Phanta (A1a.4) and the use of Anupana and Yogavahi (A1a.6) are statements about what modern pharmacy calls bioavailability.

Established. For individual constituents, bioavailability can be measured and improved. Piperine raised the bioavailability of curcumin in human volunteers [Shoba 1998] and of propranolol and theophylline in another human study [Bano 1991].

Not established. A meaningful bioequivalence between two brands of a multi-ingredient Ayurvedic formulation. A Kwatha or an Arishta contains hundreds of compounds, and the "active" ones are usually unknown. Bioequivalence in the strict sense cannot yet be shown. What can be compared today is quality: identity, marker compounds, chromatographic fingerprints and dissolution of markers. Two brands of the same classical medicine should therefore not be assumed to be interchangeable merely because they carry the same name.

A1c.3 Drug interactions

> A **drug interaction** occurs when the effect of one drug is changed by the presence of another drug, a food, a herb, a drink or an environmental chemical given before or at the same time.

Interactions may be harmful, useful or of no importance. The **object drug** is the one whose effect changes; the **precipitant drug** is the one that causes the change. Interactions matter most when the object drug has a narrow therapeutic index (warfarin, digoxin, lithium, phenytoin, theophylline, oral hypoglycaemics). They are classified by mechanism (Figure A1c.5).

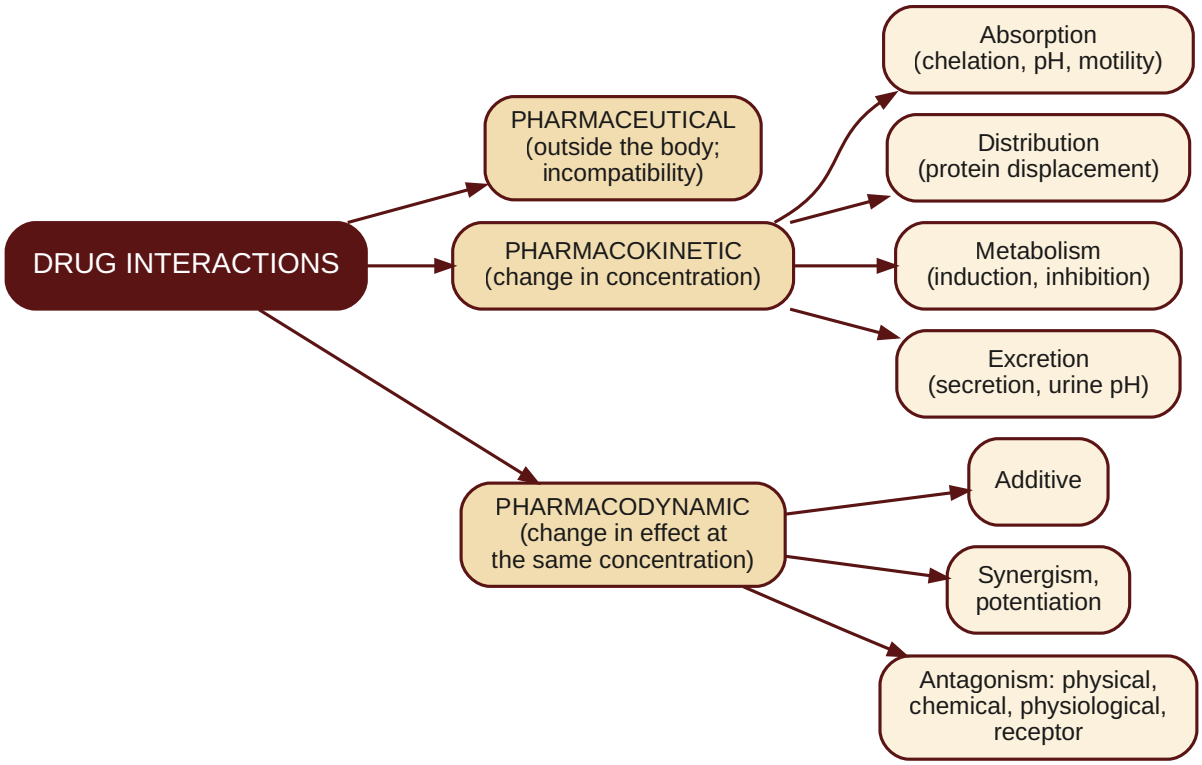


Figure A1c.5. Classification of drug interactions by mechanism.

A1c.3.1 Pharmaceutical interactions (incompatibility)

These occur outside the body, when drugs are mixed in a syringe, an infusion bag or a mixture, and react chemically or physically.

Example	What happens
Penicillin and gentamicin mixed in one syringe	Gentamicin is inactivated
Phenytoin added to a dextrose infusion	Phenytoin precipitates
Calcium salts mixed with sodium bicarbonate	Calcium carbonate precipitates

Pharmaceutical incompatibility is also a concern in compounding. In Bhaishajya Kalpana, the order of adding ingredients and the choice of media are rules that serve, among other things, to avoid incompatibility.

A1c.3.2 Pharmacokinetic interactions

One drug changes the absorption, distribution, metabolism or excretion of another.

Stage	Mechanism	Example
Absorption	Chelation or binding in the gut	Tetracyclines and fluoroquinolones with calcium, iron, magnesium, aluminium (antacids, milk) — absorption falls
	Change in gastric pH	Proton pump inhibitors reduce absorption of ketoconazole and iron
	Change in gut motility	Metoclopramide speeds, anticholinergics slow, the absorption of other drugs

Stage	Mechanism	Example
	Adsorption	Activated charcoal and cholestyramine bind other drugs
Distribution	Displacement from plasma proteins	Sulphonamides displace bilirubin in the newborn; valproate displaces phenytoin
Metabolism	Enzyme induction	Rifampicin, carbamazepine and phenytoin lower the levels of oral contraceptives, warfarin and many others
	Enzyme inhibition	Ketoconazole, erythromycin, cimetidine raise the levels of other drugs; grapefruit juice raises the levels of some calcium channel blockers
Excretion	Competition for tubular secretion	Probenecid prolongs penicillin; aspirin reduces methotrexate excretion
	Change in urine pH	Alkalinisation speeds salicylate excretion
	Change in sodium balance	Thiazides raise lithium levels

A1c.3.3 Pharmacodynamic interactions

The two drugs act on the same or related systems, without changing each other's concentration.

Type	Meaning	Example
Additive (summation)	Combined effect equals the sum of the separate effects (1 + 1 = 2)	Two analgesics; two diuretics
Synergism (supra-additive)	Combined effect is greater than the sum (1 + 1 = 3)	Sulfamethoxazole with trimethoprim
Potentiation	A drug with no effect of its own increases the effect of another (0 + 1 = 2)	Carbidopa increases the effect of levodopa
Physical antagonism	Based on a physical property	Charcoal adsorbing alkaloids in poisoning
Chemical antagonism	Direct chemical reaction	Chelating agents with heavy metals; protamine with heparin
Physiological (functional) antagonism	Opposite effects on the same function by different mechanisms	Adrenaline (on β_2) against histamine (on H_1) in bronchospasm
Receptor antagonism	Competitive or non-competitive block at the same receptor	Naloxone with morphine; atropine with acetylcholine

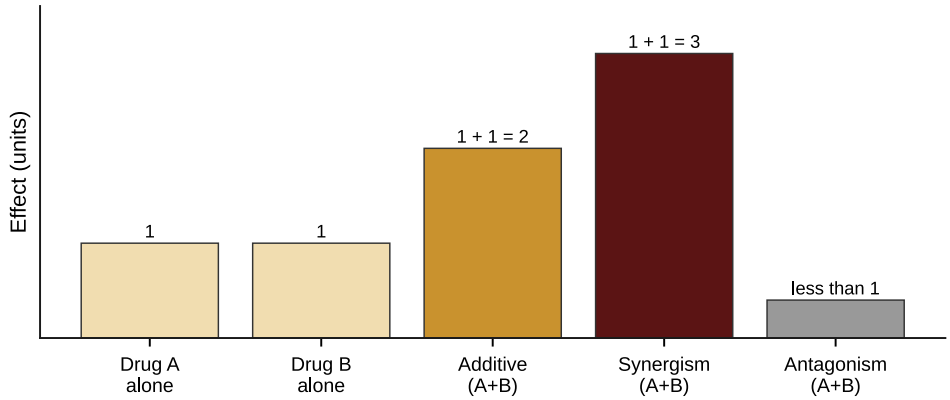


Figure A1c.6. Additive, synergistic and antagonistic interactions (schematic).

A1c.3.4 Useful interactions

Interactions are not always harmful. Many combinations are deliberate.

Combination	Purpose
Levodopa with carbidopa	Carbidopa prevents peripheral breakdown of levodopa
Amoxicillin with clavulanic acid	Clavulanate inhibits bacterial beta-lactamase
Sulfamethoxazole with trimethoprim	Sequential block of folate synthesis
Naloxone in opioid overdose	Receptor antagonism
Atropine with pralidoxime in organophosphate poisoning	Combined antidote action

A1c.3.5 Drug–food interactions

Grapefruit juice inhibits intestinal CYP3A4 and raises the levels of several drugs. Milk and calcium reduce the absorption of tetracyclines. Foods rich in tyramine (aged cheese, fermented foods) can cause a dangerous rise in blood pressure in patients on monoamine oxidase inhibitors. Green leafy vegetables rich in vitamin K reduce the effect of warfarin.

A1c.3.6 Herb–drug interactions relevant to Ayurvedic practice

Many patients take Ayurvedic medicines and modern drugs at the same time, often without telling either doctor. Herbs contain active compounds, and they interact by the same mechanisms as drugs. The following examples are taken from published human studies or case reports.

Ayurvedic drug	Modern drug	What was observed	Mechanism	Source
Piperine (from Maricha, Pippali; in Trikatu)	Propranolol, theophylline	Higher blood levels in healthy volunteers	Inhibition of metabolism	[Bano 1991]
Piperine	Curcumin	About 20-fold higher bioavailability	Inhibition of glucuronidation	[Shoba 1998]
Gugulipid (from Guggulu)	Propranolol, diltiazem	Lower peak level and AUC in healthy volunteers after a single dose	Reduced absorption	[Dalvi 1994]
Shankhapushpi preparation	Phenytoin	Reported as a clinically important interaction, with loss of seizure control	Lower phenytoin levels	[Dandekar 1992]
Karela (*Momordica charantia*)	Chlorpropamide	Altered blood glucose control in a reported case	Additive glucose lowering	[Aslam 1979]

Other interactions are often listed in reviews on theoretical grounds, for example between Haridra and anticoagulants, or between Ashwagandha and sedatives. A theoretical interaction is a reason for caution and monitoring, not a proven fact.

Practical rules for the Ayurvedic physician

1. Always ask every patient what other medicines, herbs and supplements they take, and record it.
2. Be most careful with patients on narrow-therapeutic-index drugs: warfarin, digoxin, lithium, phenytoin and other antiepileptics, theophylline, insulin and oral hypoglycaemics, immunosuppressants, and anticancer drugs.
3. Watch for additive effects: hypoglycaemic herbs with antidiabetic drugs; hypotensive herbs with antihypertensives; sedative herbs with sedatives.
4. Remember that a bioenhancer such as Trikatu enhances modern drugs too.
5. Report any suspected interaction to a pharmacovigilance centre (Chapter A6).

AYURVEDA CORRELATION: VIRUDDHA

Charaka describes Viruddha Ahara, combinations of foods that become harmful when taken together, by quantity, by processing, by time, by place and by other factors (Charaka Samhita, Sutra Sthana 26) [R26]. Milk with fish is the best-known example. The concept is close to the modern idea of an interaction: two substances, each harmless alone, become harmful in combination.

Established. That incompatibility by combination, by dose and by processing is a recognised classical principle, and that modern pharmacology confirms the general principle.

Not established. That each classical Viruddha pair produces measurable harm. Most pairs have never been tested, and some may reflect custom rather than toxicity.

A1c.4 Adverse drug reactions

A1c.4.1 Definitions

> **Adverse drug reaction (ADR)** — WHO: a response to a drug that is noxious and unintended, and that occurs at doses normally used in man for prophylaxis, diagnosis or therapy of disease, or for the modification of physiological function.

Edwards and Aronson proposed a definition that also covers the action to be taken: an appreciably harmful or unpleasant reaction, resulting from an intervention related to the use of a medicinal product, which predicts hazard from future administration and warrants prevention, specific treatment, alteration of the dosage regimen or withdrawal of the product [Edwards 2000].

Several related terms are often confused.

Term	Meaning	Example
Adverse event	Any untoward medical occurrence in a patient taking a drug, whether or not caused by the drug	A fall during a trial; the drug may or may not be the cause
Adverse drug reaction	An adverse event for which a causal relationship with the drug is at least reasonably possible	Rash after amoxicillin
Side effect	An unwanted but often predictable effect at a therapeutic dose, arising from the drug's own pharmacology	Dry mouth with atropine; drowsiness with antihistamines
Secondary effect	An indirect consequence of the primary action	Superinfection after broad-spectrum antibiotics
Toxic effect	An effect of excessive dose or excessive drug level	Respiratory depression with morphine overdose
Serious adverse reaction	One that results in death, is life-threatening, needs or prolongs hospital stay, causes persistent disability, or is a congenital anomaly	Anaphylaxis; Stevens–Johnson syndrome
Medication error	A preventable failure in prescribing, dispensing or administering	Tenfold dose error in a child

A1c.4.2 Classification of ADRs

The classic division is into **Type A** (augmented, predictable) and **Type B** (bizarre, unpredictable). It was later extended to six types (Figure A1c.7).

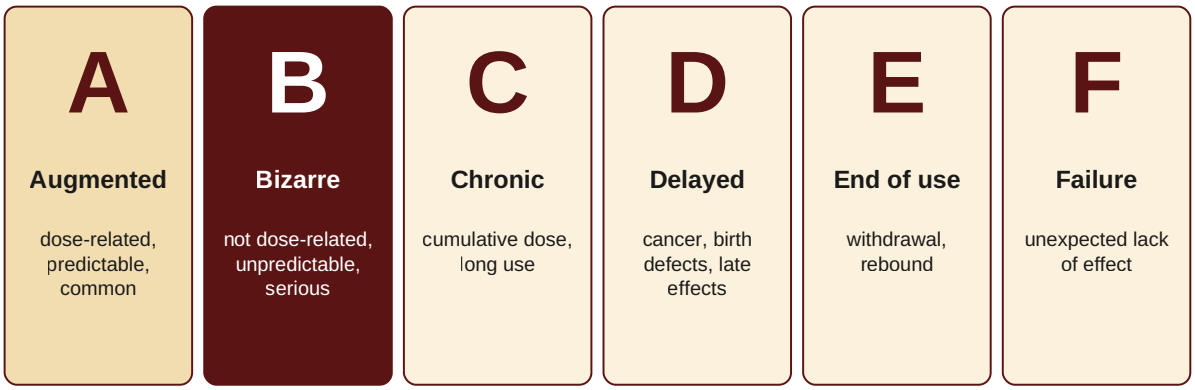


Figure A1c.7. Classification of adverse drug reactions, types A to F.

Type	Name	Features	Examples	Management
A	Augmented	Dose-related; predictable from pharmacology; common; low mortality	Hypoglycaemia with insulin; bleeding with warfarin; bradycardia with β-blockers	Reduce the dose
B	Bizarre	Not dose-related; unpredictable; uncommon; often serious	Anaphylaxis to penicillin; idiosyncratic hepatitis; malignant hyperthermia	Stop the drug; avoid in future
C	Chronic (continuing)	Related to cumulative dose and long use	Osteoporosis and adrenal suppression with corticosteroids	Reduce or withdraw slowly
D	Delayed	Appear long after exposure	Carcinogenicity; teratogenicity; tardive dyskinesia	Often hard to prevent
E	End of use (withdrawal)	Occur when the drug is stopped	Rebound hypertension after clonidine; withdrawal seizures after benzodiazepines	Taper slowly
F	Failure of therapy	Unexpected lack of effect, often from interaction	Contraceptive failure with rifampicin	Adjust dose; avoid the interaction

A more detailed system, **DoTS**, classifies an ADR by its relation to **Dose**, its **Time course** and the **Susceptibility** of the patient [Edwards 2000].

A1c.4.3 Severity

Grade	Meaning
Mild	No change in treatment needed; no antidote; no longer hospital stay
Moderate	Change in treatment needed; may need treatment or prolong stay
Severe	Life-threatening, or causes disability, or needs intensive treatment
Lethal	Directly or indirectly contributes to death

A1c.4.4 Drug allergy (hypersensitivity)

Drug allergy is an immunological reaction to a drug or its metabolite, which acts as an antigen or as a hapten joined to a body protein. It needs prior sensitisation, is not related to the dose, and recurs on re-exposure, often more severely. Allergic reactions are classified by Gell and Coombs into four types (Figure A1c.8).

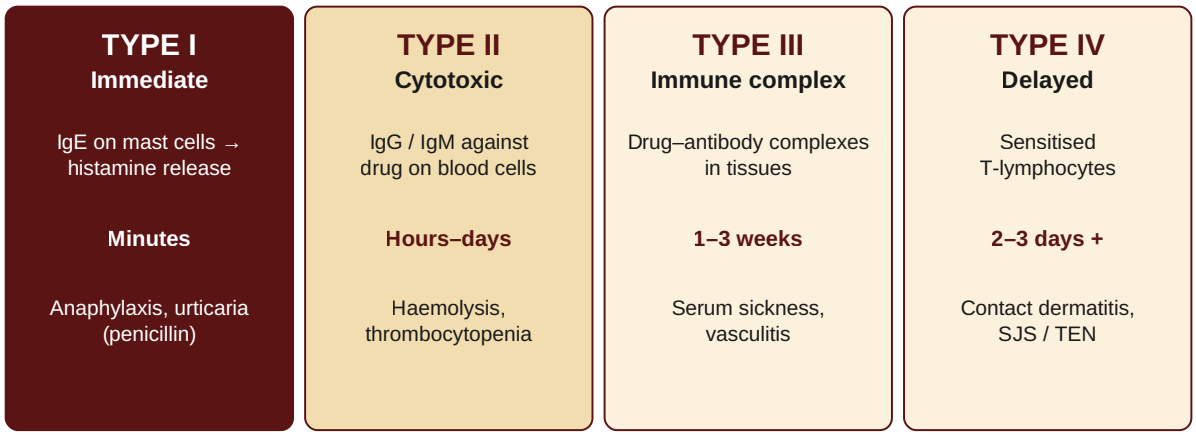


Figure A1c.8. The four types of drug allergy (Gell and Coombs).

Type	Mechanism	Time of onset	Drug examples	Clinical picture
I — Immediate (anaphylactic)	IgE on mast cells; histamine and other mediators released	Minutes	Penicillins, cephalosporins, sera, NSAIDs (some are non-allergic)	Urticaria, angioedema, bronchospasm, anaphylactic shock
II — Cytotoxic	IgG or IgM against drug bound to blood cells; complement	Hours to days	Methyldopa, quinidine, penicillin	Haemolytic anaemia, thrombocytopenia, agranulocytosis
III — Immune complex	Drug–antibody complexes deposited in tissues	1–3 weeks	Penicillins, sulphonamides, foreign serum	Serum sickness: fever, rash, joint pain; vasculitis
IV — Delayed (cell-mediated)	Sensitised T-lymphocytes	2–3 days or more	Topical drugs, neomycin, nickel; severe skin reactions	Contact dermatitis; Stevens–Johnson syndrome and toxic epidermal necrolysis

Anaphylaxis is the most urgent ADR. Treatment: adrenaline intramuscularly at once (repeat if needed), lay the patient flat with legs raised, oxygen, intravenous fluids, then antihistamines and corticosteroids as supportive treatment.

A1c.4.5 Other special types of ADR

Type	Meaning	Example
Intolerance	A characteristic effect at an unusually low dose	Tinnitus with a few tablets of aspirin in a sensitive person
Idiosyncrasy	A genetically determined abnormal reaction	Haemolysis with primaquine in G6PD deficiency
Photosensitivity	Reaction of the skin to drug plus sunlight	Tetracyclines, sulphonamides, psoralens (as in Bakuchi)
Drug dependence	Compulsive use; withdrawal on stopping	Opioids, benzodiazepines, alcohol
Iatrogenic disease	Disease caused by medical treatment	Drug-induced Parkinsonism; drug-induced peptic ulcer
Organ toxicity	Damage to a particular organ	Liver (paracetamol overdose, isoniazid), kidney (aminoglycosides, NSAIDs), ear (aminoglycosides), heart (doxorubicin)
Carcinogenicity and mutagenicity	Cancer or genetic damage, often after long latency	Some anticancer drugs; arsenic (Chapter A4)

A1c.4.6 Teratogenicity

A **teratogen** is a drug that causes structural or functional defects in the developing foetus. The thalidomide disaster of the late 1950s and early 1960s, in which thousands of children were born with limb defects, changed drug regulation throughout the world. The effect of a teratogen depends strongly on the **stage of pregnancy** at which the foetus is exposed (Figure A1c.9).

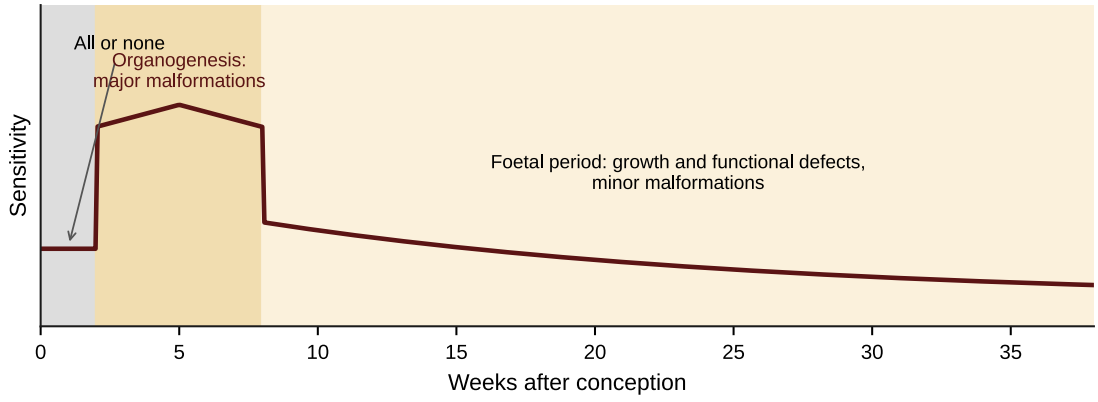


Figure A1c.9. Sensitivity of the developing embryo and foetus to teratogens by stage of pregnancy (schematic).

Stage (weeks after conception)	Effect of a harmful drug
0–2 (pre-embryonic)	"All or none": either the embryo dies or it recovers fully
3–8 (embryonic; organogenesis)	Major structural malformations; the most dangerous period
9 to birth (foetal)	Growth retardation, functional defects, minor malformations

Known teratogens include thalidomide, isotretinoin, valproate, phenytoin, warfarin, methotrexate, ACE inhibitors (later pregnancy), tetracyclines (teeth), and alcohol. Since 2015, drug labels in the United States describe pregnancy risk in narrative form, and the older letter categories (A, B, C, D, X) have been withdrawn. The letter categories are still found in many textbooks.

A1c.4.7 Prevention of ADRs

1. Prescribe only when needed; avoid unnecessary combinations.
2. Take a drug and allergy history, including Ayurvedic and other herbal medicines.
3. Choose the dose for the patient: age, weight, kidney and liver function, pregnancy.
4. Know the common and serious ADRs of every drug you prescribe.
5. Monitor drugs with a narrow therapeutic index.
6. Warn the patient what to watch for.
7. Report suspected ADRs to the pharmacovigilance programme (Chapter A6).

A1c.5 Drug toxicity

A1c.5.1 Definitions

> **Toxicity** is the capacity of a substance to cause harm to a living organism. **Drug toxicity** means harmful effects of a drug, usually from excessive dose, excessive level or prolonged use. **Poisoning** is harm caused by a substance taken, accidentally or deliberately, in a harmful amount.

Kind of toxicity	Exposure	Example
Acute	Single large dose or short exposure	Paracetamol overdose; aconite poisoning
Subacute (sub-chronic)	Repeated exposure over weeks to a few months	Toxicity in a 28- or 90-day animal study
Chronic	Repeated exposure over months or years	Lead or arsenic toxicity; analgesic nephropathy
Local	At the site of contact	Corrosive burns
Systemic	After absorption, in distant organs	Mercury in the kidney and brain

How toxicity is measured in animals (acute, repeated-dose, reproductive, genotoxicity and carcinogenicity studies) is described in Unit A1(d). Heavy-metal toxicity has its own chapter (A4), and herbo-mineral compounds theirs (A5).

A1c.5.2 Principles of managing acute poisoning

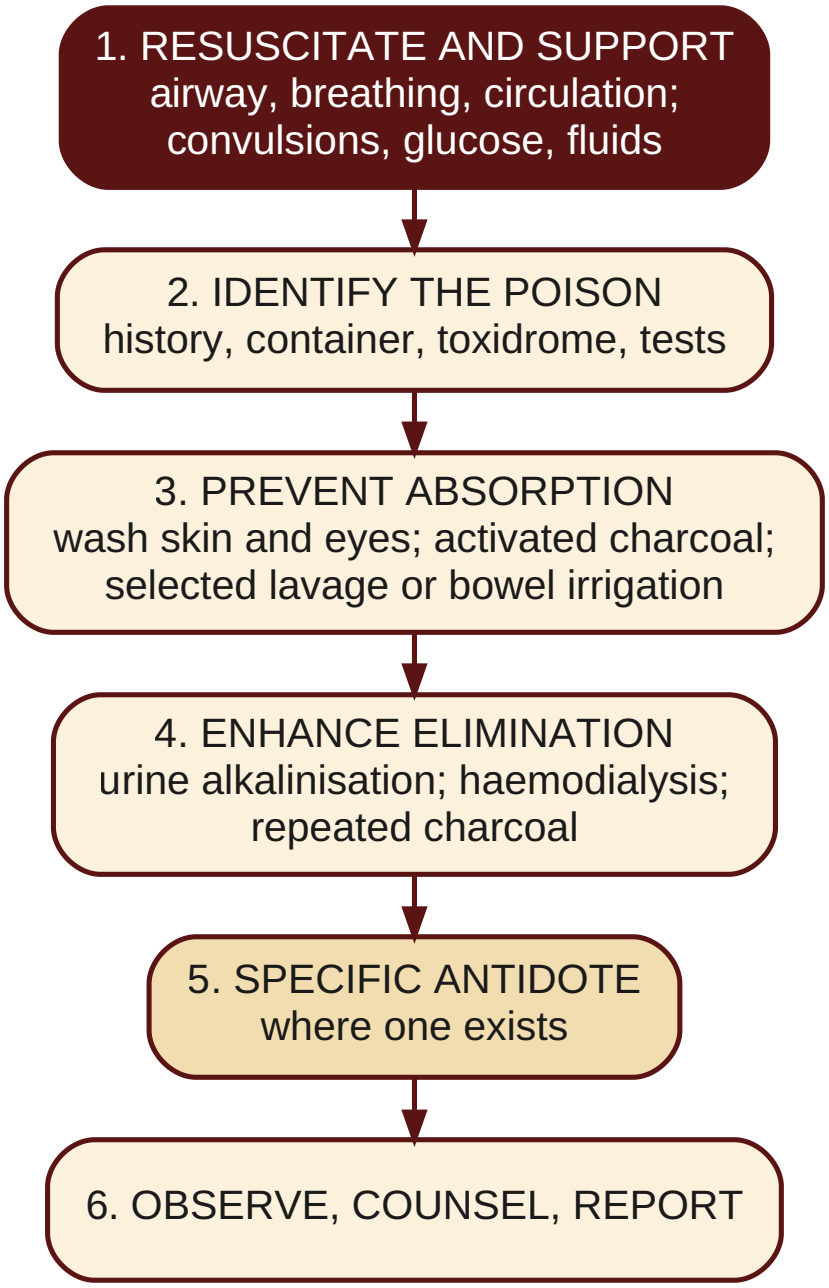


Figure A1c.10. Principles of managing acute poisoning.

1. **Resuscitation and supportive care** — airway, breathing, circulation (ABC); control of convulsions; correction of blood sugar, fluids and electrolytes. Supportive care saves more lives than antidotes.
2. **Identify the poison** — history, containers, smell, clinical picture (toxidromes), laboratory tests.
3. **Prevent further absorption** — remove from the source; wash the skin and eyes; **activated charcoal** by mouth within about an hour for many ingested poisons (not for heavy metals, alcohols, acids, alkalis or hydrocarbons); gastric lavage only in selected cases within an hour; whole-bowel irrigation for some sustained-release preparations and iron.
4. **Enhance elimination** — alkalinisation of urine (salicylates, phenobarbitone); haemodialysis (methanol, lithium, salicylates); repeated doses of activated charcoal for some drugs.
5. **Give a specific antidote** where one exists.

Poison	Antidote	How it works
Paracetamol	N-acetylcysteine	Restores glutathione to detoxify NAPQI
Opioids	Naloxone	Opioid receptor antagonist
Benzodiazepines	Flumazenil (with caution)	Benzodiazepine receptor antagonist
Organophosphates	Atropine and pralidoxime	Muscarinic block; reactivation of cholinesterase
Heparin	Protamine sulphate	Chemical neutralisation
Warfarin	Vitamin K; prothrombin complex concentrate	Restores clotting factors
Iron	Deferoxamine	Chelation
Lead, mercury, arsenic	Dimercaprol, succimer, calcium disodium EDTA, penicillamine (Chapter A4)	Chelation
Digoxin	Digoxin-specific antibody fragments (Fab)	Binding
Methanol, ethylene glycol	Fomepizole or ethanol	Block alcohol dehydrogenase
Cyanide	Hydroxocobalamin; nitrites with thiosulphate	Binding; conversion to thiocyanate
Atropine and anticholinergics	Physostigmine (selected cases)	Cholinesterase inhibition

AYURVEDA CORRELATION: VISHA, DUSHI VISHA AND TOXICITY OF RASADRAVYA

Agada Tantra, the branch of toxicology in Ayurveda, describes the nature of poisons and their treatment (Charaka Samhita, Chikitsa Sthana 23) [R26]. Several of its concepts correspond to modern toxicology.

Classical concept	Modern parallel
Visha (poison), Sthavara (plant and mineral) and Jangama (animal)	Toxicants of plant, mineral and animal origin
Dushi Visha: poison that remains in the body in a weakened form and causes disease later	Chronic, cumulative toxicity; storage in tissues
Gara Visha: harmful combinations or prepared poisons	Toxic combinations; interactions
Ashuddha, Apakva Rasadravya Sevanajanya Vikara: disorders caused by unpurified or improperly processed mineral drugs	Toxicity due to impurities, incomplete processing and free metal
Vikara Shanti Upaya: measures to counter these disorders	Antidotes and supportive treatment

The classical tradition of Rasashastra insists on Shodhana, Marana and Bhasma Pariksha precisely because it recognised the toxicity of raw metals and minerals.

Established. The classical texts describe acute and chronic poisoning, toxic combinations, and the toxicity of improperly processed mineral drugs, in terms that match modern concepts.

Not established for each product. That a particular Bhasma or Rasa preparation, made by a particular manufacturer, is free of toxicity at its classical dose. That depends on the raw material, the process and the final product, and must be shown by testing (Chapters A4 and A5).

KEY POINTS TO REMEMBER

- Drug effect is modified by the drug (dose, form, route, time), the patient (age, sex, weight, pregnancy, genes, disease, mind) and the environment (food, smoking, alcohol, time of day).
- Newborns: immature glucuronidation and kidney function (grey baby syndrome). Elderly: reduced renal function, polypharmacy, more sensitivity.
- Pharmacogenetics: acetylator status, CYP2D6, CYP2C19, G6PD deficiency, pseudocholinesterase deficiency, HLA-B*15:02 with carbamazepine.
- Dashavidha Pariksha (Prakriti, Vikriti, Sara, Samhanana, Pramana, Satmya, Satva, Ahara Shakti, Vyayama Shakti, Vaya) parallels the modern patient factors.
- Bioavailability F = fraction of dose reaching the systemic circulation; absolute F compares with IV; relative F compares two formulations.
- Bioequivalence: crossover study; 90% CI of test/reference ratio of AUC and Cmax within 80–125%.
- Interactions: pharmaceutical, pharmacokinetic (absorption, distribution, metabolism, excretion) and pharmacodynamic (additive, synergism, potentiation, antagonism).
- Herb–drug interactions with human evidence: piperine raises propranolol, theophylline and curcumin levels; gugulipid lowers propranolol and diltiazem levels; Shankhapushpi with phenytoin.
- ADR types A (augmented) to F (failure); Type A is dose-related and common; Type B is bizarre and often serious.
- Gell and Coombs: I immediate (IgE), II cytotoxic, III immune complex, IV delayed (T cell).
- Weeks 3 to 8 after conception are the most dangerous for teratogens.
- Poisoning: ABC first; identify; prevent absorption (charcoal); enhance elimination; antidote.

EXAM CORNER — UNIT A1(c)

Likely long questions (20 marks)

1. Discuss adverse drug reactions and drug toxicity. Explain the role of pharmacovigilance in the safety monitoring of Ayurvedic drugs *(set in March 2026; the pharmacovigilance part is in Chapter A6).*
2. Describe the factors modifying drug effect, with examples. Correlate them with the classical Dashavidha Pariksha.
3. Define bioavailability and bioequivalence. Describe the methods of assessing them and their importance for generic and Ayurvedic medicines.
4. Classify drug interactions with examples. Discuss herb–drug interactions relevant to Ayurvedic practice.
5. Define and classify adverse drug reactions. Describe drug allergy and teratogenicity.

Likely short notes (5 marks)

1. Factors modifying drug effect *(set in January 2024)*.
2. Drug–drug interactions, with examples relevant to Ayurvedic practice *(set in March 2026)*.
3. Drug interaction and adverse drug reaction *(set in July 2018)*.
4. Pharmacogenetics.
5. Placebo and nocebo.
6. Tolerance and tachyphylaxis.
7. Absolute and relative bioavailability.
8. Bioequivalence study design.
9. Enzyme induction and inhibition as causes of interaction.
10. Synergism and potentiation.
11. Types of adverse drug reaction (A to F).
12. Drug allergy.
13. Teratogenicity.
14. Principles of management of poisoning; antidotes.
15. Dushi Visha and Gara Visha as classical concepts of toxicity.

How to attempt

- **Factors modifying drug effect** is best answered as a classification table: drug factors, patient factors, environmental factors, each with an example. Add the Dashavidha Pariksha table for the Ayurvedic link.
- **ADR and toxicity** (20 marks): define, classify (A to F table), allergy (Gell and Coombs table), teratogenicity (stage diagram), toxicity types, management of poisoning with antidote table, and a section on Ayurvedic drugs (Dushi Visha, Ashuddha Rasadravya, need for testing).
- **Interactions "relevant to Ayurvedic practice"**: give the mechanism-wise classification first, then the herb–drug table with the source of each example, then the practical rules.

Answer frame: "Discuss ADR and drug toxicity. Explain the role of pharmacovigilance in the safety monitoring of Ayurvedic drugs." (20 marks)

Part	What to write	Suggested marks
Definitions	ADR (WHO), adverse event, side effect, toxicity	2
Classification of ADR	Types A to F (table); severity; seriousness	3
Special ADRs	Allergy (Gell and Coombs); idiosyncrasy; teratogenicity	3
Drug toxicity	Acute, subacute, chronic; organ toxicity; management of poisoning; antidotes	4
Pharmacovigilance	Definition; aims; reporting; causality; the national programme for ASU&H drugs (Chapter A6)	6
Ayurvedic perspective	Dushi Visha, Ashuddha Rasadravya; why testing and reporting are needed	2

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **26, 29, 30, 32**.

- **[R26]** Charaka Samhita: Vimana Sthana 8 (Dashavidha Pariksha); Chikitsa Sthana 30/298 (Bheshaja Kala); Sutra Sthana 26 (Viruddha Ahara); Chikitsa Sthana 23 (Visha). Verse numbers verified in Charak Samhita New Edition, carakasamhitaonline.com (accessed 16 October 2026).
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R30]** Brown MJ, Sharma P, Mir FA, Bennett PN. *Clinical Pharmacology*. 12th ed. Elsevier; 2018. (Listed in the syllabus as "Clinical Pharmacology, Lawrence Benette".)
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

Additional verified sources

- **[Edwards 2000]** Edwards IR, Aronson JK. Adverse drug reactions: definitions, diagnosis, and management. *Lancet*. 2000;356(9237):1255–1259.
- **[Shoba 1998]** Shoba G, Joy D, Joseph T, Majeed M, Rajendran R, Srinivas PS. Influence of piperine on the pharmacokinetics of curcumin in animals and human volunteers. *Planta Med*. 1998;64(4):353–356.
- **[Bano 1991]** Bano G, Raina RK, Zutshi U, Bedi KL, Johri RK, Sharma SC. Effect of piperine on bioavailability and pharmacokinetics of propranolol and theophylline in healthy volunteers. *Eur J Clin Pharmacol*. 1991;41:615–617.
- **[Dalvi 1994]** Dalvi SS and colleagues. Effect of gugulipid on bioavailability of diltiazem and propranolol. 1994. Indexed in IMSEAR (WHO South-East Asia Index Medicus), record sea-95588.
- **[Dandekar 1992]** Dandekar UP, Chandra RS, Dalvi SS, Joshi MV, Gokhale PC, Sharma AV, Shah PU, Kshirsagar NA. Analysis of a clinically important interaction between phenytoin and Shankhapushpi, an Ayurvedic preparation. *J Ethnopharmacol*. 1992;35(3):285–288.
- **[Aslam 1979]** Aslam M, Stockley IH. Interaction between curry ingredient (karela) and drug (chlorpropamide). *Lancet*. 1979;1:607.

UNIT A1(d)

PRECLINICAL EVALUATION AND ANIMAL ETHICS

SYLLABUS

Preclinical evaluation: experimental pharmacology [bioassay, in vitro, in vivo, cell line studies] animal ethics.

LEARNING OUTCOMES

After studying this unit you should be able to:

1. Place preclinical evaluation within the whole process of drug development.
2. List the pharmacological and toxicological studies required before a new drug is tested in human beings, with the relevant OECD and ICH guidelines.
3. Explain bioassay: its principles, types and designs.
4. Describe in vitro methods, including isolated tissues and cell line studies, with their uses and limits.
5. Design an in vivo study: choice of species and model, groups, controls, dose, randomisation, blinding, end-points and statistics.
6. Design a preclinical study of an Ayurvedic preparation, such as the immunomodulatory effect of Swarna Bhasma.
7. Explain the principles and the law of animal ethics in India: the 3Rs, CCSEA and the Institutional Animal Ethics Committee.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	A, Q.2	20	Explain the phases and protocol of Preclinical and Clinical evaluation of a new chemical entity.
Jan 2024	A, Q.1	20	Explain Pre – Clinical evaluation of drugs with its importance.
Mar 2026	A, Q.1	20	Explain in detail the animal model and study design to evaluate the in vivo immunomodulatory effect of Swarna Bhasma.

Every time this unit has been asked, it has been a 20-mark question. The March 2026 question is a model of what is coming: not “describe preclinical evaluation”, but “**design** the study”. Section A1d.6 works through that exact question.

A1d.1 Where preclinical evaluation fits

A new drug passes through a long sequence of stages before it reaches patients (Figure A1d.1). Most candidate molecules fail along the way, many of them in preclinical testing. The whole process commonly takes ten years or more.

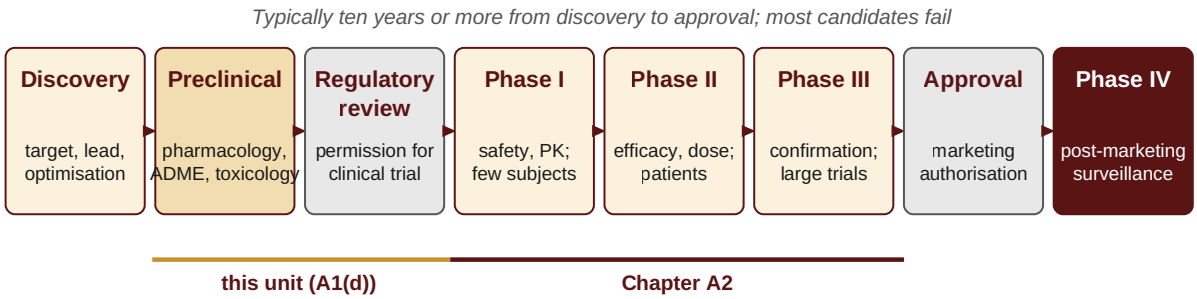


Figure A1d.1. Stages of drug development, from discovery to post-marketing surveillance.

> **Preclinical evaluation** is the testing of a new drug in the laboratory and in animals, before it is given to human beings, to establish its pharmacological activity, its pharmacokinetics and its safety, and to choose a safe starting dose for human trials.

Preclinical evaluation answers four questions.

1. **Does it work?** Primary pharmacodynamics: activity in relevant models.
2. **How does it work, and what else does it do?** Mechanism of action; secondary pharmacodynamics; safety pharmacology.
3. **What happens to it in the body?** Absorption, distribution, metabolism and excretion in animals.
4. **Is it safe enough to give to humans, and at what dose?** Toxicology, leading to the NOAEL and the starting dose (A1a.4.3).

A1d.2 Components of preclinical evaluation

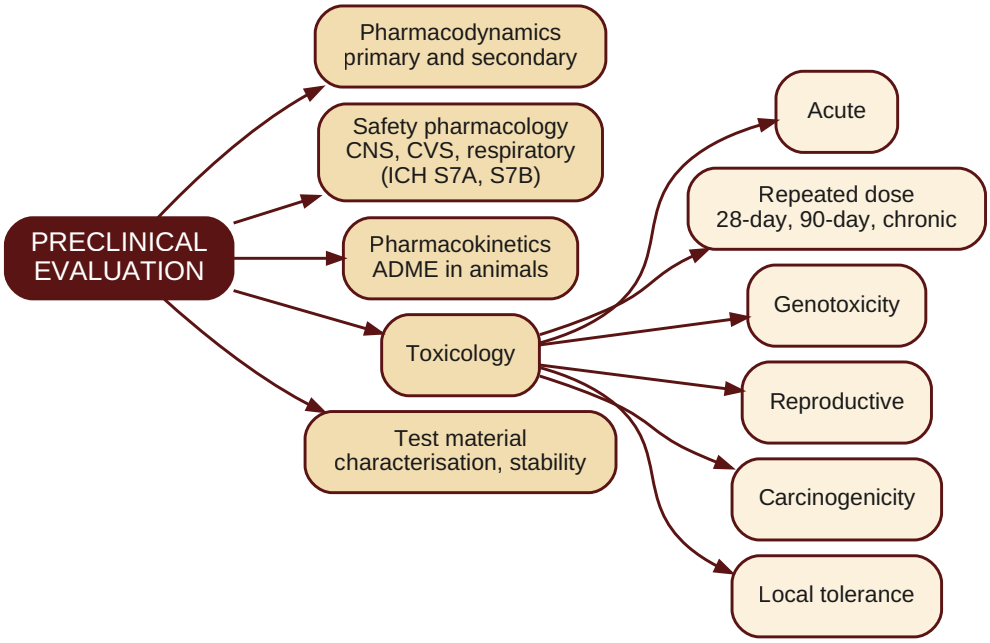


Figure A1d.2. Components of preclinical evaluation.

Area	Studies	Purpose
Primary pharmacodynamics	Screening and confirmation, using in vitro and in vivo models of the disease	Show the intended effect; find the effective dose range
Secondary pharmacodynamics	Effects on other targets and systems	Find unintended actions
Safety pharmacology	Core battery on the central nervous, cardiovascular and respiratory systems (ICH S7A); QT prolongation (ICH S7B)	Detect effects that could be dangerous in a first human dose
Pharmacokinetics (ADME)	Absorption, plasma levels, distribution, metabolites, excretion, protein binding; in two species	Link dose to exposure; choose species; predict human pharmacokinetics
Toxicology	Single-dose (acute), repeated-dose (subacute, subchronic, chronic), genotoxicity, reproductive toxicity, carcinogenicity, local tolerance, immunotoxicity	Identify target organs of toxicity, reversibility, and the NOAEL
Pharmaceutical development	Characterisation, stability and quality of the test material	Ensure that the material tested is the material that will be used

The timing of these studies relative to clinical trials is set out in the international guideline ICH M3(R2). Not every study is needed before the first human dose; carcinogenicity studies, for example, are usually done later, while the drug is in larger trials.

A1d.2.1 Good Laboratory Practice

Toxicology studies meant for regulatory submission must be done under **Good Laboratory Practice (GLP)**. GLP is a quality system covering the organisation of a laboratory and the conditions under which studies are planned, performed, monitored, recorded, archived and reported. Its purpose is to make the data reliable and traceable. The OECD Principles of GLP are the international standard. In India, laboratories are certified for GLP compliance by the National GLP Compliance Monitoring Authority under the Department of Science and Technology.

A1d.3 Experimental pharmacology

The syllabus names four methods: bioassay, in vitro, in vivo and cell line studies. They are not alternatives; a programme of study usually moves from the simplest and cheapest to the most complex.

Level	System	Strengths	Limits
In silico	Computer models, docking	Fast, cheap, no animals	Predictions only; must be confirmed
In vitro (cell-free)	Enzymes, receptors, chemical assays	Mechanism; high throughput	No absorption, metabolism or whole-body response
Cell lines	Cultured cells	Human cells possible; mechanism; cytotoxicity	Single cell type; artificial conditions
Isolated tissues	Organ bath (ileum, aorta, uterus)	Integrated tissue response; classic bioassay	Animal tissue; short-lived
In vivo	Whole animal	Absorption, metabolism, whole-body effects, toxicity	Species differences; cost; ethics

A1d.3.1 Bioassay

> A **bioassay** (biological assay) is the estimation of the potency of a substance by measuring its biological response in a living system, and comparing that response with the response to a standard preparation.

When a bioassay is needed

- When the active principle has no adequate chemical or physical assay;
- When the chemical assay does not reflect biological activity (for example, isomers or a mixture of related compounds);
- When the substance is present in very small amounts (hormones, autacoids in tissues);
- When a product is standardised in biological units (international units of insulin, heparin, vitamins, some vaccines and antibiotics).

Principles

1. The test preparation is always compared with a **standard preparation** of known potency, usually an international or national standard.
2. Test and standard must act by the **same mechanism**, so that their log dose–response lines are **parallel**.
3. Doses are chosen on the **linear part** of the log dose–response curve.
4. The order of doses is **randomised**, and enough responses are recorded to estimate error.
5. The result is expressed as the **relative potency** of the test against the standard, with its confidence limits.

Types

Type	Response	Example
Quantal (all-or-none)	Percentage of animals responding	Convulsions in mice for insulin (historical); lethality
Graded	Size of the response	Contraction of guinea pig ileum to histamine; fall in blood sugar in rabbits for insulin

Designs of graded bioassay

Design	Method
Matching (bracketing)	A dose of the test is matched against doses of the standard that give a slightly larger and a slightly smaller response. Simple; least precise.
Interpolation	A standard log dose–response line is drawn; the test response is read off it.
Three-point (2 + 1)	Two doses of standard and one of test.
Four-point (2 + 2)	Two doses each of standard and test, in a fixed ratio; parallel lines; potency from the horizontal distance between them (Figure A1d.3). The most used precise design.
Six-point (3 + 3)	Three doses of each; tests both parallelism and linearity.

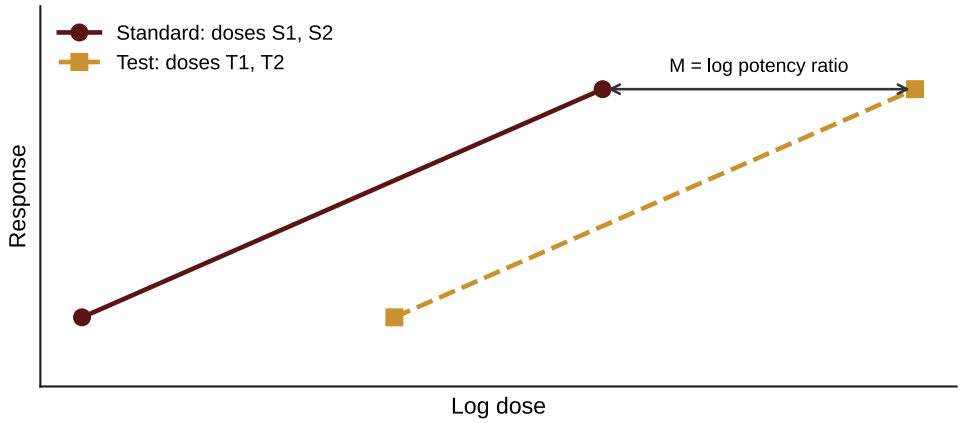


Figure A1d.3. Four-point bioassay: parallel log dose–response lines of standard and test; the horizontal distance gives the relative potency (schematic).

Classical tissue preparations used in bioassay and in isolated-tissue pharmacology are listed below. The tissue hangs in a warm, oxygenated physiological solution in an **organ bath**, and its contraction or relaxation is recorded (Figure A1d.4).

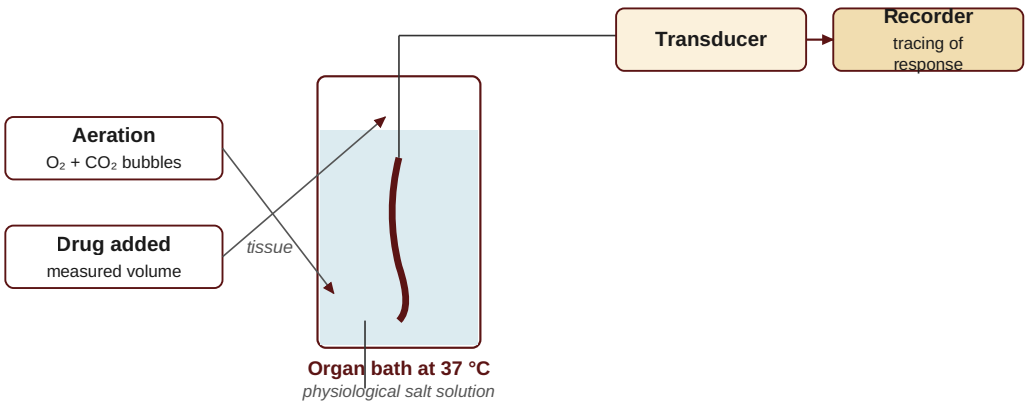


Figure A1d.4. The isolated organ bath.

Tissue	Agonist assayed or studied
Guinea pig ileum	Histamine, acetylcholine; antihistaminic and antispasmodic activity
Frog rectus abdominis	Acetylcholine
Rat uterus	Oxytocin
Rat or rabbit aorta	Vasoconstrictors and vasodilators
Rat fundus strip	5-hydroxytryptamine
Isolated heart (Langendorff)	Effects on heart rate and force

A1d.3.2 In vitro methods

In vitro (Latin, “in glass”) means outside a living organism. The main in vitro methods are:

Method	What is measured	Example
Enzyme assays	Inhibition or activation of a purified enzyme	Alpha-glucosidase or alpha-amylase inhibition (antidiabetic screening); COX inhibition; acetylcholinesterase inhibition
Receptor binding assays	Affinity of a compound for a receptor, often by displacement of a labelled ligand	Opioid or benzodiazepine receptor binding
Antimicrobial tests	Minimum inhibitory concentration (MIC) by broth dilution; zone of inhibition by agar diffusion	Antibacterial activity of a Kwatha or an essential oil
Antioxidant assays	Free-radical scavenging (DPPH, ABTS), reducing power, metal chelation	Comparing extracts or Bhasma samples
Isolated tissue studies	Contraction, relaxation	As in A1d.3.1
Dissolution and permeability	Release from a dosage form; passage across an artificial membrane or Caco-2 monolayer	Pre-formulation; bioavailability prediction

A CAUTION ABOUT IN VITRO SCREENS

A positive result in a test tube is the beginning of a question, not its answer. Antioxidant assays such as DPPH are easy and almost every plant extract is “active” in them, because plant extracts are rich in phenolic compounds. Activity at a high concentration in a test tube does not show that the same concentration can be reached in the body, nor that the effect occurs there. Many published papers on herbal drugs stop at this stage. Their conclusions should be read as *hypotheses*.

A1d.3.3 Cell line studies

A **cell line** is a population of cells that can be grown continuously in culture. Most come originally from a tumour or have been made “immortal” in the laboratory. **Primary cells**, taken directly from a tissue, are closer to the body’s own cells but survive only a short time.

Cell line	Origin	Common use
HepG2	Human liver cancer	Hepatotoxicity; drug metabolism (limited)
Caco-2	Human colon cancer; forms a gut-like monolayer	Intestinal permeability; absorption prediction
RAW 264.7	Mouse macrophage	Inflammation; immune activation
HeLa	Human cervical cancer (the first human cell line, 1951)	General cytotoxicity
MCF-7	Human breast cancer	Anticancer screening
Vero	African green monkey kidney	Toxicity; virology
HEK 293	Human embryonic kidney	Expression of receptors and transporters
3T3 fibroblasts	Mouse	Phototoxicity testing (3T3 NRU test)

A typical cytotoxicity study (Figure A1d.5): cells are seeded in a 96-well plate, exposed to a range of concentrations of the test substance, and after a fixed time their viability is measured, most often by the **MTT assay**, in which living cells reduce a yellow tetrazolium dye to a purple formazan [Mosmann 1983]. The concentration that reduces viability by half is the **IC50**.

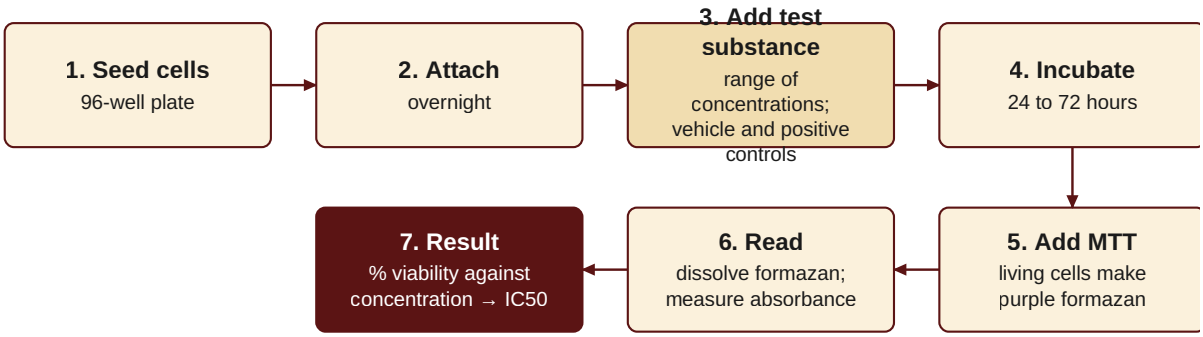


Figure A1d.5. Workflow of a cell-based cytotoxicity (MTT) assay.

Other cell-based end-points include proliferation, apoptosis (annexin V staining, caspase activity), cell migration, cytokine release, reactive oxygen species and gene or protein expression.

Good cell culture practice

- Confirm the identity of the cell line by short tandem repeat (STR) profiling. Many published studies have used misidentified or cross-contaminated lines.
- Test regularly for mycoplasma contamination.
- Record passage number; very high passages drift.
- Include vehicle and positive controls on every plate.
- Report a **selectivity index**: the IC50 on normal cells divided by the IC50 on the target cells. A substance that kills cancer cells and normal cells equally is a poison, not a lead.
- Ask whether the effective concentration could ever be reached in the blood or tissue of a patient.

A1d.3.4 In vivo methods

In vivo means in the living animal. In vivo studies are essential because they include absorption, distribution, metabolism, excretion and the response of the whole organism. They are also the most costly, slow and ethically sensitive.

Choice of species

Species	Typical use	Notes
Rat (Wistar, Sprague-Dawley)	Most pharmacology and toxicology; the standard rodent	Well characterised; enough blood for repeated sampling
Mouse (Swiss albino, BALB/c, C57BL/6)	Screening, immunology, genetic models	Small; cheap; many transgenic strains
Guinea pig	Allergy, histamine, respiratory and skin sensitisation studies	Sensitive to histamine
Rabbit	Pyrogen test (now largely replaced), eye and skin irritation (historical), teratology	
Dog, minipig, monkey	Non-rodent species for regulatory toxicology	Required for many new chemical entities
Zebrafish	Rapid toxicity and developmental screening	Embryos transparent; increasingly used

Common animal models for screening

Activity sought	Common models
Anti-inflammatory	Carrageenan-induced rat paw oedema (acute); cotton pellet granuloma (sub-acute); adjuvant arthritis (chronic)
Analgesic	Acetic acid writhing (peripheral); hot plate and tail flick (central)
Antipyretic	Brewer’s yeast-induced fever in rats
Antidiabetic	Alloxan- or streptozotocin-induced diabetes; high-fat diet with low-dose streptozotocin
Hepatoprotective	Carbon tetrachloride, paracetamol or alcohol-induced liver injury
Antiulcer	Pylorus ligation; ethanol-induced ulcer; aspirin-induced ulcer
Anticonvulsant	Maximal electroshock (MES); pentylenetetrazole (PTZ) seizures
Anxiolytic and nootropic	Elevated plus maze; Morris water maze; passive avoidance
Antihyperlipidaemic	High-fat diet; Triton WR-1339
Diuretic	Lipschitz test (urine volume and electrolytes in rats)
Immunomodulatory	Humoral response to sheep red blood cells; delayed-type hypersensitivity; carbon clearance; cyclophosphamide-induced immunosuppression

These and many others are described in standard reference works [R31].

Principles of good study design

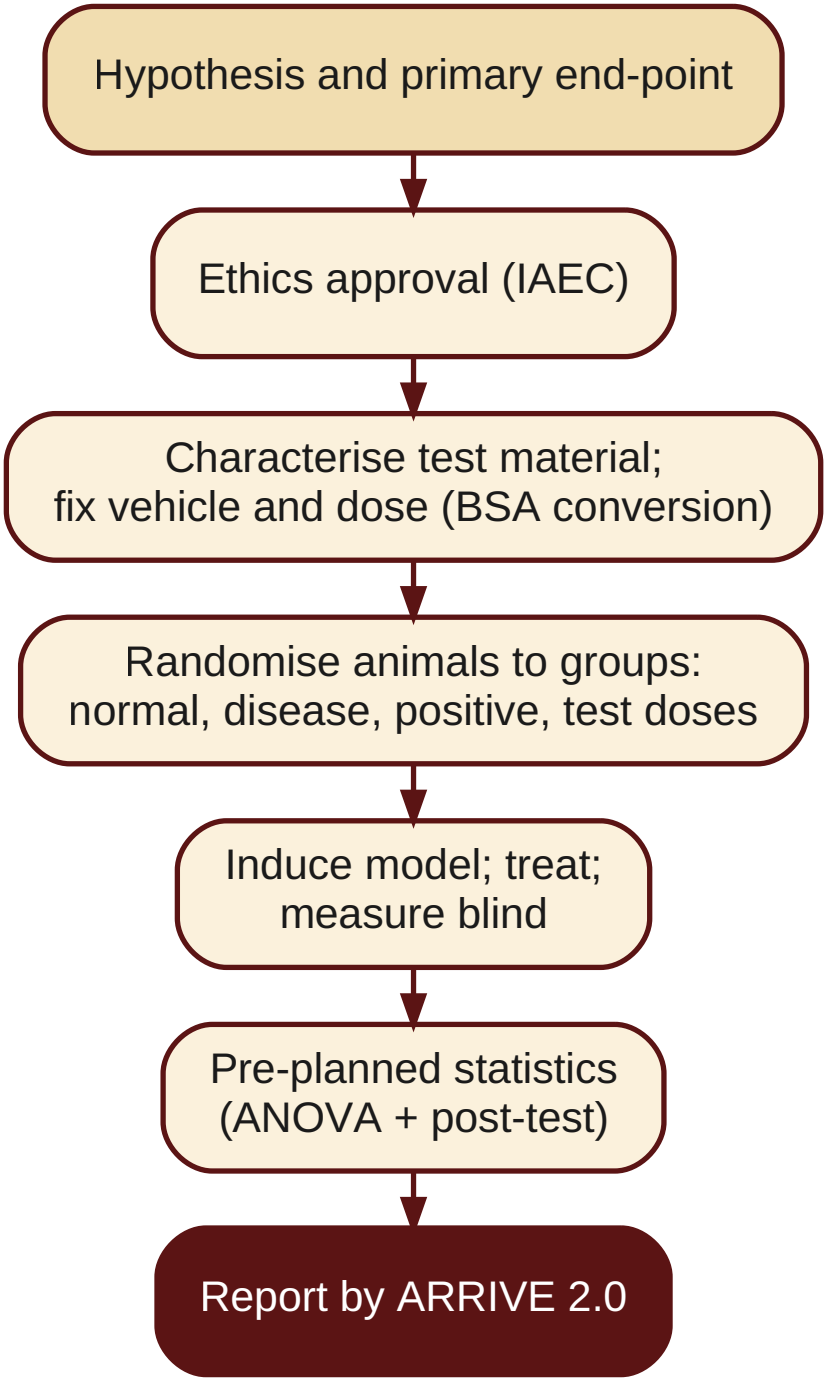


Figure A1d.6. Steps in the design of an in vivo pharmacological study.

Principle	What it means
A clear hypothesis	One primary question and one primary end-point, decided in advance
Controls	**Normal control** (no disease, vehicle only); **disease or vehicle control** (disease, vehicle only); **positive control** (disease plus a standard drug); **test groups** (disease plus test drug at two or three doses)
Characterised test material	Identity, composition and stability of the test drug are known and recorded; the same batch is used throughout

Principle	What it means
Dose selection	Based on the human dose converted by body surface area (A1a.4.3), with lower and higher multiples
Randomisation	Animals allocated to groups at random, not by convenience
Blinding	The person who measures the outcome does not know the group
Sample size	Justified by a power calculation, or at least by the “resource equation” for exploratory work; usually 6 or more per group
Statistics	Chosen in advance; for several groups, analysis of variance with a suitable post-test, not repeated t-tests
Reporting	Follow the ARRIVE 2.0 guidelines for reporting animal research [Percie du Sert 2020]

A1d.4 Toxicity studies

Toxicity studies identify the organs a drug harms, the dose at which harm begins, and whether the harm reverses. They proceed from single doses to long exposures (Figure A1d.7). Internationally accepted methods are the **OECD Guidelines for the Testing of Chemicals** [OECD TG].

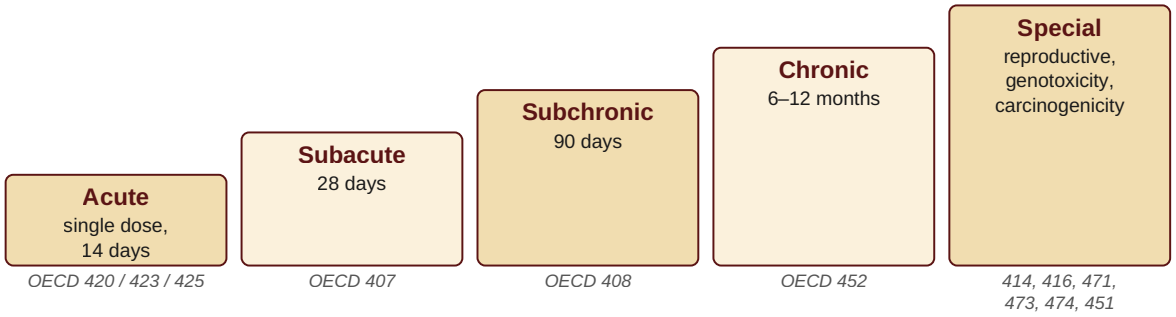


Figure A1d.7. The sequence of toxicity studies, from single dose to lifetime exposure.

Study	Design in brief	OECD Test Guideline
Acute oral toxicity	A single dose, observation for 14 days. Modern methods use few animals and estimate a toxicity class instead of an exact LD50. A “limit test” at 2000 mg/kg is used for substances expected to be of low toxicity.	420 (Fixed Dose Procedure), 423 (Acute Toxic Class), 425 (Up-and-Down Procedure)
Repeated-dose 28-day (subacute)	Daily dosing for 28 days in rodents, three dose levels and control	407
Repeated-dose 90-day (subchronic)	Daily dosing for 90 days in rodents; with a recovery group	408
Chronic toxicity	Usually 6 to 12 months or longer	452

Study	Design in brief	OECD Test Guideline
Reproductive and developmental	Effects on fertility, embryo-foetal development and offspring	414 (prenatal developmental), 416 (two-generation), 421 and 422 (screening)
Genotoxicity (a battery)	Bacterial reverse mutation (Ames) test; chromosomal aberration in cells; micronucleus test in animals	471, 473, 474
Carcinogenicity	Lifetime dosing (about two years in rats)	451
Local tolerance	Skin and eye irritation, sensitisation	404, 405, 406 and others

The classical LD50 test, which used many animals to calculate an exact lethal dose, was deleted from the OECD guidelines in 2002 and replaced by the methods above.

End-points recorded in repeated-dose studies: clinical signs and mortality; body weight and food intake; haematology; serum biochemistry (liver and kidney function, lipids, glucose, electrolytes); urine analysis; organ weights; gross pathology; histopathology of all major organs. The highest dose without adverse effect is the **NOAEL**.

AYURVEDA CORRELATION: PRECLINICAL SAFETY OF AYURVEDIC AND HERBO-MINERAL DRUGS

For Ayurvedic drugs the regulatory requirement for safety data depends on the category of the medicine. A classical formulation made as described in an authoritative book is treated differently from a new proprietary medicine or one that contains a drug from Schedule E(1). These requirements are set out in Rule 158-B of the Drugs Rules, 1945 and are explained in Chapter B8(i).

Herbo-mineral preparations need some special end-points beyond the routine ones:

- measurement of the metal in blood, urine and organs (liver, kidney, brain) by ICP-MS or atomic absorption, to see whether it accumulates;
- markers of oxidative stress and of kidney and liver function;
- histopathology of the kidney, liver and brain, the target organs of mercury, lead and arsenic;
- a recovery group, to see whether any change reverses after stopping;
- full characterisation of the test material (identity, Bhasma Pariksha, XRD, elemental analysis, particle size), because two Bhasma samples of the same name may differ.

Established. The methods of preclinical safety testing apply fully to Ayurvedic and herbo-mineral drugs, and many such studies have been published (Chapter A5).

To be remembered. A study shows the safety of the sample tested, made by that process, at those doses, for that duration. It does not show the safety of every product sold under the same name.

A1d.5 Preclinical evaluation of an Ayurvedic preparation: general approach

Ayurvedic drugs differ from a new chemical entity in three ways, and a preclinical plan must account for each.

Difference	Consequence for the study
They are used in humans already, often for centuries (“experience first”)	Human use is a source of hypotheses and of a dose; this is the basis of the **reverse pharmacology** approach, in which documented clinical experience leads to laboratory study rather than the reverse
They are complex mixtures, often with unknown active principles	The test material must be standardised and characterised; the same batch must be used throughout; markers or fingerprints recorded
They are given with specific Anupana and by specific methods	The vehicle and route in the animal should match clinical use as far as possible, and a vehicle-only group is essential

A1d.6 Worked design: the in vivo immunomodulatory effect of Swarna Bhasma

This is the question set in March 2026. It asks for two things: the **animal model** and the **study design**. The plan below shows how a complete answer is built. Any real study needs approval from the Institutional Animal Ethics Committee before it begins.

A1d.6.1 Background

Swarna Bhasma is a calcined preparation of gold used in Ayurveda as a Rasayana, among other indications, to promote strength and resistance to disease. Laboratory work on gold preparations used in Indian systems of medicine has reported augmentation of non-specific immunity in mice [Bajaj 2001]. A well-designed study tests this claim with standard models of immune function, a characterised sample, appropriate controls and safety end-points.

A1d.6.2 Objectives

- **Primary**: to evaluate the effect of Swarna Bhasma on humoral and cell-mediated immune responses in an immunosuppressed rodent model.
- **Secondary**: to assess its effect on phagocytic (non-specific) immunity, on haematological recovery, and on gold accumulation and organ safety.

A1d.6.3 Test material

Item	What to record
Source and batch	Prepared in-house by a stated classical method, or a commercial batch with manufacturer, batch number and date
Classical tests	Bhasma Pariksha (Rekhapurnatva, Varitaratva, Nishchandratva, Apunarbhava and others)
Modern characterisation	Gold content (ICP-OES or AAS), other elements including heavy metals, XRD phase, particle size (SEM or DLS)
Vehicle (Anupana)	The vehicle used clinically, for example honey or ghee, or a neutral suspending agent; the same vehicle in the control groups

A1d.6.4 Animals and dose

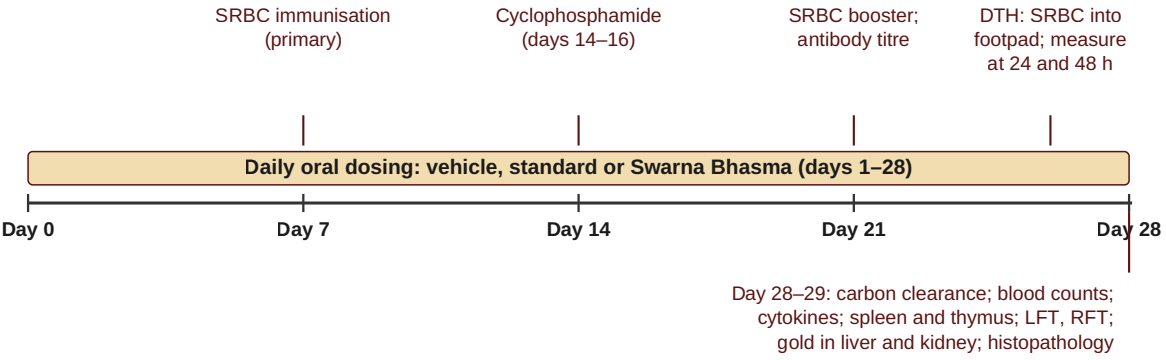
- **Species**: Wistar rats (or Swiss albino mice), one sex or both sexes analysed separately, of a stated weight range, from a CCSEA-registered breeder.
- **Housing**: standard conditions as laid down by CCSEA; acclimatisation for at least 7 days.
- **Dose**: the classical human dose converted to a rat dose by body surface area. For a human dose of *D* mg per day for a 60 kg adult: rat dose (mg/kg) = (*D* ÷ 60) × 6.2 (A1a.4.3). Use this as the middle dose, with one lower and one higher dose (for example half and twice).
- **Route and duration**: oral, by gavage, once daily for 28 days.

A1d.6.5 Groups

Group	Treatment (n = 6 or more each, by power calculation)
I	Normal control: vehicle only
II	Immunosuppressed control: vehicle + cyclophosphamide
III	Positive control: a standard immunomodulator + cyclophosphamide
IV	Swarna Bhasma, low dose + cyclophosphamide
V	Swarna Bhasma, medium dose (human equivalent) + cyclophosphamide
VI	Swarna Bhasma, high dose + cyclophosphamide
VII (optional)	Swarna Bhasma, medium dose alone (effect in healthy animals)

Cyclophosphamide is used because its immunosuppressive effect is well characterised; the model tests whether the drug restores suppressed immunity. A design without immunosuppression tests whether the drug enhances normal immunity. State which question the study asks.

A1d.6.6 Timeline and immune tests



An example design only: timings vary between published protocols.

Figure A1d.8. Timeline of a 28-day immunomodulation study in rats (an example design).

Immune function	Test	What is measured
Humoral (antibody) immunity	Immunisation with sheep red blood cells (SRBC); haemagglutination antibody titre	Highest dilution of serum that agglutinates SRBC
	Plaque-forming cell assay (optional)	Antibody-forming cells in the spleen
Cell-mediated immunity	Delayed-type hypersensitivity (DTH): SRBC challenge in the footpad	Increase in footpad thickness at 24 and 48 hours
Non-specific (phagocytic) immunity	Carbon clearance test	Rate of removal of injected colloidal carbon from blood; phagocytic index
Bone marrow and blood	Total and differential leucocyte counts; recovery from cyclophosphamide-induced myelosuppression	Neutrophils, lymphocytes
Lymphoid organs	Weight and histology of spleen and thymus	Cellularity; germinal centres
Molecular (where facilities exist)	Serum cytokines by ELISA (for example IL-2, IFN-γ, IL-4, TNF-α); lymphocyte subsets (CD4+, CD8+) by flow cytometry	Th1 / Th2 balance; T-cell subsets

A1d.6.7 Safety end-points

Body weight; food intake; liver function tests; kidney function tests (urea, creatinine); haematology; histopathology of liver, kidney and spleen; **gold content of liver and kidney** by ICP-MS, to look for accumulation.

A1d.6.8 Analysis, ethics and reporting

- Outcomes analysed by one-way analysis of variance with a post-hoc test (for example Dunnett's against the immunosuppressed control), or non-parametric tests where data are not normally distributed; significance at $p < 0.05$.
- Randomisation and blinding of the person measuring titres and footpad thickness.
- IAEC approval before the study; humane end-points; euthanasia by an approved method.
- Report by ARRIVE 2.0 [Percie du Sert 2020].

A1d.6.9 What a positive result would and would not show

A positive result would show	It would not show
That this sample, at these doses, restored immune measures in this rat model	That Swarna Bhasma prevents infection or improves immunity in patients
A plausible dose range for further work	The mechanism, unless studied specifically
Safety over 28 days at these doses, for the end-points measured	Long-term safety, or safety of other samples

The next steps would be mechanistic work, a longer safety study and, finally, a properly designed clinical trial (Chapter A2).

A1d.7 Animal ethics

A1d.7.1 Why animal ethics matters

Animals used in research can feel pain and distress. Their use is justified only when the knowledge sought is important, cannot be obtained without animals, and is obtained with the least possible harm. Poorly designed experiments are also unethical: if a study is too small, badly controlled or unblinded, the animals have suffered for results that cannot be trusted.

A1d.7.2 The 3Rs

In 1959 Russell and Burch set out the principle of the **three Rs**, which remains the foundation of animal ethics worldwide [Russell 1959] (Figure A1d.9).

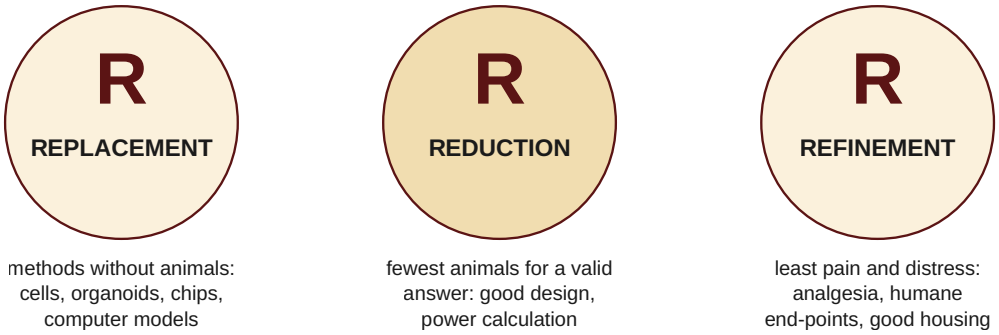


Figure A1d.9. The 3Rs of animal research.

R	Meaning	Examples
Replacement	Use methods that avoid animals where possible	Cell lines, organoids, organs-on-chips, computer models, lower organisms
Reduction	Use the minimum number of animals that gives valid results	Good design; power calculation; sharing of data and controls; fixed-dose and up-and-down methods instead of the classical LD50
Refinement	Reduce pain and distress, and improve welfare	Anaesthesia and analgesia; humane end-points; good housing and handling; trained staff

A1d.7.3 The law in India

> **LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)**
>
> The Committee for the Purpose of Control and Supervision of Experiments on Animals (**CPCSEA**), named in older books, was renamed the Committee for Control and Supervision of Experiments on Animals (**CCSEA**) in January 2023. Its functions continue.

Instrument	What it provides
Prevention of Cruelty to Animals Act, 1960	The parent Act. Chapter IV deals with experimentation on animals and provides for a committee to control and supervise experiments.
Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998 (with later amendments)	Registration of establishments that breed or use animals; Institutional Animal Ethics Committees; standards of housing and care; records
CCSEA	The Committee for Control and Supervision of Experiments on Animals (formerly CPCSEA; renamed in January 2023), under the Ministry of Fisheries, Animal Husbandry and Dairying. It registers establishments, nominates members to institutional committees, inspects facilities, and issues guidelines.
New Drugs and Clinical Trials Rules, 2019	Set out the preclinical data required before clinical trials of new drugs. An amendment in March 2023 was reported to allow validated non-animal, human-relevant methods (such as organoids and organs-on-chips) in place of animal studies where suitable.

A1d.7.4 The Institutional Animal Ethics Committee (IAEC)

Every establishment that uses animals for experiments must register with CCSEA and constitute an IAEC. No experiment may begin without its approval (Figure A1d.10).

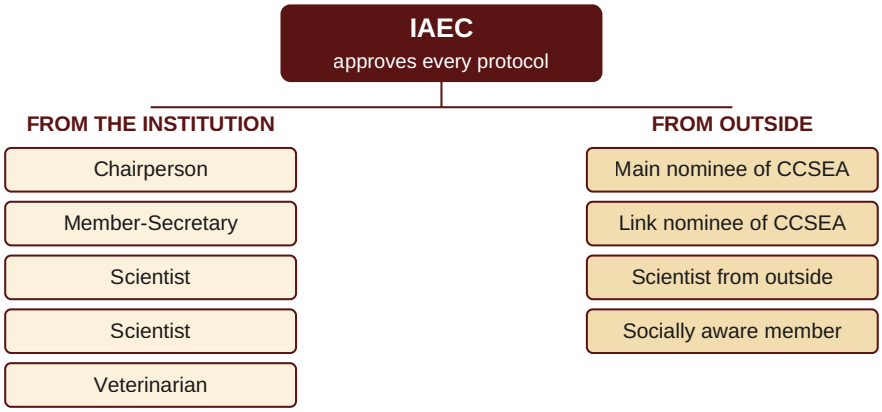


Figure A1d.10. Composition of an Institutional Animal Ethics Committee, as commonly constituted under CCSEA.

Members from within the institution	Members from outside
Chairperson	Main nominee of CCSEA
Member-Secretary	Link nominee of CCSEA
Two scientists	A scientist from outside the institution
A veterinarian	A socially aware member (non-scientist)

Functions of the IAEC

1. Review and approve, modify or reject every protocol before it starts, considering the purpose, the necessity of animals, the species, the numbers, the procedures, pain relief and humane end-points.
2. Inspect the animal house and the experiments.
3. Ensure that the investigators are trained.
4. Maintain records and report to CCSEA.

Large animals (such as dogs and monkeys) and certain other categories of experiment need approval from CCSEA itself, not only from the IAEC.

A1d.7.5 Welfare in practice

- Housing appropriate to the species: space, temperature, humidity, light–dark cycle, bedding, food and water.
- Handling and dosing by trained persons; the smallest gavage volume that is practical.
- Anaesthesia and analgesia for painful procedures.
- **Humane end-points**: an animal that reaches a defined severity of illness is removed from the study and euthanised, rather than left to die.
- **Euthanasia** by an approved method, such as an overdose of an anaesthetic agent.
- Proper disposal of carcasses and biological waste.

A1d.7.6 Moving beyond animals

Regulators are moving towards non-animal methods. In the United States, the FDA Modernization Act 2.0 of 2022 removed the legal requirement that every new drug be tested in animals before human trials. In India, the 2023 amendment of the New Drugs and Clinical Trials Rules was reported to open the same door. Animal studies will remain part of drug development for many years, but a postgraduate scholar should know the alternatives and use them where they are valid.

AYURVEDA CORRELATION

Ayurvedic ethics places Ahimsa (non-injury) among the duties of the physician, and the classical texts lay great weight on compassion for all living beings. The 3Rs express the same value in practical terms: injury to animals is accepted only when necessary, only as much as necessary, and with all possible care.

Practical point. In Ayurvedic research the “replacement” R has special force. For many classical formulations, long human use, careful clinical documentation and well-designed observational studies can supply information that would otherwise be sought in animals. This is not a licence to skip safety testing where it is needed, especially for herbo-mineral drugs. It is a reason to design every animal study so that it answers a question that cannot be answered another way.

KEY POINTS TO REMEMBER

- Preclinical evaluation establishes activity, pharmacokinetics and safety, and gives the NOAEL from which the first human dose is calculated.
- Components: primary and secondary pharmacodynamics, safety pharmacology (ICH S7A, S7B), ADME, toxicology, and characterisation of the test material. Regulatory toxicology is done under GLP.

- Bioassay compares a test with a standard in a living system; lines must be parallel; the four-point (2 + 2) design is the most used.
- In vitro: enzyme, receptor, antimicrobial (MIC), antioxidant and isolated tissue studies. A positive in vitro result is a hypothesis.
- Cell lines: HepG2, Caco-2, RAW 264.7, HeLa, MCF-7, Vero; MTT gives IC50; check identity (STR), mycoplasma and selectivity.
- In vivo design: hypothesis, normal, disease, positive and test controls, dose by body surface area, randomisation, blinding, sample size, ANOVA, ARRIVE 2.0.
- Toxicity: acute (OECD 420, 423, 425), 28-day (407), 90-day (408), chronic (452), reproductive (414, 416, 421), genotoxicity (471, 473, 474), carcinogenicity (451). The classical LD50 test was deleted in 2002.
- Herbo-mineral drugs need metal estimation in organs, kidney and liver end-points, recovery groups and full characterisation of the sample.
- Swarna Bhasma immunomodulation: cyclophosphamide model; SRBC antibody titre (humoral), DTH (cell-mediated), carbon clearance (phagocytic), blood counts, lymphoid organs, cytokines; gold in organs.
- 3Rs: Replacement, Reduction, Refinement (Russell and Burch, 1959). Law: PCA Act 1960; 1998 Rules; CCSEA; IAEC.

EXAM CORNER — UNIT A1(d)

Likely long questions (20 marks)

1. Explain preclinical evaluation of drugs with its importance *(set in January 2024)*.
2. Explain the phases and protocol of preclinical and clinical evaluation of a new chemical entity *(set in September 2018; the clinical part is in Chapter A2)*.
3. Explain in detail the animal model and study design to evaluate the in vivo immunomodulatory effect of Swarna Bhasma *(set in March 2026)*.
4. Describe the toxicity studies required for a new drug. Add a note on the special requirements for herbo-mineral preparations.
5. Discuss experimental pharmacology: bioassay, in vitro, in vivo and cell line studies, with their merits and limitations.
6. Discuss animal ethics: the 3Rs, the law in India, and the constitution and functions of the IAEC.
7. Design a preclinical study to evaluate the hepatoprotective (or anti-inflammatory, or antidiabetic) activity of an Ayurvedic formulation.

Likely short notes (5 marks)

1. Principles of bioassay.
2. Four-point bioassay.
3. Isolated organ bath.
4. MTT assay and IC50.
5. Caco-2 cell model.
6. Acute oral toxicity: OECD 423 or 425.
7. Repeated-dose toxicity studies.
8. Genotoxicity battery.

9. Good Laboratory Practice.
10. The 3Rs.
11. CCSEA and the IAEC.
12. Reverse pharmacology.
13. Carbon clearance test.
14. Delayed-type hypersensitivity test.

How to attempt

- For **"preclinical evaluation with its importance"**: begin with the drug development diagram (Figure A1d. 1), define preclinical evaluation, list its four aims, then give the components table, the toxicity table with OECD numbers, a paragraph on GLP, and a paragraph on animal ethics. Close with the importance: safety of the first human dose, selection of dose, prediction of target organs, regulatory requirement, and the saving of human subjects from avoidable harm.
- For a **study design question**, write it as a protocol, under headings: title, objectives, test material and its characterisation, animals, ethics approval, dose and its calculation, groups (table), procedure and timeline (diagram), parameters (table), safety end-points, statistics, expected outcome and limitations. Examiners reward a design that could actually be carried out.

Answer frame: "Animal model and study design for the in vivo immunomodulatory effect of Swarna Bhasma." (20 marks)

Part	What to write	Suggested marks
Introduction	Swarna Bhasma as Rasayana; immunomodulation defined; rationale	1
Objectives	Primary and secondary	1
Test material	Preparation or source, Bhasma Pariksha, ICP/XRD/particle size, vehicle	2
Animals and ethics	Species, housing, IAEC approval under CCSEA	2
Dose	Human-to-rat conversion by body surface area; three doses	2
Groups	Normal, immunosuppressed, positive control, three test doses (table)	3
Model and tests	Cyclophosphamide; SRBC titre, DTH, carbon clearance, blood counts, organs, cytokines (table and timeline)	5
Safety	LFT, RFT, histopathology, gold in organs	1
Statistics and reporting	ANOVA with Dunnett's test; blinding; ARRIVE 2.0	1
Interpretation	What a positive result would and would not show	2

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **29, 31, 32**.

- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R31]** Vogel HG, editor. *Drug Discovery and Evaluation: Pharmacological Assays*. 3rd ed. Berlin: Springer; 2008. (A 4th edition, edited by Hock FJ, was published by Springer in 2016.)

- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

Additional verified sources

- **[OECD TG]** OECD. *Guidelines for the Testing of Chemicals, Section 4: Health Effects*. Paris: OECD Publishing (individual Test Guidelines as numbered in the text).
- **[Mosmann 1983]** Mosmann T. Rapid colorimetric assay for cellular growth and survival: application to proliferation and cytotoxicity assays. *J Immunol Methods*. 1983;65(1–2):55–63.
- **[Percie du Sert 2020]** Percie du Sert N, Hurst V, Ahluwalia A, et al. The ARRIVE guidelines 2.0: updated guidelines for reporting animal research. *PLoS Biol*. 2020;18(7):e3000410.
- **[Russell 1959]** Russell WMS, Burch RL. *The Principles of Humane Experimental Technique*. London: Methuen; 1959.
- **[Bajaj 2001]** Bajaj S, Ahmad I, Raisuddin S, Vohora SB. Augmentation of non-specific immunity in mice by gold preparations used in traditional systems of medicine. *Indian J Med Res*. 2001;113:192 ff.
- ICH guidelines S7A (safety pharmacology), S7B (QT prolongation) and M3(R2) (timing of non-clinical studies). International Council for Harmonisation.
- The March 2023 amendment of the New Drugs and Clinical Trials Rules, 2019 is cited as reported in the press (accessed 16 October 2026); the official notification should be consulted for its exact wording.

Viva pointers — Chapter A1

These are the questions most often asked at the viva voce on General Pharmacology. Answer each in one or two sentences.

On principles and pharmacokinetics

- What is the difference between pharmacokinetics and pharmacodynamics?
- Define a drug. Is water a drug? Is a Bhasma a drug?
- What does LADME stand for? Which step decides the fate of a tablet?
- Why is aspirin absorbed from the stomach even though most of it is absorbed in the intestine?
- What is first-pass metabolism? Name a drug that is given sublingually to avoid it.
- A drug has a volume of distribution of 500 litres. What does that tell you?
- Which plasma protein binds acidic drugs?
- Why does thiopentone act for only a few minutes?
- Name two enzyme inducers and two enzyme inhibitors.
- What is a prodrug? Give an example.
- Which conjugation reaction is commonest? Why is chloramphenicol dangerous in the newborn?
- What is enterohepatic circulation?
- How many half-lives does it take to reach steady state?
- Which drugs follow zero-order kinetics?
- How do you calculate a loading dose?

On pharmacodynamics and dose

- What is the difference between affinity and intrinsic activity?
- Give an example of a partial agonist and of an inverse agonist.
- Name the four families of receptors. Which is the fastest?
- What is the second messenger of a β receptor?
- What is tachyphylaxis? Give an example.
- What is the difference between potency and efficacy? Which matters more?
- Draw a log dose–response curve and mark EC50.
- What happens to the agonist curve in the presence of a competitive antagonist?
- Define therapeutic index. Name three drugs with a narrow therapeutic index.
- How is the first human dose of a new drug calculated from animal data?
- State Young’s formula.
- What is a pharmacophore? What is the rule of five?
- How does Shodhana reduce the toxicity of Vatsanabha?
- What is a Yogavahi? Is there evidence for it?

On routes of administration

- Why does magnesium sulphate act differently by mouth and by injection?
- Which route has 100 per cent bioavailability?

- Why is glyceryl trinitrate given under the tongue?
- Why does the rectal route only partly avoid first-pass metabolism?
- Why is the upper outer quadrant chosen for gluteal injections?
- What particle size reaches the bronchioles?
- What properties must a drug have to be given transdermally? Name three transdermal drugs.
- How can eye drops slow the heart?
- What is the modern counterpart of Nasya, of Basti, of Dhumapana? What is known about absorption by each?

On modifying factors, interactions and ADRs

- Why is chloramphenicol dangerous in the newborn?
- What is pharmacogenetics? Give two examples.
- Name the ten factors of Dashavidha Pariksha.
- Define bioavailability. How is absolute bioavailability calculated?
- What are the acceptance limits for bioequivalence?
- Can two brands of an Arishta be called bioequivalent? Why not?
- Give an example of an enzyme-induction interaction and of an enzyme-inhibition interaction.
- Name one herb–drug interaction shown in a human study.
- What is the difference between synergism and potentiation?
- Define an adverse drug reaction (WHO). What are Type A and Type B reactions?
- Name the four types of drug allergy with one example each.
- Which weeks of pregnancy are most dangerous for teratogens?
- What is the antidote for paracetamol, for opioids, for organophosphates, for iron?
- What is Dushi Visha?

On preclinical evaluation and animal ethics

- What are the aims of preclinical evaluation?
- What is safety pharmacology? Which systems are in the core battery?
- What is GLP? Who certifies GLP laboratories in India?
- What is a bioassay? Why must the lines of standard and test be parallel?
- Name three tissues used in an organ bath and the agonists studied on them.
- What does the MTT assay measure?
- Why is the selectivity index important in cell line studies?
- Name the OECD guidelines for acute oral toxicity.
- What is the NOAEL?
- Which models would you use to test an immunomodulator in rats?
- What are the 3Rs? Who proposed them?
- What does CCSEA stand for? Who are the external members of an IAEC?

PART A · PHARMACOLOGY · CHAPTER A2

A2 · CLINICAL PHARMACOLOGY

SYLLABUS POINT (PART A, 2)

Clinical pharmacology: Evaluation of New Chemical Entity – phases and methods of clinical research. Ethics involved in human research.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A2	Clinical Pharmacology	3	2	Jul 2018, Sep 2018, Sep 2020, Aug 2021, Aug 2022

Why this chapter matters to you. Clinical research is the only way to know whether a medicine helps patients. Every claim made for an Ayurvedic formulation, however old, is tested in the end by a clinical study. Your own MD dissertation may be one. This chapter has been set in five of the seven papers examined, three times as a 20-mark question.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define clinical pharmacology and describe the pathway by which a new chemical entity is evaluated and approved in India.
2. Describe the phases of clinical research (0 to IV) with their aims, subjects, size and design.
3. Classify clinical study designs and describe the randomised controlled trial with its key elements.
4. List the contents of a clinical trial protocol.
5. Plan a clinical study of a new Ayurvedic formulation, with its special methodological and regulatory requirements.
6. Explain the ethics of human research: its history, principles, informed consent, ethics committees, and the Indian rules and guidelines.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	A, Q.3a	5	Ethics involved in human research.
Sep 2018	A, Q.2	20	Explain the phases and protocol of Preclinical and Clinical evaluation of a new chemical entity.
Sep 2020	A, Q.3b	5	Phases of clinical research
Aug 2021	A, Q.2	20	Explain in detail about Phases and Methods of Clinical Research and brief about ethics involved in human research
Aug 2022	A, Q.1	20	Explain phases and methods of Clinical research for newer formulation.

A2.1 Clinical pharmacology

> **Clinical pharmacology** is the scientific study of drugs in human beings, healthy and ill. It covers pharmacokinetics and pharmacodynamics in humans, the evaluation of new drugs in clinical trials, the optimal use of drugs in patients, adverse effects, interactions and the economics of drug use.

The syllabus names two parts of it: the **evaluation of a new chemical entity (NCE)**, through the phases and methods of clinical research, and the **ethics** of research on human beings.

A **new chemical entity** is a molecule not previously approved as a drug. Under Indian law, a “new drug” also includes an already-approved drug proposed for a new claim, a new indication, a new dosage form, a new route or a new fixed-dose combination.

A2.2 Evaluation of a new chemical entity: the pathway in India

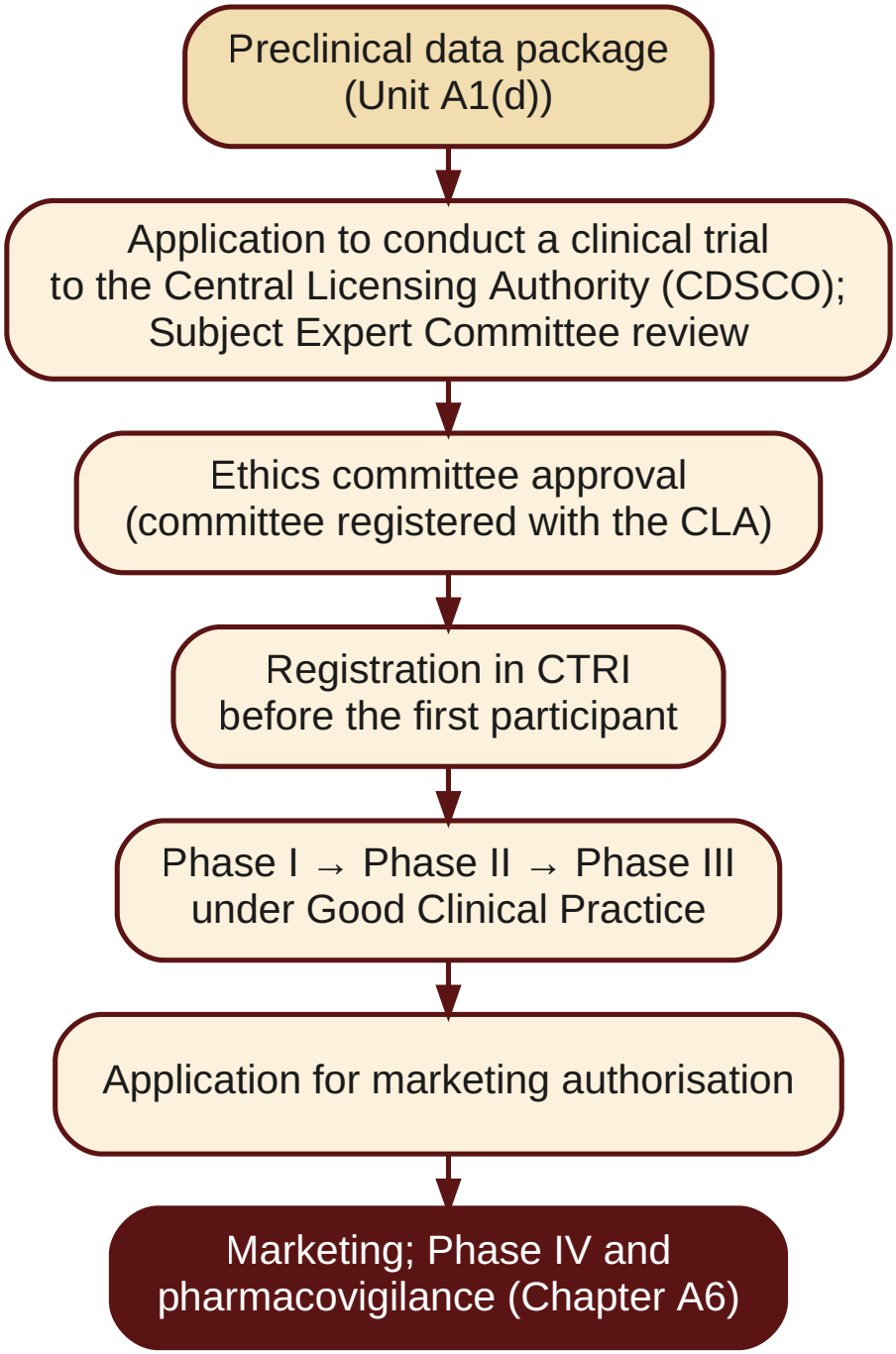


Figure A2.1. Regulatory pathway for clinical evaluation of a new drug in India.

Step	What happens	Authority
1. Preclinical package	Pharmacology, ADME and toxicology data (Unit A1(d))	Sponsor
2. Application for permission to conduct a clinical trial	Protocol, investigator's brochure, preclinical data, informed consent documents, ethics committee approval	Central Licensing Authority (the Drugs Controller General of India, CDSCO), with advice from a Subject Expert Committee
3. Ethics committee approval	Review of protocol, consent, investigators and site	Ethics committee registered with the Central Licensing Authority
4. Trial registration	Public registration before the first participant is enrolled	Clinical Trials Registry – India (CTRI)
5. Phases I to III	Conduct under Good Clinical Practice; reporting of serious adverse events	Sponsor, investigators, ethics committee, regulator
6. Application for marketing authorisation	Full data on quality, safety and efficacy	Central Licensing Authority
7. Phase IV and pharmacovigilance	Post-marketing studies and surveillance (Chapter A6)	Marketing authorisation holder, regulator

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

Older books describe clinical trials in India under Schedule Y of the Drugs and Cosmetics Rules. Since 19 March 2019, new drugs and clinical trials are governed by the **New Drugs and Clinical Trials Rules, 2019**, which replaced Part XA and Schedule Y of the Drugs and Cosmetics Rules [CAM 2019].

The governing law is the **New Drugs and Clinical Trials Rules, 2019**, made under the Drugs and Cosmetics Act, 1940 and notified on 19 March 2019 (G.S.R. 227(E)). Some of its features:

- For a new drug discovered, or researched and developed, in India and to be manufactured and marketed in India, the Central Licensing Authority must decide an application to conduct a clinical trial within **30 working days**; if it does not, permission is deemed to have been granted (Rule 23) [NDCT 2019].
- Ethics committees that review clinical trials must be **registered with the Central Licensing Authority**; registration is valid for five years [NDCT 2019].
- A participant injured in a trial is entitled to **free medical management** for as long as required, and to **compensation** for trial-related injury or death, calculated by a formula given in the **Seventh Schedule** (Chapter VI) [NDCT 2019].

A2.3 Phases of clinical research

A new drug is tested in humans in phases. Each phase asks a different question, uses a different number and kind of participant, and must succeed before the next begins (Figure A2.2).

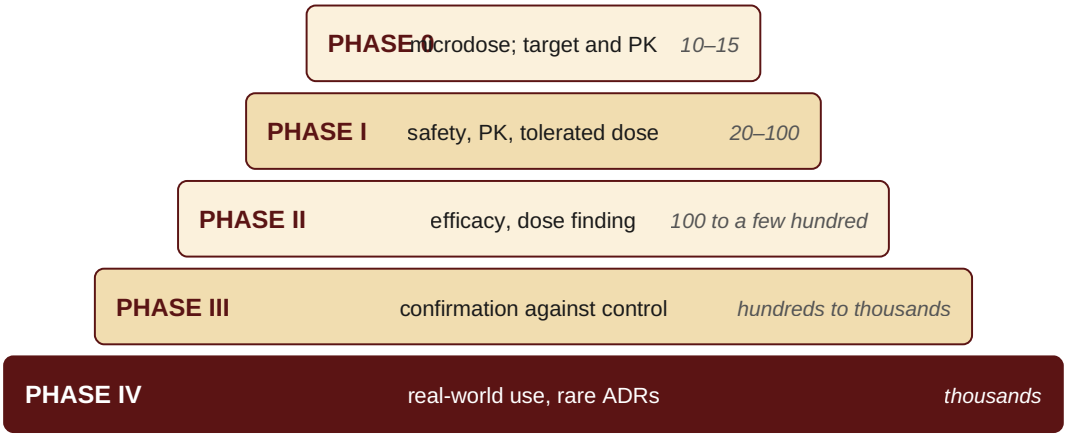


Figure A2.2. The phases of clinical research: fewer participants and simpler questions at the start; more participants and harder questions later.

Phase	Main question	Participants	Typical number	Typical design	Typical duration
0 (exploratory)	Does the drug reach its target, and how is it handled, at a tiny dose?	Healthy volunteers or patients	10–15	Microdose (a very small fraction of the expected active dose); no therapeutic intent	Days
I	Is it safe? How is it absorbed, distributed and eliminated? What dose is tolerated?	Healthy volunteers (patients for cytotoxic drugs)	20–100	Open or placebo-controlled; single and multiple ascending doses	Months
II	Does it work? At what dose?	Patients with the disease	100 to a few hundred	Randomised, controlled; dose-ranging (IIa: proof of concept; IIb: dose finding)	Months to 2 years
III	Is it better than, or as good as, existing treatment, in a large and varied population?	Patients	Several hundred to several thousand	Randomised, double-blind, multicentre; superiority or non-inferiority	1–4 years
IV	What happens in real use: rare and late adverse effects, new uses, special groups?	Patients in routine care	Thousands	Observational studies, surveillance, further trials	Continues for the life of the drug

A2.3.1 Phase 0

Phase 0 studies give a **microdose**, too small to have any effect, to a few people, to check pharmacokinetics or target binding early. They help to drop unpromising molecules before costly trials. They do not test safety or efficacy.

A2.3.2 Phase I: first in human

- **Aims**: safety and tolerability; pharmacokinetics; pharmacodynamic markers; the maximum tolerated dose.

- **Subjects**: usually healthy adult volunteers. For anticancer drugs, which are too toxic for healthy people, patients are used.
- **Starting dose**: the maximum recommended starting dose calculated from animal data (A1a.4.3).
- **Designs**: **single ascending dose (SAD)** cohorts, then **multiple ascending dose (MAD)** cohorts; food-effect and interaction studies.
- **Dose escalation**: in anticancer trials the classic **3 + 3 design** is used (Figure A2.3).
- **Safety measures**: sentinel dosing (one or two subjects dosed first, the rest only after a safety interval); close monitoring in a clinical pharmacology unit. These measures became standard after a 2006 first-in-human trial in London in which all six volunteers given an experimental antibody fell gravely ill at the same time.

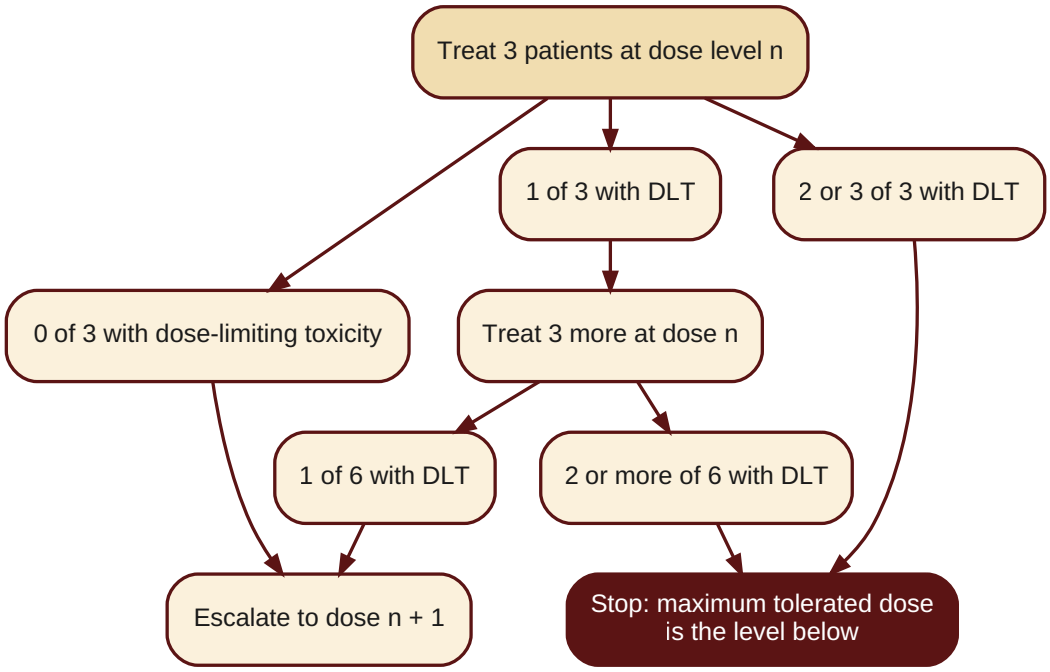


Figure A2.3. The 3 + 3 dose-escalation design used in Phase I trials of anticancer drugs.

A2.3.3 Phase II: does it work?

- **Phase IIa** (proof of concept): a small trial to see whether the drug has the intended effect in patients.
- **Phase IIb** (dose finding): several doses compared, often with placebo, to choose the dose for Phase III.
- End-points are often **surrogate markers** (blood pressure, HbA1c, viral load) rather than hard outcomes.

A2.3.4 Phase III: confirmatory trials

Phase III trials are the large, randomised, usually double-blind and multicentre trials on which approval rests. They compare the new drug with placebo or with the standard treatment. They aim for **clinical end-points** (cure, survival, symptom relief, quality of life). Three kinds of comparison are used:

Trial type	Question	Use
Superiority	Is the new drug better?	Against placebo, or when a real gain is expected
Non-inferiority	Is it not worse than the standard by more than an agreed margin?	When it offers another advantage: cost, safety, convenience
Equivalence	Is it neither better nor worse, within a margin?	Bioequivalence; biosimilars

Phase III is sometimes divided into **IIIa** (before the marketing application) and **IIIb** (after submission, before approval, often for additional uses).

A2.3.5 Phase IV: after approval

Phase IV begins once the drug is on the market. Even the largest Phase III programme includes only a few thousand people for a limited time. An adverse reaction that occurs in 1 in 10,000 patients will almost certainly be missed. As a rough guide (the "rule of three"), to have a 95 per cent chance of seeing at least one case of a reaction that occurs once in n patients, about $3 \times n$ patients must be observed. So rare reactions appear only after marketing. Phase IV includes pharmacovigilance (Chapter A6), post-marketing studies required by the regulator, studies in special groups, and new indications.

> **WHERE DRUGS FAIL**
>
> Most drugs that enter clinical trials are never approved. The commonest reasons are lack of efficacy in Phase II or III, and unacceptable toxicity. This is why claims based only on animal or test-tube studies, however impressive, must be treated as hypotheses.

A2.4 Methods of clinical research

A2.4.1 Classification of study designs

Clinical studies are first divided by whether the investigator **assigns** the treatment (experimental) or only **observes** what happens (observational) (Figure A2.4).

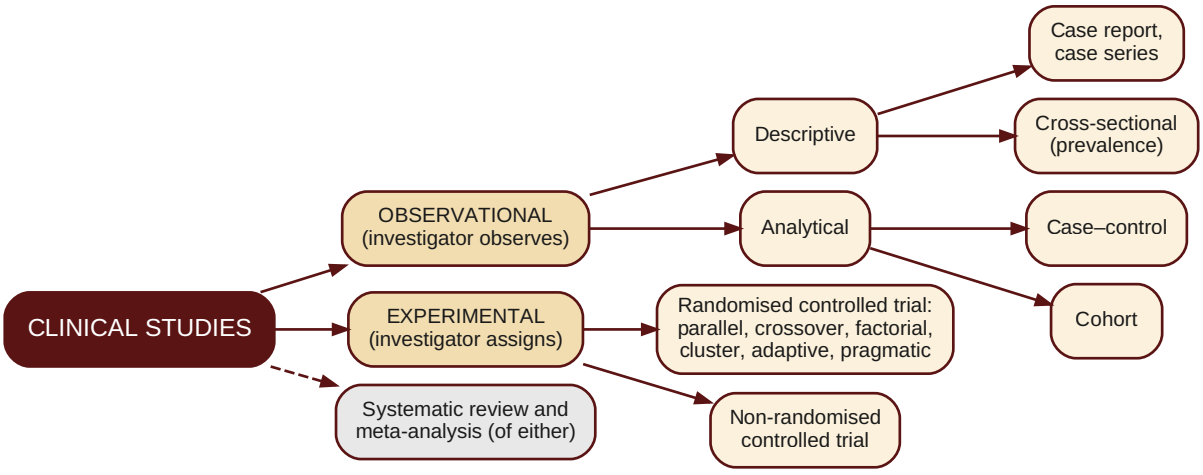


Figure A2.4. Classification of clinical study designs.

Design	What is done	Strength	Weakness	Example question
Case report, case series	Description of one or a few patients	Detects new effects and ADRs quickly	No comparison group; anecdotal	A new rash with a new drug
Cross-sectional	Exposure and outcome measured at one time in a population	Quick; gives prevalence	Cannot show which came first	How many patients use herbal medicines?
Case–control	People with the disease compared with people without, for past exposure	Good for rare diseases; quick	Recall bias; selection of controls	Is a liver injury linked to a herbal product?
Cohort (prospective or retrospective)	Exposed and unexposed people followed for the outcome	Shows sequence; gives incidence and risk	Slow, costly; confounding	Do long-term users of a drug develop cataract?
Randomised controlled trial (RCT)	Participants allocated at random to treatments	Controls known and unknown confounders; the strongest single study	Costly; strict conditions; ethical limits	Is drug A better than placebo?
Systematic review and meta-analysis	All relevant studies found and combined by explicit methods	Highest level of evidence when the studies are good	Only as good as the studies included	What do all trials of drug A show?

These designs form a **hierarchy of evidence** (Figure A2.5). A well-conducted RCT gives stronger evidence about a treatment's effect than an observational study, because randomisation balances the groups. But observational studies remain essential for rare adverse effects, long-term outcomes and questions that cannot be tested by randomisation.

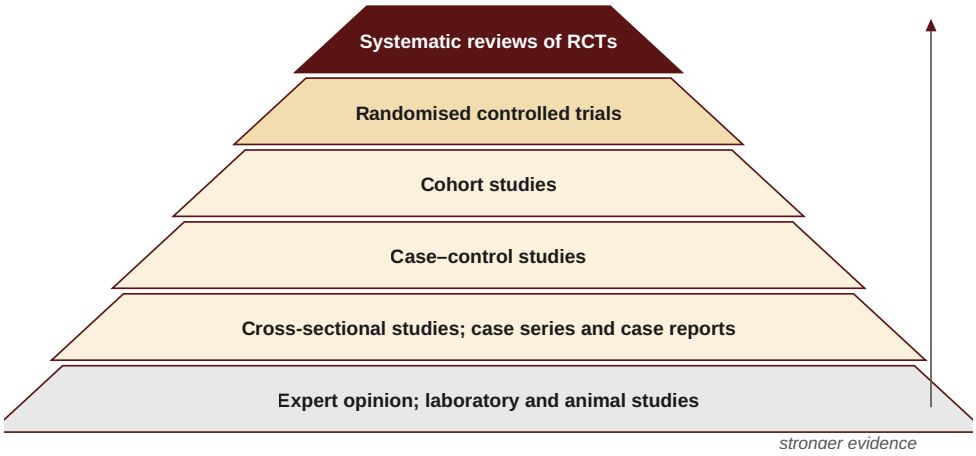


Figure A2.5. The hierarchy of evidence for questions about the effect of a treatment.

A2.4.2 The randomised controlled trial

The **randomised controlled trial** is the standard method for testing whether a treatment works. Its flow is shown in Figure A2.6, in the style of the CONSORT diagram used when reporting trials [CONSORT 2025].

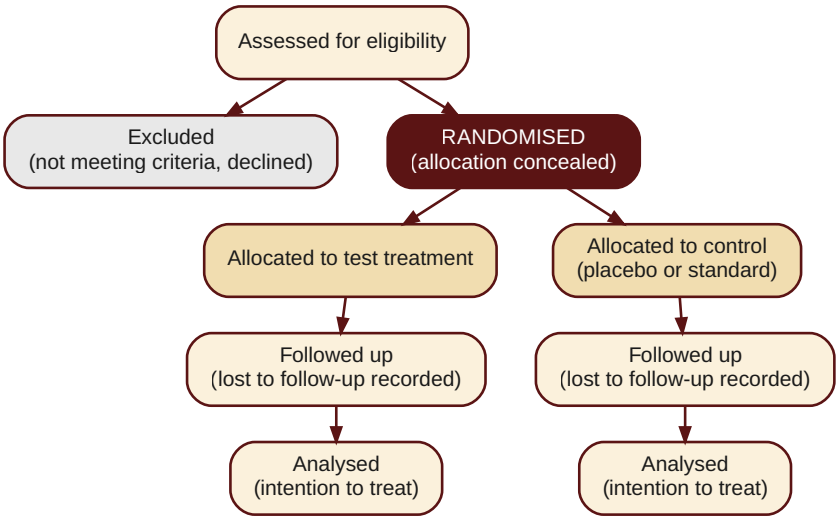


Figure A2.6. Flow of participants through a two-arm randomised controlled trial.

Key elements

Element	What it means	Why it matters
Eligibility criteria	Inclusion and exclusion criteria defined in advance	Decide who the results apply to
Randomisation	Allocation by chance: simple, block (keeps group sizes equal), stratified (balances key factors such as age or severity), or by minimisation	Makes groups comparable for known and unknown factors
Allocation concealment	The person enrolling a participant cannot know or influence the next allocation (sealed opaque envelopes; central telephone or web randomisation)	Prevents selection bias; without it, randomisation fails
Blinding (masking)	Single (participant), double (participant and investigator), triple (also the analyst)	Prevents bias in reporting and assessing outcomes
Control	Placebo; active comparator (standard treatment); add-on (both groups get standard care, one also gets the test drug); no treatment	Shows what would have happened without the test drug
Outcomes	One pre-specified primary outcome; secondary outcomes; safety outcomes	Prevents “fishing” for a significant result
Sample size	Calculated from the expected effect, the variability, the significance level (α, usually 0.05) and the power (1 − β, usually 80 or 90 per cent)	Too small a trial misses real effects and wastes participants
Analysis	Intention-to-treat (all randomised participants analysed in their allocated group) as the main analysis; per-protocol as supportive	Preserves the benefit of randomisation

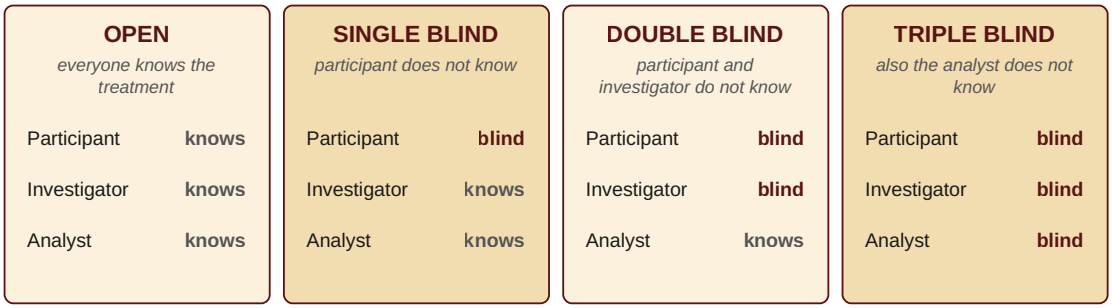


Figure A2.7. Levels of blinding in a clinical trial.

When two treatments look different (for example a tablet and a decoction), a **double-dummy** design keeps blinding: each participant takes one active and one placebo, matched to each form.

Variants of the RCT

Variant	Description	Use
Parallel group	Each participant receives one treatment throughout	The usual design
Crossover	Each participant receives both treatments in sequence, with a washout	Stable chronic conditions; bioequivalence
Factorial	Two or more treatments tested at once in combinations (A, B, A+B, neither)	Efficient testing of two interventions
Cluster	Groups (villages, clinics) rather than individuals are randomised	Public health and service interventions
N-of-1	One patient receives treatments in random sequence repeatedly	Individualised treatment decisions
Adaptive	Pre-planned changes based on interim results (dropping arms, changing sample size)	Efficient dose finding; uncertain effects
Pragmatic	Tests a treatment under usual care, broad eligibility, real-world outcomes	Effectiveness rather than efficacy; well suited to whole-system Ayurvedic care

Bias in clinical research

Bias	Source	Protection
Selection bias	Groups differ at the start	Randomisation with concealment
Performance bias	Groups treated differently apart from the intervention	Blinding of participants and carers
Detection bias	Outcomes measured differently	Blinding of assessors
Attrition bias	Dropouts differ between groups	Intention-to-treat; follow-up of all
Reporting bias	Only favourable outcomes or trials published	Trial registration; pre-specified outcomes; publishing all results

A2.4.3 The clinical trial protocol

The protocol is the plan of a trial, written and approved before the first participant is enrolled. Its contents are set out in the Indian rules, in ICH Good Clinical Practice and in the SPIRIT guidance for protocols.

Section	Contents
Administrative	Title; trial registration number; protocol version and date; sponsor; investigators; sites
Background and rationale	What is known; why the trial is needed; preclinical and earlier clinical data
Objectives	Primary and secondary objectives; hypotheses
Design	Type of trial; phase; arms; randomisation; blinding
Participants	Inclusion and exclusion criteria; setting; recruitment
Interventions	Test product (with its standardisation), comparator, dose, route, schedule, duration; concomitant treatment; compliance
Outcomes	Primary and secondary outcomes; how and when measured
Schedule	Visit-by-visit plan of assessments
Sample size	Calculation and assumptions
Allocation	Sequence generation; concealment; who enrols and assigns
Safety	Adverse event definitions, recording and reporting; stopping rules; data safety monitoring board
Statistics	Analysis plan; populations (intention-to-treat, per-protocol); handling of missing data
Ethics	Ethics committee approval; informed consent process; compensation and insurance
Data management and monitoring	Case record forms; quality control; audits
Dissemination	Publication policy; data sharing

A2.4.4 Reporting

Results of randomised trials should be reported according to the **CONSORT** statement, which was updated in 2025 [CONSORT 2025]. It requires, among other things, a flow diagram of participants and complete reporting of all pre-specified outcomes. In India, every clinical trial must be registered in the **Clinical Trials Registry – India (CTRI)** before enrolment of the first participant; registration has been mandatory since 2009.

A2.5 Clinical research on a new Ayurvedic formulation

The August 2022 paper asked for the *"phases and methods of clinical research for newer formulation"*. For an Ayurvedic formulation, the general principles above apply in full, with additions in three areas: the law, the product, and the method.

A2.5.1 The regulatory category decides what is required

Category of Ayurvedic product	Clinical data generally required (summary)
Classical formulation, made exactly as in an authoritative book, for the indications in that book	No new clinical trial is required for licensing; the book is the basis of the claim
Patent or proprietary medicine made from classical ingredients	Evidence of effectiveness, typically from a pilot clinical study, and safety data as prescribed
Product for a new indication, or containing ingredients of Schedule E(1)	Safety studies and clinical evidence as prescribed

This is a simplified summary of Rule 158-B of the Drugs Rules, 1945, which is explained in full, with its current text, in Chapter B8(i).

Two further documents govern Ayurvedic clinical research:

- **Good Clinical Practice guidelines for clinical trials in Ayurveda, Siddha and Unani medicine** (GCP-ASU), issued by the Department of AYUSH in 2013 [GCP-ASU 2013].
- The ICMR's **Ethical Requirements for Research in Integrative Medicine** (March 2025), an addendum to its 2017 national guidelines. It requires an ethics committee reviewing integrative research to co-opt two experts from the relevant Ayush system, at least one from outside the institution. It states that products already approved for clinical Ayush practice do not need additional safety evidence from animal studies or Phase I/II trials, while non-codified traditional medicines must complete the full regulatory process [ICMR 2025].

A2.5.2 The product must be defined

A trial of an Ayurvedic formulation is meaningless unless the product tested can be identified and reproduced.

- Authenticated raw materials (botanical identity, part used, source), with voucher specimens.
- A classical or documented method of preparation, followed by an SOP.
- Pharmacopoeial quality tests (API standards where they exist), heavy metals, microbial load, pesticide residues and aflatoxins.
- Chemical markers or a chromatographic fingerprint (HPTLC or HPLC); for herbo-mineral products, elemental analysis and Bhasma Pariksha.
- Stability over the trial period.
- A matching placebo, which is often the hardest part: colour, taste, smell and texture of a Churna or Kwatha must be imitated.

A2.5.3 The method must respect Ayurvedic practice without losing rigour

Issue	A possible solution
Ayurvedic treatment is individualised by Prakriti and Dosha	Stratify randomisation by Prakriti; or use a pragmatic trial of a whole treatment protocol
Treatment is a package (medicine, Pathya-Apathya, procedures)	Decide whether the trial tests the medicine alone or the whole package, and design accordingly
Blinding is difficult (taste, procedures)	Double-dummy designs; blinded outcome assessors even when participants cannot be blinded
Outcomes include Ayurvedic assessments	Use validated modern outcomes as primary; record Ayurvedic parameters as secondary, with defined scoring
Herbo-mineral products raise metal-safety questions	Blood and urine metal levels, liver and kidney function at baseline and during treatment
Long human use exists	Use it as a source of hypotheses and dose (reverse pharmacology), not as a substitute for a controlled trial

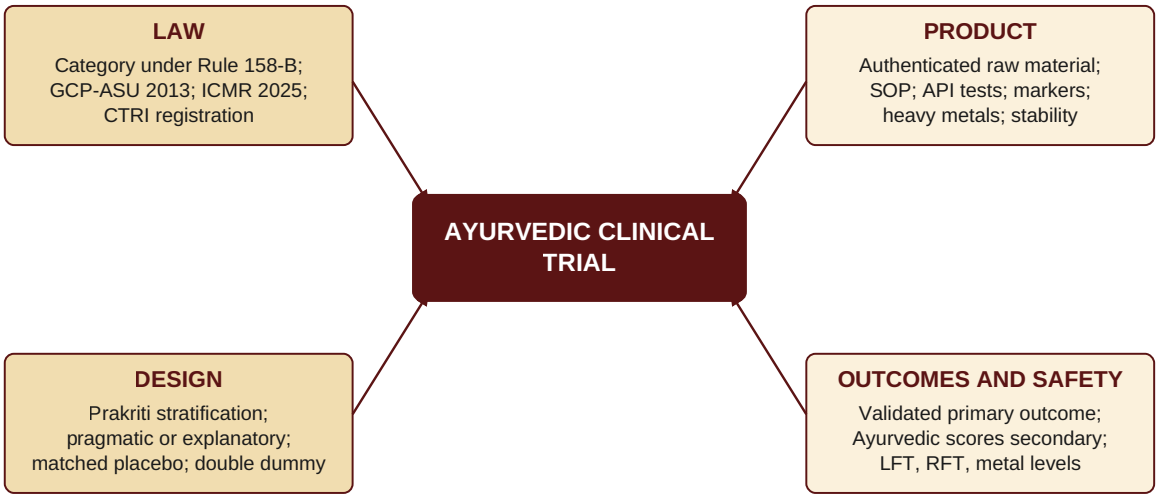


Figure A2.8. Special considerations in clinical trials of Ayurvedic formulations.

A2.5.4 Phases for an Ayurvedic formulation

Not every phase is needed for every product, and their content changes.

Phase	For a new Ayurvedic formulation
Preclinical	Quality and standardisation; acute and repeated-dose toxicity as required by category; activity in relevant models (Unit A1(d))
Phase I	Often not done as a separate phase when the ingredients have long human use; may be needed for a new ingredient, a new route, a much higher dose or a Schedule E(1) drug
Phase II	Pilot or exploratory study in patients: feasibility, dose, signals of efficacy and safety
Phase III	Randomised controlled trial against placebo or standard treatment, adequately powered, multicentre where possible
Phase IV	Pharmacovigilance through the national programme for ASU&H drugs (Chapter A6); observational studies of real-world use

AYURVEDA CORRELATION: PRAMANA AND THE METHOD OF RESEARCH

Charaka describes four means of valid knowledge, the Pramanas: Aptopadesha (authoritative testimony), Pratyaksha (direct perception), Anumana (inference) and Yukti (reasoning from the combination of many factors) (Charaka Samhita, Sutra Sthana 11) [R26]. A clinical trial can be seen as a disciplined form of Pratyaksha and Anumana: direct, controlled observation from which a general inference is drawn, with chance and bias kept out by design.

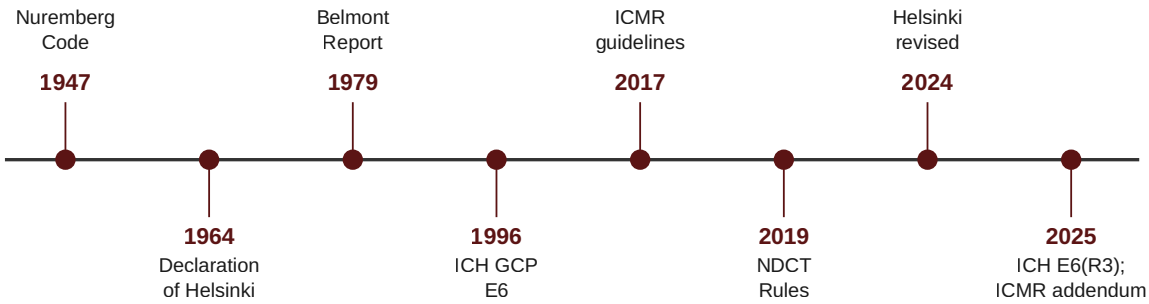
Established. The classical texts value careful observation and reasoning, and Aptopadesha is not meant to replace them.

To be remembered. Classical authority tells us what to test. A controlled trial tells us whether it works in today's patients, with today's products. Both are needed; neither replaces the other.

A2.6 Ethics involved in human research

A2.6.1 Why research ethics exists

Research ethics grew out of abuse. Its milestones are a record of what went wrong and of what was done to prevent it happening again (Figure A2.9).



Tuskegee study 1932–1972 and the thalidomide disaster 1957–1962 drove many of these reforms.

Figure A2.9. Milestones in the ethics of research on human beings.

Year	Event or document	Lesson
1932–1972	Tuskegee syphilis study (USA)	Hundreds of African American men with syphilis were observed without treatment, even after penicillin was available. Deception and denial of treatment.
1947	**Nuremberg Code**	Written after the trials of Nazi doctors. Voluntary consent is absolutely essential; research must be necessary, avoid suffering, and be stopped if harm appears.
1957–1962	Thalidomide disaster	Inadequate testing before marketing; led to stronger drug laws worldwide
1964	**Declaration of Helsinki** (World Medical Association)	Ethical principles for medical research; the interests of the participant come before those of science and society. Revised many times, most recently in **October 2024** [WMA 2024].
1979	**Belmont Report** (USA)	Three principles: respect for persons, beneficence, justice
1996 onwards	**ICH Good Clinical Practice** (E6)	International standard for conducting trials; the current version **E6(R3)** was finalised on 6 January 2025 [ICH E6(R3)]
2016	CIOMS International Ethical Guidelines	Guidance for research in low- and middle-income settings
2017	**ICMR National Ethical Guidelines for Biomedical and Health Research Involving Human Participants**	The Indian national guidelines [ICMR 2017]
2019	**New Drugs and Clinical Trials Rules**	The Indian law governing clinical trials [NDCT 2019]
2025	ICMR addendum on integrative medicine research	Requirements for research combining Ayush and modern medicine [ICMR 2025]

A2.6.2 The basic principles

The Belmont Report's three principles remain the simplest summary.

Principle	Meaning	Applied as
Respect for persons	Individuals are autonomous; those with diminished autonomy deserve protection	Informed consent; extra protection for vulnerable groups
Beneficence	Do no harm; maximise benefits and minimise risks	Scientific validity; favourable risk–benefit ratio; safety monitoring
Justice	Fair distribution of the burdens and benefits of research	Fair selection of participants; no exploitation of the poor or the powerless

The ICMR's 2017 guidelines set out **twelve general principles** [ICMR 2017]:

No.	Principle	No.	Principle
1	Essentiality	7	Professional competence
2	Voluntariness	8	Maximisation of benefit
3	Non-exploitation	9	Institutional arrangements
4	Social responsibility	10	Transparency and accountability
5	Privacy and confidentiality	11	Totality of responsibility
6	Risk minimisation	12	Environmental protection

A2.6.3 Informed consent

> **Informed consent** is the voluntary agreement of a competent person to take part in research, given after he or she has received and understood all the information relevant to the decision.

Consent is a **process**, not a signature. It begins before enrolment and continues throughout the study (Figure A2.10).

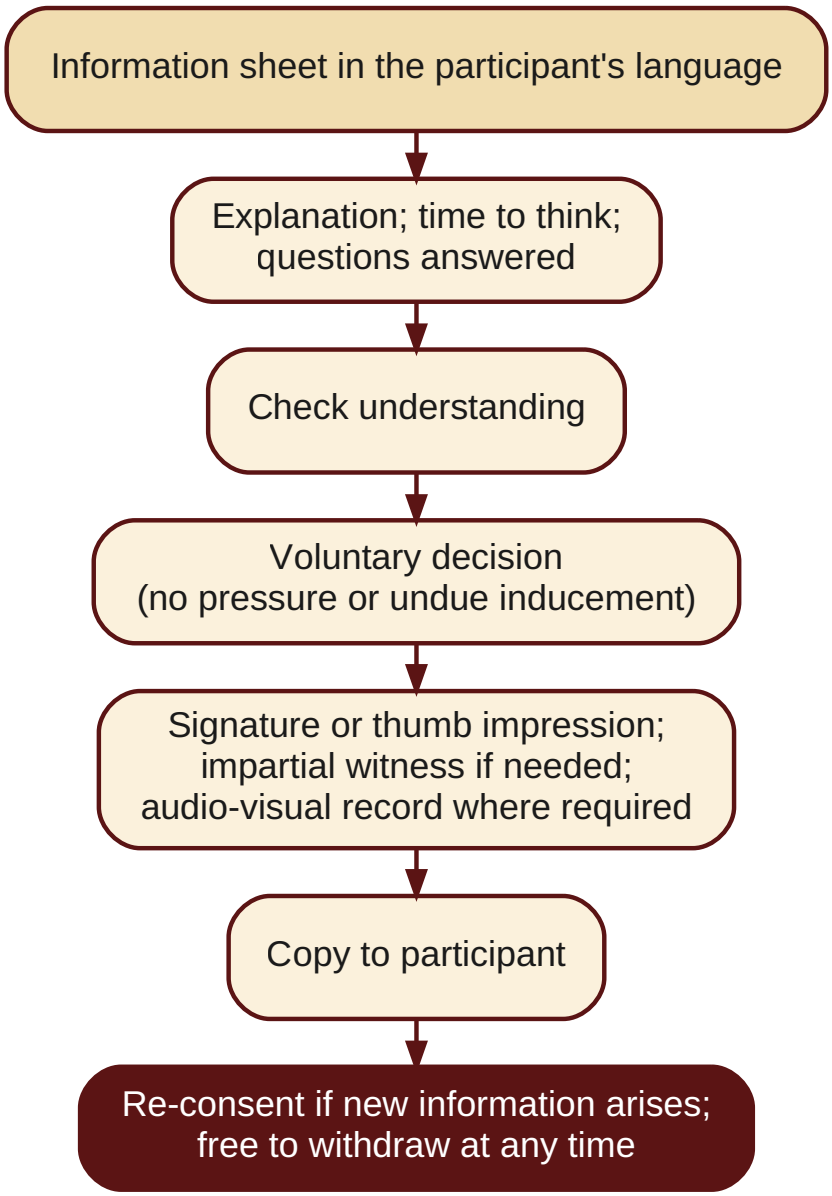


Figure A2.10. The informed consent process.

Elements of the information given to a participant

1. That the study is research, its purpose and its duration.
2. What will be done, and which parts are experimental.
3. Foreseeable risks and discomforts.
4. Expected benefits, to the participant and to others; or that there may be none.
5. Alternative treatments available.
6. Confidentiality of records.
7. Compensation and free medical management for research-related injury.
8. Whom to contact with questions or in case of injury.
9. That participation is voluntary, and that the participant may withdraw at any time without penalty or loss of benefits.
10. Any payment for participation, and that it is not an inducement.

Practical requirements: the information sheet in a language the participant understands; time to think and to ask questions; signature or thumb impression, with an impartial literate witness when the participant cannot read; a copy for the participant. Indian rules require **audio-visual recording** of the consent process in certain categories of trial. For a child, consent is given by a parent or guardian and **assent** is taken from the child where possible. For a person who cannot consent, a legally acceptable representative consents.

A2.6.4 Vulnerable participants

Some people are less able to protect their own interests and need extra safeguards: children; pregnant and nursing women; persons with mental illness or cognitive impairment; prisoners; students and employees of the investigator's institution; the very poor and the illiterate; patients with incurable or terminal illness; and people in emergencies. They may be included only when the research cannot be done without them and is relevant to their needs.

A2.6.5 The ethics committee

No research on human beings may begin without approval from an ethics committee. The committee is independent of the investigators and the sponsor. Its duty is to protect the rights, safety and well-being of participants.

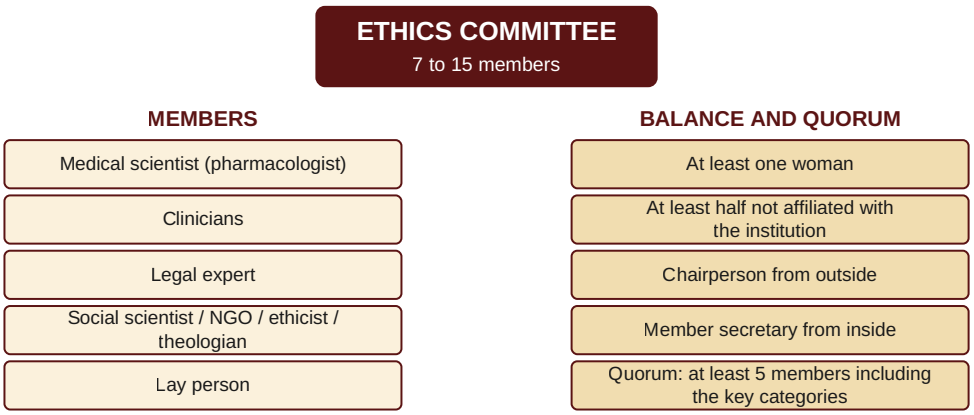


Figure A2.11. Composition of an ethics committee reviewing clinical trials under the New Drugs and Clinical Trials Rules, 2019.

Feature	Requirement for an ethics committee reviewing clinical trials
Size	7 to 15 members
Members	Medical scientists and clinicians (with postgraduate qualification); a legal expert; a social scientist or representative of a non-governmental organisation or an ethicist or theologian; a lay person
Balance	At least one woman; at least half the members not affiliated with the institution; the chairperson from outside the institution; the member secretary from within
Quorum	At least five members, including a medical scientist (preferably a pharmacologist), a clinician, a legal expert, a social scientist or equivalent, and a lay person
Conflict of interest	Members with a conflict do not take part in the decision
Registration	With the Central Licensing Authority, for five years

As set out in the New Drugs and Clinical Trials Rules, 2019 and summarised in registered committees' documents [NDCT 2019]. Ethics committees for biomedical research other than drug trials register with the Department of Health Research. For integrative research, two Ayush experts are co-opted [ICMR 2025].

Functions of the ethics committee

1. Review the protocol, the investigator's brochure, the consent documents and the investigators' qualifications.
2. Weigh the risks against the expected benefits.
3. Approve, require changes to, or reject the study.
4. Monitor the study: progress reports, amendments, protocol deviations.
5. Review serious adverse events and the compensation due.
6. Suspend or stop a study that endangers participants.

A2.6.6 Responsibilities in Good Clinical Practice

Party	Main responsibilities
Sponsor	Design and funding; regulatory permissions; investigational product of assured quality; monitoring; safety reporting; compensation and insurance; data integrity
Investigator	Conduct according to the protocol; informed consent; medical care of participants; accurate records; reporting of serious adverse events
Ethics committee	Independent review and continuing oversight
Regulator	Permission; inspection; action on safety
Participant	Informed, voluntary participation; may withdraw at any time

The 2025 revision, ICH E6(R3), emphasises **quality by design** and a **proportionate, risk-based** approach: the effort put into monitoring and control should match the risks to participants and to the reliability of the results [ICH E6(R3)].

A2.6.7 Other ethical issues

- **Placebo**: acceptable when no proven treatment exists, or when withholding it causes no serious or irreversible harm; otherwise the new treatment is compared with the best proven treatment.
- **Conflict of interest**: financial or other interests of investigators must be declared.
- **Privacy**: personal data must be protected. In India, the Digital Personal Data Protection Act, 2023 now applies alongside the research guidelines.
- **Publication**: all trials should be registered before they start and their results made public, whether favourable or not.
- **Post-trial access**: where a treatment proves beneficial, plans for continued access should be considered.

AYURVEDA CORRELATION: ETHICS OF THE PHYSICIAN

Charaka lays down a strict code of conduct for the student of medicine at initiation: to act for the good of all living beings, to avoid causing harm, to respect the patient's confidence, and never to treat a patient for gain alone (Charaka Samhita, Vimana Sthana 8) [R26]. The four pillars of treatment, the Chikitsa Chatushpada

(physician, medicine, attendant and patient) (Sutra Sthana 9), place the patient at the centre of the therapeutic relationship.

Established. The ethical ideals of the classical physician (non-harm, confidentiality, duty to the patient) are close to the modern principles of beneficence, respect and privacy.

To be remembered. Modern research ethics adds requirements that the classical texts did not foresee: written informed consent for research, independent ethics review, registration, compensation and public reporting. An Ayurvedic trial must meet them in full.

KEY POINTS TO REMEMBER

- Clinical pharmacology studies drugs in humans. A new chemical entity is evaluated in phases after preclinical testing.
- India: New Drugs and Clinical Trials Rules, 2019 (notified 19 March 2019). CDSCO grants permission; ethics committees are registered with the Central Licensing Authority; CTRI registration before enrolment; free medical management and compensation (Seventh Schedule).
- Phase 0: microdose. Phase I: safety, PK, maximum tolerated dose, 20–100 healthy volunteers. Phase II: efficacy and dose in patients. Phase III: large confirmatory RCTs. Phase IV: post-marketing surveillance.
- RCT elements: randomisation, allocation concealment, blinding, control, pre-specified primary outcome, sample size, intention-to-treat analysis. Report by CONSORT 2025.
- Pragmatic and Prakriti-stratified designs suit Ayurvedic interventions; the product must be standardised and the placebo matched.
- GCP-ASU (2013) and the ICMR integrative medicine addendum (March 2025) apply to Ayurvedic research.
- Ethics milestones: Nuremberg Code 1947; Declaration of Helsinki 1964 (revised October 2024); Belmont Report 1979; ICH E6(R3) January 2025; ICMR guidelines 2017.
- Belmont principles: respect for persons, beneficence, justice. ICMR: twelve general principles.
- Informed consent is a process; voluntary; may withdraw at any time; audio-visual recording in certain trials.
- Ethics committee: 7–15 members; legal expert, lay person, social scientist, woman member; at least half from outside; quorum of five with specified categories.

EXAM CORNER — CHAPTER A2

Likely long questions (20 marks)

1. Explain the phases and methods of clinical research and brief about the ethics involved in human research *(set in August 2021)*.
2. Explain the phases and methods of clinical research for a newer formulation *(set in August 2022)*.
3. Explain the phases and protocol of preclinical and clinical evaluation of a new chemical entity *(set in September 2018)*.
4. Describe the design of a randomised controlled trial. Discuss the special problems of conducting such trials for Ayurvedic formulations.
5. Discuss the ethical principles of research on human beings, with the Indian regulations.

Likely short notes (5 marks)

1. Phases of clinical research *(set in September 2020)*.
2. Ethics involved in human research *(set in July 2018)*.
3. Phase I clinical trial.
4. Phase IV and post-marketing surveillance.
5. Randomisation and allocation concealment.
6. Blinding.
7. Placebo.
8. Clinical trial protocol.
9. Informed consent.
10. Ethics committee: composition and functions.
11. Declaration of Helsinki.
12. CTRI.
13. Good Clinical Practice.
14. New Drugs and Clinical Trials Rules, 2019.

How to attempt

- Start the phases answer with the pathway diagram (Figure A2.1) and the phases diagram (Figure A2.2). Then give the phase table: aim, subjects, number, design, duration. Examiners look for this table.
- For "methods", draw the design classification (Figure A2.4), then describe the RCT under the headings of A2.4.2.
- For "newer formulation", add the three Ayurvedic sections: regulatory category, product definition, method.
- For ethics, write the history table briefly, the principles, informed consent, the ethics committee table and the Indian documents.

Answer frame: "Explain phases and methods of clinical research for newer formulation." (20 marks)

Part	What to write	Suggested marks
Introduction	Clinical research defined; why needed; pathway diagram	2
Phases	Phase 0 to IV table, with a paragraph on each	6
Methods	Design classification; RCT elements (randomisation, concealment, blinding, control, outcomes, sample size, analysis)	5
Protocol	Main headings of a protocol	2
Newer Ayurvedic formulation	Regulatory category (Rule 158-B); standardisation and placebo; Prakriti stratification, pragmatic design; safety monitoring; GCP-ASU, ICMR 2025	4
Ethics and registration	EC approval, informed consent, CTRI	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **26, 29, 30, 32**.

• **[R26]** Charaka Samhita: Sutra Sthana 9 and 11; Vimana Sthana 8.

- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R30]** Brown MJ, Sharma P, Mir FA, Bennett PN. *Clinical Pharmacology*. 12th ed. Elsevier; 2018.
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

Additional verified sources (all accessed 16 October 2026)

- **[NDCT 2019]** Ministry of Health and Family Welfare, Government of India. New Drugs and Clinical Trials Rules, 2019. G.S.R. 227(E), 19 March 2019. (Rule 23; Chapter VI and Seventh Schedule; provisions on ethics committees.)
- **[CAM 2019]** Cyril Amarchand Mangaldas. "New Drugs and Clinical Trials Rules, 2019: regulations in India." corporate.cyrilamarchandblogs.com, July 2019 (accessed 16 October 2026).
- **[ICMR 2017]** Indian Council of Medical Research. *National Ethical Guidelines for Biomedical and Health Research Involving Human Participants*. New Delhi: ICMR; 2017.
- **[ICMR 2025]** Indian Council of Medical Research. *Ethical Requirements for Research in Integrative Medicine*: an addendum to the National Ethical Guidelines, 2017. New Delhi: ICMR; March 2025.
- **[GCP-ASU 2013]** Department of AYUSH, Government of India. *Good Clinical Practice Guidelines for Clinical Trials in Ayurveda, Siddha and Unani Medicine*. 2013.
- **[ICH E6(R3)]** International Council for Harmonisation. *Guideline for Good Clinical Practice E6(R3)*. Step 4, 6 January 2025; effective in the European Union from 23 July 2025.
- **[WMA 2024]** World Medical Association. *Declaration of Helsinki: Ethical Principles for Medical Research Involving Human Participants*. Revised at the 75th General Assembly, Helsinki, October 2024.
- **[CONSORT 2025]** Hopewell S and colleagues. CONSORT 2025 statement: updated guideline for reporting randomised trials. Published simultaneously in several journals, 2025.
- National Commission for the Protection of Human Subjects of Biomedical and Behavioral Research (USA). *The Belmont Report*. 1979.

Viva pointers — Chapter A2

Answer each in one or two sentences.

On clinical pharmacology and the phases

- Define clinical pharmacology. How does it differ from experimental pharmacology?
- Which law governs clinical trials in India today? When was it notified?
- Who grants permission for a clinical trial in India? What is CTRI, and when must a trial be registered?
- What is a Phase 0 study? Why is it called a microdose study?
- What is the main aim of Phase I? Who are the participants?
- What is the difference between Phase II and Phase III?
- Why is Phase IV necessary when a drug has already passed Phase III?
- What is the maximum tolerated dose?

On methods

- Name the levels of evidence, from case report to meta-analysis.
- Why do we randomise? Why do we blind?
- What is the difference between single-blind, double-blind and open-label trials?
- What is a placebo? When is a placebo control unethical?
- What is an intention-to-treat analysis?
- What is a crossover design? When is it unsuitable?
- What is a non-inferiority trial?
- What is CONSORT? What is its flow diagram?
- Name four sections of a trial protocol.
- Why is it difficult to make a placebo for an Ayurvedic formulation? How would you solve it?
- How can Prakriti be used in the design of a trial?
- What is a pragmatic trial? Why does it suit Ayurvedic practice?

On ethics

- What were the Nuremberg Code and the Declaration of Helsinki? When was Helsinki last revised?
- Name the three principles of the Belmont Report.
- What are the essential elements of informed consent? What is audio-visual recording of consent?
- Who are vulnerable participants? Give four examples.
- What is the composition of an ethics committee? What is its quorum?
- What is a serious adverse event? Who must be told, and within what time?
- What is a participant entitled to if injured in a trial?
- What is GCP? What is GCP-ASU?
- What does the ICMR integrative medicine addendum of 2025 require?
- Which classical text lays down a code of conduct for the physician, and where?

PART A · PHARMACOLOGY · CHAPTER A3

A3 · ELEMENTAL CONSTITUENTS OF THE HUMAN BODY

SYLLABUS POINT (PART A, 3)

Elemental constituents of human body and its physiological importance. Deficiencies and excess of various elements (micro-nutrients).

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A3	Elemental Constituents of the Human Body	1	4	Sep 2018, Sep 2020, Aug 2022, Jan 2024, Mar 2026

Why this chapter matters to you. Rasashastra is, in large part, the pharmaceutics of elements: iron, zinc, copper, calcium, gold, silver, mercury, lead, tin and sulphur. A scholar who prescribes Loha Bhasma or Yashada Bhasma should know what iron and zinc do in the body, what happens when they are lacking, and what happens when they are in excess. This chapter has been set in five of the seven papers examined.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Describe the elemental composition of the human body and classify the elements as bulk, macro, trace and ultra-trace, and as essential, possibly essential and toxic.
2. Explain the principle of essentiality and the dose–response curve of an essential element (the basis of “bio-elementology”).
3. Describe the physiological role, sources, requirement, deficiency and excess of each important macro-mineral and trace element.
4. Explain the factors that affect the absorption of minerals, and the interactions between them.
5. Relate the elements to Rasashastra drugs, and state what is established and what is not about Bhasmas as sources of elements.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	A, Q.3a	5	Write the physiological importance and symptoms of deficiency of Iron and Zinc on human body.
Sep 2020	A, Q.3c	5	Elemental constituents of human body and its physiological importance
Aug 2022	A, Q.2	20	Explain elemental constituents of human body and explain its physiological importance.
Jan 2024	A, Q.3c	5	Elemental constituents of human body. (Theory of Bio – elementology)
Mar 2026	A, Q.3c	5	Explain the importance of micro-nutrients and the effects of their deficiency

Most questions on this chapter are short notes. The January 2024 paper used the phrase *"Theory of Bio-elementology"*. This is not a standard textbook term; it is best answered as the study of elements in living systems, their classification and the principle of essentiality (A3.2).

A3.1 Elemental composition of the human body

About 99 per cent of the mass of the human body is made of six elements: oxygen, carbon, hydrogen, nitrogen, calcium and phosphorus. Most of the oxygen and hydrogen are present as water. A further five elements (potassium, sulphur, sodium, chlorine and magnesium) make up most of the remaining one per cent. The rest, though present in tiny amounts, are no less vital.

Element	Approximate share of body mass	Main location or form
Oxygen (O)	65%	Water; all organic molecules
Carbon (C)	18%	All organic molecules
Hydrogen (H)	10%	Water; organic molecules
Nitrogen (N)	3%	Proteins, nucleic acids
Calcium (Ca)	1.5%	Bones and teeth (about 99% of body calcium)
Phosphorus (P)	1%	Bones; nucleic acids; ATP; phospholipids
Potassium (K)	0.35%	Inside cells
Sulphur (S)	0.25%	Amino acids (cysteine, methionine)
Sodium (Na)	0.15%	Extracellular fluid
Chlorine (Cl)	0.15%	Extracellular fluid
Magnesium (Mg)	0.05%	Bone; inside cells
All trace elements together	less than 0.01%	Enzymes, hormones, transport proteins

Figures are rounded values commonly given in physiology texts.

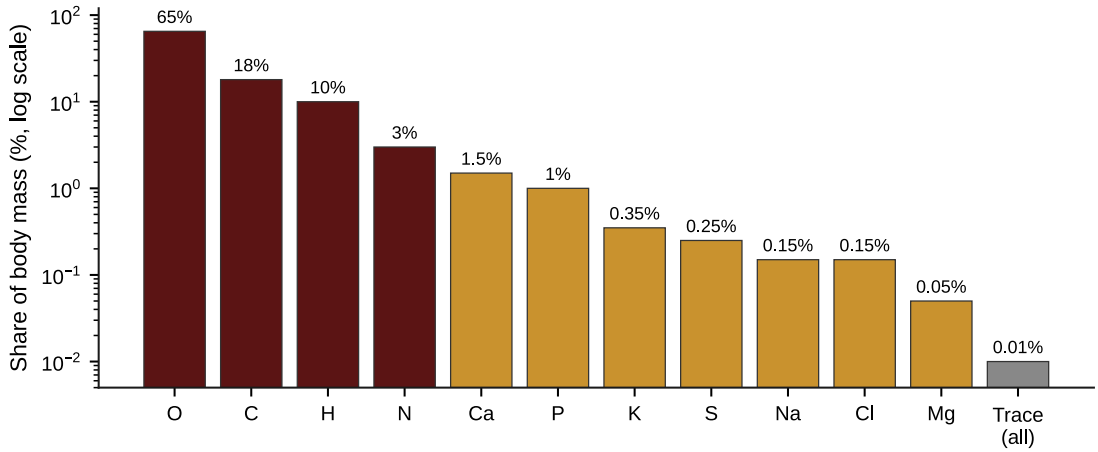


Figure A3.1. Elemental composition of the human body by mass (rounded values).

A3.1.1 Classification by amount

Group	Daily need	Elements
Bulk (structural) elements	Supplied by food, water and air	Oxygen, carbon, hydrogen, nitrogen
Macro-minerals (major minerals)	More than about 100 mg per day	Calcium, phosphorus, magnesium, sodium, potassium, chloride, sulphur
Trace elements	Less than about 100 mg per day (milligram amounts)	Iron, zinc, copper, manganese, iodine, selenium, molybdenum, chromium, fluoride, cobalt (as vitamin B_{12})
Ultra-trace elements	Microgram amounts or less	Boron, silicon, nickel, vanadium, lithium and others

A3.1.2 Classification by biological role

A widely used classification, from the WHO report on trace elements in human nutrition, divides the trace elements into three groups [WHO 1996].

Group	Elements
Essential	Iodine, zinc, selenium, copper, molybdenum, chromium (and iron, cobalt, which are also essential)
Probably essential	Manganese, silicon, nickel, boron, vanadium
Potentially toxic, some possibly essential at very low levels	Fluoride, lead, cadmium, mercury, arsenic, aluminium, lithium, tin

The third group is the one that concerns Rasashastra most. Mercury, lead and arsenic have **no known essential role** in the human body. Their toxicity is the subject of Chapter A4.

A3.2 Bio-elementology: the principle of essentiality

The study of chemical elements in living systems, their distribution, function, requirement, deficiency and toxicity, is sometimes called bio-elementology, or more commonly **bioinorganic chemistry** and **trace element nutrition**.

A3.2.1 When is an element essential?

An element is regarded as essential when:

1. it is present in all healthy tissues, within a fairly constant range;
2. its deficiency, produced by diet or other means, consistently causes a structural or functional abnormality;
3. the abnormality is prevented or cured by giving the element, and only that element;
4. a specific biochemical function can be identified, for example as part of an enzyme or hormone.

A3.2.2 The dose–response curve of an essential element

For a drug, the dose–response curve rises from no effect to a maximum (A1a.4). For an **essential element**, the curve has three zones (Figure A3.2). Too little causes deficiency. A middle range is optimal; within it the body keeps the level steady by **homeostasis** (controlling absorption, storage and excretion). Too much causes toxicity. This curve is often associated with the French scientist Gabriel Bertrand, who described it early in the twentieth century.

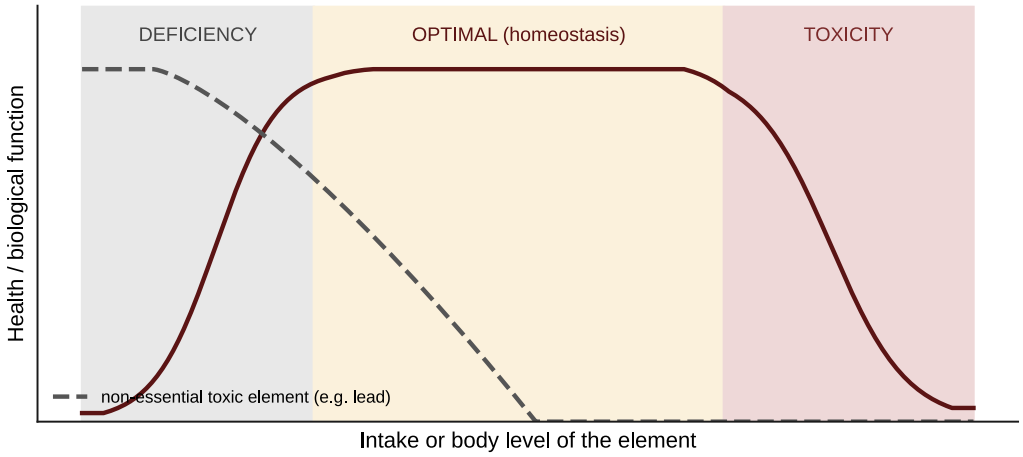


Figure A3.2. The dose–response curve of an essential element: deficiency, optimal range and toxicity (schematic).

For a **non-essential toxic element** such as lead or mercury, there is no deficiency zone and no optimal range: any increase in exposure brings the body closer to harm.

The practical lessons are three.

- The **width of the optimal range** differs between elements. It is wide for zinc, narrow for selenium and fluoride.
- **Homeostasis** protects the body only within limits. A supplement taken for long periods can overwhelm it.
- A substance that supplies an element is safe only if the dose keeps the body within the optimal range. This applies equally to a tablet of ferrous sulphate and to a Bhasma.

A3.2.3 How elements work in the body

Role	Examples
Structural	Calcium, phosphorus and fluoride in bone and teeth; sulphur in proteins
Electrical and osmotic	Sodium, potassium and chloride in nerve impulses and fluid balance
Cofactor of enzymes (metalloenzymes)	Zinc in carbonic anhydrase and alkaline phosphatase; copper in cytochrome c oxidase and superoxide dismutase; selenium in glutathione peroxidase; manganese in arginase; molybdenum in xanthine oxidase
Transport and storage	Iron in haemoglobin, myoglobin, ferritin, transferrin
Part of hormones and vitamins	Iodine in thyroid hormones; cobalt in vitamin B_{12}
Signalling	Calcium as a second messenger; zinc in gene regulation (“zinc fingers”)

A3.3 Macro-minerals

Element	Main functions	Main food sources	Deficiency	Excess
Calcium	Bone and teeth; muscle contraction; nerve conduction; blood clotting; second messenger	Milk and milk products, ragi, sesame, green leafy vegetables, small fish	Rickets (children), osteomalacia and osteoporosis (adults); tetany when blood calcium falls acutely	Hypercalcaemia: constipation, kidney stones, confusion; usually from excess vitamin D or supplements
Phosphorus	Bone; ATP; nucleic acids; phospholipids; buffer	Milk, meat, cereals, pulses	Rare; weakness, bone pain (in refeeding, with excess antacids)	In kidney failure: raised phosphate, vascular calcification
Magnesium	Cofactor of over 300 enzymes, especially those using ATP; nerve and muscle stability	Whole grains, nuts, green vegetables	Muscle cramps, tremor, arrhythmias; low calcium and potassium that do not correct until magnesium is given	Rare with normal kidneys; with kidney failure, weakness, low blood pressure
Sodium	Main cation of extracellular fluid; fluid balance; nerve impulse	Salt, processed foods	Hyponatraemia: confusion, convulsions (vomiting, diarrhoea, diuretics)	Hypertension (chronic high intake); hypernatraemia in dehydration
Potassium	Main cation inside cells; nerve and muscle excitability; heart rhythm	Fruits (banana), vegetables, pulses, coconut water	Hypokalaemia: weakness, cramps, arrhythmias (diuretics, diarrhoea)	Hyperkalaemia: dangerous arrhythmias (kidney failure, some drugs)
Chloride	Fluid balance; gastric acid	Salt	Alkalosis (with prolonged vomiting)	With sodium excess
Sulphur	Sulphur amino acids; keratin; glutathione; detoxification	Proteins (methionine, cysteine)	Linked to protein deficiency	—

Calcium requirement. The ICMR–National Institute of Nutrition report *Nutrient Requirements for Indians* (2020) recommends **1000 mg of calcium per day** for adult men and women [ICMR-NIN 2020].

A3.4 Trace elements

A3.4.1 Iron

Iron is the most common nutritional deficiency in the world and in India.

Aspect	Details
Body content	About 3 to 4 g in an adult; about two-thirds in haemoglobin
Functions	Oxygen transport (haemoglobin), oxygen storage in muscle (myoglobin), electron transport (cytochromes), many enzymes
Absorption	In the duodenum and upper jejunum. **Haem iron** (meat, fish) is absorbed well; **non-haem iron** (cereals, pulses, greens) is absorbed poorly. Absorption rises when stores are low. The hormone hepcidin controls it.
Enhancers of absorption	Vitamin C; meat; acidic medium; ferrous form (Fe^{2+})
Inhibitors of absorption	Phytates (cereals, pulses), tannins (tea, coffee), calcium, antacids, tetracyclines
Transport and storage	Transferrin in plasma; ferritin and haemosiderin in liver, spleen, bone marrow
Requirement (ICMR-NIN 2020)	**19 mg per day for adult men; 29 mg per day for adult women** [ICMR-NIN 2020]
Deficiency	Iron-deficiency anaemia: microcytic hypochromic red cells; fatigue, breathlessness, pallor, koilonychia, pica; poor growth and learning in children; poor outcome of pregnancy
Laboratory signs	Low haemoglobin, low serum ferritin (the best single test of stores), low serum iron, high total iron-binding capacity
Excess	Acute poisoning in children (vomiting, shock, liver failure; antidote deferoxamine); chronic overload (haemochromatosis, repeated transfusion): liver, heart, pancreas damage

In India, the **Anaemia Mukt Bharat** programme provides iron and folic acid supplementation, deworming and screening for children, adolescents and women.

A3.4.2 Zinc

Aspect	Details
Body content	About 2 to 3 g; mostly in muscle and bone
Functions	Part of more than 300 enzymes (carbonic anhydrase, alkaline phosphatase, alcohol dehydrogenase, superoxide dismutase); DNA and protein synthesis; growth; immunity; wound healing; taste; insulin storage; “zinc finger” proteins that regulate genes
Sources	Meat, fish, eggs, milk, nuts, whole grains (but phytates reduce absorption)
Requirement (ICMR-NIN 2020)	**17 mg per day for adult men; 13.2 mg per day for adult women** [ICMR-NIN 2020]
Deficiency	Growth retardation and delayed puberty; hypogonadism; poor wound healing; loss of taste and appetite; skin lesions; diarrhoea; poor immunity. Acrodermatitis enteropathica is an inherited form.
Use in therapy	Zinc with oral rehydration solution shortens diarrhoea in children
Excess	Nausea and vomiting (acute); with long use of high doses, **copper deficiency** (zinc blocks copper absorption), causing anaemia and neutropenia

A3.4.3 Copper

Aspect	Details
Functions	Cytochrome c oxidase (energy), superoxide dismutase (antioxidant), lysyl oxidase (collagen and elastin), ceruloplasmin (iron metabolism), tyrosinase (melanin), dopamine β-hydroxylase
Sources	Liver, shellfish, nuts, seeds, whole grains
Deficiency	Anaemia not corrected by iron; neutropenia; bone abnormalities; neurological damage. Menkes disease is an inherited defect of copper transport.
Excess	Acute: vomiting, haemolysis, liver and kidney damage. Chronic: liver cirrhosis. **Wilson’s disease** is an inherited defect of copper excretion: copper builds up in liver and brain; Kayser–Fleischer rings; treated with chelating agents (penicillamine, trientine) and zinc.

A3.4.4 Iodine

Aspect	Details
Function	Part of the thyroid hormones thyroxine (T_4) and tri-iodothyronine (T_3), which control metabolism, growth and brain development
Sources	Iodised salt, seafood; soil and water in many inland and hilly regions are poor in iodine
Requirement	About 150 µg per day for adults (more in pregnancy and lactation)
Deficiency (iodine deficiency disorders)	Goitre; hypothyroidism; in pregnancy, miscarriage and **cretinism** (severe intellectual disability and stunted growth); lower average intelligence in deficient populations
Excess	Thyroid dysfunction (hypo- or hyperthyroidism) in susceptible people

India’s programme for the control of iodine deficiency disorders is based on universal iodisation of salt.

A3.4.5 Selenium

Aspect	Details
Functions	Glutathione peroxidase (antioxidant); thyroid hormone deiodinases; selenoproteins in immunity and reproduction
Deficiency	Keshan disease (a cardiomyopathy, described in selenium-poor regions of China); Kashin–Beck disease (bone and joint disease)
Excess (selenosis)	Hair and nail loss, skin lesions, garlic odour of breath, nerve damage. The optimal range is narrow.
Interaction	Selenium binds mercury and may reduce its toxicity

A3.4.6 Fluoride

Aspect	Details
Function	Strengthens tooth enamel (fluorapatite); protects against dental caries
Deficiency	More dental caries
Excess (fluorosis)	**Dental fluorosis** (mottled enamel) and **skeletal fluorosis** (bone pain, stiffness, deformity). Endemic in several Indian states where groundwater fluoride is high; India has a national programme for the prevention and control of fluorosis.

A3.4.7 Other trace elements

Element	Function	Deficiency	Excess
Manganese	Arginase, pyruvate carboxylase, manganese superoxide dismutase; bone formation	Rare in humans	Neurotoxicity resembling Parkinsonism (miners; inhalation)
Molybdenum	Xanthine oxidase, sulphite oxidase, aldehyde oxidase	Rare; seen in long-term intravenous feeding	Rare; gout-like symptoms have been described
Chromium	Thought to enhance the action of insulin	Impaired glucose tolerance reported in long-term intravenous feeding	Hexavalent chromium (industrial) is toxic and carcinogenic; trivalent dietary chromium is of low toxicity
Cobalt	Only as part of vitamin B_{12}	Megaloblastic anaemia and neurological damage (as B_{12} deficiency)	Cardiomyopathy (historical "beer drinkers'" outbreak); release from metal joint implants

A3.5 Absorption and interactions of minerals

Most minerals are absorbed in the small intestine by specific carriers, and absorption is regulated by the body's need. Several minerals compete for the same carriers, and many substances in food bind them (Figure A3.3).

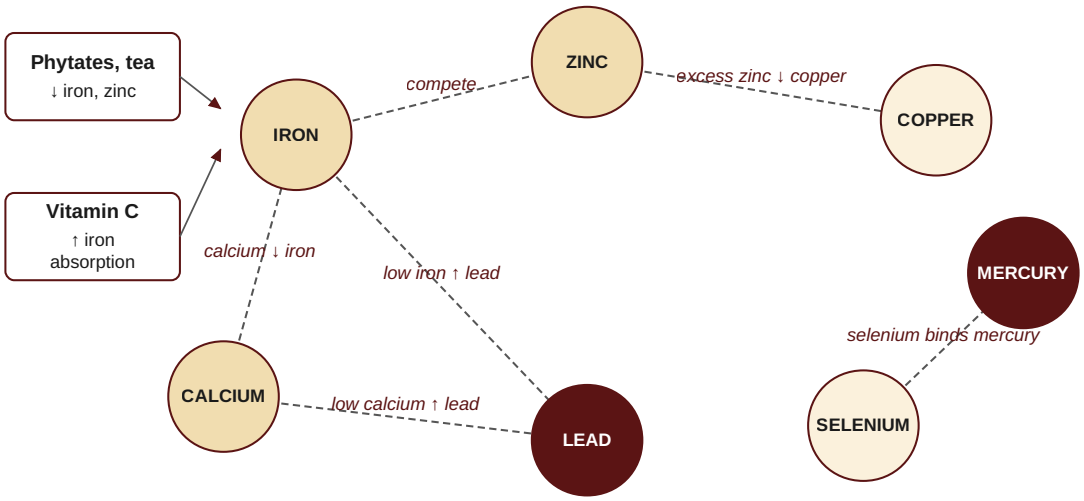


Figure A3.3. Important interactions between minerals and with food components.

Interaction	Effect
Zinc and copper	High zinc intake reduces copper absorption (used to treat Wilson's disease)
Iron and zinc	Large doses of one can reduce absorption of the other
Calcium and iron	Calcium reduces iron absorption when taken together
Vitamin C and iron	Vitamin C increases non-haem iron absorption
Phytates and polyphenols with iron and zinc	Phytates (cereals, pulses) and tannins (tea) reduce absorption
Vitamin D and calcium	Vitamin D increases calcium absorption
Selenium and mercury	Selenium binds mercury and may reduce its toxicity
Lead with calcium and iron	Deficiency of calcium or iron increases lead absorption (Chapter A4)

The last interaction has a practical lesson: a malnourished, iron-deficient child absorbs more lead from a contaminated source than a well-fed adult.

A3.6 Deficiency and excess: general principles

Causes of deficiency

1. Low intake: poverty, monotonous cereal diets, soils poor in an element (iodine, selenium).
2. Poor absorption: phytates, tannins, gut disease, drugs (antacids, proton pump inhibitors).
3. Increased loss: bleeding (iron), diarrhoea (zinc, potassium), diuretics (potassium, magnesium).
4. Increased need: growth, pregnancy, lactation.
5. Interaction: excess of one element causing deficiency of another (zinc causing copper deficiency).

Causes of excess

1. Environmental: groundwater (fluoride, arsenic), soil, industry and mining.
2. Occupational exposure.
3. Supplements and medicines taken in excess or for too long.

4. Genetic defects of excretion (Wilson's disease, haemochromatosis).
5. Kidney failure (potassium, phosphate, magnesium, aluminium).

Assessment combines diet history, clinical signs and laboratory tests: haemoglobin and serum ferritin for iron; serum zinc; serum copper and ceruloplasmin; urinary iodine (for populations) and thyroid function; serum electrolytes for sodium, potassium and magnesium.

A3.7 Elements in Rasashastra drugs

Many Rasaushadhi are, chemically, compounds of the elements described above. Characterisation studies of Bhasmas by X-ray diffraction, electron microscopy and elemental analysis have been reviewed in the literature [Pal 2014]. The table summarises the main element in common preparations, what is known about that element's role in the body, and how far the classical use is supported.

Preparation	Main element and usual reported form	Physiological role of the element	Classical use (examples)	Status of the link
Loha Bhasma, Mandura Bhasma	Iron, as iron oxides	Essential (haemoglobin)	Pandu (anaemia)	**Plausible and partly supported**: iron oxides can supply iron; the amount absorbed from a given Bhasma needs measurement
Yashada Bhasma	Zinc, as zinc oxide	Essential (enzymes, growth, immunity)	Prameha, eye disorders, as a Rasayana	**Plausible**: zinc oxide is a recognised source of zinc; clinical effect needs trials
Tamra Bhasma	Copper compounds	Essential, with a narrow margin	Amlapitta, Shula, liver and spleen disorders	**Caution**: copper is toxic in excess; dose and Shodhana critical
Praval Pishti, Mukta Pishti, Shankha, Shukti, Kapardika, Kukkutanda Twak Bhasma	Calcium, as calcium carbonate (Pishti) or calcium oxide and carbonate (Bhasma)	Essential (bone, nerve, muscle)	Amlapitta, Asthikshaya, as calcium sources	**Supported for calcium content**: these are calcium salts; antacid action of carbonates is established
Godanti Bhasma	Calcium, as calcium sulphate	Essential (calcium)	Jwara, Shirahshula	Calcium source; other claimed actions need study
Abhraka Bhasma	Silicates of iron, magnesium, aluminium, potassium	Silicon and magnesium possibly beneficial	Rasayana, respiratory disorders	**Not established**: mechanism of the claimed actions unknown
Swarna Bhasma	Gold, as fine metallic particles	**No known essential role**	Rasayana, immunity, nervous disorders	**Not established**: gold is not a nutrient; effects, if any, are pharmacological, not nutritional
Rajata Bhasma	Silver (metallic or as silver chloride)	No essential role	Nervous disorders, Rasayana	Not established

Preparation	Main element and usual reported form	Physiological role of the element	Classical use (examples)	Status of the link
Vanga Bhasma	Tin, as tin oxide	No established essential role	Prameha, genito-urinary disorders	Not established
Naga Bhasma	Lead compounds	**No essential role; toxic**	Prameha	Classical use stands against modern toxicology; see Chapters A4 and A5
Rasa Sindura, Kajjali	Mercury, as mercuric sulphide	**No essential role; toxic** in most forms	Many; Yogavahi	See Chapters A4 and A5
Gandhaka (in many preparations)	Sulphur	Sulphur is essential, but supplied by dietary protein	Skin disorders, Rasayana	Not established as a nutritional source

The usual reported form varies with the method of preparation; two Bhasmas of the same name may differ in phase and particle size [Pal 2014].

Three conclusions follow from the table.

1. **For iron, zinc and calcium preparations**, a nutritional explanation is plausible, because the element is essential and the preparation contains it in a form the body can use. What still needs measurement is how much is absorbed, and whether clinical outcomes match those of standard supplements.
2. **For gold and silver preparations**, there is no nutritional explanation, because these elements have no essential role. Any effect must be pharmacological and must be shown by experiment and trial.
3. **For mercury, lead and arsenic preparations**, the element is toxic, and the safety of each product depends on its chemical form, its dose and the duration of use (Chapters A4 and A5).

AYURVEDA CORRELATION: PANCHAMAHABHUTA, PARTHIVA DRAVYA AND DHATU

In Ayurveda every substance is made of the five Mahabhutas, and minerals and metals are predominantly **Parthiva** (earth-dominant). The body's tissues, the seven **Dhatus**, are nourished in sequence from the essence of digested food. A deficiency of a Dhatu (Dhatukshaya) is treated with substances similar in nature (the principle of Samanya), and many such substances are mineral.

Classical concept	Modern parallel
Asthi Dhatu and its nourishment; Asthikshaya	Bone mineral (calcium, phosphorus); osteoporosis
Rakta Dhatu; Pandu	Haemoglobin and iron; anaemia
Loha and Mandura as Raktavardhaka (blood-promoting)	Iron as a haematinic
Shankha, Praval and Shukti for Amlapitta	Calcium carbonate as an antacid
Samanya principle (like increases like)	Replacement therapy (iron for iron deficiency)

Established. The use of iron preparations for Pandu and of calcium-containing preparations for acid and bone disorders matches the modern pharmacology of iron and calcium.

Not established. That a Bhasma is a better source of an element than a modern salt; that Bhasmas of non-essential metals act by supplying an element; and that the Mahabhuta composition of a mineral corresponds to its chemical composition. These are claims to be tested, and they should be presented as such.

KEY POINTS TO REMEMBER

- Six elements (O, C, H, N, Ca, P) make up about 99 per cent of body mass; K, S, Na, Cl and Mg most of the rest.
- Macro-minerals: need more than about 100 mg per day (Ca, P, Mg, Na, K, Cl, S). Trace elements: less (Fe, Zn, Cu, I, Se, Mn, Mo, Cr, F, Co).
- WHO 1996: essential (I, Zn, Se, Cu, Mo, Cr), probably essential (Mn, Si, Ni, B, V), potentially toxic (F, Pb, Cd, Hg, As, Al, Li, Sn).
- An essential element shows deficiency, optimal range and toxicity; a toxic element has no safe benefit range.
- ICMR-NIN 2020: calcium 1000 mg per day; iron 19 mg (men) and 29 mg (women); zinc 17 mg (men) and 13.2 mg (women).
- Iron: haemoglobin; deficiency anaemia; ferritin the best test of stores; phytates and tea reduce, vitamin C increases absorption.
- Zinc: over 300 enzymes; growth, immunity, taste, wound healing; excess causes copper deficiency.
- Copper: cytochrome oxidase, superoxide dismutase, ceruloplasmin; Wilson’s disease is copper overload.
- Iodine: thyroid hormones; deficiency causes goitre and cretinism; salt iodisation.
- Selenium: glutathione peroxidase; Keshan disease; narrow margin. Fluoride: caries versus fluorosis.
- Loha, Mandura, Yashada and calcium Bhasmas carry essential elements; gold, silver, tin, lead and mercury preparations do not act by supplying a nutrient.

EXAM CORNER — CHAPTER A3

Likely long questions (20 marks)

1. Explain the elemental constituents of the human body and their physiological importance *(set in August 2022)*.
2. Classify the elements of the human body. Describe the physiological importance, deficiency and excess of any four trace elements.
3. Describe the role of trace elements in health and disease. Discuss Rasaushadhi as sources of elements, with the evidence.

Likely short notes (5 marks)

1. Elemental constituents of the human body (theory of bio-elementology) *(set in January 2024)*.
2. Elemental constituents of the human body and their physiological importance *(set in September 2020)*.
3. Physiological importance and symptoms of deficiency of iron and zinc *(set in September 2018)*.
4. Importance of micro-nutrients and the effects of their deficiency *(set in March 2026)*.

5. Iodine deficiency disorders.
6. Wilson’s disease.
7. Fluorosis.
8. Selenium in health and disease.
9. Interactions between minerals.
10. The dose–response curve of an essential element.

How to attempt

- Begin with the composition table or chart (Figure A3.1) and the classification by amount and by role.
- For each element asked, use the same row of headings: functions, sources, requirement, deficiency, excess. A table is the best form.
- Add the essentiality curve (Figure A3.2); it scores well in “bio-elementology” answers.
- Close with the Rasashastra table, keeping the status column honest.

Answer frame: “Write the physiological importance and symptoms of deficiency of iron and zinc.” (5 marks)

Part	What to write	Suggested marks
Iron	Body content; functions; absorption (enhancers and inhibitors); requirement; deficiency signs and tests	2.5
Zinc	Functions (enzymes, growth, immunity, taste); requirement; deficiency features	2
Link	Loha or Mandura Bhasma and Yashada Bhasma, with the evidence boundary	0.5

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **29, 32**.

- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman’s The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

Additional verified sources (accessed 16 October 2026)

- **[WHO 1996]** World Health Organization. *Trace Elements in Human Nutrition and Health*. Geneva: WHO; 1996 (prepared with FAO and IAEA).
- **[ICMR-NIN 2020]** ICMR–National Institute of Nutrition. *Nutrient Requirements for Indians: Recommended Dietary Allowances and Estimated Average Requirements*. Hyderabad: ICMR-NIN; 2020. (Values for calcium, iron and zinc as reported from the 2020 report.)
- **[Pal 2014]** Pal D, Sahu CK, Haldar A. Bhasma: the ancient Indian nanomedicine. *J Adv Pharm Technol Res*. 2014;5(1):4–12.

Viva pointers — Chapter A3

Answer each in one or two sentences.

- Which six elements make up about 99 per cent of body mass?
- What is the difference between a macro-mineral and a trace element? Give the cut-off.
- Name four essential trace elements and two potentially toxic ones.
- State the criteria for calling an element essential.
- Draw the dose–response curve of an essential element. Name its three zones.
- How does the curve of lead differ from that of zinc?
- What is homeostasis of an element? Give an example.
- What is the daily calcium requirement for an adult Indian (ICMR-NIN 2020)?
- What is the iron requirement for an adult woman? Why is it higher than for a man?
- Which form of iron is absorbed better: haem or non-haem? Name two enhancers and two inhibitors.
- Which single test best reflects iron stores?
- Name three zinc-containing enzymes. What is acrodermatitis enteropathica?
- Why can long-term zinc supplements cause anaemia?
- What is Wilson’s disease? What are Kayser–Fleischer rings? How is it treated?
- What is cretinism? How is iodine deficiency prevented in India?
- What is Keshan disease?
- What is the difference between dental and skeletal fluorosis?
- Which mineral interaction explains why an iron-deficient child absorbs more lead?
- Which Bhasmas supply an essential element? Which do not?
- Is a Bhasma proven to be a better source of iron than ferrous sulphate?
- Explain the Samanya principle with a mineral example.

PART A · PHARMACOLOGY · CHAPTER A4

A4 · TOXICITY OF HEAVY METALS AND CHELATION THERAPY

SYLLABUS POINT (PART A, 4)

Toxicity of heavy metals and chelation therapy.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A4	Toxicity of Heavy Metals and Chelation Therapy	2	3	Sep 2018, Sep 2020, Aug 2021, Aug 2022, Jan 2024

Why this chapter matters to you. No other subject in Ayurveda is asked about heavy metals as often as Rasashastra, and no scholar of Rasashastra can answer critics, or protect patients, without knowing exactly how lead, mercury and arsenic harm the body. This chapter was set in five of the seven papers examined, twice as a 20-mark question. The August 2021 paper asked for *"the Ayurvedic and modern aspects"* together: section A4.8 is written for that.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define heavy metals and explain the general mechanisms by which toxic metals damage the body.
2. Explain why the chemical form, dose, duration, route, age and nutrition change the toxicity of a metal.
3. Describe the sources, absorption, mechanism, clinical features, diagnosis and treatment of lead, mercury and arsenic toxicity, and outline cadmium and other metals.
4. Explain chelation therapy: principles, the main chelating agents, their indications and dangers.
5. Discuss the published evidence on heavy metals in Ayurvedic medicines, the Indian limits and testing requirements, and the classical view of metal toxicity.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	A, Q.3b	5	Explain the chronic toxicity of Arsenic. What is Chelation therapy? Explain its mode of action in short.
Sep 2020	A, Q.2	20	Write down about Heavy metal toxicity and Chelation therapy
Aug 2021	A, Q.3B	5	Discuss the Ayurvedic and Modern aspects in connection with Heavy Metal Toxicity and Chelation Therapy
Aug 2022	A, Q.3a	5	Define heavy metal toxicity and explain Chelation therapy.
Jan 2024	A, Q.2	20	Define toxicity and explain toxicity of heavy metals.

A4.1 What are heavy metals?

The term **heavy metal** is commonly used for metals and metalloids of high density (usually above about 5 g/cm^3) that are toxic at low concentrations: lead, mercury, cadmium and arsenic (strictly a metalloid) are the main ones, with chromium, nickel, thallium and others. Chemists criticise the term because "heavy" has no exact definition, and some light elements (beryllium) are toxic while some heavy ones (gold, in the metallic form) are not. In toxicology the more precise phrase is **toxic metals**. The term "heavy metals" is kept here because the syllabus uses it.

> **Toxicity** is the capacity of a substance to cause harm to a living organism (A1c.5). For metals, toxicity depends less on the element itself than on its **chemical form** (species), the **dose**, the **duration** and the **route** of exposure.

A4.2 General mechanisms of metal toxicity

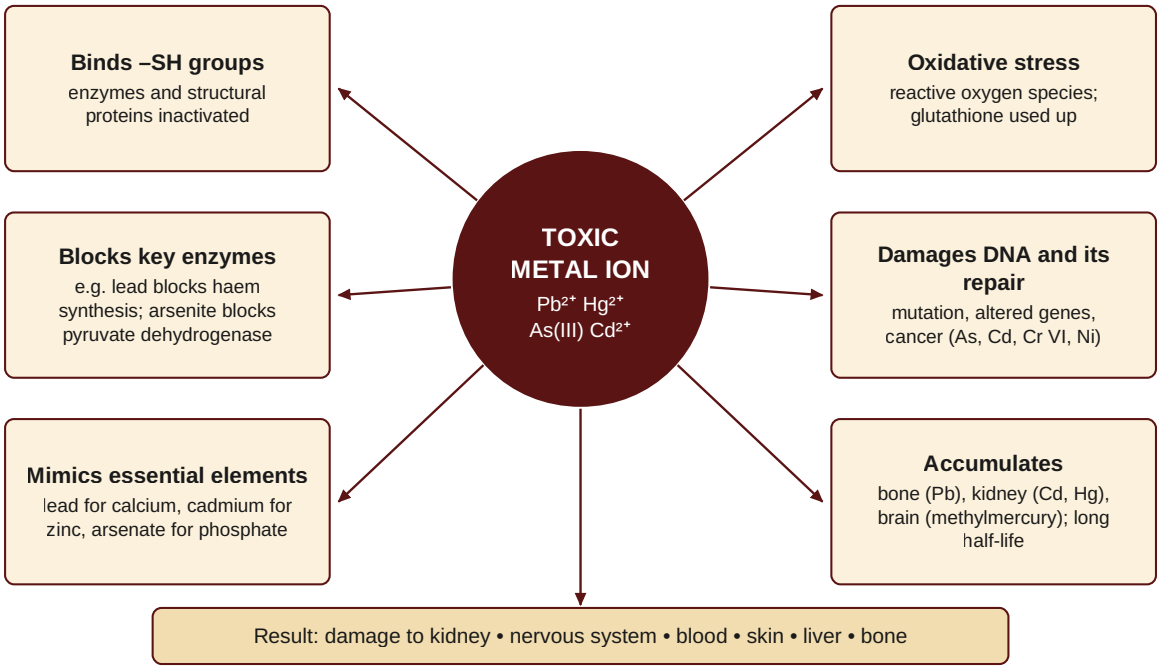


Figure A4.1. General mechanisms by which toxic metals damage cells.

Mechanism	Explanation	Examples
Binding to sulphydryl (–SH) groups	Metals bind the –SH groups of enzymes and structural proteins, inactivating them	Mercury, arsenic (III), lead, cadmium
Inhibition of specific enzymes	A metal blocks one key step of a pathway, so the product falls and the substrate builds up	Lead inhibits δ-aminolaevulinic acid dehydratase and ferrochelatase (haem synthesis); arsenite inhibits pyruvate dehydrogenase
Mimicry of essential elements	The toxic metal takes the place of an essential one in transporters, enzymes or bone	Lead mimics calcium; cadmium mimics zinc; arsenate mimics phosphate
Oxidative stress	Generation of reactive oxygen species; depletion of glutathione	Most toxic metals
Damage to DNA and its repair	Mutation, chromosome damage, altered gene expression	Arsenic, cadmium, chromium (VI), nickel: classed as human carcinogens
Accumulation in tissues	Very slow excretion; long biological half-lives	Lead in bone (decades); cadmium in kidney (decades); methylmercury in brain
Disturbance of calcium signalling and neurotransmission	The metal enters through calcium channels or alters release of transmitters	Lead in the developing brain; methylmercury

Target organs follow from where each metal accumulates: the **kidney** (mercury, cadmium, lead), the **nervous system** (lead, mercury, arsenic), the **blood** (lead), the **skin** (arsenic), the **liver** (arsenic, copper) and the **bone** (lead, cadmium, fluoride).

A4.3 Factors that change toxicity

Factor	Effect	Example
Chemical form (speciation)	The same element can be almost harmless or extremely toxic	Methylmercury is absorbed almost completely and enters the brain; cinnabar (mercuric sulphide) is very poorly absorbed (A4.5)
Solubility	Soluble salts are absorbed; insoluble compounds mostly pass through the gut	Mercuric chloride versus mercuric sulphide
Valency	Different oxidation states differ in toxicity	Arsenite (As III) is more toxic than arsenate (As V); chromium (VI) far more toxic than chromium (III)
Route	Inhalation, ingestion and skin give different absorption	Mercury vapour is absorbed well by the lungs, but liquid mercury is hardly absorbed from the gut
Dose and duration	Acute high doses and chronic low doses produce different illnesses	Acute arsenic poisoning versus arsenicosis from groundwater

Factor	Effect	Example
Age	Children absorb a larger fraction of ingested lead than adults, and their developing brain is more sensitive	Childhood lead poisoning
Nutrition	Deficiency of iron, calcium or zinc increases absorption of lead and cadmium	A3.5
Pregnancy	Many metals cross the placenta	Methylmercury and lead harm the foetal brain
Genetics	Differences in metabolism	Arsenic methylation capacity varies

A4.4 Lead

A4.4.1 Sources

- Old lead-based paints and their dust; lead in petrol (now phased out in India).
- Battery manufacture and recycling; smelting; soldering; printing; ceramics glazed with lead.
- Contaminated food, spices and water from lead pipes.
- Cosmetics and traditional products: some Surma and Kajal, some Sindoor.
- Medicines: Ayurvedic and other traditional medicines containing lead deliberately (Naga Bhasma) or as a contaminant. In the United States, the CDC reported 12 cases of lead poisoning in adults associated with Ayurvedic medicines, from five states, between 2000 and 2003 [CDC 2004].

A4.4.2 Absorption, distribution and excretion

Stage	Details
Absorption	From the gut (more in children, in fasting, and in iron and calcium deficiency) and the lungs (fumes, dust)
Distribution	Blood (in red cells), soft tissues, and especially **bone**, which holds most of the body burden in adults
Half-life	About a month in blood; years to decades in bone. Bone lead is released in pregnancy, lactation, fractures and osteoporosis, causing renewed exposure
Excretion	Mainly urine; some in bile; crosses the placenta and enters milk

The bone store explains two clinical facts: lead poisoning can return months after exposure has stopped, and a pregnant woman can pass lead absorbed years earlier to her baby (Figure A4.2).

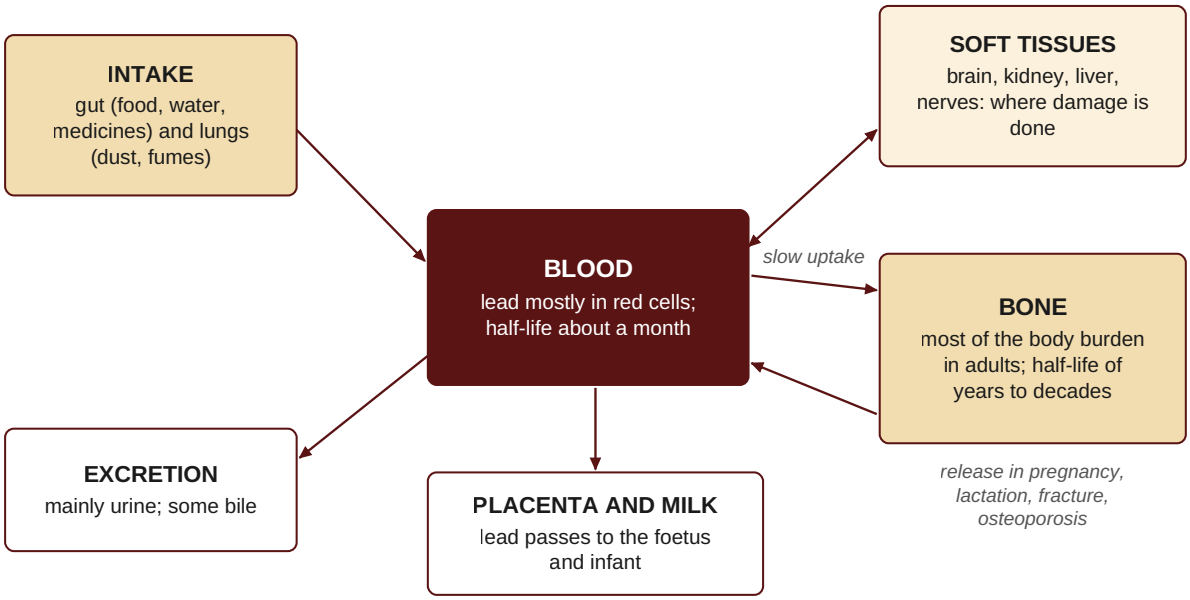


Figure A4.2. Where lead goes in the body: blood, soft tissues and bone. Bone holds most of the body burden and releases it slowly.

A4.4.3 Mechanisms

Lead inhibits two enzymes of haem synthesis, **δ-aminolaevulinic acid dehydratase (ALAD)** and **ferrochelatase**, so haem formation falls and its precursors (ALA in urine, zinc protoporphyrin in red cells) rise (Figure A4.3). Lead also mimics calcium, disturbing nerve signalling and the development of the brain; damages the kidney tubules; and raises blood pressure.

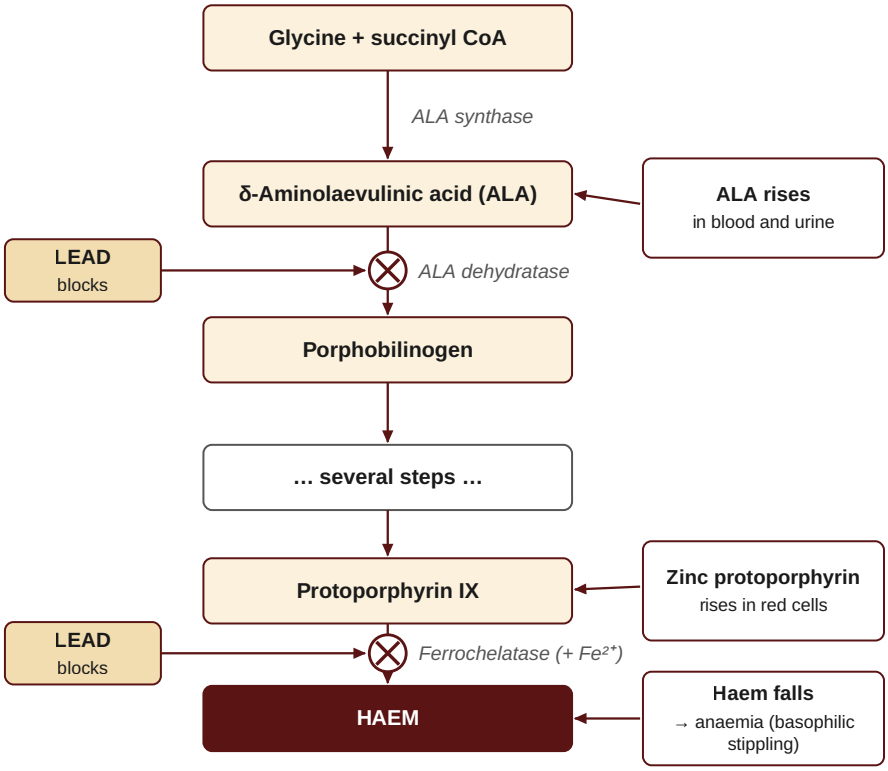


Figure A4.3. Lead blocks two steps of haem synthesis, causing anaemia and the build-up of precursors.

A4.4.4 Clinical features

System	Features
Nervous system (children)	Impaired cognitive development, lower IQ, behavioural problems, at low blood levels. Lead encephalopathy at high levels: irritability, ataxia, convulsions, coma
Nervous system (adults)	Peripheral motor neuropathy (wrist drop, foot drop); headache, poor concentration
Blood	Microcytic anaemia; **basophilic stippling** of red cells
Gastrointestinal	**Lead colic** (severe abdominal pain), constipation, anorexia; a blue-black **Burton's line** on the gums
Kidney	Chronic interstitial nephritis; reduced uric acid excretion ("saturnine gout")
Cardiovascular	Hypertension
Reproductive	Reduced fertility; miscarriage; low birth weight
Bones (children)	"Lead lines" at the growing ends of long bones on X-ray

There is no known safe level of lead exposure, especially for children. In 2021 the CDC lowered its blood lead reference value for children to **3.5 µg/dL**: a child above this level is among the most exposed in the population and needs action [CDC 2021].

A4.4.5 Diagnosis

Blood lead level (the key test); haemoglobin and peripheral smear (basophilic stippling); zinc protoporphyrin; urinary ALA; X-ray of long bones in children; abdominal X-ray for ingested lead fragments; history of the source, including every medicine and cosmetic used.

A4.4.6 Treatment

1. **Remove the source.** This matters more than anything else. Stop the suspected medicine; test it.
2. Supportive care; treat convulsions; correct iron and calcium deficiency.
3. **Chelation** according to the blood level and symptoms (A4.7): oral **succimer (DMSA)** for moderate levels, especially in children; **calcium disodium EDTA**, with **dimercaprol (BAL)** added in encephalopathy or very high levels.
4. Follow-up blood levels, because lead returns to the blood from bone after chelation (rebound).

A4.5 Mercury

A4.5.1 Three forms, three different poisons

Mercury is the clearest example of the rule that **chemical form decides toxicity**. The three forms differ in how they enter the body, where they go and what they damage (Figure A4.4).

Feature	Elemental (metallic) mercury	Inorganic mercury salts	Organic mercury
Examples	Liquid mercury (Parada); mercury vapour	Mercuric chloride ($HgCl_2$); mercurous chloride (calomel); mercuric sulphide (cinnabar, Hingula)	Methylmercury (in fish); ethylmercury (thiomersal); phenylmercury
Main route of harm	**Inhalation of vapour**	Ingestion; skin (creams)	Ingestion (fish, contaminated grain)
Absorption	About 80% of inhaled vapour; very little of swallowed liquid metal	Soluble salts about 7–15% from the gut; cinnabar about 0.2%	More than 95% from the gut
Main target	Brain, kidney, lung	**Kidney**, gut	**Brain** (including the foetal brain)
Typical illness	Tremor, erethism, gingivitis; pneumonitis after heavy exposure	Corrosive gastroenteritis, acute kidney injury; nephrotic syndrome; acrodynia in children	Paraesthesia, ataxia, narrowed visual fields, deafness; cerebral palsy-like damage in the foetus

Absorption figures for the oral forms are those reported by Liu and colleagues [Liu 2008]; the figure for vapour is the value usually given in pharmacology and toxicology texts [R32].

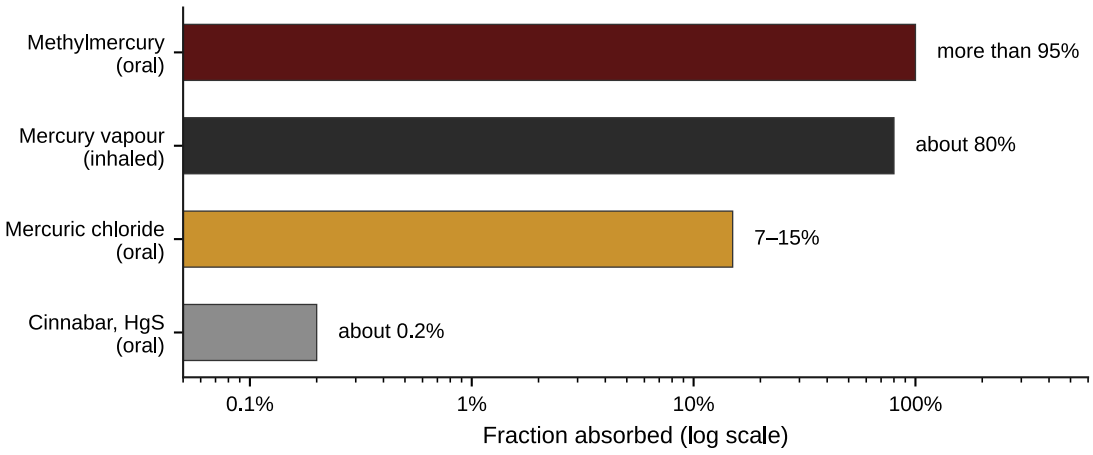


Figure A4.4. Approximate absorption of different forms of mercury. Note the logarithmic scale.

A4.5.2 Sources

- Occupational: chlor-alkali plants, dental amalgam work, broken thermometers and sphygmomanometers, fluorescent lamps, artisanal gold mining (mercury is used to extract gold and then burnt off).
- Food: **methylmercury in large predatory fish**. Mercury released into water is converted to methylmercury by microbes and concentrated up the food chain.
- Medicines and cosmetics: skin-lightening creams containing mercury; some traditional medicines; calomel (formerly in teething powders); thiomersal (preservative, ethylmercury).

A4.5.3 Mechanism

Mercury ions bind strongly to **sulphydryl groups**, inactivating many enzymes and membrane proteins. Elemental mercury vapour is lipid-soluble: it crosses the alveoli and the blood–brain barrier, and inside cells it is oxidised to Hg^{2+}, which is then trapped. Methylmercury binds cysteine and is carried into the brain by an amino-acid transporter.

A4.5.4 Clinical features

Form and exposure	Features
Elemental vapour, acute heavy	Cough, chest pain, chemical pneumonitis; may be fatal
Elemental vapour, chronic	**Erethism** (shyness, irritability, emotional instability, insomnia), intention **tremor**, gingivitis and salivation; kidney damage
Inorganic salt, acute ingestion	Metallic taste, burning pain, bloody diarrhoea, shock; **acute tubular necrosis** within days
Inorganic, chronic	Kidney damage (proteinuria, nephrotic syndrome); in children, **acrodynia** ("pink disease": painful pink hands and feet, irritability)
Organic (methylmercury)	Delayed onset; paraesthesia around mouth and of hands and feet, ataxia, constriction of visual fields, deafness, dysarthria; severe foetal brain damage

Two disasters are often asked about. At **Minamata**, Japan, in the 1950s, a factory released mercury waste into the bay; methylmercury in fish poisoned the fishing community and caused severe brain damage in babies born to exposed mothers. In **Iraq** in the early 1970s, seed grain treated with a methylmercury fungicide was made into bread, causing widespread poisoning.

A4.5.5 Diagnosis

Test	Best for
Blood mercury	Recent exposure; methylmercury (which stays mainly in red cells)
Urine mercury (24-hour or creatinine-corrected)	Chronic exposure to elemental and inorganic mercury
Hair mercury	Long-term methylmercury exposure (grows with the hair)
Kidney tests	Urine protein, serum creatinine

A4.5.6 Treatment

1. Remove the patient from the source (vapour, cream, medicine, fish).
2. Supportive care: fluids, treatment of shock and kidney failure.
3. Chelation (A4.9): **dimercaprol (BAL)** by intramuscular injection early in acute inorganic poisoning, or **DMPS**; **succimer (DMSA)** by mouth for chronic and milder cases.
4. **BAL is avoided in methylmercury and in elemental mercury poisoning**, because it can carry mercury into the brain. DMSA is preferred.

CINNABAR IS NOT MERCURIC CHLORIDE

Rasashastra uses mercury mostly as **mercuric sulphide**: Hingula (natural cinnabar), Kajjali (black mercuric sulphide made by triturating mercury with sulphur), Rasa Sindura (red mercuric sulphide made in Kupipakva). Liu and colleagues reviewed the evidence comparing cinnabar with other mercurials [Liu 2008]:

- Cinnabar is very poorly soluble and **poorly absorbed** (about 0.2% from the gut, against 7–15% for mercuric chloride and more than 95% for methylmercury).
- Absorbed mercury from cinnabar accumulates mainly in the **kidney**. Neurotoxicity in animals required doses far higher than those of methylmercury.
- The authors described cinnabar as "chemically inert with a relatively low toxic potential when taken orally", **but** warned that prolonged use may still cause **kidney damage**.

A later review by the same group concluded that the chemical forms of mercury and arsenic are "a major determinant" of their disposition and toxicity, and that the use of total mercury or arsenic content alone is insufficient for judging risk [Liu 2018].

What this does and does not mean. It means that a cinnabar-based medicine cannot be judged as if it were mercuric chloride. It does **not** mean that such medicines are safe in any dose or for any duration: lower toxicity is not zero toxicity. Most of the data come from animals. The safety of long-term use in human patients needs studies that measure blood and urine mercury and kidney function.

A4.6 Arsenic

A4.6.1 Forms

Form	Examples	Toxicity
Inorganic trivalent (As III)	Arsenic trioxide ("white arsenic"; **Sankhiya** in Rasashastra); sodium arsenite	**Most toxic** of the common forms
Inorganic pentavalent (As V)	Arsenates	Less toxic; partly reduced to As III in the body
Arsenic sulphides	**Realgar** (As_4S_4; **Manahshila**); **orpiment** (As_2S_3; **Haratala**)	Poorly soluble; toxicity depends on solubility and on contamination with arsenic trioxide
Organic arsenic in seafood	Arsenobetaine	Practically non-toxic; excreted unchanged
Arsine gas (AsH_3)	Industrial accident	Severe **haemolysis**, kidney failure

A4.6.2 Sources

The largest source worldwide is **groundwater**. In India, arsenic in tube-well water is a major problem in West Bengal and parts of the Ganga–Brahmaputra plain, and neighbouring Bangladesh is badly affected. The WHO guideline value for arsenic in drinking water is **10 µg per litre**. Other sources are pesticides and wood preservatives (older), smelting, glass and semiconductor industries, and medicines.

A4.6.3 Mechanism

- **Trivalent arsenic** binds sulphydryl groups, especially the paired –SH groups of **lipoic acid** in pyruvate dehydrogenase. The citric acid cycle and energy production fail.
- **Pentavalent arsenic** resembles phosphate. It replaces phosphate in ATP formation ("arsenolysis"), so energy is wasted.
- Arsenic also causes oxidative stress and disturbs DNA repair and gene expression; inorganic arsenic is a **human carcinogen**.
- The body methylates inorganic arsenic in the liver and excretes it in urine.

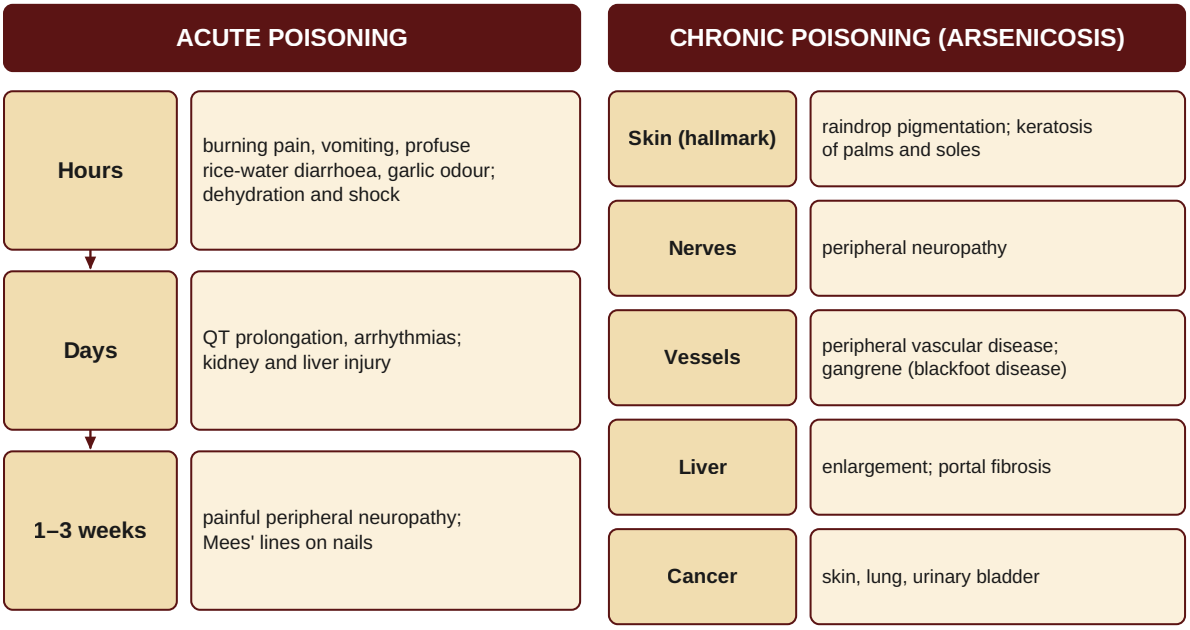


Figure A4.5. Acute and chronic arsenic poisoning: the main features.

A4.6.4 Clinical features

Acute poisoning (usually suicidal, homicidal or accidental ingestion):

- Within hours: burning pain, vomiting, profuse **"rice-water" diarrhoea**, garlic odour of breath; dehydration and shock.
- Heart: QT prolongation and dangerous arrhythmias.
- Survivors: after one to three weeks, a painful **peripheral neuropathy** (sensory then motor), and white transverse bands on the nails (**Mees' lines**).

Chronic poisoning (arsenicosis), usually from drinking water over years:

System	Features
Skin (the hallmark)	**Pigmentation** with “raindrop” spots, especially on the trunk; **keratosis** of palms and soles; Bowen’s disease
Nerves	Peripheral neuropathy
Blood vessels	Peripheral vascular disease; gangrene (“blackfoot disease”, described in Taiwan)
Liver	Enlargement; non-cirrhotic portal fibrosis
Cancer	Skin, lung and urinary bladder cancers
Others	Anaemia; chronic lung disease; raised risk of diabetes

A4.6.5 Diagnosis

Urine arsenic (24-hour), ideally with **speciation** to separate toxic inorganic arsenic from harmless seafood arsenobetaine (or after avoiding seafood for a few days); hair and nail arsenic for chronic exposure; ECG; nerve conduction studies.

A4.6.6 Treatment

1. Acute: decontamination, intensive fluid replacement, cardiac monitoring.
2. Chelation in acute symptomatic poisoning: **BAL** intramuscularly, or **DMPS**; then **DMSA** by mouth. Chelation is most useful early.
3. Arsine: exchange transfusion and haemodialysis; chelators are of little help.
4. Chronic arsenicosis: the main measure is a **safe water supply**. Chelation has a limited role.

A POISON THAT BECAME A DRUG

Arsenic trioxide is a standard treatment for **acute promyelocytic leukaemia**. In a randomised trial in low- and intermediate-risk patients, all-trans retinoic acid with arsenic trioxide was at least as good as, and probably better than, retinoic acid with chemotherapy (two-year event-free survival 97% against 86%) [Lo-Coco 2013]. The same trial recorded more liver toxicity and QT prolongation with arsenic, so it is given under close monitoring.

This is a clear modern case of a toxic metalloid used deliberately as a drug, in a pure form, at a measured dose, for a defined disease, with monitoring. It is worth quoting in an answer. It does **not**, by itself, show that classical arsenical preparations are safe or effective for their classical indications; that needs its own evidence (Chapter A5).

A4.7 Cadmium and other metals

A4.7.1 Cadmium

Aspect	Details
Sources	Smoking (tobacco concentrates cadmium); rice and vegetables grown on contaminated soil; nickel–cadmium batteries; pigments; smelting and welding fumes
Retention	Bound to metallothionein; stored in **kidney** and liver; half-life of decades
Kidney	Damage to the **proximal tubule**: loss of low-molecular-weight proteins (β_2-microglobulin), glucose, amino acids; kidney stones
Bone	Osteomalacia and osteoporosis. **Itai-itai disease** (“ouch-ouch” disease) affected women along the Jinzu River in Japan, from cadmium-contaminated rice
Lung	Inhaled fumes: acute pneumonitis; chronic: emphysema; lung cancer
Treatment	Stop exposure; supportive care. **Chelation is not useful in chronic cadmium poisoning** and may increase kidney damage

A4.7.2 Other metals of interest to Rasashastra

Metal	Rasashastra drug	Main toxic effects of the metal or its compounds
Copper	Tamra Bhasma	Acute: vomiting, haemolysis, liver and kidney injury (copper sulphate poisoning). Chronic overload: liver damage (as in Wilson’s disease)
Zinc	Yashada Bhasma	Fume fever (inhaled oxide in welders); excess intake causes copper deficiency (A3.4.2)
Tin	Vanga Bhasma	Inorganic tin of low toxicity by mouth; organotin compounds (industrial) are neurotoxic
Silver	Rajata Bhasma	**Argyria**: permanent blue-grey discolouration of skin, from long use of silver preparations such as colloidal silver
Gold	Swarna Bhasma	Gold **salts** once used in rheumatoid arthritis caused dermatitis, kidney damage and blood disorders. Metallic gold particles are a different chemical form; their toxicity must be studied separately
Iron	Loha, Mandura Bhasma	Acute iron poisoning in children; chronic overload (A3.4.1)
Thallium	—	Hair loss, painful neuropathy (rat poison); antidote Prussian blue
Aluminium	—	Encephalopathy and bone disease in patients on dialysis; deferoxamine
Chromium (VI), nickel	—	Skin allergy; lung cancer (industrial)

A4.8 Heavy metals in Ayurvedic medicines: the published evidence

This section is what examiners look for when a question asks for “Ayurvedic and modern aspects”. Know the studies, and know how to interpret them fairly.

A4.8.1 The key studies

Study	What was done	What was found
CDC, MMWR 2004 [CDC 2004]	Case reports collected by state surveillance in the United States	12 adults with lead poisoning associated with Ayurvedic medicines, from five states, 2000–2003
Saper and colleagues, JAMA 2004 [Saper 2004]	70 Ayurvedic herbal medicine products made in South Asia, bought in shops in the Boston area, tested for lead, mercury and arsenic	**14 of 70 (20%)** contained heavy metals: lead in 13, mercury in 6, arsenic in 6. If taken as recommended, each could give intakes above regulatory standards
Saper and colleagues, JAMA 2008 [Saper 2008]	193 Ayurvedic products bought over the internet, made in India or the USA	**20.7%** contained detectable lead, mercury or arsenic. **40.6% of Rasa Shastra products** against 17.1% of non-Rasa Shastra products. All the metal-containing products exceeded at least one standard for acceptable daily intake

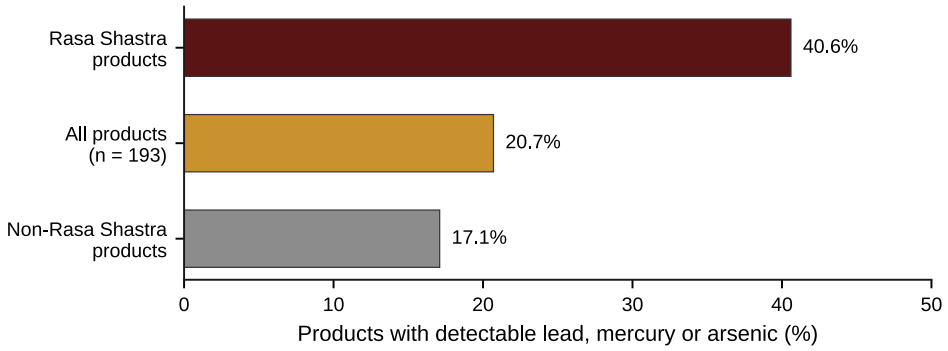


Figure A4.6. Proportion of products containing detectable lead, mercury or arsenic in the 2008 internet study, by type of product. Data from Saper and colleagues (2008).

A4.8.2 How to interpret these studies

1. **Two different problems are mixed together.** In a **herbal** product, any heavy metal is a **contaminant** from soil, water, processing or adulteration; finding it is a failure of quality control. In a **Rasaushadhi**, the metal is an **intended ingredient**; finding it is expected. The real question for a Rasaushadhi is whether its chemical form, dose and duration of use make it safe.
2. **Total content is not the whole story.** These studies measured total metal. As shown above, the chemical form decides absorption and toxicity [Liu 2008; Liu 2018]. A high total mercury content in Kajjali is not equivalent to the same amount of mercuric chloride.
3. **But total content cannot be dismissed either.** Intake standards exist for good reasons. The case reports are of real patients with raised blood lead and symptoms. Lead, in particular, has no known safe level.
4. **The honest position** for a Rasashastra scholar is: these medicines contain metals by design; their safety depends on correct Shodhana and Marana, on dose and duration, and on supervision; and the evidence for long-term human safety, with measured blood and urine levels, is still limited.

A4.8.3 Limits and testing in India

The Government of India informed Parliament in 2009 that testing for heavy metals had been made **mandatory for the export of purely herbal Ayurveda, Siddha and Unani medicines from 1 January**

2006, with limits of **lead 10 ppm, mercury 1 ppm and arsenic 3 ppm**, described as WHO limits. The same answer stated that metals in herbo-mineral preparations are permitted only after due purification as laid down in the classical texts [Lok Sabha 2009].

Element	Limit stated for export of herbal ASU medicines [Lok Sabha 2009]
Lead	10 ppm
Mercury	1 ppm
Arsenic	3 ppm
Cadmium	Not stated in that answer. A limit of 0.3 ppm for starting materials is reported from WHO guidance of 1998 [Kazi 2023]

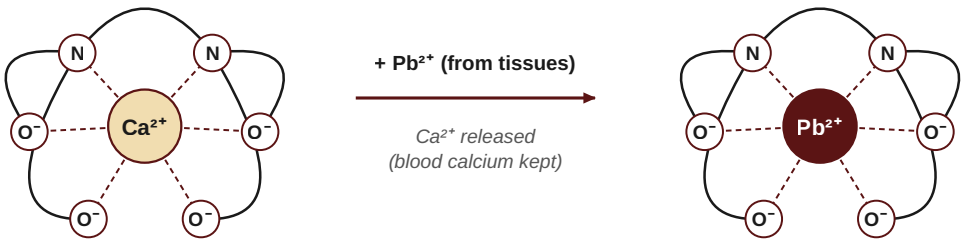
The Ayurvedic Pharmacopoeia of India gives its own limits for heavy metals in its monographs and appendices. The author could not verify the current appendix text directly for this edition. Consult the latest volume of the API for the official figures. Labelling requirements for medicines containing heavy metals are discussed with the Drugs and Cosmetics Rules in Chapter B8.

A4.9 Chelation therapy

A4.9.1 Definition and principle

The word **chelate** comes from the Greek *chele*, a crab's claw. A **chelating agent** is a molecule with two or more donor atoms (sulphur, nitrogen or oxygen) that grip a metal ion from several sides at once, like a claw, forming a stable ring-shaped complex called a **chelate** (Figure A4.7).

> **Chelation therapy** is the use of a chelating agent to bind a toxic metal in the body and form a stable, water-soluble, less toxic complex that is excreted in the urine or bile.



Calcium disodium EDTA
(calcium held in the claw)

Lead–EDTA chelate
stable, water-soluble, excreted in urine

Six donor atoms (2 N, 4 O) grip the metal from all sides, like a claw (Greek chele). Dashed lines: coordinate bonds.

Figure A4.7. A chelating agent grips a metal ion like a claw. Calcium disodium EDTA wraps around lead with six donor atoms; the lead takes the place of the calcium.

A4.9.2 Mode of action, step by step

1. The chelator is given by injection or by mouth and reaches the tissue fluids.

2. Its donor atoms bind the toxic metal **more strongly** than the body's own ligands (enzyme –SH groups, proteins) do. The metal is pulled off its target.
3. The enzyme or protein that was blocked can work again, if it is not permanently damaged.
4. The chelate is a **stable, water-soluble** complex. It does not release the metal on the way out.
5. The chelate is **excreted**, mostly by the kidney (some through bile).

Why sulphur chelators for mercury, arsenic and lead, and oxygen chelators for iron? Metals differ in the donor atoms they prefer. "Soft" metal ions such as mercury, lead, cadmium and arsenic bind strongly to sulphur, so **thiol** chelators (BAL, DMSA, DMPS, penicillamine) are used for them. "Hard" ions such as iron (Fe^{3+}) and aluminium prefer oxygen, so **deferoxamine**, with oxygen donor groups, is used for them.

A4.9.3 Properties of an ideal chelating agent

Property	Why it matters
Higher affinity for the toxic metal than the body's ligands have	So it can pull the metal off its target
Low affinity for essential metals (calcium, zinc, copper)	So it does not cause deficiency
Water-soluble, and reaches the site where the metal is stored	So it can reach the metal (bone, brain, kidney)
Forms a stable, non-toxic, readily excreted complex	So the metal leaves the body and is not redistributed
Little metabolism; low toxicity of its own	Safety
Effective by mouth	Convenience, especially in children and in chronic exposure

No chelating agent has all these properties. That is why the choice depends on the metal, the form and the patient.

A4.9.4 The chelating agents

Agent	Route	Main uses	Important adverse effects and cautions
Dimercaprol (BAL, British anti-Lewisite)	Deep IM injection, in oil	Arsenic; inorganic mercury (acute); lead encephalopathy, with $CaNa_2EDTA$; gold	Rise in blood pressure, tachycardia, nausea, burning of lips and eyes, fever in children; haemolysis in G6PD deficiency. **Not for** methylmercury or elemental mercury (moves mercury into brain), iron or cadmium (the complexes are toxic)
Succimer (DMSA, dimercaptosuccinic acid)	Oral	Lead (especially children); arsenic; mercury	Mild: nausea, rash, rise in liver enzymes. The usual first choice for moderate lead poisoning
Unithiol (DMPS)	Oral or IV	Mercury, arsenic	Allergic skin reactions; not available in every country
Calcium disodium edetate ($CaNa_2EDTA$)	IV (or IM)	**Lead**	**Kidney toxicity**; zinc loss. In encephalopathy, give BAL first, then $CaNa_2EDTA$

Agent	Route	Main uses	Important adverse effects and cautions
Penicillamine	Oral	Copper (**Wilson's disease**); lead (second line); also cystinuria, rheumatoid arthritis	Rash, loss of taste, proteinuria, bone-marrow suppression, autoimmune reactions; give pyridoxine
Trientine	Oral	Wilson's disease when penicillamine is not tolerated	Iron deficiency anaemia
Deferoxamine (desferrioxamine)	IV, IM or SC	**Iron** (acute poisoning; chronic transfusion overload); aluminium	Low blood pressure if given fast; visual and hearing changes with long use; infection with *Yersinia*
Deferiprone	Oral	Iron overload	**Agranulocytosis**; weekly blood counts
Deferasirox	Oral, once daily	Iron overload	Kidney and liver toxicity
Prussian blue	Oral	Thallium; radioactive caesium	Constipation. Strictly not a chelator: it binds the metal in the gut and stops it being reabsorbed

NEVER CONFUSE THE TWO EDTAS

Calcium disodium edetate ($CaNa_2EDTA$) is used for lead poisoning. Its calcium is exchanged for lead, so blood calcium does not fall.

Disodium edetate (Na_2EDTA) has no calcium. It binds the patient's own calcium and can cause sudden, fatal **hypocalcaemia**. The CDC reported three deaths from hypocalcaemia during chelation therapy, two of them children, in the United States between 2003 and 2005; in at least two, Na_2EDTA had been given. The CDC advised that **Na_2EDTA should never be used for lead or other heavy-metal poisoning in children** [CDC 2006].

A4.9.5 Choosing a chelator

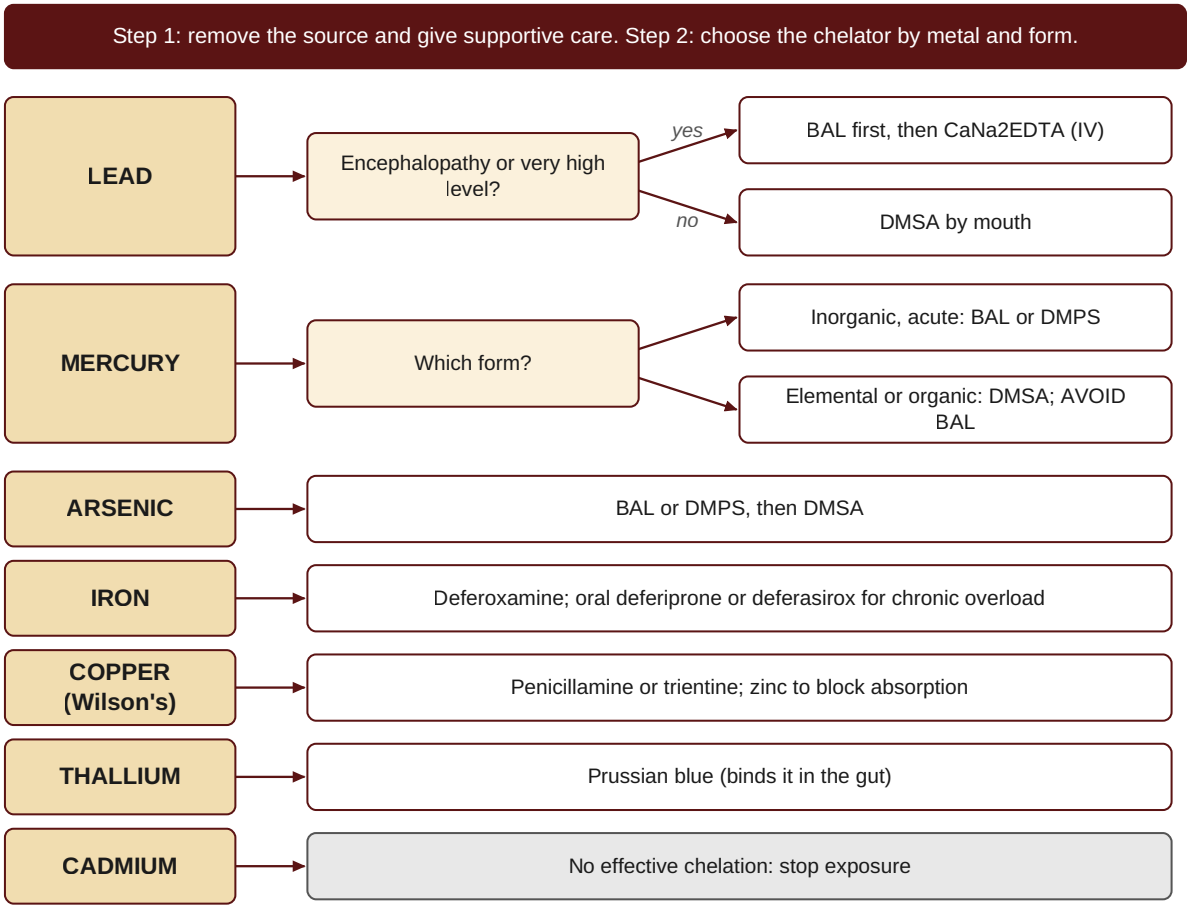


Figure A4.8. Choosing a chelating agent: a simplified guide for the commonest toxic metals.

Metal	First choice	Alternatives and notes
Lead, mild to moderate	**DMSA** (oral)	Penicillamine
Lead, severe or encephalopathy	**BAL + $CaNa_2EDTA$**	BAL is started first
Mercury, inorganic, acute	**BAL** or **DMPS**	Then DMSA
Mercury, elemental or organic	**DMSA**	**Avoid BAL**
Arsenic, acute	**BAL** or **DMPS**	Then DMSA
Copper (Wilson’s disease)	**Penicillamine** or **trientine**	Zinc to block absorption
Iron	**Deferoxamine**	Deferiprone, deferasirox (chronic overload)
Thallium	**Prussian blue**	—
Cadmium	**No effective chelation** in chronic poisoning	Stop exposure

A4.9.6 Limits and dangers of chelation

1. **Removing the source comes first.** Chelation without stopping exposure is useless.
2. Chelation removes only the metal it can reach. Metal in bone is released slowly, so blood levels **rebound** after a course.
3. A chelator can **redistribute** metal to a more dangerous place (BAL with methylmercury; chelation of lead before encephalopathy is controlled).
4. Chelators also remove **essential metals** (zinc, copper, calcium).
5. Chelation does **not reverse damage** that has already been done to the brain, nerves or kidneys.
6. The chelators have their own toxicity (kidney with EDTA; marrow with deferiprone and penicillamine).

A4.9.7 Unproven uses of chelation

Chelation is sometimes promoted for “detoxification”, for autism and for coronary artery disease. These uses are **not established**.

- In the first Trial to Assess Chelation Therapy (TACT), an EDTA-based regimen in patients after myocardial infarction gave a modest reduction in cardiovascular events, with the apparent benefit concentrated in patients with diabetes [Lamas 2013].
- A second, larger trial designed to test that finding in patients with diabetes (TACT2, about 1000 patients) found **no reduction** in cardiovascular events, even though chelation lowered blood lead by about 60% [Lamas 2024].

The lesson for a scholar: a plausible mechanism and a promising first trial are not enough. The claim must survive replication.

A4.10 Heavy metal toxicity: Ayurvedic and modern aspects

A4.10.1 The classical view

Ayurveda knew metals and minerals as poisons long before modern toxicology.

- **Sushruta** lists the mineral poisons, **Dhatu Visha**, among the sites of inanimate (Sthavara) poison, naming **Haritala** (orpiment, an arsenic sulphide) and Phenashma Bhasma among them (Sushruta Samhita, Kalpa Sthana 2/5) [R26].
- **Dushi Visha** is a poison that remains in the body in a weakened form, is repeatedly provoked by unsuitable place, season, food and day-sleep, and is so named because it vitiates the Dhatus (Sushruta Samhita, Kalpa Sthana 2/33) [R26]. Charaka describes Dushi Visha as vitiating the blood and producing skin disorders such as Kitibha and Kotha (Charaka Samhita, Chikitsa Sthana 23/31) [R26].
- **Gara Visha** is a poison prepared artificially from combinations of substances; it acts after a delay and does not kill at once (Charaka Samhita, Chikitsa Sthana 23/14) [R26].
- Charaka gives **twenty-four measures (Chaturvimshati Upakrama)** for the treatment of poisoning, including Vamana, Virechana, Raktamokshana, Prativisha and Hridayavarana (Charaka Samhita, Chikitsa Sthana 23/35–37) [R26].
- **Rasashastra** goes further. Its texts warn that mercury and metals used without proper purification and incineration (**Ashuddha** or **Apakva**) cause serious diseases and even death. The whole apparatus of **Shodhana** (purification), **Marana** (incineration to Bhasma), **Amritikarana** and **Bhasma Pariksha**

exists to prevent that toxicity. Classical texts also describe specific remedies for the ill effects of improperly processed metals (Rasatarangini) [R7].

A4.10.2 Classical safeguards and their modern reading

Classical safeguard	Purpose in the tradition	Modern reading	Status
Shodhana of the raw metal or mineral	Remove Doshas (impurities and harmful qualities)	Removes some impurities; may change the chemical form and surface	Shown for some materials in laboratory studies; effect on human toxicity not measured for most
Marana to Bhasma	Convert the metal to a fine, non-metallic, assimilable form	Converts metals to oxides, sulphides or other compounds of small particle size	Phase changes are well documented by XRD and electron microscopy
Bhasma Pariksha: Rekhapurnata, Varitara, Nishchandrata, **Niruttha**, **Apunarbhava**	Confirm fineness and **absence of free metal** that could revert	Qualitative tests of particle size and of conversion to a non-metallic form	Useful screening; do not measure soluble or absorbable metal
Matra (dose in Ratti or less) and fixed duration	Keep the dose small and the course limited	Total intake = dose × duration; the modern risk concept	Agrees with modern principles; safe limits need measurement for each product
Anupana and **Pathya**	Guide action; protect the patient	May alter absorption; diet affects metal absorption (iron, calcium)	Plausible; rarely tested
Prescription by a qualified physician; observation	Detect ill effects early	Pharmacovigilance; monitoring blood and urine levels	Monitoring of metal levels is a modern addition worth adopting (Chapter A6)

A4.10.3 Comparing the two views

Aspect	Ayurvedic view	Modern view
Cause of toxicity	Ashuddha or Apakva metal; Doshas of Parada and metals; wrong dose or use	Free or soluble metal ions binding –SH groups and mimicking essential elements
Chronic accumulation	Dushi Visha: weak poison lingering in the Dhatus and flaring later	Storage in bone (lead), kidney (cadmium), brain (methylmercury); release later
Dependence on form	Same metal is poison when raw, medicine when properly processed	Chemical form (speciation) decides absorption and toxicity
Prevention	Shodhana, Marana, Bhasma Pariksha, Matra, Anupana, Pathya	Purity limits, speciation, GMP, testing, labelling
Treatment	Stop the drug; Visha Chikitsa; specific remedies	Remove source; supportive care; chelation
Monitoring	Clinical signs	Blood and urine metal levels; kidney, liver and blood tests

Established. Raw metals and their soluble compounds are toxic. The chemical form decides absorption and toxicity, and cinnabar is far less absorbed than soluble mercury salts or methylmercury (animal and pharmacokinetic data) [Liu 2008; Liu 2018]. A significant share of Ayurvedic products sold abroad contained lead, mercury or arsenic at levels above intake standards [Saper 2004; Saper 2008], and cases of lead poisoning linked to such products are documented [CDC 2004]. Arsenic trioxide is an effective, approved drug for acute promyelocytic leukaemia [Lo-Coco 2013].

Not established. That Shodhana and Marana make every Rasaushadhi safe for long-term use; that Bhasma particles are not absorbed or are wholly excreted; that passing the classical Pariksha guarantees the absence of toxicity; and that any classical procedure removes metals from the body as a chelating agent does. These need measured human data: blood and urine metal levels, kidney function and follow-up.

KEY POINTS TO REMEMBER

- "Heavy metals" is a loose term; toxicologists prefer "toxic metals". The main ones are lead, mercury, arsenic and cadmium.
- General mechanisms: binding to –SH groups, enzyme inhibition, mimicry of essential elements, oxidative stress, DNA damage, accumulation.
- **Chemical form decides toxicity.** Oral absorption: methylmercury more than 95%; mercuric chloride 7–15%; cinnabar about 0.2% [Liu 2008].
- Lead: blocks ALA dehydratase and ferrochelatase; anaemia with basophilic stippling; colic; Burton's line; wrist drop; brain damage in children; stored in bone. CDC reference value 3.5 µg/dL (2021). No safe level.
- Mercury: vapour → brain (erethism, tremor); inorganic salts → kidney; methylmercury → brain and foetus (Minamata). BAL is avoided in organic and elemental mercury poisoning.
- Arsenic: As III binds lipoic acid; acute rice-water diarrhoea, QT prolongation, neuropathy, Mees' lines; chronic raindrop pigmentation, palmoplantar keratosis, cancers. WHO water guideline 10 µg/L. Arsenic trioxide treats APL.
- Cadmium: kidney tubules and bone (itai-itai); no useful chelation.

- Chelate = claw. Thiol chelators (BAL, DMSA, DMPS, penicillamine) for soft metals; deferoxamine for iron.
- **$CaNa_2EDTA$ for lead; never Na_2EDTA** (fatal hypocalcaemia).
- Remove the source first. Chelation causes rebound, redistribution and loss of essential metals, and does not reverse damage.
- TACT2 (2024): no cardiovascular benefit from EDTA chelation in diabetes after MI.
- Saper 2004: 20% of 70 products had metals. Saper 2008: 20.7% of 193; 40.6% of Rasa Shastra products.
- India: heavy-metal testing mandatory for export of herbal ASU medicines from 1 January 2006 (Pb 10, Hg 1, As 3 ppm).
- Classical: Dhatu Visha (Haritala), Dushi Visha, Gara Visha, 24 Upakrama; Rasashastra's Shodhana, Marana and Bhasma Pariksha exist to prevent metal toxicity, but long-term human safety needs measured data.

EXAM CORNER — CHAPTER A4

Likely long questions (20 marks)

1. Define toxicity and explain the toxicity of heavy metals *(set in January 2024)*.
2. Write about heavy metal toxicity and chelation therapy *(set in September 2020)*.
3. Describe the sources, mechanism, clinical features and management of lead poisoning. Add a note on lead in Ayurvedic medicines.
4. Classify chelating agents. Describe their mode of action, uses and adverse effects.
5. Discuss mercury toxicity in relation to the chemical form of mercury. How does this bear on the use of Parada in Rasashastra?

Likely short notes (5 marks)

1. Chronic toxicity of arsenic; chelation therapy and its mode of action *(set in September 2018)*.
2. Ayurvedic and modern aspects of heavy metal toxicity and chelation therapy *(set in August 2021)*.
3. Define heavy metal toxicity and explain chelation therapy *(set in August 2022)*.
4. Dimercaprol (BAL).
5. Calcium disodium EDTA. Why not disodium EDTA?
6. Minamata disease.
7. Itai-itai disease.
8. Dushi Visha and its modern correlation.
9. Heavy metals in Ayurvedic medicines: the published evidence.
10. Arsenic trioxide as a drug.

How to attempt

- **Start with definitions:** toxicity (A1c.5), heavy metals, chelation. Then draw the mechanisms figure (Figure A4.1).
- For each metal use the **same row of headings**: forms and sources, absorption and storage, mechanism, clinical features (acute and chronic), diagnosis, treatment. A table per metal is the clearest answer.
- For chelation, draw the claw (Figure A4.7) and give the table of agents with route, uses and adverse effects.

- In any question that mentions Ayurveda, add the comparison table (A4.10.3) and close with the **Established / Not established** lines. A balanced closing paragraph scores better than a defensive one.

Answer frame: "Define toxicity and explain toxicity of heavy metals." (20 marks)

Part	What to write	Suggested marks
Definitions	Toxicity; heavy metal (note the term's weakness); acute and chronic toxicity	2
General mechanisms	–SH binding, enzyme inhibition, mimicry, oxidative stress, DNA damage, accumulation; figure	3
Factors	Chemical form, solubility, valency, route, dose, age, nutrition	2
Lead	Sources, haem pathway figure, clinical features table, blood lead, treatment	4
Mercury	Three forms table; clinical features; Minamata; treatment	3
Arsenic	Forms; mechanism; acute and chronic features; treatment	3
Cadmium and others	Itai-itai; brief table	1
Ayurvedic relevance	Studies (Saper 2004, 2008); chemical form (cinnabar); limits; Established / Not established	2

Answer frame: "Define heavy metal toxicity and explain chelation therapy." (5 marks)

Part	What to write	Suggested marks
Definition	Heavy metal toxicity in two lines, with the main metals and mechanisms	1
Chelation: principle	Claw figure; mode of action in five steps	1.5
Agents	Table: BAL, DMSA, $CaNa_2EDTA$, penicillamine, deferoxamine with uses	1.5
Cautions	Remove source; Na_2EDTA danger; BAL and methylmercury; rebound	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **7, 26, 29, 32, 36**.

- **[R7]** Sharma S, Shastri HD. *Rasatarangini*. 11th ed. Delhi: Motilal Banarasidas; 1979.
- **[R26]** Charaka Samhita, Chikitsa Sthana 23/14, 23/31 and 23/35–37 (verse numbers verified in Charak Samhita New Edition, carakasamhitaonline.com, accessed 16 October 2026). Sushruta Samhita, Kalpa Sthana 2/5 and 2/33 (verse numbers as in the 1938 Nirnaya Sagar edition, checked against the Sushruta Project provisional edition of the Kalpasthana, saktumiva.org, accessed 16 October 2026).
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024. (Chelating agents.)
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023. (Heavy metals and chelation.)
- **[R36]** *The Ayurvedic Pharmacopoeia of India*. New Delhi: Ministry of AYUSH, Government of India. (Limits for heavy metals: consult the current volume.)

Additional verified sources (accessed 16 October 2026)

- **[CDC 2004]** Centers for Disease Control and Prevention. Lead poisoning associated with Ayurvedic medications — five states, 2000–2003. *MMWR Morb Mortal Wkly Rep*. 2004;53(26):582–584.
- **[CDC 2006]** Centers for Disease Control and Prevention. Deaths associated with hypocalcemia from chelation therapy — Texas, Pennsylvania, and Oregon, 2003–2005. *MMWR Morb Mortal Wkly Rep*. 2006;55(8):204–207.
- **[CDC 2021]** Centers for Disease Control and Prevention. Blood lead reference value updated to 3.5 µg/dL (October 2021). cdc.gov.
- **[Saper 2004]** Saper RB, Kales SN, Paquin J, Burns MJ, Eisenberg DM, Davis RB, Phillips RS. Heavy metal content of Ayurvedic herbal medicine products. *JAMA*. 2004;292(23):2868–2873.
- **[Saper 2008]** Saper RB, Phillips RS, Sehgal A, Khouri N, Davis RB, Paquin J, Thuppil V, Kales SN. Lead, mercury, and arsenic in US- and Indian-manufactured Ayurvedic medicines sold via the Internet. *JAMA*. 2008;300(8):915–923.
- **[Liu 2008]** Liu J, Shi JZ, Yu LM, Goyer RA, Waalkes MP. Mercury in traditional medicines: is cinnabar toxicologically similar to common mercurials? *Exp Biol Med (Maywood)*. 2008;233(7):810–817.
- **[Liu 2018]** Liu J, Wei LX, Wang Q, Lu YF, Zhang F, Shi JZ, Li C, Cherian MG. A review of cinnabar (HgS) and/or realgar (As_4S_4)-containing traditional medicines. *J Ethnopharmacol*. 2018;210:340–350. doi:10.1016/j.jep.2017.08.037.
- **[Lo-Coco 2013]** Lo-Coco F, Avvisati G, Vignetti M, et al. Retinoic acid and arsenic trioxide for acute promyelocytic leukemia. *N Engl J Med*. 2013;369(2):111–121.
- **[Lamas 2013]** Lamas GA, Goertz C, Boineau R, et al. Effect of disodium EDTA chelation regimen on cardiovascular events in patients with previous myocardial infarction: the TACT randomized trial. *JAMA*. 2013;309(12):1241–1250.
- **[Lamas 2024]** Lamas GA, Anstrom KJ, Navas-Acien A, et al. Edetate disodium-based chelation for patients with a previous myocardial infarction and diabetes: TACT2 randomized clinical trial. *JAMA*. 2024;332(10):794–803.
- **[Lok Sabha 2009]** Lok Sabha, Government of India. Unstarred Question No. 762, answered 25 February 2009 (heavy metals in Ayurvedic medicines). eparlib.nic.in.
- **[Kazi 2023]** Kazi S, Bais S. Evaluation of potentially hazardous contaminants in anti-viral herbal products used in clinical practice. *Asian J Pharm Clin Res*. 2023;16(3):128–132. (Secondary source for the WHO 1998 cadmium limit.)

Viva pointers — Chapter A4

Answer each in one or two sentences.

- Why do toxicologists dislike the term “heavy metal”?
- Which chemical group in enzymes do mercury, arsenic and lead attack?
- Which two enzymes of haem synthesis does lead inhibit? What do you see on the blood smear?
- What is Burton’s line? What is wrist drop?
- Where is most of the body’s lead stored? Why does that matter in pregnancy?
- What is the CDC blood lead reference value for children today?
- Name the three forms of mercury. Which one damages the brain most? Which one the kidney?
- What is erethism? What is acrodynia?
- What happened at Minamata?
- Why is BAL avoided in methylmercury poisoning?
- How much cinnabar is absorbed from the gut, compared with mercuric chloride and methylmercury?
- Which is more toxic, arsenite or arsenate? Why?
- What are Mees’ lines? What is “raindrop” pigmentation?
- What is the WHO guideline value for arsenic in drinking water?
- Name an arsenic compound that is an approved anticancer drug. For which disease?
- What is itai-itai disease? Which metal causes it?
- What does “chelate” mean? Draw a chelate.
- Why are thiol chelators used for mercury and lead, and deferoxamine for iron?
- Why must you never give disodium EDTA for lead poisoning?
- In lead encephalopathy, which chelator is given first, and why?
- Name two oral iron chelators and the main danger of each.
- What is the drug of choice for Wilson’s disease?
- What did the TACT2 trial show?
- What proportion of Ayurvedic products contained heavy metals in Saper’s 2004 and 2008 studies?
- What are the export limits for lead, mercury and arsenic in herbal ASU medicines?
- What is Dushi Visha? Give its modern parallel.
- Which Bhasma Pariksha tests aim to show the absence of free metal?
- Does a Bhasma that passes Apunarbhava and Niruttha tests have proven long-term safety? Why not?

PART A · PHARMACOLOGY · CHAPTER A5

A5 · TOXICITY AND PHARMACOLOGICAL ACTIVITIES OF HERBO-MINERAL COMPOUNDS

SYLLABUS POINT (PART A, 5)

Knowledge of toxicity and pharmacological activities of herbo-mineral compounds.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A5	Toxicity and Pharmacological Activities of Herbo-mineral Compounds	0	1	Jul 2018

Why this chapter matters to you. Every Rasashastra scholar will be asked, sooner or later, by a patient, a colleague from modern medicine or a regulator: *"Is this medicine safe, and does it work?"* This chapter teaches you to answer that question for herbo-mineral medicines honestly, with the evidence that exists, and with a clear statement of what is not yet known. It builds on Chapter A4 (heavy metals) and leads into Chapter A6 (pharmacovigilance). It was set as a short note in the July 2018 paper.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define herbo-mineral compounds and classify the main groups of Rasaushadhi.
2. Describe their physico-chemical nature as shown by modern analysis.
3. Explain the proposed ways in which herbo-mineral compounds may act, and separate what is established from what is proposed.
4. Summarise the published evidence on the pharmacological activities and the toxicity of important herbo-mineral preparations.
5. Read an animal toxicity study of a Rasaushadhi critically.
6. State the practical rules for the safe use of herbo-mineral medicines.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	A, Q.3b	5	Toxicity and pharmacological activities of herbo-mineral compounds.

A5.1 What are herbo-mineral compounds?

> A **herbo-mineral compound** (herbo-mineral or herbo-metallic formulation) is a medicine that contains one or more **metals or minerals**, processed by the methods of Rasashastra, together with **herbal** ingredients, either mixed in or used as media for trituration (Bhavana), or both.

In Rasashastra such medicines are called **Rasaushadhi** or **Rasayoga**. Strictly, a Rasaushadhi is any medicine in which a Rasa Dravya (mercury, a metal or a mineral) is a principal ingredient. Almost all are herbo-mineral in the modern sense, because the processing (Shodhana, Marana, Bhavana) uses plant juices, decoctions and other herbal materials, and many finished formulations also contain herbal powders.

A5.1.1 Groups of Rasaushadhi

Group	How it is made	Examples
Bhasma	Shodhana, then repeated incineration (Puta) with herbal media until a fine, non-metallic powder results	Loha, Mandura, Yashada (Jasada), Tamra, Swarna, Rajata, Abhraka Bhasma
Pishti	Trituration of a mineral with rose water or herbal juices, without heat	Praval Pishti, Mukta Pishti
Kharaliya Rasayana	Trituration in a mortar (Kharala), usually starting from **Kajjali** (mercury with sulphur), with other Bhasmas, herbs and Bhavana	Arogyavardhini Vati, Tribhuvana Kirti Rasa, Sutashekhara Rasa, Shwasa Kasa Chintamani Rasa
Parpati	Kajjali (with or without other drugs) melted and pressed into thin flakes	Rasa Parpati, Swarna Parpati
Kupipakva Rasayana	Heating in a glass bottle (Kupi) in a sand bath; the product sublimes to the neck or settles at the bottom	Rasa Sindura, Makaradhwaja (Siddha Makaradhwaja), Mallasindura, Talasindura
Pottali	Ingredients bound into a hard bolus and cooked in a medium	Hemagarbha Pottali
Herbo-mineral compound formulations	A Bhasma or mineral combined with herbal powders or extracts	Punarnavadi Mandura (Mandura Bhasma with herbs), Chandraprabha Vati (contains Loha Bhasma, Shilajatu and herbs)

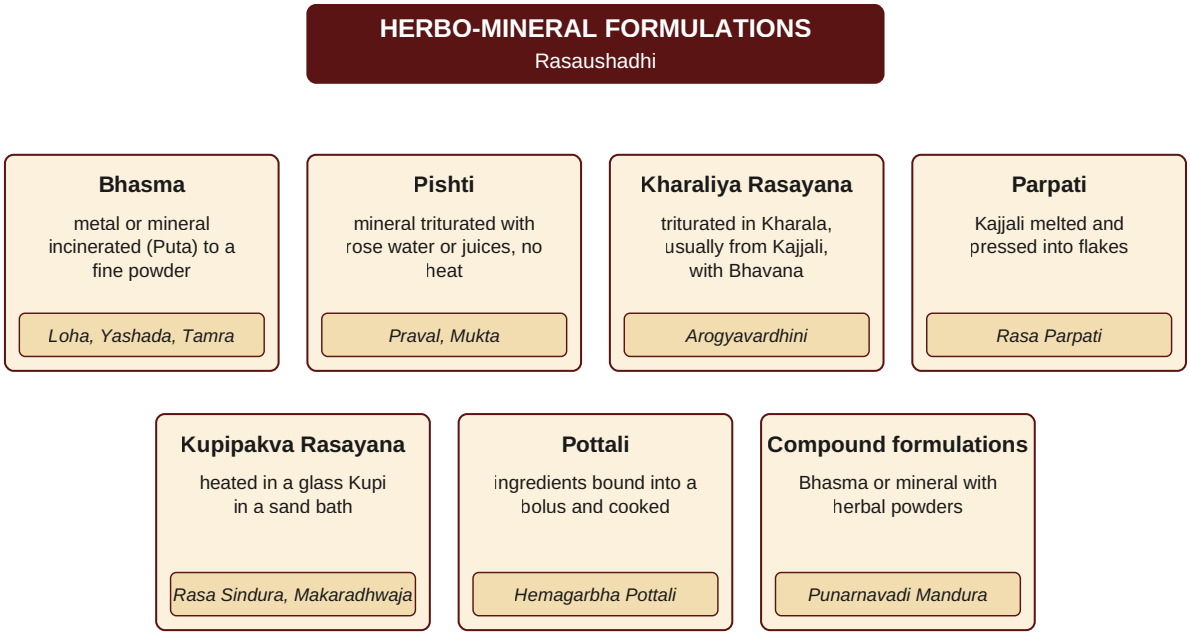


Figure A5.1. Main groups of herbo-mineral formulations in Rasashastra.

A5.1.2 Herbo-mineral compounds by principal element

Principal element	Mineral or metal (Rasashastra name)	Example formulations
Mercury	Parada, as Kajjali, Rasa Sindura or Hingula (mercuric sulphide)	Arogyavardhini Vati, Rasa Sindura, Makaradhwaja, Rasa Parpati
Arsenic	Sankhiya or Malla (arsenic trioxide), Haratala (orpiment), Manahshila (realgar)	Mallasindura, Talasindura, Rasamanikya
Iron	Loha, Mandura (iron oxides)	Punarnavadi Mandura, Navayasa Loha, Dhatri Loha
Zinc	Yashada (Jasada)	Yashada Bhasma
Copper	Tamra	Tamra Bhasma
Gold	Swarna	Swarna Bhasma, Makaradhwaja (gold used in its preparation)
Calcium	Praval, Mukta, Shankha, Shukti, Kapardika, Godanti	Praval Pishti, Shankha Bhasma, Kamadudha Rasa
Silicates	Abhraka (mica)	Abhraka Bhasma
Lead and tin	Naga, Vanga	Naga Bhasma, Vanga Bhasma

A5.2 Why herbo-mineral medicines? The classical rationale

Rasashastra teaching gives three reasons why Rasaushadhi are preferred to purely herbal medicines: they are effective in **a small dose**, they are **not unpleasant** to take, and they **act quickly**. To these the tradition adds further claims.

Classical claim	What it means	Modern reading	Status
Small dose (Alpa Matra)	Usually 125 to 250 mg or less	A low dose by weight; the dose of the element may still be high (A5.6)	True as a description; safety depends on the element
No distaste	Small, tasteless tablets or powders	Better acceptance	True as a description
Quick action (Kshipra Arogya)	Rapid relief	Not systematically measured	Not established
Yogavahi	The Rasa carries and enhances the action of the drugs it is combined with	Possible effects on absorption or delivery of other ingredients	**Not established**; no mechanism shown
Long shelf life	Bhasmas and Kupipakva products do not deteriorate easily	Inorganic compounds are chemically stable	Plausible; shelf-life rules are discussed in Chapter B8
Rasayana	Rejuvenation; prevention of ageing	Antioxidant and immunological effects are proposed	Animal and laboratory data only, for most preparations

A5.3 What are they, chemically?

Modern analysis has answered one question clearly: **what a Bhasma or a Kupipakva product is made of**. Techniques used include X-ray diffraction (XRD, for the crystal phase), scanning and transmission electron microscopy (SEM, TEM, for particle size and shape), energy-dispersive X-ray analysis (EDAX), inductively coupled plasma analysis (ICP-OES, ICP-MS) and neutron activation analysis (for elements), and thermal analysis. Pharmacopoeial standards and tests are introduced in Chapter B10.

Preparation	Reported composition	Particle size reported	Source
Kajjali	Mercuric sulphide as **metacinnabar** (cubic, black HgS) with free sulphur; Hg about 89%, S about 11% by EDAX	About 60 nm to 2 µm	[Sharma 2018]
Siddha Makaradhwaja	Stoichiometrically **HgS**	—	[Kumar 2006]
Jasada (Yashada) Bhasma	**Zinc oxide**, hexagonal (wurtzite) crystals	200 to 500 nm	[Umrani 2013]
Bhasmas in general	Oxides, sulphides, carbonates or other compounds of the metal; size from nanometres to microns; composition varies with method	Nano to micron	[Pal 2014]

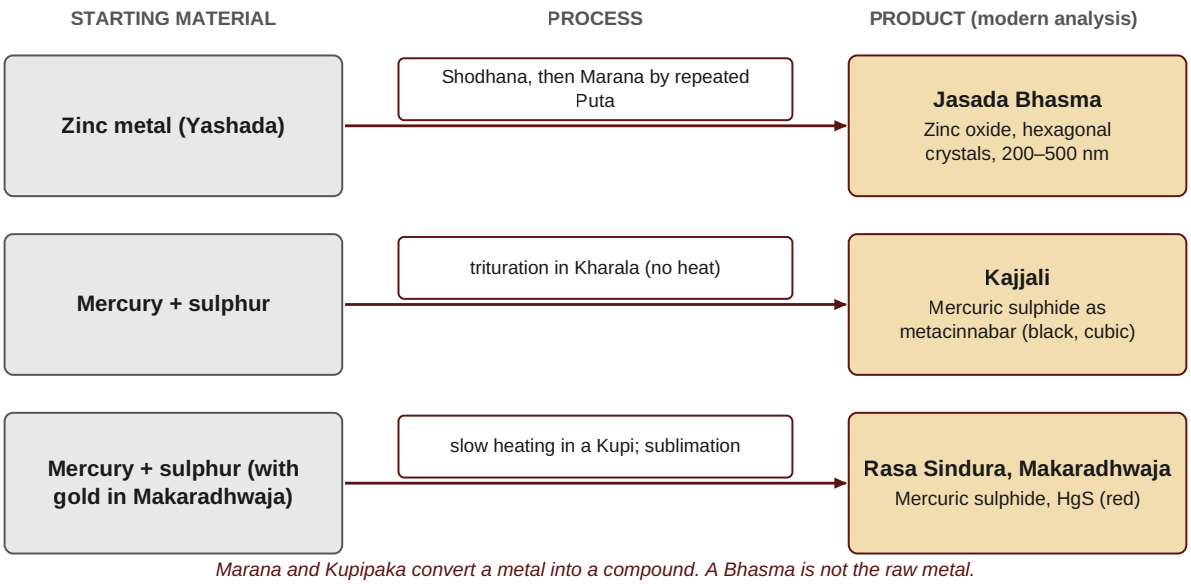


Figure A5.2. What modern analysis shows: the change of form from raw metal to Bhasma or Kupipakva product.

Three lessons follow.

1. **Marana is real chemistry.** A metal is converted into a compound: zinc into zinc oxide, mercury into mercuric sulphide. This is why a Bhasma cannot be judged as if it were the raw metal (A4.5).
2. **The products are often fine particles**, some in the nanometre range. This has led to the description of Bhasmas as “ancient nanomedicine” [Pal 2014]. Particle size is a measured fact; whether it improves absorption or action is a separate question that needs its own evidence.
3. **Composition varies.** Two products of the same name made by different methods may differ in phase, purity and size. Standardisation (Chapters B4 and B10) is therefore essential before any result on one batch is applied to another.

A5.4 How might herbo-mineral compounds act?

Several mechanisms have been proposed. They are not alternatives; more than one may apply to the same preparation.

Proposed mechanism	Explanation	Examples	Status
Supply of an essential element	The preparation releases an element the body needs	Iron from Loha and Mandura; zinc from Jasada; calcium from Praval and Shankha	**Plausible and partly supported** (A3.7); serum zinc rose after Jasada Bhasma in rats [Umrani 2013]

Proposed mechanism	Explanation	Examples	Status
Action of the element as a drug	The element or its ion acts on a target, as a modern drug does	Arsenic compounds in leukaemia [Lo-Coco 2013; Zhu 2013]; zinc on glucose regulation [Umrani 2013]	Established for arsenic in APL; animal data for zinc
Antacid and adsorbent action	Carbonates and oxides neutralise acid; fine particles adsorb	Shankha, Praval, Kapardika in Amlapitta	Established for carbonates as antacids
Particle effects	Small particles have large surfaces; they may be taken up by cells or interact with the immune system	Bhasmas generally	**Proposed**; not shown for most Bhasmas in humans
Action of herbal ingredients	The herbs in the formula or used for Bhavana have their own pharmacology	Katuki in Arogyavardhini; Punarnava in Punarnavadi Mandura	Depends on each herb; combination effects rarely studied
Yogavahi effect	The Rasa increases the absorption or action of other drugs	Kajjali in Kharaliya Rasayana	**Not established**

Established. Bhasmas are compounds of the metal, not the raw metal. Calcium carbonate preparations are antacids. Iron and zinc preparations can supply these elements. Arsenic compounds can be effective, approved drugs for acute promyelocytic leukaemia.

Not established. The Yogavahi action; the claim that particle size explains the action of a Bhasma; and the claimed actions of gold, silver, mica and mercury preparations in humans. These are hypotheses to be tested, not facts to be asserted.

A5.5 Pharmacological activities: what the evidence shows

Hundreds of papers report activities of Rasaushadhi. Most are laboratory or animal studies of one batch; few are controlled trials in patients. The table below gives representative studies whose details have been verified. Read the **level of evidence** column first: it tells you how much weight each finding can bear.

Preparation	Activity studied	Kind of study	Main finding	Level of evidence
Punarnavadi Mandura, alone and with Drakshavaleha	Iron-deficiency anaemia in women of reproductive age	Multicentre Phase III randomised controlled trial, nearly 4000 women (CCRAS and ICMR)	Reported therapeutically **equivalent** to standard iron–folic acid therapy	**Human RCT**; findings presented in May 2026; full peer-reviewed report awaited [CCRAS 2026]
Realgar–Indigo naturalis formula (Chinese medicine; contains realgar, As_4S_4)	Acute promyelocytic leukaemia	Phase III randomised non-inferiority trial, 242 patients	Oral formula with retinoic acid was **non-inferior** to intravenous arsenic trioxide (two-year disease-free survival 98.1% against 95.5%)	**Human RCT** [Zhu 2013]
Siddha Makaradhwaja (after Ashwagandha powder)	Rheumatoid arthritis	Open, uncontrolled pilot study, 86 patients enrolled, 7 weeks	ACR20 response in 56.4% of those completing; no change in liver or kidney tests; urinary mercury rose about five-fold	**Uncontrolled human pilot**; cannot prove efficacy [Kumar 2015]
Jasada Bhasma (zinc oxide, 200–500 nm)	Diabetes	Rats (type 1 and type 2 models), 4 weeks	Blood glucose fell 20–33%; glucose tolerance improved; serum zinc rose 3.5-fold after one dose	**Animal**; no human trials reported as of the authors' 2015 review [Umrani 2013; Umrani 2015]
Swarna Bhasma	Free-radical scavenging (antioxidant) activity and chemical constituents	Laboratory and animal	Antioxidant activity studied	**Laboratory/ animal** [Mitra 2002]
Arsenic trioxide (modern drug; chemically the same compound as purified Sankhiya)	Acute promyelocytic leukaemia	Phase III randomised trial, 156 patients	With retinoic acid, at least as good as chemotherapy; two-year event-free survival 97% against 86%	**Human RCT** [Lo-Coco 2013]

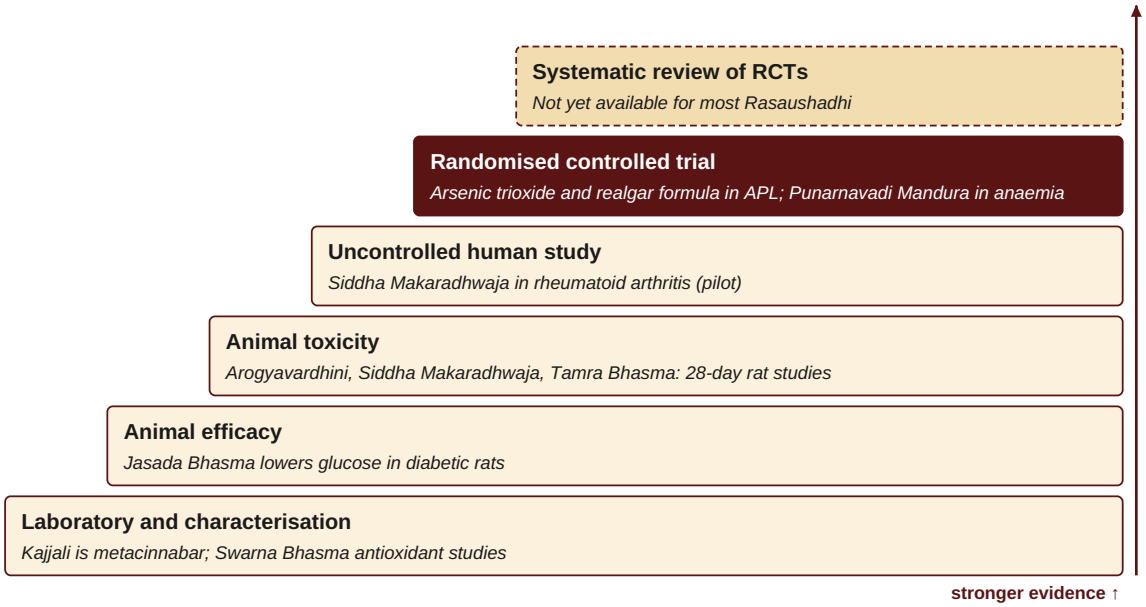


Figure A5.3. Levels of evidence for herbo-mineral medicines, with examples from this chapter.

What the table teaches.

1. Good evidence is possible. Two mineral-containing formulations, one Indian and one Chinese, have been tested in large randomised trials, and an arsenic compound is now standard treatment for a leukaemia.
2. For most Rasaushadhi, the evidence stops at the animal stage. An effect in rats is a reason to do a human trial, not a substitute for one.
3. An uncontrolled study can show that a treatment is tolerated over a short period. It cannot show that the improvement was caused by the treatment; patients with rheumatoid arthritis fluctuate, and many improve with any attention.

A5.6 Toxicity of herbo-mineral compounds

A5.6.1 Where can toxicity come from?

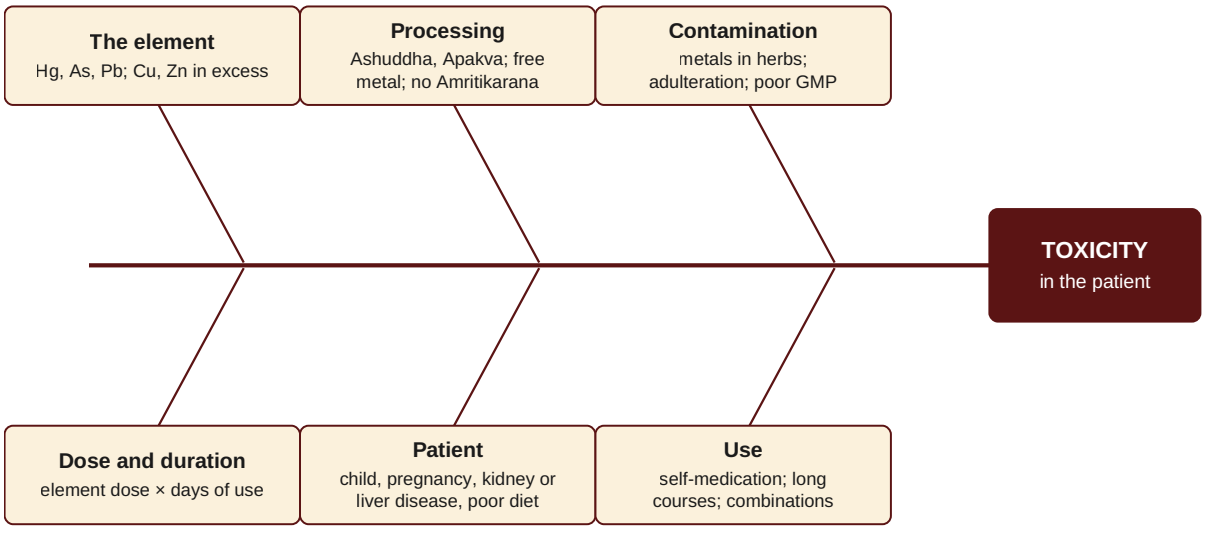


Figure A5.4. Sources of toxicity in a herbo-mineral medicine.

Source	Explanation	Example
The element itself	Mercury, arsenic and lead are toxic in most forms; copper and zinc in excess	Mercury accumulation in tissues even from HgS (A5.6.3)
Incomplete processing	Ashuddha or Apakva material; free metal remaining; wrong phase formed	Tamra Bhasma made without Amritikarana was more toxic in rats [Chaudhari 2016]
Contamination and adulteration	Metals in herbal raw materials; substitution; poor GMP	Metals found in herbal products [Saper 2004]
Wrong dose or duration	Dose of the **element** = dose of the product × its metal content × days of use	A 125 mg tablet of a mercury-rich product taken for months
Patient factors	Children, pregnancy, kidney or liver disease, poor nutrition	Iron and calcium deficiency increase lead absorption (A3.5)
Interactions	With other medicines; with diet	Not well studied
Self-medication and over-the-counter use	No supervision; no limit to duration	Cases of lead poisoning abroad [CDC 2004; Breeher 2015]

A5.6.2 Evidence of harm

Study	Finding
Saper 2004 [Saper 2004]	14 of 70 (20%) Ayurvedic herbal products bought in Boston contained lead, mercury or arsenic
Saper 2008 [Saper 2008]	20.7% of 193 products bought on the internet contained metals; 40.6% of Rasa Shastra products
CDC 2004 [CDC 2004]	12 adults with lead poisoning associated with Ayurvedic medicines in five US states, 2000–2003
Breeher 2015 [Breeher 2015]	A cluster of lead and mercury toxicity identified in 2011 among users of Ayurvedic medicines: **46 of 115 people screened (40%) had raised blood lead**
Bhalla and Pannu 2022 [Bhalla 2022]	42 over-the-counter Ayurvedic preparations bought from licensed shops in Chandigarh were analysed: metals were found in all; mercury exceeded FAO/WHO limits for herbal medicines in 29, arsenic in 6, lead in 2

Limits designed for herbal medicines are not the right yardstick for a Rasaushadhi that contains a metal by design. But the excess is real, and the patient's total intake is what matters for safety.

A5.6.3 Evidence from animal toxicity studies

Preparation	Design	Main finding	Points to note
Arogyavardhini Vati [Kumar 2012]	Rats; 1, 5 and 10 times the human-equivalent dose; 28 days; mercuric chloride as a comparison	No change in behaviour, liver or kidney tests; normal histology	**Mercury increased in brain, liver and kidney** at all doses
Siddha Makaradhwaja [Kumar 2014]	Rats; 1, 5 and 10 times the human-equivalent dose; 28 days	No change in behaviour or liver and kidney tests at any dose. At 10 times: oxidative stress in liver and kidney and mild changes in brain, liver and kidney tissue	**Mercury rose in brain, liver and kidney, dose-dependently**, though less than with mercuric chloride
Tamra Bhasma with and without Amritikarana [Chaudhari 2016]	Rats; 1, 5 and 10 times the therapeutic equivalent dose; 28 days	No toxicity at 1 and 5 times. Mild toxicity to liver, kidney, heart and thymus at 10 times; **greater without Amritikarana**	Supports the role of a classical processing step
Jasada Bhasma [Umrani 2013]	Rats; acute and sub-acute	Safe up to 300 mg/kg	About 100 times the effective dose in rats

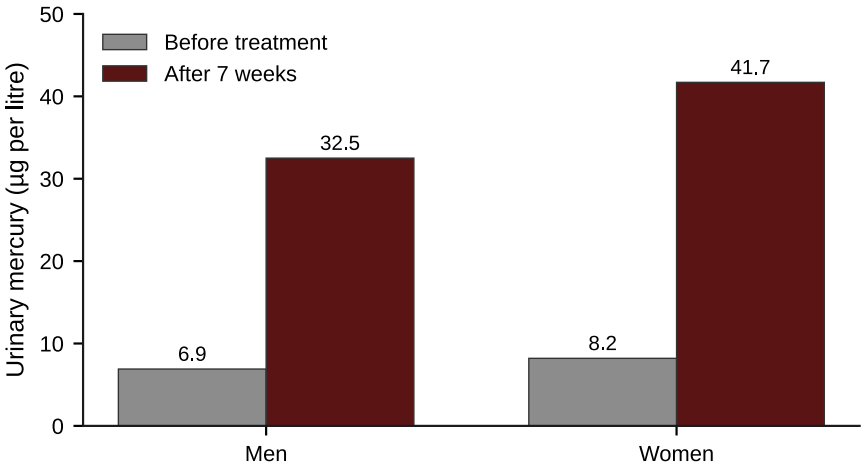


Figure A5.5. Urinary mercury before and after treatment (Ashwagandha for 3 weeks, then Siddha Makaradhwaja 100 mg daily for 4 weeks) in an uncontrolled pilot study (mean values; data from Kumar and colleagues, 2015).

A5.6.4 How to read an animal toxicity study of a Rasaushadhi

A student must be able to judge such a paper, in an examination and in practice. Ask these questions.

Question	Why it matters
How long was the study?	Most are 28 days. Many Rasaushadhi are taken for months; chronic (90-day or longer) studies are needed
Were tissue metal levels measured?	Normal blood tests can coexist with accumulation in brain and kidney [Kumar 2012; Kumar 2014]
What doses were tested, and how were they scaled from humans?	Body-surface-area scaling (A1a.5) should be stated
Was the batch characterised?	Without XRD, particle size and metal content, the result cannot be applied to another batch
Was a positive control used?	A comparison with mercuric chloride shows whether the assay can detect toxicity
Were the animals healthy?	Patients may have kidney or liver disease; animals usually do not
What did the authors conclude, and does the data support it?	"No toxicity" is sometimes claimed when the highest dose did show changes

A5.6.5 Is a Rasaushadhi safe? A balanced statement

Established. Many Rasaushadhi contain toxic metals by design, in amounts far above limits set for herbal medicines. Mercury from HgS-based preparations is absorbed to some extent and accumulates in tissues in animals and appears in human urine [Kumar 2012; Kumar 2014; Kumar 2015]. Lead poisoning from Ayurvedic products is documented [CDC 2004; Breeher 2015]. Proper processing can reduce toxicity, as shown for Amritikarana of Tamra Bhasma in rats [Chaudhari 2016]. In short animal studies, several preparations at their therapeutic-equivalent dose showed no functional toxicity.

Not established. Long-term safety in humans; the safe maximum duration of use for any mercury, arsenic or lead preparation; safety in children, pregnancy and kidney disease; and the safety of products that have not been characterised and tested.

A5.7 Safe use of herbo-mineral medicines

The safety of a Rasaushadhi is decided at four points: in the **pharmacy**, by the **prescriber**, by the **patient**, and by **monitoring**. A failure at any one point can harm the patient (Figure A5.6).

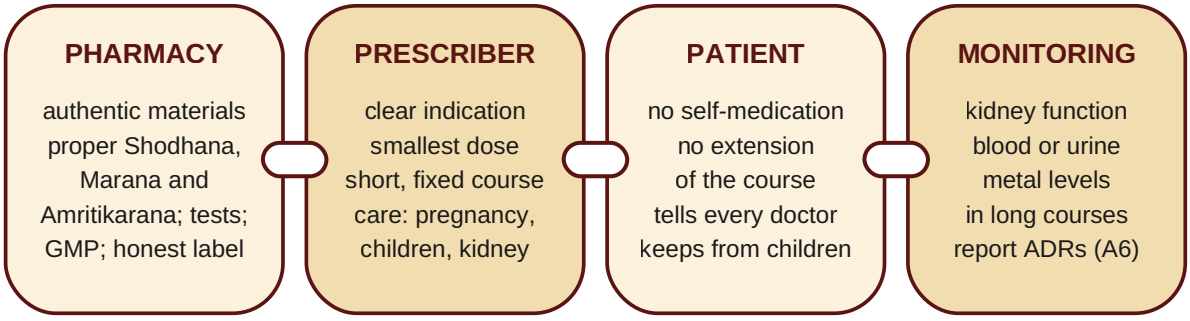


Figure A5.6. The chain of safety for a herbo-mineral medicine. Every link must hold.

Point	What must be done
Pharmacy	Authenticated raw materials; correct Shodhana, Marana and Amritikarana; classical Bhasma Pariksha **and** modern tests (XRD phase, particle size, metal content, free metal where relevant); GMP (Chapter B4); accurate labelling, including the caution required for medicines containing heavy metals (Chapter B8)
Prescriber	A clear indication; the smallest effective dose; a fixed and short duration; Anupana and Pathya; avoid or take special care in pregnancy, lactation, young children, and kidney or liver disease; no two mercury- or arsenic-based products together unless clearly intended; record the product name, manufacturer and batch number
Patient	No self-medication; no extension of the course without advice; tell every doctor about the medicine; keep it away from children
Monitoring	For long courses of mercury, arsenic or lead preparations: kidney function, and, where possible, blood or urine metal levels at the start and during treatment; report suspected adverse reactions to the pharmacovigilance programme (Chapter A6)

CALCULATE THE DOSE OF THE ELEMENT, NOT ONLY THE DOSE OF THE TABLET

Suppose a product contains 25% mercury by weight, and the dose is 125 mg twice a day for 60 days.

- Mercury per day = 125 mg × 2 × 0.25 = **62.5 mg**
- Mercury over the course = 62.5 mg × 60 = **3.75 g**

This is a large amount of a toxic element, even if most of it is not absorbed. The calculation shows why dose, duration and chemical form must all be considered together, and why “small dose” (Alpa Matra) of the product does not mean a small dose of the metal. The figures here are an illustration only; the metal content of each product must be measured.

A5.8 Classical safeguards: the Ayurvedic view of toxicity of Rasaushadhi

Rasashastra itself is a long effort to make toxic substances usable. Its texts warn that Parada and metals used without proper processing cause severe disease and death, and they prescribe Shodhana, Marana, Amritikarana and Bhasma Pariksha to prevent this (Rasatarangini) [R7]. The classical safeguards and their modern reading are set out in Chapter A4 (A4.10.2).

AYURVEDA CORRELATION: CLASSICAL SAFEGUARDS AND MODERN EVIDENCE

Classical principle	Modern evidence
Shodhana and Marana change the nature of the metal	Confirmed: metals are converted to oxides and sulphides of defined crystal phase [Sharma 2018; Umrani 2013]
Amritikarana removes residual harmful qualities	Supported in one rat study for Tamra Bhasma: less toxicity at the high dose with Amritikarana [Chaudhari 2016]
Bhasma Pariksha (Niruttha, Apunarbhava) shows absence of free metal	Qualitative screening only; modern assays are needed for free metal and soluble metal
Small dose and limited duration	Agrees with the modern rule that total intake decides toxicity
Rasaushadhi are safe when properly prepared	Short animal studies at therapeutic-equivalent doses support short-term safety for several preparations; tissue accumulation of mercury is seen [Kumar 2012; Kumar 2014]; long-term human data are lacking
Rasaushadhi act quickly and powerfully	Shown in randomised trials for an arsenic compound and a realgar formula in APL [Lo-Coco 2013; Zhu 2013] and reported for Punarnavadi Mandura in anaemia [CCRAS 2026]; not shown for most others

Established. Classical processing changes the chemistry of metals; mineral-containing formulations can be validated by modern trials; several preparations are tolerated in short animal studies.

Not established. That classical processing alone guarantees safety; the long-term human safety of mercury, arsenic and lead preparations; the clinical efficacy of most Rasaushadhi for their classical indications.

KEY POINTS TO REMEMBER

- Herbo-mineral compound: metal or mineral processed by Rasashastra methods, with herbal ingredients or herbal processing media.
- Groups: Bhasma, Pishti, Kharaliya Rasayana, Parpati, Kupipakva Rasayana, Pottali, compound herbo-mineral formulations.
- Classical reasons for preference: small dose, no distaste, quick action; plus Yogavahi (not established).
- Modern analysis: Kajjali is metacinnabar (black HgS); Siddha Makaradhwaja is HgS; Jasada Bhasma is zinc oxide (200–500 nm).
- Proposed mechanisms: element supply, element as drug, antacid action, particle effects, herbal pharmacology, Yogavahi.
- Human RCTs: arsenic trioxide in APL (Lo-Coco 2013); realgar–Indigo naturalis formula non-inferior to IV arsenic trioxide (Zhu 2013); Punarnavadi Mandura equivalent to iron–folic acid in anaemia (CCRAS–ICMR, presented 2026).
- Animal: Jasada Bhasma lowers glucose in diabetic rats; Arogyavardhini and Siddha Makaradhwaja: no functional toxicity in 28 days, but mercury accumulates in brain, liver and kidney.
- Tamra Bhasma without Amritikarana is more toxic at high dose in rats.

- Siddha Makaradhwaja in RA (uncontrolled): urinary mercury rose about five-fold; liver and kidney tests unchanged over 7 weeks.
- Harm: Saper 2004 and 2008; CDC 2004; Breeher 2015 (40% of 115 screened had raised blood lead); Bhalla and Pannu 2022.
- Safety chain: pharmacy, prescriber, patient, monitoring. Calculate the dose of the element.

EXAM CORNER — CHAPTER A5

Likely long questions (20 marks)

1. Discuss the toxicity and pharmacological activities of herbo-mineral compounds, with suitable examples and the published evidence.
2. Classify Rasaushadhi. Describe their physico-chemical nature as shown by modern analytical techniques and its bearing on their safety.
3. "A Rasaushadhi prepared as per classical texts is safe." Critically evaluate this statement in the light of published studies.

Likely short notes (5 marks)

1. Toxicity and pharmacological activities of herbo-mineral compounds *(set in July 2018)*.
2. Yogavahi: concept and evidence.
3. Bhasma as nanomedicine.
4. Amritikarana and toxicity: evidence from Tamra Bhasma.
5. Arsenic in modern medicine and in Rasashastra.
6. Critical reading of an animal toxicity study of a Rasaushadhi.
7. Rules for the safe prescription of mercury-containing preparations.

How to attempt

- Begin with the definition and the table of groups.
- Divide the answer into **pharmacological activities** and **toxicity**, as the question does.
- For activities, give examples at different levels of evidence: one human RCT, one animal study, one laboratory finding. Name the level each time.
- For toxicity, list the sources (element, processing, contamination, dose, patient) and give one study of harm and one animal study.
- End with the safe-use chain and the Established / Not established lines.

Answer frame: "Toxicity and pharmacological activities of herbo-mineral compounds." (5 marks)

Part	What to write	Suggested marks
Definition and examples	Herbo-mineral compound; two or three groups with examples	1
Pharmacological activities	Proposed mechanisms (element supply, element as drug, antacid); one RCT example (arsenic in APL or Punarnavadi Mandura); one animal example (Jasada Bhasma)	2
Toxicity	Sources; mercury accumulation in rat tissues; Amritikarana study; market studies	1.5
Conclusion	Safe-use rules; evidence boundary	0.5

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **7, 29, 32**.

- **[R7]** Sharma S, Shastri HD. *Rasatarangini*. 11th ed. Delhi: Motilal Banarasidas; 1979.
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024.
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.

Additional verified sources (accessed 16 October 2026)

- **[CCRAS 2026]** Central Council for Research in Ayurvedic Sciences. *CCRAS Newsletter*, May–June 2026 (report of the 1st ICMR Annual Clinical Trial Meet, 20 May 2026: Phase III RCT of Punarnavadi Mandura, alone and with Drakshavaleha, against iron–folic acid in nearly 4000 women). ccras.nic.in.
- **[Zhu 2013]** Zhu HH, Wu DP, Jin J, et al. Oral tetra-arsenic tetra-sulfide formula versus intravenous arsenic trioxide as first-line treatment of acute promyelocytic leukemia: a multicenter randomized controlled trial. *J Clin Oncol*. 2013;31:4215–4221.
- **[Lo-Coco 2013]** Lo-Coco F, Avvisati G, Vignetti M, et al. Retinoic acid and arsenic trioxide for acute promyelocytic leukemia. *N Engl J Med*. 2013;369(2):111–121.
- **[Kumar 2015]** Kumar G, Srivastava A, Sharma SK, Rao TD, Gupta YK. Efficacy and safety evaluation of Ayurvedic treatment (Ashwagandha powder and Sidh Makardhwaj) in rheumatoid arthritis patients: a pilot prospective study. *Indian J Med Res*. 2015;141(1):100–106.
- **[Kumar 2014]** Kumar G, Srivastava A, Sharma SK, Gupta YK. Safety evaluation of mercury based Ayurvedic formulation (Sidh Makardhwaj) on brain cerebrum, liver and kidney in rats. *Indian J Med Res*. 2014;139(4):610–618.
- **[Kumar 2012]** Kumar G, Srivastava A, Sharma SK, Gupta YK. Safety evaluation of an Ayurvedic medicine, Arogyavardhini vati on brain, liver and kidney in rats. *J Ethnopharmacol*. 2012;140(1):151–160. doi:10.1016/j.jep.2012.01.004.
- **[Chaudhari 2016]** Chaudhari SY, Nariya MB, Galib R, Prajapati PK. Acute and subchronic toxicity study of Tamra Bhasma (incinerated copper) prepared with and without Amritikarana. *J Ayurveda Integr Med*. 2016;7(1):23–29. doi:10.1016/j.jaim.2015.11.001.
- **[Umrani 2013]** Umrani RD, Agrawal DS, Paknikar KM. Anti-diabetic activity and safety assessment of Ayurvedic medicine, Jasada bhasma (zinc ash) in rats. *Indian J Exp Biol*. 2013;51(10):811–822.
- **[Umrani 2015]** Umrani RD, Paknikar KM. Jasada bhasma, a zinc-based Ayurvedic preparation: contemporary evidence of antidiabetic activity inspires development of a nanomedicine. *Evid Based Complement Alternat Med*. 2015;2015:193156. doi:10.1155/2015/193156.

- **[Mitra 2002]** Mitra A, Chakraborty S, Auddy B, Tripathi P, Sen S, Saha AV, Mukherjee B. Evaluation of chemical constituents and free-radical scavenging activity of Swarnabhasma (gold ash), an Ayurvedic drug. *J Ethnopharmacol*. 2002;80:147–153.
- **[Sharma 2018]** Sharma V, Samal AK, Pandey S, Chaudhary AK, Srivastava RK. Characterization of Hg-based Ayurvedic drug Kajjali: classical and contemporary approaches. *Curr Sci*. 2018;115(6):1174–1178. doi:10.18520/cs/v115/i6/1174-1178.
- **[Kumar 2006]** Kumar A, Nair AGC, Reddy AVR, Garg AN. Availability of essential elements in bhasmas: analysis of Ayurvedic metallic preparations by INAA. *J Radioanal Nucl Chem*. 2006;270(1):173–180. doi:10.1007/s10967-006-0326-z.
- **[Pal 2014]** Pal D, Sahu CK, Haldar A. Bhasma: the ancient Indian nanomedicine. *J Adv Pharm Technol Res*. 2014;5(1):4–12.
- **[Saper 2004]**, **[Saper 2008]**, **[CDC 2004]**: full citations in Chapter A4.
- **[Breeher 2015]** Breeher L, Mikulski MA, Czeczok T, Leinenkugel K, Fuortes LJ. A cluster of lead poisoning among consumers of Ayurvedic medicine. *Int J Occup Environ Health*. 2015;21(4):303–307. doi:10.1179/2049396715Y.0000000009.
- **[Bhalla 2022]** Bhalla A, Pannu AK. Are Ayurvedic medications store house of heavy metals? *Toxicol Res (Camb)*. 2022;11(1):179–183. doi:10.1093/toxres/tfab124.

Viva pointers — Chapter A5

Answer each in one or two sentences.

- Define a herbo-mineral compound. Is Praval Pishti herbo-mineral?
- Name the groups of Rasaushadhi with one example each.
- What is the difference between a Kharaliya Rasayana and a Kupipakva Rasayana?
- Give the three classical reasons why a Rasaushadhi is preferred to a herbal medicine.
- What is Yogavahi? Has it been proved?
- What is Kajjali chemically? What is Rasa Sindura chemically?
- What is Jasada Bhasma chemically? What size are its particles?
- Why is a Bhasma called an “ancient nanomedicine”? Does small particle size prove better action?
- Name four techniques used to characterise a Bhasma.
- Which arsenic compound is a standard modern drug? For which disease?
- What did the trial of the realgar–Indigo naturalis formula show?
- What was compared in the CCRAS–ICMR anaemia trial, and what was reported?
- What happened to urinary mercury in patients given Siddha Makaradhwaja?
- In rat studies of Arogyavardhini Vati, what was normal and what was not?
- What did the Tamra Bhasma study show about Amritikarana?
- Name six sources of toxicity in a herbo-mineral medicine.
- A tablet of 125 mg contains 25% mercury. How much mercury does a patient take in 60 days at two tablets a day?
- What questions would you ask of an animal toxicity study of a Rasaushadhi?
- Name four precautions for prescribing a mercury-containing preparation.
- Which classical tests aim to confirm the absence of free metal in a Bhasma?

PART A · PHARMACOLOGY · CHAPTER A6

A6 · PHARMACOVIGILANCE

SYLLABUS POINT (PART A, 6)

Detailed Knowledge of Pharmacovigilance – National and International Scenario. Pharmacovigilance of Ayurvedic Drugs

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
A6	Pharmacovigilance	2	5	Jul 2018, Sep 2018, Sep 2020, Aug 2021, Aug 2022, Jan 2024, Mar 2026

Why this chapter matters to you. Pharmacovigilance is the single most frequently examined topic in Part A. It was set in **every one of the seven papers** examined, twice as a 20-mark question. The questions ask for the definition, the need, the national and international programmes, and the pharmacovigilance of Ayurvedic drugs in particular. This chapter covers all four, with tables and figures you can reproduce in the examination hall.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define pharmacovigilance and its key terms, and explain why it is needed after a drug is marketed.
2. Describe the methods of pharmacovigilance and the steps from detection of an adverse event to regulatory action.
3. Assess causality, severity and preventability of a suspected adverse drug reaction using standard scales.
4. Describe the international pharmacovigilance system (WHO, ICH, major regulators).
5. Describe the Pharmacovigilance Programme of India and the pharmacovigilance programme for Ayurveda, Siddha, Unani and Homoeopathy drugs.
6. Explain the special needs and challenges of pharmacovigilance of Ayurvedic drugs, and the role of the Rasashastra scholar in it.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	A, Q.2	20	Pharmacovigilance of Ayurvedic drugs and its national program.
Sep 2018	A, Q.3c	5	What is Pharmacovigilance? Explain its need and current scenario in case of ASU drugs.
Sep 2020	A, Q.3a	5	Short note: Pharmacovigilance
Aug 2021	A, Q.3A	5	Write in detail about National and International Scenario of Pharmacovigilance of Ayurvedic Drugs
Aug 2022	A, Q.3c	5	Pharmacovigilance of Ayurveda Drugs.
Jan 2024	A, Q.3b	5	Pharmacovigilance of Ayurvedic drugs.
Mar 2026	A, Q.2	20	Discuss Adverse Drug Reactions (ADR) and drug toxicity. Explain the role of Pharmacovigilance in the safety monitoring of Ayurvedic drugs.

A6.1 Definition and scope

> **Pharmacovigilance** is "the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug-related problem" (World Health Organization, 2002) [WHO 2002].

The word comes from the Greek *pharmakon* (drug) and the Latin *vigilare* (to keep watch). Pharmacovigilance is therefore **watchfulness over medicines** throughout their life on the market.

A6.1.1 Scope

The WHO definition is deliberately broad. Pharmacovigilance covers not only adverse reactions but every drug-related problem.

Within the scope	Example
Adverse drug reactions	Rash with an antibiotic; hepatotoxicity with a herbal product
Medication errors	Wrong drug, wrong dose, wrong route
Substandard and falsified medicines	A Bhasma that fails Apunarbhava; a counterfeit tablet
Lack of efficacy	A product that does not work as claimed
Misuse and abuse	Self-medication beyond the advised course
Drug interactions	Herb–drug interactions (A1c.3)
Poisoning and overdose	Accidental ingestion by a child
Use in pregnancy and lactation	Exposure of the foetus or infant

The products covered are equally broad: modern medicines, **herbal and traditional medicines**, vaccines, blood products, biologicals and medical devices. In India, separate programmes cover blood (haemovigilance) and devices (materiovigilance), and the Ministry of Ayush runs a programme for Ayurveda, Siddha, Unani and Homoeopathy drugs (A6.8).

A6.1.2 Key terms

Term	Meaning
Adverse event (AE)	Any untoward medical occurrence in a patient given a drug, **whether or not** it is caused by the drug
Adverse drug reaction (ADR)	A response to a drug that is noxious and unintended, occurring at doses normally used in humans (A1c.4)
Serious ADR or AE	One that results in death, is life-threatening, requires or prolongs hospitalisation, causes persistent or significant disability, causes a congenital anomaly, or is otherwise medically important [ICH E2A]
Unexpected ADR	One whose nature or severity is not consistent with the product information
Signal	Reported information on a possible causal relationship between an adverse event and a drug, the relationship being unknown or incompletely documented previously
ICSR	Individual Case Safety Report: the report of one patient's suspected reaction
PSUR / PBRER	Periodic Safety Update Report; Periodic Benefit-Risk Evaluation Report: the manufacturer's periodic review of a product's safety [ICH E2C]
Risk management plan (RMP)	A plan describing the known and potential risks of a product and how they will be monitored and minimised

Serious is not the same as severe. "Severe" describes intensity (a severe headache); "serious" describes outcome (a reaction that threatens life). A severe headache is not serious; a mild-looking reaction that leads to hospital admission is.

A6.2 Why pharmacovigilance is needed

A drug is approved after clinical trials (A2.3). But trials, however good, cannot tell us everything about safety.

Limitation of pre-marketing trials	Consequence
Small numbers: a few hundred to a few thousand patients	Rare reactions are missed. To have a 95% chance of seeing at least one case of a reaction that occurs in 1 in 10,000 patients, about **30,000** patients must be observed (the "rule of three")
Short duration: weeks to months	Reactions after long use (cancer, organ damage, accumulation) are missed
Selected patients	Children, the elderly, pregnant women and people with many diseases are usually excluded
Controlled conditions	In real life, drugs are taken with other drugs, foods and herbs, by patients who miss doses or take extra
Single indication	Off-label use and misuse are not studied

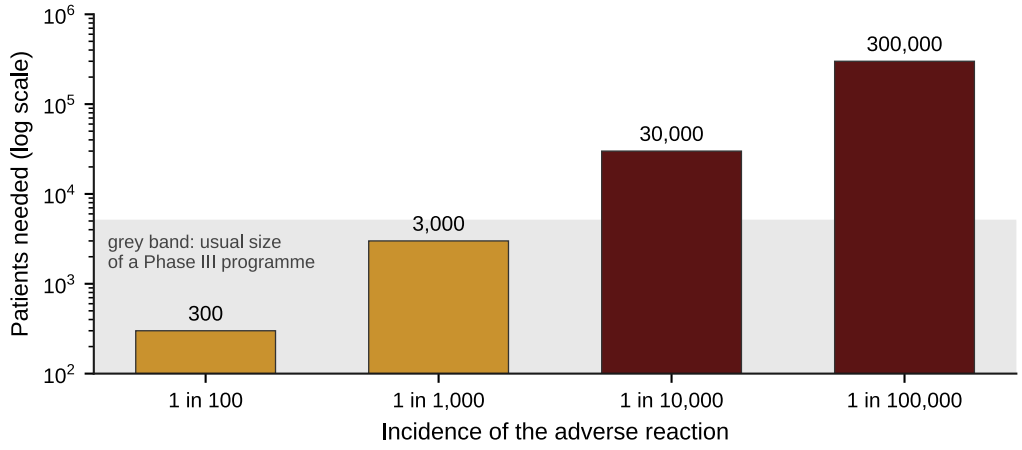


Figure A6.1. Why a clinical trial cannot detect rare adverse reactions: the number of patients needed to see one case rises steeply as the reaction becomes rarer (rule of three: about 3 ÷ incidence).

History gives the hardest lessons.

Event	What happened	Lesson
Thalidomide (1957–1961)	A sedative given in pregnancy caused severe limb defects (phocomelia) in thousands of babies. The link was reported in 1961 [McBride 1961]	Gave birth to modern drug regulation and to spontaneous reporting systems
Rofecoxib (withdrawn 2004)	An anti-inflammatory drug, used by millions, was withdrawn after it was linked to increased cardiovascular events	Even widely used, approved drugs may carry risks that appear only after years of use
Lead and mercury in traditional medicines	Poisoning cases and market studies (Chapters A4 and A5)	Traditional medicines need pharmacovigilance as much as modern drugs

A6.2.1 Aims of pharmacovigilance

1. Improve patient care and safety in the use of medicines.
2. Detect new ADRs, especially serious and rare ones, early.
3. Identify risk factors and the mechanisms of ADRs.
4. Estimate the frequency of ADRs.
5. Assess the benefits and risks of medicines and support regulatory decisions.
6. Promote rational and safe use, and communicate risks to professionals and the public.
7. Contribute to education and training.

A6.3 Milestones

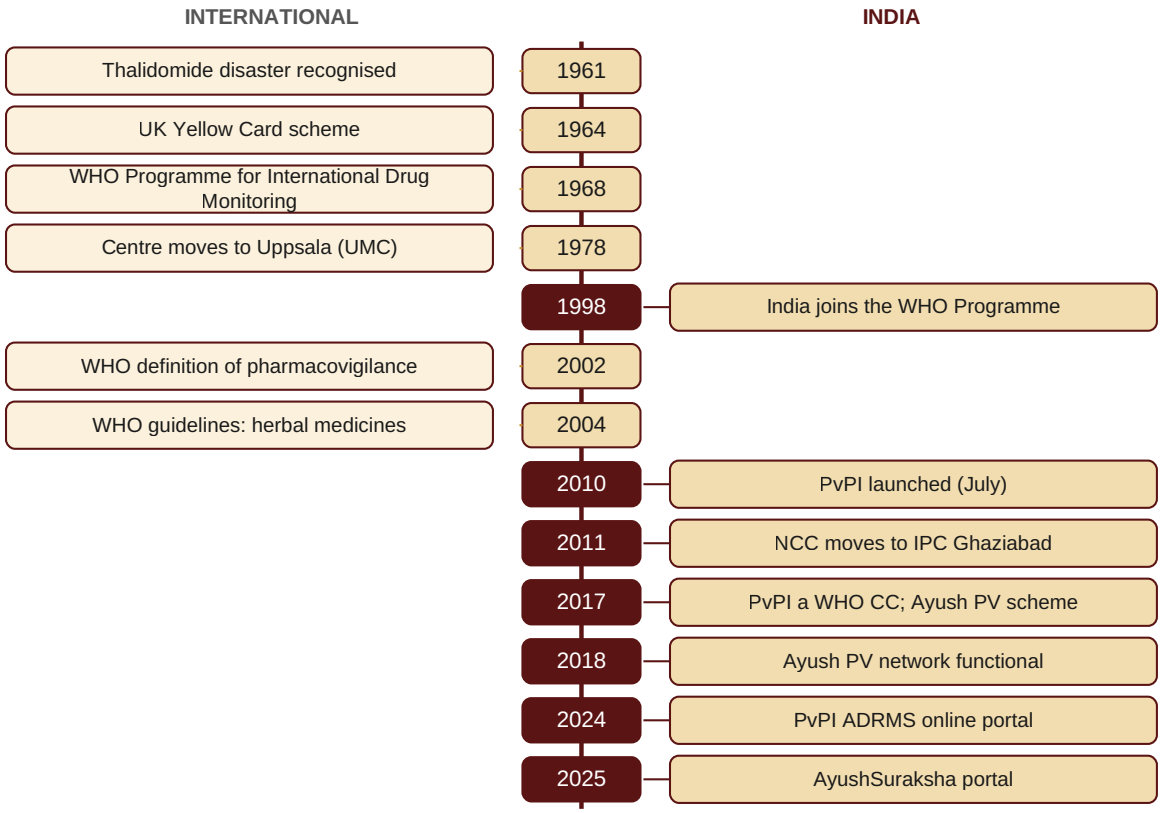


Figure A6.2. Milestones in pharmacovigilance, international and Indian.

Year	International	India
1961	Thalidomide disaster recognised [McBride 1961]	—
1964	Yellow Card scheme begins in the United Kingdom	—
1968	**WHO Programme for International Drug Monitoring** begins, with 10 pilot member countries [UMC 2026]	—
1978	WHO Collaborating Centre for International Drug Monitoring moves to Uppsala, Sweden (later the **Uppsala Monitoring Centre**)	—
1998	—	**India joins the WHO Programme** [UMC 2026]
2002	WHO publishes *The Importance of Pharmacovigilance* with the current definition [WHO 2002]	—
2004	**WHO guidelines on safety monitoring of herbal medicines in pharmacovigilance systems** [WHO 2004]	—
2010	—	**Pharmacovigilance Programme of India (PvPI)** begins, July 2010, with AIIMS New Delhi as the first coordinating centre [PvPI 2025]

Year	International	India
2011	—	National Coordination Centre moves to the **Indian Pharmacopoeia Commission, Ghaziabad**, 15 April 2011 [PvPI 2025]
2017	—	PvPI designated a **WHO Collaborating Centre** (18 July 2017); Ministry of Ayush starts the **Central Sector Scheme for pharmacovigilance of ASU&H drugs** (December 2017) [PvPI 2025; Lok Sabha 2020]
2018	—	Ayush pharmacovigilance network becomes functional (August 2018) [Lok Sabha 2020]
2019	—	New Drugs and Clinical Trials Rules, 2019 [NDCT 2019]
2024	—	PvPI launches the **ADRMS online reporting portal** (19 August 2024) [PvPI 2025]
2025	—	Ministry of Ayush launches the **AyushSuraksha** portal for adverse reactions and misleading advertisements (May 2025) [AIR 2025]

A6.4 Methods of pharmacovigilance

Method	How it works	Strengths	Limitations
Spontaneous (passive) reporting	Doctors, pharmacists, nurses, patients and manufacturers report suspected ADRs voluntarily to a national centre	Cheap; covers all drugs and all patients for the whole life of the drug; best method for detecting **new, rare** reactions	**Under-reporting** (often most reactions go unreported); no denominator, so incidence cannot be calculated; reporting bias
Stimulated reporting	Reporting is actively encouraged for a period or for a product (newly launched drugs, campaigns)	More reports	Effect is temporary
Active surveillance	A defined group is followed deliberately: sentinel sites, **cohort event monitoring**, prescription event monitoring, registries	Gives a denominator and incidence; more complete	Costly; limited in size and time
Comparative observational studies	Case-control, cohort and cross-sectional studies, often using health records	Tests a hypothesis; estimates relative risk	Confounding; needs good data
Targeted clinical investigations	Studies designed for a specific safety question (special populations, interactions)	Answers a defined question	Narrow
Drug utilisation studies	How drugs are prescribed and used in practice	Shows exposure and misuse	Do not measure harm directly
Data mining of large databases	Statistical search for disproportionate reporting in databases such as VigiBase	Finds signals early	Generates hypotheses only

A6.4.1 Under-reporting

Under-reporting is the weakness of every spontaneous system. Inman described the attitudes of doctors that lead to it, often remembered as “the seven deadly sins”:

Attitude	What the doctor thinks
Complacency	Only safe drugs are allowed on the market
Fear	Of legal action or of being blamed
Guilt	For having harmed the patient
Ambition	To collect cases and publish them personally
Ignorance	Of how and what to report
Diffidence	Fear of looking foolish by reporting a mere suspicion
Lethargy	Lack of time or interest; the form is not at hand

For Ayurvedic medicines, there are additional reasons: the belief that herbal and traditional medicines are always safe, the feeling that reporting an ADR harms the reputation of Ayurveda, and the lack of a habit of reporting (A6.8.2).

A6.5 The pharmacovigilance process

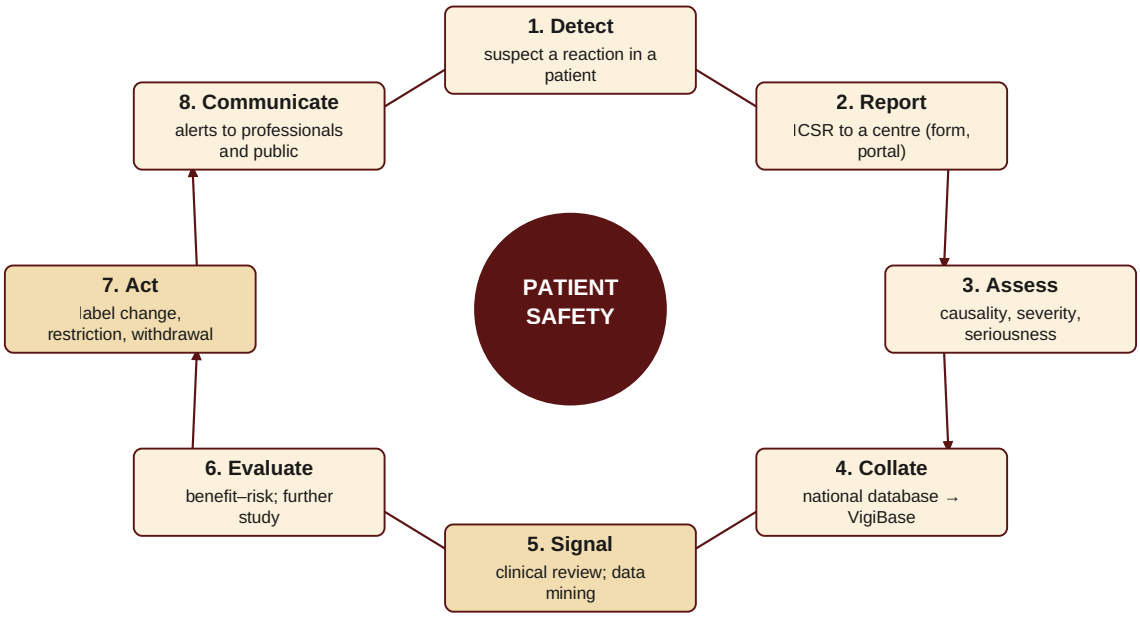


Figure A6.3. The pharmacovigilance process: from detection of a suspected reaction to action and communication.

A6.5.1 Step 1: detection and reporting

Who can report? Any healthcare professional (doctor, Vaidya, pharmacist, nurse) and, increasingly, patients and consumers. Manufacturers are obliged to report under the law.

What should be reported? All suspected reactions, especially:

- reactions to **new drugs**;
- **serious** reactions to any drug;
- **unexpected** reactions not in the product information;
- reactions in special groups (children, pregnancy, elderly);
- suspected **interactions**, including herb–drug interactions;
- **lack of efficacy** and **quality defects**.

A report is valid even if the reporter is not sure the drug caused the reaction. **Suspicion is enough.**

Minimum information for a valid report. Four elements:

1. an identifiable **patient** (initials, age or sex);
2. an identifiable **reporter**;
3. a **suspected drug**;
4. a **suspected reaction**.

A6.5.2 The reporting form

India uses the *Suspected Adverse Drug Reaction Reporting Form* for healthcare professionals (version 1.4) and a separate *Medicines Side-Effect Reporting Form* for consumers, available in ten Indian languages; reports can also be made online through the ADRMS portal [PvPI 2025].

Section	Information
A. Patient information	Initials, age or date of birth, sex, weight
B. Suspected adverse reaction	Date of onset; date of recovery; description; relevant tests; outcome (recovered, recovering, not recovered, fatal, unknown); seriousness criteria
C. Suspected medication(s)	Name (brand and generic), **manufacturer, batch number**, expiry date, dose, route, frequency, dates started and stopped, indication; whether the reaction abated on stopping (**dechallenge**) and reappeared on reintroduction (**rechallenge**)
D. Concomitant medicines	Including herbal and traditional medicines and self-medication
E. Relevant history	Diseases, allergies, pregnancy, liver or kidney disease, alcohol, smoking
F. Reporter details	Name, address, profession, contact, signature, date

The reporter's identity is kept confidential. Submitting a report does not mean that the reporter or the doctor admits that the drug caused the reaction.

A6.5.3 Step 2: assessment of the report

Each report is assessed for **causality** (did the drug cause it?), **severity** (how bad was it?), **seriousness** (what was the outcome?), **expectedness** and **preventability**.

Causality: the WHO-UMC system

The WHO-Uppsala Monitoring Centre system places each case in one of six categories [WHO-UMC].

Category	Criteria (all must be met)
Certain	Plausible time relationship to drug intake; **cannot** be explained by disease or other drugs; response to withdrawal plausible; event definitive pharmacologically or phenomenologically; **rechallenge satisfactory**, if necessary
Probable / Likely	Reasonable time relationship; **unlikely** to be due to disease or other drugs; response to withdrawal clinically reasonable; rechallenge not required
Possible	Reasonable time relationship; **could also be explained** by disease or other drugs; information on withdrawal may be lacking or unclear
Unlikely	Time relationship makes a causal link improbable (but not impossible); disease or other drugs give a plausible explanation
Conditional / Unclassified	More data are essential for a proper assessment, or additional data are under examination
Unassessable / Unclassifiable	Information is insufficient or contradictory and cannot be supplemented or verified

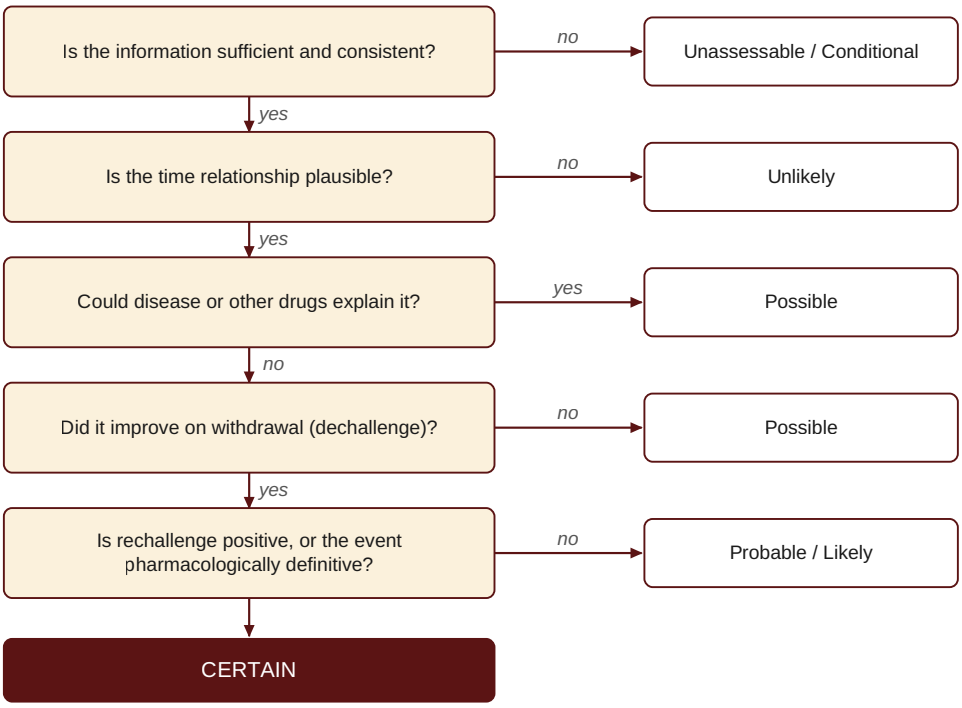


Figure A6.4. WHO-UMC causality assessment as a decision path (simplified).

Causality: the Naranjo algorithm

The Naranjo scale asks ten questions and adds the scores [Naranjo 1981].

No.	Question	Yes	No	Do not know
1	Are there previous conclusive reports on this reaction?	+1	0	0
2	Did the adverse event appear after the suspected drug was given?	+2	−1	0
3	Did the reaction improve when the drug was stopped or a specific antagonist was given?	+1	0	0
4	Did the reaction reappear when the drug was given again?	+2	−1	0
5	Are there alternative causes that could on their own have caused the reaction?	−1	+2	0
6	Did the reaction reappear when a placebo was given?	−1	+1	0
7	Was the drug detected in blood or other fluids in toxic concentrations?	+1	0	0
8	Was the reaction more severe when the dose was increased, or less severe when it was decreased?	+1	0	0
9	Did the patient have a similar reaction to the same or similar drugs before?	+1	0	0
10	Was the adverse event confirmed by objective evidence?	+1	0	0

Total score	Category
9 or more	**Definite**
5 to 8	**Probable**
1 to 4	**Possible**
0 or less	**Doubtful**

CAUSALITY WITH AYURVEDIC MEDICINES: SPECIAL DIFFICULTIES

- **Many ingredients.** A polyherbal or herbo-mineral formulation may contain ten or more ingredients. Which one caused the reaction?
- **Product identity.** The same name may be made by many manufacturers by different methods. Record the **manufacturer and batch number** and keep a sample.
- **Quality.** The reaction may be due to a contaminant, an adulterant or improper processing rather than the intended ingredients. Testing the retained sample may be the only way to know.
- **Concomitant use.** Patients often take modern drugs and Ayurvedic medicines together without telling either doctor.
- **Question 7 of Naranjo** (blood levels) is rarely available, though for metal-containing products blood or urine metal levels can be measured (A4.4.5, A4.5.5).

Severity: the Hartwig scale

Level	Description	Grade
1	ADR occurred but required no change in treatment with the suspected drug	Mild
2	The suspected drug was withheld, stopped or changed; no antidote or other treatment; no increase in hospital stay	Mild
3	The suspected drug was withheld, stopped or changed, **and/or** an antidote or other treatment was needed; no increase in hospital stay	Moderate
4	Any level 3 ADR that increased hospital stay by at least one day, **or** the ADR was the reason for admission	Moderate
5	Any level 4 ADR that required intensive medical care	Severe
6	The ADR caused permanent harm	Severe
7	The ADR led directly or indirectly to death	Severe

Source: Hartwig and colleagues [Hartwig 1992].

Preventability: the Schumock and Thornton criteria

An ADR is considered **preventable** if the answer to any of these questions is "yes" [Schumock 1992]:

1. Was the drug not appropriate for the patient's clinical condition?
2. Were the dose, route or frequency not appropriate for the patient's age, weight or disease?
3. Was required drug monitoring or another necessary laboratory test not done?
4. Was there a history of allergy or previous reaction to the drug?
5. Was a drug interaction involved?
6. Was a toxic serum concentration (or abnormal monitoring test) documented?
7. Was poor compliance involved?

For an Ayurvedic medicine, add two questions of the same spirit: was the product of assured quality from a licensed manufacturer, and were the classical rules of dose, duration, Anupana and Pathya followed?

A6.5.4 Step 3: signal detection

A single report rarely proves anything. When several reports of the same reaction with the same drug accumulate, a **signal** may arise. Signals are found by **clinical review** of case series and by **statistical data mining** of large databases.

The simplest statistical method compares how often a reaction is reported with one drug against how often it is reported with all other drugs.

	Reaction Y	All other reactions
Drug X	a	b
All other drugs	c	d

- **Proportional reporting ratio (PRR)** = [a ÷ (a + b)] ÷ [c ÷ (c + d)]
- **Reporting odds ratio (ROR)** = (a ÷ b) ÷ (c ÷ d) = ad ÷ bc

A commonly used rule treats PRR ≥ 2, with at least three cases and a chi-square of 4 or more, as a signal worth review [Evans 2001]. The Uppsala Monitoring Centre uses a Bayesian measure, the **information component (IC)**, for VigiBase.

WORKED EXAMPLE (ILLUSTRATIVE NUMBERS)

In a database, there are 40 reports of liver injury among 1000 reports for a product X, and 2000 reports of liver injury among 200,000 reports for all other products.

- Proportion for X = 40 ÷ 1000 = 0.04
- Proportion for others = 2000 ÷ 200,000 = 0.01
- **PRR = 0.04 ÷ 0.01 = 4**

Liver injury is reported four times more often, in proportion, with X than with other products. This is a **signal**, not proof. It must be confirmed by case review and, if needed, by a formal study.

A6.5.5 Step 4: action and communication

Possible regulatory action	Example
Change in product information	New warning, contraindication or interaction added to the label and package insert
Restriction of use	Limited to specialists; limited indications; limited duration
Communication to professionals	“Dear Healthcare Professional” letters; drug safety alerts
Communication to the public	Public notices; patient information leaflets
Further study	Post-authorisation safety study required of the manufacturer
Suspension or cancellation of licence	Product withdrawn from the market
Action on quality	Recall of a batch; action against the manufacturer

The cycle then begins again: the effect of the action is itself monitored.

A6.6 The international scenario

A6.6.1 The WHO Programme for International Drug Monitoring

The WHO Programme for International Drug Monitoring (PIDM) began in **1968** with ten pilot member countries. It now has **162 full members and 20 associate members**; India has been a member since **1998** [UMC 2026].

Component	Role
WHO headquarters, Geneva	Policy, guidelines, technical support to countries
Uppsala Monitoring Centre (UMC), Sweden	WHO Collaborating Centre that runs the programme's scientific and technical operations
VigiBase	The WHO global database of individual case safety reports, maintained by UMC; national centres send their reports to it
VigiFlow	A web-based system that national centres can use to manage their reports and send them to VigiBase
VigiLyze	A tool for national centres to search and analyse VigiBase
VigiAccess	Public web access to summary data from VigiBase
WHODrug	A dictionary of medicinal products, including herbal medicines, used to code drug names
National pharmacovigilance centres	Collect reports in each member country and forward them

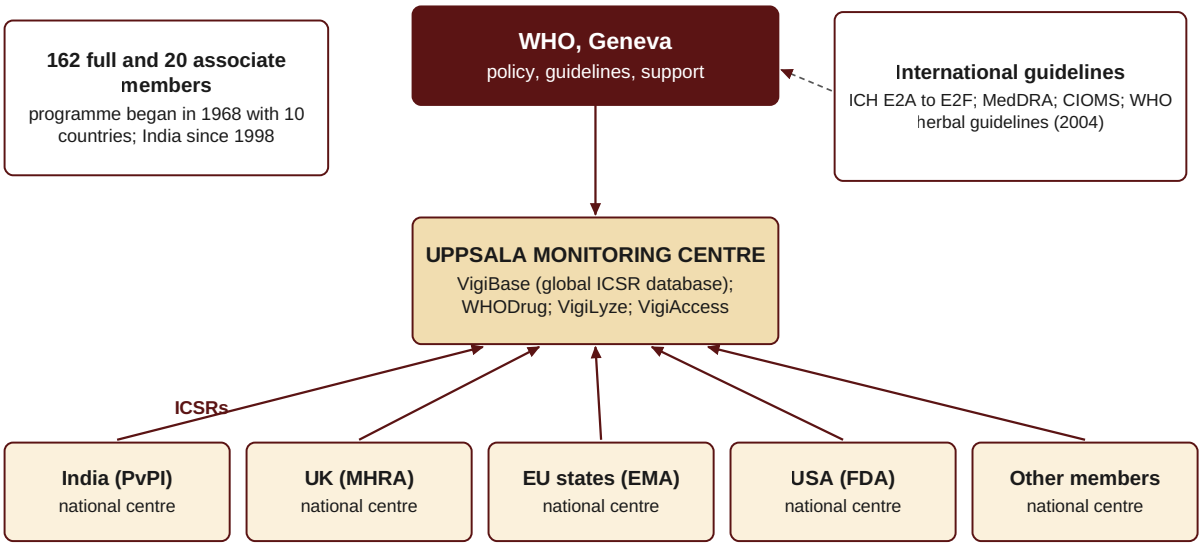


Figure A6.5. The international pharmacovigilance system: national centres feed the WHO global database, and international guidelines shape national practice.

A6.6.2 International guidelines

Guideline	Subject
ICH E2A	Definitions and standards for expedited reporting of safety data (seriousness, expectedness)
ICH E2B	Electronic format of individual case safety reports
ICH E2C	Periodic benefit-risk evaluation report (PBRER)
ICH E2D	Management of post-approval safety data
ICH E2E	Pharmacovigilance planning
ICH E2F	Development safety update report (during clinical trials)
MedDRA	Standard medical terminology for coding adverse events (maintained under ICH)
CIOMS working group reports	Practical guidance on many aspects of pharmacovigilance
WHO guidelines on safety monitoring of herbal medicines (2004)	How to include herbal medicines in national pharmacovigilance systems [WHO 2004]

A6.6.3 Pharmacovigilance in some major regulatory systems

Country or region	Regulator and system	Notes
United States	Food and Drug Administration (FDA): **MedWatch** reporting; FDA Adverse Event Reporting System (**FAERS**) database	Dietary supplements, including many herbal products, are regulated differently from drugs
European Union	European Medicines Agency (EMA): **EudraVigilance** database; Pharmacovigilance Risk Assessment Committee (**PRAC**); Good Pharmacovigilance Practices	Herbal medicinal products, including traditional herbal medicines, are covered by the same pharmacovigilance rules
United Kingdom	Medicines and Healthcare products Regulatory Agency (MHRA): **Yellow Card** scheme (since 1964)	Herbal medicines and unlicensed herbal remedies can be reported
China	National centre for ADR monitoring	Traditional Chinese medicines are included in national monitoring

A6.6.4 Pharmacovigilance of herbal medicines internationally

The WHO 2004 guidelines recognised that herbal medicines pose special problems for pharmacovigilance and asked member states to include them in their national systems [WHO 2004]. The main points are:

1. Herbal medicines should be monitored within the national pharmacovigilance system, not separately from it.
2. The **product must be identified** precisely: botanical name, part used, preparation, manufacturer and batch.

3. **Quality problems** (wrong species, adulteration, contamination with heavy metals, pesticides or microbes, addition of undeclared synthetic drugs) must be considered in every report.
4. Many reactions arise from **concomitant use** with other medicines.
5. Reporting by patients and by traditional practitioners should be encouraged.
6. Causality assessment needs adaptation for multi-ingredient products.

The examples from abroad in this book, such as lead poisoning in users of Ayurvedic products in the United States (A4.8, A5.6), came to light through such systems. Ayurveda's place in the world market depends on India answering these concerns with data of its own.

A6.7 The national scenario: Pharmacovigilance Programme of India

A6.7.1 Structure

The **Pharmacovigilance Programme of India (PvPI)** was started by the Ministry of Health and Family Welfare in **July 2010**, with AIIMS, New Delhi as the first National Coordination Centre. On **15 April 2011** the National Coordination Centre moved to the **Indian Pharmacopoeia Commission (IPC), Ghaziabad**, where it remains [PvPI 2025].

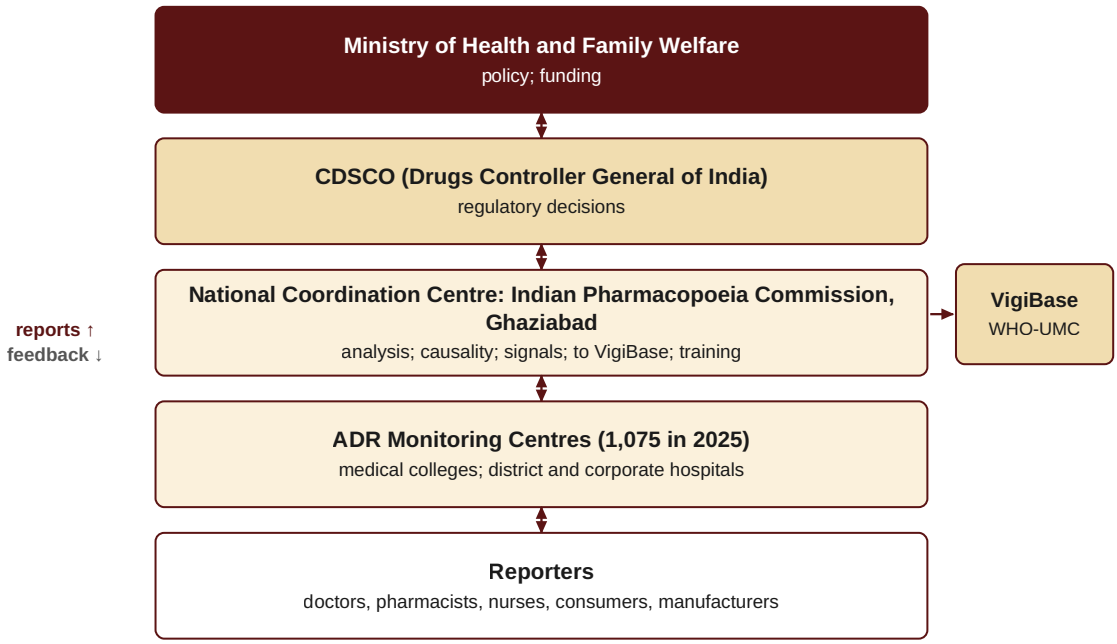


Figure A6.6. Structure of the Pharmacovigilance Programme of India and the flow of reports.

Level	Body	Role
Ministry	Ministry of Health and Family Welfare	Policy and funding
Regulator	Central Drugs Standard Control Organisation (CDSCO), headed by the Drugs Controller General (India)	Regulatory decisions on the basis of PvPI's recommendations
National Coordination Centre	Indian Pharmacopoeia Commission, Ghaziabad	Collects and analyses reports; causality assessment; signal detection; sends reports to VigiBase; training; recommends action to CDSCO
ADR Monitoring Centres (AMCs)	Medical colleges, district hospitals, corporate hospitals	Collect reports from healthcare professionals and patients; assess and forward them. The programme began with 22 AMCs; the number reported in September 2025 was **1,075** [PvPI 2025]
Reporters	Doctors, pharmacists, nurses, other healthcare professionals, consumers, manufacturers	Report suspected ADRs

A6.7.2 Achievements and tools

Item	Detail [PvPI 2025]
International standing	PvPI designated a **WHO Collaborating Centre for Pharmacovigilance in Public Health Programmes and Regulatory Services** on 18 July 2017, and since redesignated
Contribution to VigiBase	India is described as the **eighth largest reporter** of individual case safety reports
Reporting forms	Suspected ADR Reporting Form for healthcare professionals (version 1.4); Medicines Side-Effect Reporting Form for consumers in ten Indian languages; forms for medical devices and for adverse events following immunisation
Online reporting	ADRMS online portal, launched 19 August 2024
Awareness	**National Pharmacovigilance Week**, 17 to 23 September every year, since 2021
Scope	The National Coordination Centre also collects reports on **herbal products**

India also runs related programmes for blood products (haemovigilance) and medical devices (materiovigilance) under the Indian Pharmacopoeia Commission, and monitors adverse events following immunisation under the immunisation programme.

A6.7.3 Legal obligations

Under the **New Drugs and Clinical Trials Rules, 2019**, the holder of a permission to market a new drug must report serious adverse events and submit **periodic safety update reports** to the Central Licensing Authority [NDCT 2019]. The Drugs and Cosmetics Act and Rules, and the rules on misleading advertisements, are discussed in Chapter B8.

A6.8 Pharmacovigilance of Ayurvedic drugs

A6.8.1 Why Ayurvedic drugs need pharmacovigilance

Reason	Explanation
Wide and growing use	Ayurvedic medicines are used by very large numbers of people in India and abroad, often without supervision
The belief that “natural means safe”	Patients and even practitioners may not suspect a herbal or traditional medicine as the cause of a reaction
Metals and minerals	Rasaushadhi contain mercury, arsenic, lead and other metals by design; improper processing or long use can harm (Chapters A4, A5)
Toxic plants	Vatsanabha, Kuchala, Bhallataka, Jayapala and others need Shodhana and careful dosing
Quality problems	Wrong species, adulteration, contamination, undeclared modern drugs, poor GMP
Over-the-counter sale and self-medication	Many products are bought without a prescription and taken for long periods
Herb–drug interactions	Patients take Ayurvedic and modern medicines together (A1c.3)
New formulations and new uses	Proprietary products and new indications have no long classical record
International acceptance	Export markets and foreign regulators demand safety data
Misleading advertisements	Exaggerated claims lead to misuse and delay of effective treatment

A6.8.2 Challenges peculiar to Ayurvedic drugs

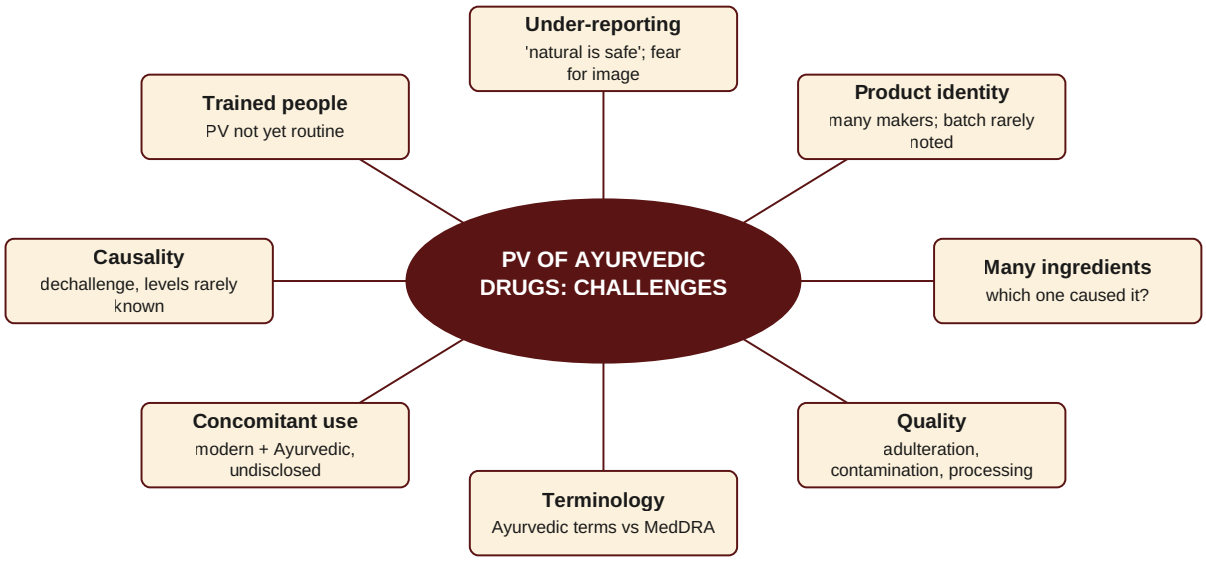


Figure A6.7. Challenges in the pharmacovigilance of Ayurvedic drugs.

Challenge	Detail	Possible solution
Under-reporting	Fear of harming the image of Ayurveda; lack of awareness; no reporting culture	Training; simple forms in local languages; online reporting; assurance that reporting strengthens Ayurveda
Product identification	Same name, many manufacturers and methods; classical and proprietary products	Record manufacturer, licence number, batch number; keep a sample
Multi-ingredient products	Hard to find the culprit ingredient	Analysis of retained samples; signal detection across products sharing an ingredient
Quality versus intrinsic toxicity	A reaction may be due to contamination, not the formulation	Laboratory testing of the suspected batch
Terminology	Ayurvedic diagnoses and effects do not map easily to MedDRA terms	Standardised Ayurvedic terminologies; dual coding
Expected purification effects	Some effects of Shodhana therapy (Vamana, Virechana) are intended, not adverse	Clear distinction between therapeutic effects and Vyapat (complications)
Concomitant use	Patients do not tell doctors about Ayurvedic medicines, and vice versa	Ask every patient; record all products
Causality	Dechallenge and rechallenge rarely documented; blood levels rarely available	Better documentation; metal levels for Rasaushadhi
Few trained personnel	Pharmacovigilance not yet a routine part of Ayurvedic practice	Inclusion in curricula (as in this syllabus) and continuing education

A6.8.3 The national programme for ASU&H drugs

An earlier **National Pharmacovigilance Programme for Ayurveda, Siddha and Unani drugs** was the first organised effort in this field. The present system is a **Central Sector Scheme for Pharmacovigilance of Ayurveda, Siddha, Unani and Homoeopathy (ASU&H) drugs**, implemented by the **Ministry of Ayush since December 2017**. The network became functional in **August 2018** [Lok Sabha 2020].

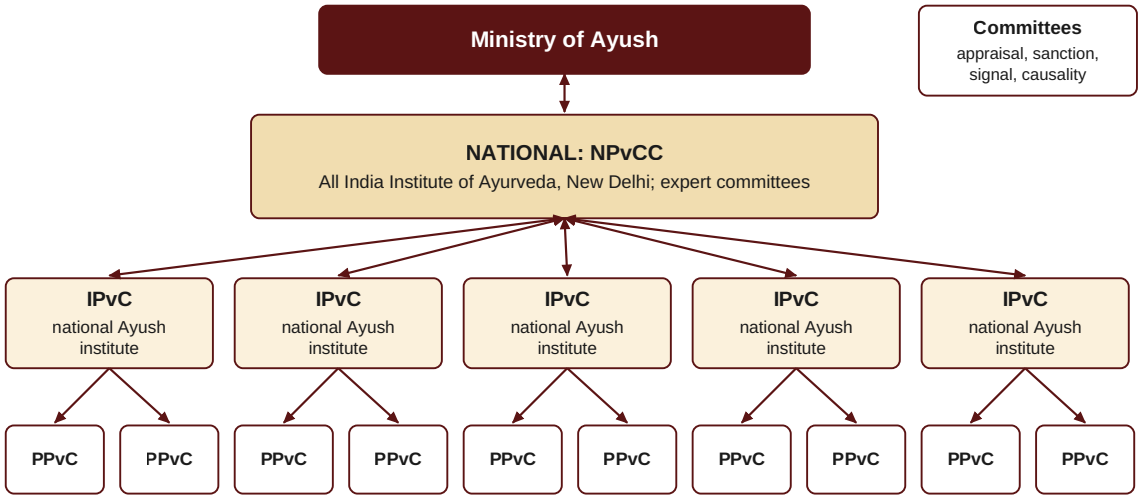


Figure A6.8. Three-tier structure of the pharmacovigilance network for ASU&H drugs.

Tier	Centre	Role
National	**National Pharmacovigilance Coordination Centre (NPvCC)**, **All India Institute of Ayurveda (AIIA), New Delhi**	Coordinates the network; maintains the database; causality assessment and signal detection through expert committees; reports to the Ministry
Intermediary	**Five Intermediary Pharmacovigilance Centres (IPvCs)** at national Ayush institutes	Supervise and train peripheral centres in their region; review reports
Peripheral	**Peripheral Pharmacovigilance Centres (PPvCs)**, at Ayush colleges and institutions: **74** as reported in September 2020 (63 at first, then 11 more)	Collect reports from practitioners and patients; awareness; monitor advertisements
Committees	Project Appraisal, Project Sanctioning, **Signal Detection** and **Causality Assessment** committees	Technical review

Figures as stated in the Lok Sabha answer of 23 September 2020 [Lok Sabha 2020]. The network may have grown since; check the current figures on the AIIA and Ministry of Ayush websites.

Two mandates. The scheme has two jobs:

1. **Reporting of suspected adverse drug events** for ASU&H drugs. Up to August 2020, 463 suspected adverse drug events had been reported: 186 for Ayurveda, 70 for Siddha, 151 for Unani and 56 for Homoeopathy [Lok Sabha 2020].
2. **Monitoring of misleading advertisements** of ASU&H drugs. Up to August 2020, 8,203 cases of misleading advertisements had been identified [Lok Sabha 2020]. These are reported to the state licensing authorities for action under the Drugs and Magic Remedies (Objectionable Advertisements) Act and the Drugs and Cosmetics Rules (Chapter B8).

AyushSuraksha. On 30 May 2025 the Ministry of Ayush launched the **AyushSuraksha** portal to track adverse drug reactions and misleading advertisements relating to Ayush medicines [AIR 2025].

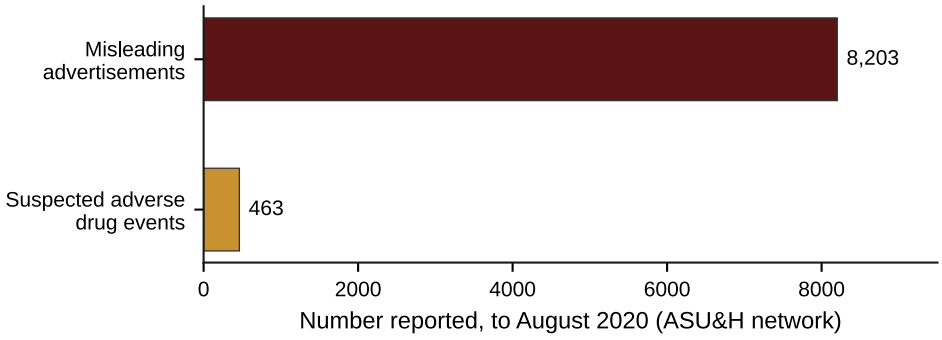


Figure A6.9. The ratio tells its own story: misleading advertisements identified far outnumbered adverse events reported (to August 2020).

The small number of adverse event reports, compared with the very large use of these medicines, is itself a finding. It reflects **under-reporting** far more than safety.

A6.8.4 What the Ayurvedic practitioner should do

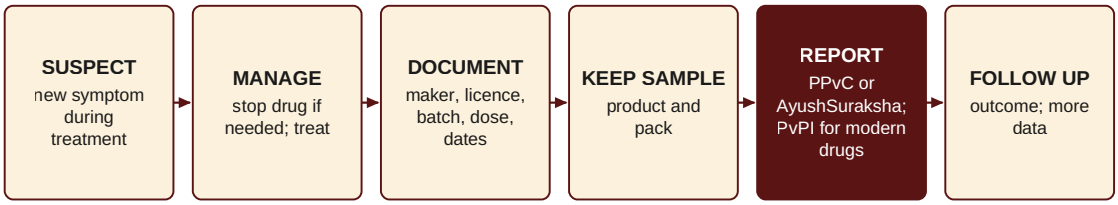


Figure A6.10. Steps for an Ayurvedic practitioner who suspects an adverse reaction.

1. **Suspect.** Think of the medicine whenever a new symptom appears during treatment. Ask about every product taken, including modern drugs and home remedies.
2. **Manage the patient.** Stop the suspected product if appropriate; treat the reaction.
3. **Document.** Product name, **manufacturer, licence number and batch number**, dose, duration, dates, Anupana, concomitant medicines, dechallenge and rechallenge, investigations.
4. **Keep a sample** of the product, and the pack.
5. **Report** to the nearest peripheral pharmacovigilance centre for ASU&H drugs, or through the AyushSuraksha portal; for modern drugs, to the nearest ADR monitoring centre or the PvPI portal.
6. **Follow up** the outcome and send additional information.

A6.8.5 Role of the Rasashastra and Bhaishajya Kalpana scholar

The Rasashastra and Bhaishajya Kalpana scholar stands at the point where many problems begin: the pharmacy.

Area	Role
Manufacture	Correct Shodhana, Marana and Amritikarana; classical and modern quality tests; GMP (Chapter B4)
Records	Batch manufacturing records that let a reported batch be traced to its raw materials
Retained samples	Keep samples of each batch so that a suspect batch can be tested
Complaints	A register of complaints and adverse reaction reports from patients and doctors; investigation of each
Labelling	Accurate composition, dose, cautions and warnings; caution for heavy-metal products (Chapter B8)
Teaching and research	Train students and practitioners to report; study the safety of Rasaushadhi with modern methods (A5.6.4)
Advertisements	Avoid and report misleading claims

AYURVEDA CORRELATION: CLASSICAL ROOTS OF PHARMACOVIGILANCE

Ayurveda was always watchful about the harm medicines can do.

Classical concept	Meaning	Pharmacovigilance parallel
A poison wisely used becomes a medicine, and a medicine wrongly used becomes a poison (Charaka Samhita, Sutra Sthana 1) [R26]	The same substance can heal or harm	Benefit–risk balance
Samyak Yoga, Ayoga, Atiyoga of Vamana and Virechana, and their **Vyapat** (complications) with treatment (Charaka Samhita, Siddhi Sthana 6) [R26]	Recognition and management of adverse outcomes of therapy	Recognition, classification and management of ADRs
Viruddha Ahara (Charaka Samhita, Sutra Sthana 26) [R26]	Harmful combinations	Interactions
Dashavidha Pariksha before deciding the dose (Charaka Samhita, Vimana Sthana 8) [R26]	Patient factors decide dose and safety	Risk factors for ADRs
Shodhana, Marana, Bhasma Pariksha (Rasatarangini) [R7]	Make toxic substances safe before use	Quality assurance and risk minimisation
Warnings on Ashuddha and Apakva Rasa Dravya [R7]	Harm from improperly processed drugs	Quality-related ADRs

Established. Ayurveda contains a clear awareness of drug-related harm and principles for preventing it, and these match the aims of modern pharmacovigilance.

Not established. That classical safeguards alone make formal reporting unnecessary. A tradition that is careful in principle still needs a system that records and counts what happens in practice.

KEY POINTS TO REMEMBER

- Pharmacovigilance (WHO 2002): the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug-related problem.
- Scope: ADRs, medication errors, substandard and falsified products, lack of efficacy, misuse, interactions; herbal and traditional medicines included.
- Serious (outcome) is not the same as severe (intensity). Serious: death, life-threatening, hospitalisation, disability, congenital anomaly, medically important.
- Trials miss rare reactions: rule of three, about 30,000 patients to see one case of a 1-in-10,000 reaction.
- Methods: spontaneous reporting (main method; under-reporting), active surveillance, observational studies, data mining.
- Valid report: identifiable patient, reporter, suspected drug, suspected reaction. Suspicion is enough.

- Causality: WHO-UMC (certain, probable, possible, unlikely, conditional, unassessable); Naranjo (≥9 definite, 5–8 probable, 1–4 possible, ≤0 doubtful).
- Severity: Hartwig (levels 1–7; mild, moderate, severe). Preventability: Schumock and Thornton.
- Signal: PRR = [a/(a+b)] ÷ [c/(c+d)]; PRR ≥ 2 with ≥ 3 cases is a common screening rule. UMC uses the information component.
- WHO PIDM: 1968; 162 full and 20 associate members; India since 1998; UMC Uppsala; VigiBase.
- ICH E2A–E2F; MedDRA. FDA MedWatch and FAERS; EMA EudraVigilance and PRAC; MHRA Yellow Card. WHO herbal PV guidelines 2004.
- PvPI: July 2010; NCC at IPC Ghaziabad since 15 April 2011; 22 AMCs at start, 1,075 in 2025; WHO Collaborating Centre since 2017; ADRMS portal 2024; Pharmacovigilance Week 17–23 September.
- Ayush: Central Sector Scheme since December 2017; functional August 2018; NPvCC at AIIA New Delhi; 5 IPvCs; 74 PPvCs (2020); ADRs and misleading advertisements; AyushSuraksha portal 2025.
- Practitioner: suspect, manage, document (manufacturer, batch), keep sample, report, follow up.

EXAM CORNER — CHAPTER A6

Likely long questions (20 marks)

1. Pharmacovigilance of Ayurvedic drugs and its national programme *(set in July 2018)*.
2. Discuss adverse drug reactions and drug toxicity. Explain the role of pharmacovigilance in the safety monitoring of Ayurvedic drugs *(set in March 2026; see also A1c)*.
3. Define pharmacovigilance. Describe its methods and the process from detection of an ADR to regulatory action. Add a note on causality assessment.
4. Describe the national and international scenario of pharmacovigilance, with special reference to herbal and Ayurvedic medicines.

Likely short notes (5 marks)

1. What is pharmacovigilance? Explain its need and current scenario in the case of ASU drugs *(set in September 2018)*.
2. Pharmacovigilance *(set in September 2020)*.
3. National and international scenario of pharmacovigilance of Ayurvedic drugs *(set in August 2021)*.
4. Pharmacovigilance of Ayurveda drugs *(set in August 2022 and January 2024)*.
5. Causality assessment of ADRs: WHO-UMC and Naranjo.
6. Pharmacovigilance Programme of India.
7. Signal detection.
8. Challenges in the pharmacovigilance of Ayurvedic drugs.
9. Role of the Rasashastra pharmacist in pharmacovigilance.

How to attempt

- **Every** pharmacovigilance answer should open with the WHO definition, word for word.
- Then “why”: the table of limitations of trials and the rule of three.
- For “national programme”, draw the two structures: PvPI (Figure A6.6) and the three-tier Ayush network (Figure A6.8). Examiners give marks for a correct diagram.
- For “international scenario”, give the WHO programme (with numbers), ICH guidelines and two or three regulators, and the WHO herbal guidelines of 2004.

- For “Ayurvedic drugs”, give the need table, the challenges table and the practitioner’s steps.
- Close with the classical roots and an honest line on under-reporting.

Answer frame: “Pharmacovigilance of Ayurvedic drugs and its national programme.” (20 marks)

Part	What to write	Suggested marks
Definition and scope	WHO 2002 definition; scope table; key terms (AE, ADR, serious, signal)	2
Need	Limits of trials; rule of three; history (thalidomide); need table for Ayurvedic drugs	3
Process	Reporting (who, what, minimum data); causality (WHO-UMC table); severity; signal	4
National programme: PvPI	History, structure diagram, numbers, tools	3
National programme: ASU&H	2017 scheme; three-tier diagram; committees; ADR and advertisement data; AyushSuraksha	4
Challenges and solutions	Table	2
Classical roots and conclusion	Ayurveda correlation; role of the Rasashastra scholar	2

Answer frame: “National and international scenario of pharmacovigilance of Ayurvedic drugs.” (5 marks)

Part	What to write	Suggested marks
Definition	WHO 2002	0.5
International	WHO PIDM (1968; 162 full members; UMC; VigiBase); ICH E2 series; WHO herbal guidelines 2004; one regulator	2
National	PvPI (2010; IPC; AMCs); Ayush scheme (2017; AIIA; IPvCs; PPvCs); AyushSuraksha	2
Comment	Under-reporting; challenges; way forward	0.5

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **7, 26, 29, 32, 43**.

- **[R7]** Sharma S, Shastri HD. *Rasatarangini*. 11th ed. Delhi: Motilal Banarasidas; 1979.
- **[R26]** Charaka Samhita: Sutra Sthana 1 and 26; Vimana Sthana 8; Siddhi Sthana 6.
- **[R29]** Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024. (Adverse drug effects; pharmacovigilance.)
- **[R32]** Brunton LL, Knollmann BC, editors. *Goodman & Gilman’s The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023.
- **[R43]** The Drugs and Cosmetics Act, 1940 and Rules, 1945, as amended (Chapter B8).

Additional verified sources (accessed 16 October 2026)

- **[WHO 2002]** World Health Organization. *The Importance of Pharmacovigilance: Safety Monitoring of Medicinal Products*. Geneva: WHO; 2002.
- **[WHO 2004]** World Health Organization. *WHO Guidelines on Safety Monitoring of Herbal Medicines in Pharmacovigilance Systems*. Geneva: WHO; 2004.
- **[WHO-UMC]** Uppsala Monitoring Centre. *The use of the WHO-UMC system for standardised case causality assessment*. Uppsala: UMC.
- **[UMC 2026]** Uppsala Monitoring Centre. Members of the WHO Programme for International Drug Monitoring. who-umc.org (accessed 16 October 2026: 162 full and 20 associate members).
- **[ICH E2A]** International Council for Harmonisation. *E2A: Clinical Safety Data Management: Definitions and Standards for Expedited Reporting*. 1994.
- **[ICH E2C]** International Council for Harmonisation. *E2C(R2): Periodic Benefit-Risk Evaluation Report*. 2012.
- **[PvPI 2025]** Indian Pharmacopoeia Commission. *Pharmacovigilance Programme of India: Profile*. Ghaziabad: IPC; 19 September 2025. ipc.gov.in.
- **[Lok Sabha 2020]** Lok Sabha, Government of India. Unstarred Question No. 2119, answered 23 September 2020 (pharmacovigilance of ASU&H drugs). eparlib.nic.in.
- **[AIR 2025]** All India Radio News. “AyushSuraksha portal to safeguard Ayush systems against misleading advertisements.” newsonair.gov.in, 30 May 2025.
- **[NDCT 2019]** Ministry of Health and Family Welfare, Government of India. New Drugs and Clinical Trials Rules, 2019. G.S.R. 227(E), 19 March 2019.
- **[McBride 1961]** McBride WG. Thalidomide and congenital abnormalities. *Lancet*. 1961;2(7216):1358.
- **[Naranjo 1981]** Naranjo CA, Busto U, Sellers EM, et al. A method for estimating the probability of adverse drug reactions. *Clin Pharmacol Ther*. 1981;30(2):239–245.
- **[Hartwig 1992]** Hartwig SC, Siegel J, Schneider PJ. Preventability and severity assessment in reporting adverse drug reactions. *Am J Hosp Pharm*. 1992;49(9):2229–2232.
- **[Schumock 1992]** Schumock GT, Thornton JP. Focusing on the preventability of adverse drug reactions. *Hosp Pharm*. 1992;27(6):538.
- **[Evans 2001]** Evans SJW, Waller PC, Davis S. Use of proportional reporting ratios (PRRs) for signal generation from spontaneous adverse drug reaction reports. *Pharmacoepidemiol Drug Saf*. 2001;10(6):483–486.

Viva pointers — Chapter A6

Answer each in one or two sentences.

On principles

- Give the WHO definition of pharmacovigilance. Who gave it, and when?
- What is the difference between an adverse event and an adverse drug reaction?
- What is the difference between a serious and a severe reaction?
- What makes an ADR “unexpected”?
- What is a signal?
- What is an ICSR? What is a PSUR?
- Why can a clinical trial of 3000 patients not detect a reaction that occurs in 1 in 10,000?
- What is the rule of three?
- Which disaster gave birth to modern pharmacovigilance?

On methods and assessment

- Which is the commonest method of pharmacovigilance? What is its main weakness?
- What is cohort event monitoring?
- What are the four minimum elements of a valid ADR report?
- Must the reporter be sure that the drug caused the reaction?
- What are dechallenge and rechallenge?
- Name the six WHO-UMC causality categories.
- How many questions does the Naranjo scale have? What score means “probable”?
- What is the Hartwig scale used for? What level is a reaction that caused admission to hospital?
- Name three questions of the Schumock and Thornton criteria.
- How is the proportional reporting ratio calculated?

On the national and international scene

- When did the WHO Programme for International Drug Monitoring begin? Where is the Uppsala Monitoring Centre?
- What is VigiBase? What is VigiAccess?
- When did India join the WHO Programme?
- When was PvPI launched? Where is its National Coordination Centre?
- What is an AMC? About how many are there?
- When is National Pharmacovigilance Week?
- Name three ICH guidelines on pharmacovigilance.
- What is the Yellow Card scheme? What is EudraVigilance? What is MedWatch?
- What did the WHO guidelines of 2004 say about herbal medicines?

On Ayurvedic drugs

- Why do Ayurvedic drugs need pharmacovigilance? Give four reasons.
- Which institute is the National Pharmacovigilance Coordination Centre for ASU&H drugs?

- Describe the three tiers of the Ayush pharmacovigilance network.
- What are the two mandates of the Ayush pharmacovigilance scheme?
- What is AyushSuraksha?
- Why is the number of reported ADRs for Ayurvedic drugs so low?
- Which three details of the product must an Ayurvedic practitioner record when reporting?
- Which classical chapter describes the complications of Vamana and Virechana?
- What can the Rasashastra pharmacist do to support pharmacovigilance?

PART A · PHARMACOLOGY

QUESTION BANK — PART A

Part A of the syllabus is examined in **Section A** of the paper. This question bank gathers every Section A question set in the seven papers examined (July 2018 to March 2026), shows which topics are asked most often, and adds practice questions, model papers and multiple-choice questions for revision.

A-QB.1 Section A questions, paper by paper

16 July 2018

Q	Marks	Question as set	Chapter
Q.1	20	Routes of Drug administration.	A1
Q.2	20	Pharmacovigilance of Ayurvedic drugs and its national program.	A6
Q.3a	5	Ethics involved in human research.	A2
Q.3b	5	Toxicity and pharmacological activities of herbo-mineral compounds.	A5
Q.3c	5	Drug interaction and adverse drug reaction.	A1

29 September 2018

Q	Marks	Question as set	Chapter
Q.1	20	Explain the general principles of Pharmacology. Write Pharmacodynamics and Pharmacokinetics in detail. Explain any 2 routes of administration of Drug.	A1
Q.2	20	Explain the phases and protocol of Preclinical and Clinical evaluation of a new chemical entity.	A1, A2
Q.3a	5	Write the physiological importance and symptoms of deficiency of Iron and Zinc on human body.	A3
Q.3b	5	Explain the chronic toxicity of Arsenic. What is Chelation therapy? Explain its mode of action in short.	A4
Q.3c	5	What is Pharmacovigilance? Explain its need and current scenario in case of ASU drugs.	A6

30 September 2020

Q	Marks	Question as set	Chapter
Q.1	20	Write down principles of Pharmacology, Pharmacodynamics & Pharmacokinetics	A1
Q.2	20	Write down about Heavy metal toxicity and Chelation therapy	A4
Q.3a	5	Short note: Pharmacovigilance	A6
Q.3b	5	Phases of clinical research	A2
Q.3c	5	Elemental constituents of human body and its physiological importance	A3

31 August 2021

Q	Marks	Question as set	Chapter
Q.1	20	Explain in detail regarding Principles of Pharmacodynamics and Pharmacokinetics by illustrating with any Ayurvedic Formulation	A1
Q.2	20	Explain in detail about Phases and Methods of Clinical Research and brief about ethics involved in human research	A2
Q.3A	5	Write in detail about National and International Scenario of Pharmacovigilance of Ayurvedic Drugs	A6
Q.3B	5	Discuss the Ayurvedic and Modern aspects in connection with Heavy Metal Toxicity and Chelation Therapy	A4
Q.3C	5	Brief about Routes of Drug Administration	A1

29 August 2022

Q	Marks	Question as set	Chapter
Q.1	20	Explain phases and methods of Clinical research for newer formulation.	A2
Q.2	20	Explain elemental constituents of human body and explain its physiological importance.	A3
Q.3a	5	Define heavy metal toxicity and explain Chelation therapy.	A4
Q.3b	5	Explain routes of drug administration with their importance.	A1
Q.3c	5	Pharmacovigilance of Ayurveda Drugs.	A6

4 January 2024

Q	Marks	Question as set	Chapter
Q.1	20	Explain Pre – Clinical evaluation of drugs with its importance.	A1
Q.2	20	Define toxicity and explain toxicity of heavy metals.	A4
Q.3a	5	Mention the factors modifying drug effect.	A1
Q.3b	5	Pharmacovigilance of Ayurvedic drugs.	A6
Q.3c	5	Elemental constituents of human body. (Theory of Bio – elementology)	A3

6 March 2026

Q	Marks	Question as set	Chapter
Q.1	20	Explain in detail the animal model and study design to evaluate the in vivo immunomodulatory effect of Swarna Bhasma.	A1
Q.2	20	Discuss Adverse Drug Reactions (ADR) and drug toxicity. Explain the role of Pharmacovigilance in the safety monitoring of Ayurvedic drugs.	A1, A6
Q.3a	5	Define “Dose Response Relationship” and “Structure-Activity Relationship”.	A1
Q.3b	5	What are drug-drug interactions? Give examples relevant to Ayurvedic practice.	A1
Q.3c	5	Explain the importance of micro-nutrients and the effects of their deficiency	A3

A-QB.2 Which topics are asked most often

Chapter	Topic	Long (20)	Short (5)	Papers (of 7)
A1	General Pharmacology (principles, routes, factors, ADR, toxicity, experimental pharmacology)	8	6	**7**
A6	Pharmacovigilance	2	5	**7**
A2	Clinical Pharmacology: new chemical entity, clinical research, ethics	3	2	5
A3	Elemental constituents of the human body	1	4	5
A4	Toxicity of heavy metals and chelation therapy	2	3	5
A5	Toxicity and pharmacological activities of herbo-mineral compounds	0	1	1

A question that draws on two chapters is counted under both. Counts are from the seven Paper IV question papers examined (Parul University, July 2018 to March 2026).

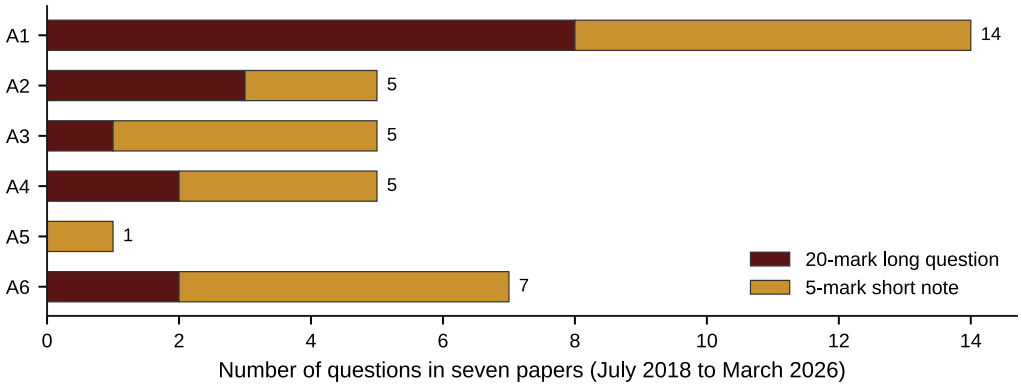


Figure QBA.1. Section A: how often each chapter was asked, as 20-mark and 5-mark questions.

What the record tells you.

1. **Pharmacovigilance and general pharmacology appear in every paper.** No scholar should enter the hall without a full answer ready on each, with the diagrams.

2. **General pharmacology carries the heaviest weight**: eight of the fourteen 20-mark questions in Section A drew on it. Routes of administration, pharmacokinetics, pharmacodynamics, factors modifying drug action, ADRs and experimental pharmacology each deserve a prepared answer.
3. Heavy metals, elements and clinical research appear in about two papers out of three.
4. Herbo-mineral compounds have been asked only once, but they are the natural partner of heavy-metal toxicity, and a combined question is likely.
5. The same question often returns in almost the same words. The answer frames in each chapter's exam corner are written for those exact wordings.

A-QB.3 Practice questions, chapter by chapter

Questions marked ★ have been set in a university paper (the year is given). The others are written in the same style, to cover the rest of each syllabus point. The section of this book that answers each question is given in brackets.

Chapter A1 — General Pharmacology

Long questions (20 marks)

1. ★ Explain the general principles of pharmacology. Write pharmacodynamics and pharmacokinetics in detail. Explain any two routes of administration of drugs *(September 2018)* (A1a, A1b).
2. ★ Write down the principles of pharmacology, pharmacodynamics and pharmacokinetics *(September 2020)* (A1a).
3. ★ Explain in detail the principles of pharmacodynamics and pharmacokinetics by illustrating with any Ayurvedic formulation *(August 2021)* (A1a.2, A1a.3, A1a.6).
4. ★ Routes of drug administration *(July 2018)* (A1b).
5. ★ Explain pre-clinical evaluation of drugs with its importance *(January 2024)* (A1d).
6. ★ Explain in detail the animal model and study design to evaluate the in vivo immunomodulatory effect of Swarna Bhasma *(March 2026)* (A1d.6).
7. ★ Discuss adverse drug reactions and drug toxicity. Explain the role of pharmacovigilance in the safety monitoring of Ayurvedic drugs *(March 2026)* (A1c, A6).
8. Describe the absorption, distribution, metabolism and excretion of drugs. Explain how Ayurvedic concepts of Agni and Srotas can be related to them.
9. Classify drug receptors. Describe the mechanisms of signal transduction with diagrams.
10. Describe the factors that modify drug action. Discuss bioavailability and bioequivalence.
11. Describe the methods of acute, sub-acute and chronic toxicity testing according to OECD guidelines. How would you apply them to a Bhasma?

Short notes (5 marks)

1. ★ Drug interaction and adverse drug reaction *(July 2018)* (A1c).
2. ★ Brief about routes of drug administration *(August 2021)*; explain routes of drug administration with their importance *(August 2022)* (A1b).
3. ★ Mention the factors modifying drug effect *(January 2024)* (A1c.1).
4. ★ Define "dose–response relationship" and "structure–activity relationship" *(March 2026)* (A1a.4, A1a.5).

5. ★ What are drug–drug interactions? Give examples relevant to Ayurvedic practice *(March 2026)* (A1c. 3).
6. First-pass metabolism.
7. Plasma half-life and its clinical importance.
8. Therapeutic index.
9. Enzyme induction and inhibition.
10. G-protein coupled receptors.
11. Competitive and non-competitive antagonism.
12. Transdermal drug delivery.
13. Bioavailability and bioequivalence.
14. Herb–drug interactions.
15. Type A and Type B adverse drug reactions.
16. Teratogenicity.
17. Antidotes.
18. The 3Rs in animal experimentation.
19. Calculation of the animal dose from the human dose.
20. Yogavahi and bioenhancers (Trikatu, piperine).

Chapter A2 — Clinical Pharmacology

Long questions (20 marks)

1. ★ Explain the phases and protocol of pre-clinical and clinical evaluation of a new chemical entity *(September 2018)* (A1d, A2.2, A2.3).
2. ★ Explain in detail the phases and methods of clinical research and brief about ethics involved in human research *(August 2021)* (A2.3, A2.4, A2.6).
3. ★ Explain the phases and methods of clinical research for a newer formulation *(August 2022)* (A2.3–A2.5).
4. Describe the design of a randomised controlled trial of an Ayurvedic formulation. How will you handle placebo, blinding and Prakriti?
5. Describe the ethical principles of research on human participants, with the role of the ethics committee and informed consent.

Short notes (5 marks)

1. ★ Ethics involved in human research *(July 2018)* (A2.6).
2. ★ Phases of clinical research *(September 2020)* (A2.3).
3. Phase 0 (microdosing) studies.
4. Phase IV studies.
5. Randomisation and blinding.
6. Informed consent.
7. Composition and functions of an ethics committee.
8. New Drugs and Clinical Trials Rules, 2019.
9. Clinical Trials Registry – India.
10. Declaration of Helsinki.

11. Good Clinical Practice for ASU drugs.
12. Levels of evidence.

Chapter A3 — Elemental Constituents of the Human Body

Long questions (20 marks)

1. ★ Explain the elemental constituents of the human body and their physiological importance *(August 2022)* (A3.1–A3.4).
2. Classify the elements of the human body. Describe the physiological importance, deficiency and excess of any four trace elements.
3. Describe the role of trace elements in health and disease. Discuss Rasaushadhi as sources of elements, with the evidence.

Short notes (5 marks)

1. ★ Physiological importance and deficiency of iron and zinc *(September 2018)* (A3.4.1, A3.4.2).
2. ★ Elemental constituents of the human body and their physiological importance *(September 2020)* (A3.1).
3. ★ Elemental constituents of the human body (theory of bio-elementology) *(January 2024)* (A3.2).
4. ★ Importance of micro-nutrients and the effects of their deficiency *(March 2026)* (A3.4, A3.6).
5. Calcium metabolism and its disorders.
6. Iodine deficiency disorders.
7. Wilson’s disease.
8. Selenium.
9. Fluorosis.
10. Interactions between minerals.

Chapter A4 — Toxicity of Heavy Metals and Chelation Therapy

Long questions (20 marks)

1. ★ Write about heavy metal toxicity and chelation therapy *(September 2020)* (A4.2–A4.9).
2. ★ Define toxicity and explain the toxicity of heavy metals *(January 2024)* (A4.1–A4.7).
3. Describe the sources, mechanism, clinical features and management of lead poisoning. Add a note on lead in Ayurvedic medicines.
4. Discuss mercury toxicity in relation to the chemical form of mercury. How does this bear on the use of Parada in Rasashastra?
5. Classify chelating agents. Describe their mode of action, uses and adverse effects.

Short notes (5 marks)

1. ★ Chronic toxicity of arsenic; chelation therapy and its mode of action *(September 2018)* (A4.6, A4.9).
2. ★ Ayurvedic and modern aspects of heavy metal toxicity and chelation therapy *(August 2021)* (A4.10).
3. ★ Define heavy metal toxicity and explain chelation therapy *(August 2022)* (A4.1, A4.9).
4. Dimercaprol.
5. Calcium disodium EDTA.
6. Minamata disease.

7. Itai-itai disease.
8. Dushi Visha.
9. Heavy metals in Ayurvedic medicines: the published studies.
10. Arsenic trioxide in acute promyelocytic leukaemia.

Chapter A5 — Toxicity and Pharmacological Activities of Herbo-mineral Compounds

Long questions (20 marks)

1. Discuss the toxicity and pharmacological activities of herbo-mineral compounds, with examples and the published evidence (A5).
2. "A Rasaushadhi prepared as per classical texts is safe." Critically evaluate this statement (A5.6, A5.8).

Short notes (5 marks)

1. ★ Toxicity and pharmacological activities of herbo-mineral compounds *(July 2018)* (A5).
2. Classification of Rasaushadhi.
3. Modern characterisation of Bhasma.
4. Yogavahi.
5. Amritikarana and toxicity.
6. Safe prescription of mercury-containing preparations.

Chapter A6 — Pharmacovigilance

Long questions (20 marks)

1. ★ Pharmacovigilance of Ayurvedic drugs and its national programme *(July 2018)* (A6.7, A6.8).
2. ★ Discuss ADRs and drug toxicity; role of pharmacovigilance in the safety monitoring of Ayurvedic drugs *(March 2026)* (A1c, A6).
3. Define pharmacovigilance. Describe its methods and the process from reporting to regulatory action, with causality assessment.
4. Describe the national and international scenario of pharmacovigilance with special reference to herbal medicines.

Short notes (5 marks)

1. ★ What is pharmacovigilance? Its need and current scenario for ASU drugs *(September 2018)* (A6.1, A6.2, A6.8).
2. ★ Pharmacovigilance *(September 2020)* (A6.1–A6.5).
3. ★ National and international scenario of pharmacovigilance of Ayurvedic drugs *(August 2021)* (A6.6–A6.8).
4. ★ Pharmacovigilance of Ayurveda drugs *(August 2022, January 2024)* (A6.8).
5. Causality assessment.
6. Pharmacovigilance Programme of India.
7. Signal detection.
8. Under-reporting.
9. WHO Programme for International Drug Monitoring.
10. Role of the Rasashastra pharmacist in pharmacovigilance.

A-QB.4 Model question papers: Section A

Each model paper follows the university pattern for one section: two long questions of 20 marks and three short notes, of which any two are to be answered (5 marks each), for 50 marks. Allow about **85 minutes** for a section: 30 minutes for each long question, 12 minutes for each short note, and a few minutes to read the paper and check your answers.

MODEL PAPER 1 — SECTION A (50 MARKS)

Q.1 Explain the principles of pharmacokinetics. Describe the absorption, distribution, metabolism and excretion of drugs, and illustrate each with an Ayurvedic formulation. **(20)**

Q.2 What is pharmacovigilance? Describe the national and international pharmacovigilance programmes, and discuss the challenges in the pharmacovigilance of Ayurvedic drugs. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. Chelation therapy.

b. Informed consent.

c. Physiological importance of iodine and selenium.

MODEL PAPER 2 — SECTION A (50 MARKS)

Q.1 Describe the phases of clinical research. Explain how you would plan a clinical trial of a new Ayurvedic herbo-mineral formulation, including its ethical aspects. **(20)**

Q.2 Define toxicity. Describe the toxicity of lead, mercury and arsenic, and discuss heavy metals in Ayurvedic medicines in the light of published studies. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. Routes of drug administration with their importance.

b. Therapeutic index.

c. Causality assessment of adverse drug reactions.

MODEL PAPER 3 — SECTION A (50 MARKS)

Q.1 Explain the pre-clinical evaluation of drugs. Design an animal study to evaluate the hepatoprotective activity and the sub-acute toxicity of a Bhasma. **(20)**

Q.2 Explain the elemental constituents of the human body and their physiological importance. Discuss Rasaushadhi as sources of essential elements. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. Toxicity and pharmacological activities of herbo-mineral compounds.

b. Factors modifying drug effect.

c. Herb–drug interactions.

Guidance for Model paper 1, Q.1

The table shows how a 20-mark answer to Q.1 of Model paper 1 can be planned. The same planning applies to every long question.

Part of the answer	Content	Book section	Suggested marks
Introduction	Define pharmacokinetics; the LADME scheme with a diagram	A1a.2	2
Absorption	Mechanisms of transport; factors; bioavailability; first-pass effect	A1a.2, A1c.2	4
Distribution	Volume of distribution; plasma protein binding; barriers; redistribution	A1a.2	3
Metabolism	Phase I and II; CYP enzymes; induction and inhibition; prodrugs	A1a.2	4
Excretion	Renal (filtration, secretion, reabsorption); biliary; half-life; clearance; steady state	A1a.2	3
Ayurvedic illustrations	One example for each stage (Haridra with Maricha for absorption; Sneha as a vehicle)	A1a.6	3
Conclusion	Clinical relevance; evidence boundary	—	1

A-QB.5 Multiple-choice questions for self-assessment

The university paper has no multiple-choice questions, but these sixty questions are a quick test of the facts you must have at your fingertips for long answers, viva voce, and entrance and eligibility examinations. Choose one answer for each. The key, with a one-line reason and the section to revise, follows the questions.

Chapter A1

1. The fraction of an administered dose that reaches the systemic circulation unchanged is called: (a) clearance (b) bioavailability (c) volume of distribution (d) half-life
2. With repeated dosing at a constant rate, steady state is reached after about: (a) one half-life (b) two half-lives (c) four to five half-lives (d) ten half-lives
3. Which drug shows saturation (zero-order) kinetics at higher therapeutic doses? (a) phenytoin (b) amoxicillin (c) atenolol (d) metformin
4. The second messenger of the β-adrenergic receptor is: (a) inositol trisphosphate (b) cyclic AMP (c) cyclic GMP (d) nitric oxide
5. The therapeutic index is: (a) ED50 ÷ LD50 (b) LD50 ÷ ED50 (c) ED50 × LD50 (d) LD50 − ED50
6. A competitive antagonist shifts the log dose–response curve of an agonist: (a) to the right, with the maximum unchanged (b) downward, with the maximum reduced (c) to the left (d) not at all
7. The sublingual route avoids: (a) first-pass metabolism in the liver (b) renal excretion (c) plasma protein binding (d) distribution to tissues
8. Which is a Type B (bizarre) adverse reaction? (a) hypoglycaemia with insulin (b) anaphylaxis with penicillin (c) bradycardia with propranolol (d) bleeding with warfarin
9. Piperine of Maricha raises the bioavailability of curcumin mainly by: (a) inhibiting its metabolism (glucuronidation) (b) increasing its renal excretion (c) increasing its protein binding (d) blocking its receptor
10. The 3Rs of animal experimentation are: (a) replacement, reduction, refinement (b) review, record, report (c) randomise, replicate, repeat (d) recruit, retain, release

Chapter A2

11. Phase I clinical studies are usually carried out in: (a) healthy volunteers (b) large numbers of patients (c) animals (d) the general population after marketing
12. A study in which a sub-therapeutic microdose is given to a few people is a: (a) Phase 0 study (b) Phase II study (c) Phase III study (d) Phase IV study
13. Post-marketing surveillance belongs to: (a) Phase I (b) Phase II (c) Phase III (d) Phase IV
14. Clinical trials in India are now governed by the: (a) New Drugs and Clinical Trials Rules, 2019 (b) Pharmacy Act, 1948 (c) Drugs and Magic Remedies Act, 1954 (d) Consumer Protection Act, 2019
15. In India, clinical trials are registered with: (a) CTRI (b) TKDL (c) PvPI (d) PLIM
16. The three principles of the Belmont Report are: (a) respect for persons, beneficence, justice (b) autonomy, secrecy, profit (c) safety, efficacy, quality (d) consent, compensation, confidentiality
17. The Declaration of Helsinki was first adopted in: (a) 1947 (b) 1964 (c) 1979 (d) 2017
18. In a double-blind trial: (a) neither the participant nor the investigator knows the treatment allocation (b) only the participant knows (c) only the investigator knows (d) two placebos are used
19. The main purpose of randomisation is to: (a) balance known and unknown factors between groups and prevent selection bias (b) increase the sample size (c) shorten the trial (d) avoid informed consent
20. In an intention-to-treat analysis, participants are analysed: (a) in the groups to which they were randomised (b) according to the treatment they actually took (c) only if they completed the trial (d) only if they responded

Chapter A3

21. The most abundant element in the human body by mass is: (a) carbon (b) oxygen (c) hydrogen (d) calcium
22. About 99% of body calcium is found in: (a) blood (b) muscle (c) bones and teeth (d) liver
23. The ICMR–NIN (2020) recommended calcium intake for adults is: (a) 400 mg (b) 600 mg (c) 1000 mg (d) 2000 mg per day
24. The best single test of the body’s iron stores is: (a) serum iron (b) serum ferritin (c) haemoglobin (d) total iron-binding capacity
25. Kayser–Fleischer rings are seen in: (a) iron overload (b) Wilson’s disease (c) zinc deficiency (d) lead poisoning
26. Keshan disease is associated with deficiency of: (a) iodine (b) selenium (c) zinc (d) copper
27. Acrodermatitis enteropathica is an inherited disorder of: (a) iron (b) zinc (c) copper (d) manganese
28. Cretinism results from deficiency of: (a) iodine (b) fluoride (c) iron (d) calcium
29. Absorption of non-haem iron is increased by: (a) tea (b) phytates (c) vitamin C (d) calcium
30. Long-term high doses of zinc can cause deficiency of: (a) iron (b) copper (c) calcium (d) iodine

Chapter A4

31. Lead inhibits which two enzymes of haem synthesis? (a) ALA dehydratase and ferrochelatase (b) ALA synthase and catalase (c) cytochrome oxidase and catalase (d) carbonic anhydrase and ferrochelatase
32. Basophilic stippling of red cells is typical of poisoning with: (a) mercury (b) lead (c) arsenic (d) cadmium

33. The CDC blood lead reference value for children, set in 2021, is: (a) 3.5 µg/dL (b) 5 µg/dL (c) 10 µg/dL (d) 25 µg/dL
34. Minamata disease was caused by: (a) cadmium (b) methylmercury (c) arsenic (d) lead
35. Itai-itai disease was caused by: (a) cadmium (b) mercury (c) thallium (d) lead
36. Mees' lines and "raindrop" pigmentation are features of: (a) arsenic poisoning (b) lead poisoning (c) copper overload (d) iron poisoning
37. Dimercaprol (BAL) should be avoided in poisoning with: (a) arsenic (b) methylmercury (c) inorganic mercury (d) lead encephalopathy, together with EDTA
38. In lead encephalopathy, BAL is combined with: (a) calcium disodium EDTA (b) disodium EDTA (c) deferoxamine (d) Prussian blue
39. The chelating agent used in acute iron poisoning is: (a) penicillamine (b) deferoxamine (c) succimer (d) dimercaprol
40. The oral absorption of cinnabar (mercuric sulphide) reported by Liu and colleagues is about: (a) 0.2% (b) 10% (c) 50% (d) 95%

Chapter A5

41. X-ray diffraction shows that Kajjali consists mainly of: (a) metacinnabar (black HgS) (b) free mercury (c) mercuric chloride (d) mercuric oxide
42. Jasada (Yashada) Bhasma consists mainly of: (a) zinc sulphate (b) zinc oxide (c) metallic zinc (d) zinc carbonate
43. Which is a Kupipakva Rasayana? (a) Arogyavardhini Vati (b) Rasa Sindura (c) Praval Pishti (d) Rasa Parpati
44. Arsenic trioxide is an established treatment for: (a) acute promyelocytic leukaemia (b) psoriasis (c) malaria (d) tuberculosis
45. Realgar (Manahshila) is chemically: (a) As_2O_3 (b) As_2S_3 (c) As_4S_4 (d) HgS
46. Haratala (orpiment) is chemically: (a) As_2S_3 (b) As_4S_4 (c) PbO (d) HgS
47. In a 28-day rat study of Tamra Bhasma, greater toxicity at the highest dose was seen with Bhasma prepared: (a) without Amritikarana (b) with Amritikarana (c) with Bhavana of Kumari (d) by Kupipaka
48. In 28-day rat studies of Arogyavardhini Vati, the important finding despite normal liver and kidney tests was: (a) mercury accumulation in brain, liver and kidney (b) weight loss (c) anaemia (d) death at the human dose
49. In Saper's 2008 study, the proportion of Rasa Shastra products with detectable lead, mercury or arsenic was: (a) 10.4% (b) 20.7% (c) 40.6% (d) 75%
50. The Yogavahi property of Rasa is: (a) proved by randomised trials (b) a classical claim not yet established (c) a pharmacopoeial test (d) a regulatory requirement

Chapter A6

51. The WHO definition of pharmacovigilance now in use was published in: (a) 1968 (b) 1998 (c) 2002 (d) 2010
52. The WHO Programme for International Drug Monitoring began in: (a) 1961 (b) 1968 (c) 1978 (d) 1998
53. The Uppsala Monitoring Centre is in: (a) Geneva (b) London (c) Uppsala, Sweden (d) New Delhi
54. India joined the WHO Programme for International Drug Monitoring in: (a) 1986 (b) 1998 (c) 2010 (d) 2017

55. The National Coordination Centre of PvPI is at the: (a) Indian Pharmacopoeia Commission, Ghaziabad (b) AIIMS, New Delhi (c) CDSCO headquarters (d) AIIA, New Delhi
56. A Naranjo score of 6 means the reaction is: (a) definite (b) probable (c) possible (d) doubtful
57. The National Pharmacovigilance Coordination Centre for ASU&H drugs is the: (a) All India Institute of Ayurveda, New Delhi (b) Indian Pharmacopoeia Commission (c) PLIM, Ghaziabad (d) CCRAS headquarters
58. National Pharmacovigilance Week is observed from: (a) 1 to 7 July (b) 17 to 23 September (c) 1 to 7 December (d) 7 to 13 April
59. A reaction that leads to admission to hospital is classed as: (a) serious (b) severe (c) mild (d) unexpected
60. The main limitation of spontaneous reporting is: (a) under-reporting (b) high cost (c) short duration (d) need for randomisation

Answer key

Q	Ans	Reason (section)	Q	Ans	Reason (section)
1	b	Definition of bioavailability (A1c.2)	31	a	Haem pathway (A4.4.3)
2	c	About 97% of steady state after five half-lives (A1a.2)	32	b	Lead: anaemia with stippling (A4.4.4)
3	a	Phenytoin metabolism saturates (A1a.2)	33	a	CDC 2021 (A4.4.4)
4	b	β receptors act through Gs and cAMP (A1a.3)	34	b	Methylmercury in fish (A4.5.4)
5	b	LD50 ÷ ED50 (A1a.4)	35	a	Cadmium in rice (A4.7.1)
6	a	Surmountable antagonism (A1a.3)	36	a	Arsenic (A4.6.4)
7	a	Sublingual veins drain into the systemic circulation (A1b.4)	37	b	BAL moves mercury into the brain (A4.5.6)
8	b	Type B: unpredictable, not dose-related (A1c.4)	38	a	Never Na_2EDTA (A4.9.4)
9	a	Inhibition of glucuronidation (A1a.6)	39	b	Deferoxamine binds iron (A4.9.4)
10	a	The 3Rs (A1d.7)	40	a	Liu 2008 (A4.5.1)
11	a	First in human, healthy volunteers (A2.3.2)	41	a	Sharma 2018 (A5.3)
12	a	Microdose (A2.3.1)	42	b	Umrani 2013 (A5.3)
13	d	After approval (A2.3.5)	43	b	Kupipakva (A5.1.1)
14	a	NDCT Rules, 2019 (A2.2)	44	a	Lo-Coco 2013 (A4.6, A5.5)
15	a	Clinical Trials Registry – India (A2.2)	45	c	Realgar As_4S_4 (A4.6.1)
16	a	Belmont Report, 1979 (A2.6.2)	46	a	Orpiment As_2S_3 (A4.6.1)
17	b	WMA, 1964 (A2.6.1)	47	a	Chaudhari 2016 (A5.6.3)
18	a	Blinding (A2.4.2)	48	a	Kumar 2012 (A5.6.3)
19	a	Purpose of randomisation (A2.4.2)	49	c	Saper 2008 (A4.8.1)
20	a	Intention to treat (A2.4.2)	50	b	Not established (A5.2)

Q	Ans	Reason (section)	Q	Ans	Reason (section)
21	b	About 65% of body mass (A3.1)	51	c	WHO 2002 (A6.1)
22	c	Skeleton (A3.1)	52	b	Ten pilot countries (A6.3)
23	c	ICMR-NIN 2020 (A3.3)	53	c	UMC (A6.6.1)
24	b	Ferritin reflects stores (A3.4.1)	54	b	India, 1998 (A6.6.1)
25	b	Copper in the cornea (A3.4.3)	55	a	Since 15 April 2011 (A6.7.1)
26	b	Selenium (A3.4.5)	56	b	Score 5 to 8 (A6.5.3)
27	b	Zinc (A3.4.2)	57	a	AIIA (A6.8.3)
28	a	Iodine (A3.4.4)	58	b	Since 2021 (A6.7.2)
29	c	Vitamin C (A3.4.1)	59	a	Seriousness is by outcome (A6.1.2)
30	b	Zinc blocks copper absorption (A3.4.2)	60	a	Under-reporting (A6.4.1)

PART B

PHARMACY MANAGEMENT

Paper IV · Section B of the university question paper (50 marks)

Chapter	Syllabus point
B1	Scope and Evolution of Pharmacy; Information Resources
B2	Pharmaceutical Dosage Form Design (Pre-formulation)
B3	Packaging Materials and Labelling
B4	Pharmacy Management and Good Manufacturing Practices
B5	Pharmaceutical Marketing, Product Release and Withdrawals
B6	Hospital, Dispensing and Community Pharmacy
B7	Patenting and Intellectual Property Rights
B8	Laws Governing Ayurvedic Drugs
B9	Regulatory Affairs in International Trade of Ayurvedic Drugs
B10	Ayurvedic Pharmacopoeia and Formulary of India
B11	Indian, British and United States Pharmacopoeias and the Pharmaceutical Codex
B12	Traditional Knowledge Digital Library

PART B · PHARMACY MANAGEMENT · CHAPTER B1

B1 · SCOPE AND EVOLUTION OF PHARMACY; INFORMATION RESOURCES

SYLLABUS POINT (PART B, 1)

Scope and evolution of pharmacy. Information resources in pharmacy and pharmaceutical Science.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B1	Scope and Evolution of Pharmacy; Information Resources	0	1	Jul 2018

Why this chapter matters to you. A postgraduate in Bhaishajya Kalpana is a pharmacist of Ayurveda. This chapter places that profession within the larger history and scope of pharmacy, and teaches the skill that every later chapter assumes: finding, judging and citing reliable information. It was set as a short note in the July 2018 paper.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define pharmacy and describe its evolution, in the world and in India, including the place of Bhaishajya Kalpana and Rasashastra.
2. Describe the scope of pharmacy and of Ayurvedic pharmacy today.
3. Classify information resources in pharmacy and pharmaceutical science, with examples.
4. Search the literature systematically and judge the reliability of a source.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	B, Q.3b	5	Information resources in pharmacy and pharmaceutical science.

B1.1 What is pharmacy?

> **Pharmacy** is the science and profession concerned with the discovery, preparation, standardisation, preservation, dispensing and safe and effective use of medicines. The word also means the place where medicines are prepared or dispensed.

The word comes from the Greek *pharmakon*, which meant both a remedy and a poison. The double meaning is apt: the pharmacist's task is to make sure that a substance acts as a remedy and not as a poison. Charaka said the same: a poison rightly used becomes a fine medicine, and a medicine wrongly used becomes a poison (Charaka Samhita, Sutra Sthana 1) [R26].

In Ayurveda, the branch that deals with the preparation of medicines is **Bhaishajya Kalpana** (*Bhaishajya*, medicine; *Kalpana*, preparation or design). **Rasashastra** deals with the processing and use of metals, minerals and mercury. Together they form the pharmaceutics of Ayurveda.

B1.1.1 Pharmacy in the classical texts

Classical idea	Source	Meaning for pharmacy
Chikitsa Chatushpada: physician, drug, attendant and patient are the four pillars of treatment	Charaka Samhita, Sutra Sthana 9 [R26]	The **drug** is one of the four pillars; its quality decides the outcome as much as the physician's skill
Four qualities of a good drug: abundance, effectiveness, suitability for many preparations, good quality (Bahuta, Yogyatva, Anekavidha Kalpana, Sampat)	Charaka Samhita, Sutra Sthana 9 [R26]	Availability, efficacy, formulability and quality: the first questions of pharmaceutics (A1a.1)
Four qualities of the attendant who prepares and gives medicines: knowledge of nursing and preparation (Upacharajnata), skill (Dakshya), devotion (Anuraga), cleanliness (Shaucha)	Charaka Samhita, Sutra Sthana 9 [R26]	The qualities of a good pharmacist and pharmacy technician
The five basic preparations (Panchavidha Kashaya Kalpana) and their derivatives	Charaka Samhita, Sutra Sthana 4; systematised in Sharangadhara Samhita [R11]	The classical system of dosage forms
The design of the **Rasashala**, the pharmacy for Rasa preparations, with its sections and equipment	Rasaratna Samuchchaya [R3]	An early description of a pharmaceutical facility; the idea behind modern GMP layouts (Chapter B4)

B1.2 Evolution of pharmacy

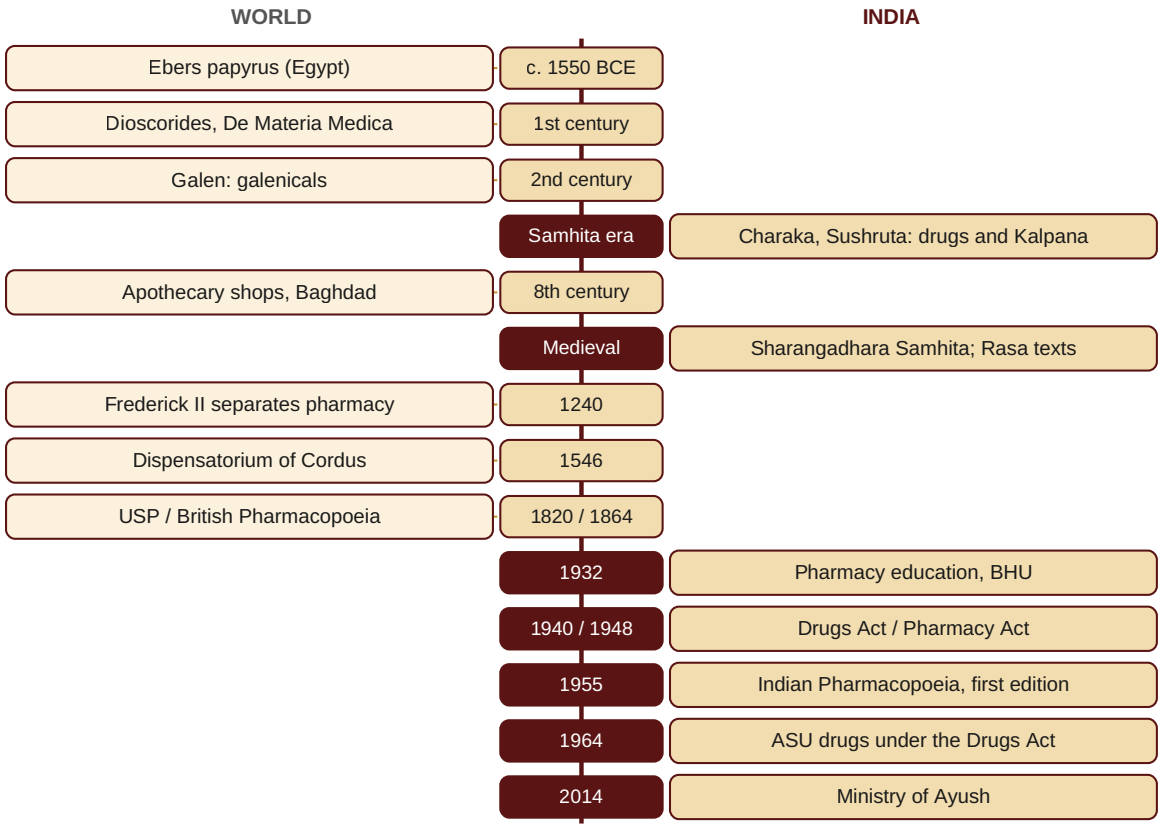


Figure B1.1. Milestones in the evolution of pharmacy, in the world and in India.

B1.2.1 In the world

Period	Development
Ancient civilisations	Mesopotamian clay tablets and the Egyptian **Ebers papyrus** (about 1550 BCE) record hundreds of remedies and their preparation
Greece and Rome	Hippocrates' rational medicine; **Dioscorides'** *De Materia Medica* (1st century CE) describes about 600 plants and remained a standard for 1500 years; **Galen** (2nd century CE) gave his name to "galenicals", preparations made from crude drugs
The Arab world	Pharmacy became a separate profession; privately owned apothecary shops are recorded in Baghdad from the eighth century; Ibn Sina's *Canon of Medicine*
Medieval Europe	An edict of Emperor Frederick II (1240) separated pharmacy from medicine and regulated the apothecary
16th–17th centuries	Official pharmacopoeias appear: the *Dispensatorium* of Valerius Cordus (Nuremberg, 1546); the *London Pharmacopoeia* (1618)
19th century	Isolation of pure active principles (morphine, quinine); the beginnings of the pharmaceutical industry; national pharmacopoeias (*United States Pharmacopeia*, 1820; *British Pharmacopoeia*, 1864)
20th century	Antibiotics, vaccines, synthetic drugs; drug regulation after disasters (thalidomide, A6.2); dosage form science (biopharmaceutics); clinical pharmacy

Period	Development
21st century	Biotechnology products; novel delivery systems; pharmacogenomics; digital health; regulatory harmonisation (ICH)

B1.2.2 In India

Period	Development
Vedic period	Plants and their uses in the Rigveda and Atharvaveda
Samhita period	Charaka and Sushruta Samhitas: drugs, their collection, storage and preparation; the five Kashaya Kalpana
Medieval period	**Sharangadhara Samhita** systematises dosage forms, doses and shelf life [R11]; **Rasashastra** flourishes, with Rasahridaya Tantra, Rasarnava, Rasaratna Samuchchaya and later Rasatarangini [R1, R2, R3, R7]
Colonial period	Western pharmacy introduced through hospitals and dispensaries; the modern Indian pharmaceutical industry begins in the early twentieth century
Early 20th century	Organised Ayurvedic pharmacies begin to manufacture classical medicines on a commercial scale
1932	Pharmacy education begins at Banaras Hindu University under Professor M. L. Schroff, often called the father of pharmacy education in India
1940	**Drugs Act, 1940** (later the Drugs and Cosmetics Act); Ayurvedic, Siddha and Unani drugs brought within it by amendment in 1964 (Chapter B8)
1948	**Pharmacy Act, 1948** regulates the profession of pharmacy (Chapter B8)
1955	First edition of the *Indian Pharmacopoeia*
1970s onward	*Ayurvedic Formulary of India* and *Ayurvedic Pharmacopoeia of India* published (Chapter B10)
2000s onward	GMP for ASU drugs (Schedule T, Chapter B4); Pharmacopoeia Commission for Indian Medicine and Homoeopathy; Ministry of Ayush (2014)

The dates of the pharmacopoeial publications and regulatory milestones for Ayurveda are given, with sources, in Chapters B4, B8 and B10.

B1.3 Scope of pharmacy

Pharmacy today is a family of professions. The same is true of Ayurvedic pharmacy.

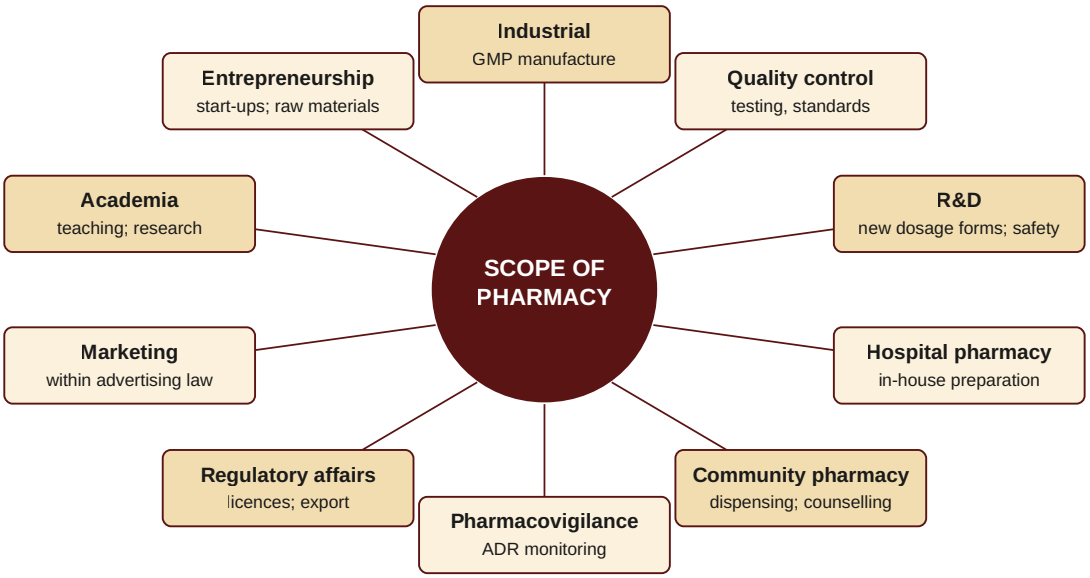


Figure B1.2. The scope of pharmacy, with the corresponding Ayurvedic fields.

Field	Modern pharmacy	Ayurvedic pharmacy
Industrial pharmacy	Formulation, production, scale-up	Manufacture of classical and proprietary ASU medicines in licensed GMP units
Quality assurance and control	Analytical testing; validation	Pharmacognosy, physico-chemical and analytical testing of raw materials and finished products; drug testing laboratories
Research and development	New drugs and dosage forms	Standardisation of classical preparations; new dosage forms (tablets, capsules, syrups, granules from Kwatha); safety studies of Rasaushadhi
Hospital pharmacy	Procurement, storage, dispensing in hospitals	Pharmacy of Ayurvedic teaching hospitals; in-house preparation of medicines for Panchakarma and inpatients (Chapter B6)
Community pharmacy	Retail dispensing and counselling	Ayurvedic medical stores; counselling on dose, Anupana and Pathya (Chapter B6)
Clinical pharmacy and pharmacovigilance	Rational use; ADR monitoring	Pharmacovigilance of ASU&H drugs (Chapter A6)
Regulatory affairs	Licensing, registration, compliance	Licensing under the Drugs and Cosmetics Rules; export registration; dossiers (Chapters B8, B9)
Marketing	Product management; medical representatives	Marketing of ASU products within the advertising laws (Chapters B5, B8)
Academia	Teaching and research	Teaching of Rasashastra and Bhaishajya Kalpana; research guidance
Entrepreneurship	Start-ups; contract manufacturing	Ayurvedic start-ups; contract and loan licence manufacturing; herbal raw material supply

Field	Modern pharmacy	Ayurvedic pharmacy
Allied sectors	Nutraceuticals, cosmetics, veterinary products	Ayurvedic nutraceuticals and health supplements (under food law, Chapter B8); cosmetics

B1.4 Information resources in pharmacy and pharmaceutical science

Good pharmacy rests on good information. A scholar must know **where** information is kept, **how** to search for it, and **how far** to trust it.

B1.4.1 Classification of information sources

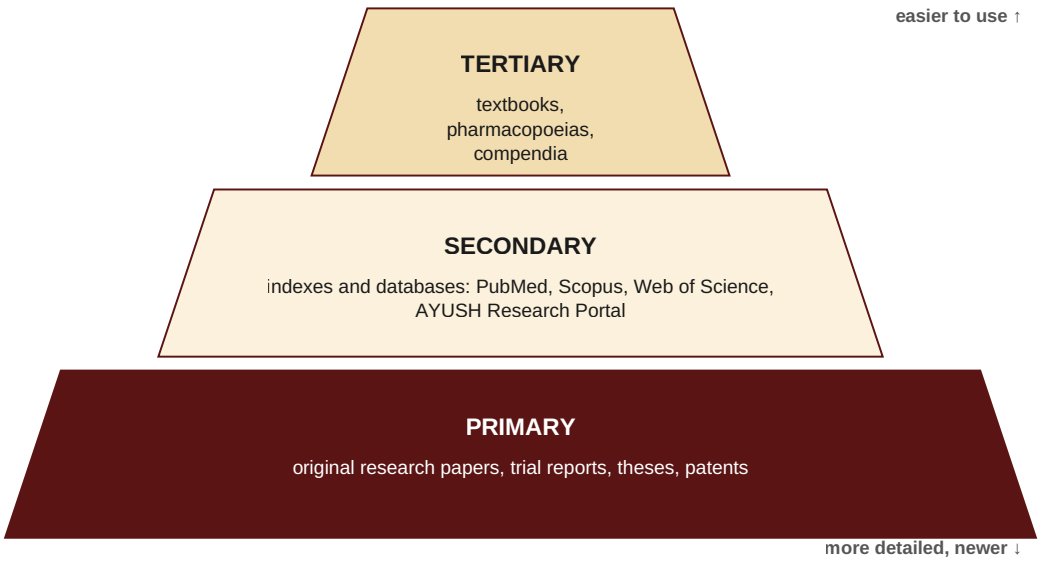


Figure B1.3. Primary, secondary and tertiary sources: the information pyramid.

Type	What it is	Examples	Strengths and limits
Primary	The original report of new work	Research papers in journals; clinical trial reports; theses; patents; conference papers; official monographs that set a standard for the first time	Most up to date and detailed; must be judged critically; one study is never the last word
Secondary	Tools that index, abstract or summarise primary literature	Bibliographic databases (PubMed/ MEDLINE, Scopus, Web of Science, Embase, CAB Abstracts); abstracting services; citation indexes	Lead the reader to the primary literature; do not themselves contain the full evidence
Tertiary	Compiled, digested knowledge	Textbooks; pharmacopoeias and formularies (API, AFI, IP, BP, USP); drug compendia (Martindale); encyclopaedias; handbooks; systematic reviews and guidelines	Easy to use and authoritative; may lag behind new findings

Systematic reviews are often placed with secondary sources; what matters is that a good systematic review summarises all the primary studies on a question by an explicit method.

B1.4.2 Resources for Ayurvedic pharmacy

Need	Resource
Classical texts	Charaka, Sushruta and Ashtanga texts; Sharangadhara Samhita; Rasa texts listed in the syllabus [R1–R26]; online editions such as the Charak Samhita New Edition (carakasamhitaonline.com)
Official standards	*Ayurvedic Pharmacopoeia of India* and *Ayurvedic Formulary of India* (Chapter B10); *Indian Pharmacopoeia* (Chapter B11)
Laws and regulations	Drugs and Cosmetics Act and Rules; Gazette notifications; websites of the Ministry of Ayush, the Central Drugs Standard Control Organisation and the Pharmacopoeia Commission for Indian Medicine and Homoeopathy (Chapter B8)
Research literature	PubMed; Scopus; Google Scholar; DOAJ (open-access journals); AYUSH Research Portal (Ministry of Ayush); Shodhganga (Indian theses, INFLIBNET)
Plant names	**World Flora Online**, which has superseded *The Plant List* as the reference for accepted botanical names; Plants of the World Online; International Plant Names Index
Phytochemistry	IMPPAT database of Indian medicinal plants, phytochemistry and therapeutics [Mohanraj 2018]; PubChem; ChemSpider
Medicinal plant data	Database of Medicinal Plants (CCRAS) [R33]; *Quality Standards of Indian Medicinal Plants* (ICMR) [R34]
Traditional knowledge	Traditional Knowledge Digital Library (Chapter B12)
Terminology and morbidity codes	NAMASTE portal (National AYUSH Morbidity and Standardized Terminologies Electronic)
Clinical trials	Clinical Trials Registry – India (CTRI); ClinicalTrials.gov; WHO International Clinical Trials Registry Platform
Drug safety	PvPI and the Ayush pharmacovigilance programme (Chapter A6); VigiAccess

B1.4.3 How to search the literature

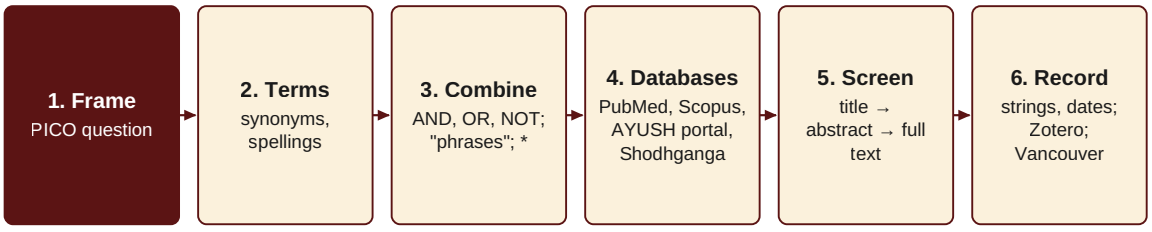


Figure B1.4. A systematic literature search in six steps.

1. **Frame the question.** For a clinical question use **PICO**: Population, Intervention, Comparison, Outcome. Example: in adults with iron-deficiency anaemia (P), does Punarnavadi Mandura (I), compared with ferrous sulphate (C), raise haemoglobin (O)?
2. **List the terms**, with synonyms and spellings: Mandura, Manduram, “iron oxide”, “Punarnavadi”; anaemia, anemia.

3. **Combine with Boolean operators**: AND narrows (Mandura AND anaemia); OR widens (anaemia OR anemia); NOT excludes. Use quotation marks for phrases and an asterisk for word endings (Ayurved* finds Ayurveda, Ayurvedic).
4. **Choose databases**: PubMed for medicine, Scopus or Google Scholar for breadth, AYUSH Research Portal and Shodhganga for Indian work.
5. **Filter and screen**: by year, study type, language; read titles, then abstracts, then full texts.
6. **Record and manage**: keep the search strings and dates; use reference software (Zotero, Mendeley); cite in a standard style (Vancouver is used in this book).

B1.4.4 Judging a source

Question	Good sign	Warning sign
Who wrote it?	Named authors with relevant affiliation	Anonymous; commercial interest undisclosed
Where was it published?	Peer-reviewed journal indexed in major databases; official website	Journal that promises acceptance in days for a fee; no clear peer review
Is it current?	Recent, or the classic source on the point	Outdated law or standard quoted as current
What kind of evidence?	Randomised trial, systematic review, pharmacopoeial standard	Testimonials, advertisements, single case claims
Can it be checked?	Methods and data given; references verifiable	Vague methods; no references
Is the claim proportionate?	Conclusions match the data	“Cures”, “no side effects”, “100% safe”

Predatory journals. Some journals publish anything for a fee without real peer review. In February 2025 the University Grants Commission discontinued its UGC-CARE list of journals and issued suggestive parameters to help institutions choose peer-reviewed journals [UGC 2025]. A scholar should check that a journal is indexed in a major database, has a transparent peer-review process and a named editorial board, and does not guarantee acceptance.

B1.4.5 Citing sources

Every statement of fact in a thesis, a paper or a textbook should be traceable to its source. Use one style consistently. This book uses the **Vancouver style**: authors, title, journal, year, volume, issue and pages. For a classical text, give the text, the section, the chapter and the verse, and the edition used. Never cite a source you have not seen, and never invent a reference.

AYURVEDA CORRELATION: APTOPADESHA AND THE HIERARCHY OF SOURCES

Ayurveda ranks authoritative testimony (**Aptopadesha**) first among the means of valid knowledge, and defines an Apta as a person free from passion and ignorance whose word is reliable (Charaka Samhita, Sutra Sthana 11) [R26]. The modern scholar faces the same problem in another form: which sources can be trusted?

Classical idea	Modern parallel
Aptopadesha: the word of the reliable expert	Authoritative tertiary sources: pharmacopoeias, standard textbooks, guidelines
Pratyaksha (direct observation) and Anumana (inference)	Primary research: observation and experiment
Yukti (reasoning from many factors together)	Systematic review and synthesis of evidence
Tantra Yukti: the methods for reading and interpreting a text	Methods for critical appraisal of a paper

Established. The classical texts set out explicit rules for judging knowledge and for reading texts.

Not established. That a classical statement on a matter of fact (the effect of a drug, its safety) needs no further testing. In science, the reliability of a source is judged by the evidence it presents, and every claim stays open to new observation.

KEY POINTS TO REMEMBER

- Pharmacy: the science and profession of discovering, preparing, standardising, preserving, dispensing and ensuring the safe use of medicines.
- Bhaishajya Kalpana and Rasashastra are the pharmaceutics of Ayurveda.
- Charaka: the drug is one of the four pillars of treatment; four qualities of a drug; four qualities of the attendant (Upacharajnata, Dakshya, Anuraga, Shaucha).
- World milestones: Ebers papyrus; Dioscorides; Galen; Arab apothecaries; Frederick II's edict (1240); first pharmacopoeias (1546, 1618); USP 1820; BP 1864.
- India: Sharangadhara Samhita; Rasashastra texts; BHU pharmacy education 1932 (M. L. Schroff); Drugs Act 1940; Pharmacy Act 1948; IP 1955; ASU drugs under the Act from 1964; Ministry of Ayush 2014.
- Scope: industrial, QA/QC, R&D, hospital, community, clinical, regulatory, marketing, academia, entrepreneurship.
- Sources: primary (original research), secondary (indexes and databases), tertiary (textbooks, pharmacopoeias, compendia).
- World Flora Online has superseded The Plant List. UGC discontinued the UGC-CARE list in February 2025.
- Search: PICO; synonyms; Boolean AND, OR, NOT; databases; screening; reference management.

EXAM CORNER — CHAPTER B1

Likely long questions (20 marks)

1. Describe the scope and evolution of pharmacy, with special reference to Ayurvedic pharmacy in India.
2. Classify information resources in pharmacy and pharmaceutical science. Describe the important resources for an Ayurvedic pharmacist and how to search them.

Likely short notes (5 marks)

1. Information resources in pharmacy and pharmaceutical science *(set in July 2018)*.

2. Evolution of pharmacy in India.
3. Scope of Ayurvedic pharmacy.
4. Primary, secondary and tertiary sources of information.
5. Literature search strategy.
6. Predatory journals.

How to attempt

- Start with a definition and the classical four pillars.
- For information resources, the three-tier table (primary, secondary, tertiary) with examples earns most of the marks; add the pyramid figure.
- Add Ayurveda-specific resources and one line on judging a source.

Answer frame: "Information resources in pharmacy and pharmaceutical science." (5 marks)

Part	What to write	Suggested marks
Introduction	Why information matters; definition of information resource	0.5
Classification	Primary, secondary, tertiary: table with examples; pyramid	2
Ayurvedic resources	Classical texts, API/AFI, Ayush portals, databases, TKDL	1.5
Search and evaluation	PICO and Boolean; signs of a reliable source	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **1, 2, 3, 7, 11, 26, 27, 33, 34, 49**.

- **[R1]** *Rasahridaya Tantra*; **[R2]** *Rasarnava*; **[R3]** *Rasaratna Samuchchaya*; **[R7]** *Rasatarangini*; **[R11]** *Sharangadhara Samhita*. Classical texts named in the syllabus; cited here for their place in the history of Indian pharmacy. Editions are listed in the Principal Sources at the end of the book.
- **[R26]** Charaka Samhita: Sutra Sthana 1, 4, 9 and 11.
- **[R27]** *Remington: The Science and Practice of Pharmacy* (history and scope of pharmacy).
- **[R33]** CCRAS, *Database on Medicinal Plants Used in Ayurveda*.
- **[R34]** ICMR, *Quality Standards of Indian Medicinal Plants*.
- **[R49]** The syllabus entry "Brief information on the peer reviewed journals, official websites and other official search engines along with their links" is a topic, not a citation. Section B1.4 answers it.

Additional verified sources (accessed 16 October 2026)

- **[Mohanraj 2018]** Mohanraj K, Karthikeyan BS, Vivek-Ananth RP, et al. IMPPAT: a curated database of Indian Medicinal Plants, Phytochemistry And Therapeutics. *Sci Rep*. 2018;8:4329.
- **[UGC 2025]** University Grants Commission. Discontinuation of the UGC-CARE list and suggestive parameters for choosing peer-reviewed journals. February 2025.

Viva pointers — Chapter B1

Answer each in one or two sentences.

- Define pharmacy. What did the Greek word *pharmakon* mean?
- What are the four pillars of treatment? Where are they described?
- Name the four qualities of a drug and of an attendant according to Charaka.
- What is the Ebers papyrus? What is *De Materia Medica*?
- What are galenicals?
- Which edict separated pharmacy from medicine in Europe?
- Name the first official pharmacopoeia.
- Who is called the father of pharmacy education in India? Where did it begin?
- When were Ayurvedic drugs brought under the Drugs and Cosmetics Act?
- Give five fields of work for a postgraduate in Bhaishajya Kalpana.
- Distinguish primary, secondary and tertiary sources with one example each.
- Is a pharmacopoeia a primary or a tertiary source?
- What is PICO?
- What do the Boolean operators AND, OR and NOT do in a search?
- Which database should you use to check the accepted botanical name of a plant?
- What is Shodhganga? What is the AYUSH Research Portal?
- How do you recognise a predatory journal?
- What is Aptopadesha? How does it relate to evaluating sources?

PART B · PHARMACY MANAGEMENT · CHAPTER B2

B2 · PHARMACEUTICAL DOSAGE FORM DESIGN (PRE-FORMULATION)

SYLLABUS POINT (PART B, 2)

Pharmaceutical dosage form design (Pre-formulation)

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B2	Pharmaceutical Dosage Form Design (Pre-formulation)	2	3	Sep 2018, Sep 2020, Aug 2022, Jan 2024, Mar 2026

Why this chapter matters to you. Every Kalpana of Bhaishajya Kalpana is a dosage form, and every decision in its making (the solvent, the particle size, the binder, the heat, the container) is a decision of dosage form design. This chapter gives you the modern language of that design, **pre-formulation**, and shows how closely it matches the classical rules. It has been set in five of the seven papers, twice as a 20-mark question; the March 2026 paper asked you to *correlate classical concepts with modern pre-formulation parameters* (B2.5).

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define a dosage form, explain why dosage forms are needed, and classify modern and Ayurvedic dosage forms.
2. Describe the stages and considerations of dosage form design.
3. Define pre-formulation and describe its parameters, the methods used to study them and their significance.
4. Explain the special problems of pre-formulation of herbal and herbo-mineral products.
5. Correlate classical concepts of Bhaishajya Kalpana with modern pre-formulation parameters.
6. Describe the approaches to drug design, including reverse pharmacology.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	B, Q.3c	5	Pharmaceutical Dosage Form Design.
Sep 2020	B, Q.2	20	Pre-Formulation – Pharmaceutical dosage form design
Aug 2022	B, Q.3c	5	Designing of Pharmaceutical dosage forms.
Jan 2024	B, Q.3a	5	Drug designing.
Mar 2026	B, Q.2	20	Discuss pharmaceutical dosage form design with special reference to Pre-formulation Studies. Correlate classical concepts with modern pre-formulation parameters.

B2.1 Dosage forms

> A **dosage form** is the physical form in which a drug is presented for use: the drug (active ingredient) combined with other substances (excipients) and made into a product, such as a tablet, syrup, ointment or injection, that delivers an accurate dose by a chosen route, safely and acceptably.

A drug substance is seldom given as such. A few milligrams of a potent drug cannot be weighed by a patient; a bitter herb is hard to swallow; a volatile oil evaporates; a drug destroyed by stomach acid must be protected. The dosage form solves these problems.

B2.1.1 Why dosage forms are needed

Need	How the dosage form meets it	Classical example
Accurate dose	Divides the drug into measured units	Vati of fixed weight; Ratti-based dose of Bhasma
Protection from the environment	Coating, packing, low moisture	Churna kept in airtight containers; Ghrita protects lipid-soluble actives
Protection from the stomach	Enteric coating; delayed release	—
Masking of taste and odour	Sugar, honey, flavouring, capsules	Avaleha with sugar or jaggery; honey as Anupana
Convenience and acceptance	Easy to carry, swallow, store	Vati in place of Kwatha
Controlled or sustained release	Matrix, coating, depot	—
Delivery to a particular site	Topical, nasal, rectal, eye forms	Lepa, Nasya, Basti, Anjana
Stability and shelf life	Drying, preservatives, self-preservation by alcohol	Asava and Arishta preserved by self-generated alcohol; Bhasma as stable inorganic compounds
Combination of drugs	Several drugs in one unit	Polyherbal Vati and Churna

B2.1.2 Classification of dosage forms

Basis	Classes	Examples
Physical state	Solid; semi-solid; liquid; gaseous	Tablets, capsules, powders; ointments, creams, gels; solutions, syrups, suspensions, emulsions; aerosols, inhalations
Route	Oral; parenteral; topical; rectal; vaginal; nasal; ophthalmic; otic; inhalation (A1b)	—
Release	Immediate; modified (delayed, extended, sustained, controlled)	—
Ayurvedic	Primary Kalpana and their derivatives (below)	—

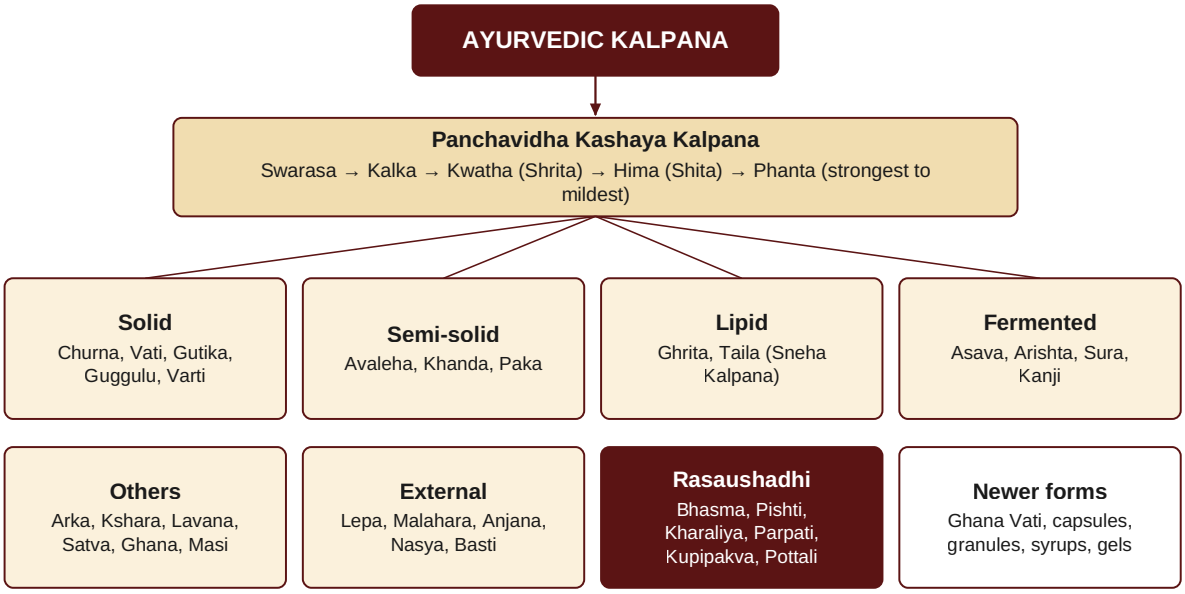


Figure B2.1. Classification of Ayurvedic dosage forms (Kalpana), from the five basic forms to their derivatives and Rasaushadhi.

Group	Kalpana	Modern equivalent
Panchavidha Kashaya Kalpana (five basic forms)	Swarasa (expressed juice), Kalka (paste), Kwatha or Shrita (decoction), Hima or Shita (cold infusion), Phanta (hot infusion)	Expressed juice; paste; decoction; cold and hot infusion
Solid oral	Churna (powder), Vati and Gutika (pills), Guggulu Kalpana, Varti (wick or suppository)	Powders, tablets, pills, suppositories
Semi-solid oral	Avaleha or Leha (confection), Khanda, Paka	Confections, electuaries, jams
Lipid-based	Sneha Kalpana: Ghrita, Taila	Medicated ghee and oils; lipid-based delivery
Fermented	Sandhana Kalpana: Asava, Arishta, Sura, Kanji, Shukta	Self-generated hydro-alcoholic preparations

Group	Kalpana	Modern equivalent
Others	Arka (distillate), Kshara (alkali), Lavana, Satva (extract), Masi (carbonised), Ghana (dried extract)	Aromatic waters; alkalis; extracts
External	Lepa, Malahara, Upanaha, Anjana, Nasya, Basti preparations	Pastes, ointments, poultices, eye preparations, nasal drops, enemas
Rasaushadhi	Bhasma, Pishti, Kharaliya, Parpati, Kupipakva, Pottali (A5.1.1)	Inorganic and organo-metallic preparations
Newer Ayurvedic forms	Ghana Vati and tablets from extracts, capsules, granules, syrups, gels, creams, nasal sprays	Modern dosage forms using classical formulations

B2.2 Dosage form design

> **Dosage form design** is the planned process of turning a drug substance into a product that delivers the right dose, by the right route, at the right rate, safely, stably and acceptably, and that can be manufactured reproducibly at scale.

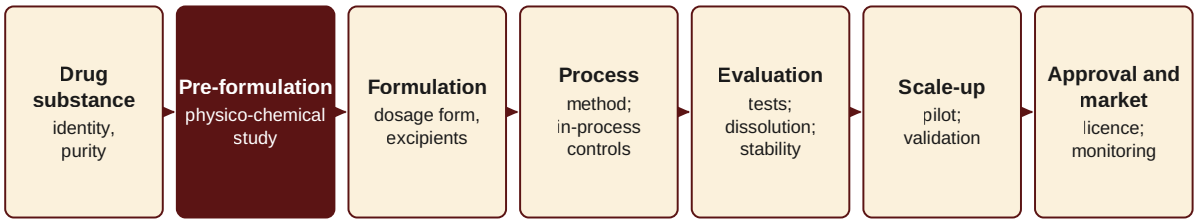


Figure B2.2. The stages of dosage form design, from drug substance to marketed product.

B2.2.1 Considerations in dosage form design

Consideration	Questions to ask
Therapeutic	Which disease? Local or systemic action? How fast should it act and for how long? Which route suits the patient (child, elderly, unconscious)?
Biopharmaceutical	Will the drug dissolve and be absorbed? Is it destroyed in the gut or by first-pass metabolism? (A1a.2, A1c.2)
Physico-chemical (pre-formulation)	Solubility, particle size, polymorphism, stability, hygroscopicity, compatibility with excipients (B2.3)
Manufacturing	Can it be made reproducibly at scale with available equipment? Flow, compressibility, mixing
Stability and packaging	How long will it last, and in what container and conditions? (Chapter B3)
Patient factors	Taste, size, frequency of dosing, cost, cultural acceptability
Regulatory	Pharmacopoeial standards; licensing requirements; labelling (Chapters B8, B10)

B2.2.2 The design sequence

1. **Pre-formulation**: characterise the drug substance (B2.3).
2. **Choice of dosage form and route**, from the therapeutic and pre-formulation data.
3. **Formulation development**: choose excipients and their amounts; make trial batches.
4. **Process development**: choose and optimise the manufacturing method.
5. **Evaluation**: in-process and finished product tests; dissolution; stability studies.
6. **Scale-up** from laboratory to pilot to production batch; process validation.
7. **Regulatory submission** and **post-marketing** monitoring of quality and safety.

B2.3 Pre-formulation

> **Pre-formulation** is the study of the physical and chemical properties of a drug substance, alone and in combination with excipients, carried out before formulation begins, in order to develop a stable, safe, effective and manufacturable dosage form.

B2.3.1 Objectives

1. To establish the **identity, purity and physico-chemical character** of the drug substance.
2. To find the properties that will decide **bioavailability** (solubility, dissolution, permeability).
3. To find the properties that will decide **stability** (degradation pathways and rates).
4. To find the properties that will decide **manufacture** (flow, compressibility, particle size).
5. To identify **compatible excipients** and packaging.
6. To guide the choice of **dosage form and route**, and to save time and money by avoiding failures later.

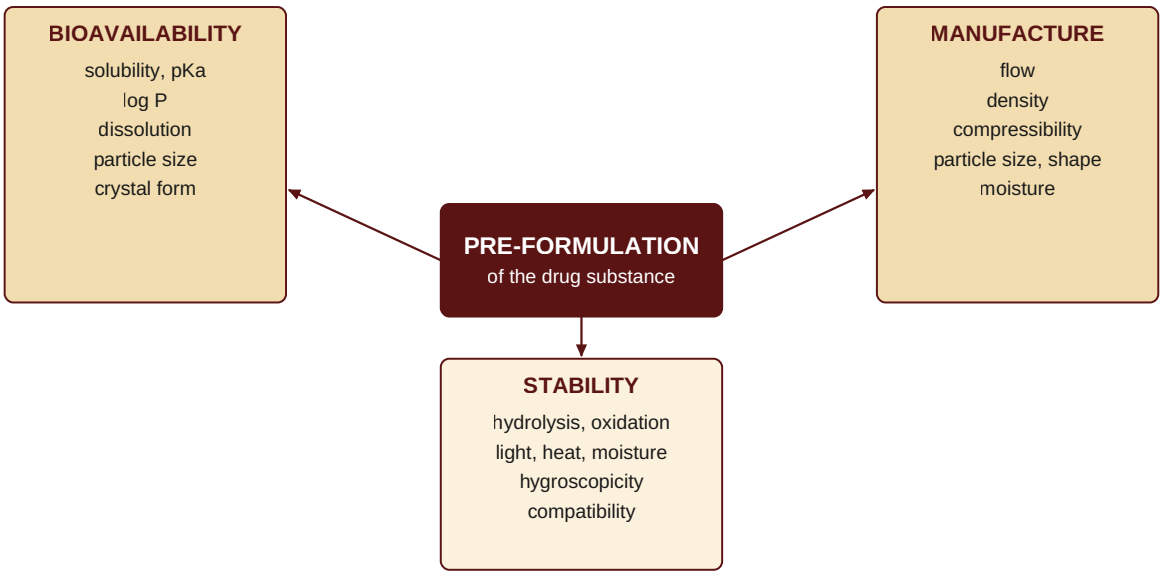


Figure B2.3. Pre-formulation parameters grouped by what they decide: bioavailability, stability and manufacturability.

B2.3.2 Parameters of pre-formulation

Group	Parameter	Methods	Why it matters
Organoleptic	Colour, odour, taste, appearance	Description	Identity; patient acceptance; change on storage signals degradation
Bulk properties	**Particle size and shape**; surface area	Sieving, microscopy, laser diffraction, light scattering, SEM; gas adsorption (BET) for surface area	Dissolution rate; content uniformity; flow; texture
	Flow: angle of repose, bulk and tapped density, Carr's index, Hausner ratio	Funnel method; tapped density tester	Uniform filling of dies and capsules; weight variation
	Crystal form: **polymorphism**, amorphous form, hydrates and solvates	X-ray diffraction, DSC, microscopy, FTIR	Solubility, dissolution, stability can differ between forms
	Hygroscopicity and moisture content	Weight gain at defined humidity; loss on drying; Karl Fischer titration	Caking, microbial growth, chemical degradation
	Melting point; thermal behaviour	DSC, thermogravimetry	Purity; polymorphs; processing temperatures
Solubility	**Solubility** in water and other solvents; pH–solubility profile; pKa	Shake-flask method; potentiometric titration	Absorption; choice of solvent and dosage form
	Partition coefficient (log P)	Octanol–water partition	Membrane permeability; lipid versus water solubility
	Dissolution rate; intrinsic dissolution	Dissolution apparatus; constant-surface disc	Rate-limiting step for poorly soluble drugs
Stability	Solid-state and solution stability; pH–rate profile; light, heat, oxygen, moisture	Stress (forced degradation) studies; HPLC, TLC	Shelf life; storage conditions; packaging
Compatibility	Drug–excipient interaction	Binary mixtures under stress; DSC, FTIR, HPLC	Choice of excipients
Biological	Permeability; first-pass metabolism	Cell models (Caco-2); animal studies	Route and dosage form
Microbiological	Microbial load	Pharmacopoeial microbial limit tests	Need for preservatives; safety

B2.3.3 Particle size

Smaller particles have a larger surface area, so they dissolve faster. The **Noyes–Whitney equation** describes this:

$$dC/dt = (D \times A \div h) \times (C_s - C)$$

where dC/dt is the rate of dissolution, D the diffusion coefficient, A the surface area of the particles, h the thickness of the diffusion layer, C_s the saturation solubility and C the concentration already in solution.

Reducing particle size increases **A**, and therefore the rate of dissolution. This is the scientific basis of fine grinding (Sukshma Churna) and of the repeated trituration of Bhasma and Pishti (B2.5). But very fine powders may flow poorly, become electrostatically charged and absorb more moisture.

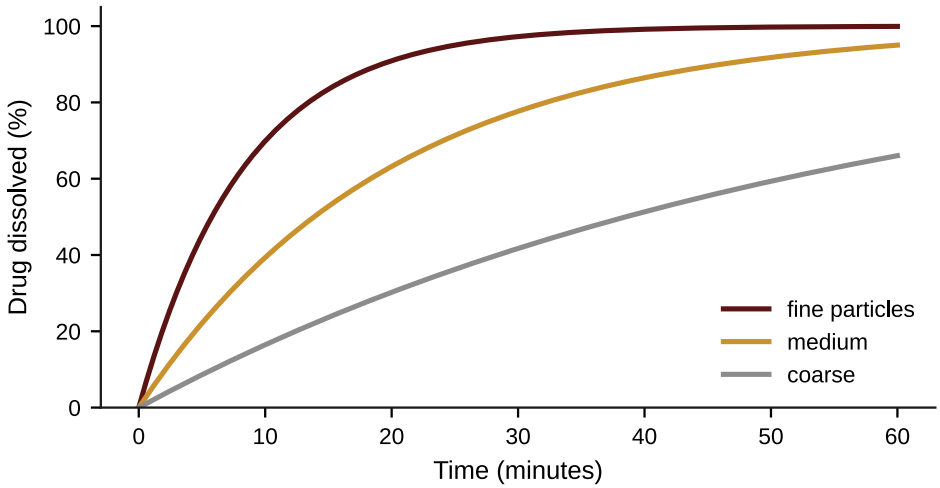


Figure B2.4. Effect of particle size on the rate of dissolution (schematic, illustrating the Noyes–Whitney relationship).

B2.3.4 Flow properties

Powders for Churna packing, capsule filling and tablet compression must flow evenly. Flow is assessed by three simple tests.

Test	Formula or method
Angle of repose (θ)	Powder poured through a funnel forms a cone; tan θ = height ÷ radius
Carr’s compressibility index (%)	(Tapped density − Bulk density) ÷ Tapped density × 100
Hausner ratio	Tapped density ÷ Bulk density

Flow character	Angle of repose (degrees)	Compressibility index (%)	Hausner ratio
Excellent	25–30	10 or less	1.00–1.11
Good	31–35	11–15	1.12–1.18
Fair	36–40	16–20	1.19–1.25
Passable	41–45	21–25	1.26–1.34
Poor	46–55	26–31	1.35–1.45
Very poor	56–65	32–37	1.46–1.59
Very, very poor	more than 66	more than 38	more than 1.60

Scale of flowability as given in the United States Pharmacopeia general chapter on powder flow [R40].

Many herbal Churnas, especially those rich in extracts, sugars or fibres, flow poorly. Remedies include granulation, glidants (colloidal silica, talc) and control of moisture.

B2.3.5 Polymorphism

A substance that can exist in more than one crystal form is **polymorphic**. Polymorphs have the same chemical composition but different arrangements of molecules, and so they may differ in solubility, dissolution rate, stability and bioavailability. An **amorphous** form has no crystal order and usually dissolves faster but is less stable.

Two classic examples show why this matters.

- **Chloramphenicol palmitate** exists in forms that differ greatly in bioavailability; only one form gives adequate blood levels.
- **Ritonavir**, an anti-HIV drug, unexpectedly appeared in a new, less soluble crystal form after marketing; the capsules failed dissolution and had to be reformulated.

For Rasaushadhi, the **crystal phase** (for example, metacinnabar in Kajjali against cinnabar in Rasa Sindura; zinc oxide in Jasada Bhasma) is the equivalent question, studied by X-ray diffraction (A5.3).

B2.3.6 Solubility

Solubility is the maximum amount of a substance that dissolves in a given amount of solvent at a stated temperature. Pharmacopoeias use descriptive terms.

Descriptive term	Parts of solvent needed to dissolve 1 part of solute
Very soluble	less than 1
Freely soluble	1 to 10
Soluble	10 to 30
Sparingly soluble	30 to 100
Slightly soluble	100 to 1000
Very slightly soluble	1000 to 10,000
Practically insoluble	10,000 or more

Descriptive terms as used in the Indian, British and United States Pharmacopoeias [R38–R40].

Factors affecting solubility: temperature; pH and pKa (weak acids dissolve better at high pH, weak bases at low pH, as described by the Henderson–Hasselbalch equation); particle size; crystal form; the solvent and co-solvents; complexation; surfactants.

Partition coefficient (P) is the ratio of the concentrations of a drug in octanol and in water at equilibrium; it is usually expressed as **log P**. A drug must be soluble enough in water to dissolve in the gut and soluble enough in lipid to cross membranes. Lipinski's rule of five (A1a.5) uses log P as one of its criteria.

B2.3.7 The Biopharmaceutics Classification System

The Biopharmaceutics Classification System (BCS) groups drugs by **solubility** and **intestinal permeability**, the two properties that chiefly decide oral absorption [Amidon 1995].

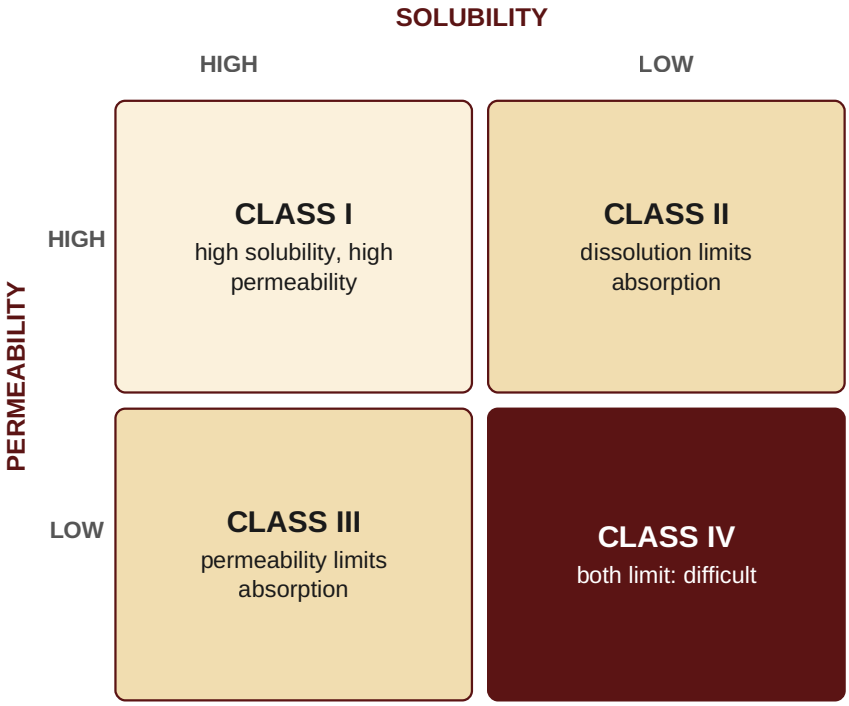


Figure B2.5. The Biopharmaceutics Classification System.

Class	Solubility	Permeability	Rate-limiting step	Formulation approach
I	High	High	Gastric emptying	Simple immediate-release forms
II	Low	High	**Dissolution**	Particle size reduction; solid dispersions; lipid systems
III	High	Low	**Permeability**	Permeation enhancers; ensure release in the absorptive region
IV	Low	Low	Both	Difficult; novel delivery systems

Curcumin, poorly soluble in water and extensively metabolised, is a well-known example of a natural product with poor oral bioavailability. The classical practice of giving Haridra with fat or with Maricha is discussed in A1a.6.

B2.3.8 Stability

Drugs degrade chiefly by **hydrolysis** (esters, amides), **oxidation** (unsaturated oils, phenols), **photolysis** (light-sensitive compounds) and, in solids, by moisture-driven reactions and physical changes (caking, colour change, polymorphic change).

The effect of temperature on the rate of degradation is described by the **Arrhenius equation**:

$$k = A \times e^{-Ea/RT} \text{ or } \log k = \log A - Ea \div (2.303\ RT)$$

where k is the rate constant, A is the frequency factor, Ea the activation energy, R the gas constant and T the absolute temperature.

Because the rate rises with temperature in a predictable way, products can be stored at raised temperature and humidity to **predict** shelf life (accelerated stability testing). Real-time studies at the intended storage conditions confirm it. Stability testing conditions and shelf life are discussed further in Chapter B3.

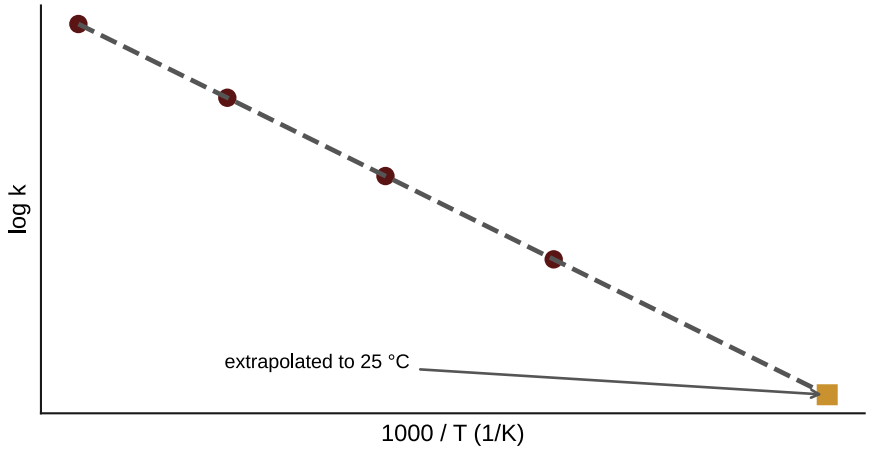


Figure B2.6. Accelerated stability testing: the logarithm of the degradation rate constant plotted against the reciprocal of temperature gives a straight line that can be extrapolated to room temperature (schematic).

B2.3.9 Drug–excipient compatibility

Excipients are not inert in every case. Mixtures of the drug with each proposed excipient are stored under stress and examined by DSC, FTIR and chromatography. Well-known incompatibilities include:

- **lactose with primary amines** (the Maillard browning reaction);
- **magnesium stearate with aspirin** (hydrolysis);
- **peroxide impurities in povidone** oxidising sensitive drugs;
- in herbal products, **tannins with iron salts** (dark complexes) and with alkaloids (precipitates).

B2.3.10 Pre-formulation of herbal and herbo-mineral products

A single chemical entity can be characterised completely. A herbal or herbo-mineral product cannot, and pre-formulation must be adapted.

Problem	Pre-formulation approach
Many constituents, often unknown	Identity by pharmacognosy and chromatographic fingerprint (HPTLC, HPLC); **marker compounds** for assay
Natural variation between batches	Standardised raw materials (Grahya lakshana, pharmacopoeial standards); fixed extract ratio
Hygroscopic, sticky extracts (Ghana)	Moisture studies; drying; adsorbents; granulation; moisture-proof packing
Poor flow of Churna	Particle size control; granulation; glidants
Bitter or unpleasant taste	Taste masking: coating, sweeteners, honey or jaggery bases
Volatile constituents	Low processing temperatures; closed containers
Microbial load	Microbial limit tests; hygiene; preservatives where allowed
Contaminants	Heavy metals, pesticide residues, aflatoxins (Chapter B10)
Interactions within the formula	Stability of the combined product, not only of each herb
Bhasma and other Rasaushadhi	Crystal phase (XRD), particle size (SEM, laser diffraction), elemental content, free metal; compatibility with Anupana (honey, ghee)

B2.4 Dosage form design in Bhaishajya Kalpana

The classical texts do not use the word “pre-formulation”, but they apply its logic at every step: selecting raw material, choosing the form that suits the drug and the patient, controlling the process, testing the end point and fixing the period of potency.

B2.4.1 Why there are many Kalpana

The **Panchavidha Kashaya Kalpana** are arranged in order of strength (Guru to Laghu): Swarasa, Kalka, Shrita (Kwatha), Shita (Hima), Phanta (Charaka Samhita, Sutra Sthana 4) [R26]. The choice among them depends on the strength of the disease and of the patient. This is dosage form selection on therapeutic and patient grounds.

The derivative Kalpana solve specific pharmaceutical problems:

Problem	Classical solution	Modern principle
Swarasa and Kwatha spoil quickly	Avaleha (concentration with sugar), Sandhana (fermentation to alcohol), Ghana (drying to a solid extract)	Preservation by low water activity, by alcohol, by drying
Lipid-soluble actives not extracted by water	**Sneha Kalpana**: herbs processed in ghee or oil with water-based media	Solvent selection by solubility; lipid-based delivery
Water-soluble and alcohol-soluble actives	**Asava, Arishta**: hydro-alcoholic extraction by self-generated alcohol	Co-solvent extraction; preservation
Volatile actives lost on boiling	**Hima** (cold infusion), **Phanta** (hot infusion, short contact), **Arka** (distillation)	Temperature control for thermolabile and volatile constituents

Problem	Classical solution	Modern principle
Bulky, bitter Churna	**Vati** with binders (honey, jaggery, Guggulu); coating with sugar or silver foil	Tablets; binders; coating; taste masking
Unstable, toxic metals	**Shodhana and Marana** to Bhasma; Kupipakva	Conversion to stable compounds of defined phase
Need for local action	Lepa, Nasya, Basti, Anjana, Varti	Topical and site-specific dosage forms

B2.5 Classical concepts and modern pre-formulation parameters

This section answers the March 2026 question directly.

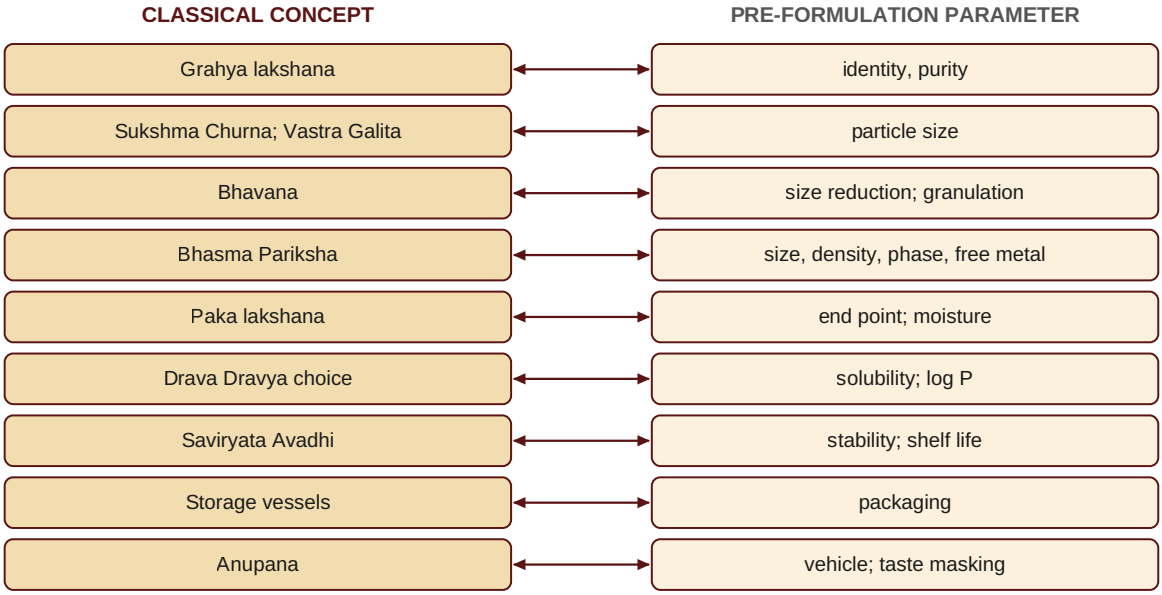


Figure B2.7. Classical concepts of Bhaishajya Kalpana mapped to modern pre-formulation parameters.

Classical concept	What the text requires	Modern pre-formulation parameter	Comment
Grahya and Agrahya lakshana; collection rules (season, part, age of plant)	Use only authentic, mature, undamaged material collected at the right time	Identity, purity, assay; control of natural variation	Pharmacognosy and markers give a measurable version of the same rule
Shodhana of raw materials	Purify and detoxify before use	Purity; removal of impurities; change in chemistry	Shown for some materials; must be measured case by case (A5)
Sukshma Churna; filtering through cloth (Vastra Galita)	Powder must be fine and uniform	**Particle size** and size distribution	Noyes–Whitney: finer particles dissolve faster
Bhavana (trituration with liquids)	Repeated wet grinding with juices or decoctions	Particle size reduction; wet granulation; incorporation of actives; surface modification	Effects on composition and particle size have been measured for some Bhasmas

Classical concept	What the text requires	Modern pre-formulation parameter	Comment
Bhasma Pariksha: Rekhapurnata, Varitara, Nishchandrata, Apunarbhava, Niruttha	Fine, light, lustre-free, no reversion to metal	Particle size; density; crystal phase; absence of free metal	Qualitative versions of size, density and phase tests (A4.10.2)
Paka of Sneha (Mridu, Madhya, Khara) and its signs (Sneha Siddhi Lakshana)	Stop heating at the right stage	Process end point; **moisture content**; extraction efficiency	Measurable by moisture, acid value and chromatographic profile
Avaleha Paka lakshana (thread formation, sinking in water, not spreading)	Correct consistency	Viscosity; total solids; water activity	Same aim as modern in-process controls
Sandhana time and signs of completion	Fermentation complete; sealed vessel	Alcohol content; pH; total sugars	Self-generated alcohol content is a pharmacopoeial parameter (Chapter B10)
Choice of **Drava Dravya** (water, milk, oil, alcohol)	Medium suited to the drug	**Solubility**; partition between water and lipid	Kwatha for water-soluble, Sneha for lipid-soluble constituents
Saviryata Avadhi (period of potency) for each Kalpana	Use within the stated period [R11]	**Stability**; shelf life	Legal shelf life of ASU drugs is now laid down in the Drugs and Cosmetics Rules (Chapter B8)
Containers: glass, earthen, metal; airtight storage; cool, dry place	Proper storage	Packaging; storage conditions; protection from moisture and light	Chapter B3
Anupana (vehicle)	Medium with which the drug is taken	Vehicle; solubility; taste masking; possible absorption effects	Plausible; rarely measured
Matra (dose) by Kala, Agni, Vaya, Bala	Dose suited to patient	Dose and dosage regimen; patient factors	A1a.4, A1c.1
Prakriti, Agni, Koshtha	Patient-based choice of Kalpana	Biopharmaceutical and patient factors in dosage form choice	Parallel in principle

Established. The classical rules aim at the same goals as modern pre-formulation: identity, purity, suitable particle size, suitable solvent, controlled process, stability and proper storage. Several classical end points correspond to measurable parameters.

Not established. That the classical tests alone are sufficient to guarantee uniform quality, bioavailability or stability of a modern commercial batch. For that, the classical signs need to be supported by measured values, which is the work of standardisation (Chapter B10).

B2.6 Drug design

The January 2024 paper asked for a note on **drug designing**.

> **Drug design** is the inventive process of finding new medicines on the basis of knowledge of a biological target or of molecules known to act on it.

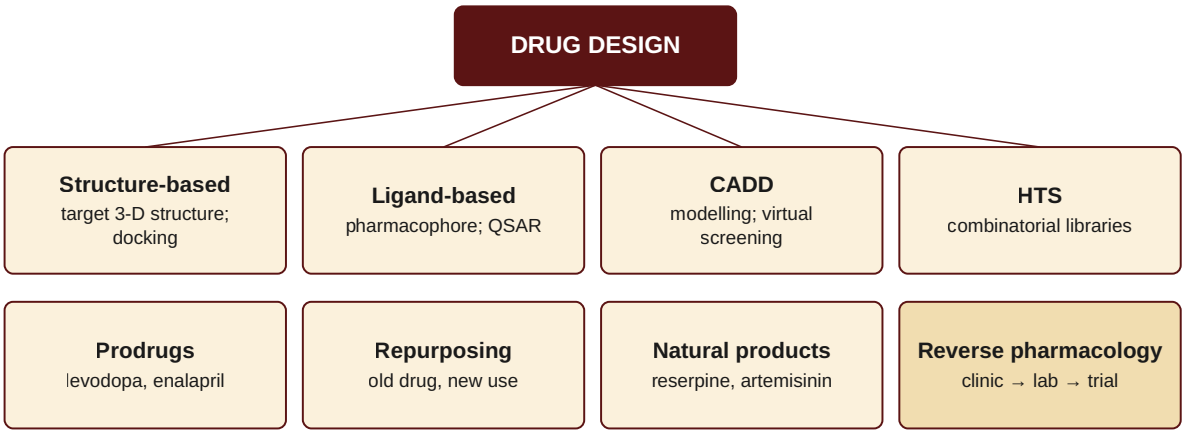


Figure B2.8. Approaches to drug design and discovery.

Approach	Principle	Example
Structure-based (target-based) design	The three-dimensional structure of the target (enzyme, receptor) is known; molecules are designed or docked to fit its binding site	HIV protease inhibitors
Ligand-based design	The target structure is unknown, but active molecules are known; their common features (pharmacophore) and quantitative structure–activity relationships (**QSAR**) guide new designs	Analogues of a natural lead
Lead from natural peptides	A natural molecule is modified into a drug	Captopril, designed from a peptide in snake venom
Computer-aided drug design (CADD)	Molecular modelling, docking, virtual screening, molecular dynamics	Widely used to screen libraries before synthesis
Combinatorial chemistry and **high-throughput screening**	Large libraries of compounds tested rapidly against a target	—
Prodrug design	An inactive form converted to the active drug in the body, to improve absorption, stability or targeting	Levodopa, enalapril
Bioisosterism	Replacing a group with another of similar properties to improve the drug	—
Drug repurposing	Finding new uses for existing drugs	Thalidomide for multiple myeloma
Natural product leads	Active principles from plants and other organisms	Reserpine from Sarpagandha; artemisinin from *Artemisia annua*

Approach	Principle	Example
Reverse pharmacology	Starts from documented clinical experience (for example, in Ayurveda), then moves to experimental studies and controlled trials to confirm safety and efficacy and identify actives [Patwardhan 2010]	Programmes built on classical Ayurvedic formulations
Network pharmacology	Studies how many constituents act on many targets together	Used to study polyherbal formulations

Rule of five and drug-likeness. Designed molecules are screened for properties that favour oral absorption: Lipinski's rule of five (molecular weight, log P, hydrogen bond donors and acceptors) (A1a.5).

Reverse pharmacology and Ayurveda. In the conventional path, a molecule goes from the laboratory to the clinic. In reverse pharmacology, a remedy with a long history of human use goes from the clinic (documented experience) back to the laboratory and then forward to formal trials. Its advantage is that safety and efficacy signals come from human use from the start; its risk is that the "experience" may be anecdotal and must be tested rigorously [Patwardhan 2010].

AYURVEDA CORRELATION: YUKTI IN PHARMACEUTICS

Charaka holds that the success of a drug depends on its right use (Yukti): the right substance, in the right form and dose, for the right patient (Charaka Samhita, Sutra Sthana 1 and 2) [R26]. Sharangadhara gave the doses, methods and periods of potency of each Kalpana in a systematic form [R11]. Modern dosage form design is Yukti made measurable.

Established. The logic of classical Kalpana design matches modern pharmaceutics at many points (B2.4, B2.5).

Not established. That a classical Kalpana is always the optimal dosage form for a given drug. Each comparison (for example, Kwatha against a tablet made from its extract) must be tested for equivalent composition and effect.

KEY POINTS TO REMEMBER

- Dosage form: the drug with excipients in a usable form that delivers an accurate dose by a chosen route, safely and acceptably.
- Panchavidha Kashaya Kalpana: Swarasa, Kalka, Kwatha, Hima, Phanta (strongest to mildest); derivatives solve specific pharmaceutical problems.
- Design considerations: therapeutic, biopharmaceutical, physico-chemical, manufacturing, stability and packaging, patient, regulatory.
- Pre-formulation: study of the drug substance's physical and chemical properties, alone and with excipients, before formulation.
- Parameters: organoleptic; particle size; flow (angle of repose, Carr's index, Hausner ratio); polymorphism; hygroscopicity; solubility, pKa, log P; dissolution; stability; compatibility.
- Noyes–Whitney: dissolution rate rises with surface area. Arrhenius: degradation rate rises with temperature; basis of accelerated stability.
- Carr's index ≤ 10 and Hausner ratio ≤ 1.11: excellent flow. Angle of repose 25–30 degrees: excellent.

- BCS: I high solubility/high permeability; II low/high; III high/low; IV low/low.
- Polymorphism: chloramphenicol palmitate; ritonavir. For Rasaushadhi, crystal phase by XRD.
- Herbal pre-formulation: fingerprint and markers; hygroscopic extracts; poor flow; taste; contaminants; for Bhasma, phase, size and free metal.
- Classical correlations: Grahya lakshana (identity), Sukshma Churna (particle size), Bhavana (granulation, size reduction), Paka lakshana (end point, moisture), Drava Dravya (solubility), Saviryata Avadhi (stability), Anupana (vehicle).
- Drug design: structure-based, ligand-based (QSAR), CADD, HTS, prodrugs, repurposing, natural products, reverse pharmacology.

EXAM CORNER — CHAPTER B2

Likely long questions (20 marks)

1. Pre-formulation: pharmaceutical dosage form design *(set in September 2020)*.
2. Discuss pharmaceutical dosage form design with special reference to pre-formulation studies. Correlate classical concepts with modern pre-formulation parameters *(set in March 2026)*.
3. Classify dosage forms. Describe the considerations in designing a dosage form, with examples from Bhaishajya Kalpana.

Likely short notes (5 marks)

1. Pharmaceutical dosage form design *(set in September 2018)*.
2. Designing of pharmaceutical dosage forms *(set in August 2022)*.
3. Drug designing *(set in January 2024)*.
4. Flow properties of powders.
5. Polymorphism.
6. Biopharmaceutics Classification System.
7. Noyes–Whitney equation.
8. Accelerated stability testing.
9. Drug–excipient compatibility.
10. Reverse pharmacology.

How to attempt

- Define dosage form, dosage form design and pre-formulation in the first lines.
- Draw the design flow (Figure B2.2) and the parameter map (Figure B2.3).
- Give the parameter table, then write a paragraph each on particle size, flow (with the scale), polymorphism, solubility (with the descriptive terms) and stability.
- For the March 2026 type of question, the correlation table (B2.5) is the core of the answer; close with the Established / Not established lines.

Answer frame: "Discuss pharmaceutical dosage form design with special reference to pre-formulation. Correlate classical concepts with modern pre-formulation parameters." (20 marks)

Part	What to write	Suggested marks
Definitions	Dosage form; dosage form design; pre-formulation	2
Need and classification	Why dosage forms; modern and Ayurvedic classification	2
Design	Considerations table; stages figure	3
Pre-formulation	Objectives; parameter table; particle size (Noyes–Whitney); flow (scale); polymorphism; solubility; stability (Arrhenius); compatibility	7
Classical correlation	Table B2.5 with at least eight pairs	5
Conclusion	Evidence boundary; need for standardisation	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **11, 26, 27, 28, 38, 39, 40**.

- **[R11]** *Sharangadhara Samhita* (dosage forms; Saviryata Avadhi).
- **[R26]** Charaka Samhita: Sutra Sthana 1, 2 and 4.
- **[R27]** *Remington: The Science and Practice of Pharmacy* (dosage forms; pre-formulation; stability).
- **[R28]** Lachman L, Lieberman HA, Kanig JL. *The Theory and Practice of Industrial Pharmacy* (pre-formulation; tablets; stability).
- **[R38]** *Indian Pharmacopoeia*; **[R39]** *British Pharmacopoeia*; **[R40]** *United States Pharmacopeia* (solubility terms; powder flow).

Additional verified sources (accessed 16 October 2026)

- **[Amidon 1995]** Amidon GL, Lennernäs H, Shah VP, Crison JR. A theoretical basis for a biopharmaceutic drug classification: the correlation of in vitro drug product dissolution and in vivo bioavailability. *Pharm Res*. 1995;12(3):413–420.
- **[Patwardhan 2010]** Patwardhan B, Vaidya ADB. Natural products drug discovery: accelerating the clinical candidate development using reverse pharmacology approaches. *Indian J Exp Biol*. 2010;48(3):220–227.

Viva pointers — Chapter B2

Answer each in one or two sentences.

- Define a dosage form. Why can a drug not usually be given as such?
- Name the Panchavidha Kashaya Kalpana in order of strength.
- Why was Sandhana Kalpana developed? Why Sneha Kalpana?
- What is pre-formulation? Name five of its parameters.
- State the Noyes–Whitney equation. Which term does fine grinding change?
- How do you measure the angle of repose? What value means excellent flow?
- Define Carr’s index and Hausner ratio.
- What is polymorphism? Give one example where it mattered clinically.
- How would you identify the crystal phase of a Bhasma?
- What does “sparingly soluble” mean?
- What is log P? Why does it matter?
- Name the four BCS classes. In which class is dissolution rate-limiting?
- State the Arrhenius equation. How is it used in stability testing?
- Name two well-known drug–excipient incompatibilities.
- Why is pre-formulation of a polyherbal product harder than that of a single chemical?
- What is the modern parallel of Sneha Siddhi Lakshana?
- What is the modern parallel of Vastra Galita Churna?
- What is Saviryata Avadhi?
- Distinguish structure-based from ligand-based drug design.
- What is QSAR?
- What is a prodrug? Give an example.
- What is reverse pharmacology?

PART B · PHARMACY MANAGEMENT · CHAPTER B3

B3 · PACKAGING MATERIALS AND LABELLING

SYLLABUS POINT (PART B, 3)

Packaging materials and Labeling

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B3	Packaging Materials and Labelling	0	4	Sep 2018, Sep 2020, Aug 2021, Mar 2026

Why this chapter matters to you. A well-made medicine can be ruined by a poor container, and a correct medicine can harm if its label is wrong. Packaging and labelling are the last steps of manufacture and the first thing the patient sees. This chapter was set in four of the seven papers as a short note: on packaging materials, on the contents of an ideal label, on the rules for packaging and labelling of Ayurvedic drugs, and, in March 2026, on the shelf life and storage of herbo-mineral preparations.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define packaging and describe its functions and levels.
2. Describe the main packaging materials (glass, plastics, metals, paper and board, rubber, laminates), their merits, limitations and tests.
3. Select suitable packaging for each Ayurvedic dosage form.
4. Explain shelf life, stability testing and storage conditions, with special reference to herbo-mineral preparations.
5. State the contents of an ideal label and the legal requirements for labelling and shelf life of Ayurvedic drugs.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	B, Q.3a	5	Examples of Packaging material and contents of an Ideal Label.
Sep 2020	B, Q.3c	5	Packaging Materials and Labeling
Aug 2021	B, Q.3B	5	Discuss regarding the rules pertaining to Packaging and Labelling of Ayurvedic Drugs
Mar 2026	B, Q.3a	5	Explain the significance of shelf-life and storage conditions for herbo-mineral preparations.

B3.1 Packaging

> **Packaging** is the science, art and technology of enclosing or protecting products for distribution, storage, sale and use. For medicines, the package must also keep the product stable, identifiable and safe until it is used.

B3.1.1 Functions of a pharmaceutical package

Function	Meaning	Example
Containment	Holds the product without leakage or loss	Leak-proof bottle for Asava
Protection	Against moisture, oxygen, light, heat, microbes, insects and mechanical damage	Moisture-proof container for hygroscopic Churna; amber glass for light-sensitive products
Preservation	Keeps the product stable for its shelf life	Airtight packing of Bhasma
Identification and information	Label: name, composition, dose, batch, dates, cautions	Chapter B3.5
Presentation	Attractive, recognisable appearance	Cartons; brand design
Convenience and compliance	Easy to open, use, dose and carry; calendar packs	Strip packs of Vati
Safety	Tamper evidence; child-resistant closures	Sealed caps

B3.1.2 Levels of packaging

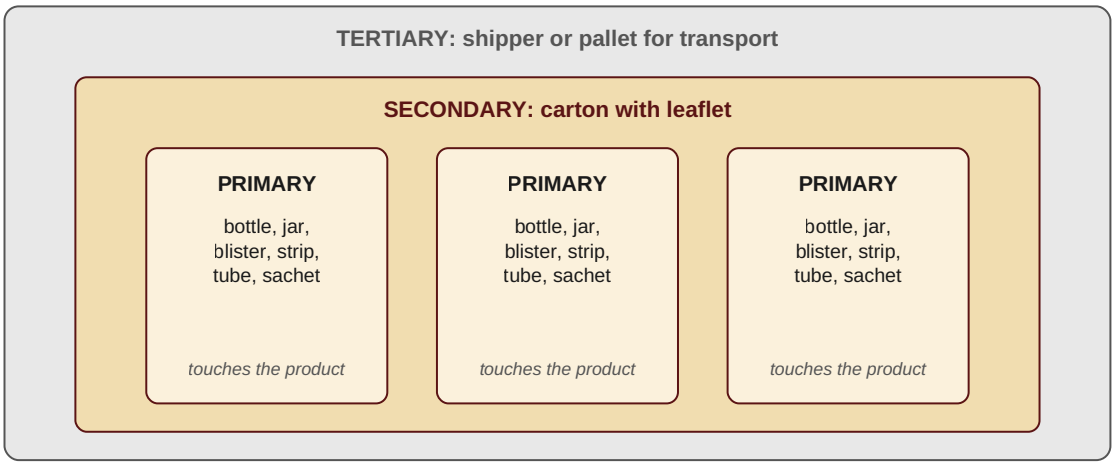


Figure B3.1. Primary, secondary and tertiary packaging.

Level	Definition	Examples
Primary	In direct contact with the product	Bottle, jar, blister, strip, tube, sachet
Secondary	Encloses one or more primary packs	Carton with leaflet
Tertiary	For bulk handling, storage and transport	Shipper (corrugated box), pallet

The primary package matters most for stability, because it is in contact with the product. It must not react with, absorb from, or release substances into the product.

B3.2 Packaging materials

B3.2.1 Glass

Glass is chemically resistant, impermeable to gases and moisture, transparent, easy to clean and to sterilise, and can be coloured (amber) to protect from light. Its disadvantages are weight and fragility, and the release of alkali from some types.

Type (pharmacopoeial)	Composition	Use
Type I	Borosilicate glass: high resistance to water attack	Parenteral and sensitive preparations
Type II	Soda-lime glass, surface-treated to reduce alkali release	Some aqueous preparations
Type III	Ordinary soda-lime glass	Oral liquids, powders, Asava and Arishta, oils
Non-parenteral (NP)	General-purpose soda-lime glass	Oral and topical products

Tests: hydrolytic resistance (powdered glass test; surface test, which distinguishes the types), light transmission for coloured glass, arsenic (in some pharmacopoeias), and physical tests (thermal shock, internal pressure) [R38].

Amber glass absorbs ultraviolet and blue light and protects light-sensitive products.

B3.2.2 Plastics

Plastic	Properties	Uses
Low-density polyethylene (LDPE)	Flexible; good moisture barrier; poor oxygen barrier	Squeeze bottles, liners, pouches
High-density polyethylene (HDPE)	Rigid; good moisture barrier; opaque	Jars and bottles for tablets, Churna, Bhasma
Polypropylene (PP)	Rigid; heat resistant; good chemical resistance	Caps, closures, containers
Polyvinyl chloride (PVC)	Clear; easily formed; moderate barrier	Blister packs
Polyvinylidene chloride (PVDC) coating	Very good moisture and oxygen barrier	Coating on PVC blisters for hygroscopic products
Polyethylene terephthalate (PET)	Clear, strong, good gas barrier	Bottles for syrups and oral liquids
Polystyrene	Clear, rigid, brittle	Some jars

Problems with plastics: **permeation** (gases and moisture pass slowly through the wall), **leaching** (additives, plasticisers or monomers pass into the product), **sorption** (the product's constituents, especially volatile oils and preservatives, are absorbed into the plastic), and **deformation** by heat or solvents. Volatile oils and alcohol-containing products need careful choice of plastic, or glass.

Tests: leakage, water vapour permeability, light transmission, collapsibility (for squeeze containers), and extractables and biological tests as required by the pharmacopoeia.

B3.2.3 Metals

Metal	Uses	Notes
Aluminium	Collapsible tubes for ointments and creams; foils for strips and blisters (Alu-Alu packs); caps	Excellent barrier to moisture, gas and light; tubes may be lacquered inside
Tin plate (tinned steel)	Containers for oils and ghee in bulk; aerosol cans	Lacquered to prevent reaction
Stainless steel	Bulk storage and processing vessels	Not usually a retail pack

B3.2.4 Paper and board

Paper and paperboard are used for labels, leaflets, cartons and corrugated shippers. They give no barrier to moisture unless laminated or coated, so they serve as secondary and tertiary packaging.

B3.2.5 Rubber and elastomers

Used for stoppers, closures and droppers. They can absorb preservatives and release substances (leachables), so they are tested for compatibility.

B3.2.6 Laminates, strips and blisters

Laminates combine layers (paper, aluminium foil, plastic films) to unite their barrier properties: for example, sachets of paper/foil/polyethylene for Churna and granules. **Strip packs** seal each tablet between two layers of foil or laminate. **Blister packs** hold each unit in a formed cavity (PVC, PVC/PVDC or cold-formed aluminium) sealed with lidding foil; Alu-Alu blisters give the best moisture protection.

B3.2.7 Closures

The closure is part of the container system. Screw caps (with liners), snap caps, child-resistant caps, tamper-evident seals and induction seals are used. The liner must be compatible with the product and give a tight seal.

B3.2.8 Traditional containers

The classical texts direct the storage of preparations in suitable vessels of glass, earthenware, metal, horn or wood, closed and kept in a protected store; Sandhana preparations are fermented and stored in earthen or wooden vessels or in glazed jars sealed during fermentation. These choices reflect the same aims as modern packaging: chemical inertness, protection from moisture, air and light, and prevention of contamination. Modern GMP prefers inert, cleanable and impermeable materials (glass, food-grade plastics, stainless steel) for these purposes.

B3.3 Selecting packaging for Ayurvedic dosage forms

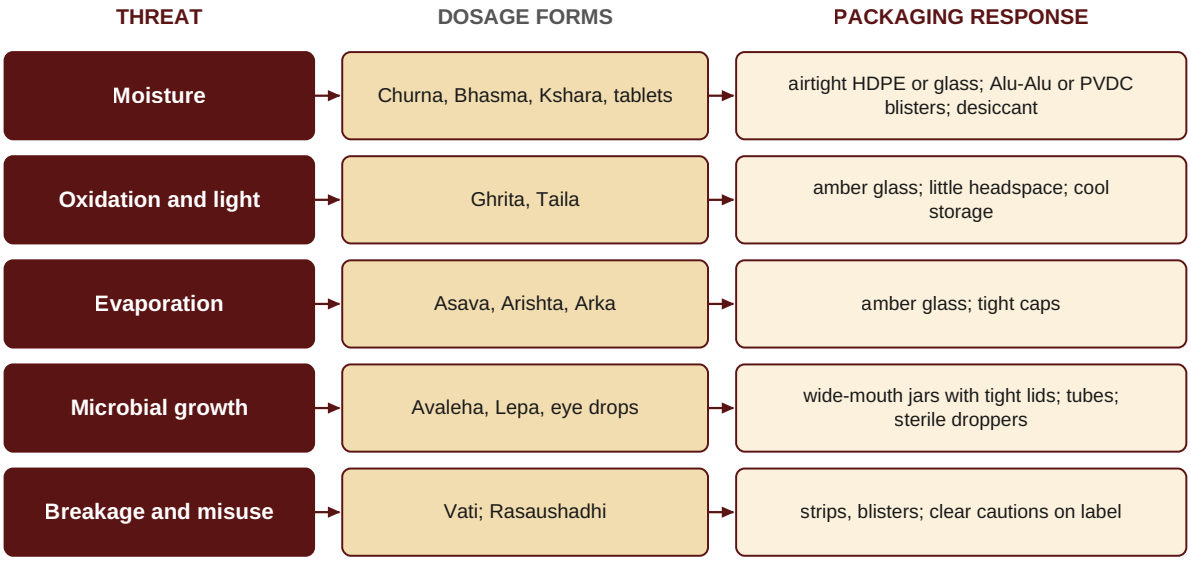


Figure B3.2. Choice of primary packaging for Ayurvedic dosage forms, by the main threat to quality.

Dosage form	Main threats	Suitable primary pack
Churna	Moisture (caking, microbial growth), loss of volatile oils, insects	HDPE or glass jars with tight caps and liners; laminated pouches; desiccant where needed
Vati, Gutika, tablets	Moisture, breakage	HDPE bottles; strip or blister packs (PVC/PVDC or Alu-Alu for hygroscopic tablets)

Dosage form	Main threats	Suitable primary pack
Guggulu preparations	Moisture, sticking	HDPE or glass containers; blisters
Avaleha	Moisture uptake, fermentation, microbial growth	Wide-mouth glass or PET jars with tight lids
Asava, Arishta	Evaporation of alcohol, light, leakage	Amber glass bottles with tight caps; PET only if compatible with alcohol and constituents
Ghrita, Taila	Oxidation (rancidity), light, heat	Glass (amber) or compatible food-grade containers; tin for bulk; filled to reduce headspace
Bhasma, Pishti	Moisture	Airtight glass or HDPE containers; sachets of laminate
Kupipakva and other Rasaushadhi	Moisture; misuse	Airtight glass or HDPE containers; clear labelling and cautions
Kshara, Lavana	Moisture (deliquescent)	Airtight glass with moisture-proof closures; desiccant
Arka	Loss of volatile constituents	Glass bottles, tightly closed
Lepa, Malahara (ointments, creams)	Drying, contamination	Collapsible aluminium or laminated tubes; jars
Eye and nasal preparations	Microbial contamination	Sterile dropper bottles as required

B3.4 Shelf life, stability and storage

> **Shelf life** is the period during which a product, stored under the conditions stated on its label, remains within its specification for identity, strength, quality and purity. The **date of expiry** marks the end of the shelf life; after it, the product must not be sold or used.

B3.4.1 Factors that limit shelf life

Factor	Effect	Most affected Ayurvedic forms
Moisture	Caking, microbial growth, hydrolysis, loss of actives	Churna, Kshara, Lavana, herbal Vati, Ghana
Oxygen	Oxidation, rancidity	Ghrita, Taila
Light	Photodegradation; colour change	Asava, Arishta, Taila, coloured preparations
Heat	Faster reactions (Arrhenius, B2.3.8); evaporation	Products with volatile oils; Arka
Microbes	Spoilage; toxins	Avaleha, Swarasa, Kwatha, water-based liquids
Insects	Infestation	Churna, crude drugs
Interaction with the container	Leaching, sorption	Products in unsuitable plastics

B3.4.2 Stability testing

Real-time (long-term) studies keep the product under the intended storage conditions and test it at intervals until it fails. **Accelerated studies** keep it at raised temperature and humidity to predict the shelf life sooner. International guidelines give the standard conditions: long-term at 25 °C/60% RH, 30 °C/65% RH or 30 °C/75% RH according to the climatic zone, and accelerated at 40 °C ± 2 °C/75% RH ± 5% RH [WHO 2018].

For ASU drugs, **Rule 161B** of the Drugs and Cosmetics Rules requires that applicants for a manufacturing licence submit shelf-life data based on **real-time stability studies**, following the stability guidelines given in the *Ayurvedic Pharmacopoeia of India*, Part I, Volume VIII [Rule 161B]. Under the 2026 amendment, accelerated studies of three or six months may support an initial shelf life of one or two years, with real-time data to follow [Corpseed 2026; Lexplosion 2026].

Tests during a stability study of an Ayurvedic product include organoleptic properties, physico-chemical parameters (pH, loss on drying, ash, extractives, viscosity, alcohol content as applicable), chromatographic fingerprint and marker content, and microbial load.

B3.4.3 Shelf life of ASU drugs under the law

Rule 161B(8) gives the shelf life of each category of Ayurvedic medicine, counted from the date of manufacture, unless scientific data justify a different period [Rule 161B].

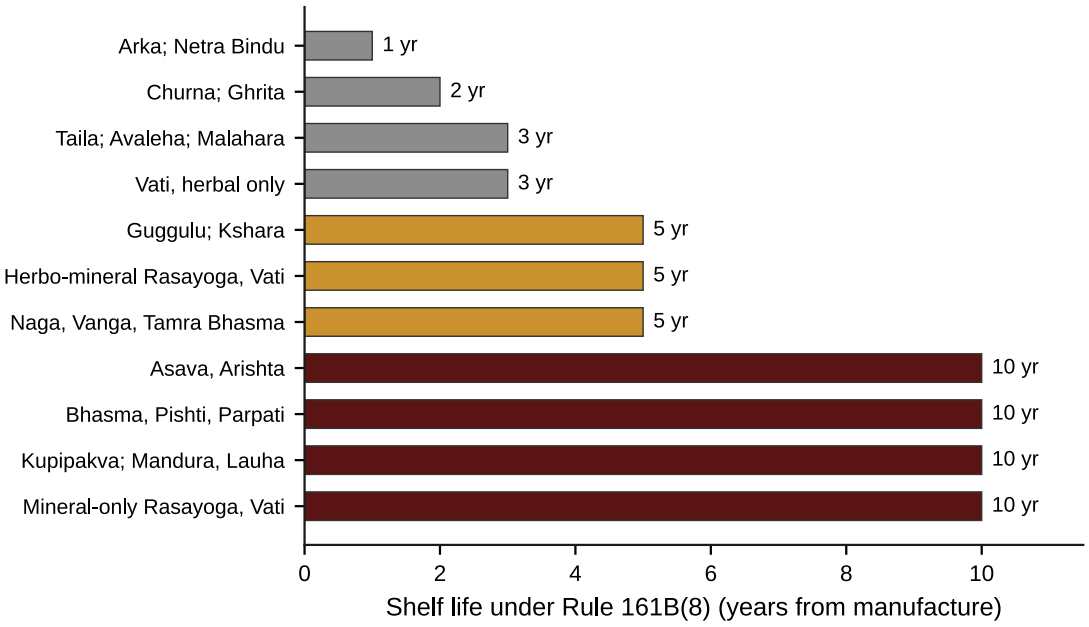


Figure B3.3. Shelf life of selected categories of Ayurvedic medicines under Rule 161B(8). Herbo-mineral and mineral preparations have the longest periods, except those containing Naga, Vanga and Tamra Bhasma.

Category (Ayurveda)	Shelf life
Anjana: from Kashthaushadhi only	1 year
Anjana: Kashthaushadhi with Rasa, Uparasa or Bhasma	2 years

Category (Ayurveda)	Shelf life
Anjana: only Rasa, Uparasa or Bhasma	3 years
Arka	1 year
Asava and Arishta	**10 years**
Avaleha, Khanda, Paka, Guda	3 years
Churna, Kwatha Churna, Lepa Churna, Danta Manjana (Churna)	2 years
Dhoopana	2 years
Dravaka, Lavana, Kshara	5 years
Ghrita	2 years
Guggulu	5 years
Gutika or Vati: Kashthaushadhi with Rasa, Uparasa, Bhasma or Guggulu (including Lepa Gutika and Ghana Vati)	5 years
Gutika or Vati: only Kashthaushadhi (including Lepa Gutika and Ghana Vati)	3 years
Gutika or Vati: only Rasa, Uparasa or Bhasma, except Naga, Vanga and Tamra Bhasma	10 years
Karna and Nasa Bindu	2 years
Kupipakva Rasayana	**10 years**
Malahara	3 years
Mandura and Lauha	**10 years**
Naga, Vanga and Tamra Bhasma	**5 years**
Netra Bindu	1 year
Parpati	**10 years**

Category (Ayurveda)	Shelf life
Pishti and Bhasma, except Naga, Vanga and Tamra Bhasma	**10 years**
Pravahi Kwatha	3 years
Rasayoga with only Rasa, Uparasa or Bhasma, except Naga, Vanga and Tamra	**10 years**
Rasayoga with Rasa, Uparasa or Bhasma plus Kashthaushadhi or Guggulu	5 years
Satva (from medicinal plants)	2 years
Sharkara, Panaka, Sharbat	3 years
Shweta Parpati	2 years
Taila	3 years
Varti	2 years

Rule 161B as substituted by G.S.R. 789(E), 12 August 2016, sub-rule (8), Ayurveda table, as reproduced on Indian Kanoon (accessed 16 October 2026). Separate tables are given for Siddha and Unani medicines. Check the current Gazette text for any later change.

B3.4.4 Shelf life and storage of herbo-mineral preparations

The March 2026 paper asked for the significance of shelf life and storage conditions for herbo-mineral preparations.

1. **Why mineral preparations last longer.** Bhasmas, Pishti, Parpati and Kupipakva products are inorganic compounds (oxides, sulphides, carbonates) that are chemically stable and contain little water. This is why the law allows them ten years, and why the classical texts regard Rasaushadhi as long-lasting.
2. **Why some do not.** Naga, Vanga and Tamra Bhasma are given only five years. Preparations that combine minerals with herbal material (Rasayoga with Kashthaushadhi, Vati with herbs and Bhasma) are given five years, because the **herbal part** degrades: it absorbs moisture, supports microbial growth and loses actives.
3. **Moisture is the main enemy.** Many Bhasmas and Kharaliya products are hygroscopic or are triturated with herbal juices (Bhavana) that leave organic residues. Moisture causes caking, microbial growth and chemical change.
4. **Storage conditions** therefore matter as much as the product: airtight glass or HDPE containers, a desiccant where needed, a cool and dry place, protection from light, and closed containers after each use.
5. **Labelling** must state the batch, date of manufacture and expiry, storage conditions and, for products containing Schedule E(1) substances, the caution required by law (B3.5).

6. **Stability data** are now required: the legal shelf life is a default, and manufacturers must support their shelf-life claims with stability studies.

B3.5 Labelling

> A **label** is any written, printed or graphic matter on or attached to a container or package. **Labelling** includes the label and any leaflet or other material accompanying the product.

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

Two changes of 2026 affect labels. The **Drugs (Eleventh Amendment) Rules, 2026** require a specific product code and allow label information to be given through a QR code, GTIN or barcode, and allow accelerated stability data to support an initial shelf life of one or two years. The **Drugs (Thirteenth Amendment) Rules, 2026** substituted **Schedule E(1)**, the list of poisonous ingredients whose presence requires the caution "To be taken under medical supervision". See B3.5.2, B3.5.3 and Chapter B8(i).

B3.5.1 Contents of an ideal label

An ideal label is **accurate, complete, legible, durable and understandable**. It identifies the product beyond doubt, tells the user how to use it safely, and allows the batch to be traced.

Element	Purpose
Name of the product	Identity
Composition: each ingredient with its quantity; botanical names and parts used	Identity; safety (allergies, toxic ingredients)
Reference to the classical text (for classical products)	Authenticity of the formula
Dosage form and net content	Correct use; value
Dose, method of use, Anupana	Correct use
Indications (as permitted by law)	Correct use, without misleading claims
Cautions and warnings	Safety: "Caution: To be taken under medical supervision"; "For external use only"; "Keep out of reach of children"
Storage conditions	Stability
Batch number; date of manufacture; date of expiry	Traceability; safety
Manufacturing licence number	Legal status
Name and address of the manufacturer	Accountability
Category ("Ayurvedic medicine")	Legal identity
Price	Consumer information
Product code; QR code or barcode	Traceability and access to full information

B3.5.2 Legal requirements for labelling of ASU drugs (Rule 161)

Rule 161 of the Drugs and Cosmetics Rules lays down the particulars that must appear on the labels of Ayurvedic, Siddha and Unani drugs [Rule 161; Damodaran 2021].

Particular	Where
Name of the drug	Innermost container and outer covering
Net content (weight, measure or number)	Innermost container and outer covering
Name and address of the manufacturer	Innermost container and outer covering
Manufacturing licence number; batch number; date of manufacture; date of expiry	Innermost container and outer covering
The words **"Ayurvedic medicine"** (or "Siddha medicine" or "Unani medicine")	Innermost container and outer covering
"For external use only", for drugs meant for external application	Innermost container and outer covering
"Physician's sample. Not to be sold", overprinted on free samples	Samples
A list of all ingredients, with **botanical names** of plant ingredients, the part used, the form and the **quantity**	Innermost container and outer covering
A reference to the method of preparation in the authoritative books listed in the First Schedule to the Act (for classical, not proprietary, drugs)	Innermost container and outer covering
"Caution: To be taken under medical supervision", in English and Hindi, for drugs for internal use containing a substance listed in **Schedule E(1)**	Conspicuously on the container

Further points:

- Coverings used only for transport (transparent covers, wrappers) need not carry these particulars.
- Standard abbreviations from the official pharmacopoeias and formularies may be used for parts and forms of ingredients.
- If the list of ingredients is too long, it may be printed on a separate leaflet enclosed with the pack and referred to on the label; under the 2026 amendment, label information may also be provided through a **QR code, GTIN or barcode**, and a **specific product code** must appear on the label within two years [Corpseed 2026; Lexplosion 2026].
- **Rule 161A**: drugs meant for export may be labelled to meet the requirements of the importing country.
- The Central Consumer Protection Authority advised e-commerce platforms in July 2022 that ASU drugs containing Schedule E(1) substances may be sold online only against a valid prescription uploaded by the user [CCPA 2022].

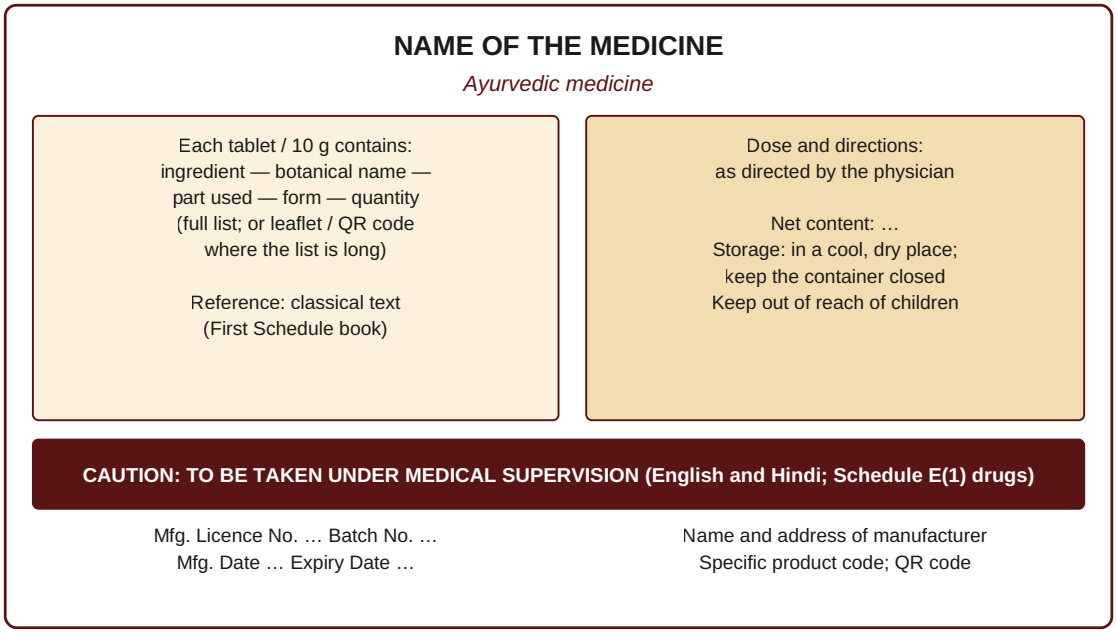


Figure B3.4. A specimen label for a classical Ayurvedic product, showing the legally required particulars.

B3.5.3 Schedule E(1)

Schedule E(1) lists poisonous substances under the Ayurvedic (including Siddha), Unani and, since 2026, Sowa-Rigpa systems: substances of plant, animal and mineral origin, such as Vatsanabha, Kuchala, Bhallataka and Jayapala among plants, and arsenic, mercury and lead compounds among minerals. Drugs for internal use containing them must carry the caution above. Schedule E(1) was substituted by the **Drugs (Thirteenth Amendment) Rules, 2026** (notification dated 21 August 2026), which expanded the mineral entries and provides that any ingredient containing arsenic or mercury as a chemical constituent is treated as a Schedule E(1) substance even if not listed by name [Gazette 2026b; Indian Practitioner 2026]. The details are given in Chapter B8.

AYURVEDA CORRELATION: SAVIRYATA AVADHI

Sharangadhara gave the period of potency (**Saviryata Avadhi**) for each class of preparation, and the Rasa texts regard well-prepared Rasaushadhi as long-lasting [R11]. The classical periods for herbal preparations are generally shorter than those now allowed by law; for example, a study comparing them found that the classical period for Churna is two months, while Rule 161B allows two years [Unnikrishnan 2016]. The difference reflects modern packaging, drying and storage, and the law's requirement that shelf life be supported by stability data.

Established. Inorganic Rasaushadhi are chemically stable; herbal preparations degrade faster; moisture, heat and light shorten shelf life.

Not established. That every classical period of potency, or every legal default period, is right for every product. Each manufacturer's product needs its own stability data [Badal 2024].

KEY POINTS TO REMEMBER

- Functions of packaging: containment, protection, preservation, identification, presentation, convenience, safety.

- Levels: primary (touches the product), secondary (carton), tertiary (shipper).
- Glass types I (borosilicate), II (treated soda-lime), III (soda-lime), NP; amber for light protection.
- Plastics: LDPE, HDPE, PP, PVC, PVDC, PET; problems of permeation, leaching, sorption.
- Aluminium tubes and foils; laminates; strips and blisters (Alu-Alu best for moisture).
- Choose the pack by the main threat: moisture (Churna, Bhasma, Kshara), oxidation and light (Ghrita, Taila), evaporation (Asava, Arka), microbes (Avaleha).
- Stability: long-term 25/60, 30/65 or 30/75; accelerated 40 °C/75% RH. Rule 161B requires real-time stability data; the 2026 amendment allows accelerated data for an initial one or two years.
- Rule 161B(8): Asava–Arishta, Kupipakva, Parpati, Pishti and Bhasma, Mandura–Lauha, mineral-only Rasayoga and Vati: 10 years; Naga, Vanga, Tamra Bhasma: 5 years; Churna and Ghrita: 2 years; Taila, Avaleha: 3 years; Arka, Netra Bindu: 1 year.
- Rule 161 label: name, net content, manufacturer, licence number, batch, dates, “Ayurvedic medicine”, ingredients with botanical names and quantities, classical reference, “For external use only”, “Caution: To be taken under medical supervision” for Schedule E(1) drugs.
- 2026: specific product code; QR code, GTIN or barcode; Schedule E(1) substituted (any arsenic- or mercury-containing ingredient included).

EXAM CORNER — CHAPTER B3

Likely long questions (20 marks)

1. Describe packaging materials used for pharmaceutical products and their selection for Ayurvedic dosage forms. Add a note on the labelling requirements of Ayurvedic drugs.
2. Discuss shelf life and stability of Ayurvedic drugs, with the legal provisions.

Likely short notes (5 marks)

1. Examples of packaging material and contents of an ideal label *(set in September 2018)*.
2. Packaging materials and labelling *(set in September 2020)*.
3. Rules pertaining to packaging and labelling of Ayurvedic drugs *(set in August 2021)*.
4. Significance of shelf-life and storage conditions for herbo-mineral preparations *(set in March 2026)*.
5. Types of glass.
6. Plastics as packaging materials.
7. Rule 161B.
8. Schedule E(1).

How to attempt

- Define packaging and give its functions and levels.
- Give the materials in a table (merits, limits, uses) and the selection table for Ayurvedic forms.
- For labelling, list the Rule 161 particulars; for shelf life, give selected Rule 161B entries and the reasons why mineral preparations last longer.

Answer frame: “Significance of shelf-life and storage conditions for herbo-mineral preparations.” (5 marks)

Part	What to write	Suggested marks
Definitions	Shelf life; expiry; Saviryata Avadhi	0.5
Legal periods	Rule 161B entries for Bhasma, Pishti, Kupipakva, Parpati, Rasayoga (10 years), Naga, Vanga, Tamra Bhasma and mixed herbo-mineral (5 years)	1.5
Reasons	Inorganic stability versus herbal degradation; moisture	1
Storage	Containers, desiccants, cool dry place, light protection	1
Stability studies and labelling	Real-time and accelerated data; label particulars; Schedule E(1) caution	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **11, 27, 28, 36, 38, 43**.

- **[R11]** *Sharangadhara Samhita* (Saviryata Avadhi). **[R27]** *Remington*; **[R28]** Lachman, *Theory and Practice of Industrial Pharmacy* (packaging; stability). **[R36]** *Ayurvedic Pharmacopoeia of India*, Part I, Volume VIII (stability guidelines). **[R38]** *Indian Pharmacopoeia* (containers; glass).
- **[R43]** Drugs and Cosmetics Rules, 1945: **[Rule 161]** labelling of ASU drugs; **[Rule 161B]** shelf life, as substituted by G.S.R. 789(E), 12 August 2016 (text as reproduced on Indian Kanoon, accessed 16 October 2026).

Additional verified sources (accessed 16 October 2026)

- **[WHO 2018]** World Health Organization. Stability testing of active pharmaceutical ingredients and finished pharmaceutical products. WHO Technical Report Series 1010, Annex 10; 2018.
- **[Damodaran 2021]** Damodaran D. Labelling requirements of Ayurvedic drugs. Mondaq; 3 March 2021.
- **[CCPA 2022]** Central Consumer Protection Authority. Advisory on sale of Ayurvedic, Siddha and Unani drugs containing Schedule E(1) substances by e-commerce platforms. File J-25/64/2022-CCPA, 14 July 2022.
- **[Corpseed 2026]**, **[Lexplosion 2026]**: summaries of the Drugs (Eleventh Amendment) Rules, 2026 (full citations in Chapter B4).
- **[Gazette 2026b]** Ministry of Ayush. Drugs (Thirteenth Amendment) Rules, 2026 (substitution of Schedule E(1)). Notification dated 21 August 2026; Gazette ID CG-DL-E-08102026-276954.
- **[Indian Practitioner 2026]** “Ministry of Ayush updates list of poisonous substances to include additional minerals.” *The Indian Practitioner*; 13 October 2026.
- **[Unnikrishnan 2016]** Unnikrishnan V, Nishteswar K, Patel BR. Shelf life evaluation and comparative HPTLC profile of Hridya Yoga Churna. *Pharmacogn J*. 2016;8(3):234–238.
- **[Badal 2024]** Badal R, Kaushik S, Yadav P, Prajapati PK. Critical appraisal of documented shelf life of Ayurvedic formulations: current status and future implications. *Indian J Pharm Sci*. 2024;86(4):1207–1217.

Viva pointers — Chapter B3

Answer each in one or two sentences.

- Define packaging. Name five of its functions.
- What is the difference between primary, secondary and tertiary packaging?
- Name the types of pharmaceutical glass. Which is used for injections?
- Why is amber glass used?
- Name four plastics used in packaging and one use of each.
- What are leaching and sorption?
- What is an Alu-Alu blister? When is it needed?
- Which pack would you choose for a hygroscopic Churna? For Ghrita? For Asava?
- Define shelf life and date of expiry.
- What are the accelerated stability conditions?
- What does Rule 161B require of a manufacturer?
- What is the legal shelf life of Asava–Arishta? Of Churna? Of Kupipakva Rasayana? Of Naga Bhasma?
- Why do Bhasmas have longer shelf lives than herbal Vati?
- List ten particulars of an ASU drug label under Rule 161.
- When must “Caution: To be taken under medical supervision” appear?
- What is Schedule E(1)? What changed in 2026?
- What does Rule 161A allow?
- What labelling changes did the 2026 amendment introduce?

PART B · PHARMACY MANAGEMENT · CHAPTER B4

B4 · PHARMACY MANAGEMENT AND GOOD MANUFACTURING PRACTICES

SYLLABUS POINT (PART B, 4)

Management of pharmacy, store and inventory management, personnel management, Good Manufacturing Practices related to Ayurvedic drug industry.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B4	Pharmacy Management and Good Manufacturing Practices	6	1	Sep 2018, Sep 2020, Aug 2021, Aug 2022, Jan 2024, Mar 2026

Why this chapter matters to you. This is the most heavily examined chapter of Part B: it was set in six of the seven papers, five times as a 20-mark question, almost always as “management of pharmacy and Good Manufacturing Practices related to the Ayurvedic drug industry”. A postgraduate in Bhaishajya Kalpana may run a pharmacy, head production or quality in an Ayurvedic company, or inspect one. This chapter gives the management principles, store, inventory and personnel management, and GMP for ASU drugs, with tables and diagrams you can reproduce in the hall.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define management and describe its principles and functions as applied to an Ayurvedic pharmacy.
2. Describe the planning, layout, organisation and quality management of an Ayurvedic manufacturing unit, including the safe handling of toxic materials.
3. Describe store management for raw materials, packing materials and finished goods.
4. Apply the techniques of inventory control (ABC, VED and related analyses; economic order quantity; reorder level).
5. Describe personnel management in a pharmaceutical unit.
6. Describe Good Manufacturing Practices for ASU drugs under Schedule T and related quality systems.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	B, Q.1	20	Explain Drug & Cosmetics Act with relevance to ASU drugs along with schedule T and Good Manufacturing Practices.
Sep 2020	B, Q.1	20	Write down in brief Management of pharmacy & good manufacturing practices related to Ayurvedic drug industry
Aug 2021	B, Q.2	20	Discuss in detail about Good Manufacturing Practices (GMP) related to Ayurvedic Drug Industry
Aug 2022	B, Q.1	20	Good manufacturing practices related to Ayurved industries
Jan 2024	B, Q.2	20	Explain Good manufacturing practices related to Ayurvedic drug industry.
Mar 2026	B, Q.1	20	Discuss the management of an Ayurvedic Pharmacy. Include aspects of international trade and regulatory affairs related to the export of Ayurvedic drugs.
Mar 2026	B, Q.3a	5	Explain the significance of shelf-life and storage conditions for herbo-mineral preparations.

B4.1 Management: principles and functions

> **Management** is the process of planning, organising, staffing, directing and controlling the work of people and the use of resources (men, materials, machines, money and methods) to achieve defined goals efficiently and effectively.

B4.1.1 Functions of management

The functions of management are often remembered by Luther Gulick's acronym **POSDCORB**.

Function	Meaning	In an Ayurvedic pharmacy
Planning	Deciding in advance what to do, how and when	Annual production plan; procurement of seasonal herbs; capacity planning
Organising	Arranging departments, roles and authority	Production, quality control, quality assurance, stores, maintenance, marketing
Staffing	Recruiting, training and keeping people	Qualified technical staff; trained workers for Paka, Puta and Kupipaka
Directing	Guiding, motivating and supervising	Supervision of batches; communication of SOPs
Coordinating	Making departments work together	Production plan matched to raw material availability and sales
Reporting	Keeping superiors and regulators informed	Batch records; production and stock reports; regulatory returns
Budgeting	Financial planning and control	Costing of formulations; capital for equipment and GMP upgrades

Controlling (comparing performance with plans and correcting deviations) runs through all of these.

B4.1.2 Principles of management (Fayol)

Henri Fayol's fourteen principles are still the standard list. The most useful for a pharmacy are:

Principle	Meaning	Example
Division of work	Specialisation improves skill and output	Separate teams for Churna, Vati, Sneha Paka and Rasaushadhi
Authority and responsibility	Authority must match responsibility	The QC head who is responsible for release must have authority to reject
Discipline	Respect for rules	Adherence to SOPs and gowning
Unity of command	Each worker has one superior	Clear reporting lines
Unity of direction	One plan for activities with the same objective	One production plan
Subordination of individual interest	The organisation's goal comes first	Quality before output
Remuneration	Fair pay	Wages for skilled classical work
Centralisation	Degree of central decision-making	Central approval of master formulae
Scalar chain	Line of authority	Organisation chart
Order	A place for everything and everything in its place	Store layout; labelled materials
Equity	Fairness	—
Stability of tenure	Low staff turnover	Retaining skilled Rasashastra workers
Initiative	Encouraging ideas	Suggestions for process improvement
Esprit de corps	Team spirit	—

B4.2 Management of an Ayurvedic pharmacy

B4.2.1 Types of Ayurvedic pharmacy

Type	Description
Industrial manufacturing unit	Licensed unit making products for sale; must comply with GMP (B4.6)
Loan licence manufacturing	A firm without its own unit gets products made in another licensed unit under a loan licence (Chapter B8)
Teaching pharmacy	In an Ayurvedic college, for training students and supplying the teaching hospital
Hospital pharmacy	Prepares and dispenses for a hospital (Chapter B6)
Dispensary of a practitioner	Small-scale preparation and dispensing by a Vaidya

B4.2.2 Planning a manufacturing unit

Aspect	Points to plan
Product mix	Classical and proprietary formulations; dosage forms (Churna, Vati, Asava-Arishta, Taila-Ghrita, Avaleha, Bhasma, Kupipakva); market demand
Capacity	Batch sizes; number of batches; equipment and space
Location	Away from pollution, open drains and refuse; adequate water and power; approach roads; industrial area permitted for the activity; consent of the pollution control board
Layout	One-way flow of materials (B4.2.3); separate areas for each operation; separate area for Rasaushadhi
Licences	Manufacturing licence for ASU drugs; GMP certification; pollution control consent; factory licence; fire safety (Chapter B8)
Finance	Fixed capital (land, building, machinery); working capital (raw materials, wages, stocks)
Raw material sourcing	Seasonal availability; authentic suppliers; cultivation; storage capacity

B4.2.3 Layout and flow

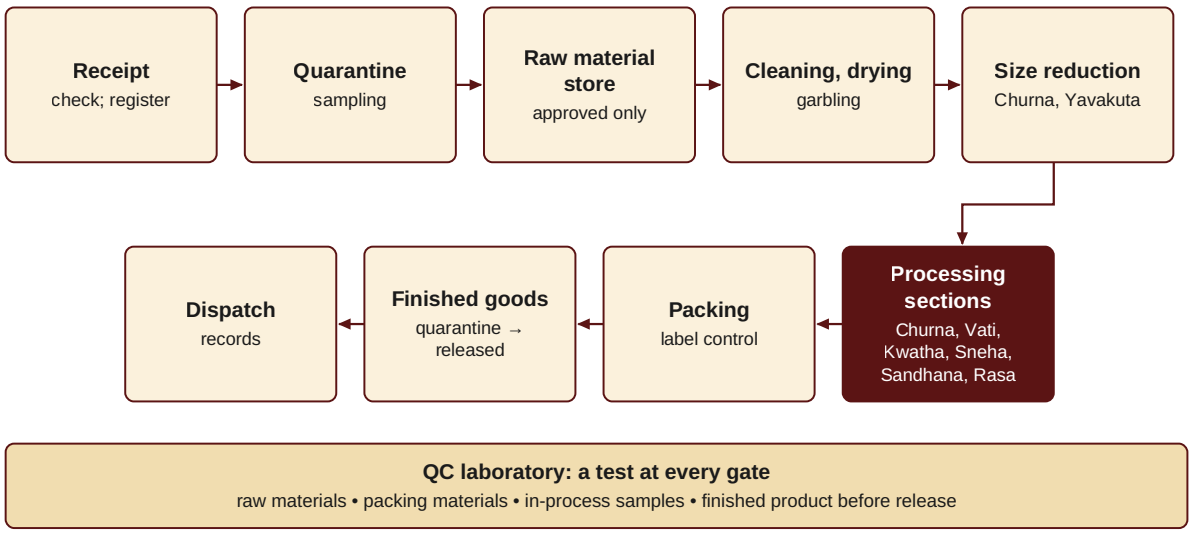


Figure B4.1. Flow of materials in an Ayurvedic manufacturing unit: one-way movement from receipt to dispatch, with quality control at each gate.

The layout should allow materials to move in **one direction**, from receipt of raw materials through processing and packing to dispatch, without crossing back. This prevents mix-ups and contamination. Separate areas are needed for raw material storage, washing and drying, size reduction, each manufacturing section, packing, quality control, finished goods, and rejected materials. The minimum areas and equipment for each section are laid down in Schedule T (B4.6).

B4.2.4 Organisation

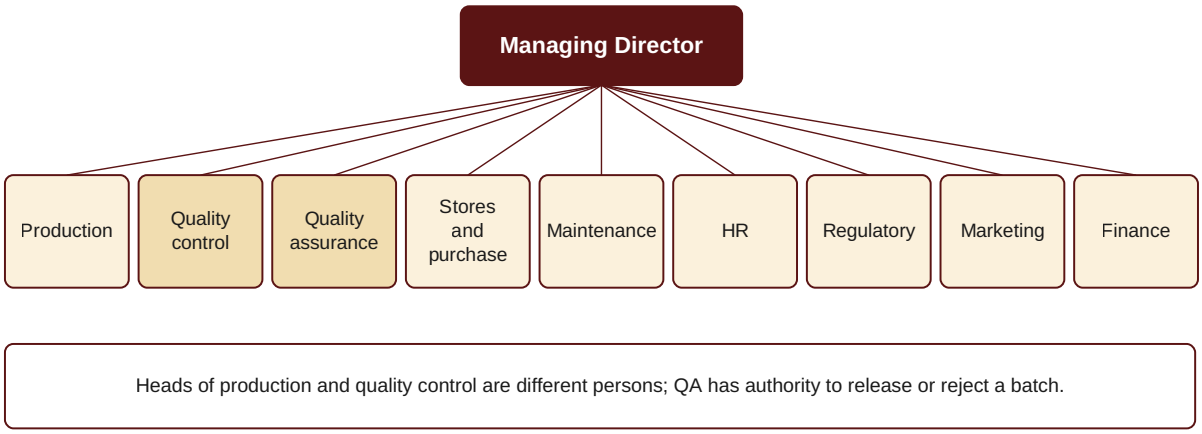


Figure B4.2. Organisation chart of a medium-sized Ayurvedic manufacturing unit.

Department	Responsibilities
Production	Manufacturing as per master formula and SOPs; batch records; equipment cleaning
Quality control	Testing of raw materials, in-process samples, packing materials and finished products
Quality assurance	The overall system: SOPs, documentation, validation, training, audits, complaints, recalls, batch release
Stores	Receipt, storage and issue of materials; inventory control
Maintenance (engineering)	Equipment, utilities, preventive maintenance, calibration
Purchase	Procurement of raw and packing materials
Human resources	Staffing, training, welfare
Regulatory affairs	Licences, product permissions, dossiers
Marketing and sales	Chapter B5
Finance and accounts	Costing, budgets, payments

B4.2.5 Production planning and control

Production planning and control (PPC) makes sure that the right product is made in the right quantity at the right time at the least cost.

Step	Meaning
Planning	Forecast demand; prepare the production schedule; check materials and capacity
Routing	Decide the sequence of operations for each product
Scheduling	Decide when each operation will be done (Sandhana needs weeks; Puta cycles need days)
Dispatching	Issue orders, materials and documents to start work
Follow-up (expediting)	Check progress; remove bottlenecks
Inspection	Ensure quality at each stage (in-process control)

Long-cycle products (Asava and Arishta, Bhasmas with many Puta, Kupipakva) need special planning, because the stock must be started months before it is needed.

B4.2.6 Costing

The cost of a product has three elements.

Element	Examples
Direct material	Herbs, minerals, metals, ghee, oil, jaggery, honey; packing materials
Direct labour	Wages of workers for the batch
Overheads	Fuel, power, water, depreciation of equipment, QC testing, administration, selling

Break-even point is the volume of sales at which total revenue equals total cost. Below it the unit makes a loss; above it, a profit.

$$\text{Break-even point (units)} = \text{Fixed costs} \div (\text{Selling price per unit} - \text{Variable cost per unit})$$

B4.2.7 Quality management

Term	Meaning
Quality control (QC)	Testing of materials and products against specifications; part of QA
Quality assurance (QA)	The whole system of planned activities that ensure products are of the quality required for their use; includes GMP, documentation, validation, training, audits
GMP	That part of QA which ensures products are consistently made and controlled to the required standards (B4.6)
Validation	Documented evidence that a process, equipment or method consistently does what it is meant to do
Change control	A system for approving and recording changes in materials, processes or equipment
CAPA	Corrective and preventive action after a deviation or complaint
Total quality management	A culture in which every worker is responsible for quality, with continuous improvement
ISO 9001	An international standard for quality management systems, which some ASU manufacturers adopt in addition to GMP

B4.2.8 Safety and environment in an Ayurvedic pharmacy

SAFETY IN A PHARMACY THAT HANDLES MERCURY, ARSENIC AND LEAD

Hazards

- **Mercury vapour** during Shodhana, Kajjali preparation, Kupipaka and Marana; spillage of liquid mercury.
- **Arsenic** compounds (Sankhiya, Haratala, Manahshila) and **lead** (Naga) during Shodhana and Marana; fumes from Puta.
- **Heat and fire**: furnaces, Puta, Kupipaka, Sneha Paka (oil and ghee fires).
- **Dust**: size reduction of herbs and minerals.
- **Chemical burns**: Kshara, acids, alkalis.
- **Glass**: Kupi breakage under heat.

Controls

- Separate, well-ventilated Rasashastra section with local exhaust over heating operations; fume hoods or chimneys for Kupipaka.
- Personal protective equipment: gloves, aprons, eye protection, respirators suited to vapours and dust; heat-resistant gloves.
- Trays to contain mercury spills; a written spill procedure; mercury spill kits; never sweep or vacuum mercury with an ordinary vacuum cleaner.
- Locked, labelled storage of Visha and Upavisha Dravya and Rasa Dravya; issue against records.
- No eating, drinking or smoking in work areas; hand washing; work clothes kept at work.
- Fire extinguishers suited to oil fires; trained staff; clear exits.
- Health surveillance of exposed workers (periodic medical examination; blood or urine metal levels where exposure is significant).
- Disposal of mercury- and arsenic-containing waste through authorised hazardous waste channels; no discharge to drains.
- Consent and compliance with the State Pollution Control Board for emissions from furnaces and Puta.

B4.3 Store management

> **Store management** is the receipt, safe storage, preservation, issue and accounting of materials (raw materials, packing materials, intermediate products and finished goods) so that the right material, of the right quality, is available when needed, at the least cost and with no loss.

B4.3.1 Types of store

Store	Contents	Special needs
Raw material store (crude drugs)	Dried herbs, roots, barks, seeds, resins	Dry, cool, ventilated; protection from insects, rodents and fungi; on pallets, off the floor and away from walls
Rasa Dravya and Visha store	Mercury, sulphur, metals, minerals, arsenicals, Vatsanabha, Kuchala and other poisonous drugs	**Locked**, separate, labelled; access and issue recorded

Store	Contents	Special needs
Liquid and fat store	Ghee, oils, honey, jaggery, alcohol, volatile oils	Cool; closed containers; protection from light; fire safety for oils and alcohol; excise rules for alcohol
Animal-origin materials	Shells, horns, pearls, coral, Gorochana, musk substitutes	Legal origin; separate storage
Packing material store	Bottles, caps, labels, cartons, foils	Labels kept under lock (mix-ups of labels are a major cause of recall)
Quarantine area	Materials awaiting QC approval	Clearly marked and separated
Finished goods store	Released products awaiting dispatch	Storage as labelled; FEFO
Rejected materials area	Failed materials awaiting return or destruction	Locked; clearly marked
Cold store	Heat-sensitive items	Temperature monitored and recorded

B4.3.2 Receipt of materials

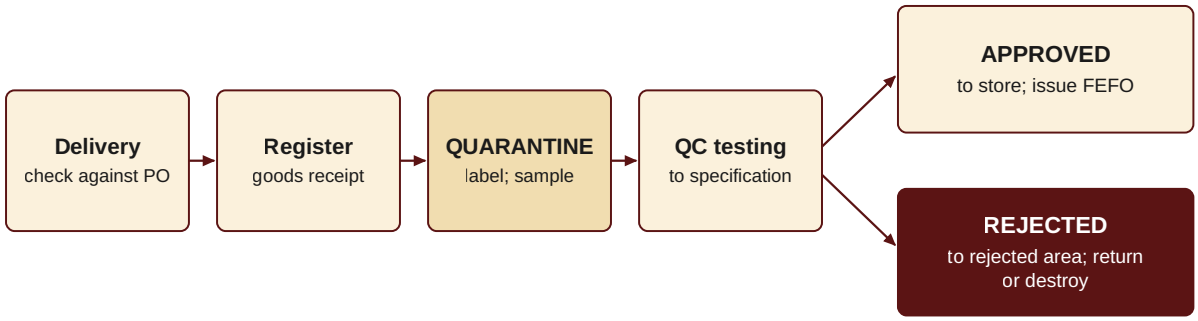


Figure B4.3. Receipt and status control of incoming materials.

1. **Check the delivery** against the purchase order: name, quantity, supplier, condition of containers, labels.
2. **Record** in the goods receipt register (goods receipt note); give an internal receipt number.
3. **Clean** the outside of containers; move to the **quarantine** area with a “QUARANTINE” label.
4. **Sample** by quality control under defined procedures.
5. **Test** against the specification (identity, purity, foreign matter, moisture, ash values, heavy metals where required).
6. **Status label**: “APPROVED” (released for use) or “REJECTED”. Colour-coded labels are commonly used.
7. Move approved material to the store; rejected material to the rejected area.

B4.3.3 Storage of crude drugs

Crude herbal drugs are living or once-living materials. They deteriorate by:

- **moisture**: fungal growth, loss of actives, caking;
- **insects and rodents**: weevils in seeds and roots; contamination;
- **heat and light**: loss of volatile oils; oxidation;
- **air**: oxidation of oils and resins;

- **time**: gradual loss of potency.

Good practice: dry material to a safe moisture level before storage; store in clean, closed containers or bags on pallets; use **FIFO or FEFO**; regular inspection; **pest control** by approved methods that do not leave residues in the drugs; keep records of the date of collection and receipt. The classical texts also direct that collected drugs be kept in clean, closed containers in a store protected from smoke, rain, wind and dust (Charaka Samhita, Kalpa Sthana 1) [R26].

B4.3.4 Issue of materials

Materials are issued only against a written **requisition** (for production, the batch manufacturing record's material list), in the quantity approved, and only from **approved** stock. The issue is recorded on the **bin card** (kept with the material) and in the **stock register**, so that the balance is always known. Weighing and dispensing of raw materials for a batch are done in a designated area, checked by a second person and recorded.

B4.3.5 Stock verification

Physical stock is compared with records periodically (**stock-taking**). Differences are investigated. Expired or deteriorated materials are identified and disposed of with records. **Perpetual inventory**, in which some items are checked every day, avoids closing the store for a full count.

B4.4 Inventory management

> **Inventory management** (inventory control) is the planning and control of the quantities of materials held, so that production never stops for want of material, while the money locked in stock and the cost of holding it are kept as low as possible.

B4.4.1 Costs of inventory

Cost	Meaning
Ordering cost	Cost of placing and receiving an order: paperwork, transport, inspection
Carrying (holding) cost	Cost of keeping stock: interest on money, storage space, deterioration, insurance, pilferage
Stock-out cost	Cost of running out: lost production, lost sales, emergency purchase at high price

The aim is to balance these. Large, infrequent orders reduce ordering cost but raise carrying cost; small, frequent orders do the reverse.

B4.4.2 Selective control techniques

Not every item deserves the same attention. Selective control techniques classify items so that effort is spent where it matters.

Technique	Basis	Classes	Use
ABC	Annual consumption **value**	A: few items, most of the value; B: middle; C: many items, little value	Tight control and frequent review of A items
VED	**Criticality** for production or patient care	Vital, Essential, Desirable	Never let V items run out
FSN	Rate of **movement**	Fast, Slow, Non-moving	Identify dead stock
HML	**Unit price**	High, Medium, Low	Purchase decisions
SDE	**Availability** in the market	Scarce, Difficult, Easy	Plan ahead for scarce and seasonal items
GOLF	**Source**	Government, Open market, Local, Foreign	Lead time and procedures

ABC analysis commonly shows a pattern like this: about 10 to 20 per cent of items account for about 70 to 80 per cent of the annual value (class A); about 20 to 30 per cent of items account for about 15 to 25 per cent (class B); and the remaining items, the majority by number, account for only about 5 to 10 per cent (class C). The exact cut-offs are chosen by each organisation.

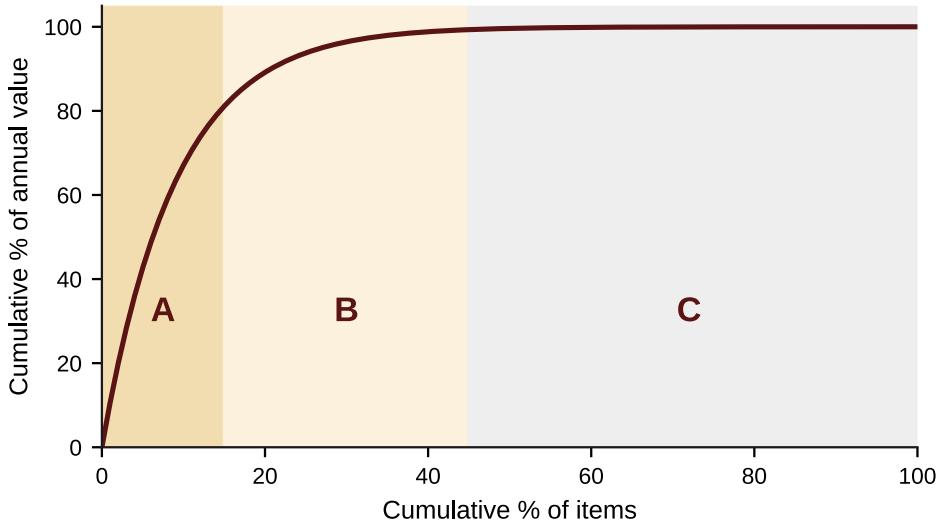


Figure B4.4. ABC analysis: a few items account for most of the value (illustrative cumulative curve).

For an Ayurvedic pharmacy, the classes often look like this:

Class	Typical items
A (high value)	Swarna, Rajata, Kesara (saffron), Mukta, Praval, cow's ghee, costly imported or scarce herbs
B	Common Bhasma raw materials, honey, sesame oil, jaggery
C (low value)	Common herbs in bulk, packing consumables
V (vital)	Ingredients without which a major product cannot be made (for example, Shuddha Parada and Gandhaka for Kharaliya products)
S (scarce)	Seasonal fresh herbs; herbs from the wild; materials under legal restriction

ABC–VED matrix. Combining the two gives nine groups. Items that are both A and V need the closest control; items that are C and D need the least.

B4.4.3 Economic order quantity

The **economic order quantity** (EOQ) is the order size that minimises the total of ordering and carrying costs.

$$EOQ = \sqrt{(2 \times D \times S \div H)}$$

where D is the annual demand (units), S the cost of placing one order, and H the cost of carrying one unit in stock for a year.

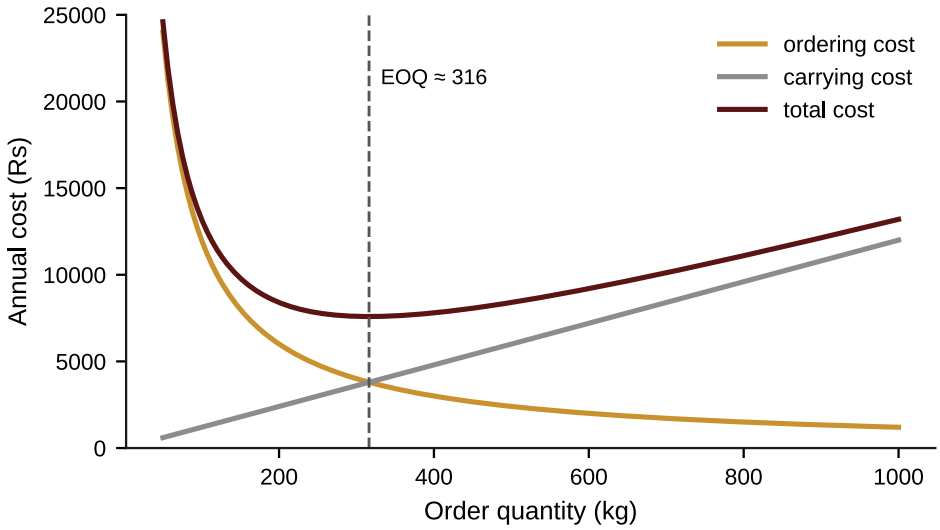


Figure B4.5. Economic order quantity: the total cost is lowest where ordering cost and carrying cost are equal.

WORKED EXAMPLE (ILLUSTRATIVE NUMBERS)

A pharmacy uses 2400 kg of a herb a year. Each order costs Rs 500 to place. Carrying one kilogram for a year costs Rs 24.

EOQ = √(2 × 2400 × 500 ÷ 24) = √100,000 ≈ **316 kg** per order, about 7 or 8 orders a year.

For seasonal herbs that can be collected only once a year, EOQ cannot be applied directly: the whole year's need must be bought in season and stored well.

B4.4.4 Stock levels

Level	Formula or meaning
Reorder level	Maximum usage per day × maximum lead time (days), or average usage × lead time + safety stock. When stock falls to this level, a new order is placed
Safety (buffer) stock	Extra stock kept to cover delays and unexpected demand
Minimum level	Reorder level − (normal usage × normal lead time); stock should not fall below this
Maximum level	Reorder level + reorder quantity − (minimum usage × minimum lead time); stock should not exceed this
Lead time	Time between placing an order and receiving the material

Just-in-time (JIT) inventory keeps stocks very low and relies on frequent, reliable deliveries. It suits steady supplies of standard materials, but not seasonal herbs or materials with long and uncertain lead times.

B4.5 Personnel management

> **Personnel management** (human resource management) is the part of management concerned with people at work: their recruitment, training, development, motivation, welfare and relations with the organisation.

B4.5.1 Functions

Function	In a pharmaceutical unit
Manpower planning	Number and type of staff needed for production, QC, QA, stores and maintenance
Recruitment and selection	Qualified technical staff as required by the licensing rules; skilled workers for classical processes
Training	Induction; GMP; SOPs; hygiene; safety (especially for Rasashastra operations); refresher training; records of all training
Job description	Written duties and responsibilities for each post
Performance appraisal	Regular review of work against agreed standards
Wages and salaries	Fair, as per law; incentives
Motivation	Recognition, career growth, good working conditions
Welfare and health	Medical examination at joining and periodically; canteen, rest rooms, protective clothing (Factories Act and the labour codes, Chapter B8)
Industrial relations	Grievance handling; communication; discipline
Records	Personal files; training records; medical records

B4.5.2 Motivation

Two classic theories are useful.

Theory	Idea	Application
Maslow's hierarchy of needs	People satisfy needs in order: physiological, safety, social, esteem, self-actualisation	Fair wages and safe work first; then team spirit, recognition and opportunities to grow
Herzberg's two-factor theory	Hygiene factors (pay, conditions, supervision) prevent dissatisfaction; motivators (achievement, recognition, responsibility, growth) create satisfaction	Good conditions alone do not motivate; recognise skill and give responsibility

B4.5.3 Personnel requirements under GMP

GMP requires that:

1. there are enough staff with the right **qualifications and experience** for each function;
2. responsibilities are written down, and the heads of **production** and **quality control** are different persons;
3. all staff receive **training** in GMP and in their own tasks, with records;
4. staff are **medically examined** before employment and periodically, and anyone with an infectious disease or open lesions is kept away from products;
5. staff follow **hygiene** rules: clean clothing, head covers, hand washing; no eating, drinking or smoking in production areas.

The specific requirements for ASU manufacturing units are given in Schedule T (B4.6).

B4.6 Good Manufacturing Practices for the Ayurvedic drug industry

> **Good Manufacturing Practice** (GMP) is that part of quality management which ensures that products are consistently produced and controlled according to the quality standards appropriate to their intended use and as required by the marketing authorisation (World Health Organization).

GMP is built on a simple idea: **quality cannot be tested into a product; it must be built into every step of its making.** A final test examines only a sample. GMP controls the people, premises, equipment, materials, methods and records so that every unit of every batch is right.

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

Schedule T was substantially rewritten by the **Drugs (Eleventh Amendment) Rules, 2026**, published in the Gazette in July 2026. The new Schedule T covers Ayurveda, Siddha, Sowa-Rigpa and Unani medicines, and existing manufacturers must comply by **31 July 2029**. Until then most units still work to the earlier Schedule T, so both are taught here (B4.6.3 and B4.6.4).

B4.6.1 Why GMP is essential for ASU drugs

1. Herbal raw materials vary naturally and may be adulterated or contaminated (Chapter B2.3.10).
2. Classical processes (Paka, Puta, Sandhana, Kupipaka) depend on skill and judgement; without written procedures and records, batches vary.
3. Rasaushadhi involve toxic metals and minerals; errors can harm patients and workers (Chapters A4, A5).
4. Market and export studies have found contamination and excess metals in some products (A4.8).

5. GMP compliance is a **legal condition** of the manufacturing licence and a requirement for export certification.

B4.6.2 The ten principles of GMP

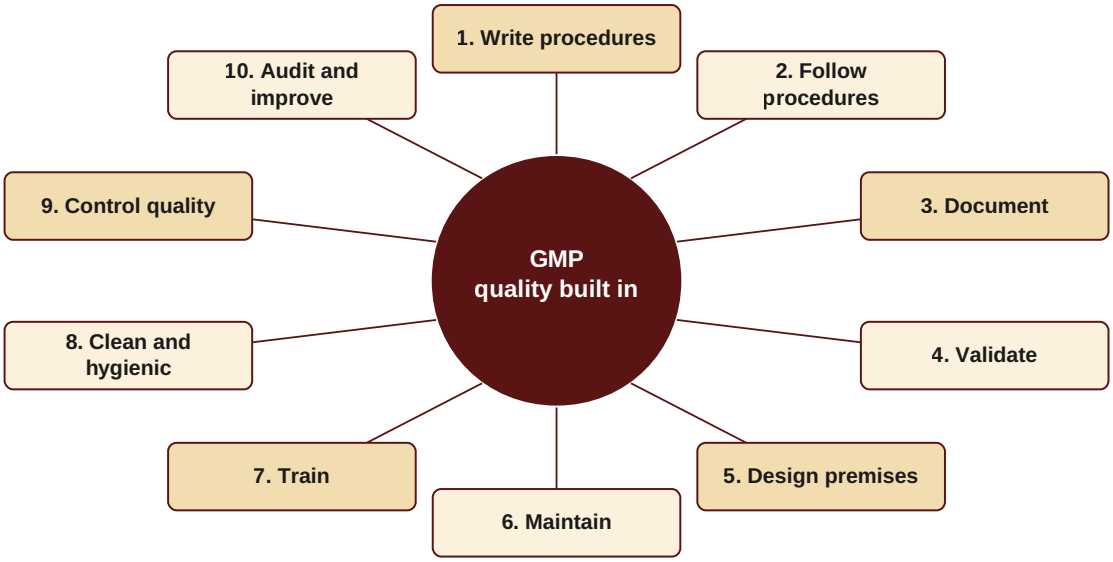


Figure B4.6. The building blocks of GMP.

Principle	Meaning
1. Write procedures	Standard operating procedures (SOPs) for every operation
2. Follow procedures	Work exactly as written
3. Document the work	Record what was done, when and by whom, at the time (“if it is not written, it was not done”)
4. Validate	Prove that processes, equipment, cleaning and methods work consistently
5. Design premises and equipment for the work	Suitable layout, materials and capacity; easy to clean
6. Maintain	Preventive maintenance and calibration
7. Train	Qualified, trained people for every job
8. Keep clean and hygienic	Sanitation of premises and personal hygiene
9. Control quality	Test materials, in-process samples and products; release only approved material
10. Audit and improve	Self-inspection; complaints; recalls; corrective and preventive action

B4.6.3 Schedule T: GMP for ASU medicines

Under the Drugs and Cosmetics Rules, 1945, a manufacturing licence for Ayurvedic, Siddha and Unani drugs is granted only if the unit complies with the GMP laid down in **Schedule T**. Schedule T was introduced in 2000 and has since been revised; a substantially rewritten Schedule T was notified in July 2026 (B4.6.4).

The structure of Schedule T as it stood before the 2026 revision, still the basis of most existing units, is summarised below [Schedule T].

Part I: Good Manufacturing Practice: general requirements

Head	Main requirements
A. Location and surroundings	Away from open drains, sewage, refuse, and anything that may contaminate the product
B. Buildings	Suitable for the operations; protection from insects, rodents, dust; walls and floors smooth and washable; adequate light and ventilation
C. Water supply	Potable water for manufacture and washing
D. Disposal of waste	Waste and effluents disposed of without contamination; as per pollution control requirements
E. Container cleaning	Facilities to clean containers before use
F. Stores	Separate stores for raw materials, packaging materials and finished goods; proper storage conditions; quarantine and approval
G. Working space	Adequate space for each operation to avoid mix-ups and contamination
H. Health, clothing, sanitation and hygiene of workers	Medical examination; clean uniforms, head covers; hand washing; no eating or smoking in work areas
I. Medical services	First aid; medical examination of workers
J. Machinery and equipment	Suitable, clean, maintained; records
K. Batch manufacturing record	Records of each batch: materials, quantities, process steps, in-process checks, yield, signatures
L. Distribution records	Records of sale to allow recall
M. Record of market complaints	Complaint register; investigation; action; adverse reaction records
N. Quality control	In-house QC section, or testing at an approved laboratory; specifications; records
Sterile products	Additional requirements for manufacturing areas and precautions against contamination and mix-ups

Part II: machinery, equipment and minimum premises

Part II lists, for each category of medicine (Churna, Vati, Avaleha, Asava-Arishta, Sneha, Bhasma, Kupipakva and others, for Ayurveda and Siddha; and the Unani categories), the recommended machinery and equipment and the minimum manufacturing area. It also lists the equipment recommended for the in-house quality control section (chemistry and pharmacognosy) and gives **supplementary guidelines for the manufacture of Rasaushadhi** (Rasamarunthukal in Siddha and Kushtajat in Unani), covering manufacturing areas, quality control, in-process control, product recall and self-inspection [Schedule T].

B4.6.4 The 2026 revision of Schedule T

A draft revision was published on 14 February 2025 (G.S.R. 140(E)) [Draft 2025]. The final rules, the **Drugs (Eleventh Amendment) Rules, 2026**, were published in the Gazette in July 2026 (Gazette ID CG-DL-E-25072026-274789) [Gazette 2026a]. The new Schedule T covers Ayurveda, Siddha, **Sowa-Rigpa** and

Unani medicines, and existing manufacturers are given until **31 July 2029** to comply [Corpseed 2026; Lexplosion 2026].

Area	Changes reported in summaries of the final rules
Premises and workflow	Detailed controls on premises, separation of man and material movement, prevention of cross-contamination, storage and sanitation
Equipment	Qualification of equipment; detailed controls
Personnel	Annual medical examination of personnel; personal hygiene
Rasaushadhi and herbo-mineral-metallic products	Additional safeguards: segregated manufacturing, **emission controls**, adverse-event investigation, product recall, **employee health surveillance**, controlled product release
Technical staff	Revised qualifications for testing staff, including a qualified microbiologist
Inspections	Risk-based inspections in place of a fixed periodic schedule
Licences	Manufacturing and loan licences made valid perpetually, subject to continued compliance
Labelling and shelf life	Specific product code; QR code, GTIN or barcode for label information; accelerated stability data allowed to support an initial shelf life of one or two years (Chapters B3, B8)

The table is based on secondary summaries of the Gazette notification and on the published draft. Before relying on any detail, read the notification itself in the Gazette of India (egazette.gov.in) or on the Ministry of Ayush website.

B4.6.5 Documentation under GMP

Document	Purpose
Master formula record	The approved formula, method, equipment and in-process controls for each product
Batch manufacturing record (BMR)	The record of what was actually done for each batch
Standard operating procedures	How each operation is done (receipt, sampling, cleaning, each manufacturing step, QC tests, release, recall)
Specifications	Standards for raw materials, packing materials and finished products (API, AFI, in-house)
Analytical records	Test results with raw data
Equipment logbooks	Use, cleaning, maintenance, calibration
Stock and distribution records	Receipt, issue, sale
Complaint and recall records	Investigation and action
Training records	Who was trained in what, and when
Deviation and change control records	Departures from procedure and approved changes

B4.6.6 Quality certifications beyond Schedule T

Certification	What it is
WHO-GMP and **Certificate of Pharmaceutical Product (COPP)**	For export: CDSCO issues the COPP for Ayurvedic products intended for export, after a joint inspection by CDSCO, the Ministry of Ayush and the State Licensing Authority [Lok Sabha 2024]
AYUSH Mark (AYUSH Standard Mark and AYUSH Premium Mark)	Voluntary product certification operated by the Quality Council of India, started in 2009; the Standard Mark is based on Schedule T GMP and the Premium Mark on WHO GMP guidelines for herbal medicines [Lok Sabha 2024; Ayush 2022]
ISO 9001; other standards	Quality management systems adopted voluntarily

B4.6.7 GMP in the Rasashala: special points

1. **Segregation**: separate, well-ventilated areas for Shodhana, Marana and Kupipaka of mercury, arsenic and lead, with exhaust and emission control (B4.2.8).
2. **Controlled stores** for Rasa Dravya and Visha Dravya; issue against batch records; reconciliation of quantities.
3. **Process records** for every Puta: number, temperature or fuel used, duration, weight before and after; Bhasma Pariksha results for each batch.
4. **Equipment** dedicated to toxic materials where possible; validated cleaning.
5. **Worker health surveillance**, including periodic medical examination and, where exposure is significant, blood or urine metal levels.
6. **Testing** of finished Rasaushadhi for identity, phase and heavy metal content where required; retained samples.
7. **Labelling** with the cautions required for Schedule E(1) ingredients (Chapter B8).
8. **Waste**: metal-containing residues handled as hazardous waste.

AYURVEDA CORRELATION: THE RASASHALA AND THE DRUG STORE IN THE CLASSICS

The Rasa texts describe the **Rasashala**: its site, its building and the placement of equipment for different operations, and the qualities of the teacher, the pupil and the attendants who work in it (Rasaratna Samuchchaya) [R3]. Charaka directs that collected drugs be stored in clean, closed containers in a store protected from smoke, rain, wind and dust (Charaka Samhita, Kalpa Sthana 1) [R26]. These are the ancestors of the GMP requirements for premises, storage and personnel.

Established. The classical texts lay down rules for the site, layout and conduct of the pharmacy and for storage of drugs.

To be remembered. Classical rules did not include written batch records, validation or independent quality control. Modern GMP adds these, and they are what make quality consistent from batch to batch.

KEY POINTS TO REMEMBER

- Management: planning, organising, staffing, directing, coordinating, reporting, budgeting (POSDCORB) and controlling. Fayol’s fourteen principles.

- Ayurvedic unit: plan product mix, capacity, location, layout (one-way flow), licences, finance, raw material sourcing; departments of production, QC, QA, stores, maintenance, HR, regulatory, marketing, finance.
- PPC: planning, routing, scheduling, dispatching, follow-up, inspection. Long-cycle products (Asava, Bhasma, Kupipakva) need advance planning.
- Safety in the Rasashala: exhaust, PPE, mercury spill procedure, locked stores, fire safety, health surveillance, hazardous waste.
- Stores: receipt → quarantine → sampling → QC → approved or rejected; FEFO; bin cards and stock registers; pest control; locked Rasa and Visha store.
- Inventory: ordering, carrying and stock-out costs; ABC (value), VED (criticality), FSN, HML, SDE, GOLF; EOQ = √(2DS/H); reorder level, safety stock, maximum and minimum levels.
- Personnel: planning, recruitment, training (with records), appraisal, motivation (Maslow, Herzberg), welfare, medical examination.
- GMP: quality built in, not tested in. Write, follow, document, validate, design, maintain, train, clean, control, audit.
- Schedule T (pre-2026): Part I general requirements A–N; Part II machinery, minimum areas, QC equipment, supplementary guidelines for Rasaushadhi.
- Drugs (Eleventh Amendment) Rules, 2026: rewritten Schedule T for Ayurveda, Siddha, Sowa-Rigpa and Unani; compliance by 31 July 2029; stronger Rasaushadhi safeguards; risk-based inspection; perpetual licences.
- COPP for export after joint inspection; AYUSH Standard and Premium Marks (QCI, voluntary, since 2009).

EXAM CORNER — CHAPTER B4

Likely long questions (20 marks)

1. Write in brief the management of pharmacy and Good Manufacturing Practices related to the Ayurvedic drug industry *(set in September 2020)*.
2. Discuss in detail Good Manufacturing Practices related to the Ayurvedic drug industry *(set in August 2021, August 2022 and January 2024)*.
3. Explain the Drugs and Cosmetics Act with relevance to ASU drugs, along with Schedule T and GMP *(set in September 2018; see also Chapter B8)*.
4. Discuss the management of an Ayurvedic pharmacy, including aspects of international trade and regulatory affairs related to export *(set in March 2026; see also Chapter B9)*.
5. Describe store and inventory management in an Ayurvedic pharmacy.

Likely short notes (5 marks)

1. Significance of shelf-life and storage conditions for herbo-mineral preparations *(set in March 2026; see Chapter B3)*.
2. ABC and VED analysis.
3. Economic order quantity.
4. Batch manufacturing record.
5. Schedule T.
6. Personnel management in a pharmaceutical unit.

7. Safety in a Rasashala.
8. AYUSH Mark.

How to attempt

- Begin with definitions of management and of GMP.
- For "management of pharmacy", cover planning, layout (with the flow diagram), organisation chart, stores, inventory (ABC–VED, EOQ) and personnel, briefly but in tables.
- For GMP, give the principles, then Schedule T Part I heads in a table, then Part II, then the 2026 revision and the special points for Rasaushadhi.
- Add documentation and certifications (COPP, AYUSH Mark).

Answer frame: "Explain Good Manufacturing Practices related to the Ayurvedic drug industry." (20 marks)

Part	What to write	Suggested marks
Definition and need	WHO definition; "quality built in"; why GMP matters for ASU drugs	2
Principles	The ten principles; figure	3
Schedule T, Part I	Table of heads A–N with requirements	5
Schedule T, Part II	Machinery and minimum areas; QC section; Rasaushadhi guidelines	3
2026 revision	Key changes; compliance deadline	2
Documentation	Master formula, BMR, SOPs, records	2
Rasashala GMP and safety	Special points	2
Certifications	COPP; AYUSH Mark	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **3, 26, 27, 28, 42, 43**.

- **[R3]** *Rasaratna Samuchchaya* (Rasashala). **[R26]** Charaka Samhita, Kalpa Sthana 1.
- **[R27]** *Remington: The Science and Practice of Pharmacy*; **[R28]** Lachman L, Lieberman HA, Kanig JL. *The Theory and Practice of Industrial Pharmacy* (production management, GMP).
- **[R42]** Current Good Manufacturing Practices. **[R43]** Drugs and Cosmetics Act, 1940 and Rules, 1945: **[Schedule T]** Good Manufacturing Practices for Ayurvedic, Siddha and Unani medicines (pre-2026 text).

Additional verified sources (accessed 16 October 2026)

- **[Draft 2025]** Ministry of Ayush. Draft Drugs (Amendment) Rules, 2025 (revision of Schedule T). G.S.R. 140(E), 14 February 2025 (as summarised by Pharmadocx Consultants).
- **[Gazette 2026a]** Ministry of Ayush. Drugs (Eleventh Amendment) Rules, 2026. Gazette of India, Extraordinary; Gazette ID CG-DL-E-25072026-274789 (listing date 25 July 2026, gazettetracker.com).
- **[Corpseed 2026]** Corpseed. "Ayush Drugs Eleventh Amendment Rules 2026 introduce new compliance requirements." 29 July 2026. (Secondary summary.)
- **[Lexplosion 2026]** Lexplosion Solutions. "Drugs (Eleventh Amendment) Rules, 2026 revise product identification, labelling, shelf-life and GMP requirements for ASU drugs." 2026. (Secondary summary.)
- **[Lok Sabha 2024]** Lok Sabha, Government of India. Unstarred Question No. 1318, answered 9 February 2024 (production and distribution of AYUSH medicines).

- **[Ayush 2022]** Ministry of Ayush. Notification on the Ayush Mark certification scheme, 22 July 2022 (as reported by TeamLease RegTech).
- World Health Organization. *WHO Guidelines on Good Manufacturing Practices (GMP) for Herbal Medicines*. Geneva: WHO; 2007.

Viva pointers — Chapter B4

Answer each in one or two sentences.

- Define management. What does POSDCORB stand for?
- Name five of Fayol’s principles with an example from a pharmacy.
- Why must material flow in one direction in a manufacturing unit?
- Which departments of a pharmaceutical unit must be headed by different persons?
- What is production planning and control? Name its steps.
- What is break-even point?
- Distinguish QC from QA.
- What is validation? What is CAPA?
- Name four hazards in a Rasashala and how to control them.
- What happens to a raw material from receipt to approval?
- What do the colours of status labels mean?
- What is a bin card?
- What is ABC analysis? What is VED analysis? Combine them.
- State the EOQ formula. Why does it not suit seasonal herbs?
- How is the reorder level calculated?
- What is safety stock?
- Name the steps of personnel management.
- What is Maslow’s hierarchy? What are Herzberg’s hygiene factors?
- Define GMP. Why can quality not be tested into a product?
- Under which Schedule are GMP for ASU drugs laid down?
- Name the heads of Part I of Schedule T.
- What does Part II of Schedule T contain?
- What changed in Schedule T in 2026? By when must existing units comply?
- What is a batch manufacturing record?
- What is the AYUSH Premium Mark? Who operates it?
- Who issues the COPP for Ayurvedic exports?

PART B · PHARMACY MANAGEMENT · CHAPTER B5

B5 · PHARMACEUTICAL MARKETING, PRODUCT RELEASE AND WITHDRAWALS

SYLLABUS POINT (PART B, 5)

Pharmaceutical Marketing, product release and withdrawals.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B5	Pharmaceutical Marketing, Product Release and Withdrawals	0	1	Jul 2018

Why this chapter matters to you. A medicine helps no one until it reaches the patient, and it harms many if it reaches them on false claims or in a defective batch. This chapter covers the three gates a product passes through after manufacture: its **marketing**, its **release** for sale, and, when something goes wrong, its **withdrawal** from the market. It was set as a short note in the July 2018 paper.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define pharmaceutical marketing and describe its elements, channels and ethics, with special reference to Ayurvedic products.
2. Describe the process of batch release of a pharmaceutical product, and the launch of a new product.
3. Describe product withdrawal and recall: types, classes, procedure and responsibilities.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	B, Q.3c	5	Product release and withdrawals.

B5.1 Pharmaceutical marketing

> **Pharmaceutical marketing** is the set of activities by which a company identifies the needs of patients and prescribers, develops products to meet them, prices and distributes them, and informs professionals and the public about them, within the limits set by law and ethics.

Medicines are not ordinary goods. The patient usually does not choose the product; the doctor does. The patient cannot judge quality by looking at it. A false claim may cost a life. Pharmaceutical marketing is therefore more tightly regulated than marketing of most other goods.

B5.1.1 The marketing mix (the four Ps)

Element	Meaning	In Ayurvedic pharmacy
Product	What is sold: the drug, dosage form, pack, brand, quality	Classical formulations (sold under the classical name) and proprietary formulations (brand names); dosage form choice (B2); packing (B3)
Price	What the customer pays	Cost of raw materials (some very costly: Swarna, Kesara, Kasturi substitutes); competition; affordability
Place	How the product reaches the customer	Distribution channels; hospitals; Ayurvedic stores; online sales; export
Promotion	How the customer is informed	Medical representatives, literature, conferences, samples, advertisements, digital media: all within the advertising laws (Chapter B8)

Modern marketing adds **People**, **Process** and **Physical evidence** for services, and many writers place the **Patient** at the centre.

B5.1.2 Market research and segmentation

Before a product is launched, the company studies the market: the disease burden, existing products, prescribers' habits, patients' preferences and the regulatory position. The market is divided into **segments** (by therapeutic area, by prescriber group, by region, by income) and the product is **positioned** for one or more of them.

B5.1.3 The product life cycle

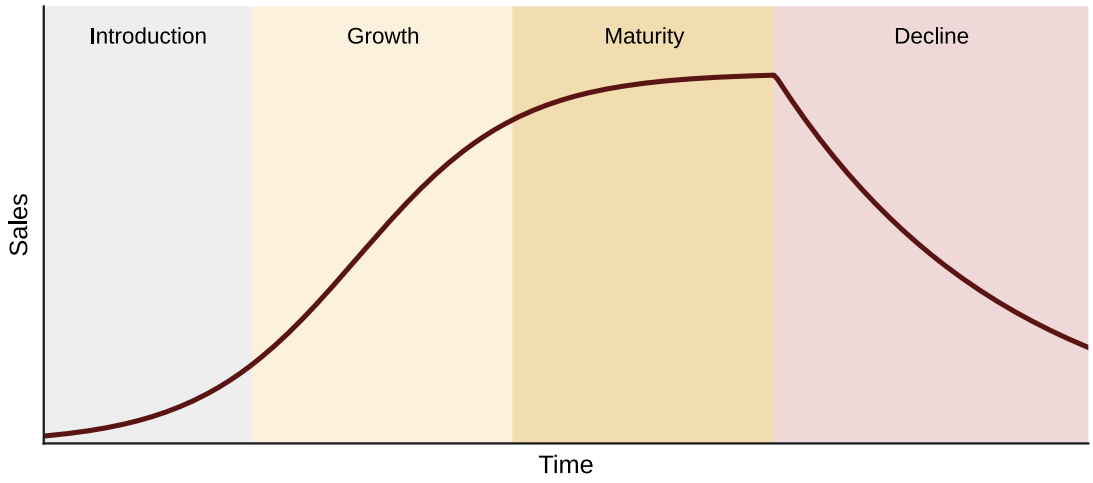


Figure B5.1. The product life cycle, with typical marketing activity at each stage.

Stage	Sales	Marketing focus
Introduction	Low, rising slowly	Awareness among prescribers; samples; scientific literature
Growth	Rising fast	Wider distribution; brand building; new indications or strengths
Maturity	Stable, high	Defend share; line extensions (new pack sizes, dosage forms)
Decline	Falling	Reduce cost; withdraw or reposition

Many classical Ayurvedic formulations have been in the "maturity" stage for centuries, which is unusual for a pharmaceutical product. Proprietary Ayurvedic products follow the ordinary life cycle.

B5.1.4 Distribution channels

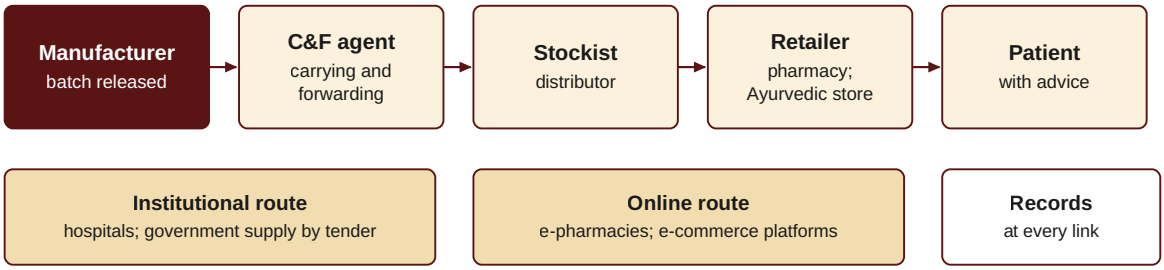


Figure B5.2. Distribution channel for medicines in India.

A typical channel is: **manufacturer → carrying and forwarding agent → stockist or distributor → retailer (pharmacy) → patient**. Hospitals and government supply systems buy directly or by tender. Online pharmacies and e-commerce platforms are a growing channel. Every link must store the product under the labelled conditions and keep records, so that a batch can be traced and recalled if needed (B5.3).

B5.1.5 Promotion and its ethics

Tool	Use	Ethical limits
Medical representatives	Inform doctors about products	Accurate information; no inducements
Scientific literature	Product monographs, published studies	Claims must be supported by evidence; no selective quotation
Continuing medical education, conferences	Education	Must not be disguised promotion; transparent sponsorship
Free samples	Trial of a product	Limited quantities; not for sale
Advertisements	To professionals; to the public where allowed	No advertisement to the public for the diseases listed in the advertising law; no misleading claims (Chapter B8)
Digital media	Websites, social media, influencers	Same rules apply; endorsements must not mislead

For modern medicines, India's Department of Pharmaceuticals issued the **Uniform Code for Pharmaceutical Marketing Practices, 2024**, which restricts gifts, travel and hospitality to healthcare professionals and requires that promotional claims be accurate and supported by evidence [UCPMP 2024].

For **Ayurvedic, Siddha and Unani** products, the main controls on promotion are the **Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954**, the provisions on advertisements in the **Drugs and Cosmetics Rules**, and consumer protection law on misleading advertisements (Chapter B8). The Ayush pharmacovigilance network monitors misleading advertisements (A6.8.3).

CLAIMS TO AVOID

- "Cures diabetes", "cures cancer", "permanent cure": prohibited for the listed diseases and unsupported.
- "100% safe", "no side effects", "because it is natural": unsupported and misleading (Chapters A4–A6).
- "Clinically proven" without a published, controlled study.
- Testimonials presented as evidence.
- Use of the name or image of a government body to suggest approval.

B5.1.6 Special features of Ayurvedic marketing

1. **Classical and proprietary products.** Classical formulations are sold under their textual names (for example, Arogyavardhini Vati), so companies compete on quality and trust rather than on a unique product. Proprietary products carry brand names and need evidence for their claims.
2. **Over-the-counter sale.** Many Ayurvedic products are bought without prescription. Labels, leaflets and pharmacists' advice therefore carry more weight.
3. **Practitioner prescribing.** Ayurvedic physicians prescribe both classical and proprietary products; many also dispense.
4. **Export.** International markets require product registration, quality documentation and compliance with foreign rules on herbal products and heavy metals (Chapter B9).
5. **Health supplements.** Some products are marketed as foods or nutraceuticals under food law, with different rules on claims (Chapter B8).

B5.2 Product release

The term "release" is used in two senses: the **release of a batch** for sale after quality control, and the **launch** of a new product on the market.

B5.2.1 Batch release

A batch of medicine may be sold only after it has been **released** by the quality function of the company. Release is the final quality decision on a batch.

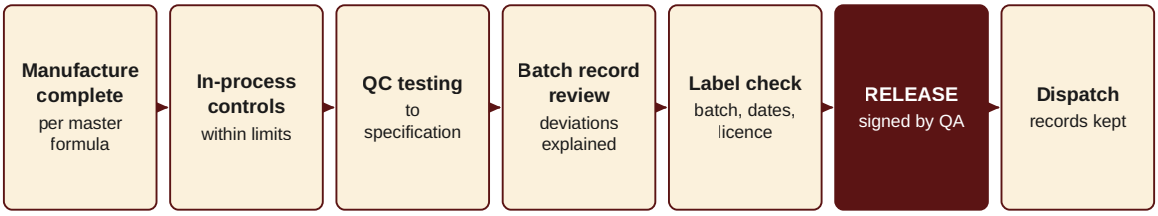


Figure B5.3. Batch release: the steps from completion of manufacture to dispatch.

Step	What is checked
1. Manufacturing complete	All stages done as per the master formula and standard operating procedures
2. In-process controls	Results within limits (weight, hardness, Paka lakshana, alcohol content as applicable)
3. Finished product testing	Quality control tests to the specification: identity, assay or markers, physico-chemical tests, microbial limits, heavy metals where required
4. Batch record review	The batch manufacturing record (BMR) and packing record are complete, signed and free of unexplained deviations
5. Labelling check	Correct labels, batch number, dates, licence number and cautions (Chapter B8)
6. Retained sample	A sample of the batch kept for reference
7. Release decision	Authorised quality person signs the release; status changed from “quarantine” to “released”
8. Dispatch	Stored and transported under labelled conditions; distribution records kept

Until release, the batch is held under **quarantine**. A batch that fails is **rejected**, and either reprocessed (if permitted and justified) or destroyed, with records. The GMP requirements behind each step are described in Chapter B4.

B5.2.2 Launch of a new product

1. Regulatory approval or licence for the product (Chapter B8).
2. Validation of the manufacturing process; stability data supporting the shelf life (Chapters B3, B4).
3. Market research, positioning, pricing.
4. Training of the sales force; preparation of literature that complies with the law.
5. Stocking of the distribution chain before the launch date.
6. Launch activities (scientific meetings, information to prescribers).
7. Post-launch monitoring of sales, complaints and adverse reactions (Chapter A6).

B5.3 Product withdrawal and recall

> A **recall** is the action taken to remove from the market, or from use, a batch or a product that is defective, unsafe, or in breach of the law. A **market withdrawal** is the removal of a product for reasons that do not

involve a violation of the law or a risk to safety (for example, a commercial decision), though the terms are often used loosely.

B5.3.1 Reasons for recall

- Failure of a quality test after release (not of standard quality).
- Contamination (microbial, heavy metal, foreign matter).
- Mislabelling, wrong batch number or wrong expiry date; mix-up of products.
- Spurious or adulterated product found in the market.
- Stability failure before the expiry date.
- New safety information (serious adverse reactions, Chapter A6).
- Regulatory action: cancellation of a licence or prohibition of a product.

B5.3.2 Classes of recall

Regulators commonly classify recalls by the risk to health. India's Central Drugs Standard Control Organisation (CDSCO) has issued guidelines on recall and a rapid alert system for drugs [CDSCO Recall].

Class	Risk	Example	Urgency
I	Use may cause serious harm or death	Wrong drug in the pack; dangerous contamination	Immediate; down to the patient level
II	Use may cause temporary or reversible harm, or harm is unlikely to be serious	Sub-potent product for a non-critical condition; microbial limit exceeded	Rapid; to the retail level
III	Use is unlikely to cause harm	Minor labelling defect; packaging defect without effect on quality	Within a reasonable time; to the wholesale level

The exact time limits and levels are laid down in the regulator's guidelines; consult the current version.

B5.3.3 Types of recall

Type	Initiated by
Voluntary recall	The manufacturer, on its own finding (complaint, test failure)
Statutory (mandatory) recall	The regulator, by order
Recall depth	Wholesale level; retail level; patient (consumer) level, according to the class

B5.3.4 The recall procedure

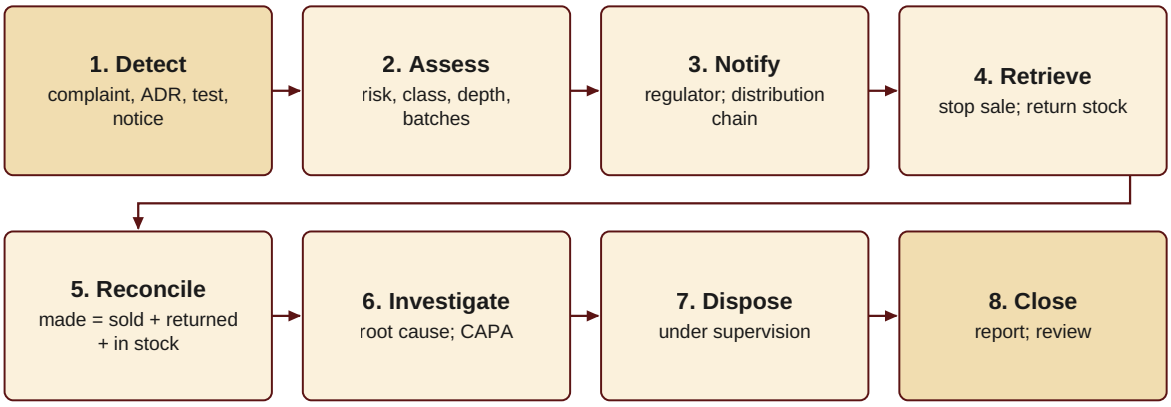


Figure B5.4. Steps in a product recall.

1. **Detection** of the problem: complaint, adverse reaction report, test result, regulatory notice.
2. **Assessment** by a recall committee: risk, class, depth, batches affected.
3. **Decision** and **notification** to the regulator (the State Licensing Authority for ASU drugs) and to the distribution chain.
4. **Retrieval**: stockists and retailers stop sale, quarantine and return stock; public notice if needed.
5. **Reconciliation**: quantity made, quantity distributed, quantity returned; records of every consignee.
6. **Investigation and corrective action** (CAPA): root cause; prevention of recurrence.
7. **Disposal** of recalled stock under supervision, with records.
8. **Closure report** to the regulator; review of effectiveness.

Prerequisites for an effective recall are written recall procedures, complete **distribution records**, batch numbering on every pack, a designated recall officer, and periodic **mock recalls** to test the system. These are part of GMP (Chapter B4).

AYURVEDA CORRELATION

The classical physician was usually also the maker of the medicine, so the questions of marketing and recall did not arise in their modern form. But the texts insist that only a drug of good quality (Sampat) should be used (Charaka Samhita, Sutra Sthana 9) [R26], and that a drug past its period of potency (Saviryata Avadhi) loses its value [R11]. Modern batch release and recall serve the same ends: no defective medicine should reach a patient, and one that has reached the market should be taken back quickly.

Established. Batch release and recall are legal and GMP requirements for licensed ASU manufacturers.

To be remembered. Recall works only if distribution records are kept. A product sold without batch records cannot be recalled.

KEY POINTS TO REMEMBER

- Pharmaceutical marketing is more tightly regulated than ordinary marketing: the prescriber chooses, the patient cannot judge quality, and false claims harm.

- Marketing mix: product, price, place, promotion (plus people, process, physical evidence; the patient at the centre).
- Product life cycle: introduction, growth, maturity, decline.
- Channel: manufacturer → C&F agent → stockist → retailer → patient; records at every step.
- Modern medicines: Uniform Code for Pharmaceutical Marketing Practices, 2024. ASU products: DMR Act 1954, Drugs and Cosmetics Rules, consumer protection law (Chapter B8).
- Batch release: in-process controls, finished product tests, batch record review, labelling check, retained sample, signed release; quarantine until then.
- Recall: removal of a defective or unsafe product. Classes I (serious harm), II (temporary or unlikely serious harm), III (unlikely harm). Voluntary or statutory.
- Recall steps: detect, assess, decide and notify, retrieve, reconcile, investigate (CAPA), dispose, close. Mock recalls test the system.

EXAM CORNER — CHAPTER B5

Likely long questions (20 marks)

1. Describe pharmaceutical marketing with special reference to Ayurvedic products. Add notes on product release and withdrawal.
2. What is a product recall? Describe the classes, types and procedure of recall, and the GMP requirements that make it possible.

Likely short notes (5 marks)

1. Product release and withdrawals *(set in July 2018)*.
2. Marketing mix.
3. Product life cycle.
4. Batch release.
5. Classes of recall.
6. Ethics of pharmaceutical promotion.

How to attempt

- Define each term (marketing, release, recall) first.
- For “product release and withdrawals”, give the batch release table and the recall classes table, with the recall steps as a flow chart.
- Add one line on the laws governing advertisement of Ayurvedic products.

Answer frame: “Product release and withdrawals.” (5 marks)

Part	What to write	Suggested marks
Definitions	Batch release; product launch; recall; market withdrawal	1
Release	Steps of batch release (table or flow); quarantine; authorised person	1.5
Withdrawal	Reasons; classes I–III; voluntary and statutory	1.5
Procedure	Recall steps; distribution records; mock recall	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **11, 26, 27, 42, 43, 44**.

- **[R11]** *Sharangadhara Samhita* (Saviryata Avadhi). **[R26]** Charaka Samhita, Sutra Sthana 9.
- **[R27]** *Remington: The Science and Practice of Pharmacy* (pharmaceutical marketing).
- **[R42]** Current Good Manufacturing Practices (Chapter B4). **[R43]** Drugs and Cosmetics Act, 1940 and Rules, 1945. **[R44]** Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954 (Chapter B8).

Additional verified sources (accessed 16 October 2026)

- **[UCPMP 2024]** Department of Pharmaceuticals, Government of India. *Uniform Code for Pharmaceutical Marketing Practices, 2024*.
- **[CDSCO Recall]** Central Drugs Standard Control Organisation. *Guidelines on Recall and Rapid Alert System for Drugs (including Biologicals and Vaccines)*. cdsco.gov.in.

Viva pointers — Chapter B5

Answer each in one or two sentences.

- Why is pharmaceutical marketing more regulated than marketing of soaps?
- Name the four Ps of marketing.
- Describe the product life cycle. At which stage are most classical formulations?
- Name the links in the distribution channel for medicines.
- What does the Uniform Code for Pharmaceutical Marketing Practices, 2024 restrict?
- Which law prohibits advertisements claiming to cure certain diseases?
- What is the difference between a classical and a proprietary Ayurvedic product in marketing?
- What is batch release? Who signs it?
- What does “quarantine” mean in a pharmacy store?
- What is a retained sample?
- Define recall. How does it differ from a market withdrawal?
- Describe the three classes of recall with an example each.
- What is a mock recall? Why is it done?
- Which records make a recall possible?
- What is CAPA?

PART B · PHARMACY MANAGEMENT · CHAPTER B6

B6 · HOSPITAL, DISPENSING AND COMMUNITY PHARMACY

SYLLABUS POINT (PART B, 6)

Hospital, Dispensing and Community pharmacy.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B6	Hospital, Dispensing and Community Pharmacy	0	1	Jul 2018

Why this chapter matters to you. Most Ayurvedic medicines reach patients through a hospital pharmacy, a dispensary or a community pharmacy. The postgraduate in Bhaishajya Kalpana may head the pharmacy of a teaching hospital, supervise in-house preparation for Panchakarma, or advise a community pharmacy. This chapter describes how each works. Community pharmacy was set as a short note in the July 2018 paper.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Describe the organisation, functions and drug distribution systems of a hospital pharmacy, including an Ayurvedic hospital pharmacy.
2. Describe the dispensing process, the parts of a prescription, dispensing errors and their prevention, and the special features of dispensing Ayurvedic medicines.
3. Describe the roles of a community pharmacy and good pharmacy practice.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	B, Q.3a	5	Short note: community Pharmacy.

B6.1 Hospital pharmacy

> **Hospital pharmacy** is the department of a hospital, under the direction of a qualified pharmacist, that is responsible for the selection, procurement, storage, preparation, dispensing and safe use of medicines and related supplies for inpatients and outpatients.

B6.1.1 Objectives

1. Ensure that the right medicine, of assured quality, is available to the right patient at the right time and in the right dose.
2. Procure and store medicines economically and safely.
3. Dispense accurately and counsel patients.
4. Prepare medicines that are not available commercially (extemporaneous and in-house preparations).
5. Provide drug information to doctors, nurses and patients.
6. Promote rational use of medicines; take part in pharmacovigilance.
7. Train students and staff; support research.

B6.1.2 Organisation and layout

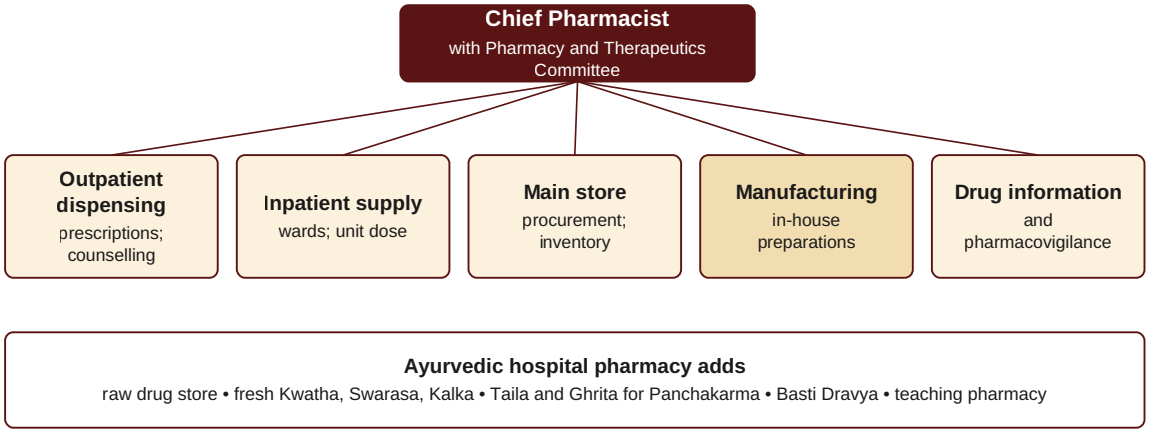


Figure B6.1. Organisation of a hospital pharmacy.

Element	Requirement
Location	Easily reached by outpatients and by wards; near the receiving area for supplies
Sections	Outpatient dispensing; inpatient (ward) supply; main store; manufacturing section; drug information; office and records
Staff	Chief pharmacist; pharmacists; pharmacy assistants; store keeper; clerical staff
Equipment	Shelving and racks; refrigerator for cold-chain items; balances; counting and packing aids; computer system for stock and prescriptions
Records	Purchase, stock, issue, expiry, narcotic and Schedule registers, adverse reaction reports

B6.1.3 Functions

Function	Description
Selection	A **hospital formulary** (list of approved medicines) prepared by a **Pharmacy and Therapeutics Committee** of doctors, pharmacists and nurses
Procurement	Purchase from licensed suppliers or by tender; quality and price both considered
Storage and inventory control	Proper conditions; FEFO (first expiry, first out); ABC and VED analysis; reorder levels (Chapter B4)
Dispensing	Outpatient and inpatient (B6.2)
In-house preparation	Preparations not available commercially; in an Ayurvedic hospital, Kwatha, Taila, Ghrita, Basti preparations, Lepa and other preparations for Panchakarma and daily use
Drug information	Answering questions on dose, interactions, availability, safety
Pharmacovigilance	Collecting and forwarding ADR reports (Chapter A6)
Education and research	Training of students; drug use studies

B6.1.4 Drug distribution systems for inpatients

System	How it works	Advantages	Disadvantages
Individual prescription order	Each patient's prescription is dispensed separately	Pharmacist reviews every order; little waste	Slow; more pharmacy work
Floor (ward) stock	Commonly used medicines kept on the ward	Quick availability	More errors, pilferage and expired stock; no pharmacist check
Combination	Floor stock for common items; individual orders for others	Balance of speed and control	—
Unit dose	Each dose packed and labelled for a patient, usually for 24 hours	Fewer errors; less waste; better control	Needs more staff and packaging
Automated dispensing	Cabinets or robots controlled by computer	Accuracy; records	Cost

B6.1.5 The Ayurvedic hospital pharmacy

The pharmacy of an Ayurvedic hospital has features of its own.

1. **In-house preparation** is common and essential: freshly prepared Kwatha, Swarasa, Kalka, Basti Dravya, Taila for Abhyanga, and medicines for special procedures cannot always be bought.
2. **Raw drug store**: authenticated, properly stored crude drugs, often in large quantities; protection from moisture, insects and fungi.
3. **Equipment** for classical preparations: Khalva, vessels for Kwatha and Sneha Paka, heating facilities, and for teaching, equipment for Rasashastra preparations.

4. **Quality control**: identity and quality of raw drugs; in-process checks (Paka lakshana); records of each batch.
5. **Panchakarma supplies**: medicated oils and ghee in bulk, with records of preparation and use.
6. **Integration with teaching** in Ayurvedic colleges: the hospital pharmacy and the teaching pharmacy support the training of students in Bhaishajya Kalpana.

Medicines prepared in-house for the hospital's own patients must still meet the standards of quality and the record-keeping that apply to any preparation (Chapters B4 and B8).

B6.2 Dispensing

> **Dispensing** is the preparation and supply of medicines to a patient on a prescription (or on request, for medicines that may be sold without prescription), together with the information and advice needed for their safe and effective use.

B6.2.1 The prescription

A prescription is a written order from a registered practitioner to a pharmacist to supply a medicine to a named patient, with directions for its use.

Part	Content
Date	Date of writing
Patient details	Name, age, sex, address; weight for children
Superscription	The symbol Rx ("take thou")
Inscription	Name and quantity (strength) of each medicine
Subscription	Directions to the pharmacist: dosage form and quantity to dispense
Signa (transcription)	Directions to the patient: dose, frequency, time, route, duration, Anupana
Prescriber details	Name, qualification, registration number, address, signature
Refill and other notes	Number of repeats, cautions

An Ayurvedic prescription adds the **Anupana** (vehicle), the **Bheshaja Kala** (time of administration: before, with or after food, at bedtime) and, often, **Pathya** (diet and conduct).

B6.2.2 Steps in dispensing

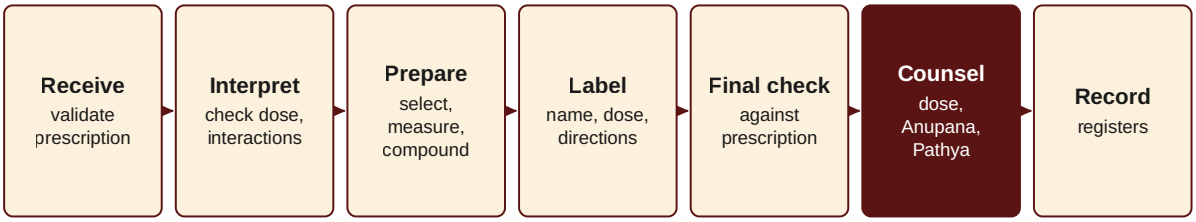


Figure B6.2. The dispensing process.

1. **Receive and validate** the prescription: is it legible, complete, genuine, and from a registered practitioner?
2. **Interpret and check**: medicine, dose, frequency, duration, interactions, contraindications; clarify doubts with the prescriber.
3. **Prepare**: select the product; check name, strength, batch and expiry; count, measure or compound; for Ayurvedic preparations, prepare extemporaneous forms (Kwatha from Yavakuta, mixing of Churna) where required.
4. **Label**: patient's name, medicine, dose and directions, date, pharmacy name, cautions ("For external use only", "Shake well before use").
5. **Final check**: a second check of the medicine against the prescription.
6. **Hand over and counsel**: explain the dose, time, Anupana, duration, possible side effects, storage, and what to do if a dose is missed; confirm understanding.
7. **Record**: prescription register; special registers for controlled drugs.

B6.2.3 Dispensing errors and their prevention

Type of error	Example	Prevention
Wrong medicine	Look-alike or sound-alike names	Separate storage; read the label three times; tall-man lettering
Wrong strength or dose	250 mg instead of 500 mg; Ratti and gram confused	Check units; standard conversion tables
Wrong patient	Two patients with similar names	Confirm name and age at hand-over
Wrong directions	Labels unclear	Standard, simple labels in the patient's language
Expired or damaged product	Old stock	FEFO; regular expiry checks
Missed interaction	Herb–drug interaction	Ask about all medicines taken (A1c.3)

B6.2.4 Weights and measures in Ayurvedic dispensing

Classical texts state doses in units such as Ratti, Masha, Tola and Pala. For practical work these are converted to metric units. The conversions adopted in the *Ayurvedic Formulary of India* are used in dispensing and manufacture [R37].

Classical unit	Metric equivalent
1 Ratti (Gunja)	125 mg
1 Masha	1 g
1 Tola (Karsha)	12 g
1 Pala	48 g
1 Kudava	192 g
1 Prastha	768 g

Check these values against the edition of the Ayurvedic Formulary of India in use; for liquids, some texts double the measure (Dravadvaiguna). The rule should be followed as laid down in the formulary.

B6.2.5 Counselling for Ayurvedic medicines

Point	What to tell the patient
Dose and time	Exact quantity; before or after food; at bedtime
Anupana	Warm water, honey, ghee, milk, as prescribed; how much
Duration	How long to take; not to continue beyond the advised period, especially for Rasaushadhi
Pathya	Diet and activity advice given by the physician
Storage	Airtight container, away from moisture and sunlight; shake liquids
Other medicines	Tell every doctor about all medicines; do not stop modern medicines without advice
Side effects	What to watch for; whom to contact; how to report

B6.3 Community pharmacy

> A **community pharmacy** (retail pharmacy) is a pharmacy that serves the general public directly, supplying medicines on prescription and without prescription, and providing advice and services for health.

B6.3.1 Roles of the community pharmacist

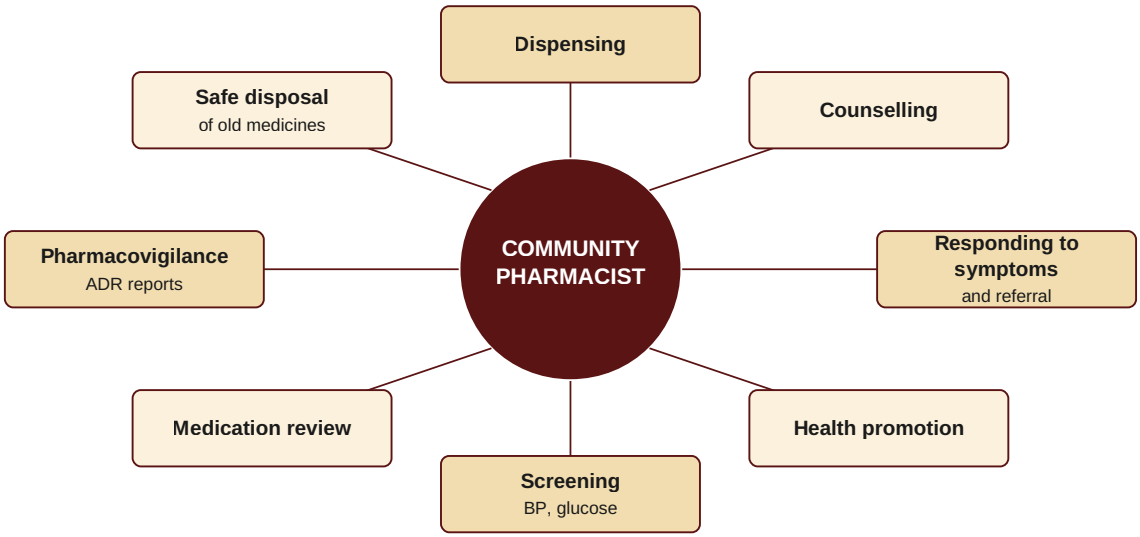


Figure B6.3. Roles of the community pharmacist.

Role	Description
Dispensing	Accurate supply on prescription
Counselling	Advice on correct use of medicines
Responding to symptoms	Advice on minor illnesses; supply of over-the-counter products; **referral** to a doctor when needed
Health promotion	Advice on diet, tobacco, alcohol, exercise; awareness campaigns
Screening	Blood pressure, blood glucose, body weight; identifying patients who need a doctor
Medication review	Checking the full list of medicines of chronic patients for duplication and interactions
Pharmacovigilance	Receiving and forwarding ADR reports (Chapter A6)
Safe disposal	Taking back expired and unused medicines
Supply management	Stocking, storage, expiry control, records

B6.3.2 Good pharmacy practice

Good pharmacy practice (GPP) means practice that meets the needs of the people who use pharmacy services, by providing optimal, evidence-based care. WHO and the International Pharmaceutical Federation have published joint guidelines on GPP. Its main elements are:

1. Promotion of health and prevention of disease.
2. Supply and use of medicines and health products.
3. Self-care: helping people to treat minor ailments safely.
4. Improvement of prescribing and medicine use.

B6.3.3 Premises and records

A community pharmacy must have clean premises of adequate size, protection from heat, light and moisture, a refrigerator where needed, proper shelving, and a separate area for counselling where possible. It must keep purchase and sale records, prescription records where required, and special registers for controlled substances. Licensing requirements for sale of drugs, and the role of the registered pharmacist under the Pharmacy Act, 1948, are described in Chapter B8.

B6.3.4 The Ayurvedic community pharmacy

Feature	Practice point
Many over-the-counter purchases	Counsel on correct use; refer when symptoms are serious, persistent or unclear
Classical and proprietary products	Know the composition, especially of Rasaushadhi and products containing toxic plants
Dispensing by the physician	Many Vaidyas dispense their own medicines; the same standards of labelling and records apply
Storage	Hygroscopic Churna, Avaleha and Sandhana products need care (Chapter B3)
Spurious and substandard products	Buy only from licensed manufacturers and distributors; check labels and batch numbers
Advertisements and claims	Do not repeat misleading claims; report them (Chapter A6)

AYURVEDA CORRELATION: THE PARICHARAKA AND THE PHARMACIST

Charaka lists the qualities of the attendant (Paricharaka or Upasthata), one of the four pillars of treatment: knowledge of nursing and of the preparation of medicines (Upacharajnata), skill (Dakshya), devotion to the patient (Anuraga) and cleanliness (Shaucha) (Charaka Samhita, Sutra Sthana 9) [R26]. These are the qualities of a good dispensing pharmacist today.

The time of giving medicine (Bheshaja Kala) and the vehicle (Anupana) are part of the classical prescription (Charaka Samhita, Chikitsa Sthana 30) [R26], and the dispensing pharmacist must pass them on to the patient.

Established. Correct dispensing and counselling reduce errors and improve adherence.

To be remembered. Pharmacy practice for Ayurveda is still largely informal. Written labels, counselling and records are as necessary for an Ayurvedic medicine as for any other.

KEY POINTS TO REMEMBER

- Hospital pharmacy: selection (formulary; Pharmacy and Therapeutics Committee), procurement, storage, dispensing, in-house preparation, drug information, pharmacovigilance, education.
- Inpatient distribution: individual order, floor stock, combination, unit dose (fewest errors), automated.
- Ayurvedic hospital pharmacy: in-house Kwatha, Taila, Ghrita, Basti preparations; raw drug store; batch records.
- Prescription: superscription (Rx), inscription, subscription, signa; plus Anupana, Bheshaja Kala and Pathya in Ayurveda.

- Dispensing steps: receive and validate, interpret and check, prepare, label, final check, counsel, record.
- AFI conversions: Ratti 125 mg; Masha 1 g; Tola 12 g; Pala 48 g; Kudava 192 g; Prastha 768 g.
- Community pharmacist: dispensing, counselling, responding to symptoms and referral, health promotion, screening, medication review, pharmacovigilance, disposal.
- Good pharmacy practice (WHO and FIP): health promotion; supply and use; self-care; improving prescribing.
- Charaka's four qualities of the attendant: Upacharajnata, Dakshya, Anuraga, Shaucha.

EXAM CORNER — CHAPTER B6

Likely long questions (20 marks)

1. Describe the organisation and functions of a hospital pharmacy, with special reference to the pharmacy of an Ayurvedic teaching hospital.
2. Describe dispensing and community pharmacy. Add a note on counselling for Ayurvedic medicines.

Likely short notes (5 marks)

1. Community pharmacy *(set in July 2018).*
2. Parts of a prescription.
3. Unit dose dispensing.
4. Pharmacy and Therapeutics Committee and the hospital formulary.
5. Dispensing errors.
6. Good pharmacy practice.
7. Classical weights and their metric equivalents.

How to attempt

- Define the term; give the roles or functions in a table; draw the flow (dispensing) or the organisation chart (hospital pharmacy).
- Add the Ayurvedic features in a separate paragraph or table: examiners look for them.

Answer frame: "Community pharmacy." (5 marks)

Part	What to write	Suggested marks
Definition	Community (retail) pharmacy	0.5
Roles	Table of roles of the community pharmacist	2
Good pharmacy practice	Four elements	1
Premises and records	Requirements; registered pharmacist (Pharmacy Act)	0.5
Ayurvedic community pharmacy	Counselling on Anupana and duration; OTC; quality of products	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **26, 27, 37, 47**.

- **[R26]** Charaka Samhita: Sutra Sthana 9; Chikitsa Sthana 30.
- **[R27]** *Remington: The Science and Practice of Pharmacy* (hospital and community pharmacy; dispensing).
- **[R37]** *The Ayurvedic Formulary of India* (weights and measures).
- **[R47]** Factories and Pharmacy Acts (Chapter B8).

Additional source

- World Health Organization and International Pharmaceutical Federation. *Joint FIP/WHO Guidelines on Good Pharmacy Practice: Standards for Quality of Pharmacy Services*. WHO Technical Report Series 961, Annex 8; 2011.

Viva pointers — Chapter B6

Answer each in one or two sentences.

- Define hospital pharmacy. Name five of its functions.
- What is a hospital formulary? Who prepares it?
- Compare floor stock and unit dose systems.
- What is FEFO?
- What does an Ayurvedic hospital pharmacy prepare in-house?
- Name the parts of a prescription.
- What extra instructions does an Ayurvedic prescription carry?
- List the steps in dispensing.
- Give three causes of dispensing errors and how to prevent them.
- How many milligrams is one Ratti? One Tola?
- What is Dravadvaiguna?
- Define community pharmacy. Name six roles of the community pharmacist.
- What is good pharmacy practice?
- When should a community pharmacist refer a patient to a doctor?
- What are Charaka’s four qualities of the attendant?

PART B · PHARMACY MANAGEMENT · CHAPTER B7

B7 · PATENTING AND INTELLECTUAL PROPERTY RIGHTS

SYLLABUS POINT (PART B, 7)

Patenting and Intellectual Property Rights.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B7	Patenting and Intellectual Property Rights	1	3	Sep 2018, Aug 2021, Aug 2022, Jan 2024

Why this chapter matters to you. Ayurveda is a vast store of traditional knowledge, and much of it has been claimed in patents abroad. A Rasashastra and Bhaishajya Kalpana scholar who develops a new formulation, process or dosage form needs to know what can and cannot be protected, and how to protect India's traditional knowledge from misappropriation. This chapter was set in four of the seven papers, once as a 20-mark question.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define intellectual property and describe its kinds and the laws that protect each in India.
2. Describe the Indian patent system: patentability, exclusions (including traditional knowledge), procedure, rights and duration.
3. Explain the international framework (TRIPS, the Paris Convention, the Patent Cooperation Treaty, the Convention on Biological Diversity and the Nagoya Protocol).
4. Explain the protection of traditional knowledge, with the turmeric, neem and basmati cases.
5. Explain the need for IPR awareness in Ayurveda and the patentability of Ayurvedic inventions.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2018	B, Q.3b	5	Need for awareness about Patenting and Intellectual Property Rights.
Aug 2021	B, Q.1	20	Explain in detail about the Patenting and Intellectual Property Rights
Aug 2022	B, Q.3b	5	Explain necessity and importance of IPR.
Jan 2024	B, Q.3c	5	Intellectual Property Rights.

B7.1 Intellectual property

> **Intellectual property** (IP) means creations of the mind (inventions, literary and artistic works, designs, symbols, names and images used in commerce) over which the law gives the creator exclusive rights for a limited period. **Intellectual property rights** (IPR) are those legal rights.

The purpose of IPR is to reward and encourage creativity and investment, while making the knowledge available to the public in return: a patent, for example, must disclose the invention fully, and after the patent expires anyone may use it.

B7.1.1 Kinds of intellectual property and the Indian laws

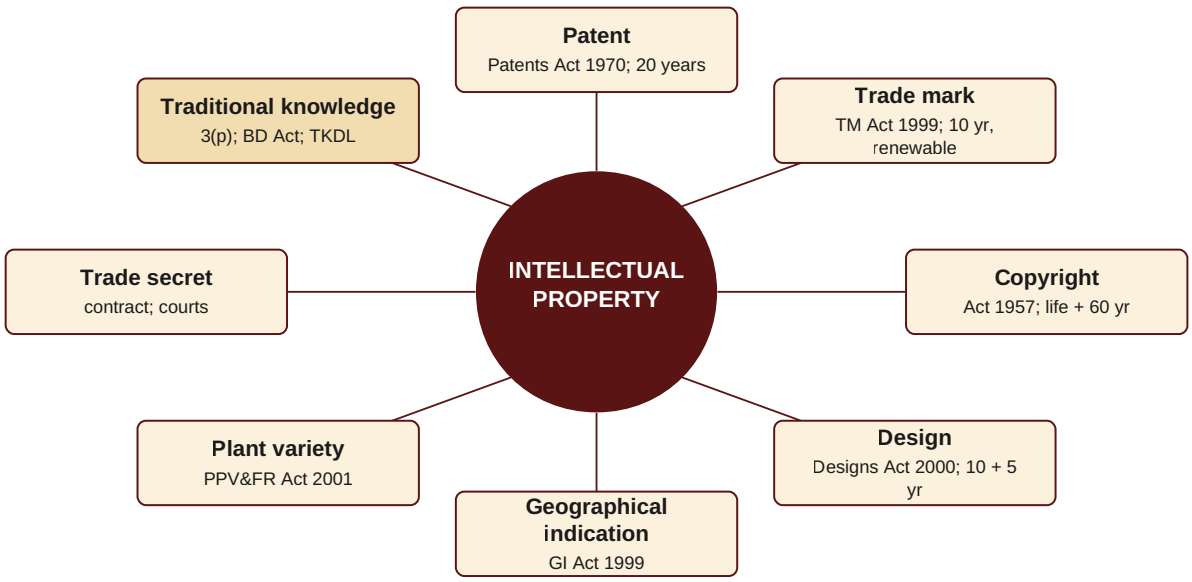


Figure B7.1. Kinds of intellectual property and the Indian laws that protect them.

Kind	What it protects	Indian law	Duration (general rule)
Patent	A new, inventive and industrially applicable product or process	Patents Act, 1970	20 years from the date of filing
Trade mark	A mark that distinguishes goods or services of one trader from another	Trade Marks Act, 1999	10 years, renewable indefinitely

Kind	What it protects	Indian law	Duration (general rule)
Copyright	Literary, artistic, musical and dramatic works; software; films	Copyright Act, 1957	Generally the author's life plus 60 years
Industrial design	The shape, pattern or ornament of an article (its appearance)	Designs Act, 2000	10 years, extendable by 5
Geographical indication (GI)	A name identifying goods as coming from a particular place, whose quality or reputation is due to that place	Geographical Indications of Goods (Registration and Protection) Act, 1999	10 years, renewable
Plant variety	New plant varieties; farmers' rights	Protection of Plant Varieties and Farmers' Rights Act, 2001	Varies by crop type
Layout design of integrated circuits	Semiconductor layouts	Semiconductor Integrated Circuits Layout-Design Act, 2000	10 years
Trade secret	Confidential business information (formulae, processes)	No specific statute; protected by contract and by the courts	As long as it remains secret
Traditional knowledge	Knowledge, practices and innovations of communities	No separate IP statute; defensive protection through the Patents Act (Section 3(p)), the Biological Diversity Act, 2002 and the TKDL (B7.5, Chapter B12)	—

B7.1.2 IPR in the Ayurvedic drug industry

Asset	Possible protection
A new process to make a Bhasma with less time or fuel	Patent (if new, inventive and useful)
A new herbal combination with proven, unexpected (synergistic) effect	Possibly a patent; must clear Section 3(e) and 3(p)
A brand name for a proprietary product	Trade mark
Shape of a bottle or tablet	Design
Product literature, labels, books, software	Copyright
Know-how of a family pharmacy	Trade secret
Classical formulations and their known uses	Not patentable; protected defensively through TKDL and the law
Regional products (for example, a medicinal rice or a regional herb)	Geographical indication

B7.2 The Indian patent system

B7.2.1 History

Year	Event
1856	First Indian legislation on protection of inventions (under British rule)
1911	Indian Patents and Designs Act
1970	**Patents Act, 1970** (in force 1972): only **process** patents for food, medicines and chemicals; this allowed the Indian generic industry to grow
1995	India joins the World Trade Organization; the **TRIPS Agreement** requires product patents in all fields of technology, with transition periods
1999, 2002	Amendments: “mailbox” applications for pharmaceutical products; patent term made 20 years; Section 3 expanded, including traditional knowledge (3(p))
2005	**Patents (Amendment) Act, 2005**: **product patents** for pharmaceuticals, food and chemicals; Section 3(d); pre-grant and post-grant opposition
2013	**Novartis v. Union of India**: the Supreme Court upholds the refusal of a patent for a new form of imatinib under Section 3(d)

B7.2.2 What is patentable

An invention is patentable in India if it is:

1. **Novel** (new): not published or used anywhere in the world before the date of filing;
2. **Inventive** (involves an inventive step): a technical advance or economic significance, not obvious to a person skilled in the art; and
3. **Capable of industrial application**: can be made or used in an industry;

and it is **not excluded** by Sections 3 and 4 of the Act.

B7.2.3 What is not patentable (Section 3, selected clauses)

Clause	Excluded subject matter	Relevance to Ayurveda
3(c)	Mere discovery of a scientific principle; discovery of a living thing or non-living substance occurring in nature	Discovering that a plant contains a compound is not an invention
3(d)	Mere discovery of a **new form of a known substance** that does not enhance its known **efficacy**; mere discovery of a new property or new use of a known substance	A new salt or crystal form of a known compound, without better efficacy, is not patentable
3(e)	A substance obtained by a **mere admixture** resulting only in the aggregation of the properties of the components	A mixture of herbs whose effect is only the sum of their known effects
3(h)	A method of agriculture or horticulture	—
3(i)	Any process for the medicinal, surgical or other **treatment** of human beings or animals	Methods of treatment such as Panchakarma procedures

Clause	Excluded subject matter	Relevance to Ayurveda
3(j)	Plants and animals in whole or in part, other than micro-organisms	A medicinal plant itself
3(p)	An invention which, in effect, is **traditional knowledge** or an aggregation or duplication of known properties of traditionally known components	Classical Ayurvedic formulations and their known uses

B7.2.4 Procedure for a patent in India

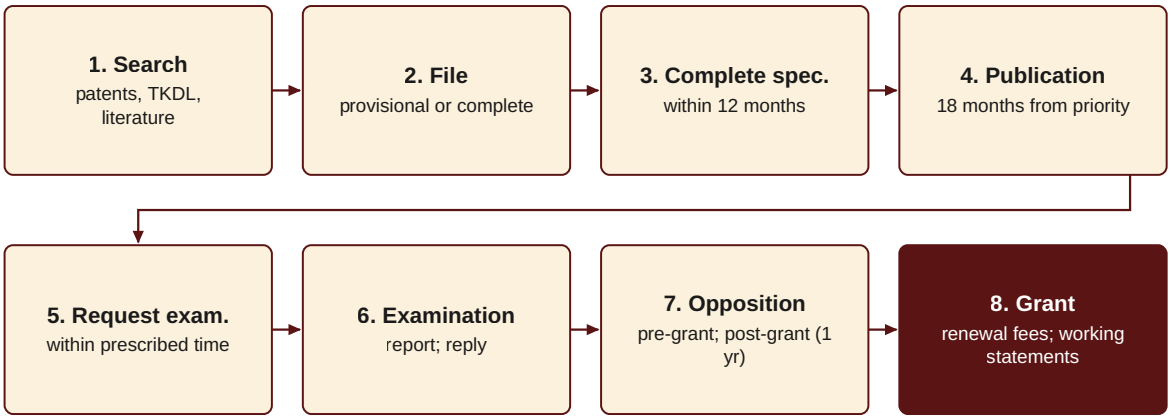


Figure B7.2. Steps from filing to grant of an Indian patent.

Step	What happens
1. Search	Search patent databases, TKDL and the literature to check novelty
2. Filing	Application with a **provisional** specification (to secure an early date) or a **complete** specification, at the appropriate Patent Office (Kolkata, Delhi, Mumbai, Chennai)
3. Complete specification	If a provisional was filed, the complete specification follows within 12 months
4. Publication	The application is published, ordinarily 18 months from the priority date (earlier on request)
5. Request for examination	The applicant must request examination within the period prescribed in the Patents Rules
6. Examination	The examiner issues a report; the applicant replies to objections
7. Opposition	**Pre-grant opposition** by any person after publication; **post-grant opposition** within one year of grant
8. Grant	Patent granted and published; renewal fees payable every year to keep it in force
9. Working	The patentee files statements on the working of the patent in India

Disclosure of biological material. When an invention uses biological material, the specification must disclose the source and geographical origin of the material (Section 10). For inventions based on biological resources or associated knowledge obtained from India, approval of the **National Biodiversity Authority** is required under the Biological Diversity Act, 2002 (B7.4).

B7.2.5 Rights of the patentee and limits

The patentee may prevent others from making, using, offering for sale, selling or importing the patented product (or the product of a patented process) in India for the term of the patent. The rights are limited by:

- **compulsory licences** (Section 84) when the reasonable requirements of the public are not met, the invention is not available at a reasonably affordable price, or it is not worked in India;
- the **research exemption** and the **Bolar provision** (Section 107A), which allow use for research and for obtaining regulatory approval;
- **revocation** on grounds such as lack of novelty or wrongful obtaining.

India's first compulsory licence was granted in 2012 for sorafenib, an anticancer drug.

B7.3 The international framework

Instrument	Year	Essence	India
Paris Convention for the Protection of Industrial Property	1883	National treatment; **right of priority** (12 months for patents)	Member since 1998
World Intellectual Property Organization (WIPO), Geneva	1967	UN agency administering IP treaties	Member
Patent Cooperation Treaty (PCT)	1970	One international application, later entering many countries (national phase)	Member since 1998
TRIPS Agreement (World Trade Organization)	1995	Minimum standards of IP protection for all WTO members; patents in all fields of technology; 20-year term	Bound since 1995; full product patents from 2005
Convention on Biological Diversity (CBD)	1992	Sovereign rights of states over their biological resources; access and benefit sharing; respect for traditional knowledge	Party; Biological Diversity Act, 2002
Nagoya Protocol on Access and Benefit-sharing	2010 (in force 2014)	Prior informed consent and mutually agreed terms for access to genetic resources and associated traditional knowledge	Party
WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge	2024	Patent applicants must disclose the origin or source of genetic resources and associated traditional knowledge on which a claimed invention is based [WIPO 2024]	—

B7.4 The Biological Diversity Act, 2002

The Biological Diversity Act, 2002 gives effect to the CBD in India. Its three objectives are the **conservation** of biological diversity, its **sustainable use**, and the **fair and equitable sharing of benefits** arising from the use of biological resources and associated knowledge.

Body	Level	Role
National Biodiversity Authority (NBA), Chennai	National	Approves access by foreign persons and entities; approves IPR applications based on Indian biological resources; benefit sharing
State Biodiversity Boards	State	Regulate access by Indian entities for commercial use
Biodiversity Management Committees	Local body	Document local biodiversity and knowledge in **People's Biodiversity Registers**

The Act was amended in 2023. For the Ayurvedic industry, the points to know are that biological resources used for medicines are subject to the access and benefit-sharing provisions, that an application for a patent or other IPR based on Indian biological resources or associated knowledge needs the approval of the NBA, and that the amendment changed the conditions for several categories of users and resources. Read the current text of the Act and its rules before any commercial use.

B7.5 Traditional knowledge and biopiracy

Traditional knowledge (TK) is the knowledge, innovations and practices of communities, developed over generations and often transmitted orally or through classical texts. Ayurveda is codified TK: it is written in texts many centuries old.

Biopiracy is the appropriation of biological resources or traditional knowledge, usually through patents, without the consent of, or benefit to, the communities or the country of origin.

B7.5.1 Landmark cases

Case	What happened	Outcome	Lesson
Turmeric (Haridra)	A United States patent granted in 1995 claimed the use of turmeric powder for healing wounds	India's Council of Scientific and Industrial Research challenged it, citing ancient texts and published literature; the patent was **revoked in 1997**	Documented prior art defeats a patent on traditional use
Neem (Nimba)	A European patent for a fungicidal use of neem oil extract	Opposed by India and others; **revoked** by the European Patent Office in 2000, upheld on appeal in 2005	Traditional use known in India is prior art
Basmati rice	A United States patent (1997) on rice lines and grains, using the name "Basmati"	After India's challenge, most claims were withdrawn or rejected; the patent was narrowed	Protection of names through geographical indications
Jeevani (Kani tribe, Kerala)	A tonic developed from *Trichopus zeylanicus* (Arogyappacha), whose use was learnt from the Kani tribe	The research institute agreed to share licence fees and royalties with the tribe through a trust	A model of benefit sharing

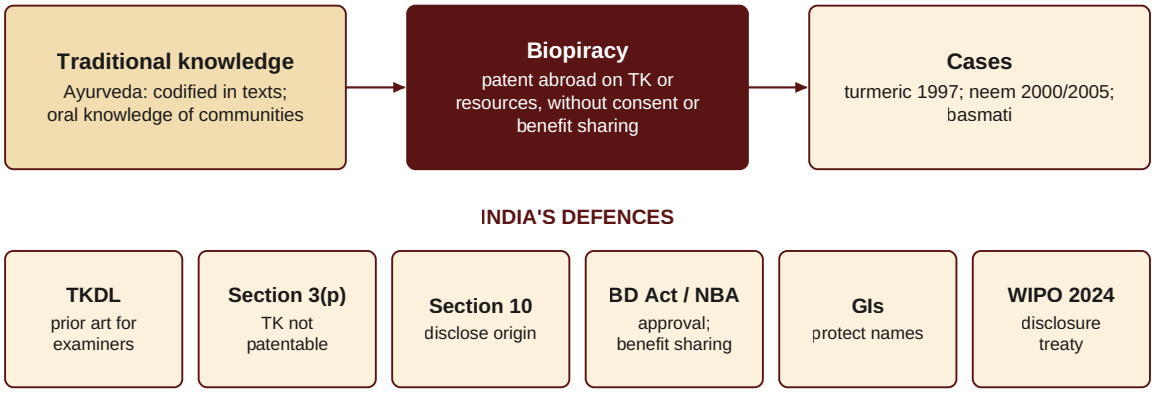


Figure B7.3. The problem of biopiracy and India's defences.

B7.5.2 How India protects its traditional knowledge

1. **Defensive protection**: making TK available as **prior art** to patent examiners, so that patents claiming it are not granted. The **Traditional Knowledge Digital Library** (Chapter B12) does this for Ayurveda, Unani, Siddha, Sowa Rigpa and Yoga.
2. **Legal exclusion**: Section 3(p) of the Patents Act excludes TK from patentability in India.
3. **Disclosure**: Section 10 of the Patents Act requires disclosure of the source and origin of biological material.
4. **Access and benefit sharing**: the Biological Diversity Act and the Nagoya Protocol.
5. **Geographical indications** for regional products.
6. **International advocacy**: at WIPO and the WTO, for disclosure requirements and TK protection, leading to the 2024 WIPO treaty.

B7.6 IPR and the Ayurvedic scholar

B7.6.1 Why awareness of IPR is needed

The September 2018 and August 2022 papers asked about the need and importance of IPR awareness.

Reason	Explanation
To protect one's own innovations	New processes, dosage forms and devices developed in research can be patented; without awareness, early publication may destroy novelty
To avoid infringing others' rights	Brand names, packaging, processes and devices may be protected by others
To protect India's traditional knowledge	Recognise and oppose patents that claim TK
To use patent literature	Patents are a rich source of technical information (Chapter B1)
Commercialisation	Universities and institutes can license inventions; start-ups need IP strategy
Collaboration	Agreements with industry need clear terms on IP ownership
Access and benefit sharing	Use of biological resources and community knowledge requires approvals and benefit sharing
Ethics	Respect for the rights of communities and of other researchers

B7.6.2 Is an Ayurvedic invention patentable?

Situation	Usually patentable?	Reason
A classical formulation for its classical use	No	Section 3(p); not novel
A new use of a known herb	No (in India)	Section 3(d): new use of a known substance
A simple mixture of herbs, effects only additive	No	Section 3(e)
A combination with a demonstrated synergistic (more than additive) effect	Possibly	Must show the unexpected effect with data
A new process (for example, a faster, cleaner method of Marana) with technical advantage	Possibly	Novel and inventive process
A new dosage form with demonstrated better efficacy (for example, improved bioavailability)	Possibly	Must overcome Section 3(d) with efficacy data
A device for Panchakarma or pharmacy	Possibly	New, inventive device; or design protection for its appearance
A method of treatment	No	Section 3(i)

Practical rule for a scholar: before you publish or present a potentially valuable finding, discuss it with your institution's IPR cell. Publication first may make it impossible to obtain a patent later.

AYURVEDA CORRELATION: KNOWLEDGE AS A SHARED HERITAGE

The classical tradition passed knowledge from teacher to pupil and treated it as a trust to be used for the welfare of all living beings; Charaka's code for the physician emphasises service rather than gain (Charaka Samhita, Vimana Sthana 8) [R26]. The classical texts are therefore **public knowledge**: they cannot be patented by anyone, Indian or foreign. Modern IPR does not conflict with this; it protects new contributions while the classical heritage remains free to all, and the law now guards that heritage against misappropriation.

Established. Classical Ayurvedic knowledge is prior art and is excluded from patentability in India; documented prior art has defeated foreign patents on turmeric and neem.

To be remembered. Protection is defensive. It works only if knowledge is documented and accessible to examiners (Chapter B12).

KEY POINTS TO REMEMBER

- IP: patents, trade marks, copyright, designs, geographical indications, plant varieties, layout designs, trade secrets; traditional knowledge protected defensively.
- Indian laws: Patents Act 1970; Trade Marks Act 1999; Copyright Act 1957; Designs Act 2000; GI Act 1999; PPV&FR Act 2001; Biological Diversity Act 2002.
- Patent: novel, inventive, industrially applicable, not excluded; 20 years from filing.
- Section 3: (d) new forms or new uses of known substances without enhanced efficacy; (e) mere admixture; (i) methods of treatment; (j) plants; (p) traditional knowledge.

- Procedure: search, file (provisional or complete), publication at 18 months, request for examination, examination, pre- and post-grant opposition, grant, renewal, working statements.
- Limits: compulsory licence (Section 84; sorafenib 2012), research and Bolar exemptions (Section 107A).
- International: Paris Convention 1883 and PCT 1970 (India 1998); TRIPS 1995; CBD 1992; Nagoya Protocol 2010; WIPO GRATK Treaty 2024 (disclosure of origin).
- Biopiracy cases: turmeric (revoked 1997), neem (revoked 2000, upheld 2005), basmati (narrowed); benefit sharing: Jeevani and the Kani tribe.
- India's defences: TKDL, Section 3(p), Section 10 disclosure, NBA approval, GIs, international advocacy.
- Classical formulations are not patentable; new processes, synergistic combinations with data, improved dosage forms and devices may be.

EXAM CORNER — CHAPTER B7

Likely long questions (20 marks)

1. Explain in detail patenting and intellectual property rights *(set in August 2021)*.
2. What is biopiracy? Describe the protection of traditional knowledge in India with landmark cases.
3. Describe the procedure for obtaining a patent in India. Discuss the patentability of Ayurvedic inventions.

Likely short notes (5 marks)

1. Need for awareness about patenting and IPR *(set in September 2018)*.
2. Necessity and importance of IPR *(set in August 2022)*.
3. Intellectual property rights *(set in January 2024)*.
4. Section 3(d) and 3(p) of the Patents Act.
5. TRIPS Agreement.
6. Turmeric patent case.
7. Geographical indications.
8. Biological Diversity Act, 2002.

How to attempt

- Define IP and IPR; give the table of kinds with laws and durations.
- For patents: criteria, exclusions (Section 3 table), procedure (flow chart).
- Add the international framework and TK protection with at least two cases.
- End with the relevance to Ayurveda: what can and cannot be patented.

Answer frame: "Explain in detail patenting and intellectual property rights." (20 marks)

Part	What to write	Suggested marks
Definitions	IP; IPR; purpose	1
Kinds of IP	Table with laws and durations	3
Patent: criteria and exclusions	Novelty, inventive step, industrial application; Section 3 table	4
Patent procedure	Flow chart and table; opposition; rights and limits	4
International framework	Paris, PCT, TRIPS, CBD, Nagoya, WIPO 2024	3
Traditional knowledge	Biopiracy; turmeric and neem; TKDL; Section 3(p)	3
Relevance to Ayurveda	Patentability table; need for awareness	2

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **26**. The syllabus lists no specific source on IPR; the following official sources were used.

- **[R26]** Charaka Samhita, Vimana Sthana 8.

Additional sources (accessed 16 October 2026)

- Government of India. *The Patents Act, 1970* (as amended), and *The Patents Rules, 2003* (as amended). Office of the Controller General of Patents, Designs and Trade Marks, ipindia.gov.in.
- Government of India. *The Biological Diversity Act, 2002* (as amended in 2023). National Biodiversity Authority, nbaindia.org.
- World Trade Organization. *Agreement on Trade-Related Aspects of Intellectual Property Rights*. 1994 (in force 1995).
- **[WIPO 2024]** World Intellectual Property Organization. *WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge*. Adopted May 2024. wipo.int.
- Supreme Court of India. *Novartis AG v. Union of India*, judgment of 1 April 2013.

Viva pointers — Chapter B7

Answer each in one or two sentences.

- Define intellectual property. Name six kinds.
- What is the term of a patent in India? Of a trade mark? Of copyright?
- What are the three criteria of patentability?
- What does Section 3(d) exclude? Which case made it famous?
- What does Section 3(p) exclude?
- Is a classical Ayurvedic formulation patentable? Why not?
- What is a provisional specification?
- When is a patent application published?
- What are pre-grant and post-grant opposition?
- What is a compulsory licence? Give the first Indian example.
- What is the Bolar provision?
- What is TRIPS? When did it come into force?
- What is the PCT? When did India join?
- What is biopiracy? Describe the turmeric case.
- Which body approves patents based on Indian biological resources?
- What are People’s Biodiversity Registers?
- What is a geographical indication? Name one Indian GI.
- What did the WIPO treaty of 2024 require?
- Why should a scholar consult the IPR cell before publishing?

PART B · PHARMACY MANAGEMENT · CHAPTER B8

B8 · LAWS GOVERNING AYURVEDIC DRUGS

SYLLABUS POINT (PART B, 8)

Laws Governing Ayurvedic drugs

- **B8(i)** Relevant regulatory provisions of Ayurvedic drugs in Drugs and Cosmetics Act, 1940 and Rules - 1945
- **B8(ii)** Laws pertaining to Drugs and Magic remedies (Objectionable Advertisement) Act - 1954.
- **B8(iii)** Prevention of Food Adulteration (PFA) act.
- **B8(iv)** Food Safety and Standards Act, 2006
- **B8(v)** Laws pertaining to Narcotics
- **B8(vi)** Factories and Pharmacy Acts
- **B8(vii)** Consumer Protection Act -1986

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B8(i)	Drugs and Cosmetics Act, 1940 and Rules, 1945	4	0	Jul 2018, Sep 2018, Aug 2022, Jan 2024
B8(ii)	Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954	2	1	Sep 2020, Aug 2022, Jan 2024
B8(iii)	Prevention of Food Adulteration Act	2	1	Aug 2022, Jan 2024, Mar 2026
B8(iv)	Food Safety and Standards Act, 2006	2	0	Aug 2022, Jan 2024
B8(v)	Laws Pertaining to Narcotics	2	0	Aug 2022, Jan 2024
B8(vi)	Factories and Pharmacy Acts	2	0	Aug 2022, Jan 2024
B8(vii)	Consumer Protection Act	2	0	Aug 2022, Jan 2024

Why this chapter matters to you. Every Ayurvedic medicine made, sold, advertised or exported in India is governed by law. This chapter was set in six of the seven papers, four times as a 20-mark question: on the Drugs and Cosmetics Act and Rules, on the laws governing Ayurvedic drugs and their necessity, on the Drugs and Magic Remedies Act, and on the Prevention of Food Adulteration Act in relation to herbal supplements. The law has changed substantially in 2024–2026; this chapter states it as it stands in October 2026, and says where to check for later changes.

Why laws are needed for Ayurvedic drugs

The August 2022 and January 2024 papers asked for the laws governing Ayurvedic drugs "with their necessity".

Need	What the law does
Quality	Sets standards (pharmacopoeia), requires GMP, licensing and testing, and punishes adulterated, misbranded and spurious drugs
Safety	Requires cautions for poisonous ingredients (Schedule E(1)), shelf life, safety evidence for new products, pharmacovigilance
Honest information	Prescribes labelling; prohibits misleading claims and advertisements for listed diseases
Protection of the consumer	Gives remedies for defective goods and unfair trade practices
Control of dangerous substances	Regulates narcotic and psychotropic ingredients and alcohol
Protection of workers	Provides for health, safety and welfare in factories
Professional standards	Regulates who may dispense and practise
Trust and trade	Gives credibility in India and abroad; supports export (Chapter B9)

The legal framework at a glance

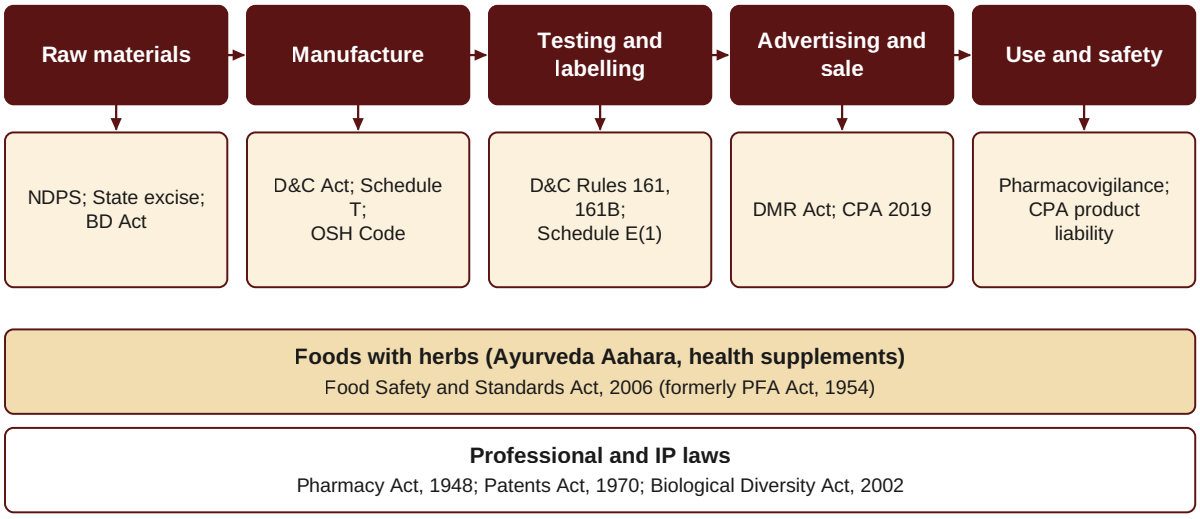


Figure B8.1. The main laws governing Ayurvedic drugs and the stages of a product's life they control.

Law	Main concern for Ayurvedic drugs	Unit
Drugs and Cosmetics Act, 1940 and **Drugs Rules, 1945**	Definition, manufacture, licensing, GMP (Schedule T), standards, labelling, shelf life, testing, inspection, penalties	B8(i)
Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954	Advertisements for listed diseases and misleading advertisements	B8(ii)
Prevention of Food Adulteration Act, 1954 (repealed)	Food adulteration (historical; replaced by FSS Act)	B8(iii)

Law	Main concern for Ayurvedic drugs	Unit
Food Safety and Standards Act, 2006	Ayurveda Aahara, health supplements and nutraceuticals, food products containing herbs	B8(iv)
Narcotic Drugs and Psychotropic Substances Act, 1985	Opium, cannabis and other controlled ingredients	B8(v)
Occupational Safety, Health and Working Conditions Code, 2020 (formerly the Factories Act, 1948); **Pharmacy Act, 1948**	Workers' safety and welfare; the profession of pharmacy	B8(vi)
Consumer Protection Act, 2019 (formerly 1986)	Consumer rights; product liability; misleading advertisements	B8(vii)
Patents Act, 1970; Biological Diversity Act, 2002	Intellectual property; access to bio-resources	Chapter B7

UNIT B8(i)

DRUGS AND COSMETICS ACT, 1940 AND RULES, 1945

SYLLABUS

Relevant regulatory provisions of Ayurvedic drugs in Drugs and Cosmetics Act, 1940 and Rules - 1945

LEARNING OUTCOMES

After studying this unit you should be able to:

1. State the definitions of Ayurvedic, Siddha and Unani drugs and of patent or proprietary medicines under the Act.
2. Describe the provisions of Chapter IV-A of the Act and the bodies it creates.
3. Describe the rules for licensing of manufacture, GMP, standards, labelling, shelf life, testing and inspection of ASU drugs.
4. Describe Schedule E(1) and the provisions on advertisement.
5. Summarise the changes made in 2024–2026.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	B, Q.1	20	Drug and cosmetics act 1940 and rules 1945.
Sep 2018	B, Q.1	20	Explain Drug & Cosmetics Act with relevance to ASU drugs along with schedule T and Good Manufacturing Practices.
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

The Act and Rules have changed several times since the syllabus was written. **Rule 170**, on advertisement of ASU drugs, was omitted on 1 July 2024; the Supreme Court's stay on the omission was vacated on 11 August 2025. The **Jan Vishwas (Amendment of Provisions) Act, 2026** replaced imprisonment by monetary penalties for some offences; its provisions on ASU drugs came into force on 1 July 2026. The **Drugs (Eleventh Amendment) Rules, 2026** made manufacturing licences perpetual, introduced a product code and QR code, GTIN or barcode on labels, and substituted a new **Schedule T** (compliance by 31 July 2029). The **Drugs (Thirteenth Amendment) Rules, 2026** substituted **Schedule E(1)**. Details and sources are in B8(i).8 and B8(i).9.

B8(i).1 The Act and its history

The **Drugs Act, 1940** was enacted to regulate the import, manufacture, distribution and sale of drugs. It became the **Drugs and Cosmetics Act** when cosmetics were brought under it, and the rules made under it, the **Drugs and Cosmetics Rules, 1945**, are now called the **Drugs Rules, 1945** in recent amending notifications. **Ayurvedic, Siddha and Unani (ASU) drugs** were brought under the Act in **1964** by the insertion of **Chapter IV-A**. Later amendments extended its provisions to Sowa-Rigpa in several places.

Year	Milestone for ASU drugs
1940	Drugs Act enacted (modern drugs)
1945	Drugs Rules framed
1964	**Chapter IV-A** inserted: ASU drugs brought under the Act
2000	**Schedule T** (GMP for ASU drugs) introduced
2005, 2009, 2016	Shelf life provisions for ASU drugs; Rule 161B substituted in 2016 (Chapter B3)
2018	Rule 170 on advertisement of ASU drugs inserted
2024	Rule 170 omitted (1 July 2024); the omission was stayed by the Supreme Court and the stay vacated on 11 August 2025 (B8(i).8)
2025	Schedule TB: guidelines for nasal sprays of ASU drugs (January 2025) [Gazette 2025]
2026	Jan Vishwas (Amendment of Provisions) Act, 2026: provisions on ASU drugs in force from 1 July 2026 [S.O. 3362(E)]; **Drugs (Eleventh Amendment) Rules, 2026**: licensing, product code, labelling, shelf life, risk-based inspection, rewritten Schedule T; **Drugs (Thirteenth Amendment) Rules, 2026**: Schedule E(1) substituted

B8(i).2 Definitions

Term	Definition (in substance)
Ayurvedic, Siddha or Unani drug (Section 3(a))	All medicines intended for internal or external use for the diagnosis, treatment, mitigation or prevention of disease or disorder in human beings or animals, **manufactured exclusively in accordance with the formulae described in the authoritative books** of these systems specified in the **First Schedule** to the Act
Patent or proprietary medicine (Section 3(h), for ASU)	A formulation containing **only such ingredients** as are mentioned in the formulae of the authoritative books, but which is **not itself a formulation included in those books**; it does not include a medicine given by parenteral route
Misbranded ASU drug (Section 33E)	Coloured, coated or polished to conceal damage or to appear better than it is; not labelled as prescribed; or labelled with false or misleading claims
Adulterated ASU drug (Section 33EE)	Containing filthy, putrid or decomposed substance; prepared or stored in insanitary conditions; in a container that may make it injurious; containing a non-permitted colour or a harmful substance; mixed with a substance that reduces its quality or strength
Spurious ASU drug (Section 33EEA)	Sold under a name belonging to another drug; an imitation or substitute; bearing the name of a fictitious or other manufacturer; or wholly or partly substituted

The **First Schedule** lists the authoritative books: for Ayurveda, the major Samhitas and the classical works on pharmacy and Rasashastra (among them Sharangadhara Samhita, Bhaishajya Ratnavali, Rasatarangini and Rasaratna Samuchchaya). A classical (non-proprietary) product must follow a formula in one of these books, and its label must give the reference (Chapter B3).

B8(i).3 Chapter IV-A of the Act

Section	Subject
33B	Application of the Chapter
33C	**Ayurvedic, Siddha and Unani Drugs Technical Advisory Board** (ASUDTAB): advises governments on technical matters
33D	**Ayurvedic, Siddha and Unani Drugs Consultative Committee** (ASUDCC): secures uniformity in administration of the Act across the country
33E, 33EE, 33EEA	Misbranded, adulterated and spurious ASU drugs
33EEB	Regulation of manufacture for sale: no ASU drug may be manufactured for sale without a licence
33EEC	Prohibition of manufacture and sale of misbranded, adulterated, spurious and unlicensed ASU drugs
33EED	Power of the Central Government to prohibit an ASU drug in the public interest
33F, 33G	Government Analysts and Inspectors
33I, 33J, 33K	Penalties, enhanced penalties for subsequent offences, confiscation
33M	Cognizance of offences
33N	Power of the Central Government to make rules (after consulting ASUDTAB)
33O	Power to amend the First Schedule

The Jan Vishwas (Amendment of Provisions) Act, 2026 (Act 8 of 2026) replaces imprisonment by monetary penalties for some offences under the Act. The Ministry of Ayush brought its provisions on ASU drugs into force on 1 July 2026 (S.O. 3362(E), 23 June 2026) [S.O. 3362(E)]; other provisions on the Act are to come into force on 30 June 2027 [NDA 2026]. Read the amended Act for the current penalties.

Retail sale. Chapter IV-A regulates the **manufacture** of ASU drugs for sale; it does not set up a system of retail sale licences for ASU drugs comparable to that for modern drugs. Sale of ASU drugs that are misbranded, adulterated or spurious is, however, an offence.

B8(i).4 Licensing of manufacture (Part XVI of the Rules)

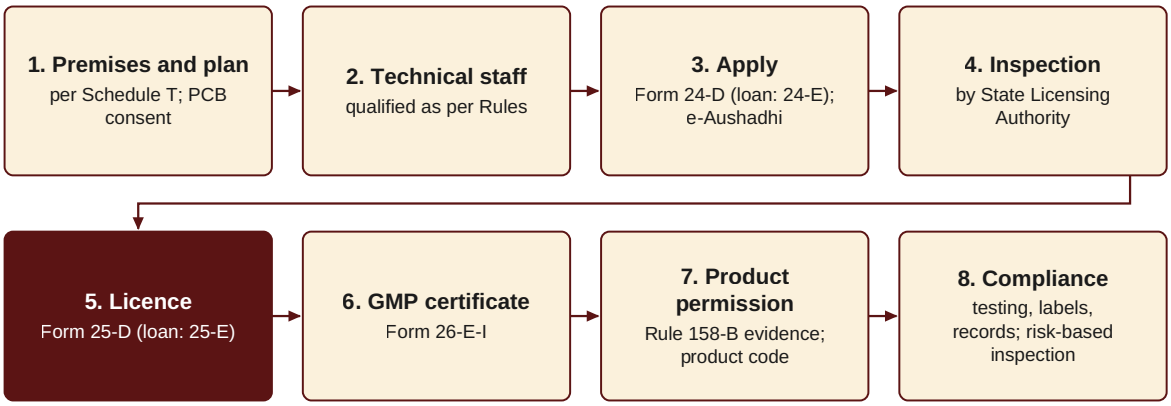


Figure B8i.1. Steps to obtain a licence to manufacture ASU drugs for sale.

Rule	Provision
152	**Licensing Authority**: appointed by the State Government (the State Licensing Authority for ASU drugs)
153	Application for a licence in **Form 24-D** (loan licence: Form 24-E), with fees; applications are now made online through the e-Aushadhi portal or the state portal [Lawrbit]
154	Grant of licence in **Form 25-D** after inspection; loan licence in **Form 25-E** (Rule 154A)
155B	Certificate of GMP in **Form 26-E-I**
156	Duration of licence: **omitted in 2026**; licences are now valid **perpetually**, subject to continued compliance [Corpseed 2026; Lexplosion 2026]
157	**Conditions** of licence: compliance with Schedule T; competent technical staff with prescribed qualifications; records; testing; adequate premises and equipment
158	Conditions of loan licence
158-B	**Guidelines for the issue of licence**: the evidence of safety and effectiveness required for different categories of ASU drugs; since 2026, a **specific product code** in a standard pattern [Corpseed 2026]

B8(i).4.1 Rule 158-B: evidence for a licence

Category	What must be shown
Classical formulation from an authoritative book	Reference to the book; the formulation as described; quality
Patent or proprietary formulation	**Experience or evidence of effectiveness**, based on textual rationale, published literature and, where these do not support the claim, a **pilot study**; safety data as specified
Products with Schedule E(1) ingredients or new indications or routes	Additional safety and effectiveness evidence as laid down

The Ministry of Ayush clarified in 2018 that the ASU provisions do not mention “clinical trials”, that licences should not be held up for clinical trial reports, and that a pilot study is needed only when textual rationale and published literature do not support the claimed use [Ayush 2018]. Research on ASU drugs that is conducted as a clinical trial follows the ethical and registration requirements described in Chapter A2.

B8(i).4.2 Qualifications of technical staff

The person in charge of manufacture must hold one of the qualifications prescribed in the Rules: for example, a degree in Ayurveda, Siddha or Unani, or in Ayurvedic Pharmacy; a diploma in these systems; a degree in pharmacy, pharmaceutical chemistry, chemistry or botany with experience in manufacturing ASU drugs; or a registered Vaidya or Hakim with experience [Lawrbit]. The 2026 amendment revised the qualifications for testing staff and added a route for a qualified microbiologist [Corpseed 2026].

B8(i).5 Good Manufacturing Practices: Schedule T

Schedule T and its 2026 revision are described in Chapter B4.6. A licence is granted only to a unit that complies with Schedule T.

B8(i).6 Standards, testing and inspection

Provision	Content
Standards (Rule 168)	ASU drugs must comply with the standards laid down in the official pharmacopoeias where available (Chapter B10)
Drug testing laboratories (Part XVI-A, Rules 160A–160J)	Approval of laboratories to test ASU drugs and raw materials on behalf of manufacturers
Government Analysts and Inspectors (Part XVIII, Rules 162–167)	Qualifications, powers and duties; sampling; reports. The 2026 amendment substituted Rule 167 (qualifications of Inspectors) and introduced **risk-based inspections** [Corpseed 2026]
Central laboratory	The appellate laboratory for ASU drugs is the Pharmacopoeia Commission for Indian Medicine and Homoeopathy, Ghaziabad (Chapter B10); the 2026 amendment allows an authorised laboratory of an Ayush autonomous body to act as the Central Drugs Laboratory for certain samples [Corpseed 2026]

B8(i).7 Labelling, packing and shelf life (Part XVII)

Rule	Content (details in Chapter B3)
161	Labelling of ASU drugs: name, contents, manufacturer, licence number, batch, dates, “Ayurvedic medicine”, ingredients with botanical names and quantities, reference to the book, “For external use only”, **“Caution: To be taken under medical supervision”** for Schedule E(1) drugs; since 2026, specific product code and QR code, GTIN or barcode
161A	Labels for export may follow the importing country’s requirements
161B	**Shelf life**: expiry date compulsory; real-time stability data; default shelf lives by category (Chapter B3.4.3); since 2026, accelerated data may support an initial shelf life of one or two years

B8(i).8 Schedule E(1) and advertisement

B8(i).8.1 Schedule E(1)

Schedule E(1) lists **poisonous substances** used in the ASU systems (and, since 2026, Sowa-Rigpa). A drug for internal use containing any of them must carry the caution "To be taken under medical supervision" in English and Hindi (Rule 161), and e-commerce platforms may sell such drugs only against a prescription [CCPA 2022].

The **Drugs (Thirteenth Amendment) Rules, 2026** (notification dated 21 August 2026) substituted Schedule E(1) with a revised list [Gazette 2026b]. As reported [Indian Practitioner 2026]:

Change	Detail
Mineral substances	Expanded, including compounds of mercury, copper and lead, tin and brass; examples include arsenic trioxide, realgar, cinnabar, litharge, stibnite, galena, copper sulphate and yellow oxide of mercury
General rule for arsenic and mercury	**Any ingredient containing arsenic or mercury as a chemical constituent** is treated as a Schedule E(1) substance even if not named
Plant substances (Ayurveda)	Increased from 13 to 16 entries
Mineral substances (Ayurveda)	Increased from 7 to 20 entries
Other systems	Siddha, Unani and Sowa-Rigpa lists revised or added
Animal substances	Snake poison and specified Mylabris (blister beetles)

Read the Gazette notification for the full list before relying on any entry.

B8(i).8.2 Rule 170: advertisement of ASU drugs

Rule 170, inserted in 2018, prohibited advertisement of ASU drugs without the approval of the State Licensing Authority and listed the kinds of advertisement that would not be approved. The Ministry of Ayush **omitted** Rule 170 by a notification of 1 July 2024. In *Indian Medical Association v. Union of India*, the Supreme Court **stayed** the omission on 27 August 2024 and, on **11 August 2025**, closed the petition and **vacated the stay**, leaving the omission in effect and reserving liberty to aggrieved persons to seek remedies in law [LiveLaw 2025].

Advertisements of ASU drugs therefore remain controlled chiefly by the **Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954** (B8(ii)), by the **Consumer Protection Act, 2019** (B8(vii)), and by the misbranding provisions of the Drugs and Cosmetics Act.

B8(i).9 The 2026 amendments in brief

Instrument	Main changes for ASU drugs
Jan Vishwas (Amendment of Provisions) Act, 2026	Monetary penalties in place of imprisonment for some offences; ASU provisions in force from 1 July 2026
Drugs (Eleventh Amendment) Rules, 2026 (Gazette, July 2026)	Perpetual licences; Rules 156 and 156A omitted; specific product code (Rule 158-B); QR code, GTIN or barcode on labels; accelerated stability data for initial shelf life (Rule 161B); risk-based inspection (Rule 162); revised Inspector and testing staff qualifications; rewritten **Schedule T** for Ayurveda, Siddha, Sowa-Rigpa and Unani with compliance by **31 July 2029**; stronger Rasaushadhi safeguards
Drugs (Thirteenth Amendment) Rules, 2026	**Schedule E(1)** substituted

The 2026 amendments are summarised from the Gazette listings and from secondary summaries [Gazette 2026a; Gazette 2026b; Corpseed 2026; Lexplosion 2026; Indian Practitioner 2026]. Laws change; consult the current text on the Ministry of Ayush website or in the Gazette of India before any regulatory decision.

KEY POINTS TO REMEMBER

- ASU drugs came under the Act in 1964 (Chapter IV-A, Sections 33B–33O).
- ASU drug: made exclusively per formulae of the authoritative books in the First Schedule. Patent or proprietary: only ingredients from those formulae, but not itself a book formulation; not parenteral.
- Misbranded (33E), adulterated (33EE), spurious (33EEA); licence for manufacture (33EEB).
- ASUDTAB (33C) advises; ASUDCC (33D) secures uniformity.
- Licence: Form 24-D application; Form 25-D licence; loan licence Form 25-E; GMP certificate Form 26-E-I; State Licensing Authority; perpetual validity since 2026.
- Rule 158-B: evidence of safety and effectiveness; for proprietary products, textual rationale, literature and, if needed, pilot study.
- Schedule T (GMP); Rule 161 (labelling); 161A (export labels); 161B (shelf life); Rules 160A–J (testing labs); Rules 162–167 (analysts, inspectors).
- Schedule E(1): poisonous substances; caution on label; substituted in 2026 (any arsenic- or mercury-containing ingredient included).
- Rule 170 omitted (2024); Supreme Court stay vacated 11 August 2025.

EXAM CORNER — UNIT B8(i)

Likely long questions (20 marks)

1. Drugs and Cosmetics Act, 1940 and Rules, 1945 *(set in July 2018)*.
2. Explain the Drugs and Cosmetics Act with relevance to ASU drugs, along with Schedule T and GMP *(set in September 2018)*.
3. Explain the laws governing Ayurvedic drugs in detail, with examples *(set in August 2022)*; with their necessity *(set in January 2024)*.

Likely short notes (5 marks)

1. Chapter IV-A of the Drugs and Cosmetics Act.

2. Misbranded, adulterated and spurious ASU drugs.
3. Rule 158-B.
4. Schedule E(1).
5. Licensing of an Ayurvedic manufacturing unit.
6. Rule 170 and its omission.

Answer frame: "Drugs and Cosmetics Act, 1940 and Rules, 1945 (with reference to ASU drugs)." (20 marks)

Part	What to write	Suggested marks
History and objectives	1940, 1945, 1964 (Chapter IV-A), recent amendments	2
Definitions	ASU drug; patent or proprietary; misbranded, adulterated, spurious	4
Chapter IV-A	Table of sections; ASUDTAB, ASUDCC	3
Licensing	Forms, authority, conditions, Rule 158-B	4
GMP, standards, testing	Schedule T; labs; analysts and inspectors	3
Labelling and shelf life	Rules 161, 161A, 161B; Schedule E(1)	3
Recent changes	2024–2026	1

REFERENCES USED IN THIS UNIT

Syllabus reference number for this unit: **43**.

- **[R43]** *The Drugs and Cosmetics Act, 1940 and the Drugs and Cosmetics Rules, 1945*, with amendments (Chapter IV-A; Parts XVI, XVI-A, XVII, XVIII; Schedules E(1), T).

Additional verified sources (accessed 16 October 2026)

- **[S.O. 3362(E)]** Ministry of Ayush. Notification S.O. 3362(E), 23 June 2026, appointing 1 July 2026 for the commencement of provisions of the Jan Vishwas (Amendment of Provisions) Act, 2026 (8 of 2026) relating to the Drugs and Cosmetics Act, 1940. egazette.gov.in.
- **[NDA 2026]** Nishith Desai Associates. Mid-year updates in the pharmaceutical and healthcare industry. 2026 (Jan Vishwas commencement on 30 June 2027 for other provisions).
- **[Gazette 2025]** Ministry of Ayush. Notification dated 10 January 2025 introducing Schedule TB (guidelines for nasal sprays of ASU drugs). Gazette ID CG-DL-E-14012025-260208.
- **[Gazette 2026a]**, **[Corpseed 2026]**, **[Lexplosion 2026]**: Drugs (Eleventh Amendment) Rules, 2026 (citations in Chapter B4).
- **[Gazette 2026b]**, **[Indian Practitioner 2026]**: Drugs (Thirteenth Amendment) Rules, 2026 (citations in Chapter B3).
- **[Lawrbit]** Lawrbit. Licence to manufacture Ayurvedic, Siddha, Sowa-Rigpa and Unani drugs. lawrbit.com (accessed 16 October 2026).
- **[Ayush 2018]** Ministry of AYUSH. Letter to State Licensing Authorities on interpretation of Rule 158-B (pilot study; no clinical trial requirement), 2018 (reported by Business Medical Dialogues, 11 July 2018).
- **[CCPA 2022]** Central Consumer Protection Authority. Advisory on sale of ASU drugs by e-commerce platforms, 14 July 2022.

- **[LiveLaw 2025]** “Supreme Court vacates stay on Ayush Ministry’s notification that permitted advertisements of Ayurvedic, Unani and Siddha drugs.” LiveLaw; August 2025 (*Indian Medical Association v. Union of India*, W.P. (C) 645/2022, order of 11 August 2025).

UNIT B8(ii)

DRUGS AND MAGIC REMEDIES (OBJECTIONABLE ADVERTISEMENTS) ACT, 1954

SYLLABUS

Laws pertaining to Drugs and Magic remedies (Objectionable Advertisement) Act - 1954.

LEARNING OUTCOMES

After studying this unit you should be able to describe the objectives, definitions, prohibitions, exceptions, penalties and enforcement of the Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954, and explain its importance for the marketing of Ayurvedic drugs.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2020	B, Q.3b	5	Drugs and magic remedy act 1954
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

B8(ii).1 Objective

The **Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954** controls the advertisement of drugs in certain cases, prohibits the advertisement of remedies alleged to possess magic qualities, and provides for related matters. Its purpose is to stop people from being misled into self-treatment of serious diseases, or into delaying proper treatment, by advertisements. It was amended substantially in 1963, when the Schedule of diseases was added [DMR Act].

B8(ii).2 Definitions (Section 2)

Term	Meaning
Advertisement	Any notice, circular, label, wrapper or other document, and any announcement made orally or by means of light, sound or smoke
Drug	Includes medicines for human or animal use, substances for diagnosis or treatment, and articles intended to affect the structure or function of the body
Magic remedy	Any talisman, mantra, kavacha or other charm claimed to have miraculous powers to cure, diagnose, prevent or treat disease, or to affect the structure or function of the body
Registered medical practitioner	A practitioner registered under the relevant law, including practitioners of Indian systems of medicine

B8(ii).3 What is prohibited

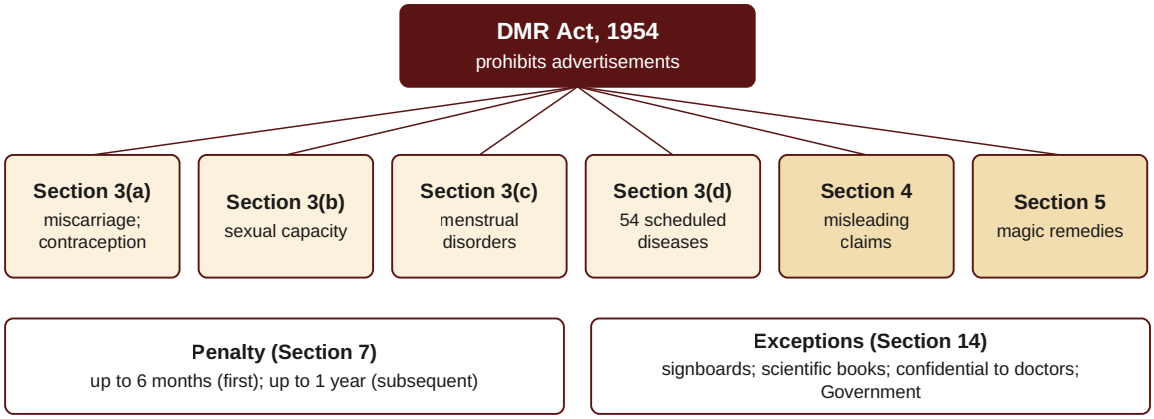


Figure B8ii.1. What the Drugs and Magic Remedies Act prohibits.

Section	Prohibition
3	Advertising a drug in terms that suggest or are calculated to lead to its use for: (a) procuring miscarriage or preventing conception; (b) maintaining or improving sexual capacity; (c) correcting menstrual disorders; (d) diagnosing, curing, mitigating, treating or preventing any **disease or condition in the Schedule** (or added by rules)
4	**Misleading advertisements**: any advertisement that gives a false impression of the true character of the drug, makes a false claim, or is otherwise false or misleading in a material particular
5	Advertising **magic remedies** for any of the purposes in Section 3
6	Import into or export from India of documents containing such advertisements

B8(ii).3.1 The Schedule

The Schedule lists **54 entries** of diseases, disorders and conditions for which drugs may not be advertised [DMR Act]. Examples include appendicitis, arteriosclerosis, blindness, cancer, cataract, diabetes, epilepsy, heart diseases, leprosy and tuberculosis. These are conditions that need timely treatment by a qualified practitioner, or for which no reliable cure is generally accepted.

B8(ii).4 Exceptions (Section 14)

The Act does not apply to:

- signboards or notices displayed by a registered medical practitioner at his premises indicating that treatment is given for any disease;
- bona fide scientific or social treatises or books dealing with any of the matters;
- advertisements sent confidentially only to registered medical practitioners;
- advertisements published by the Government;
- certain advertisements published with the previous sanction of the Government.

B8(ii).5 Penalties and enforcement

Section	Provision
7	**First conviction**: imprisonment up to **six months**, or fine, or both. **Subsequent conviction**: imprisonment up to **one year**, or fine, or both
8	A **Gazetted Officer** authorised by the State Government may enter and search, and seize advertisements and related material; seizures are reported to a Magistrate

B8(ii).5.1 The Supreme Court's directions of 2025

In the case filed by the Indian Medical Association on misleading health advertisements, the Supreme Court, on **26 March 2025**, directed among other things that States appoint Gazetted Officers under Section 8, set up grievance redressal mechanisms for complaints about misleading advertisements, sensitise the police, publicise the complaint mechanisms, and that the Union maintain a dashboard of action taken [LiveLaw 2025].

B8(ii).6 Relevance to Ayurvedic drugs

1. Advertisements of Ayurvedic drugs claiming to **cure** diabetes, cancer, heart disease, epilepsy, obesity, infertility or any other scheduled condition are **prohibited**, whatever the evidence the manufacturer believes it has.
2. Claims of increased **sexual capacity** are prohibited (Section 3(b)), a frequent violation by "vajikarana" product advertisements.
3. **Misleading** advertisements (Section 4) are prohibited for any condition.
4. With **Rule 170 omitted** (B8(i).8.2), the DMR Act and the Consumer Protection Act, 2019 (B8(vii)) are the main legal controls on advertisements of ASU drugs.
5. The Ayush pharmacovigilance network and the **AyushSuraksha** portal record and refer misleading advertisements (Chapter A6.8.3).

AYURVEDA CORRELATION

Charaka warns against the false physician who boasts, advertises cures and deceives patients (the Chadmachara), and contrasts him with the true physician (Charaka Samhita, Sutra Sthana 11 and 29) [R26]. The DMR Act gives legal force to the same principle: medicine must not be sold by false promises.

KEY POINTS TO REMEMBER

- DMR Act 1954; amended 1963 (Schedule added).
- Section 3: no advertisement for miscarriage or contraception, sexual capacity, menstrual disorders, or the 54 scheduled diseases. Section 4: misleading advertisements. Section 5: magic remedies.
- Exceptions (Section 14): practitioners' signboards, scientific books, confidential communications to doctors, Government advertisements.
- Penalties (Section 7): up to 6 months (first), up to 1 year (subsequent). Section 8: search and seizure by authorised Gazetted Officer.
- Supreme Court directions of 26 March 2025: Section 8 officers, grievance mechanisms, dashboard.

EXAM CORNER — UNIT B8(ii)

Likely short notes (5 marks)

1. Drugs and Magic Remedies Act, 1954 *(set in September 2020)*.
2. Objectionable advertisements of Ayurvedic drugs.
3. Magic remedies.

Answer frame: “Drugs and Magic Remedies Act, 1954.” (5 marks)

Part	What to write	Suggested marks
Objective and definitions	Advertisement; magic remedy	1
Prohibitions	Sections 3, 4, 5, 6; Schedule with examples	2
Exceptions	Section 14	0.5
Penalties and enforcement	Sections 7, 8; 2025 directions	1
Relevance to Ayurveda	Examples of prohibited claims; Rule 170 position	0.5

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **26, 44**.

- **[R44]** *The Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954*. **[DMR Act]** Text as published on India Code (indiacode.nic.in), with amendments up to Act 34 of 2019 (accessed 16 October 2026).
- **[R26]** Charaka Samhita, Sutra Sthana 11 and 29.
- **[LiveLaw 2025]** Reports of the Supreme Court’s orders in *Indian Medical Association v. Union of India*, W.P.(C) 645/2022: directions of 26 March 2025 and order of 11 August 2025. LiveLaw (accessed 16 October 2026).

UNIT B8(iii)

PREVENTION OF FOOD ADULTERATION ACT

SYLLABUS

Prevention of Food Adulteration (PFA) act.

LEARNING OUTCOMES

After studying this unit you should be able to describe the objectives and main provisions of the Prevention of Food Adulteration Act, 1954, explain why it was replaced, and discuss its relevance to herbal supplements.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.
Mar 2026	B, Q.3b	5	Discuss the “Prevention of Food Adulteration (PFA) Act” in relation to herbal supplements.

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

The **Prevention of Food Adulteration (PFA) Act, 1954** has been **repealed**. It was replaced by the **Food Safety and Standards Act, 2006** (B8(iv)); the repeal took effect on **5 August 2011** under Section 97 of the 2006 Act, with savings for proceedings already pending [LiveLaw 2023]. It is studied because the syllabus names it and because its concepts (adulteration, misbranding, standards, analysts, inspectors) carry over into the present law.

B8(iii).1 Objectives

The PFA Act aimed to protect the public from adulterated, misbranded and unsafe food, to lay down standards of food quality, and to punish those who made or sold adulterated food.

B8(iii).2 Main provisions

Provision	Content
Definition of **food**	Any article used as food or drink for human consumption (other than drugs and water), including ingredients, flavouring and colouring matters

Provision	Content
Definition of **adulterated** food	Food was adulterated if, among other grounds: it was not of the nature, substance or quality demanded; a cheaper or inferior substance had been substituted; a constituent had been abstracted; it was prepared or stored in insanitary conditions; it was filthy, putrid, decomposed or insect-infested; it came from a diseased animal; it contained a poisonous or harmful ingredient or a prohibited colour or preservative; or its quality fell below the prescribed standard
Definition of **misbranded** food	Falsely labelled, imitation sold under another name, or with false or misleading claims
Central Committee for Food Standards	Advised governments on food standards
Central Food Laboratory	Final analysis of disputed samples
Public Analysts and **Food Inspectors**	Analysis of samples; inspection and sampling
Prohibitions	Manufacture, storage, sale or distribution of adulterated or misbranded food, or food in breach of a licence
Penalties	Imprisonment and fine, more severe where the adulterant was injurious to health
PFA Rules, 1955	Detailed standards for foods, colours, preservatives, additives, labelling and packing

B8(iii).3 Why it was replaced

The food sector was governed by many separate laws and orders, with overlapping authorities and standards. The PFA Act was criminal in approach and focused on adulteration rather than on the overall safety of the food chain. The Food Safety and Standards Act, 2006 consolidated eight laws and orders into one, set up a single authority, adopted a science-based, risk-based approach, and created a separate framework for health supplements and nutraceuticals (B8(iv)).

B8(iii).4 The PFA Act and herbal supplements

The March 2026 paper asked for the PFA Act “in relation to herbal supplements”.

1. **The grey area.** Under the PFA regime there was no specific category for herbal or health supplements. Herbal products marketed as foods were judged by the general definitions of food and adulteration and by any standards that existed for the food type. Some products were sold as foods to avoid the licensing and standards that apply to drugs.
2. **Adulteration principles still apply.** A herbal supplement adulterated with a cheaper plant, contaminated with heavy metals or microbes, containing an undeclared drug, or mislabelled was adulterated or misbranded within the PFA’s definitions.
3. **The boundary between food and drug.** A product that claims to diagnose, cure, mitigate, treat or prevent disease is a **drug** under the Drugs and Cosmetics Act; an ASU product made per the authoritative books is an **ASU drug**, whatever it is called on the label.

4. **The present law.** Today herbal supplements, nutraceuticals and Ayurveda Aahara are regulated under the **Food Safety and Standards Act, 2006** and its regulations (B8(iv)), which define these categories, list permitted ingredients and limit claims.

KEY POINTS TO REMEMBER

- PFA Act 1954; PFA Rules 1955; repealed with effect from 5 August 2011 by Section 97 of the FSS Act 2006.
- Adulterated food: wrong nature or quality; substitution; abstraction; insanitary preparation; filthy or decomposed; diseased animal; poisonous ingredient; prohibited colour or preservative; below standard.
- Misbranded food: false labels; imitations; misleading claims.
- Bodies: Central Committee for Food Standards; Central Food Laboratory; Public Analysts; Food Inspectors.
- Herbal supplements had no specific category under PFA; the FSS Act now regulates them.
- A product claiming to treat disease is a drug, not a food.

EXAM CORNER — UNIT B8(iii)

Likely short notes (5 marks)

1. Prevention of Food Adulteration Act in relation to herbal supplements *(set in March 2026)*.
2. Adulteration of food: definition and types.
3. Why was the PFA Act replaced?

Answer frame: “Discuss the PFA Act in relation to herbal supplements.” (5 marks)

Part	What to write	Suggested marks
Status	Repealed; replaced by the FSS Act 2006 from 5 August 2011	0.5
Provisions	Definitions of food, adulterated, misbranded; bodies; penalties	2
Herbal supplements	Grey area under PFA; adulteration and misbranding principles; food–drug boundary	1.5
Present law	FSS Act; health supplements and nutraceuticals; Ayurveda Aahara	1

REFERENCES USED IN THIS UNIT

Syllabus reference number for this unit: **45**.

- **[R45]** *The Prevention of Food Adulteration Act, 1954* and *Rules, 1955* (repealed).
- **[LiveLaw 2023]** Report of *Manik Hiru Jhangiani v. State of Madhya Pradesh*, Supreme Court, 14 December 2023 (repeal of the PFA Act with effect from 5 August 2011 under Section 97 of the FSS Act). LiveLaw (accessed 16 October 2026).
- *The Food Safety and Standards Act, 2006*, Section 97 and the Second Schedule.

UNIT B8(iv)

FOOD SAFETY AND STANDARDS ACT, 2006

SYLLABUS

Food Safety and Standards Act, 2006

LEARNING OUTCOMES

After studying this unit you should be able to describe the objectives, authorities, key definitions, licensing, enforcement and penalties of the Food Safety and Standards Act, 2006, and explain how it applies to Ayurveda Aahara, health supplements and nutraceuticals.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

B8(iv).1 Objectives

The **Food Safety and Standards Act, 2006** consolidates the laws relating to food and establishes the **Food Safety and Standards Authority of India** (FSSAI) to lay down science-based standards for articles of food and to regulate their manufacture, storage, distribution, sale and import, so as to ensure the availability of safe and wholesome food for human consumption.

It replaced eight earlier laws and orders, among them the Prevention of Food Adulteration Act, 1954, the Fruit Products Order, 1955, the Meat Food Products Order, 1973, and the Milk and Milk Products Order, 1992 (Second Schedule to the Act).

B8(iv).2 Structure of the food safety system

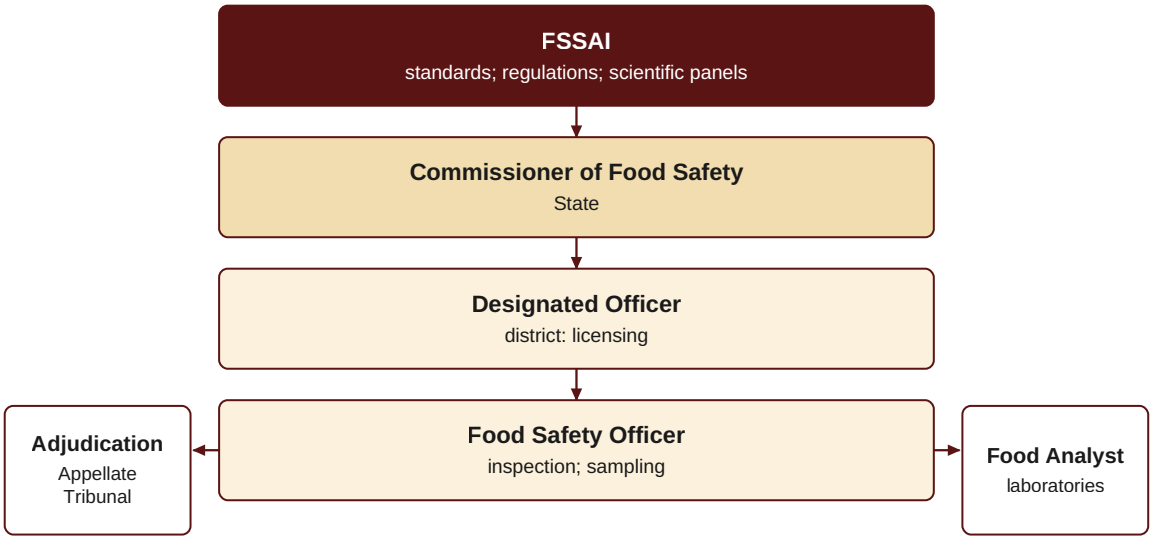


Figure B8iv.1. Structure of food safety administration under the FSS Act, 2006.

Level	Body or officer	Function
National	**FSSAI** (Chairperson; Chief Executive Officer), with a Central Advisory Committee, Scientific Panels and a Scientific Committee	Standards and regulations; guidelines for accreditation of laboratories; coordination
State	**Commissioner of Food Safety**	Enforcement in the State
District	**Designated Officer**	Licensing; prosecution sanction
Local	**Food Safety Officer**	Inspection; sampling
Laboratories	Food Analysts; notified laboratories; referral laboratories	Analysis
Adjudication and appeal	Adjudicating Officer; Food Safety Appellate Tribunal	Penalties; appeals

B8(iv).3 Key definitions

Term	Meaning (in substance)
Food	Any substance intended for human consumption, processed, partially processed or unprocessed, including primary food, but not drugs or cosmetics
Unsafe food	Food whose nature, substance or quality is so affected as to render it injurious to health (for example, by adulterants, contaminants, pesticides, insanitary processing, or toxic substances)
Adulterant	Any material used to make food unsafe, sub-standard, misbranded or containing extraneous matter
Sub-standard	Food that does not meet the specified standards but is not unsafe
Misbranded	Food with false, misleading or deceptive labelling or claims

B8(iv).4 Special foods: Section 22

Section 22 deals with **nutraceuticals, functional foods, health supplements, foods for special dietary use** and similar categories, which may be manufactured or sold only as provided under the Act and the regulations. FSSAI has framed regulations for these categories (first notified in 2016 and since revised), which define them, list permitted ingredients (including many botanicals) and their limits, and restrict the **claims** that may be made.

B8(iv).5 Ayurveda Aahara

The **Food Safety and Standards (Ayurveda Aahara) Regulations, 2022** were published in the Gazette of India in May 2022 [USDA 2022]. They cover foods prepared according to the recipes, ingredients or processes described in the authoritative books of Ayurveda listed in their Schedule A. Ayurveda Aahara is **food**, not medicine: it is licensed under the food law and must not be presented as a treatment for disease.

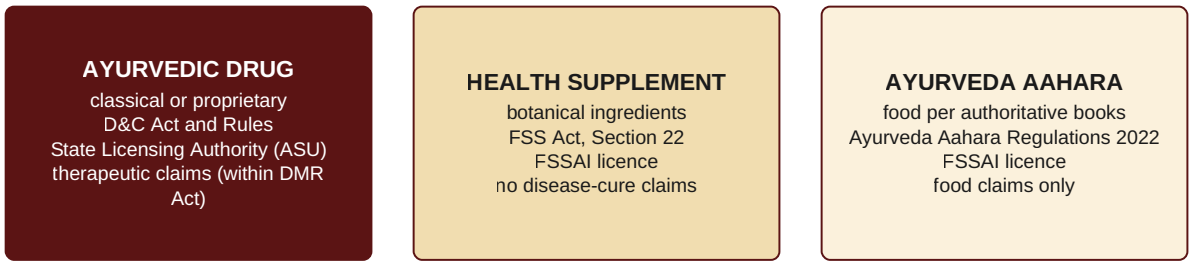


Figure B8iv.2. The boundary between an Ayurvedic drug, a health supplement and Ayurveda Aahara.

Product	Law	Licence from	Claims allowed
Ayurvedic drug (classical or proprietary)	Drugs and Cosmetics Act and Rules	State Licensing Authority (ASU)	Therapeutic indications, within the advertisement laws
Health supplement / nutraceutical with botanical ingredients	FSS Act, Section 22 and regulations	FSSAI / State food authority	Only as permitted (no disease cure or prevention claims)
Ayurveda Aahara	FSS Act; Ayurveda Aahara Regulations, 2022	FSSAI / State food authority	As permitted for foods

B8(iv).6 Licensing, enforcement and penalties

- Every food business operator must hold a **licence** or **registration** (Section 31), according to scale.
- Food Safety Officers inspect and take samples; samples are analysed by Food Analysts.
- Penalties are graded: monetary penalties for sub-standard and misbranded food and misleading advertisements, imposed by Adjudicating Officers; imprisonment and fine for unsafe food, increasing with the harm caused, up to life imprisonment where unsafe food causes death.

KEY POINTS TO REMEMBER

- FSS Act 2006; FSSAI; repealed PFA and seven other laws and orders.
- Officers: Commissioner of Food Safety (State), Designated Officer (district), Food Safety Officer; Food Analysts; Appellate Tribunal.
- Unsafe, sub-standard and misbranded food; adulterant.
- Section 22: nutraceuticals, functional foods, health supplements; regulations since 2016.
- Ayurveda Aahara Regulations, 2022: foods per authoritative Ayurveda books; food, not medicine.
- A product claiming to treat disease is a drug.

Likely short notes (5 marks)

1. Food Safety and Standards Act, 2006.
2. FSSAI.
3. Ayurveda Aahara.
4. Health supplements and nutraceuticals: regulation.
5. Distinguish an Ayurvedic drug from a health supplement.

REFERENCES USED IN THIS UNIT

Syllabus reference numbers for this unit: **45** (the syllabus names the FSS Act but lists no separate reference for it).

- *The Food Safety and Standards Act, 2006* (Sections 3, 22, 31, 97; Second Schedule).
- **[USDA 2022]** United States Department of Agriculture, Foreign Agricultural Service. “India’s FSSAI notifies final standards for Ayurveda foods.” GAIN report IN2022-0054, May 2022 (Gazette publication 6 May 2022).

UNIT B8(v)

LAWS PERTAINING TO NARCOTICS

SYLLABUS

Laws pertaining to Narcotics

LEARNING OUTCOMES

After studying this unit you should be able to describe the laws pertaining to narcotic drugs and psychotropic substances in India, and explain their relevance to Ayurvedic drugs containing opium, cannabis and other controlled ingredients, and to products containing alcohol.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

B8(v).1 The international framework

India's narcotics law implements three United Nations conventions: the **Single Convention on Narcotic Drugs, 1961**; the **Convention on Psychotropic Substances, 1971**; and the **Convention against Illicit Traffic in Narcotic Drugs and Psychotropic Substances, 1988**.

B8(v).2 The Narcotic Drugs and Psychotropic Substances Act, 1985

The **NDPS Act, 1985** consolidated the earlier laws (the Opium Acts of 1857 and 1878 and the Dangerous Drugs Act, 1930). It prohibits the cultivation, production, manufacture, possession, sale, purchase, transport, storage, consumption, import and export of narcotic drugs and psychotropic substances **except for medical or scientific purposes** and in the manner permitted by the Act and the rules.

Term	Meaning (in substance)
Narcotic drug	Coca leaf, cannabis (hemp), opium, poppy straw, and all manufactured drugs derived from them
Cannabis (hemp)	**Charas** (the separated resin) and **ganja** (the flowering or fruiting tops of the cannabis plant, **excluding the seeds and leaves when not accompanied by the tops**), and mixtures of these
Opium	The coagulated juice of the opium poppy and mixtures of it
Psychotropic substance	Substances listed in the Schedule to the Act (for example, many sedatives and stimulants)
Manufactured drug	Preparations of narcotic drugs, as notified

Feature	Provision
Enforcement	**Narcotics Control Bureau** (central coordinating agency); state police and excise; customs
Licensing	Central Government controls opium poppy cultivation and manufacture of opium and its derivatives; State Governments control certain activities such as cannabis cultivation and sale, under rules (NDPS Rules, 1985, and State rules)
Penalties	Graded by quantity (small, intermediate, commercial), severe for trafficking; lighter for personal consumption; immunity for addicts who volunteer for treatment
Essential narcotic drugs	Since an amendment in 2014, certain narcotic drugs needed for pain relief (such as morphine) are regulated centrally for medical access

B8(v).3 Relevance to Ayurvedic drugs



Figure B8v.1. Ayurvedic ingredients and the laws that control them.

Ingredient	Status	Practical requirement
Ahiphena (opium)	Narcotic drug under the NDPS Act	Manufacture of preparations containing opium needs the licences and permits prescribed under the NDPS Act and Rules, and the ASU manufacturing licence; strict records
Bhanga (cannabis leaves)	**Leaves without tops are excluded** from the definition of cannabis (hemp) under the NDPS Act; controlled by **State excise** laws	State excise permits for possession and use in manufacture; ASU manufacturing licence; Schedule E(1) caution where applicable
Ganja, Charas	Narcotic drugs	Not used in licensed products except under the law
Alcohol (Asava, Arishta; tinctures)	Not a narcotic; self-generated alcohol in Sandhana preparations is controlled by State excise rules where applicable	Alcohol content stated on the label; records
Dhattura, Kuchala, Vatsanabha and other Visha and Upavisha	Not narcotics under the NDPS Act; listed in **Schedule E(1)** of the Drugs Rules	Caution on label; supervised use (B8(i).8)

The **Medicinal and Toilet Preparations (Excise Duties) Act, 1955**, which formerly levied duty on medicinal preparations containing alcohol, opium or cannabis, has been **repealed**.

B8(v).4 Practical points for an Ayurvedic pharmacy

1. Obtain the necessary licences and permits **before** procuring any controlled ingredient.
2. Store controlled ingredients under lock, with a register of receipt, issue and use, reconciled for each batch.
3. Label products as required and supply them only through permitted channels.
4. Keep the records for inspection by the excise and narcotics authorities as well as the drug authorities.

KEY POINTS TO REMEMBER

- UN conventions: 1961 (narcotics), 1971 (psychotropics), 1988 (illicit traffic).
- NDPS Act 1985; NDPS Rules 1985; Narcotics Control Bureau.
- Narcotic drugs: coca, cannabis (hemp), opium, poppy straw and derivatives. Cannabis (hemp) = charas and ganja; seeds and leaves without tops are excluded (Bhanga is controlled by State excise).
- Ahiphena preparations need NDPS licences and an ASU licence.
- Dhattura, Kuchala, Vatsanabha: Schedule E(1), not NDPS.
- Medicinal and Toilet Preparations (Excise Duties) Act, 1955: repealed.

Likely short notes (5 marks)

1. Laws pertaining to narcotics.
2. NDPS Act and Ayurvedic drugs.

3. Legal status of Bhanga and Ahiphena in Ayurvedic preparations.

REFERENCES USED IN THIS UNIT

Syllabus reference number for this unit: **46**.

- **[R46]** Laws pertaining to narcotics: *The Narcotic Drugs and Psychotropic Substances Act, 1985* and *NDPS Rules, 1985*, as amended.
- United Nations. *Single Convention on Narcotic Drugs, 1961*; *Convention on Psychotropic Substances, 1971*; *Convention against Illicit Traffic in Narcotic Drugs and Psychotropic Substances, 1988.*
- *The Medicinal and Toilet Preparations (Excise Duties) Act, 1955* (repealed).

UNIT B8(vi)

FACTORIES AND PHARMACY ACTS

SYLLABUS

Factories and Pharmacy Acts

LEARNING OUTCOMES

After studying this unit you should be able to describe the main provisions of the Factories Act, 1948, explain its replacement by the Occupational Safety, Health and Working Conditions Code, 2020, and describe the Pharmacy Act, 1948.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

B8(vi).1 The Factories Act, 1948 and the OSH Code, 2020

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

The **Occupational Safety, Health and Working Conditions Code, 2020** (OSH Code) came into force on **21 November 2025** and **repealed thirteen Acts, including the Factories Act, 1948** [SCC 2025]. The syllabus names the Factories Act; its provisions are summarised below because they form the basis of the new Code, and because the same subjects (health, safety, welfare, working hours) are regulated in the Code and its rules.

B8(vi).1.1 The Factories Act, 1948: main provisions

Chapter	Main provisions
Definitions	A **factory** was premises where 10 or more workers worked with the aid of power, or 20 or more without power
Approval, licensing, registration	Plans approved; factory licensed and registered; occupier and manager named
Health	Cleanliness; disposal of wastes and effluents; ventilation and temperature; dust and fumes; artificial humidification; overcrowding; lighting; drinking water; latrines and urinals; spittoons

Chapter	Main provisions
Safety	Fencing of machinery; work on or near machinery in motion; employment of young persons on dangerous machines; hoists, lifts, lifting machines; pressure plant; floors, stairs and means of access; pits and openings; protection of eyes; precautions against dangerous fumes; explosive or inflammable dust or gas; **precautions in case of fire**; safety officers
Hazardous processes	Special provisions, including medical examination and disclosure of information
Welfare	Washing facilities; facilities for storing and drying clothing; sitting arrangements; **first-aid**; canteens, rest rooms, crèches and welfare officers, above prescribed numbers of workers
Working hours	Weekly hours limited to 48; daily hours limited to 9; weekly holiday; intervals for rest; restrictions on night work of women and on employment of young persons
Annual leave with wages	Leave earned according to days worked
Penalties	For contraventions

B8(vi).1.2 The OSH Code, 2020

The OSH Code consolidates the laws on safety, health and working conditions of workers in factories, mines, plantations, construction, contract labour and other establishments into one law with common registration, licensing and returns.

Feature	OSH Code
Factory	Premises where **20 or more** workers work with the aid of power, or **40 or more** without power (subject to State law where different) [Taxmann 2025]
Establishment	Covers establishments with **10 or more** workers
Duties of employers	Safe workplace; free annual health examination as prescribed; safety committees and safety officers where prescribed
Subjects	Occupational safety and health standards; health and working conditions; welfare; hours of work and annual leave; special provisions for contract labour, women and other categories
Rules	Occupational Safety, Health and Working Conditions (Central) Rules, 2026, and State rules [Taxmann 2025]

B8(vi).1.3 Relevance to an Ayurvedic pharmacy

An Ayurvedic manufacturing unit is a factory or establishment under the Code. The provisions on dust and fumes, ventilation, fire safety, protective equipment, first aid, medical examination and working hours apply directly to the hazards of a Rasashala and a Sneha Paka section (Chapter B4.2.8). They complement the GMP requirements of Schedule T, which protect the **product**; the labour law protects the **worker**.

B8(vi).2 The Pharmacy Act, 1948

B8(vi).2.1 Objectives

The **Pharmacy Act, 1948** regulates the **profession and practice of pharmacy**: it sets standards of pharmacy education, provides for the registration of pharmacists, and restricts compounding and dispensing on prescription to registered pharmacists.

B8(vi).2.2 Main provisions

Provision	Content
Pharmacy Council of India (PCI)	Central body; prescribes minimum standards of education (**Education Regulations**); approves courses and institutions; maintains the Central Register of Pharmacists
State Pharmacy Councils	Register pharmacists in each State; maintain the State register
Registration	Of persons with the prescribed qualifications; register open to inspection
Section 42	**No person other than a registered pharmacist** shall compound, prepare, mix or dispense any medicine on the prescription of a medical practitioner (with stated exceptions); contravention is punishable
Falsely claiming to be registered	Punishable
Pharmacy Practice Regulations	Made by the PCI under the Act; standards of professional practice, ethics and the role of the pharmacist

B8(vi).2.3 Relevance to Ayurveda

The Pharmacy Act and the Pharmacy Council of India deal with the pharmacy profession as it relates to modern medicines. Education in Ayurvedic pharmacy and the qualifications of technical staff for ASU manufacturing are governed by other authorities and by the Drugs Rules (B8(i).4.2). A registered pharmacist working in an integrated hospital or a retail pharmacy that dispenses modern medicines must, however, comply with the Pharmacy Act, and the Act's principles (registration, competence, accountability for dispensing) are a model for Ayurvedic dispensing (Chapter B6).

KEY POINTS TO REMEMBER

- Factories Act 1948: health, safety, welfare, working hours (48 per week, 9 per day), leave; factory = 10 workers with power or 20 without.
- OSH Code 2020 in force from 21 November 2025; repealed 13 Acts including the Factories Act; factory = 20 workers with power or 40 without; establishment = 10 workers.
- Rasashala hazards (fumes, heat, fire) fall under the labour law; GMP protects the product, labour law the worker.
- Pharmacy Act 1948: PCI; State Pharmacy Councils; registration; Section 42 (only registered pharmacists dispense on prescription); Pharmacy Practice Regulations.

Likely short notes (5 marks)

1. Factories Act, 1948: health, safety and welfare provisions.

2. The Occupational Safety, Health and Working Conditions Code, 2020.
3. Pharmacy Act, 1948.
4. Pharmacy Council of India.

REFERENCES USED IN THIS UNIT

Syllabus reference number for this unit: **47**.

- **[R47]** *The Factories Act, 1948* (repealed with effect from 21 November 2025) and *The Pharmacy Act, 1948*.
- *The Occupational Safety, Health and Working Conditions Code, 2020* (Act 37 of 2020).
- **[SCC 2025]** SCC Online Blog. “Enforcement dates of labour codes: key notifications.” 26 November 2025.
- **[Taxmann 2025]** Taxmann. “Analysis: Occupational Safety, Health and Working Conditions Code, 2020.” (Definitions of factory and establishment; Central Rules, 2026.)

UNIT B8(vii)

CONSUMER PROTECTION ACT

SYLLABUS

Consumer Protection Act -1986

LEARNING OUTCOMES

After studying this unit you should be able to describe the Consumer Protection Act, 1986, its replacement by the Consumer Protection Act, 2019, the rights of consumers, the redressal machinery, product liability and the control of misleading advertisements, and their relevance to Ayurvedic products and practice.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Aug 2022	B, Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.
Jan 2024	B, Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.

LAW UPDATE (POSITION CHECKED ON 16 OCTOBER 2026)

The syllabus names the **Consumer Protection Act, 1986**. It has been **replaced** by the **Consumer Protection Act, 2019**, which came into force in July 2020 and established the **Central Consumer Protection Authority** (CCPA). This unit describes the 1986 Act briefly and the 2019 Act in detail.

B8(vii).1 The Consumer Protection Act, 1986

The 1986 Act gave consumers a simple, quick and inexpensive way to seek redress against defective goods, deficient services and unfair trade practices, through a three-tier system of **District Forums**, **State Commissions** and the **National Commission**, and set up Consumer Protection Councils to promote consumer rights. In *Indian Medical Association v. V.P. Shantha* (1995), the Supreme Court held that medical services rendered for payment are “services” under the Act, so that patients could seek compensation for deficiency in medical service.

B8(vii).2 The Consumer Protection Act, 2019

B8(vii).2.1 Rights of consumers

1. The right to be **protected** against goods and services hazardous to life and property.
2. The right to be **informed** about the quality, quantity, potency, purity, standard and price of goods.
3. The right to be **assured access** to a variety of goods at competitive prices.

4. The right to be **heard**.
5. The right to seek **redressal** against unfair trade practices and exploitation.
6. The right to **consumer awareness**.

B8(vii).2.2 Machinery

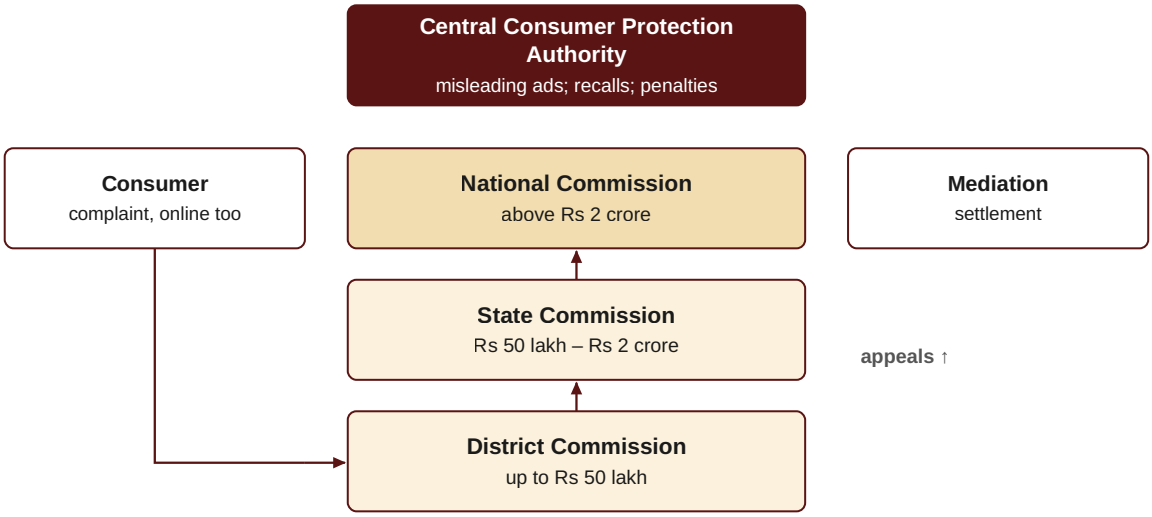


Figure B8vii.1. Consumer protection machinery under the Consumer Protection Act, 2019.

Body	Role
Central Consumer Protection Authority (CCPA)	Regulates violations of consumer rights, unfair trade practices and **false or misleading advertisements**; can investigate, order recall, discontinuation and refunds, and impose penalties
District Commission	Complaints of value up to **Rs 50 lakh**
State Commission	Above Rs 50 lakh up to **Rs 2 crore**; appeals from District Commissions
National Commission	Above **Rs 2 crore**; appeals from State Commissions
Mediation cells	Settlement by mediation

Pecuniary limits as revised by the Consumer Protection (Jurisdiction of the District Commission, the State Commission and the National Commission) Rules, 2021 [Drishti 2021]. Complaints may be filed electronically and where the complainant lives or works.

B8(vii).2.3 Product liability

The 2019 Act introduced **product liability**: a product manufacturer, product service provider or product seller is liable to compensate a consumer for harm caused by a defective product or deficient service, for example a manufacturing defect, a design defect, a deviation from specifications, or **inadequate instructions or warnings**. For an Ayurvedic manufacturer, a missing caution for a Schedule E(1) ingredient, a wrong dose on the label, or contamination can found a claim.

B8(vii).2.4 Misleading advertisements

Provision	Content
CCPA powers (Section 21)	Direct discontinuation or modification of a misleading advertisement; penalty on the manufacturer or endorser up to **Rs 10 lakh**, and up to **Rs 50 lakh** for subsequent contravention; prohibit an endorser from endorsing for up to **1 year** (up to 3 years for subsequent contravention). An endorser who has exercised due diligence is not liable [Nyaaya]
Criminal penalty (Section 89)	For false or misleading advertisement: imprisonment up to **2 years** and fine up to Rs 10 lakh; for subsequent offence, up to **5 years** and fine up to Rs 50 lakh [Nyaaya]
Adulterated and spurious products (Sections 90, 91)	Penalties graded by harm, up to life imprisonment where death results; suspension or cancellation of the licence of the manufacturer [Nyaaya]
CCPA guidelines (2022)	Guidelines for the prevention of misleading advertisements and endorsements
E-commerce	Consumer protection rules for e-commerce; the CCPA advisory of July 2022 on online sale of ASU drugs containing Schedule E(1) substances (B8(i).8.1)

B8(vii).2.5 Relevance to Ayurveda

1. **Ayurvedic products**: claims on labels, leaflets and advertisements must be true; misleading claims attract CCPA action and penalties, in addition to action under the DMR Act.
2. **Ayurvedic practitioners**: medical services for payment are “services”; deficiency in service (negligence, wrong medicine, failure to warn) can lead to compensation.
3. **Pharmacies**: sale of spurious, adulterated or expired products, or overcharging, is an unfair trade practice.
4. **Endorsers**, including celebrities and social media influencers, must verify claims before endorsing Ayurvedic products.

KEY POINTS TO REMEMBER

- CPA 1986 replaced by CPA 2019 (in force July 2020); CCPA established.
- Six consumer rights: safety, information, choice, to be heard, redressal, awareness.
- Commissions: District (up to Rs 50 lakh), State (to Rs 2 crore), National (above Rs 2 crore) (2021 rules).
- Product liability: manufacturer, service provider, seller; includes inadequate warnings.
- Misleading advertisements: CCPA penalty up to Rs 10 lakh (Rs 50 lakh repeat); endorser ban 1 (3) years; criminal 2 years (5 years repeat).
- Medical services are services (IMA v. V.P. Shantha, 1995).

Likely short notes (5 marks)

1. Consumer Protection Act, 1986.
2. Consumer Protection Act, 2019: key changes.
3. Central Consumer Protection Authority.
4. Product liability.

5. Consumer protection and Ayurvedic practice.

REFERENCES USED IN THIS UNIT

Syllabus reference number for this unit: **48**.

- **[R48]** *The Consumer Protection Act, 1986* (repealed); *The Consumer Protection Act, 2019*.
- **[Drishti 2021]** Drishti IAS. “Consumer Protection Rules, 2021.” 31 December 2021 (revised pecuniary jurisdiction).
- **[Nyaaya]** Nyaaya. “Punishments for consumer rights violations” (Sections 21, 88–91, Consumer Protection Act, 2019). nyaaya.org (accessed 16 October 2026).
- Supreme Court of India. *Indian Medical Association v. V.P. Shantha*, (1995) 6 SCC 651.

Viva pointers — Chapter B8

Answer each in one or two sentences.

Drugs and Cosmetics Act and Rules

- When were ASU drugs brought under the Drugs and Cosmetics Act? Which chapter?
- Define an Ayurvedic drug under the Act. What is a patent or proprietary ASU medicine?
- Distinguish misbranded, adulterated and spurious ASU drugs.
- What are ASUDTAB and ASUDCC?
- Which form is used to apply for a licence to manufacture ASU drugs? Which form is the licence?
- What is a loan licence?
- What does Rule 158-B require for a proprietary product?
- What is Schedule T? What is Schedule E(1)?
- What changed in Schedule E(1) in 2026?
- What is the status of Rule 170 today?
- Name three changes made by the Drugs (Eleventh Amendment) Rules, 2026.

Drugs and Magic Remedies Act

- What does Section 3 of the DMR Act prohibit?
- How many conditions are listed in its Schedule? Name five.
- What is a magic remedy?
- What are the penalties under the DMR Act?
- What did the Supreme Court direct in March 2025?

Food laws

- When was the PFA Act repealed, and by which Act?
- Define adulterated food.
- What is FSSAI? Who is the Food Safety Officer?
- Which section of the FSS Act deals with nutraceuticals and health supplements?
- What is Ayurveda Aahara? When were its regulations notified?
- How would you decide whether a herbal product is a drug or a food?

Narcotics

- Name the three UN conventions on drugs.
- What is included in “cannabis (hemp)” under the NDPS Act? Is Bhanga included?
- Which Ayurvedic ingredient needs NDPS licences?

Factories and Pharmacy Acts

- When did the OSH Code come into force? Which Act did it repeal?
- Define a factory under the Factories Act and under the OSH Code.
- What does Section 42 of the Pharmacy Act provide?

Consumer protection

- Name the six rights of consumers.
- What are the pecuniary limits of the consumer commissions?
- What penalties can the CCPA impose for misleading advertisements?
- Are medical services covered by consumer protection law? Which case decided it?

PART B · PHARMACY MANAGEMENT · CHAPTER B9

B9 · REGULATORY AFFAIRS IN INTERNATIONAL TRADE OF AYURVEDIC DRUGS

SYLLABUS POINT (PART B, 9)

Regulatory Affairs related to International Trade and Practices of Ayurvedic Drugs

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B9	Regulatory Affairs in International Trade of Ayurvedic Drugs	1	0	Mar 2026

Why this chapter matters to you. Ayurvedic products are exported to many countries, and Ayurveda as a system is practised and taught abroad. Every importing country has its own rules, and most do not recognise Ayurveda as a medical system in the way India does. A scholar who works in industry, regulation or research must understand how Ayurvedic products are classified, registered and judged abroad. The March 2026 paper combined this chapter with pharmacy management in a 20-mark question.

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Describe India’s export of Ayurvedic products and the institutions that support it.
2. Describe the Indian documentation required for export (licences, certificates, testing).
3. Describe how Ayurvedic products are regulated in major importing markets.
4. Explain the main barriers to international trade in Ayurvedic drugs and how to overcome them.
5. Describe India’s international initiatives for Ayurveda.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Mar 2026	B, Q.1	20	Discuss the management of an Ayurvedic Pharmacy. Include aspects of international trade and regulatory affairs related to the export of Ayurvedic drugs.

B9.1 India’s trade in Ayurvedic products

According to the Ministry of Commerce and Industry, exports of **Ayush and herbal products** were **US$ 689.34 million in 2024–25**, against US$ 651.17 million in 2023–24, a growth of about 5.9 per cent in value; in quantity, exports rose from about 1.06 lakh to 1.29 lakh tonnes [ISTN 2025].

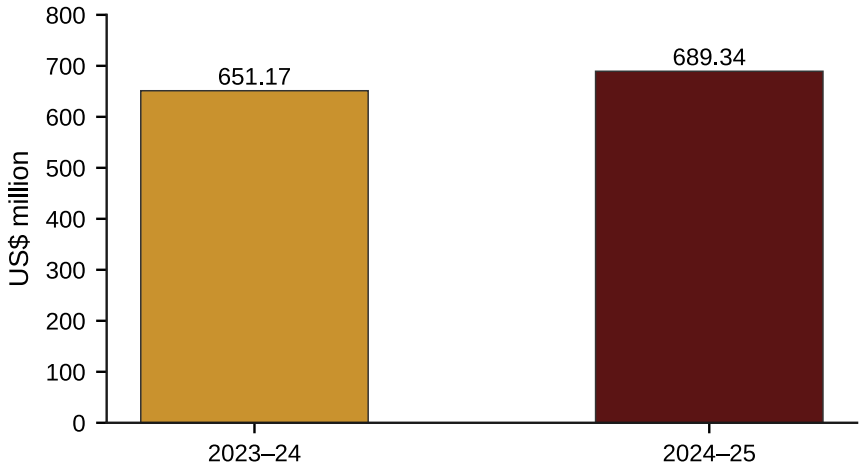


Figure B9.1. Ayush and herbal products exports, 2023–24 and 2024–25 (US$ million). Data from the Ministry of Commerce and Industry, as reported.

Institution	Role
Ministry of Ayush	Policy; international cooperation; support for registration and promotion abroad
Ministry of Commerce and Industry; Directorate General of Foreign Trade (DGFT)	Export policy; export codes; trade data
Ayush Export Promotion Council (AYUSHEXCIL)	Launched on 20 April 2022 at the Global Ayush Investment and Innovation Summit, Gandhinagar, to promote exports of Ayush products and services across Ayurveda, Siddha, Sowa-Rigpa, Unani and Homoeopathy [ISTN 2025]
Central Drugs Standard Control Organisation (CDSCO)	Issues the Certificate of Pharmaceutical Product for exports after joint inspection (Chapter B4.6.6)
State Licensing Authorities	Manufacturing licences; free sale and other certificates
Pharmacopoeia Commission for Indian Medicine and Homoeopathy	Standards (Chapter B10)

B9.2 Export documentation from India

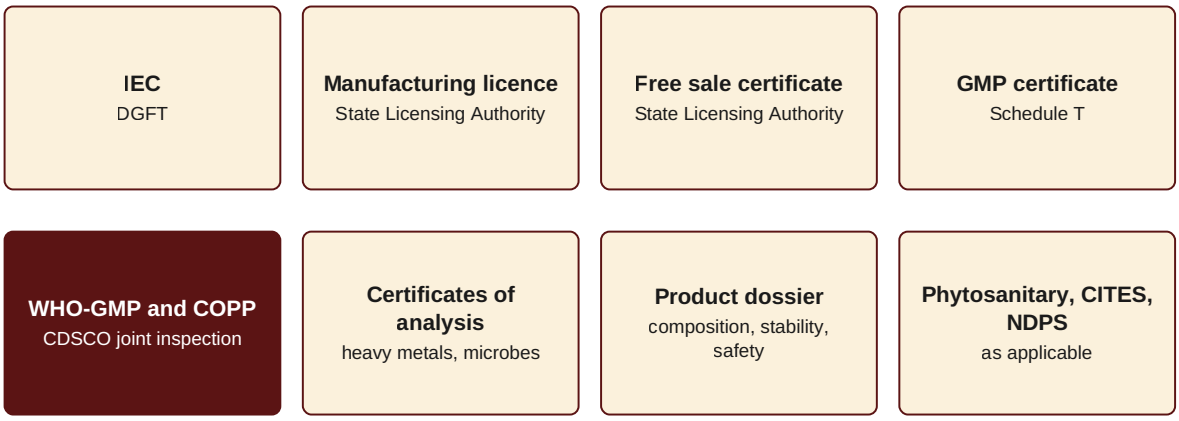


Figure B9.2. Documents commonly needed to export an Ayurvedic product.

Document	Issued by	Purpose
Importer–Exporter Code (IEC)	DGFT	Required for any export from India
Registration-cum-Membership Certificate	Export promotion council	Access to export benefits
Manufacturing licence (Form 25-D) and product permission	State Licensing Authority	Proof that the product is lawfully made
Free Sale Certificate / Market Standing Certificate	State Licensing Authority	Proof that the product is freely sold in India
GMP certificate (Schedule T)	State Licensing Authority	Proof of GMP compliance
WHO-GMP certificate and **Certificate of Pharmaceutical Product** (COPP)	CDSCO, after joint inspection with the Ministry of Ayush and the State Licensing Authority	Required by many importing countries for medicines [Lok Sabha 2024]
Certificates of analysis	Approved laboratory	Quality, including **heavy metals**, microbial load, pesticide residues and aflatoxins as required
Product dossier	Manufacturer	Composition, method of manufacture, specifications, stability, safety and efficacy data, labels
Phytosanitary certificate	Plant quarantine authority	For raw herbs and plant materials
Permits for controlled ingredients	NDPS and wildlife authorities, CITES	Opium-containing products; endangered species and their parts (Chapter B8(v))

Labels for export may be adapted to the requirements of the importing country (Rule 161A, Chapter B3). **Heavy metal testing** of purely herbal ASU medicines for export was made mandatory from 1 January 2006, with limits for lead, mercury and arsenic (Chapter A4.8.3).

B9.3 Regulation of Ayurvedic products in major markets

Most countries do not have a category called “Ayurvedic medicine”. An Ayurvedic product is fitted into whatever category the importing country uses: medicine, traditional herbal medicine, dietary or food supplement, natural health product, cosmetic or food.

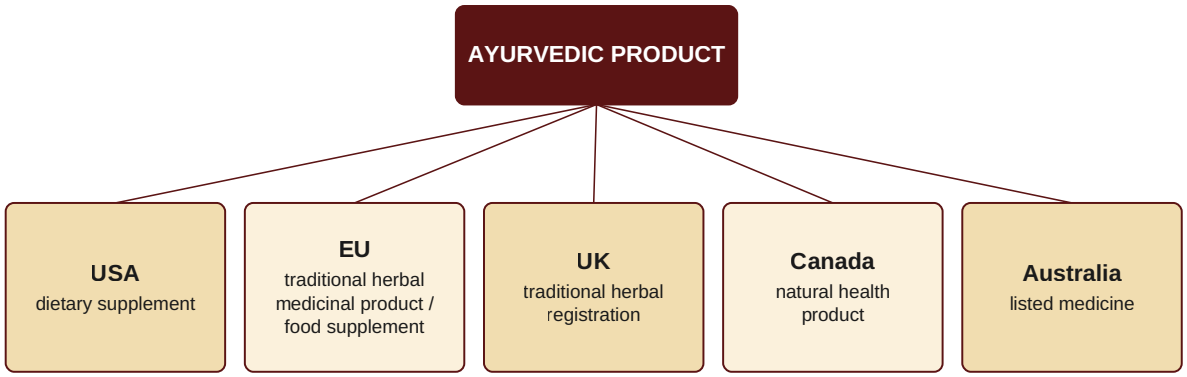


Figure B9.3. The category an Ayurvedic product falls into differs from country to country.

Market	Main route for Ayurvedic products	Key conditions
United States	Usually **dietary supplements** (Dietary Supplement Health and Education Act, 1994), regulated by the Food and Drug Administration	No claim to diagnose, treat, cure or prevent disease; structure/function claims with a disclaimer; current GMP for supplements; products with disease claims are unapproved drugs; import alerts have been issued against products with heavy metals
European Union	**Traditional herbal medicinal products** (Directive 2004/24/EC); or food supplements	Traditional use for at least 30 years, including 15 years in the EU; safety documented; quality to European standards; minerals and metals are not accepted as active ingredients of traditional herbal registrations; food supplement law for others
United Kingdom	**Traditional Herbal Registration** (MHRA); or food supplements	Similar to the EU scheme
Canada	**Natural health products** (Natural Health Products Regulations)	Product licence (NPN); evidence of safety and efficacy proportionate to the claim; site licence
Australia	**Listed** or registered complementary medicines (Therapeutic Goods Administration)	Permitted ingredients; permitted indications; GMP
Gulf, Africa, Asia	Registration as herbal or traditional medicines	Varies; COPP, free sale certificate and GMP certificates commonly required
Neighbouring and South Asian countries	Often recognise Ayurveda	Registration with national authorities

Regulations abroad change. Before exporting, consult the current law of the importing country and the guidance of AYUSHEXCIL and the Ministry of Ayush.

B9.4 Barriers to international trade and how to overcome them

Barrier	Way forward
No recognition of Ayurveda as a medical system	Government-to-government agreements; WHO initiatives; evidence
Heavy metals in Rasaushadhi; reports of poisoning (Chapters A4, A5)	Export of herbal products tested for metals; quality and safety data for herbo-mineral products; honest labelling
Lack of **clinical evidence** acceptable to foreign regulators	Well-designed trials (Chapter A2); publication in indexed journals
Quality variability; adulteration; microbial and pesticide contamination	GMP (Schedule T, WHO-GMP); pharmacopoeial standards (Chapter B10); testing
Restrictions on ingredients: endangered species (CITES), animal products, minerals	Substitutes; legal sourcing; documentation
Different labelling and claims rules	Market-specific labels (Rule 161A)
Cost and time of registration abroad	Government support schemes; council assistance
Intellectual property and biopiracy	TKDL; patents for genuine innovations (Chapters B7, B12)

B9.5 India’s international initiatives for Ayurveda

Initiative	Description
International cooperation by the Ministry of Ayush	Country-to-country and institute-level agreements; Ayush chairs in foreign universities; Ayush information cells; support for product registration and trade fairs
WHO Global Centre for Traditional Medicine	Established by the WHO with the support of India, at Jamnagar, Gujarat
Ayush Visa	Introduced on **27 July 2023** for foreigners seeking Ayush treatment in India, with Ayush (AY-1), Ayush Attendant (AY-2), e-Ayush and e-Ayush Attendant categories [Lok Sabha 2025]
Medical value travel	Facilitation of foreign patients for Ayush treatment
Standard terminologies	WHO international standard terminologies and the traditional medicine chapter of the International Classification of Diseases, developed with India’s support, help Ayurvedic diagnoses to be recorded internationally
AYUSHEXCIL	Export promotion (B9.1)

B9.6 International practice of Ayurveda

Outside India, Ayurveda is practised in many countries, but its legal status varies widely: in some it is recognised as a system of medicine, in others practitioners work under general complementary medicine or wellness rules, and in many there is no specific regulation. Products used by these practitioners must still comply with the product law of the country (B9.3). A practitioner abroad must therefore know both the professional and the product rules.

AYURVEDA CORRELATION

The classical texts describe the exchange of drugs and knowledge across regions and the adaptation of treatment to the land (Desha) and the people (Satmya) (Charaka Samhita, Vimana Sthana 1 and 8) [R26]. International trade today requires the same adaptation, with the added discipline of the importing country's law.

Established. Ayurvedic products are traded internationally under a variety of regulatory categories; quality and safety documentation is decisive.

To be remembered. A product acceptable in India may be unlawful abroad, especially if it contains metals or minerals, animal parts or narcotic ingredients, or carries disease claims.

KEY POINTS TO REMEMBER

- Ayush and herbal exports: US$ 689.34 million in 2024–25 (US$ 651.17 million in 2023–24).
- AYUSHEXCIL launched 20 April 2022, Gandhinagar.
- Export documents: IEC, manufacturing licence, free sale certificate, GMP certificate, WHO-GMP and COPP (CDSCO joint inspection), certificates of analysis (heavy metals), dossier, phytosanitary and CITES permits.
- Rule 161A: export labels may follow importing country's law.
- Markets: USA dietary supplements (DSHEA 1994); EU traditional herbal medicinal products (2004/24/EC: 30 years, 15 in EU); UK THR; Canada NHP; Australia listed medicines.
- Barriers: recognition, heavy metals, evidence, quality, ingredient restrictions, labels, cost.
- Initiatives: international cooperation; WHO GCTM, Jamnagar; Ayush Visa (27 July 2023); standard terminologies.

EXAM CORNER — CHAPTER B9

Likely long questions (20 marks)

1. Discuss the management of an Ayurvedic pharmacy, including aspects of international trade and regulatory affairs related to the export of Ayurvedic drugs *(set in March 2026; see also Chapter B4)*.
2. Describe the regulatory affairs related to international trade of Ayurvedic drugs, with the requirements of major importing countries.

Likely short notes (5 marks)

1. Export documentation for Ayurvedic products.
2. Certificate of Pharmaceutical Product.
3. Regulation of herbal products in the European Union.
4. Barriers to export of Ayurvedic drugs.
5. AYUSHEXCIL.

Answer frame: "Regulatory affairs related to international trade and practices of Ayurvedic drugs." (20 marks)

Part	What to write	Suggested marks
Introduction	Scope; export data; institutions	3
Indian documentation	Table of documents; Rule 161A; heavy metal testing	5
Foreign markets	Table: USA, EU, UK, Canada, Australia	5
Barriers and solutions	Table	4
Initiatives	International cooperation; GCTM; Ayush Visa; terminologies	2
Conclusion	Quality and evidence as the key	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **26, 36, 43**.

- **[R26]** Charaka Samhita, Vimana Sthana 1 and 8. **[R36]** *Ayurvedic Pharmacopoeia of India*. **[R43]** Drugs and Cosmetics Rules (Rule 161A).

Additional verified sources (accessed 16 October 2026)

- **[ISTN 2025]** India Seatrade News. "Ayush and herbal products exports grew 5.9 per cent in FY25." 27 May 2025 (data of the Ministry of Commerce and Industry; launch of AYUSHEXCIL).
- **[Lok Sabha 2024]** Lok Sabha Unstarred Question No. 1318, answered 9 February 2024 (COPP; AYUSH Mark).
- **[Lok Sabha 2025]** Lok Sabha Unstarred Question No. 820, answered 7 February 2025 (Ayush Visa).
- European Parliament and Council. Directive 2004/24/EC on traditional herbal medicinal products. *Official Journal of the European Union*. 2004;L136:85–90.
- United States Congress. *Dietary Supplement Health and Education Act of 1994*.

Viva pointers — Chapter B9

Answer each in one or two sentences.

- What was the value of Ayush and herbal product exports in 2024–25?
- What is AYUSHEXCIL? When was it launched?
- Name five documents needed to export an Ayurvedic product.
- What is a COPP? Who issues it for Ayurvedic exports?
- What is a free sale certificate?
- Under which category are Ayurvedic products usually sold in the United States?
- What are the conditions for a traditional herbal medicinal product in the European Union?
- Why are many Rasaushadhi difficult to export to Europe?
- What does Rule 161A allow?
- Name four barriers to export of Ayurvedic drugs.
- What is the Ayush Visa? When was it introduced?
- Where is the WHO Global Centre for Traditional Medicine?

PART B · PHARMACY MANAGEMENT · CHAPTER B10

B10 · AYURVEDIC PHARMACOPOEIA AND FORMULARY OF INDIA

SYLLABUS POINT (PART B, 10)

Introduction to Ayurvedic Pharmacopoeia of India, Ayurvedic Formulary of India.

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B10	Ayurvedic Pharmacopoeia and Formulary of India	2	3	Jul 2018, Sep 2018, Aug 2021, Aug 2022, Mar 2026

Why this chapter matters to you. The *Ayurvedic Pharmacopoeia of India* (API) and the *Ayurvedic Formulary of India* (AFI) are the official books of standards for Ayurvedic drugs. Every quality control laboratory, every manufacturer and every regulator works from them. This chapter was set in five of the seven papers, twice as a 20-mark question, including a demand to describe the structure of an ideal monograph with an example (B10.5).

LEARNING OUTCOMES

After studying this chapter you should be able to:

1. Define a pharmacopoeia and a formulary, and explain the need for them in Ayurveda.
2. Describe the history and functions of the Ayurvedic Pharmacopoeia Committee and of the Pharmacopoeia Commission for Indian Medicine and Homoeopathy.
3. List the publications of the API and AFI.
4. Describe the structure of an API monograph for a single drug and for a formulation, with an example.
5. Explain the legal status of the API and AFI.

These outcomes are set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Jul 2018	B, Q.2	20	Write down in brief regarding API & AFI.
Sep 2018	B, Q.2	20	Write the need for Ayurvedic Pharmacopoeia and Formulary of India. Enlist the functions of Ayurvedic Pharmacopoeial Committee. Enlist the publications in the form of API and AFI. Write the structure of ideal monograph with an example.
Aug 2021	B, Q.3C	5	Brief about API and AFI
Aug 2022	B, Q.3a	5	Ayurved Pharmacopeia of India.
Mar 2026	B, Q.3c	5	Write a short note on the Ayurvedic Pharmacopoeia of India (API) and Ayurvedic Formulary of India (AFI).

B10.1 Pharmacopoeia and formulary

> A **pharmacopoeia** is an official book of **standards** for drugs: it describes their identity, purity and strength, and the methods of testing them, so that a drug can be judged as of standard quality or not. A **formulary** is an official book of **formulations**: it gives their composition and method of preparation, and often their dose and uses.

Feature	Pharmacopoeia (API)	Formulary (AFI)
Main content	Standards and tests	Composition and method
Question it answers	“Is this drug genuine and of good quality?”	“How is this formulation made, and from what?”
Users	QC laboratories, regulators, manufacturers	Manufacturers, teachers, physicians
Example	Total ash of Amalaki not more than 7 per cent	Ingredients and method for Chyavanaprasha

B10.2 Why Ayurveda needs a pharmacopoeia and formulary

The September 2018 paper asked for the need for the API and AFI.

1. **Many names for one drug and one name for many drugs**: Sanskrit names and regional names differ; substitutes and adulterants are common. A pharmacopoeia fixes the botanical identity.
2. **Different versions of one formula** in different texts and traditions. A formulary chooses one authentic version as the official reference.
3. **Classical units and methods** need translation into metric units and standard procedures.
4. **Quality control needs measurable standards**: classical signs are qualitative; modern regulation needs limits and tests.
5. **Legal enforcement**: the Drugs and Cosmetics Act needs official books against which a sample can be judged “of standard quality” or not.
6. **Safety**: limits for heavy metals, pesticide residues, microbial load and aflatoxins.

7. **Trade and export**: importing countries ask for official standards.
8. **Teaching and research**: a common reference for all.

B10.3 History and institutions

Year	Event
1962	**First Ayurvedic Pharmacopoeia Committee** constituted, under Col. Sir R. N. Chopra [API II(1)]
1972	Committee reconstituted under Prof. A. N. Namjoshi [API II(1)]
1975	**Pharmacopoeial Laboratory for Indian Medicine** (PLIM) and Homoeopathic Pharmacopoeia Laboratory established at Ghaziabad [Drishti 2020]
1978	**AFI Part I** published (revised edition 2003) [API II(1)]
1980s onward	**API Part I** (single drugs) published volume by volume [Yao 2023]
2000	AFI Part II [API II(1)]
2001, 2006	Committee reconstituted under the Department of AYUSH [API II(1)]
2007	**API Part II** (formulations), Volume I [API II(1)]
2010	**Pharmacopoeia Commission for Indian Medicine and Homoeopathy** (PCIM&H) set up as an autonomous body [Drishti 2020]
2020	Cabinet approves re-establishment of PCIM&H as a **subordinate office** of the Ministry of Ayush, with PLIM and HPL merged into it [Drishti 2020]

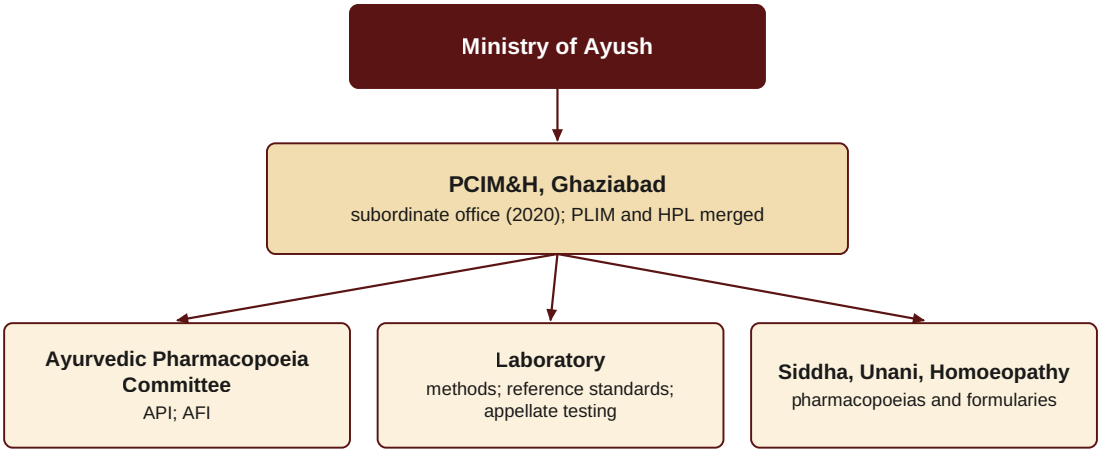


Figure B10.1. Institutions responsible for Ayurvedic pharmacopoeial standards.

B10.3.1 Functions of the Ayurvedic Pharmacopoeia Committee and PCIM&H

1. Prepare the **official formularies** (AFI) and **pharmacopoeias** (API) of Ayurvedic drugs.
2. Lay down **standards** for single drugs of plant, animal and mineral origin, for extracts and for compound formulations.
3. Develop and validate **methods of testing**: pharmacognostic, physico-chemical, chromatographic (TLC, HPTLC), and for contaminants.
4. Prescribe limits for **heavy metals, pesticide residues, microbial contamination and aflatoxins**.
5. Prepare **reference standards** and atlases (macroscopy, microscopy, chromatography).

6. Revise and update the standards; publish amendments.
7. Act as the **appellate laboratory** for testing ASU drugs (Chapter B8(i).6) [Lok Sabha 2024].
8. Train analysts and support state drug testing laboratories.

B10.4 Publications

B10.4.1 Ayurvedic Pharmacopoeia of India, Part I: single drugs

Volume	Monographs	Year
I	80	1986
II	78	1999
III	100	2001
IV	68	2004
V	92	2006
VI	101	2008
VII	21 (minerals and metals)	2008
VIII	60 (15 single drugs and their extracts)	2011
IX	45 (15 single drugs and their extracts)	2016

Volumes, monograph counts and years as tabulated by Yao and colleagues [Yao 2023]; the authors note that the first volume is variously dated. A tenth volume of Part I, on minerals and metals, has also been published by PCIM&H. Volume VIII contains the guidelines for stability studies referred to in Rule 161B (Chapter B3).

B10.4.2 Ayurvedic Pharmacopoeia of India, Part II: formulations

Part II gives standards for **compound formulations**. Volume I (first edition, 2007) contained 50 monographs on formulations grouped by dosage form: Avaleha, Churna, Ghrita, Guggulu, Vati and Gutika, Kshara and Lavana, Taila and Lepa [API II(1)]. Further volumes have followed.

B10.4.3 Ayurvedic Formulary of India

Part	Year	Formulations
I	1978 (revised 2003)	about 444
II	2000	about 191
III	Published later	—

Figures for Parts I and II from the preface to API Part II, Volume I [API II(1)].

The AFI is organised by **dosage form** (Asava and Arishta, Arka, Avaleha, Kwatha, Gutika, Ghrita, Churna, Taila, Bhasma, Rasa Yoga and others). Each formulation gives the reference text, the ingredients with botanical identity, part and quantity, and the method of preparation, and often the dose, Anupana and uses. Its appendices give the **classical weights and measures** with metric equivalents (Chapter B6.2.4), definitions of processes (Paribhasha), and lists of plants with botanical names.

B10.4.4 Other publications

Macroscopy and microscopy atlases; TLC and HPTLC atlases; and the *Protocol for Testing of Ayurvedic, Siddha and Unani Medicines* prepared at the Pharmacopoeial Laboratory for Indian Medicine [R35].

B10.5 Structure of an API monograph

B10.5.1 Single drug (Part I)



Figure B10.2. Structure of an API Part I monograph.

Heading	Content
Title	Sanskrit name of the drug and part used
Definition	Botanical source (species, family), part used, distribution
Synonyms	Sanskrit synonyms; English and regional names
Description	Macroscopic and microscopic characters (with figures in atlases)
Identity, purity and strength	Limits for foreign matter, total ash, acid-insoluble ash, alcohol-soluble and water-soluble extractives, moisture; in later volumes, TLC and assay of markers
Constituents	Main chemical constituents
Properties and action	Rasa, Guna, Virya, Vipaka, Karma
Important formulations	Classical formulations containing the drug
Therapeutic uses	Classical indications
Dose	Dose of the drug in each form

B10.5.2 Example: Amalaki (fresh fruit), API Part I, Volume I

Heading	Entry [API I(1)]
Definition	Fresh fruit pulp of *Emblica officinalis* Gaertn. (Euphorbiaceae)

Heading	Entry [API I(1)]
Synonyms	English: Emblic Myrobalan; Hindi: Amla, Aonla; and other regional names
Macroscopic	Fruit round, 2.5–3.5 cm across, fleshy, smooth, with six prominent lines; green when young, light yellow or pinkish when ripe; taste sour and astringent, then mildly sweet
Microscopic	Single-layered epidermis with thick cuticle, 2–4 hypodermal layers, mostly thin-walled parenchyma with collateral vascular bundles and clusters of silica crystals
Foreign matter	Not more than 2 per cent
Total ash	Not more than 7 per cent
Acid-insoluble ash	Not more than 2 per cent
Alcohol-soluble extractive	Not less than 40 per cent (dried basis)
Water-soluble extractive	Not less than 50 per cent
Moisture content	Not less than 80 per cent (fresh fruit)
Constituents	Ascorbic acid; tannins
Rasa	Madhura, Amla, Katu, Tikta, Kashaya
Guna	Laghu, Ruksha
Virya	Shita
Vipaka	Madhura
Karma	Tridoshahara, Rasayana, Chakshushya
Important formulation	Chyavanaprasha
Therapeutic uses	Raktapitta, Amlapitta, Prameha, Daha
Dose	Fruit pulp 10–20 g; fresh juice 5–10 ml

B10.5.3 Formulation (Part II)

A Part II monograph gives: the title (as in the AFI); definition; **formulation composition** (ingredients, parts, quantities); **method of preparation**; description of the finished product; **identification** (microscopy and physico-chemical parameters such as pH, loss on drying, ash values, extractives, TLC or HPTLC profile); **other requirements** (heavy metals, microbial contamination, pesticide residues, aflatoxins); therapeutic uses; dose; administration; and **storage** [API II(1)].

B10.5.4 The ideal monograph

An ideal monograph is **complete** (identity, purity, strength and safety), **specific** (tests that distinguish the drug from substitutes and adulterants), **quantitative** (numerical limits, marker assays), **validated** (methods tested for accuracy and reproducibility), **practical** (methods within the reach of ordinary laboratories), **current** (revised with new knowledge) and **consistent** with the classical description.

B10.6 Legal status

The API and AFI are the **official books** under the Drugs and Cosmetics Act. A classical ASU drug must follow the formula in an authoritative book (Chapter B8(i).2), and the standards in the API are the standards against which a sample is judged. The API volumes carry a legal notice stating that they are the books of standards for the drugs included in them [API II(1)].

AYURVEDA CORRELATION: GRAHYA LAKSHANA AND THE PHARMACOPOEIA

The classical texts gave detailed signs of good and bad drugs (Grahya and Agrahya lakshana), rules for collection and storage, and the recognition of substitutes (Pratinidhi Dravya). The API turns these qualitative rules into measurable standards; the AFI gives one authentic version of each classical formula.

Established. Pharmacopoeial standards improve identification and detect gross adulteration and contamination.

To be remembered. A drug that passes the pharmacopoeial tests is of standard quality; whether it is effective for a given use is a separate question that standards alone do not answer.

KEY POINTS TO REMEMBER

- Pharmacopoeia: standards and tests. Formulary: composition and method.
- Need: identity, authentic formula, metric units, measurable quality, legal enforcement, safety, export, teaching.
- APC 1962 (R. N. Chopra); reconstituted 1972 (A. N. Namjoshi), 2001, 2006. PLIM 1975. PCIM&H 2010; subordinate office with PLIM and HPL merged (2020).
- API Part I: single drugs, nine volumes (1986–2016) with a tenth on minerals and metals; Vol VII minerals and metals (21); Vol VIII, IX extracts; Vol VIII stability guidelines.
- API Part II: formulations, Vol I 2007 (50 monographs).
- AFI Part I 1978 (rev. 2003; ~444), Part II 2000 (~191), Part III.
- Monograph: title, definition, synonyms, description, identity–purity–strength, constituents, properties and action, formulations, uses, dose.
- Amalaki: foreign matter ≤2%; total ash ≤7%; acid-insoluble ash ≤2%; alcohol-soluble extractive ≥40%; water-soluble extractive ≥50%.

EXAM CORNER — CHAPTER B10

Likely long questions (20 marks)

1. Write in brief regarding API and AFI *(set in July 2018)*.
2. Write the need for the API and AFI. Enlist the functions of the Ayurvedic Pharmacopoeia Committee. Enlist the publications in the form of API and AFI. Write the structure of an ideal monograph with an example *(set in September 2018)*.

Likely short notes (5 marks)

1. API and AFI *(set in August 2021 and March 2026)*.
2. Ayurvedic Pharmacopoeia of India *(set in August 2022)*.
3. PCIM&H.

4. Structure of an API monograph.
5. Difference between a pharmacopoeia and a formulary.

Answer frame: the September 2018 question (20 marks)

Part	What to write	Suggested marks
Definitions	Pharmacopoeia; formulary; difference table	2
Need	Eight points	3
Committee and functions	History; functions list; PCIM&H	4
Publications	API Part I table; Part II; AFI table; other publications	5
Monograph	Structure table; Amalaki example	5
Legal status	Official books under the Act	1

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **35, 36, 37**.

- **[R35]** Quality control of Ayurvedic drugs, PLIM, Ghaziabad (syllabus reference 35). Cited here as: Lohar D. *Protocol for Testing of Ayurvedic, Siddha and Unani Medicines*. Department of AYUSH, Ministry of Health and Family Welfare, Government of India; 2008.
- **[R36]** *The Ayurvedic Pharmacopoeia of India*. Part I and Part II. Government of India. **[API I(1)]** Part I, Volume I, monograph on Amalaki (fresh fruit). **[API II(1)]** Part II, Volume I, first edition, 2007 (preface: history of the Committee; AFI; monograph structure; legal notices).
- **[R37]** *The Ayurvedic Formulary of India*. Parts I–III. Government of India.

Additional verified sources (accessed 16 October 2026)

- **[Yao 2023]** Yao R, Heinrich M, Zhang B, Wei X, Qi Y, Gao W. Single botanical drugs in the Ayurvedic Pharmacopoeia of India: a quantitative ethnobotanical analysis. *Front Pharmacol*. 2023;14:1136446.
- **[Drishti 2020]** Drishti IAS. “Re-establishment of PCIM&H.” 4 June 2020 (Cabinet decision).
- **[Lok Sabha 2024]** Lok Sabha Unstarred Question No. 1318, 9 February 2024 (PCIM&H as appellate laboratory; numbers of standards).

Viva pointers — Chapter B10

Answer each in one or two sentences.

- Distinguish a pharmacopoeia from a formulary.
- Give five reasons why Ayurveda needs a pharmacopoeia.
- When was the first Ayurvedic Pharmacopoeia Committee formed, and under whom?
- What is PCIM&H? What happened to PLIM in 2020?
- How many volumes does API Part I have? Which volume deals with minerals and metals?
- Which API volume contains the stability guidelines?
- When was AFI Part I first published? About how many formulations does it contain?
- Name the headings of an API Part I monograph.
- What are the limits for total ash and acid-insoluble ash for Amalaki?
- What does a Part II (formulation) monograph contain that a Part I monograph does not?
- What is the legal status of the API?
- What are the qualities of an ideal monograph?

PART B · PHARMACY MANAGEMENT · CHAPTER B11

B11 · INDIAN, BRITISH AND UNITED STATES PHARMACOPOEIAS AND THE PHARMACEUTICAL CODEX

SYLLABUS POINT (PART B, 11)

Introduction to Indian Pharmacopoeia, British and United States Pharmacopoeia, Pharmaceutical Codex

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B11	Indian, British and United States Pharmacopoeias and the Pharmaceutical Codex	0	0	—

Why this chapter matters to you. The Indian Pharmacopoeia is a legal standard in India; the British and United States Pharmacopoeias are used worldwide and by importing countries. A Rasashastra and Bhaishajya Kalpana scholar uses them for general methods of analysis, for excipients and packaging materials, and for standards of modern ingredients used in Ayurvedic products. This chapter has not yet been set in the papers examined, but it is in the syllabus and can be asked.

LEARNING OUTCOMES

After studying this chapter you should be able to describe the history, publishing body, contents, legal status and current edition of the Indian Pharmacopoeia, the British Pharmacopoeia and the United States Pharmacopeia–National Formulary, and describe the Pharmaceutical Codex.

This outcome is set by the author.

Asked in university examinations

This topic has not been asked in the seven university papers examined (July 2018 to March 2026). It remains in the syllabus and can be asked.

B11.1 Pharmacopoeias and their purpose

A pharmacopoeia is an official book of standards for drugs (Chapter B10.1). National pharmacopoeias are legally binding in their own countries; several are also recognised by other countries. Their monographs give the **identity, purity, strength** and **quality** requirements for drug substances, excipients and dosage forms,

and their **general chapters** (appendices) describe methods of analysis, such as ash values, loss on drying, chromatography, dissolution, microbial limits and heavy metals, which apply to all products.

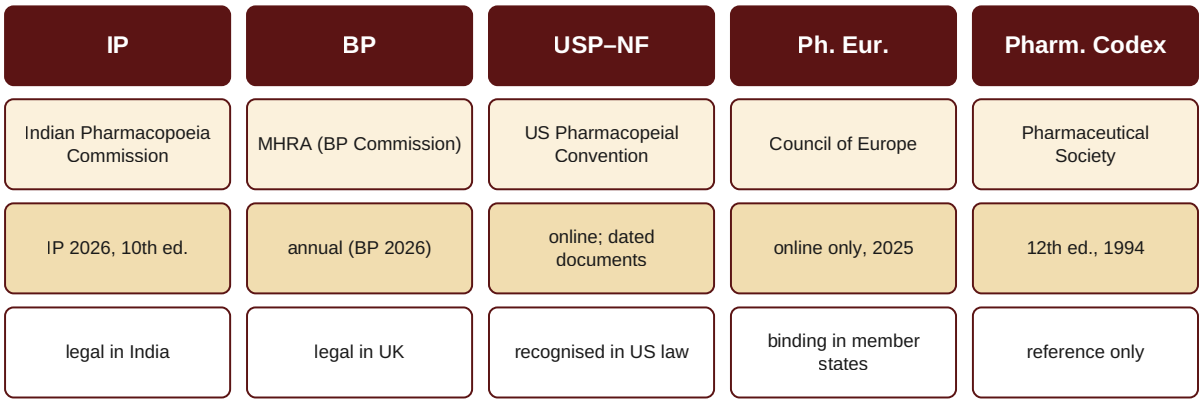


Figure B11.1. The main pharmacopoeias and compendia: publishers and current form.

B11.2 The Indian Pharmacopoeia

Aspect	Details
Publisher	**Indian Pharmacopoeia Commission** (IPC), Ghaziabad, an autonomous body of the Ministry of Health and Family Welfare
Legal status	Official book of standards for drugs under the Drugs and Cosmetics Act, 1940
Editions	1955 (first), 1966, 1985, 1996, 2007, 2010, 2014, 2018, 2022, and **2026 (tenth edition)**
Current edition	**IP 2026**, the 10th edition, released by the Union Health Minister around the start of 2026, with **3,340 monographs**, of which **121 are new** [IPC 2026]
Contents	General notices; monographs on drug substances, excipients, dosage forms, biological products, vaccines and blood products, radiopharmaceuticals, veterinary products, and **herbs and herbal products**; general chapters on methods of analysis
Other roles of IPC	National Coordination Centre of the Pharmacovigilance Programme of India (Chapter A6); reference substances

Use for the Ayurvedic pharmacist: general methods of analysis; standards for excipients, solvents, alcohol, sugars, preservatives and packaging materials; monographs on herbs and herbal products.

B11.3 The British Pharmacopoeia

Aspect	Details
Publisher	British Pharmacopoeia Commission, through the **Medicines and Healthcare products Regulatory Agency** (MHRA), United Kingdom
First edition	1864
Form	Published **annually**; each edition takes effect on 1 January of its year (for example, **BP 2026**)
Contents	Monographs on drug substances, excipients and formulated preparations; herbal drugs and herbal preparations; general notices and appendices. It incorporates the monographs of the **European Pharmacopoeia**
Status	Legally binding in the United Kingdom; used in many Commonwealth countries

B11.4 The United States Pharmacopeia–National Formulary

Aspect	Details
Publisher	**United States Pharmacopeial Convention** (USP), a non-governmental, not-for-profit organisation
First edition	USP 1820; National Formulary (NF) from 1888; published together as **USP–NF**
Contents	USP: drug substances, dosage forms, compounded preparations, **dietary supplements** and botanicals; NF: excipients
Form	Online, with standards becoming official on stated dates; the printed edition has been discontinued (USP 43–NF 38 was the last print edition), and each document now carries its own official date [ECA 2024]
Status	Recognised in United States law (Federal Food, Drug, and Cosmetic Act); used worldwide

Use for the Ayurvedic pharmacist: USP monographs on botanicals and dietary supplements are relevant for exports to the United States (Chapter B9).

B11.5 The European Pharmacopoeia

The **European Pharmacopoeia** (Ph. Eur.), published by the Council of Europe, is binding in its member states. From June 2025 it is published **online only**, as an annual edition of three issues, beginning with the 12th edition [NSF 2025]. Its herbal monographs are relevant to export of Ayurvedic products to Europe.

B11.6 The Pharmaceutical Codex

Aspect	Details
Publisher	Pharmaceutical Society of Great Britain (now the Royal Pharmaceutical Society)
History	First published in 1907 as the **British Pharmaceutical Codex** (BPC), a supplement to the BP covering drugs and preparations not in the BP; renamed **The Pharmaceutical Codex** with the 11th edition (1979)
Last edition	**12th edition, 1994**, subtitled *Principles and Practice of Pharmaceutics*
Nature	Not a book of legal standards; a **reference work** on pharmaceutical principles, dosage forms, formulation and the properties of drugs

A misprint in the syllabus wording of this item is corrected here to the **Pharmaceutical Codex** (see the list of corrections in How to Use This Book).

B11.7 Other pharmacopoeias

Pharmacopoeia	Note
International Pharmacopoeia	Published by WHO; recommended standards for countries without their own
Japanese, Chinese Pharmacopoeias	National standards; the Chinese Pharmacopoeia includes many traditional Chinese medicines
Ayurvedic, Siddha, Unani and Homoeopathic Pharmacopoeias of India	Published by PCIM&H (Chapter B10)

KEY POINTS TO REMEMBER

- IP: Indian Pharmacopoeia Commission, Ghaziabad; editions from 1955; IP 2026 is the 10th edition (3,340 monographs; 121 new); legal standard under the D&C Act.
- BP: MHRA; first 1864; annual; includes Ph. Eur. monographs.
- USP–NF: USP Convention; USP 1820, NF 1888; online; USP 43–NF 38 last printed edition.
- Ph. Eur.: online only from June 2025; annual edition of three issues.
- Pharmaceutical Codex: Pharmaceutical Society of Great Britain; BPC 1907; 12th edition 1994 (*Principles and Practice of Pharmaceutics*); a reference work, not a legal standard.

EXAM CORNER — CHAPTER B11

Likely long question (20 marks)

1. Give an introduction to the Indian, British and United States Pharmacopoeias and the Pharmaceutical Codex. Compare them with the Ayurvedic Pharmacopoeia of India.

Likely short notes (5 marks)

1. Indian Pharmacopoeia.
2. British Pharmacopoeia.

3. United States Pharmacopeia–National Formulary.
4. Pharmaceutical Codex.
5. Indian Pharmacopoeia Commission.

How to attempt

Use one table per pharmacopoeia with the same rows (publisher, history, form, contents, status), and finish with a comparison table including the API.

REFERENCES USED IN THIS CHAPTER

Syllabus reference numbers for this chapter: **38, 39, 40, 41**.

- **[R38]** *Indian Pharmacopoeia*. Ghaziabad: Indian Pharmacopoeia Commission. **[IPC 2026]** Indian Pharmacopoeia Commission. “Union Health Minister releases 10th edition of Indian Pharmacopoeia (IP 2026).” ipc.gov.in, January 2026.
- **[R39]** *British Pharmacopoeia*. London: Medicines and Healthcare products Regulatory Agency (annual).
- **[R40]** *United States Pharmacopeia–National Formulary*. Rockville: United States Pharmacopeial Convention. **[ECA 2024]** ECA Academy. “Recent changes to the USP: what you need to know.” gmp-compliance.org.
- **[R41]** The syllabus lists “Pharmacopeia Codex”. Cited here as: Lund W, editor. *The Pharmaceutical Codex: Principles and Practice of Pharmaceutics*. 12th ed. London: Pharmaceutical Press; 1994.
- **[NSF 2025]** NSF. “European Pharmacopoeia online only, June 2025.” nsf.org.

Viva pointers — Chapter B11

Answer each in one or two sentences.

- Who publishes the Indian Pharmacopoeia? Which is the current edition?
- How many monographs does IP 2026 contain?
- When was the first IP published?
- Who publishes the British Pharmacopoeia? How often?
- What do USP and NF stand for? What does the NF cover?
- What changed in the publication of USP–NF and the European Pharmacopoeia recently?
- What is the Pharmaceutical Codex? Which is its latest edition?
- Is the Pharmaceutical Codex a legal book of standards?
- Why should an Ayurvedic pharmacist consult the IP?

PART B · PHARMACY MANAGEMENT · CHAPTER B12

B12 · TRADITIONAL KNOWLEDGE DIGITAL LIBRARY

SYLLABUS POINT (PART B, 12)

Introduction to Traditional Knowledge Digital Library

Record in university examinations

Seven Paper IV question papers were examined (Parul University, July 2018 to March 2026). The table shows how often each point was set as a 20-mark long question or a 5-mark short note.

Point	Topic	Long (20)	Short (5)	Papers in which it was asked
B12	Traditional Knowledge Digital Library	0	3	Sep 2020, Aug 2021, Jan 2024

Why this chapter matters to you. The Traditional Knowledge Digital Library (TKDL) is India's main defence against the patenting of Ayurvedic knowledge abroad. It has been set as a short note in three of the seven papers. It also links Chapters B7 (intellectual property) and B9 (international trade).

LEARNING OUTCOMES

After studying this chapter you should be able to describe the origin, objectives, contents, structure, access and achievements of the TKDL, explain how it protects traditional knowledge, and state its limitations.

This outcome is set by the author.

Asked in university examinations

Paper	Section, Q	Marks	Question as set
Sep 2020	B, Q.3a	5	Short note: TKDL
Aug 2021	B, Q.3A	5	Write in detail about TKDL
Jan 2024	B, Q.3b	5	Introduction to TKDL (Traditional Knowledge Digital library)

B12.1 What is the TKDL?

> The **Traditional Knowledge Digital Library** is a **prior-art database** of Indian traditional knowledge, set up to help patent examiners worldwide find documented traditional knowledge and so prevent the grant of wrong patents on it [PIB 2022a].

B12.1.1 Origin

The turmeric and neem cases (Chapter B7.5.1) showed that patent examiners abroad could not find Indian traditional knowledge, because it was recorded in Sanskrit, Arabic, Persian, Tamil and other languages in texts that were not in their search databases. India had to spend time and money to have such patents revoked. The answer was to make the knowledge **searchable as prior art** before patents were granted.

The TKDL was established in **2001** as a joint initiative of the **Council of Scientific and Industrial Research** (CSIR) and the then Department of Indian Systems of Medicine and Homoeopathy (now the **Ministry of Ayush**) [PIB 2022a; PIB 2022b].

B12.1.2 Objectives

1. To prevent the misappropriation of Indian traditional knowledge through wrong patents.
2. To make traditional knowledge available to patent examiners in their own languages and classification systems.
3. To bridge the gap between traditional knowledge in ancient languages and modern patent examination.
4. Since 2022, to make the database available to industry, research and education for innovation and research.

B12.2 Contents and structure

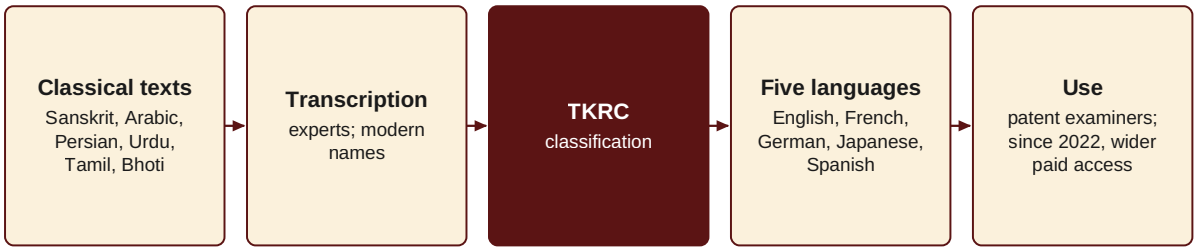


Figure B12.1. The TKDL: sources, transcription, classification and use.

Feature	Details
Systems covered	**Ayurveda, Unani, Siddha, Sowa Rigpa** and **Yoga**
Source	Classical texts in Sanskrit, Hindi, Arabic, Persian, Urdu, Tamil, Bhoti and other languages
Languages of the database	**English, French, German, Japanese and Spanish**
Classification	**Traditional Knowledge Resource Classification** (TKRC), an innovative classification of traditional knowledge, structured to work with the International Patent Classification
Formulations transcribed (as of 25 March 2022)	**418,885** in total: Ayurveda 119,269; Unani 236,399; Siddha 54,689; Yoga 4,151; Sowa Rigpa 4,377 [PIB 2022b]
Format	Each formulation transcribed from the text into a structured record (ingredients with modern names, method, uses), with reference to the source text

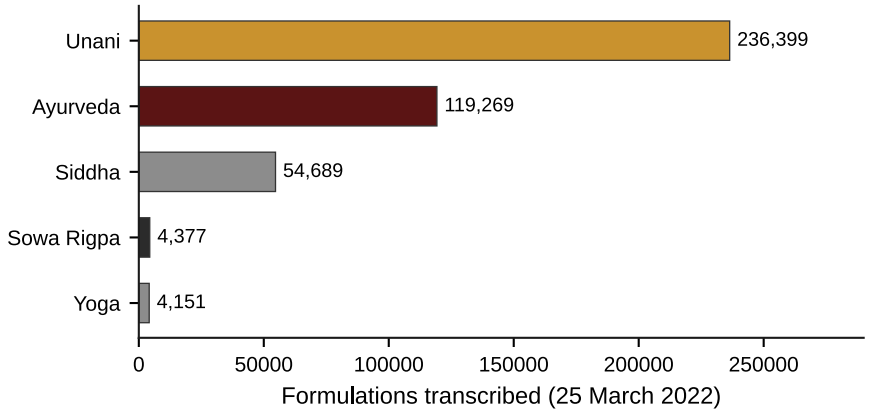


Figure B12.2. Formulations in the TKDL by system (as of 25 March 2022). Data from the Ministry of Ayush via PIB.

B12.3 Access

Phase	Access
Patent offices	Full access is given to patent offices under **non-disclosure access agreements** with CSIR, for search and examination only. **14 patent offices**, including the Indian Patent Office, had such agreements as of 2022 [PIB 2022a; PIB 2022b]
Since 2022	The Union Cabinet approved **wider access** to the database for users beyond patent offices: industry (Ayush, pharmaceuticals, nutraceuticals, personal care and others), research and educational institutions, Indian systems of medicine practitioners, knowledge holders, patentees and Government, through a **paid subscription model**, in phases, national and then international [PIB 2022a]

The non-disclosure condition protects the knowledge from being copied while still allowing examiners to cite it against patent applications.

B12.4 How the TKDL works against wrong patents

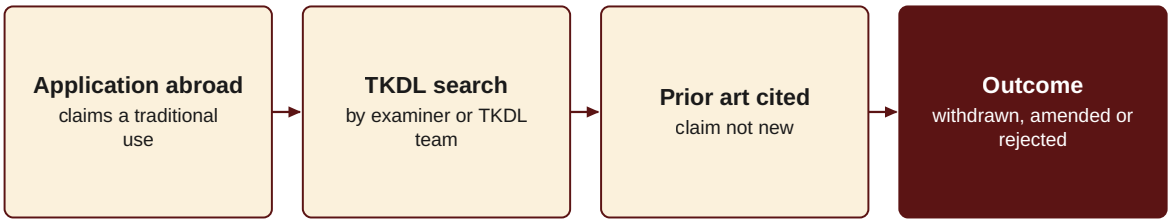


Figure B12.3. How TKDL evidence stops a wrong patent.

1. A patent application claiming a use of a traditional formulation or herb is filed at a patent office abroad.
2. The examiner (with access) searches the TKDL, or the TKDL team monitors published applications.
3. Relevant TKDL records are cited as **prior art**, showing that the claimed invention is not new.
4. The applicant amends or withdraws the claims, or the application is rejected.

Results. As of 25 March 2022, **265** patent applications had been withdrawn or deemed withdrawn, amended, or set aside on the basis of TKDL evidence [PIB 2022b].

B12.5 Significance and limitations

Significance	Limitations
Prevents biopiracy cheaply and early, before patents are granted	Defensive only: it does not give India ownership or royalties
Gives international recognition to Indian traditional knowledge as prior art	Covers mainly **codified** knowledge from texts; oral community knowledge is less covered (People's Biodiversity Registers help here, Chapter B7.4)
A model copied by other countries	Patent offices without access may still grant wrong patents
Since 2022, a resource for research, education and product development	Paid access may limit use by individual scholars
Supports India's position in international negotiations (WIPO, Chapter B7.3)	Transcription and interpretation of classical texts need constant expert review

AYURVEDA CORRELATION

The classical texts were themselves an act of preservation: knowledge passed orally was codified in Samhitas and commentaries so that it would not be lost. The TKDL continues this work in a new form: it preserves the classical record and makes it visible to the modern legal system, so that what was shared freely for centuries cannot be claimed as private property.

KEY POINTS TO REMEMBER

- TKDL: prior-art database of Indian traditional knowledge; 2001; CSIR with the Ministry of Ayush.
- Systems: Ayurveda, Unani, Siddha, Sowa Rigpa, Yoga. Languages: English, French, German, Japanese, Spanish.
- TKRC: Traditional Knowledge Resource Classification.
- 418,885 formulations (25 March 2022): Ayurveda 119,269; Unani 236,399; Siddha 54,689; Yoga 4,151; Sowa Rigpa 4,377.
- Access: 14 patent offices under non-disclosure agreements; wider paid access approved in 2022.
- 265 patent applications withdrawn, amended or set aside (to 25 March 2022).
- Defensive protection; limits: no ownership or royalties; codified knowledge mainly.

EXAM CORNER — CHAPTER B12

Likely short notes (5 marks)

1. TKDL *(set in September 2020)*.
2. Write in detail about TKDL *(set in August 2021)*.
3. Introduction to TKDL *(set in January 2024)*.
4. Biopiracy and TKDL.
5. TKRC.

Answer frame: “Write in detail about TKDL.” (5 marks)

Part	What to write	Suggested marks
Definition and origin	Prior-art database; 2001; CSIR and Ayush; turmeric and neem	1
Contents	Systems; languages; TKRC; numbers	1.5
Access	Patent offices; 2022 widening	1
Working and results	Flow; 265 applications	1
Significance and limits	Table, briefly	0.5

REFERENCES USED IN THIS CHAPTER

Syllabus reference number for this chapter: **49** (the syllabus lists no specific source on TKDL; reference 49 asks for official websites).

Additional verified sources (accessed 16 October 2026)

- **[PIB 2022a]** Prime Minister’s Office / Press Information Bureau. “Cabinet approves widening access of the Traditional Knowledge Digital Library (TKDL) database to users besides patent offices.” 17 August 2022. pmindia.gov.in.
- **[PIB 2022b]** Press Information Bureau. Background document on TKDL, September 2022 (figures as of 25 March 2022). static.pib.gov.in.

Viva pointers — Chapter B12

Answer each in one or two sentences.

- What is the TKDL? When and by whom was it set up?
- Which systems does it cover? In which languages?
- What is TKRC?
- How many formulations had been transcribed by March 2022?
- How many patent offices have access? On what terms?
- What changed in TKDL access in 2022?
- How many patent applications had been withdrawn or set aside on TKDL evidence by 2022?
- Give two limitations of the TKDL.
- Which case led to the idea of the TKDL?

PART B · PHARMACY MANAGEMENT

QUESTION BANK — PART B

Part B of the syllabus is examined in **Section B** of the paper. This question bank gathers every Section B question set in the seven papers examined (July 2018 to March 2026), shows which topics are asked most often, and adds practice questions, model papers and multiple-choice questions for revision.

B-QB.1 Section B questions, paper by paper

16 July 2018

Q	Marks	Question as set	Chapter
Q.1	20	Drug and cosmetics act 1940 and rules 1945.	B8
Q.2	20	Write down in brief regarding API & AFI.	B10
Q.3a	5	Short note: community Pharmacy.	B6
Q.3b	5	Information resources in pharmacy and pharmaceutical science.	B1
Q.3c	5	Product release and withdrawals.	B5

29 September 2018

Q	Marks	Question as set	Chapter
Q.1	20	Explain Drug & Cosmetics Act with relevance to ASU drugs along with schedule T and Good Manufacturing Practices.	B4, B8
Q.2	20	Write the need for Ayurvedic Pharmacopoeia and Formulary of India. Enlist the functions of Ayurvedic Pharmacopoeial Committee. Enlist the publications in the form of API and AFI. Write the structure of ideal monograph with an example.	B10
Q.3a	5	Examples of Packaging material and contents of an Ideal Label.	B3
Q.3b	5	Need for awareness about Patenting and Intellectual Property Rights.	B7
Q.3c	5	Pharmaceutical Dosage Form Design.	B2

30 September 2020

Q	Marks	Question as set	Chapter
Q.1	20	Write down in brief Management of pharmacy & good manufacturing practices related to Ayurvedic drug industry	B4
Q.2	20	Pre-Formulation – Pharmaceutical dosage form design	B2
Q.3a	5	Short note: TKDL	B12
Q.3b	5	Drugs and magic remedy act 1954	B8
Q.3c	5	Packaging Materials and Labeling	B3

31 August 2021

Q	Marks	Question as set	Chapter
Q.1	20	Explain in detail about the Patenting and Intellectual Property Rights	B7
Q.2	20	Discuss in detail about Good Manufacturing Practices (GMP) related to Ayurvedic Drug Industry	B4
Q.3A	5	Write in detail about TKDL	B12
Q.3B	5	Discuss regarding the rules pertaining to Packaging and Labelling of Ayurvedic Drugs	B3
Q.3C	5	Brief about API and AFI	B10

29 August 2022

Q	Marks	Question as set	Chapter
Q.1	20	Good manufacturing practices related to Ayurved industries	B4
Q.2	20	Explain Laws governing Ayurvedic drugs in detail with appropriate examples.	B8
Q.3a	5	Ayurved Pharmacopeia of India.	B10
Q.3b	5	Explain necessity and importance of IPR.	B7
Q.3c	5	Designing of Pharmaceutical dosage forms.	B2

4 January 2024

Q	Marks	Question as set	Chapter
Q.1	20	Explain laws governing Ayurvedic drugs with its necessity.	B8
Q.2	20	Explain Good manufacturing practices related to Ayurvedic drug industry.	B4
Q.3a	5	Drug designing.	B2
Q.3b	5	Introduction to TKDL (Traditional Knowledge Digital library)	B12
Q.3c	5	Intellectual Property Rights.	B7

6 March 2026

Q	Marks	Question as set	Chapter
Q.1	20	Discuss the management of an Ayurvedic Pharmacy. Include aspects of international trade and regulatory affairs related to the export of Ayurvedic drugs.	B4, B9
Q.2	20	Discuss pharmaceutical dosage form design with special reference to Pre-formulation Studies. Correlate classical concepts with modern pre-formulation parameters.	B2
Q.3a	5	Explain the significance of shelf-life and storage conditions for herbo-mineral preparations.	B3, B4
Q.3b	5	Discuss the “Prevention of Food Adulteration (PFA) Act” in relation to herbal supplements.	B8
Q.3c	5	Write a short note on the Ayurvedic Pharmacopoeia of India (API) and Ayurvedic Formulary of India (AFI).	B10

B-QB.2 Which topics are asked most often

Chapter	Topic	Long (20)	Short (5)	Papers (of 7)
B4	Pharmacy management and GMP	6	1	**6**
B8	Laws governing Ayurvedic drugs	4	2	**6**
B2	Dosage form design (pre-formulation)	2	3	5
B10	API and AFI	2	3	5
B3	Packaging and labelling	0	4	4
B7	Patenting and IPR	1	3	4
B12	TKDL	0	3	3
B1	Scope of pharmacy; information resources	0	1	1
B5	Marketing; product release and withdrawals	0	1	1
B6	Hospital, dispensing and community pharmacy	0	1	1
B9	International trade and regulatory affairs	1	0	1
B11	Indian, British and US pharmacopoeias; Codex	0	0	0

A question that draws on two chapters is counted under both. Counts are from the seven Paper IV question papers examined (Parul University, July 2018 to March 2026).

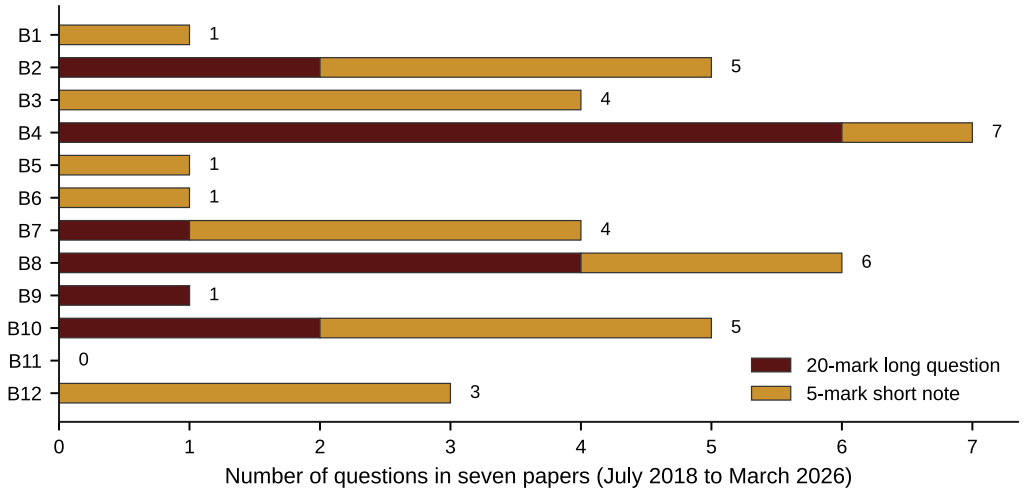


Figure QBB.1. Section B: how often each chapter was asked, as 20-mark and 5-mark questions.

What the record tells you.

1. **GMP and the laws** are the core of Section B. Between them they account for most of the 20-mark questions. Prepare full answers on Schedule T, the Drugs and Cosmetics Act and Rules for ASU drugs, and "laws governing Ayurvedic drugs with their necessity".
2. **Pre-formulation** and **API–AFI** come next; both have been set as 20-mark questions.
3. Packaging and labelling, IPR and TKDL are regular **short notes**.
4. International trade (B9) entered the paper in March 2026, combined with pharmacy management; expect it again.
5. The law changed in 2024–2026 (Rule 170, Schedule T, Schedule E(1), the labour codes, the consumer law). Examiners reward an answer that states the law as it now stands.

B-QB.3 Practice questions, chapter by chapter

Questions marked ★ have been set in a university paper (the year is given). The others are written in the same style. The section that answers each question is given in brackets.

Chapter B1 — Scope and Evolution of Pharmacy; Information Resources

Long: 1. Describe the scope and evolution of pharmacy with special reference to Ayurvedic pharmacy in India (B1.1–B1.3). 2. Classify information resources in pharmacy; describe those important to an Ayurvedic pharmacist and how to search them (B1.4).

Short: 1. ★ Information resources in pharmacy and pharmaceutical science *(July 2018)* (B1.4). 2. Evolution of pharmacy in India. 3. Primary, secondary and tertiary sources. 4. Literature search strategy. 5. Predatory journals.

Chapter B2 — Dosage Form Design (Pre-formulation)

Long: 1. ★ Pre-formulation: pharmaceutical dosage form design *(September 2020)* (B2.2, B2.3). 2. ★ Pharmaceutical dosage form design with special reference to pre-formulation studies; correlate classical concepts with modern pre-formulation parameters *(March 2026)* (B2.3, B2.5). 3. Classify dosage forms;

describe the considerations in designing a dosage form, with examples from Bhaishajya Kalpana (B2.1, B2.2, B2.4).

Short: 1. ★ Pharmaceutical dosage form design *(September 2018)*. 2. ★ Designing of pharmaceutical dosage forms *(August 2022)*. 3. ★ Drug designing *(January 2024)* (B2.6). 4. Flow properties of powders. 5. Polymorphism. 6. BCS. 7. Noyes–Whitney equation. 8. Accelerated stability testing. 9. Reverse pharmacology.

Chapter B3 — Packaging Materials and Labelling

Long: 1. Packaging materials and their selection for Ayurvedic dosage forms; labelling requirements of Ayurvedic drugs (B3.2–B3.5). 2. Shelf life and stability of Ayurvedic drugs with the legal provisions (B3.4).

Short: 1. ★ Examples of packaging material and contents of an ideal label *(September 2018)*. 2. ★ Packaging materials and labelling *(September 2020)*. 3. ★ Rules pertaining to packaging and labelling of Ayurvedic drugs *(August 2021)* (B3.5.2). 4. ★ Significance of shelf-life and storage conditions for herbo-mineral preparations *(March 2026)* (B3.4.4). 5. Types of glass. 6. Rule 161B. 7. Schedule E(1).

Chapter B4 — Pharmacy Management and GMP

Long: 1. ★ Management of pharmacy and GMP related to the Ayurvedic drug industry *(September 2020)*. 2. ★ GMP related to the Ayurvedic drug industry *(August 2021, August 2022, January 2024)* (B4.6). 3. ★ Drugs and Cosmetics Act with relevance to ASU drugs, with Schedule T and GMP *(September 2018)* (B8(i), B4.6). 4. ★ Management of an Ayurvedic pharmacy, with international trade and export regulatory affairs *(March 2026)* (B4.2, B9). 5. Store and inventory management in an Ayurvedic pharmacy (B4.3, B4.4).

Short: 1. ABC and VED analysis. 2. Economic order quantity. 3. Batch manufacturing record. 4. Schedule T. 5. Personnel management. 6. Safety in a Rasashala. 7. AYUSH Mark.

Chapter B5 — Marketing, Product Release and Withdrawals

Long: 1. Pharmaceutical marketing with special reference to Ayurvedic products; product release and withdrawal (B5). 2. Product recall: classes, types, procedure and the GMP requirements behind it (B5.3).

Short: 1. ★ Product release and withdrawals *(July 2018)*. 2. Marketing mix. 3. Product life cycle. 4. Classes of recall.

Chapter B6 — Hospital, Dispensing and Community Pharmacy

Long: 1. Organisation and functions of a hospital pharmacy, with reference to an Ayurvedic teaching hospital (B6.1). 2. Dispensing and community pharmacy; counselling for Ayurvedic medicines (B6.2, B6.3).

Short: 1. ★ Community pharmacy *(July 2018)* (B6.3). 2. Parts of a prescription. 3. Unit dose dispensing. 4. Dispensing errors. 5. Classical weights and measures.

Chapter B7 — Patenting and IPR

Long: 1. ★ Explain in detail patenting and IPR *(August 2021)* (B7). 2. Biopiracy and protection of traditional knowledge with landmark cases (B7.5). 3. Procedure for a patent in India; patentability of Ayurvedic inventions (B7.2, B7.6).

Short: 1. ★ Need for awareness about patenting and IPR *(September 2018)*. 2. ★ Necessity and importance of IPR *(August 2022)* (B7.6.1). 3. ★ Intellectual property rights *(January 2024)*. 4. Sections 3(d) and 3(p). 5. TRIPS. 6. Turmeric patent case. 7. Geographical indications.

Chapter B8 — Laws Governing Ayurvedic Drugs

Long: 1. ★ Drugs and Cosmetics Act 1940 and Rules 1945 *(July 2018)* (B8(i)). 2. ★ Laws governing Ayurvedic drugs in detail with examples *(August 2022)* (B8). 3. ★ Laws governing Ayurvedic drugs with their necessity *(January 2024)* (B8). 4. Food laws and herbal products: PFA Act, FSS Act, health supplements and Ayurveda Aahara (B8(iii), B8(iv)).

Short: 1. ★ Drugs and Magic Remedies Act 1954 *(September 2020)* (B8(ii)). 2. ★ PFA Act in relation to herbal supplements *(March 2026)* (B8(iii)). 3. Rule 158-B. 4. Schedule E(1). 5. NDPS Act and Ayurvedic drugs. 6. OSH Code 2020. 7. Pharmacy Act 1948. 8. Consumer Protection Act 2019. 9. Rule 170.

Chapter B9 — International Trade and Regulatory Affairs

Long: 1. Regulatory affairs related to international trade of Ayurvedic drugs, with requirements of major importing countries (B9).

Short: 1. Export documentation. 2. COPP. 3. Herbal products in the European Union. 4. Barriers to export. 5. AYUSHEXCIL.

Chapter B10 — API and AFI

Long: 1. ★ API and AFI in brief *(July 2018)*. 2. ★ Need for API and AFI; functions of the Ayurvedic Pharmacopoeia Committee; publications; structure of an ideal monograph with an example *(September 2018)* (B10).

Short: 1. ★ API and AFI *(August 2021, March 2026)*. 2. ★ Ayurvedic Pharmacopoeia of India *(August 2022)*. 3. PCIM&H. 4. Pharmacopoeia versus formulary.

Chapter B11 — Indian, British and US Pharmacopoeias; Pharmaceutical Codex

Long: 1. Introduction to the IP, BP, USP–NF and Pharmaceutical Codex, compared with the API (B11).

Short: 1. Indian Pharmacopoeia. 2. British Pharmacopoeia. 3. USP–NF. 4. Pharmaceutical Codex.

Chapter B12 — TKDL

Short: 1. ★ TKDL *(September 2020)*. 2. ★ Write in detail about TKDL *(August 2021)*. 3. ★ Introduction to TKDL *(January 2024)*. 4. TKRC. 5. Biopiracy and TKDL.

B-QB.4 Model question papers: Section B

Each paper follows the university pattern: two long questions of 20 marks and three short notes, any two to be answered.

MODEL PAPER 1 — SECTION B (50 MARKS)

Q.1 Discuss Good Manufacturing Practices related to the Ayurvedic drug industry, with special reference to Schedule T and its 2026 revision. **(20)**

Q.2 Write the need for the Ayurvedic Pharmacopoeia and Formulary of India. Describe their publications and the structure of a monograph with an example. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. TKDL.

b. Contents of an ideal label for an Ayurvedic drug.

c. Drugs and Magic Remedies Act, 1954.

MODEL PAPER 2 — SECTION B (50 MARKS)

Q.1 Explain the laws governing Ayurvedic drugs with their necessity. **(20)**

Q.2 Discuss pharmaceutical dosage form design with special reference to pre-formulation studies, and correlate classical concepts with modern parameters. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. Significance of shelf life for herbo-mineral preparations.

b. Necessity and importance of IPR.

c. Community pharmacy.

MODEL PAPER 3 — SECTION B (50 MARKS)

Q.1 Discuss the management of an Ayurvedic pharmacy, including store and inventory management and the regulatory requirements for export. **(20)**

Q.2 Explain patenting and intellectual property rights, with special reference to traditional knowledge. **(20)**

Q.3 Write short notes on any **two**: **(2 × 5 = 10)**

a. Product recall.

b. PFA Act in relation to herbal supplements.

c. Information resources in pharmacy.

Guidance for Model paper 2, Q.1

Part of the answer	Content	Book section	Suggested marks
Necessity	Eight needs in a table	B8 introduction	3
Framework	Table of laws and what each controls; figure	B8 introduction	2
Drugs and Cosmetics Act and Rules	Definitions; Chapter IV-A; licensing; Rule 158-B; Schedule T; Rules 161, 161B; Schedule E(1); 2026 changes	B8(i)	6
DMR Act	Sections 3–5; Schedule; penalties; Rule 170 position	B8(ii)	3
Food laws	PFA (repealed); FSS Act; health supplements; Ayurveda Aahara	B8(iii), B8(iv)	2
Other laws	NDPS; OSH Code and Pharmacy Act; Consumer Protection Act 2019	B8(v)–(vii)	3

Part of the answer	Content	Book section	Suggested marks
Conclusion	The law protects quality, safety and honesty	—	1

B-QB.5 Multiple-choice questions for self-assessment

Choose one answer for each. The key, with the section to revise, follows the questions.

Chapters B1 and B2

1. Charaka's four qualities of the attendant include all of the following EXCEPT: (a) Upacharajnata (b) Dakshya (c) Anuraga (d) Bahuta
2. Pharmacy education in India began in 1932 at: (a) Banaras Hindu University (b) Calcutta University (c) Bombay University (d) Madras Medical College
3. A pharmacopoeia is best classed as a: (a) primary source (b) secondary source (c) tertiary source (d) grey literature
4. The accepted botanical name of a plant is best checked in: (a) The Plant List (b) World Flora Online (c) PubMed (d) TKDL
5. In 2025 the UGC: (a) discontinued the UGC-CARE list (b) made UGC-CARE compulsory (c) banned open-access journals (d) created a new impact factor
6. The Noyes–Whitney equation shows that the dissolution rate rises with: (a) surface area (b) particle size (c) diffusion layer thickness (d) concentration already in solution
7. A Carr's index of 8 per cent indicates flow that is: (a) excellent (b) fair (c) poor (d) very poor
8. In BCS class II, the rate-limiting step for absorption is: (a) dissolution (b) permeability (c) gastric emptying (d) metabolism
9. "Sparingly soluble" means that one part of solute needs: (a) less than 1 part (b) 1–10 parts (c) 30–100 parts (d) more than 10,000 parts of solvent
10. The modern parallel of Sneha Siddhi Lakshana is: (a) process end point and moisture (b) particle size (c) polymorphism (d) log P

Chapters B3 and B4

11. Borosilicate glass is pharmacopoeial type: (a) I (b) II (c) III (d) NP
12. The best moisture protection among these blister packs is: (a) PVC (b) PVC/PVDC (c) Alu-Alu (d) paper strip
13. Under Rule 161B, the shelf life of Asava and Arishta is: (a) 2 years (b) 5 years (c) 10 years (d) unlimited
14. Under Rule 161B, Naga, Vanga and Tamra Bhasma have a shelf life of: (a) 3 years (b) 5 years (c) 10 years (d) 1 year
15. The caution "To be taken under medical supervision" is required for drugs containing substances listed in: (a) Schedule T (b) Schedule E(1) (c) Schedule M (d) First Schedule
16. EOQ equals: (a) √(2DS/H) (b) 2DS/H (c) D/S (d) H/2DS
17. VED analysis classifies items by: (a) value (b) criticality (c) movement (d) price
18. GMP for ASU drugs is laid down in: (a) Schedule M (b) Schedule T (c) Schedule Y (d) Schedule H
19. The revised Schedule T notified in 2026 must be complied with by existing units by: (a) 31 July 2027 (b) 31 July 2029 (c) 1 January 2026 (d) 31 March 2030

20. The AYUSH Premium Mark is based on: (a) Schedule T only (b) WHO GMP guidelines for herbal medicines (c) ISO 14001 (d) USP standards

Chapters B5, B6 and B7

21. A recall where use of the product may cause serious harm or death is: (a) Class I (b) Class II (c) Class III (d) a market withdrawal
22. Until a batch is released by QA, it is kept in: (a) quarantine (b) the finished goods store (c) the rejected area (d) the dispatch bay
23. The unit dose system of inpatient distribution: (a) reduces medication errors (b) increases floor stock (c) needs no pharmacist (d) is used only in community pharmacy
24. According to the Ayurvedic Formulary of India, one Ratti equals: (a) 125 mg (b) 250 mg (c) 1 g (d) 12 g
25. The term of a patent in India is: (a) 10 years (b) 14 years (c) 20 years from filing (d) life plus 60 years
26. Section 3(p) of the Patents Act excludes: (a) traditional knowledge (b) software (c) new salts (d) micro-organisms
27. The turmeric patent was revoked in: (a) 1995 (b) 1997 (c) 2000 (d) 2005
28. The WIPO treaty requiring disclosure of the origin of genetic resources and associated traditional knowledge was adopted in: (a) 1994 (b) 2010 (c) 2024 (d) 2026
29. Approval of the National Biodiversity Authority is needed for: (a) an IPR application based on Indian biological resources (b) every Ayurvedic prescription (c) a retail sale licence (d) a trade mark
30. India’s first compulsory licence (2012) was for: (a) imatinib (b) sorafenib (c) curcumin (d) artemisinin

Chapter B8

31. ASU drugs were brought under the Drugs and Cosmetics Act in: (a) 1940 (b) 1945 (c) 1964 (d) 2000
32. The application for a licence to manufacture ASU drugs is made in Form: (a) 24-D (b) 25-D (c) 20-B (d) 28
33. A patent or proprietary ASU medicine may NOT be: (a) given by parenteral route (b) made from ingredients of authoritative books (c) sold under a brand name (d) licensed by the State
34. Rule 170, on advertisement of ASU drugs, was: (a) omitted in 2024, with the Supreme Court’s stay vacated in 2025 (b) strengthened in 2026 (c) never notified (d) moved to the DMR Act
35. The Schedule to the DMR Act lists: (a) 54 diseases and conditions (b) 78 diseases (c) 13 poisonous plants (d) 30 dosage forms
36. The penalty for a first conviction under the DMR Act is imprisonment up to: (a) 6 months (b) 1 year (c) 3 years (d) 7 years
37. The PFA Act was repealed with effect from: (a) 2006 (b) 5 August 2011 (c) 2016 (d) 2022
38. Health supplements and nutraceuticals are dealt with in Section 22 of the: (a) FSS Act, 2006 (b) Drugs and Cosmetics Act (c) Consumer Protection Act (d) NDPS Act
39. Under the NDPS Act, cannabis (hemp) does NOT include: (a) charas (b) ganja (c) leaves and seeds when not accompanied by the tops (d) mixtures of charas and ganja
40. The OSH Code, 2020 came into force on: (a) 1 April 2021 (b) 21 November 2025 (c) 1 July 2026 (d) it is not yet in force
41. Under the Pharmacy Act, only a registered pharmacist may dispense on prescription under Section: (a) 42 (b) 3(p) (c) 33 (d) 161

42. The District Consumer Commission hears complaints of value up to: (a) Rs 20 lakh (b) Rs 50 lakh (c) Rs 1 crore (d) Rs 2 crore
43. Under the CPA 2019, the CCPA may penalise a first misleading advertisement up to: (a) Rs 1 lakh (b) Rs 10 lakh (c) Rs 50 lakh (d) Rs 1 crore
44. Medical services were held to be “services” under consumer law in: (a) IMA v. V.P. Shantha (b) Novartis v. Union of India (c) IMA v. Union of India (2025) (d) Manik Jhangiani v. State of MP
45. Schedule E(1) as substituted in 2026 treats as poisonous any ingredient containing: (a) arsenic or mercury (b) sugar (c) alcohol (d) zinc

Chapters B9 to B12

46. Ayush and herbal exports in 2024–25 were about: (a) US$ 69 million (b) US$ 689 million (c) US$ 6.9 billion (d) US$ 69 billion
47. The COPP for Ayurvedic exports is issued by: (a) CDSCO after joint inspection (b) the State Pharmacy Council (c) FSSAI (d) TKDL
48. In the United States, Ayurvedic products are usually sold as: (a) dietary supplements (b) prescription drugs (c) cosmetics only (d) devices
49. EU traditional herbal registration requires traditional use for at least: (a) 10 years (b) 15 years (c) 30 years, including 15 in the EU (d) 100 years
50. The Ayush Visa was introduced on: (a) 27 July 2023 (b) 1 January 2020 (c) 20 April 2022 (d) 30 May 2025
51. The first Ayurvedic Pharmacopoeia Committee was formed in: (a) 1955 (b) 1962 (c) 1978 (d) 2010
52. Volume VII of API Part I deals with: (a) minerals and metals (b) extracts (c) formulations (d) animal products
53. In the API, the limit for total ash of Amalaki is: (a) not more than 2% (b) not more than 7% (c) not less than 40% (d) not less than 50%
54. AFI Part I was first published in: (a) 1955 (b) 1978 (c) 2000 (d) 2011
55. The tenth edition of the Indian Pharmacopoeia is: (a) IP 2018 (b) IP 2022 (c) IP 2026 (d) IP 2030
56. The 12th edition of the Pharmaceutical Codex (1994) is subtitled: (a) Principles and Practice of Pharmaceutics (b) Martindale (c) The Extra Pharmacopoeia (d) The National Formulary
57. TKDL was established in: (a) 1995 (b) 2001 (c) 2010 (d) 2022
58. TKDL is available in all these languages EXCEPT: (a) English (b) Japanese (c) Spanish (d) Chinese
59. The number of patent offices with TKDL access agreements (2022) was: (a) 4 (b) 14 (c) 40 (d) 140
60. TKRC stands for: (a) Traditional Knowledge Resource Classification (b) Traditional Knowledge Registration Council (c) Technical Knowledge Review Committee (d) Trade Knowledge Rights Convention

Answer key

Q	Ans	Section	Q	Ans	Section	Q	Ans	Section
1	d	B1.1.1	21	a	B5.3.2	41	a	B8(vi).2
2	a	B1.2.2	22	a	B5.2.1	42	b	B8(vii).2.2
3	c	B1.4.1	23	a	B6.1.4	43	b	B8(vii).2.4

Q	Ans	Section	Q	Ans	Section	Q	Ans	Section
4	b	B1.4.2	24	a	B6.2.4	44	a	B8(vii).1
5	a	B1.4.4	25	c	B7.1.1	45	a	B8(i).8.1
6	a	B2.3.3	26	a	B7.2.3	46	b	B9.1
7	a	B2.3.4	27	b	B7.5.1	47	a	B9.2
8	a	B2.3.7	28	c	B7.3	48	a	B9.3
9	c	B2.3.6	29	a	B7.4	49	c	B9.3
10	a	B2.5	30	b	B7.2.5	50	a	B9.5
11	a	B3.2.1	31	c	B8(i).1	51	b	B10.3
12	c	B3.2.6	32	a	B8(i).4	52	a	B10.4.1
13	c	B3.4.3	33	a	B8(i).2	53	b	B10.5.2
14	b	B3.4.3	34	a	B8(i).8.2	54	b	B10.4.3
15	b	B3.5.2	35	a	B8(ii).3.1	55	c	B11.2
16	a	B4.4.3	36	a	B8(ii).5	56	a	B11.6
17	b	B4.4.2	37	b	B8(iii)	57	b	B12.1.1
18	b	B4.6.3	38	a	B8(iv).4	58	d	B12.2
19	b	B4.6.4	39	c	B8(v).2	59	b	B12.3
20	b	B4.6.6	40	b	B8(vi).1	60	a	B12.2

PRACTICALS

PRACTICAL WORK AND PRACTICAL EXAMINATION

Why this part matters to you. The practical examination carries 100 marks, as much as the theory paper. Paper IV contributes to it directly: drug testing and analysis, the reports of visits to GMP pharmacies, the R&D or production posting, and the record of clinical cases studied for efficacy and adverse drug reactions. This part gives formats, worked examples and safety rules for each, so that your journal and reports are complete and your performance on the two days is sure.

PR.1 The practical requirements of the syllabus

The syllabus lists seven practical requirements. They are reproduced below exactly, with what you should prepare and where this book helps.

Requirement (as in the syllabus)	What to prepare	Help in this book
Minimum 150 Practicals are to be performed covering all the dosage forms.	Journal of practicals with full write-ups, including the safety section	PR.3 (format), PR.4 (worked examples)
Educational Visits to minimum Five (5) GMP approved Ayurvedic Pharmacies.	Five visit reports with GMP checklist	PR.7
Ten days posting in R & D / Production unit in GMP certified Ayurvedic Pharmacy.	Daily log; certificate	PR.8
Minimum attendance in three National Conferences within three academic years.	Certificates; short notes on what was learnt	PR.10
Minimum 10 clinical cases using different dosage forms are to be studied and recorded for efficacy and ADRs (if any).	Ten case records with ADR assessment	PR.9; Chapter A6
Minimum one research paper on the core contributory work of dissertation published or accepted in peer reviewed, indexed journal.	Published paper or acceptance letter	PR.10; Chapter B1 (choosing a journal)
Examinee has to submit supporting documents in relation to above mentioned academic activities at the time of Practical examination.	A file with all certificates and reports	PR.10

A misprint in the fifth line of the printed syllabus is corrected here (see the list of corrections in How to Use This Book).

PR.2 The practical examination

The pattern of the practical examination (100 marks, two days) is set out in the Scheme of Examination at the front of this book. In brief:

Item	Marks	How to prepare
1. Identification of the specimen	10	PR.6
2. Drug processing: two major practicals (one Rasashastra, one Bhaishajya Kalpana) and two minor practicals (one each)	30	PR.3, PR.4
3. Drug testing and analysis: one Rasaushadhi, one Kashthaushadhi	10	PR.5
4. PowerPoint presentation on dissertation work (15 minutes)	10	PR.10
6. Demonstration lecture on any subject topic (10 minutes)	10	PR.10
7. Documentation: journals of at least 75 practicals (25 each from Rasashastra, Bhaishajya Kalpana and the analytical part); reports of the visits; case records	15	PR.3, PR.7, PR.9
8. Viva voce	15	Viva pointers at the end of every chapter; PR.11

The printed scheme has no item 5; the numbering is kept as printed.

PR.3 How to write up a practical

Every practical in your journal should follow the same format. Examiners check two things first: whether the **safety** section is present, and whether **observations** were recorded during the work rather than written afterwards.

Heading	Content
Title and date	Name of the preparation; date started and finished
Aim	What is to be prepared or tested
Reference	Classical text (and the Ayurvedic Formulary of India where the formulation is listed)
Principle	The pharmaceutical principle in two or three lines
Ingredients and quantities	Table: ingredient (with botanical or chemical name), part used, quantity, whether Shuddha
Equipment	Khalva, vessels, heat source, Puta, Kupi, balances, thermometer
Procedure	Numbered steps, as actually done
Observations	Table recorded at the time: time, temperature, colour, consistency, Paka lakshana, weight; photographs if allowed
Precautions	Steps that decide quality
Safety	Hazards and controls for this practical (compulsory)
Result	Yield (weight and percentage); description of the product
Tests	Classical tests (for example, Bhasma Pariksha, Sneha Siddhi Lakshana) and any modern tests done

Heading	Content
Storage and shelf life	Container, conditions, shelf life as per the rules (Chapter B8)
Label	A specimen label as it would appear on a licensed product (Chapter B3)
Inference	What was learnt; problems faced

GENERAL SAFETY RULES FOR THE PHARMACY LABORATORY

1. Wear an apron, closed shoes and, when required, gloves, goggles and a mask suited to the hazard.
2. Know where the fire extinguisher, first-aid box, eyewash and exits are.
3. Never leave a heating operation unattended. Keep oils, ghee and alcohol away from open flame when not being processed.
4. Handle mercury, arsenicals, lead, and poisonous plants (Visha and Upavisha) only under supervision, in the designated area, with records of issue and return.
5. Work with mercury over a tray; clean spills only by the approved method; never heat mercury in an open vessel outside a fume hood or chimney.
6. Do not eat, drink or apply cosmetics in the laboratory. Wash hands before leaving.
7. Label every container with contents, date and hazard.
8. Dispose of waste as instructed: hazardous metal waste separately; never into the drain.
9. Report every accident, however small.

PR.4 Worked examples of practical write-ups

The two examples below show the format of PR.3 in use: one Rasashastra practical and one Bhaishajya Kalpana practical. Quantities are illustrative laboratory quantities; follow the quantities and method prescribed by your department.

PR.4.1 Rasashastra: preparation of Kajjali

Heading	Write-up
Aim	To prepare Kajjali from Shuddha Parada and Shuddha Gandhaka in equal parts
Principle	Trituration of mercury with sulphur, without heat, converts the liquid metal into a black, lustreless mercuric sulphide (metacinnabar, A5.3), which is the base of Kharaliya Rasayana
Ingredients	Shuddha Parada 50 g; Shuddha Gandhaka 50 g (both purified earlier by the prescribed Shodhana, with records)
Equipment	Stone Khalva with pestle, placed in a tray; balance; spatula; glass jar with lid
Procedure	1. Weigh both ingredients. 2. Powder Gandhaka finely in the Khalva. 3. Add Parada and triturate continuously with moderate pressure. 4. Scrape the sides at intervals. 5. Continue until the tests in the next row are passed. 6. Weigh the product and store it
Tests for completion	**Nishchandratva**: no shining globule of mercury seen under a lens; **Rekhapurnatva**: the powder fills the lines of the fingers; **Varitaratva**: a pinch floats on still water; smooth, jet-black, fine powder

Observations

Time (hours)	Colour and texture	Free mercury globules	Remarks
0	Yellow sulphur; shining mercury	Many	Start
1	Greyish	Numerous small	—
3	Dark grey	Few	—
6	Black	Very few under lens	—
Until complete	Jet black, smooth	None	Tests passed

Record your own times; the table shows the format, not expected values.

Result. Weight of Kajjali obtained; yield as a percentage of the total weight taken; loss and its probable reason (adhesion to the Khalva, spillage).

Precautions. Use only Shuddha Parada and Gandhaka; triturate without interruption for long periods; check for free mercury under a lens before declaring completion; store in an airtight glass jar.

SAFETY: KAJJALI

- **Hazard**: mercury vapour from free mercury, especially if the room is warm; skin contact; spillage.
- Work in a well-ventilated area or under local exhaust; keep the room cool; never heat.
- Place the Khalva in a tray to catch spills. Keep a mercury spill kit at hand. Never sweep or vacuum spilled mercury; collect droplets with the method taught in the department.
- Wear gloves and an apron; avoid touching the face; wash hands thoroughly after work.
- Weigh and transfer mercury over a tray; keep stock mercury in a closed, labelled, unbreakable container in the locked Rasa Dravya store.
- Do not eat or drink in the laboratory. Pregnant persons should not take part in mercury work.
- Record quantities issued, used and returned; dispose of mercury-contaminated waste through the hazardous waste route.

Label (specimen). “Kajjali (intermediate for use in manufacture). Contains Shuddha Parada and Shuddha Gandhaka in equal parts. Batch No. … Date … Store in an airtight container. Caution: contains mercury; for manufacturing use under supervision.”

PR.4.2 Bhaishajya Kalpana: preparation of a medicated oil (Taila Paka)

Heading	Write-up
Aim	To prepare a medicated oil by the general method of Sneha Paka
Principle	Lipid-soluble and water-soluble constituents of the herbs are extracted into the oil by heating with Kalka (paste) and Drava (liquid) until the water is driven off and the Sneha Siddhi Lakshana appear (B2.4)
Ingredients	Murchhita Tila Taila 4 parts; Kalka of the prescribed herb(s) 1 part; Drava (Kwatha, Swarasa or water) 16 parts, the classical proportion for Sneha Kalpana [R11]
Equipment	Stainless steel or brass vessel; stirrer; heat source with control; thermometer; filtering cloth; glass bottles
Procedure	1. Prepare the Kalka and the Kwatha. 2. Mix oil, Kalka and Drava in the vessel. 3. Heat on low to moderate fire with constant stirring. 4. Watch for the signs of Paka. 5. Stop heating at the required stage (Mridu, Madhya or Khara). 6. Filter while warm through cloth. 7. Cool, bottle and label

Observations

Time	Temperature	Appearance	Kalka (wick test)	Sound when Kalka is put on fire	Froth
Start	—	—	—	—	—
(record at intervals)	—	—	—	—	—
End point	—	Clear oil; characteristic colour and odour	Rolls into a wick (Varti)	No crackling (water driven off)	Froth appears in Taila (it subsides in Ghrita)

Result. Volume or weight of oil obtained; yield; Paka stage reached.

Tests. Sneha Siddhi Lakshana as above; where facilities exist: acid value, saponification value, refractive index, specific gravity, chromatographic profile as per the pharmacopoeial monograph (Chapter B10).

SAFETY: SNEHA PAKA

- **Hazards**: burns from hot oil and splashes when water is still present; oil fire; slips on oily floors.
- Use a vessel of adequate size (not more than half full); stir with a long-handled stirrer; keep the face away from the vessel when adding liquids.
- Keep a lid and a fire extinguisher suitable for oil fires (never water) at hand.
- Never leave the vessel unattended; control the flame.
- Wear an apron and heat-resistant gloves when filtering hot oil.
- Clean spills immediately.

PR.4.3 Minor practicals

Minor practicals (for example, Churna, Vati, Kwatha, Arka, a single Shodhana procedure) are written in the same format, more briefly. Every one still needs **observations**, **precautions** and a **safety** line.

PR.5 Drug testing and analysis

The examination asks for testing of one Rasaushadhi and one Kashthaushadhi. The methods below are the standard pharmacopoeial procedures described in the appendices of the *Ayurvedic Pharmacopoeia of India* [R36]; always follow the current edition and the monograph for the drug.

PR.5.1 Kashthaushadhi (herbal drug or formulation)

Test	Outline of method	Significance
Organoleptic and macroscopic	Colour, odour, taste, size, shape, surface, fracture	Identity
Microscopy	Transverse section or powder, with stains	Identity; detection of adulterants
Foreign matter	Weigh a sample, spread it out, separate foreign matter, weigh it	Purity
Loss on drying	Dry a weighed sample at 105 °C to constant weight	Moisture; risk of deterioration
Total ash	Incinerate 2 to 3 g in a tared silica crucible at a temperature not exceeding 450 °C until free from carbon; cool and weigh	Inorganic matter; earthy contamination
Acid-insoluble ash	Boil the total ash with dilute hydrochloric acid; collect the insoluble part on ashless filter paper; wash; ignite; weigh	Sand and silica
Water-soluble extractive	Macerate with chloroform water in a closed flask with frequent shaking, then let stand; filter; evaporate an aliquot to dryness; weigh	Water-soluble constituents; exhausted drug gives low values
Alcohol-soluble extractive	As above, with alcohol of the strength stated	Alcohol-soluble constituents
pH	Of an aqueous solution or suspension of stated strength	Consistency between batches
Thin-layer chromatography	Fingerprint compared with the reference in the monograph	Identity; consistency

PR.5.2 Rasaushadhi (Bhasma or other mineral preparation)

Test	Outline of method	Significance
Varna, Rupa	Colour and appearance	Identity; completeness of Marana
Rekhapurnatva	Rub between thumb and finger: fills the lines	Fineness
Varitaratva	A pinch on still water floats	Fineness and lightness
Nishchandratva	No lustre under light	Absence of metallic lustre
Apunarbhava	Heated with Mitra Panchaka: does not revert to metal	Absence of reversion to metal

Test	Outline of method	Significance
Niruttha	Heated with a silver leaf: silver does not gain weight	Absence of free metal combining with silver
Loss on drying, total ash, acid-insoluble ash	As above	Moisture; inorganic content; siliceous matter
pH	Of an aqueous suspension	Consistency
Particle size (where facilities exist)	Microscopy; laser diffraction	Fineness (B2.3.3)
Crystal phase and elements (where facilities exist)	X-ray diffraction; ICP or AAS	Composition (A5.3); heavy metal content

SAFETY: DRUG TESTING

- Muffle furnace and Bunsen burner: use tongs and heat-resistant gloves; let crucibles cool in a desiccator; never touch hot silica.
- Acids: add acid to water, never water to acid; use a fume hood for boiling with hydrochloric acid; wear goggles.
- Solvents (alcohol, chloroform): no open flame nearby; work in a fume hood; keep containers closed.
- Apunarbhava and Niruttha tests of metal preparations release fumes: do them under exhaust; do not inhale.
- Mercury- and arsenic-containing samples: handle as in PR.4.1; collect residues as hazardous waste.

PR.6 Identification of specimens

Specimens may be raw drugs (herbal, mineral, metal, animal), intermediate products (Shuddha drugs, Kajjali) or finished products (Bhasma, Pishti, Kupipakva, Churna, Vati, Avaleha, Asava, Taila). A clear, ordered answer earns the marks.

Point	For a raw drug	For a finished product
Name	Sanskrit name; English name; botanical, mineral or zoological identity	Name of the formulation; reference text
Identifying features	Colour, form, texture, odour, taste (where safe), fracture, lustre, streak, hardness	Colour, form, consistency, odour; classical tests of quality
Classification	Rasa group (Maharasa, Uparasa, Sadharana Rasa, Dhatu, Ratna, Sudha Varga) or plant family	Kalpana group (A5.1.1, B2.1.2)
Shodhana or processing	Method in brief	Main steps of preparation
Chemistry	Main chemical composition (for example, HgS for Hingula; As_2S_3 for Haratala)	Reported composition where known
Dose and uses	Classical dose; main indications	Dose; Anupana; main indications
Cautions	Toxicity of the raw material; Ashuddha effects	Storage; shelf life; cautions for heavy-metal or Visha-containing products

PR.7 Reports of visits to GMP pharmacies

The syllabus requires visits to at least five GMP-approved Ayurvedic pharmacies. A report is more than a description; it should show that you observed the plant against the requirements of GMP (Chapter B4).

PR.7.1 Format of the report

1. Name and address of the pharmacy; date of visit; persons met.
2. Licences seen: manufacturing licence, GMP certificate, other certifications (Chapters B4, B8).
3. Product range and dosage forms made.
4. Layout and flow: sketch of the plant with the flow of materials (B4.2.3).
5. Observations against the checklist below.
6. Special processes seen (for example, Puta, Kupipaka, Sandhana, large-scale Sneha Paka) with notes.
7. Quality control laboratory: tests and instruments seen.
8. **Safety and environment**: protective equipment, ventilation, handling of Rasa Dravya, fire safety, waste disposal.
9. Good practices worth adopting; gaps observed (stated politely and factually).
10. Conclusion and signature of the guide.

PR.7.2 Checklist for observation

Area	Look for
Premises	Clean surroundings; separate areas; one-way flow; smooth walls and floors; drainage
Raw material store	Pallets; pest control; quarantine and approved areas; labels; locked store for Rasa and Visha Dravya
Production areas	Separate sections for each dosage form; cleanliness; status labels on equipment (“clean”, “in use”)

Area	Look for
Equipment	Suitable, clean, maintained, calibrated; records
Documentation	Master formula; batch manufacturing records; SOPs; cleaning records; training records
Personnel	Uniforms, head covers, hand washing; medical check records; trained supervisors
Quality control	Laboratory; specifications; test records; retained samples
Packing	Line clearance; label control; batch coding
Finished goods	Storage conditions; FEFO; dispatch records
Complaints and recalls	Complaint register; recall procedure
Safety	PPE; exhaust; fire extinguishers; first aid; handling of mercury and furnaces
Environment	Effluent treatment; emission control for furnaces; hazardous waste handling

PR.8 Posting in an R&D or production unit

Keep a **daily log** for the ten days of posting.

Day	Section	Work observed or done	Process parameters noted	Documents seen	Learning point	Signature
1						
2						
…						
10						

At the end, write a summary of not more than two pages: the processes learnt, one problem you saw solved, and one improvement you would suggest. Attach the certificate of posting.

PR.9 Clinical cases studied for efficacy and ADRs

The syllabus requires at least ten clinical cases, using different dosage forms, studied and recorded for efficacy and adverse drug reactions. Each case is a small, careful observational record.

PR.9.1 Format of a case record

Section	Content
Patient	Registration number; initials; age; sex; occupation
Consent	Informed consent obtained for the use of the record for academic purposes (Chapter A2)
Diagnosis	Ayurvedic diagnosis; modern diagnosis where available
Assessment	Prakriti; Agni; Koshtha; relevant examination
Treatment	Each medicine: name, **manufacturer and batch number** (or in-house batch), dosage form, dose, Anupana, Bheshaja Kala, duration; Pathya

Section	Content
Concomitant medicines	Modern and other medicines taken
Outcome measures	Subjective and objective parameters, with a scale (for example, symptom scores; laboratory values), recorded at baseline and at each follow-up
Follow-up	Dates and findings
Efficacy	Change in each parameter; overall response
Adverse events	Any new symptom or abnormal test; description; date of onset; action taken; outcome
ADR assessment	Causality (WHO-UMC and Naranjo); severity (Hartwig); seriousness; preventability (Chapter A6)
Reporting	Whether the ADR was reported to the pharmacovigilance centre; date
Discussion	What the case teaches about the dosage form, its use and its safety

PR.9.2 Points to remember

- A case record is **not** proof of efficacy (A5.5); state findings modestly.
- Record **all** adverse events, not only those you think are caused by the medicine; causality is assessed afterwards.
- For **Rasaushadhi** and long courses, note the duration, and record kidney function and, where possible, blood or urine metal levels (A5.7).
- Keep patient identity confidential in the journal.

PR.10 Presentations, conferences, publication and documentation

PR.10.1 PowerPoint presentation on dissertation work (15 minutes)

Part	Time	Content
Title	0.5 min	Title, your name, guide, institution
Introduction and need	2 min	The problem; gap in knowledge
Aims and objectives	1 min	Clear, numbered
Materials and methods	4 min	Pharmaceutical work, analytical work, experimental or clinical design; ethics approvals
Results	4 min	Tables and graphs; one message per slide
Discussion	2 min	Meaning of results; comparison with published work; limitations
Conclusion	1 min	Answer to each objective; publication status
Questions	—	Prepare for questions on methods and limitations

Use few words per slide, legible fonts, honest graphs with labelled axes, and the same units throughout.

PR.10.2 Demonstration lecture (10 minutes)

Choose a topic you can teach well in ten minutes. Begin with learning objectives; use the board or one or two slides; include one diagram or table; ask one question of the audience; end with a summary. Keep to time.

PR.10.3 National conferences and research paper

Keep certificates of attendance (at least three national conferences within three academic years) with a short note on what you learnt. For the research paper, choose a peer-reviewed, indexed journal (Chapter B1), follow its instructions, and keep the acceptance letter or the published paper.

PR.10.4 Documentation file

Arrange in this order: journals of practicals (at least 75: 25 Rasashastra, 25 Bhaishajya Kalpana, 25 analytical); reports of the five visits; posting log and certificate; ten case records; conference certificates; research paper; any other supporting documents. Number every page and add a contents list.

PR.11 Viva voce

The viva carries 15 marks. Revise the **viva pointers** at the end of each chapter of this book. Answer the question asked, in a sentence or two, and stop. If you do not know, say so; do not guess on matters of safety or law. Expect questions on your own journal, your dissertation, the specimens you identified and the practicals you performed.

APPENDIX 1

APPENDIX 1 · SPECIMEN DOCUMENTS FOR AN AYURVEDIC PHARMACY

Why this appendix is here. Chapter B4 lists the documents that GMP requires. This appendix shows what they look like. A scholar who has filled in a batch record, read a certificate of analysis and written an SOP understands GMP far better than one who has only learned the list. The specimens also give a format for the GMP visit report and the R&D or production posting in the practical record (PR.7 and PR.8).

AP1.1 How to use these specimens

READ THIS FIRST

- These are **teaching specimens** written by the author. They are **not official forms**, and they are not copied from any company.
- A licensed unit designs its own documents within its quality system. They must meet **Schedule T as in force**. Schedule T was substantially revised by the Drugs (Eleventh Amendment) Rules, 2026 (Chapter B4.6.4).
- Where an **official form** exists, always use the current official version. Examples are the licence application forms under the Drugs Rules (Chapter B8(i).4) and the suspected ADR reporting form of the national pharmacovigilance programme for ASU&H drugs (Chapter A6.8.3).
- Names, numbers and quantities in the specimens are illustrative. “[Name of the unit]” stands for the manufacturer.
- **Test limits** must be taken from the current API monograph or the approved specification. The specimens write “as per API” instead of quoting numbers that may change.

AP1.2 The documents that travel with one batch

Every batch leaves a trail of paper. If any link is missing, the batch cannot be traced, and a defect cannot be investigated or recalled (Figure AP1.1).

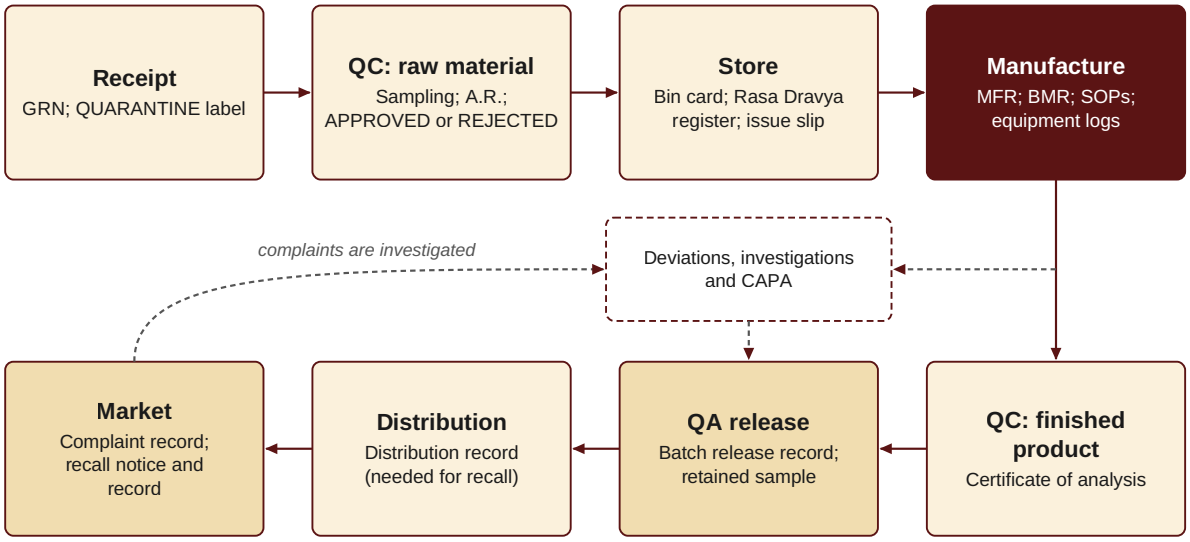


Figure AP1.1. The documents that follow one batch, from receipt of raw material to the market and back.

Document	Stage	Prepared by	Checked or approved by	Specimen
Goods receipt note (GRN)	Receipt of material	Stores	Stores in-charge	AP1.4
Status labels	Receipt to release	Stores; QC	QC	AP1.4
Bin card; Rasa Dravya register	Storage and issue	Stores	Stores in-charge; QA	AP1.5
Master formula record (MFR)	Once for each product	Production with QC	QA head	AP1.6
Batch manufacturing record (BMR)	During manufacture of each batch	Production	Production in-charge; QA	AP1.7
Standard operating procedures (SOPs)	Every routine operation	Department concerned	QA head	AP1.8 to AP1.10
Equipment log book	Every use, cleaning and repair	Operator	Supervisor	AP1.8
Certificate of analysis (CoA)	Testing of raw material and finished product	QC analyst	QC head	AP1.11
Batch release record	Before sale	QA	Authorised person	AP1.11
Deviation report	Whenever a procedure is not followed	Person who finds it	QA	AP1.12
Market complaint record	After sale	QA	QA head	AP1.13
Recall notice and recall record	When a batch must be withdrawn	QA	Management	AP1.14

The job titles are those of a typical medium-sized unit. A small unit may combine roles, but the person who approves a batch should not be the person who made it.

AP1.3 Good documentation practice

A document is evidence. In an inspection, in a complaint or in a court, what was not written down is taken as not done. The World Health Organization sums up the qualities of good records in the word **ALCOA+**: records should be **a**ttributable, **l**egible, **c**ontemporaneous, **o**riginal and **a**ccurate, and also complete, consistent, enduring and available for the whole retention period [WHO 2021].

Letter	Meaning	In practice
A	Attributable	Every entry is signed or initialled and dated by the person who did the work
L	Legible	Readable now and years later; indelible ink; no pencil
C	Contemporaneous	Written at the time the work is done, not at the end of the shift from memory
O	Original	The first record is kept; a fair copy does not replace it
A	Accurate	Values as actually observed, with units; no rounding to “make it fit”
+	Complete, consistent, enduring, available	No blank spaces; the same format across batches; durable paper or validated software; retrievable when asked for

Rules every scholar should follow in the laboratory record as well:

1. Use indelible ink. Never use pencil, correction fluid or erasers.
2. To correct an entry, strike it through with a **single line** so that it can still be read. Write the correct value beside it, with your initials, the date and the reason [WHO 2021].
3. Fill every field. Where a field does not apply, write “NA”. Strike through unused space.
4. Record the result itself, not only “OK” or “complies”.
5. Write the unit with every number: g, kg, °C, minutes.
6. Never write on loose scraps of paper and copy later.
7. Sign only for what you did or checked yourself.

> **AYURVEDA CORRELATION: THE WRITTEN RECORD**
>
> The Rasa texts describe the qualities of the person who prepares medicines and require that each process be carried out exactly as taught, with tests of completion at each stage (Siddhi Lakshana and Bhasma Pariksha). What they do not require is a written record of each batch.
>
> **To be remembered.** A classical test of completion is valuable only if the result is written down for that batch. “Rekhapurnatva: complies, 3 April, 11.20 a.m., initials” is evidence. A memory that the test was passed is not.

AP1.4 Goods receipt note and status labels

When material arrives, the stores department checks it against the purchase order, records it on a **goods receipt note** and places it under **quarantine**. Nothing may be used until QC has sampled it, tested it and approved it (Chapter B4.3.2).

Goods Receipt Note

TEACHING SPECIMEN

Field	Entry
[Name of the unit] — GRN No.	…… Date of receipt …… Time ……
Supplier (name and address)	……
Invoice or challan No. and date	……
Purchase order No.	……
Vehicle No.	……
Material: Sanskrit name; botanical name; part used	e.g. Haritaki; *Terminalia chebula* Retz.; fruit
Form	Whole / coarse / powder / other ……
Quantity ordered	…… kg
Quantity received	Number of bags …… Gross …… kg Net …… kg
Supplier’s batch or lot No.	……
Condition of containers	Intact / torn / wet / infested / mixed with foreign matter (tick)
Supplier’s certificate of analysis received	Yes / No
Internal lot No. allotted	……
Status on receipt	**QUARANTINE** label fixed on every container: Yes / No
QC informed for sampling on	…… (date and time)
Received by (sign, date)	……
Checked by, stores in-charge (sign, date)	……

Status labels. Every container carries one label showing its status at that moment. The label is changed only by the person authorised to change it.

Label	Colour often used*	Meaning	Fixed by	Next step
QUARANTINE	Yellow	Received; not yet sampled	Stores	QC samples it
UNDER TEST	Yellow or white	Sampled; testing in progress	QC	Awaits result
APPROVED	Green	Tested and found to comply	QC	May be issued for use
REJECTED	Red	Does not comply	QC	Moved to the rejected area; returned or destroyed, with records
CLEANED	Green or white	Equipment or area cleaned and ready	Production	Line clearance before the next batch
TO BE CLEANED	Red or orange	Equipment used; not yet cleaned	Production	Cleaning by SOP

**The colours are a common convention, not a legal requirement. Each unit fixes its own colours in an SOP and uses them consistently.*

Status label: APPROVED

TEACHING SPECIMEN

APPROVED — Material …… Internal lot No. …… Quantity …… Analytical report (A.R.) No. …… Date of approval …… Retest or expiry date …… Approved by (QC) ……

AP1.5 Bin card and the Rasa Dravya register

A **bin card** (stock card) is kept with each material in the store. It shows at a glance how much is in stock, and it is checked against the physical stock during stock verification (Chapter B4.3.5). The stock levels printed on it come from inventory control (Chapter B4.4.4).

Bin Card

TEACHING SPECIMEN

Material: **Haritaki (fruit, dried)** Code: RM-017 Unit: kg Location: Rack C-3 Minimum level: 40 kg Reorder level: 90 kg Maximum level: 300 kg

Date	Reference (GRN or issue slip)	Lot No.	Received	Issued	Balance	Sign
01-04	Opening balance	H-0312	—	—	120.0	……
03-04	Issue slip 0418, batch TC-0107	H-0312	—	10.0	110.0	……
07-04	Issue slip 0425, batch TC-0108	H-0312	—	10.0	100.0	……
09-04	Issue slip 0431, batch HC-0044	H-0312	—	15.0	85.0	……
09-04	Below reorder level: purchase requisition raised	—	—	—	85.0	……
16-04	GRN 0207 (approved 18-04)	H-0408	200.0	—	285.0	……

*The material from GRN 0207 is entered on the bin card only after QC approval, or is shown in a separate "quarantine" column until then. Issue follows **first expiry, first out**: the older lot H-0312 is used before H-0408.*

For **Rasa Dravya and Visha Dravya** (mercury, arsenic compounds, Vatsanabha and the like), an ordinary bin card is not enough. The quantities are small and the risks are high, so every gram is accounted for in a bound, page-numbered register. The register is kept in the locked store, and each entry is signed by two persons (Chapter B4.6.7).

Rasa Dravya Register: Parada (mercury)

TEACHING SPECIMEN

Material: **Parada (Ashodhita)** Storage: locked Rasa Dravya store, cupboard 2, in secondary container Unit: g

Date	Lot No.	Received (g)	Issued (g)	Issued for (batch and process)	Returned (g)	Balance (g)	Issued by	Received by	Verified by
02-05	P-0901	—	—	Opening balance	—	2500			……
05-05	P-0901	—	500	Shodhana, batch SP-0021	—	2000	……	……	……
12-05	P-0901	—	—	Shodhana completed; Shuddha Parada 468 g transferred to Shuddha Parada register	—	2000	……	……	……

The loss in Shodhana (here 500 g taken, 468 g obtained) is recorded in the batch record and explained: material lost with the washings, adhesion to vessels, spillage. An unexplained loss of mercury is both a quality and a safety matter and must be investigated.

AP1.6 Master formula record

The **master formula record** is the approved recipe for a product. It is written once, approved, and then followed for every batch. It answers five questions: what goes in, in what quantity, with which equipment, by what steps, and with what checks. The specimen below is for **Triphala Churna**, a simple classical Churna of three fruit rinds in equal parts, chosen because every step is easy to follow.

Master Formula Record: Triphala Churna

TEACHING SPECIMEN

1. Product details

Field	Entry
Product name	Triphala Churna
Category	Classical Ayurvedic medicine (ingredients and method as in an authoritative book listed in the First Schedule)
Reference	Ayurvedic Formulary of India, Part I, serial No. …… [R37]
Dosage form	Churna (fine powder)
MFR No.; version; effective date	MFR/TC/01; version 02; ……
Standard batch size	30.000 kg
Pack	100 g in an HDPE container with induction-sealed cap
Shelf life	2 years from the date of manufacture (Rule 161B(8), Churna; Chapter B3.4.3)
Storage	In a cool, dry place, protected from light and moisture
Licence No. under which made	……

2. Formula

No.	Ingredient	Botanical name	Part	Proportion	Quantity per batch	Specification
1	Haritaki	*Terminalia chebula* Retz.	Fruit rind (seed removed)	1 part	10.000 kg	API Part I
2	Bibhitaki	*Terminalia bellirica* (Gaertn.) Roxb.	Fruit rind (seed removed)	1 part	10.000 kg	API Part I
3	Amalaki	*Phyllanthus emblica* L.	Fruit rind (seed removed)	1 part	10.000 kg	API Part I
				Total	**30.000 kg**	

The quantities are of approved, cleaned and dried material. No overage. The botanical names are those accepted in World Flora Online (Chapter B1.4.2); older texts write Emblica officinalis Gaertn. for Amalaki.

3. Equipment

Equipment	ID No.	Use
Platform balance, 100 kg	BAL-03	Dispensing
Tray dryer	TD-01	Drying of raw material
Pulveriser with screen	PUL-02	Size reduction
Vibro-sifter with sieve as specified	VS-01	Sieving
Double-cone blender, 50 kg working capacity	DCB-01	Mixing
Moisture balance	MB-01	In-process loss on drying

4. Method of manufacture

Step	Instruction	In-process control
1	**Line clearance.** Confirm that no material, label or document of the previous product remains, and that the area and equipment carry a “Cleaned” label	Checked and signed by QA
2	**Dispensing.** Weigh each ingredient from containers bearing an “Approved” label. Record the A.R. No.	Second person checks each weighing
3	**Garbling.** Spread on clean trays; remove foreign matter, stones, insect-damaged pieces and remaining seeds	Foreign matter removed is weighed and recorded
4	**Drying.** Dry in the tray dryer at the temperature fixed in this MFR (for example, not above 60 °C) until loss on drying is within the in-house limit	Loss on drying of a sample from each tray
5	**Size reduction.** Pulverise each ingredient through the specified screen	Feed rate and screen size recorded
6	**Sieving.** Pass each powder through the sieve number stated in the specification. Return the coarse fraction to the pulveriser	Sieve integrity checked before and after use
7	**Mixing.** Load the three powders into the blender. Mix for the time fixed during validation (for example, 20 minutes)	Samples from top, middle and bottom for uniformity
8	**Unloading.** Unload into clean, labelled, lined containers. Weigh	Weight recorded; yield calculated
9	**Quarantine.** Label “Under test”. Send the sample to QC with the sample request	QC sample request No.
10	**Packing** (only after QC release). Fill, seal and label. Reconcile labels	Fill weight checked every 30 minutes

5. Yield limits

Theoretical yield: 30.000 kg. Acceptable yield: from …… % to …… %, fixed from the first three validation batches. A yield outside the limits is a deviation (AP1.12).

6. Prepared, checked and approved

	Name	Designation	Signature	Date
Prepared by		Production		
Checked by		Quality control		
Approved by		Head, quality assurance		

SAFETY: MANUFACTURE OF CHURNA

- **Dust.** Fine plant dust irritates the eyes and lungs and can cause allergy. Use dust extraction at the pulveriser, sifter and blender. Workers wear dust masks, caps and goggles.
- **Dust explosion.** A cloud of fine organic dust can ignite. Keep machines earthed. Keep naked flames, sparks and smoking out of the powder section. Clean dust from ledges and light fittings regularly.
- **Machines.** Never open a pulveriser, sifter or blender, or put a hand inside, until the machine is switched off, isolated at the mains and tagged ("lock-out, tag-out"). Keep all guards in place. Know where the emergency stop is.
- **Noise.** Pulverisers are loud. Ear protection is worn in the size-reduction room.
- **Manual handling.** Lift sacks with a straight back, or use a trolley. Do not lift more than you can safely manage.
- **Heat.** Tray dryers have hot surfaces. Use heat-resistant gloves when removing trays.

AP1.7 Batch manufacturing record

The **batch manufacturing record** is a copy of the master formula with spaces to record what was actually done for one batch. It is issued by QA, numbered, and returned complete. A BMR with a blank field cannot be released.

Batch Manufacturing Record: Triphala Churna

TEACHING SPECIMEN

Field	Entry	Field	Entry
BMR issued by QA (sign, date)	……	MFR No. and version	MFR/TC/01; version 02
Batch No.	TC-0108	Batch size	30.000 kg
Date of starting	……	Date of completion	……
Date of manufacture	……	Date of expiry	…… (2 years)

A. Line clearance

Check	Yes / No	Done by	Checked by (QA)
No material of the previous product in the area			
No labels or documents of the previous product			
Equipment carries a “Cleaned” label with date			
Balance calibrated; calibration within due date			
Temperature and humidity of the area recorded			

B. Dispensing

Ingredient	Lot No.	A.R. No.	Required (kg)	Dispensed (kg)	Dispensed by	Checked by
Haritaki			10.000			
Bibhitaki			10.000			
Amalaki			10.000			

C. Processing

Step	Start (date, time)	End (date, time)	Observations and readings	Done by	Checked by
Garbling: foreign matter removed (g)					
Drying: temperature (°C); LOD (%)					
Pulverising: screen size; machine ID					
Sieving: sieve No.; integrity before and after					
Mixing: blender ID; speed; time (min)					
Unloading: number of containers; gross and net weight					

D. In-process checks

Test	Limit	Result	Sign
Description (colour, odour)	As per specification		
Loss on drying	In-house limit		
Passes through sieve No. ……	Not less than …… %		
Uniformity: top, middle, bottom	As per specification		

E. Yield

Theoretical: 30.000 kg Actual: …… kg Yield: …… % Within limits: Yes / No Deviation No. (if any): ……

F. Packing and label reconciliation (after release)

Labels issued	Labels used	Labels damaged	Labels returned	Labels destroyed	Balance (must be 0)

G. Review

Production in-charge (sign, date) …… QA review: BMR complete, deviations closed (sign, date) …… QC certificate of analysis No. ……

WORKED EXAMPLE: YIELD

In batch TC-0108, 30.000 kg of dried material was charged and 29.250 kg of sieved, mixed powder was unloaded.

Yield = 29.250 ÷ 30.000 × 100 = **97.5 per cent**.

If the acceptable range fixed in the MFR is 96 to 99 per cent, the batch is within limits. If the yield had been 92 per cent, the difference of 2.4 kg must be explained, for example by a high coarse fraction or a spill. It is recorded as a deviation and investigated before the batch is released.

AP1.8 The standard operating procedure

A **standard operating procedure** tells a trained person exactly how to do one routine task, the same way every time. A good SOP is short, numbered and written in the imperative (“Switch off … Remove …”). It is kept at the place of work, and only the current version is in use.

Section of an SOP	What it contains
Header	Unit name; title; SOP No.; version; effective date; review date; page *x* of *y*
1. Purpose	Why the SOP exists, in one sentence
2. Scope	Where and to what it applies
3. Responsibility	Who does each part; who checks
4. Safety	Hazards and the precautions against each (always included in this subject)
5. Materials and equipment	What is needed
6. Procedure	Numbered steps, in order
7. Records	Which log book or form is filled
8. Abbreviations and references	Terms used; related SOPs
9. Revision history	Version, date and what was changed
Approval block	Prepared, checked and approved by, with signatures and dates

SOP: Operation and cleaning of the pulveriser

TEACHING SPECIMEN

[Name of the unit] SOP No. PRD-012 Version 03 Effective …… Review due …… Page 1 of 1

1. Purpose. To operate and clean the pulveriser safely and so that one product does not contaminate the next.

2. Scope. Pulveriser PUL-02 in the size-reduction room.

3. Responsibility. Operator: operation and cleaning. Production supervisor: checking. QA: line clearance.

4. Safety. Switch off, isolate and tag the machine before opening it or putting a hand near the beater or screen. Never feed material by hand beyond the hopper mouth; use the scoop. Wear a dust mask, goggles, cap and ear protection. Keep the machine earthed and the dust extraction running. Stop at once if there is unusual noise, vibration or smell of burning.

5. Materials. Screen of the size stated in the BMR; clean, lined containers; brushes; lint-free cloth; purified water; 70 per cent ethanol (for final wipe, where the SOP for cleaning agents allows).

6. Procedure.

6.1 *Before starting.* Check the “Cleaned” label and the log book. Check that the screen is of the correct size and is intact. Check that the guard and the hopper cover are in place. Switch on the dust extraction.

6.2 *Operation.* Switch on the pulveriser and allow it to reach full speed. Feed the material steadily; do not overload. Collect the powder in the lined container. Check the screen at the end of the batch.

6.3 *Cleaning between batches of the same product.* Switch off and isolate. Remove the residual powder with a brush. Wipe the outer surfaces. Record in the log book.

6.4 *Cleaning at change of product.* Switch off and isolate. Dismantle the hopper, beater chamber and screen. Brush, then wash with water and dry completely. Wipe with 70 per cent ethanol where allowed. Reassemble. Fix a “Cleaned” label with the date and signature. QA checks before the next product.

7. Records. Equipment log book (below); BMR.

9. Revision history. Version 03: step 6.4 added the ethanol wipe; …… (date).

Prepared by …… Checked by …… Approved by (QA) ……

Equipment log book: Pulveriser PUL-02

TEACHING SPECIMEN

Date	Product	Batch No.	Start	End	Activity (use, cleaning, repair)	Done by	Checked by	Remarks

AP1.9 SOP for handling mercury in the Rasashala

This SOP is given in full because mercury is the most hazardous material a Rasashastra scholar handles routinely. It applies equally to the department laboratory.

SOP: Receipt, storage, issue and handling of mercury (Parada)

TEACHING SPECIMEN

[Name of the unit] SOP No. RS-003 Version 01 Effective ……

1. Purpose. To handle mercury so that no worker is exposed to its vapour, no mercury is lost, and none reaches the environment.

2. Scope. Receipt, storage, issue, weighing, Shodhana, trituration (for example, Kajjali), spills and waste in the Rasashala.

3. Responsibility. Rasashala in-charge: custody and register. Worker: following this SOP. Safety officer: spill kit, training and health checks.

4. Hazards. Mercury gives off vapour at room temperature, and more when warm. Inhaled vapour is the main route of harm; long exposure damages the nervous system and the kidneys (Chapter A4.5). Spilled mercury breaks into tiny droplets that lodge in cracks and keep giving off vapour.

5. Controls and protective equipment.

Control	Requirement
Room	Designated room; smooth, crack-free, impervious floor and benches; no carpet; kept cool
Ventilation	Local exhaust at the work bench; exhaust to the outside, away from air intakes
Containment	All work done over a tray with raised edges, large enough to catch a spill
Heating	Only in closed apparatus under exhaust (for example, Urdhvapatana); never on an open flame in the room
Protective equipment	Nitrile gloves; apron or coat kept in the room; goggles; a respirator suitable for mercury vapour during spill cleanup
Persons	Only trained workers. Pregnant workers do not work with mercury
Hygiene	No eating, drinking or smoking in the room. Wash hands and face before leaving
Health	Periodic medical examination; urine mercury where exposure is significant (B4.6.7)

6. Procedure.

6.1 *Receipt.* Check the container for leaks over a tray. Weigh. Enter in the Rasa Dravya register (AP1.5). Label "Quarantine" until QC approves.

6.2 *Storage.* Keep in a tightly closed, unbreakable, labelled container, inside a second container, in the locked Rasa Dravya store. The key is held by the in-charge.

6.3 *Issue.* Issue only against a BMR or a practical requisition. Weigh in the presence of two persons over a tray. Both sign the register.

6.4 *Working.* Keep the Khalva or vessel in its tray throughout. Cover containers when not in use. Do not leave mercury uncovered overnight.

6.5 *Return.* Weigh the unused quantity and the product. Record both. Report any unexplained loss to the in-charge the same day.

7. Spill procedure.

1. Keep other people out of the area. Do not walk through the spill. Switch off any heating nearby. Keep the exhaust running and open windows to the outside.
2. Put on gloves and the respirator from the spill kit.
3. **Never use a broom or a vacuum cleaner.** A broom breaks mercury into finer droplets and spreads it; a vacuum cleaner throws it into the air as vapour [EPA 2025].
4. Push droplets together with a card or squeegee and draw them up with a dropper. Place them in a strong, closable container.
5. Dust the area with **sulphur powder** (or a commercial mercury-binding powder). Sulphur binds the remaining fine droplets, makes them easier to see, and reduces vapour [EPA 2025]. Collect the powder.
6. Seal all collected mercury, powder, cards and gloves in a labelled container: "Mercury waste".
7. Ventilate the room well after the cleanup. Record the spill in the incident register.

8. Waste. Never pour mercury or its washings into a drain [EPA 2025]. Collect all mercury-containing waste, including Shodhana washings and residues, in labelled, closed containers, and dispose of it through an authorised hazardous waste handler.

9. Records. Rasa Dravya register; BMR; incident register; training record; health records.

Prepared by …… Checked by …… Approved by ……

AP1.10 SOP for Puta in an electric muffle furnace

Many units now use a programmable electric muffle furnace for Puta, because the temperature can be set, recorded and repeated from batch to batch. The record of each Puta is the modern form of the classical instruction to count the Puta and judge the product.

SOP: Puta of Bhasma in the electric muffle furnace

TEACHING SPECIMEN

[Name of the unit] SOP No. RS-011 Version 02 Effective ……

1. Purpose. To carry out each Puta at the temperature and time stated in the BMR, and to record it.

2. Scope. Electric muffle furnace EMF-01 in the Marana section.

3. Responsibility. Operator: loading, programming, unloading and recording. Supervisor: checking the temperature record.

4. Safety.

- Furnace surfaces and the Sharava Samputa reach several hundred degrees. Use long tongs, heat-resistant gloves and a face shield.
- Do not open the door until the temperature has fallen to the safe level written in this SOP.
- Fumes from metal and mineral Bhasmas (for example, of lead or arsenic compounds) must go to the exhaust. Keep the furnace vent connected and the exhaust on.
- Keep the furnace earthed. Keep flammable material away. Keep a fire extinguisher suitable for electrical fires nearby.
- Workers handling lead or arsenic compounds follow the health checks in B4.6.7.

5. Procedure.

1. Check the thermocouple calibration date and the controller.
2. Weigh the sealed Sharava Samputa. Record.
3. Place it in the centre of the furnace on a refractory plate.
4. Enter the programme from the BMR: rate of rise, peak temperature, time at peak.
5. Start. Note the start time.
6. At the end of the programme, let the furnace cool on its own (Svanga Shita). Do not force cooling.
7. Open only when below the safe temperature. Remove the Samputa with tongs.
8. Weigh. Open. Note colour and texture. Do the tests stated in the BMR.
9. Fill the Puta record.

6. Records. Puta record (below); controller print-out, if available; BMR.

Puta record

TEACHING SPECIMEN

Product …… Batch No. …… Furnace ID ……

Puta No.	Date	Weight before (g)	Peak (°C)	Time at peak (min)	Weight after (g)	Colour and texture	Bhasma Pariksha done	Sign
1								
2								
3								

AP1.11 Certificate of analysis and batch release

The **certificate of analysis** reports the test results of one batch against its specification. The **batch release record** is the final check by QA that everything about the batch, not only the test results, is in order (Chapter B5.2.1).

Certificate of Analysis: Triphala Churna (finished product)

TEACHING SPECIMEN

Field	Entry	Field	Entry
A.R. No.	……	Batch No.	TC-0108
Date sample received	……	Batch size	30.000 kg
Date of manufacture	……	Date of expiry	……
Specification No.	…… (based on the API)	Date of report	……

Test	Specification	Result	Method
Description	Brown powder; characteristic odour; astringent taste		Organoleptic
Identification: microscopy	Shows the diagnostic characters of each of the three ingredients		API
Identification: TLC or HPTLC	Pattern matches the reference		API
Loss on drying at 105 °C	As per specification		API
Total ash	As per API		API
Acid-insoluble ash	As per API		API

Test	Specification	Result	Method
Alcohol-soluble extractive	As per API		API
Water-soluble extractive	As per API		API
pH (aqueous suspension)	As per specification		API
Fineness	Passes through the specified sieve		API
Heavy metals: Pb, Cd, As, Hg	As per API		API
Microbial limits: total aerobic count, yeast and mould, specified pathogens	As per API		API
Aflatoxins	As per API		API
Pesticide residues	As per API		API
Marker assay, where specified (for example, a tannin or gallic acid content)	As per specification		Validated method

Conclusion: The sample complies / does not comply with the specification.

Analysed by …… Checked by …… Approved by (head, QC) ……

The specimen deliberately gives no numerical limits. The limits for this product are in its API monograph and the API appendices, which are revised from time to time (Chapter B10). Copying a number from a book like this one, instead of from the current monograph, is a common source of error.

Batch release record

TEACHING SPECIMEN

Product …… Batch No. …… Quantity ……

Check before release	Yes / No	Initials
BMR complete; every field filled, signed and dated		
All deviations recorded, investigated and closed; impact assessed		
Yield within limits, or the difference explained		
Certificate of analysis: complies		
Packing material approved; label text checked against the approved artwork		
Label reconciliation balanced		
Retained (control) sample kept, labelled and stored		
Storage conditions during manufacture and quarantine within limits		

Decision: RELEASED for sale / REJECTED / ON HOLD pending ……

Authorised person (name, signature, date) ……

AP1.12 Deviation report

A **deviation** is any departure from an approved procedure, specification or limit. Deviations are not hidden; they are reported, investigated and closed. The worked specimen below shows how.

Deviation report (filled specimen)

TEACHING SPECIMEN

Field	Entry
Deviation No.; date	DEV-0031; ……
Product; batch; stage	Triphala Churna; TC-0109; drying (step 4 of the MFR)
What should have happened	Drying at not more than 60 °C
What happened	The dryer reached 72 °C for about 25 minutes. The operator noticed it on the dial during the hourly check
Immediate action	Dryer switched off; trays removed to a clean area; material labelled “On hold”; supervisor and QA informed
Classification	Major (possible effect on quality; no effect on safety expected)

Field	Entry
Investigation and root cause	The thermostat had failed. Its calibration was due in the following month. The hourly temperature check was the only control, and it had worked
Impact on the batch	Additional QC tests on the batch: description, LOD, extractive values and marker assay compared with three previous batches. Results within specification and within the usual range
Corrective action	Thermostat replaced and dryer re-qualified
Preventive action	Over-temperature cut-out fitted to both dryers; temperature check made every 30 minutes; SOP revised
QA decision	Batch may be processed further; decision recorded in the BMR
Closed by (QA); date	……

AP1.13 Market complaint record

Every complaint from a patient, doctor or trader is recorded, even if it seems trivial. A pattern of small complaints can reveal a serious problem. A complaint of an **adverse reaction** is also a pharmacovigilance report and is forwarded as such (Chapter A6).

Market complaint record

TEACHING SPECIMEN

Field	Entry
Complaint No.; date received	……
Received from (name, address, telephone); how (letter, phone, e-mail, representative)	……
Product; pack; batch No.; expiry	……
Nature of complaint	Quality (colour, smell, caking, foreign matter) / packing or label / short quantity / adverse reaction / other (tick)
Details in the complainant’s words	……
Complaint sample received	Yes / No
Retained sample of the same batch examined; result	……
Other complaints on the same batch	Yes / No; numbers ……
Investigation and root cause	……
If adverse reaction: forwarded for pharmacovigilance reporting on	……

Field	Entry
Action taken	Reply sent / replacement / batch put on hold / recall considered (see AP1.14)
Closed by (QA); date	……

AP1.14 Recall notice and recall record

When a batch must be withdrawn, speed matters. The notice is short, clear and says exactly what the receiver must do. The classes of recall are those of Chapter B5.3.2.

Recall notice to distributors and retailers

TEACHING SPECIMEN

URGENT: PRODUCT RECALL — CLASS ……

To: All distributors and retailers who received the batch below **From:** [Name of the unit], Licence No. …… **Date:** ……

Product	Pack	Batch No.	Date of manufacture	Date of expiry
……	……	……	……	……

Reason for recall: …… (for example, “a test result outside the specification for microbial contamination”).

What you must do now:

1. Stop selling or supplying this batch **immediately**.
2. Separate all stock of this batch and label it “Recalled — do not sell”.
3. Return the stock to …… by …… (date). The cost of return will be borne by us.
4. If you have supplied this batch to other traders, pass this notice to them at once and give us their names.
5. Fill in and return the attached reply form within …… days, **even if you hold no stock**.

Other batches of this product are not affected.

Contact: ……, telephone ……, e-mail ……

Signed …… (Head, quality assurance)

Recall record and reconciliation

TEACHING SPECIMEN

Item	Quantity (packs)
(A) Quantity manufactured and released	……
(B) Quantity in the unit's own stock at the time of recall	……
(C) Quantity distributed (A – B)	……
(D) Quantity returned from the market	……
(E) Quantity confirmed as sold to patients before the recall	……
Percentage accounted for: (D + E) ÷ C × 100	…… %

Licensing authority informed on …… Closure report sent on …… Effectiveness of recall reviewed by …… Disposal of returned stock witnessed by ……

KEY POINTS TO REMEMBER

- GMP is proved by documents. What is not written is taken as not done.
- ALCOA+: attributable, legible, contemporaneous, original, accurate; complete, consistent, enduring, available.
- Correct by a single line, with initials, date and reason. No correction fluid.
- MFR: the approved recipe, written once. BMR: what was done for one batch.
- An SOP in this subject always has a safety section.
- Mercury spill: no broom, no vacuum cleaner; collect droplets; sulphur powder; seal; ventilate; record.
- Release is by QA after review of the whole batch record, not only the test result.
- Every deviation is reported, investigated and closed with corrective and preventive action.
- A recall notice says what to do, by when, and asks for a reply even if no stock is held.

REFERENCES USED IN THIS APPENDIX

Syllabus reference number used: **37**.

- **[R37]** *The Ayurvedic Formulary of India*, Part I. Government of India (the edition in use in the unit).

Additional verified sources (accessed 16 October 2026)

- **[WHO 2021]** World Health Organization. *Guideline on data integrity*. Annex 4, WHO Technical Report Series, No. 1033. Geneva: WHO; 2021.
- **[EPA 2025]** United States Environmental Protection Agency. "What to do if a mercury thermometer breaks." epa.gov (archived snapshot of 19 January 2025).

APPENDIX 2

APPENDIX 2 · NUMERICAL EXERCISES IN PHARMACOLOGY AND PHARMACY MANAGEMENT

Why this appendix is here. Pharmacokinetics, dose determination and inventory control are understood only when a scholar has worked through the numbers. A short calculation also lifts a long answer: an examiner remembers the candidate who showed how a half-life is found from two plasma levels. Every formula used here is explained in the chapter named beside it. The drugs in the problems are imaginary unless a real one is named, so that the numbers serve the method and are not mistaken for data.

AP2.1 How to solve a numerical problem

1. **Write down the data**, each value with its unit.
2. **Write the formula** in symbols before putting in any number.
3. **Convert units** so that they match (mg with L, hours with per hour, mL per minute with mL per minute).
4. **Substitute and calculate.** Keep two or three significant figures; do not round in the middle.
5. **State the answer with its unit** and underline it.
6. **Ask whether it is sensible.** A half-life of 300 hours for a drug given three times a day, or a child's dose larger than the adult dose, means a slip somewhere.

AP2.2 Formulae at a glance

Quantity	Formula	Chapter
Elimination rate constant	$k = 0.693 \div t_{½}$	A1a.2.7
Concentration after time t (first order)	$C_t = C_0 \times e^{-kt}$, or $C_0 \times (½)^n$ where $n = t \div t_{½}$	A1a.2.7
Volume of distribution	Vd = Amount in the body ÷ Plasma concentration = Dose (IV) ÷ C_0	A1a.2.4
Clearance	$CL = k \times Vd = 0.693 \times Vd \div t_{½}$	A1a.2.7
Renal clearance	CL_R = (Urine concentration × Urine flow) ÷ Plasma concentration	A1a.2.6
AUC by the trapezoidal rule	Sum of $(C_1 + C_2) \div 2 \times (t_2 - t_1)$ for each interval	A1a.2.8
AUC from the last point to infinity	$C_{last} \div k$	A1a.2.8
Absolute bioavailability	$F = (AUC_{oral} \div Dose_{oral}) \div (AUC_{IV} \div Dose_{IV})$	A1c.2
Loading dose	LD = Target C × Vd ÷ F	A1a.2.9
Maintenance dose rate	MD rate = Target C_{ss} × CL ÷ F	A1a.2.9
Accumulation factor (repeated IV doses)	$R = 1 \div (1 - e^{-k\tau})$, where τ is the dosing interval	A1a.2.9

Quantity	Formula	Chapter
Human equivalent dose	HED (mg/kg) = Animal dose (mg/kg) × Animal Km ÷ Human Km	A1a.4.3
Child's dose by body surface area	Adult dose × BSA (m^2) ÷ 1.73	A1a.4.3
Body surface area (Mosteller)	BSA (m^2) = √(Height in cm × Weight in kg ÷ 3600)	—
Economic order quantity	EOQ = √(2DS ÷ H)	B4.4.3
Reorder level	Average usage × Lead time + Safety stock	B4.4.4
Break-even point	Fixed costs ÷ (Price − Variable cost per unit)	B4.2.6
Yield	Actual output ÷ Theoretical output × 100	AP1.7

The Mosteller formula for body surface area is the one most used in practice; it is added here because the chapter states the BSA rule but not how BSA is found [Mosteller 1987].

AP2.3 Elimination and half-life

Problem 1. Rate constant from half-life

A drug has an elimination half-life of 4 hours. Find its elimination rate constant.

> k = 0.693 ÷ t½ = 0.693 ÷ 4 = **0.173 per hour**. About 17 per cent of the drug present is removed each hour.

Problem 2. Concentration after several half-lives

After an intravenous bolus, the plasma concentration of a drug is 80 mg/L. Its half-life is 6 hours. What will the concentration be after 24 hours, if elimination is first order?

> Number of half-lives n = 24 ÷ 6 = 4.
>
> C = 80 × $(½)^4$ = 80 ÷ 16 = **5 mg/L**.
>
> So 93.75 per cent has been eliminated in four half-lives, as in the table in A1a.2.7.

Problem 3. Half-life from two plasma levels

After an intravenous dose, the plasma level is 40 mg/L at 2 hours and 10 mg/L at 8 hours. Elimination is first order. Find the half-life, the rate constant and the concentration at time zero.

> The level fell from 40 to 10 mg/L, that is, to one quarter, in 8 − 2 = 6 hours. One quarter is two halvings, so two half-lives took 6 hours.
>
> t½ = 6 ÷ 2 = **3 hours**; k = 0.693 ÷ 3 = **0.231 per hour**.
>
> To find C_0, go back 2 hours (two-thirds of a half-life) from 40 mg/L: $C_0 = 40 \times 2^{(2/3)} = 40 \times 1.587 \approx$ **63.5 mg/L**.
>
> *Short cut for the examination:* when the ratio of the two levels is a power of 2, count the halvings. Otherwise use $t½ = 0.693 \times (t_2 - t_1) \div \ln(C_1 \div C_2)$.

Problem 4. Time to eliminate 90 per cent

How many half-lives does it take to eliminate 90 per cent of a drug? How long is this for the drug in Problem 3?

> Ten per cent remains when $(½)^n = 0.1$, so n = ln 10 ÷ ln 2 = 2.303 ÷ 0.693 ≈ **3.3 half-lives**.
>
> For t½ = 3 hours: 3.3 × 3 ≈ **10 hours**.

Problem 5. First order and zero order compared

Drug X (first order, t½ = 5 hours) and drug Y (zero order, eliminated at a constant 10 mg per hour) are each present in the body at 100 mg. How much of each remains after 5 and after 10 hours?

Time	Drug X (first order)	Drug Y (zero order)
0 h	100 mg	100 mg
5 h	50 mg (half removed)	50 mg (5 × 10 mg removed)
10 h	25 mg (half of what was left)	0 mg (all removed)

> The first-order drug loses a constant **fraction**; the zero-order drug a constant **amount**. If the dose of Y is doubled to 200 mg, Y takes 20 hours to clear, while X still halves every 5 hours. This is why toxicity appears suddenly when a drug's elimination becomes saturated (A1a.2.7).

AP2.4 Volume of distribution and clearance

Problem 6. Volume of distribution

A 500 mg intravenous bolus gives an initial plasma concentration (by back-extrapolation) of 10 mg/L in a 70 kg man. Find Vd and comment.

> Vd = Dose ÷ C_0 = 500 mg ÷ 10 mg/L = **50 L**, or 50 ÷ 70 ≈ **0.71 L/kg**.
>
> Total body water in an adult man is roughly 60 per cent of body weight, about 42 L in a 70 kg man. A Vd of 50 L is a little larger than this, which suggests that the drug distributes through body water and is also taken up to some extent by tissues. A Vd of several hundred litres would mean strong tissue binding (A1a. 2.4).

Problem 7. Amount needed to reach a target level

Drug Z has a Vd of 2 L/kg. How much must be given intravenously to a 60 kg patient to reach a plasma level of 4 mg/L at once?

> Vd = 2 × 60 = 120 L. Amount = C × Vd = 4 mg/L × 120 L = **480 mg**. This is the loading dose for intravenous use (F = 1).

Problem 8. Clearance from Vd and half-life

A drug has Vd = 40 L and t½ = 7 hours. Find its total clearance.

> CL = 0.693 × Vd ÷ t½ = 0.693 × 40 ÷ 7 = **3.96 L per hour** (about 66 mL per minute).

Problem 9. Renal clearance

At steady state, the urine concentration of a drug is 60 mg/L, the urine flow is 1.5 mL per minute and the plasma concentration is 0.75 mg/L. Find the renal clearance and interpret it.

> $CL_R = (U \times V) \div P = (60 \times 1.5) \div 0.75 =$ **120 mL per minute**.
>
> This is close to a normal adult glomerular filtration rate (about 120 to 125 mL per minute). The simplest reading is that the drug is filtered and neither secreted nor reabsorbed to any net extent. A renal clearance much above the GFR means tubular secretion; much below means reabsorption or plasma protein binding (A1a.2.6).

AP2.5 Area under the curve and bioavailability

Problem 10. AUC by the trapezoidal rule

After a single oral dose, these plasma levels were measured:

Time (h)	0	1	2	4	6	8
Concentration (mg/L)	0	4	6	4	2	1

Find Cmax, Tmax, AUC from 0 to 8 hours, and AUC from 0 to infinity.

> Cmax = **6 mg/L**; Tmax = **2 hours**.
>
Interval (h)	Trapezoid	Area (mg·h/L)
0–1	(0 + 4) ÷ 2 × 1	2
1–2	(4 + 6) ÷ 2 × 1	5
2–4	(6 + 4) ÷ 2 × 2	10
4–6	(4 + 2) ÷ 2 × 2	6
6–8	(2 + 1) ÷ 2 × 2	3
	$AUC_{0–8}$	**26**
>
> The level halves from 6 to 8 hours (2 → 1 mg/L), so the terminal t½ = 2 hours and k = 0.693 ÷ 2 = 0.347 per hour.
>
> AUC from 8 hours to infinity = $C_{last} \div k = 1 \div 0.347 \approx 2.9$ mg·h/L.
>
> $AUC_{0–\infty} = 26 + 2.9 \approx$ **28.9 mg·h/L**. The extrapolated part is about 10 per cent of the total; a bioavailability study aims to keep it small by sampling long enough.

Problem 11. Absolute bioavailability

A 100 mg oral dose gives an AUC of 24 mg·h/L. A 50 mg intravenous dose of the same drug, in the same subjects, gives an AUC of 20 mg·h/L. Find the oral bioavailability.

> $F = (AUC_{oral} \div Dose_{oral}) \div (AUC_{IV} \div Dose_{IV}) = (24 \div 100) \div (20 \div 50) = 0.24 \div 0.40 =$ **0.6, or 60 per cent**.
>
> The doses differ, so each AUC must be divided by its dose before comparison.

Problem 12. Relative bioavailability and bioequivalence

For a test formulation, the geometric mean AUC is 95 mg·h/L; for the reference formulation, 100 mg·h/L. The 90 per cent confidence interval of the test/reference ratio is 88 to 103 per cent. Is the test product bioequivalent for AUC?

> Relative bioavailability = 95 ÷ 100 × 100 = **95 per cent**.
>
> The decision does not rest on the point estimate but on the confidence interval. The interval 88 to 103 per cent lies wholly within the accepted range of 80 to 125 per cent, so the products are **bioequivalent for AUC** (A1c.2). The same test must also be passed for Cmax.

AP2.6 Dosage regimens

Problem 13. Loading dose

The target plasma level of a drug is 10 mg/L. Its Vd is 35 L and its oral bioavailability is 0.7. What oral loading dose is needed?

> LD = Target C × Vd ÷ F = 10 × 35 ÷ 0.7 = **500 mg**.

Problem 14. Maintenance dose

For the same drug, CL = 2 L per hour. What oral dose every 12 hours will keep the average steady-state level at 10 mg/L?

> Dose rate = Css × CL ÷ F = 10 × 2 ÷ 0.7 = 28.6 mg per hour.
>
> Dose every 12 hours = 28.6 × 12 ≈ **343 mg**, in practice 350 mg twice a day.

Problem 15. Time to steady state and accumulation

A drug with t½ = 8 hours is given by intravenous bolus every 8 hours. (a) When is steady state reached? (b) How much higher are the steady-state peak levels than the peak after the first dose?

> a. Steady state takes four to five half-lives: 4 × 8 to 5 × 8 = **32 to 40 hours**, whatever the dose.
> b. Accumulation factor R = 1 ÷ (1 − $e^{-k\tau}$). Here τ = t½, so $e^{-k\tau}$ = ½ and R = 1 ÷ (1 − ½) = **2**. At steady state the peak is twice the first peak.
>
> If the same drug were given every 4 hours (half a half-life), R = 1 ÷ (1 − 0.707) ≈ 3.4. The more often a drug is given in relation to its half-life, the more it accumulates.

AP2.7 Dose conversion between animals and humans

Problem 16. Human dose of a Churna to a rat and a mouse dose

In a study of an Ayurvedic Churna, the human dose is 3 g per day for a 60 kg adult. Find the equivalent dose for a 200 g rat and a 25 g mouse, using body surface area scaling (Km: human 37, rat 6, mouse 3).

> Human dose = 3000 mg ÷ 60 kg = **50 mg/kg**.
>
> Animal dose (mg/kg) = Human dose (mg/kg) × Human Km ÷ Animal Km.

Species	Factor	Dose (mg/kg)	Dose per animal
Rat	37 ÷ 6 = 6.17	50 × 6.17 ≈ **308 mg/kg**	308 × 0.2 ≈ **62 mg** per 200 g rat
Mouse	37 ÷ 3 = 12.3	50 × 12.3 ≈ **617 mg/kg**	617 × 0.025 ≈ **15 mg** per 25 g mouse

This is the conversion most often used in Ayurvedic animal studies [Reagan-Shaw 2008]. Older studies sometimes used a different table (for example, a fixed ratio based on a 200 g rat and a 70 kg man), so the method used must always be stated in the paper and the thesis.

Problem 17. First human dose from a rat NOAEL

In a 90-day study in rats, the no-observed-adverse-effect level of a new extract is 500 mg/kg per day. Find the human equivalent dose and the maximum recommended starting dose for a 60 kg adult.

HED = 500 × (6 ÷ 37) = 500 × 0.162 = **81 mg/kg**, or 81 × 60 = 4.86 g per day for a 60 kg adult.

MRSD = HED ÷ safety factor of 10 = 8.1 mg/kg, or about **486 mg per day** [FDA 2005].

AP2.8 Doses for children

Problem 18. The four methods compared

The adult dose of a drug is 500 mg. A child is 6 years old, weighs 20 kg and is 115 cm tall. Calculate the child’s dose by Young’s, Clark’s and Dilling’s formulae and by body surface area.

BSA (Mosteller) = √(115 × 20 ÷ 3600) = √0.639 ≈ **0.80 m²**. Weight in pounds = 20 × 2.2 = 44 lb.

Method	Formula	Child’s dose
Young	Age ÷ (Age + 12) × Adult dose = 6 ÷ 18 × 500	167 mg
Clark	Weight (lb) ÷ 150 × Adult dose = 44 ÷ 150 × 500	147 mg
Dilling	Age ÷ 20 × Adult dose = 6 ÷ 20 × 500	150 mg
Body surface area	BSA ÷ 1.73 × Adult dose = 0.80 ÷ 1.73 × 500	231 mg

The answers differ by more than 50 per cent. Age-based formulae ignore the child’s actual size; BSA scaling follows metabolic rate more closely and is preferred (A1a.4.3). For a marketed medicine, the dose on the label or in the approved literature, or the physician’s prescription, always takes precedence over any formula.

AP2.9 Classical measures and batch calculations

Problem 19. Classical doses in metric units

Using the AFI equivalents (1 Gunja = 125 mg; 1 Masha = 1 g; 1 Tola = 12 g; 1 Pala = 48 g), convert: (a) a Bhasma dose of 2 Gunja; (b) a Churna dose of 3 Masha; (c) a Kwatha drug quantity of 1 Pala.

a. 2 × 125 mg = **250 mg**. (b) 3 × 1 g = **3 g**. (c) 1 × 48 g = **48 g**.

The conversion values come from the formulary in use (B6.2.4) [R37]. A thesis must state which values it used.

Problem 20. Scaling a classical formula to a batch

Sitopaladi Churna contains Sita (sugar candy), Vamshalochana, Pippali, Ela and Tvak in the proportion 16 : 8 : 4 : 2 : 1 (Sharangadhara Samhita, Madhyama Khanda 6) [R11]. Calculate the quantity of each for a 50 kg batch.

Total parts = 16 + 8 + 4 + 2 + 1 = 31. One part = 50 ÷ 31 = 1.6129 kg.

Ingredient	Parts	Quantity (kg)
Sita	16	25.806
Vamshalochana	8	12.903
Pippali	4	6.452
Ela	2	3.226
Tvak	1	1.613
Total	**31**	**50.000**

Check that the column adds up to the batch size. Here the rounded figures total 50.000 kg exactly. When rounding leaves a difference of a few grams, adjust the largest ingredient and record the adjustment in the MFR.

Problem 21. Yield and loss in a Bhasma

In a Marana process, 500 g of Shuddha raw material was taken. After the prescribed Puta and final grinding, 455 g of Bhasma was obtained. Express the yield and the loss.

Yield = 455 ÷ 500 × 100 = **91 per cent**; loss = **9 per cent** (45 g).

For a Bhasma, the weight can rise or fall during Marana. It may rise because oxygen or sulphur is taken up, or because ash from the Bhavana herbs is added. It may fall because volatile matter is lost, or because material is lost in handling. So the yield should be compared with the range found in earlier validated batches, not with an imagined 100 per cent.

AP2.10 Pharmacy management calculations

Problem 22. Economic order quantity

A pharmacy uses 3600 kg of a herb in a year. Placing one order costs Rs 500. Carrying 1 kg in stock for a year costs Rs 40. Find the EOQ, the number of orders a year, and show that ordering and carrying costs are equal at the EOQ.

> EOQ = √(2DS ÷ H) = √(2 × 3600 × 500 ÷ 40) = √90,000 = **300 kg**.
>
> Orders a year = 3600 ÷ 300 = **12** (one a month).
>
> Annual ordering cost = 12 × Rs 500 = Rs 6,000. Annual carrying cost = average stock × H = (300 ÷ 2) × Rs 40 = Rs 6,000. The two are equal, as Figure B4.5 shows.

Problem 23. Reorder level

A unit uses on average 10 kg of Guggulu a day. The supplier takes 15 days to deliver. The unit keeps a safety stock of 60 kg. At what stock should a new order be placed?

> Reorder level = Average usage × Lead time + Safety stock = 10 × 15 + 60 = **210 kg**.

Problem 24. ABC classification

The annual consumption of eight materials in a small Ayurvedic unit is given below. Classify them by ABC analysis.

Material	Annual use	Unit cost (Rs)	Annual value (Rs)	Cumulative % of value	Class
Kesara (saffron)	2 kg	2,50,000 per kg	5,00,000	46.7	A
Cow’s ghee	600 kg	600 per kg	3,60,000	80.4	A
Honey	400 kg	300 per kg	1,20,000	91.6	B
Sesame oil	200 kg	200 per kg	40,000	95.3	B
Jaggery	500 kg	50 per kg	25,000	97.7	C
Haritaki	300 kg	50 per kg	15,000	99.1	C
Labels	20,000	0.40 each	8,000	99.8	C
Cartons	1,000	2 each	2,000	100.0	C
Total			**10,70,000**		

> Two of eight items (25 per cent) carry about 80 per cent of the value: these are class A and get the tightest control. Note that a C item can still be V (vital) in VED analysis: without labels, no batch can be sold (B4.4.2). The prices here are illustrative only.

Problem 25. Break-even point

A new product line has fixed costs of Rs 6,00,000 a year. Each pack sells for Rs 120 and its variable cost is Rs 70. How many packs must be sold to break even?

> Break-even point = Fixed costs ÷ (Price − Variable cost) = 6,00,000 ÷ (120 − 70) = 6,00,000 ÷ 50 = **12,000 packs** a year.
>
> Each pack sold contributes Rs 50 towards fixed costs. After 12,000 packs, each further pack adds Rs 50 to profit.

AP2.11 Exercises for practice

Answers are given at the end. Work each one in full before looking.

1. A drug has t½ = 2.5 hours. Find k.
2. The plasma level of a drug falls from 64 mg/L to 8 mg/L in 9 hours. Find t½.
3. After an IV dose of 1 g, C_0 is 25 mg/L. Find Vd.
4. A drug has k = 0.1 per hour and Vd = 30 L. Find CL.
5. Oral dose 200 mg, AUC 30 mg·h/L; IV dose 100 mg, AUC 25 mg·h/L. Find F.
6. Target level 5 mg/L; Vd 60 L; F = 0.5. Find the oral loading dose.
7. A human dose is 20 mg/kg. Find the rat dose in mg/kg by BSA scaling.
8. Adult dose 300 mg; child aged 8 years. Find the dose by Young's formula.
9. Convert 4 Gunja and 2 Tola to metric units.
10. Annual demand 1000 units; ordering cost Rs 200; carrying cost Rs 10 per unit per year. Find EOQ.

ANSWERS

1. 0.277 per hour. 2. 3 hours (three halvings: 64 → 32 → 16 → 8). 3. 40 L. 4. 3 L per hour. 5. 0.6 (60 per cent). 6. 600 mg. 7. 20 × 6.17 ≈ 123 mg/kg. 8. 8 ÷ 20 × 300 = 120 mg. 9. 500 mg; 24 g. 10. √40,000 = 200 units.

REFERENCES USED IN THIS APPENDIX

Syllabus reference numbers used: **11, 37**.

- **[R11]** *Sharangadhara Samhita*, Madhyama Khanda 6 (Churna Kalpana).
- **[R37]** *The Ayurvedic Formulary of India* (weights and measures).

Additional verified sources (accessed 16 October 2026)

- **[FDA 2005]** US Food and Drug Administration, Center for Drug Evaluation and Research. *Guidance for Industry: Estimating the Maximum Safe Starting Dose in Initial Clinical Trials for Therapeutics in Adult Healthy Volunteers*. July 2005.
- **[Reagan-Shaw 2008]** Reagan-Shaw S, Nihal M, Ahmad N. Dose translation from animal to human studies revisited. *FASEB J*. 2008;22(3):659–661.
- **[Mosteller 1987]** Mosteller RD. Simplified calculation of body-surface area. *N Engl J Med*. 1987;317(17):1098.

APPENDIX 3

APPENDIX 3 · MODEL ANSWERS TO FREQUENTLY ASKED LONG QUESTIONS

Why this appendix is here. Each chapter ends with an answer frame: the parts of a long answer and the marks each part may carry. This appendix goes one step further. It writes out, in full, the long answers to the nine questions set most often in the seven papers examined. Each is the length a well-prepared scholar can write in the 30 to 35 minutes that a 20-mark question allows. Read one before you attempt the matching question in a model paper, then write your own and compare.

1. **Start with a definition** in one or two lines. It tells the examiner at once that you know what the question is about.
2. **Use the words of the question as headings**, in the order the question asks. If it says “phases and methods”, write “Phases” and then “Methods”.
3. **Put lists into tables.** A table of five rows says more, and is marked faster, than a page of prose.
4. **Draw at least one labelled diagram.** The figure to copy is named in each answer. Draw it neatly, in pencil if you wish, with a title.
5. **Add the Ayurveda correlation**, and say plainly which part is established and which is proposed. This is expected of a Rasashastra and Bhaishajya Kalpana scholar.
6. **State the law as it stands today.** Where the law changed in 2024–2026, say so in one line.
7. **End with a short conclusion** that answers the question in two sentences.

These answers are the author’s. They show one good way to answer, not the only way. Examiners mark by their own judgement.

AP3.1 Principles of pharmacology, pharmacodynamics and pharmacokinetics

Set in: September 2018 (with two routes of administration); September 2020; August 2021 (to be illustrated with an Ayurvedic formulation). **Chapter:** A1(a).

Plan (2 minutes). Definitions → general principles (table) → pharmacokinetics (ADME, kinetics, curve) → pharmacodynamics (mechanisms, receptors, dose–response) → Ayurvedic illustration → conclusion. Two diagrams: the concentration–time curve and the log dose–response curve.

1. Definitions

Pharmacology is the science of drugs: their sources, properties, actions on living systems, fate in the body, and uses. A **drug**, as defined by the WHO, is any substance or product used, or intended to be used, to modify or explore physiological systems or pathological states for the benefit of the recipient.

Pharmacokinetics (PK) is what the body does to the drug: absorption, distribution, metabolism and excretion (ADME). **Pharmacodynamics** (PD) is what the drug does to the body: its actions and how it produces them. PK decides how much drug reaches the target and for how long; PD decides what happens there (Figure A1a.1).

2. General principles of pharmacology

No.	Principle
1	A drug does not create a new function; it only increases or decreases an existing one
2	Most drugs act by binding to a specific target: receptor, enzyme, ion channel or transporter
3	The effect depends on the concentration at the site of action, which depends on dose and PK
4	The effect rises with dose up to a ceiling; beyond it only toxicity rises
5	No drug has a single effect; selectivity is relative
6	Individuals differ in response (age, weight, genes, disease, other drugs)
7	Every effect has a time course: onset, peak, duration
8	Drugs interact with other drugs and with food
9	Benefit must outweigh risk

3. Pharmacokinetics

(a) Absorption. Most drugs cross membranes by passive diffusion of the **un-ionised, lipid-soluble** form. Weak acids are less ionised in acid (stomach), weak bases in alkali. Other routes across membranes: filtration, carrier transport, active transport, endocytosis. Drugs absorbed from the gut pass through the liver first; **first-pass metabolism** reduces the fraction reaching the circulation, the **bioavailability** (F).

(b) Distribution. Measured by the **apparent volume of distribution**: Vd = amount in the body ÷ plasma concentration. Compared with plasma (about 3 L), extracellular fluid (about 14 L) and total body water (about 42 L in a 70 kg adult), it shows where the drug has gone. Only **free** drug acts and is eliminated; albumin binds acidic drugs. Barriers: blood–brain barrier, placenta.

(c) Metabolism. Mainly in the liver. **Phase I** (oxidation by cytochrome P450, reduction, hydrolysis) adds or exposes a reactive group. **Phase II** (glucuronidation, sulphation, acetylation) conjugates it, making it water-soluble. Enzyme **inducers** lower, and **inhibitors** raise, the levels of other drugs.

(d) Excretion. Mainly renal: glomerular filtration of free drug, active tubular secretion, passive reabsorption of lipid-soluble drug. Alkaline urine speeds the excretion of weak acids. Also bile, lungs, milk.

(e) Kinetics.

	First order	Zero order
Removed per unit time	Constant **fraction**	Constant **amount**
Half-life	Constant	Rises with dose
Examples	Most drugs	Ethanol; phenytoin at high dose

Key relations: t½ = 0.693 × Vd ÷ CL. Steady state, and near-complete elimination, each take 4 to 5 half-lives. Loading dose = target C × Vd ÷ F; maintenance dose rate = target C × CL ÷ F.

(f) The plasma concentration–time curve. *Draw Figure A1a.12*, marking Cmax, Tmax, AUC, minimum effective and minimum toxic concentrations, therapeutic window, onset and duration.

4. Pharmacodynamics

(a) Types of action: stimulation, depression, irritation, replacement, cytotoxic action.

(b) Mechanisms (Figure A1a.14):

Target	Example
Receptor	Adrenaline on β receptors
Enzyme	Aspirin inhibits cyclo-oxygenase
Ion channel	Local anaesthetics block sodium channels
Transporter	Digoxin inhibits Na^+/K^+-ATPase
Physical or chemical action	Antacids neutralise acid; osmotic purgatives; chelators

(c) Receptors. Affinity decides binding; **intrinsic activity** decides response. A full agonist has intrinsic activity 1, a partial agonist between 0 and 1, an antagonist 0, an inverse agonist less than 0. Four receptor families: ligand-gated ion channels (milliseconds), G-protein-coupled receptors (seconds), kinase-linked receptors (minutes to hours), nuclear receptors (hours to days).

(d) Dose–response. *Draw Figure A1a.18* (the sigmoid log dose–response curve). **Potency** is the position of the curve on the dose axis; **efficacy** is the height of its plateau. Therapeutic index = TD50 ÷ ED50; drugs with a narrow index (digoxin, lithium, warfarin) need careful dosing.

5. Illustration with an Ayurvedic formulation: Haridra with Maricha

PK: curcumin from Haridra is poorly soluble and is rapidly glucuronidated in the gut wall and liver, so little reaches the blood. Piperine of Maricha inhibits glucuronidation. In healthy volunteers, piperine raised the bioavailability of curcumin about 20-fold (Shoba and colleagues, 1998). This is a measured basis for one part of the classical idea of Trikatu as **Yogavahi**.

PD: the same idea of dose and response appears in Charaka's teaching that a poison becomes a medicine by right use, and a medicine a poison by wrong use. Rasashastra lives by it: Vatsanabha and Rasaushadhi are given in small, fixed doses after processing.

Boundary: that piperine raises blood levels of curcumin is **established**; that this alone gives better clinical results, or that every Yogavahi works this way, is **not established**.

6. Conclusion

Pharmacokinetics decides how much drug reaches its target and for how long; pharmacodynamics decides what it does there. Together they decide the dose, the dosing interval and the margin of safety, for a classical formulation as much as for a modern drug.

Why this answer scores well. It follows the question's order, puts the general principles in a table, gives two diagrams, uses one worked Ayurvedic illustration with evidence, and marks the evidence boundary. **Common mistakes:** writing only ADME without kinetics; confusing potency with efficacy; giving Ayurvedic parallels as if they were proven equivalents.

AP3.2 Phases and methods of clinical research, and ethics in human research

Set in: September 2018 (with preclinical evaluation); August 2021; August 2022 ("for newer formulation"). **Chapter:** A2.

Plan. Definition → pathway in India → phases (table) → methods (classification and RCT) → special points for an Ayurvedic formulation → ethics (milestones, principles, consent, committee) → conclusion. Diagrams: the phases (Figure A2.2) and the RCT flow (Figure A2.6).

1. Introduction

Clinical research is research on human beings to find out the safety and effectiveness of a drug, device or procedure. A new drug or formulation reaches patients only after preclinical studies and then clinical trials in phases. In India, trials of new drugs are governed by the **New Drugs and Clinical Trials Rules, 2019**: permission from the Central Licensing Authority (CDSCO), approval of a registered ethics committee, registration in the **Clinical Trials Registry – India (CTRI)** before the first participant is enrolled, and free medical management and compensation for trial-related injury.

2. Phases of clinical research

Draw Figure A2.2.

Phase	Purpose	Participants	Typical size
0	Microdose; early pharmacokinetics or target binding	Healthy volunteers or patients	Very few
I	Safety, tolerability, pharmacokinetics; maximum tolerated dose	Healthy volunteers (patients for toxic drugs)	20–100
II	Proof of efficacy (IIa) and dose finding (IIb)	Patients	Up to a few hundred
III	Confirmation of efficacy and safety against placebo or standard; basis of approval	Patients, often multicentre	Hundreds to thousands
IV	Post-marketing surveillance: rare and long-term effects, new uses	Patients in practice	Thousands or more

3. Methods of clinical research

Classification of designs (Figure A2.4): *observational* (case report and case series, cross-sectional, case–control, cohort) and *experimental* (non-randomised trials and randomised controlled trials). Systematic reviews and meta-analyses of good RCTs stand at the top of the evidence hierarchy.

The randomised controlled trial, the standard for proving efficacy, has these elements:

Element	Purpose
Clear hypothesis and pre-specified primary outcome	Avoids searching for a result
Randomisation, with allocation concealment	Makes groups comparable; prevents selection bias
Blinding (single, double, triple)	Prevents bias in treatment and assessment
Control: placebo or standard treatment	Gives the comparison
Sample size calculation	Enough power to detect a real difference
Intention-to-treat analysis	Keeps the benefit of randomisation
Reporting by CONSORT; registration in CTRI	Transparency

Draw Figure A2.6. Other designs: crossover, factorial, non-inferiority, pragmatic.

4. Clinical research on a new Ayurvedic formulation

1. Standardise the product: authenticated raw materials, GMP manufacture, fingerprint and markers, heavy metals and microbial limits, stability. The same batch, or batches shown to be equivalent, throughout.
2. Safety first: published toxicity data or preclinical studies, especially for Rasaushadhi.
3. Choose the design: pragmatic trials suit whole-system treatment; Prakriti-stratified randomisation can be used.
4. A placebo matched in colour, taste and smell; or an active comparator.
5. Valid outcome measures: both modern and Ayurvedic assessment criteria.
6. Follow GCP for ASU drugs (2013) and the ICMR guidelines.

5. Ethics involved in human research

Milestones (Figure A2.9): Nuremberg Code (1947); Declaration of Helsinki (1964, latest revision October 2024); Belmont Report (1979); ICH GCP E6 (R3, January 2025); ICMR National Ethical Guidelines (2017).

Principles (Belmont): respect for persons (autonomy, consent), beneficence (do good, avoid harm), justice (fair selection and sharing of benefits and burdens).

Informed consent is a process, not a signature: information in the participant's language; understanding; voluntary agreement; the right to withdraw at any time without loss of care; audio-visual recording in specified trials.

Ethics committee: 7 to 15 members, including medical and non-medical members, a legal expert, a social scientist or representative of an NGO, a lay person and a woman member, at least half from outside the institution. It approves the protocol, reviews progress and adverse events, and protects participants.

6. Conclusion

Clinical research moves step by step from a few volunteers to thousands of patients, and from safety to efficacy to long-term surveillance. For an Ayurvedic formulation the same rules apply, with two additions: a standardised product, and a design that respects the system. Ethics is not a final step but a condition of every step.

Why this answer scores well. It answers all three parts named in the question, uses tables for phases and RCT elements, adds the Indian rules and the Ayurvedic special points. **Common mistakes:** omitting Phase 0 and Phase IV; writing ethics as a single paragraph without consent and committee; forgetting CTRI registration.

AP3.3 Heavy metal toxicity and chelation therapy

Set in: September 2020 (“heavy metal toxicity and chelation therapy”); January 2024 (“define toxicity and explain toxicity of heavy metals”). **Chapter:** A4.

Plan. Definitions → general mechanisms → lead, mercury, arsenic, cadmium (one table) → chelation: principle, ideal agent, agents (table), limits → heavy metals in Ayurvedic medicines → classical view → conclusion. Diagrams: lead and haem synthesis (Figure A4.3) or the chelate claw (Figure A4.7).

1. Definitions

Toxicity is the capacity of a substance to cause harm to a living organism; the harm depends on dose, route, duration and chemical form. **Heavy metals** is a loose term for metals and metalloids of high density that are toxic at low levels; toxicologists prefer “toxic metals”. The main ones are lead, mercury, arsenic and cadmium.

2. General mechanisms (Figure A4.1)

1. Binding to **sulphhydryl (–SH) groups** of enzymes and proteins, inhibiting them.
2. **Mimicry** of essential elements (lead for calcium, cadmium for zinc).
3. **Oxidative stress** and damage to membranes and DNA.
4. **Accumulation** in bone, kidney and brain, with slow release.

The **chemical form decides toxicity**: methylmercury is absorbed more than 95 per cent from the gut; mercuric chloride 7 to 15 per cent; cinnabar (HgS) only about 0.2 per cent (Liu and colleagues, 2008).

3. Toxicity of the main metals

	Lead	Mercury	Arsenic	Cadmium
Main sources	Paint, batteries, contaminated products	Vapour; inorganic salts; methylmercury in fish	Groundwater; pesticides	Smoking; industry; rice from polluted soil
Key mechanism	Blocks ALA dehydratase and ferrochelatase (haem synthesis)	Binds –SH; vapour enters brain	As III binds lipoic acid; blocks cellular respiration	Accumulates in kidney tubules
Acute	Colic, encephalopathy	Vapour: pneumonitis; salts: gut corrosion, kidney failure	Rice-water diarrhoea, shock, QT prolongation	Pneumonitis (fumes)
Chronic	Anaemia with basophilic stippling; Burton’s line; wrist drop; learning deficit in children	Erethism, tremor, gingivitis; kidney damage; Minamata disease (methylmercury)	Raindrop pigmentation; palmoplantar keratosis; Mees’ lines; neuropathy; cancers	Kidney damage; bone pain and fractures (itai-itai)

	Lead	Mercury	Arsenic	Cadmium
Note	No safe blood level; CDC reference value 3.5 µg/dL (2021)	BAL avoided in organic and elemental poisoning	WHO drinking-water guideline 10 µg/L	No useful chelation

Draw Figure A4.3.

4. Chelation therapy

Principle. A chelating agent (Greek *chele*, claw) has two or more donor atoms that grip a metal ion in a ring. It binds the toxic metal more strongly than the body's ligands, pulls it off its target, and forms a stable, water-soluble complex that is excreted, mainly in urine. *Draw Figure A4.7.*

Ideal chelator: greater affinity for the toxic metal than body ligands; low affinity for essential metals; reaches the site of storage; forms a stable, non-toxic, excretable complex; effective by mouth; low toxicity. No agent has all of these.

Agent	Route	Main uses	Caution
Dimercaprol (BAL)	Deep IM	Arsenic; acute inorganic mercury; lead encephalopathy (with EDTA)	Not for methylmercury, elemental mercury, iron, cadmium
Succimer (DMSA)	Oral	Lead (children), arsenic, mercury	Mild side effects; usual first choice for moderate lead poisoning
Unithiol (DMPS)	Oral or IV	Mercury, arsenic	Allergic reactions
Calcium disodium EDTA	IV	Lead	**Never disodium EDTA** (fatal hypocalcaemia)
Penicillamine	Oral	Copper (Wilson's disease); lead (second line)	Many adverse effects
Deferoxamine	IV, SC	Iron	Oxygen donor chelator for "hard" metals

Limits. First remove the source. Chelation causes redistribution and rebound, removes essential metals, and does not reverse established damage. Chelation for unproven uses is dangerous; the TACT2 trial (2024) found no cardiovascular benefit of EDTA in diabetic patients after myocardial infarction.

5. Heavy metals and Ayurvedic medicines

Studies of products sold in the United States found detectable lead, mercury or arsenic in about one in five (Saper and colleagues, 2004 and 2008), with a higher proportion in Rasashastra products. India made heavy-metal testing mandatory for the export of purely herbal ASU medicines from 1 January 2006 (lead 10, mercury 1, arsenic 3 ppm). For Rasaushadhi, in which metals are intended ingredients, the questions are chemical form, dose, duration and measured safety.

Classical view. The texts recognise Dhatu Visha (mineral poisons such as Haritala), Dushi Visha and Gara Visha, and prescribe Shodhana, Marana and Bhasma Pariksha so that metals are used only in processed form, in small doses, for limited periods. *Boundary:* that processing changes the chemical form

is **established** for several Bhasmas; that each processed product is safe in long-term human use is **not established** and needs measured data.

6. Conclusion

Metal toxicity depends on the metal, its chemical form, the dose and the duration. Chelation helps in acute and significant poisoning when the right agent is chosen, but removal of the source comes first. For Rasashastra, the answer to toxicity is proper processing, quality control and pharmacovigilance.

Why this answer scores well. One comparative table covers four metals; chelation is explained from principle to agents to limits; the Ayurvedic part is balanced. **Common mistakes:** listing features without mechanisms; recommending BAL for every metal; confusing $CaNa_2EDTA$ with Na_2EDTA.

AP3.4 Pharmacovigilance of Ayurvedic drugs and its national programme

Set in: July 2018 (“pharmacovigilance of Ayurvedic drugs and its national program”); March 2026 (with adverse drug reactions and drug toxicity). **Chapters:** A6; A1(c).

Plan. Definition and scope → need → ADR terms and types (table) → the process (detect, assess, signal, act) → international and national scenario → Ayurvedic drugs: challenges, national programme for ASU&H drugs, role of the practitioner → conclusion. Diagrams: the pharmacovigilance cycle (Figure A6.3) and the three-tier ASU&H network (Figure A6.8).

1. Definition and scope

Pharmacovigilance (WHO, 2002) is the science and activities relating to the detection, assessment, understanding and prevention of adverse effects or any other drug-related problem. Its scope covers adverse reactions, medication errors, substandard and falsified products, lack of efficacy, misuse, abuse and interactions, and it includes herbal and traditional medicines.

An **adverse drug reaction** (WHO) is a response to a drug that is noxious and unintended, and occurs at doses normally used in humans. A **serious** reaction is one that causes death, threatens life, needs or prolongs hospital stay, causes disability or a congenital anomaly; “serious” describes the outcome, “severe” the intensity.

2. Why it is needed

Clinical trials are too small and too short to detect rare reactions. By the **rule of three**, about 30,000 patients must be observed to have a 95 per cent chance of seeing one case of a reaction that occurs in 1 in 10,000. Trials also exclude children, the elderly, pregnant women and patients on many drugs. So the full safety profile is learned only after marketing. The thalidomide disaster (1961) led to the WHO Programme for International Drug Monitoring (1968).

3. Types of adverse reaction

Type	Name	Example
A	Augmented: dose-related, predictable	Bleeding with warfarin
B	Bizarre: not dose-related, unpredictable	Anaphylaxis to penicillin
C	Chronic: cumulative dose	Adrenal suppression with steroids
D	Delayed	Teratogenicity; cancer
E	End of use: withdrawal	Rebound hypertension after clonidine
F	Failure of therapy	Contraceptive failure with rifampicin

4. The pharmacovigilance process

Draw Figure A6.3.

1. **Detection and reporting.** Spontaneous reporting is the main method. A valid report needs only four items: an identifiable patient, a reporter, a suspected drug and a suspected reaction. Suspicion is enough.
2. **Assessment.** Causality by the **WHO-UMC** system (certain, probable, possible, unlikely, conditional, unassessable) or the **Naranjo** scale (9 or more definite; 5–8 probable; 1–4 possible; 0 or less doubtful). Severity by the Hartwig scale; preventability by the Schumock and Thornton criteria.
3. **Signal detection.** A signal is information suggesting a new causal association. Disproportionality measures such as the proportional reporting ratio are used to screen databases.
4. **Action and communication.** Label changes, warnings, restriction or withdrawal, and information to doctors and patients.

5. International and national scenario

- **WHO Programme for International Drug Monitoring** (1968), run by the Uppsala Monitoring Centre; global database VigiBase. India joined in 1998. WHO guidelines on safety monitoring of herbal medicines were issued in 2004.
- **Pharmacovigilance Programme of India (PvPI)**, launched July 2010; National Coordination Centre at the **Indian Pharmacopoeia Commission**, Ghaziabad, since April 2011; adverse drug reaction monitoring centres in medical colleges and hospitals.

6. Pharmacovigilance of Ayurvedic drugs

Why needed. Wide use, often self-medication; multi-ingredient products; Rasaushadhi containing metals; adulteration and contamination; concomitant use with modern drugs; the mistaken belief that “natural is safe”.

Challenges and answers.

Challenge	Answer
Under-reporting, fear of harming Ayurveda's image	Training; simple forms; online reporting
Same product name, many makers	Record manufacturer, licence and batch; keep a sample
Multi-ingredient products	Analyse the retained sample
Contamination versus intrinsic toxicity	Test the suspected batch
Intended effects of Shodhana therapy	Distinguish therapeutic effects from Vyapat
Concomitant modern drugs	Ask every patient about every product

The national programme. *Draw Figure A6.8.* The **Central Sector Scheme for Pharmacovigilance of ASU&H drugs** has been implemented by the Ministry of Ayush since December 2017 and became functional in August 2018. It has three tiers: the **National Pharmacovigilance Coordination Centre** at the All India Institute of Ayurveda, New Delhi; **five intermediary centres** at national Ayush institutes; and **peripheral centres** at Ayush colleges (74 as reported in 2020). It has two mandates: reporting of suspected adverse events, and monitoring of **misleading advertisements**, which are reported to the state licensing authorities. In May 2025 the Ministry launched the **AyushSuraksha** portal for both.

Role of the practitioner. Suspect; manage the patient; document the product name, manufacturer and batch; keep the sample; report to the nearest centre; follow up.

7. Conclusion

Pharmacovigilance completes the safety evaluation that trials begin. For Ayurvedic drugs it protects patients and also protects the system: a reaction reported and investigated builds confidence, while one hidden damages it.

Why this answer scores well. It defines, justifies with the rule of three, shows the process as a cycle, and gives the national ASU&H structure with dates. **Common mistakes:** describing only PvPI and not the Ayush programme; leaving out the advertisement-monitoring mandate; claiming that Ayurvedic drugs have no adverse effects.

AP3.5 Preclinical evaluation of drugs and its importance

Set in: September 2018 (with clinical evaluation); January 2024; March 2026 (as a study design for Swarna Bhasma). **Chapter:** A1(d).

Plan. Definition → place in drug development → components (table) → experimental methods (bioassay, in vitro, in vivo design) → toxicity studies (table of OECD guidelines) → Ayurvedic preparation → animal ethics → importance → conclusion. Diagrams: the drug development pipeline (Figure A1d.1) and the steps of in vivo design (Figure A1d.6).

1. Definition

Preclinical evaluation is the testing of a new drug in the laboratory and in animals, before it is given to human beings, to establish its pharmacological activity, pharmacokinetics and safety, and to choose a safe starting dose for human trials. *Draw Figure A1d.1.*

2. Components

Area	What is studied
Primary pharmacodynamics	The intended effect in models of the disease; effective dose range
Secondary pharmacodynamics	Effects on other targets
Safety pharmacology	Central nervous, cardiovascular and respiratory systems (ICH S7A); QT prolongation (ICH S7B)
Pharmacokinetics	ADME, in two species
Toxicology	Acute, repeated-dose, genotoxicity, reproductive toxicity, carcinogenicity, local tolerance
Test material	Characterisation, stability and quality

Regulatory toxicology is done under **Good Laboratory Practice**.

3. Experimental methods

Bioassay compares the potency of a test preparation with a standard in a living system; the log dose–response lines must be parallel; the four-point (2 + 2) design is the most used.

In vitro methods: enzyme and receptor assays, antimicrobial tests (MIC), antioxidant assays, isolated tissues in an organ bath, and cell lines (for example, HepG2, RAW 264.7) with the MTT assay for cytotoxicity. An in vitro result is a hypothesis, not a proof of effect in the body.

In vivo design (*draw Figure A1d.6*): a clear hypothesis; the right species and model; groups (normal control, disease control, positive control with a standard drug, test groups at two or three doses); dose converted from the human dose by body surface area; randomisation; blinding of the assessor; sample size; suitable statistics (for example, ANOVA with a post-hoc test); reporting by ARRIVE 2.0.

4. Toxicity studies

Study	OECD test guideline
Acute oral toxicity	420, 423, 425 (the classical LD50 test, 401, was deleted in 2002)
Repeated dose, 28 days	407
Repeated dose, 90 days	408
Chronic toxicity	452
Reproductive and developmental	414, 416, 421
Genotoxicity	471 (Ames), 473, 474
Carcinogenicity	451

The end-point is the **no-observed-adverse-effect level** (NOAEL), from which the human starting dose is calculated (NOAEL → human equivalent dose by body surface area → divided by a safety factor of 10).

5. Preclinical evaluation of an Ayurvedic preparation

The product must be the one used in practice, made by the classical method, and characterised (fingerprint, markers, and for Bhasma the phase, particle size and metal content). For herbo-mineral drugs, add metal estimation in organs, kidney and liver end-points, and recovery groups. *Example:* the immunomodulatory effect of Swarna Bhasma can be tested in rats immunosuppressed with cyclophosphamide, measuring humoral immunity (antibody titre to sheep red blood cells), cell-mediated immunity (delayed-type hypersensitivity), phagocytosis (carbon clearance), blood counts and lymphoid organ weights, and gold in the organs.

6. Animal ethics

The **3Rs** of Russell and Burch (1959): **Replacement**, **Reduction**, **Refinement**. In India, experiments on animals are regulated under the Prevention of Cruelty to Animals Act, 1960 and the Breeding of and Experiments on Animals (Control and Supervision) Rules, 1998, by the **Committee for Control and Supervision of Experiments on Animals** (CCSEA, formerly CPCSEA). Every institution has an **Institutional Animal Ethics Committee** that approves each protocol.

7. Importance

1. Shows whether the drug is worth testing in humans at all.
2. Identifies the target organs of toxicity and whether damage is reversible.
3. Gives the NOAEL and so the safe starting dose for Phase I.
4. Links dose to exposure through pharmacokinetics.
5. Is required by law before clinical trials.
6. For Ayurvedic drugs, especially Rasaushadhi, provides the safety data that the classical texts did not record in measurable form.

8. Conclusion

Preclinical evaluation is the gate between the laboratory and the patient. It must be well designed, ethically done and honestly reported. For an Ayurvedic preparation, its value depends on testing the real, characterised product at a properly converted dose.

Why this answer scores well. Components and toxicity guidelines are in tables, the in vivo design is complete with controls, and both the Ayurvedic adaptation and the law on animal ethics are included. **Common mistakes:** naming CPCSEA instead of CCSEA; omitting the positive control; converting human to animal doses by body weight alone.

AP3.6 Good Manufacturing Practices related to the Ayurvedic drug industry

Set in: September 2018 (with the Drugs and Cosmetics Act); September 2020 (with management of pharmacy); August 2021; August 2022; January 2024. The most frequently set long question in the paper. **Chapter:** B4.

Plan. Definition → why GMP for ASU drugs → principles (table) → Schedule T: Part I heads (table) and Part II → documentation → GMP in the Rasashala → the 2026 revision → certifications → classical correlation → conclusion. Diagram: building blocks of GMP (Figure B4.6) or the one-way flow of materials (Figure B4.1).

1. Definition

Good Manufacturing Practice is that part of quality management which ensures that products are consistently produced and controlled to the quality standards appropriate to their intended use (WHO). Its central idea: **quality cannot be tested into a product; it must be built into every step of its making.**

2. Why GMP is essential for Ayurvedic drugs

1. Herbal raw materials vary naturally and may be adulterated or contaminated.
2. Classical processes (Paka, Puta, Sandhana, Kupipaka) depend on skill; without written procedures and records, batches vary.
3. Rasaushadhi involve toxic metals; errors harm patients and workers.
4. Studies have found contamination and excess metals in some marketed products.
5. GMP is a **legal condition** of the manufacturing licence and of export certification.

3. Principles of GMP

Draw Figure B4.6.

Principle	Meaning
Write procedures	SOPs for every operation
Follow them	Work exactly as written
Document	“If it is not written, it was not done”
Validate	Prove that processes and cleaning work consistently
Design premises and equipment	Suitable, easy to clean, one-way flow
Maintain and calibrate	Preventive maintenance
Train	Qualified, trained people
Keep clean	Sanitation and personal hygiene
Control quality	Test materials and products; release only approved material
Audit and improve	Self-inspection, complaints, recall, CAPA

4. Schedule T of the Drugs Rules, 1945

A licence to manufacture ASU drugs is granted only if the unit complies with **Schedule T**. Before its 2026 revision, Schedule T had two parts.

Part I: general requirements.

Head	Requirement in brief
A. Location and surroundings	Away from drains, refuse and sources of contamination
B. Buildings	Suitable; protected from insects, rodents, dust; smooth washable surfaces; light and ventilation
C. Water supply	Potable water
D. Disposal of waste	Without contamination; pollution control norms

Head	Requirement in brief
E. Container cleaning	Facilities to clean containers
F. Stores	Separate raw material, packing and finished goods stores; quarantine and approval
G. Working space	Adequate, to avoid mix-ups
H. Health, clothing, sanitation, hygiene of workers	Medical examination; uniforms; no eating or smoking in work areas
I. Medical services	First aid; medical examination
J. Machinery and equipment	Suitable, clean, maintained
K. Batch manufacturing record	Materials, quantities, steps, in-process checks, yield, signatures
L. Distribution records	To allow recall
M. Record of market complaints	Including adverse reactions
N. Quality control	In-house section or an approved laboratory

Part II lists the machinery and minimum manufacturing area for each category (Churna, Vati, Avaleha, Asava-Arishta, Sneha, Bhasma, Kupipakva and others), the equipment for the QC section, and **supplementary guidelines for the manufacture of Rasaushadhi**.

5. Documentation

Master formula record; batch manufacturing record; SOPs; specifications; analytical records; equipment logs; stock and distribution records; complaint and recall records; training records; deviation and change control records.

6. GMP in the Rasashala

Segregated, ventilated areas with exhaust and emission control for Shodhana, Marana and Kupipaka; locked stores and double-signed registers for Rasa and Visha Dravya; a record of every Puta (temperature, time, weight); Bhasma Pariksha results for each batch; dedicated equipment; worker health surveillance; testing for metals; cautionary labelling for Schedule E(1) ingredients; hazardous waste disposal.

7. The 2026 revision

The **Drugs (Eleventh Amendment) Rules, 2026**, published in July 2026, substitute a new Schedule T covering Ayurveda, Siddha, Sowa-Rigpa and Unani medicines. Existing units must comply by **31 July 2029**. Reported changes include detailed controls on premises and cross-contamination, equipment qualification, annual medical examination of staff, additional safeguards for Rasaushadhi (segregation, emission control, health surveillance, controlled release), risk-based inspection, and manufacturing licences valid perpetually subject to compliance.

8. Certifications beyond Schedule T

WHO-GMP and the Certificate of Pharmaceutical Product, issued by CDSCO after joint inspection, for export. The voluntary **AYUSH Standard Mark** (based on Schedule T) and **AYUSH Premium Mark** (based on WHO GMP for herbal medicines), operated by the Quality Council of India.

9. Ayurveda correlation

The Rasa texts describe the site, layout and equipment of the **Rasashala** and the qualities of those who work in it; Charaka directs that drugs be stored in clean, closed containers protected from smoke, rain, wind and dust. *Boundary:* the classical rules for premises and storage are **established** parallels; what they lacked, and GMP adds, is written batch records, validation and independent quality control.

10. Conclusion

GMP turns the skill of a few into the reliability of a system. For the Ayurvedic industry it is a legal duty under Schedule T, a condition of export, and the means by which a classical formulation gives the same result in every batch.

Why this answer scores well. Schedule T Part I is given head by head in a table; the Rasashala and the 2026 revision show current, subject-specific knowledge. **Common mistakes:** writing “Schedule M” for ASU drugs (Schedule M is for modern drugs); listing Schedule T heads without any explanation; ignoring the 2026 changes.

AP3.7 Laws governing Ayurvedic drugs and their necessity

Set in: August 2022 (“in detail with appropriate examples”); January 2024 (“with its necessity”). Related: the Drugs and Cosmetics Act alone (July and September 2018). **Chapter:** B8.

Plan. Introduction → necessity (table) → the framework (diagram and table) → each law briefly, the Drugs and Cosmetics Act in most detail → changes of 2024–2026 → conclusion. Diagram: Figure B8.1.

1. Introduction

Every Ayurvedic medicine made, sold, advertised or exported in India is governed by law. The laws define what an Ayurvedic drug is, who may make it and how, what its label must say, what may be claimed for it, and what remedy a consumer has if it is defective.

2. Necessity of laws

Need	What the law does	Example
Quality	Standards, GMP, licensing, testing; punishes adulterated, misbranded and spurious drugs	Schedule T; API standards
Safety	Cautions for poisonous ingredients; shelf life; evidence of safety	Schedule E(1); Rule 161B
Honest information	Labelling; ban on magic-remedy claims	Rule 161; DMR Act
Consumer protection	Remedies for defective goods; misleading advertisements	Consumer Protection Act, 2019
Control of dangerous substances	Narcotic ingredients	NDPS Act (Ahiphena); State excise law (Bhanga leaves)
Workers' safety	Health and safety in factories	OSH Code, 2020
Professional standards	Who may dispense	Pharmacy Act, 1948
Trade	Credibility for export	COPP; heavy-metal testing

3. The legal framework

Draw Figure B8.1.

(i) Drugs and Cosmetics Act, 1940 and Rules, 1945. ASU drugs were brought under the Act in **1964** (Chapter IV-A, Sections 33B to 33O).

- *Definitions:* an **Ayurvedic drug** is one made exclusively according to the formulae in the authoritative books listed in the First Schedule; a **patent or proprietary** medicine uses only ingredients from those formulae but is not itself a book formulation, and may not be given by the parenteral route.
- *Offences:* misbranded (33E), adulterated (33EE), spurious (33EEA) drugs.
- *Bodies:* the ASU Drugs Technical Advisory Board (33C) and the ASU Drugs Consultative Committee (33D).
- *Licensing:* application in **Form 24-D**, licence in **Form 25-D**, granted by the **State Licensing Authority**; loan licence in Form 25-E; evidence of safety and effectiveness under Rule 158-B.
- *GMP:* **Schedule T**. *Labelling:* **Rule 161**; export labels Rule 161A; **shelf life Rule 161B**. *Poisonous ingredients:* **Schedule E(1)**, with the caution "to be taken under medical supervision".

(ii) Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954. Prohibits advertisements claiming to cure the diseases and conditions in its Schedule (54 entries), and misleading advertisements. First conviction: up to six months' imprisonment, or fine, or both. *Example:* an advertisement claiming that a Churna cures diabetes is an offence.

(iii) Prevention of Food Adulteration Act, 1954. Repealed with effect from 5 August 2011; replaced by the Food Safety and Standards Act.

(iv) Food Safety and Standards Act, 2006. Through FSSAI, it governs health supplements and nutraceuticals (Section 22) and **Ayurveda Aahara** (regulations of 2022). *Example:* an Ayurvedic herbal food product sold as Ayurveda Aahara is licensed under this Act, not the Drugs Act.

(v) Narcotic Drugs and Psychotropic Substances Act, 1985. Controls opium, cannabis and their products; licences and records for their use in ASU manufacture. *Example:* Ahiphena-containing formulations need NDPS licences as well as the ASU licence. Bhanga leaves without the flowering tops are excluded from the Act's definition of cannabis and are controlled by State excise law.

(vi) Factories Act, 1948 (now the OSH Code, 2020) and Pharmacy Act, 1948. The **Occupational Safety, Health and Working Conditions Code, 2020**, in force from 21 November 2025, replaced the Factories Act: safety, health and welfare of workers, important in the Rasashala. The Pharmacy Act regulates the profession of pharmacy; only a registered pharmacist may dispense on prescription (Section 42).

(vii) Consumer Protection Act, 2019 (replacing the 1986 Act). Consumer rights; three-tier commissions (District, up to Rs 50 lakh); product liability; the Central Consumer Protection Authority may penalise misleading advertisements.

Related laws: Patents Act, 1970 and Biological Diversity Act, 2002 (Chapter B7).

4. Recent changes to state

Rule 170 on advertisement of ASU drugs was omitted in 2024, and the Supreme Court's stay on the omission was vacated in August 2025. In 2026, a new Schedule T was notified (compliance by 31 July 2029), Schedule E(1) was substituted (any ingredient containing arsenic or mercury is included), and manufacturing licences were made perpetual subject to compliance.

5. Conclusion

The laws exist to make sure that an Ayurvedic medicine is what it claims to be, is made properly, is labelled honestly, is advertised truthfully and can be traced if something goes wrong. Knowing them is part of the professional competence of every Rasashastra and Bhaishajya Kalpana scholar.

Why this answer scores well. The necessity is answered in a table with examples, as the question asks; each law gets its key provisions; the 2024–2026 changes show that the candidate knows the law as it stands. **Common mistakes:** writing only the Drugs and Cosmetics Act; describing the PFA Act as current; giving Schedule M instead of Schedule T.

AP3.8 Pharmaceutical dosage form design with special reference to pre-formulation

Set in: September 2020; March 2026 ("correlate classical concepts with modern pre-formulation parameters"). **Chapter:** B2.

Plan. Definitions → need for dosage forms → considerations in design → pre-formulation: objectives and parameters (table) → key laws (Noyes–Whitney, flow, BCS, stability) → herbal products → classical correlation (table) → conclusion. Diagrams: pre-formulation parameters (Figure B2.3) and the classical–modern map (Figure B2.7).

1. Definitions

A **dosage form** is the drug combined with excipients in a form that delivers an accurate dose by a chosen route, safely and acceptably. **Dosage form design** is the planned choice of that form. **Pre-formulation** is the study of the physical and chemical properties of the drug substance, alone and with excipients, before the formulation is developed.

2. Considerations in dosage form design

Therapeutic (disease, onset, duration, route); biopharmaceutical (solubility, permeability, absorption); physico-chemical (stability, particle size, flow); manufacturing (scale, equipment, cost); packaging and stability; patient (age, taste, convenience); regulatory.

3. Pre-formulation

Objectives: to know the drug well enough to choose a form, excipients, process and package that will give a stable, bioavailable, manufacturable product.

Draw Figure B2.3.

Parameter	What it decides	Test or rule
Organoleptic	Acceptability; identity	Colour, odour, taste
Particle size	Dissolution, uniformity	Sieving, microscopy; Noyes–Whitney: smaller particles dissolve faster
Flow	Uniform filling and tableting	Angle of repose (25–30° excellent); Carr's index (≤10 excellent); Hausner ratio (≤1.11 excellent)
Polymorphism	Solubility, stability	DSC, XRD; example: chloramphenicol palmitate, ritonavir
Hygroscopicity	Stability, flow	Moisture uptake
Solubility, pKa, log P	Absorption; choice of vehicle	Solubility terms of the pharmacopoeia; partition coefficient
Permeability	Absorption	BCS class I to IV
Stability	Shelf life	Stress testing; Arrhenius: degradation rises with temperature (accelerated stability)
Compatibility	Choice of excipients	Drug–excipient mixtures stored and tested

BCS: class I high solubility and high permeability; II low and high; III high and low; IV low and low. For class II, dissolution is the rate-limiting step, so particle size reduction helps.

4. Pre-formulation of herbal and herbo-mineral products

Many constituents, so identity rests on fingerprint and markers; extracts are hygroscopic and flow poorly; taste needs masking; contaminants (heavy metals, microbes, aflatoxins, pesticides) must be controlled; for Bhasma, crystal phase (XRD), particle size and absence of free metal are the key parameters.

5. Classical concepts and modern pre-formulation

Draw Figure B2.7.

Classical concept	Modern parameter
Grahya lakshana; collection rules	Identity, purity, control of natural variation
Sukshma Churna; Vastra Galita	Particle size
Bhavana	Particle size reduction; granulation; incorporation of actives
Bhasma Pariksha (Rekhapurnata, Varitara, Nishchandrata, Apunarbhava)	Fineness, low density, absence of free metal, stable phase
Sneha Siddhi Lakshana; Avaleha Paka lakshana	Process end point; moisture; consistency
Choice of Drava Dravya (water, milk, oil, alcohol)	Solubility; partition
Saviryata Avadhi	Stability and shelf life
Anupana	Vehicle; taste; possible effect on absorption
Panchavidha Kashaya Kalpana (Swarasa strongest to Phanta mildest)	Concentration and dose per unit form

Boundary: the classical rules aim at the same goals as pre-formulation, and several classical end points correspond to measurable parameters (**established** as parallels). That the classical tests alone guarantee uniform quality, bioavailability or stability of a modern batch is **not established**; they need measured values.

6. Conclusion

Pre-formulation is the scientific foundation of a good dosage form. Bhaishajya Kalpana already asked the same questions in qualitative form; modern pre-formulation answers them with numbers. The best Ayurvedic product uses both.

Why this answer scores well. The parameter table gives the test and the rule for each, and the correlation table answers the 2026 wording directly with the evidence boundary marked. **Common mistakes:** giving only a list of parameter names; forgetting herbal-specific problems; claiming exact equivalence between classical and modern tests.

AP3.9 Ayurvedic Pharmacopoeia and Formulary of India

Set in: July 2018 (“API and AFI in brief”); September 2018 (“need for API and AFI; functions of the Pharmacopoeial Committee; publications; ideal monograph with an example”). **Chapter:** B10.

Plan. Definitions and difference → need → history and institutions → functions of the committee → publications (tables) → structure of a monograph with Amalaki as example → ideal monograph → legal status → conclusion. Diagrams: institutions (Figure B10.1) and the monograph structure (Figure B10.2).

1. Definitions

A **pharmacopoeia** is an official book of **standards** for drugs: their identity, purity and strength, and the methods of testing them. A **formulary** is an official book of **formulations**: composition and method of

preparation. The *Ayurvedic Pharmacopoeia of India* (API) and the *Ayurvedic Formulary of India* (AFI) are the official books for Ayurvedic drugs.

	API	AFI
Gives	Standards and tests	Composition and method
Answers	"Is this sample of good quality?"	"How is this formulation made?"
Used by	QC laboratories, regulators	Manufacturers, teachers, physicians

2. Need

1. One drug has many names and one name many drugs; substitutes and adulterants are common.
2. One formula has different versions in different texts; one authentic version is needed.
3. Classical units and methods must be expressed in metric terms.
4. Quality control needs measurable standards.
5. The Drugs and Cosmetics Act needs official books against which samples are judged.
6. Safety: limits for heavy metals, pesticides, microbes and aflatoxins.
7. Export and trade; teaching and research.

3. History and institutions

Year	Event
1962	First Ayurvedic Pharmacopoeia Committee, under Col. Sir R. N. Chopra
1972	Committee reconstituted under Prof. A. N. Namjoshi
1975	Pharmacopoeial Laboratory for Indian Medicine (PLIM), Ghaziabad
1978	AFI Part I
1986 onward	API Part I, volume by volume
2007	API Part II, Volume I (formulations)
2010	Pharmacopoeia Commission for Indian Medicine and Homoeopathy (PCIM&H)
2020	PCIM&H made a subordinate office of the Ministry of Ayush, with PLIM and the Homoeopathic Pharmacopoeia Laboratory merged

Draw Figure B10.1.

4. Functions of the Ayurvedic Pharmacopoeia Committee (now through PCIM&H)

Prepare the API and AFI; lay down standards for single drugs, extracts and formulations; develop methods of testing (pharmacognostic, physico-chemical, TLC and HPTLC); prescribe limits for contaminants; prepare reference standards and atlases; revise standards; act as the appellate laboratory for ASU drugs; train analysts.

5. Publications

Book	Content
API Part I	Single drugs: nine volumes from 1986 to 2016 (Volume VII on 21 minerals and metals; Volumes VIII and IX on extracts), and a later volume on minerals and metals
API Part II	Formulations: Volume I (2007) with 50 monographs, and later volumes
AFI Part I	1978, revised 2003; about 444 formulations
AFI Part II	2000; about 191 formulations
AFI Part III	Published later
Others	Macroscopy, microscopy and TLC atlases; *Protocol for Testing of ASU Medicines*

6. Structure of a monograph, with an example

Draw Figure B10.2. A Part I monograph gives: title (Sanskrit name and part used); definition (botanical source, family, part); synonyms; description (macroscopy, microscopy); identity, purity and strength; constituents; properties and action (Rasa, Guna, Virya, Vipaka, Karma); important formulations; therapeutic uses; dose.

Example: Amalaki (fresh fruit), API Part I, Volume I.

Test	Limit
Foreign matter	Not more than 2 per cent
Total ash	Not more than 7 per cent
Acid-insoluble ash	Not more than 2 per cent
Alcohol-soluble extractive	Not less than 40 per cent
Water-soluble extractive	Not less than 50 per cent

An ideal monograph is complete (identity, purity, strength, safety), specific (tells the drug from its adulterants), quantitative (numerical limits, marker assay), validated, practical and current.

7. Legal status

The API and AFI are official books under the Drugs and Cosmetics Act. A classical ASU drug must follow the formula of an authoritative book, and API standards are those against which a sample is judged to be of standard quality.

8. Conclusion

The AFI tells how an Ayurvedic medicine is made; the API tells whether it has been made well. Together they turn the classical rules of Grahya lakshana into measurable, enforceable standards.

Why this answer scores well. It distinguishes pharmacopoeia from formulary in a table, gives dated history, lists publications, and supports the monograph structure with real limits for Amalaki. **Common mistakes:** mixing up API and AFI; giving wrong years; writing a monograph without numerical limits.

APPENDIX 4

APPENDIX 4 · CHRONOLOGY OF LAWS, STANDARDS AND MILESTONES

Why this appendix is here. Dates are easy to confuse, and laws are often asked "with their history". This appendix gathers, in one table, every date given in the chapters for the laws, standards, institutions and programmes that govern Ayurvedic drugs. Each entry names the chapter where it is explained and sourced. Nothing here is new; it is a revision aid.

AP4.1 The story in one picture

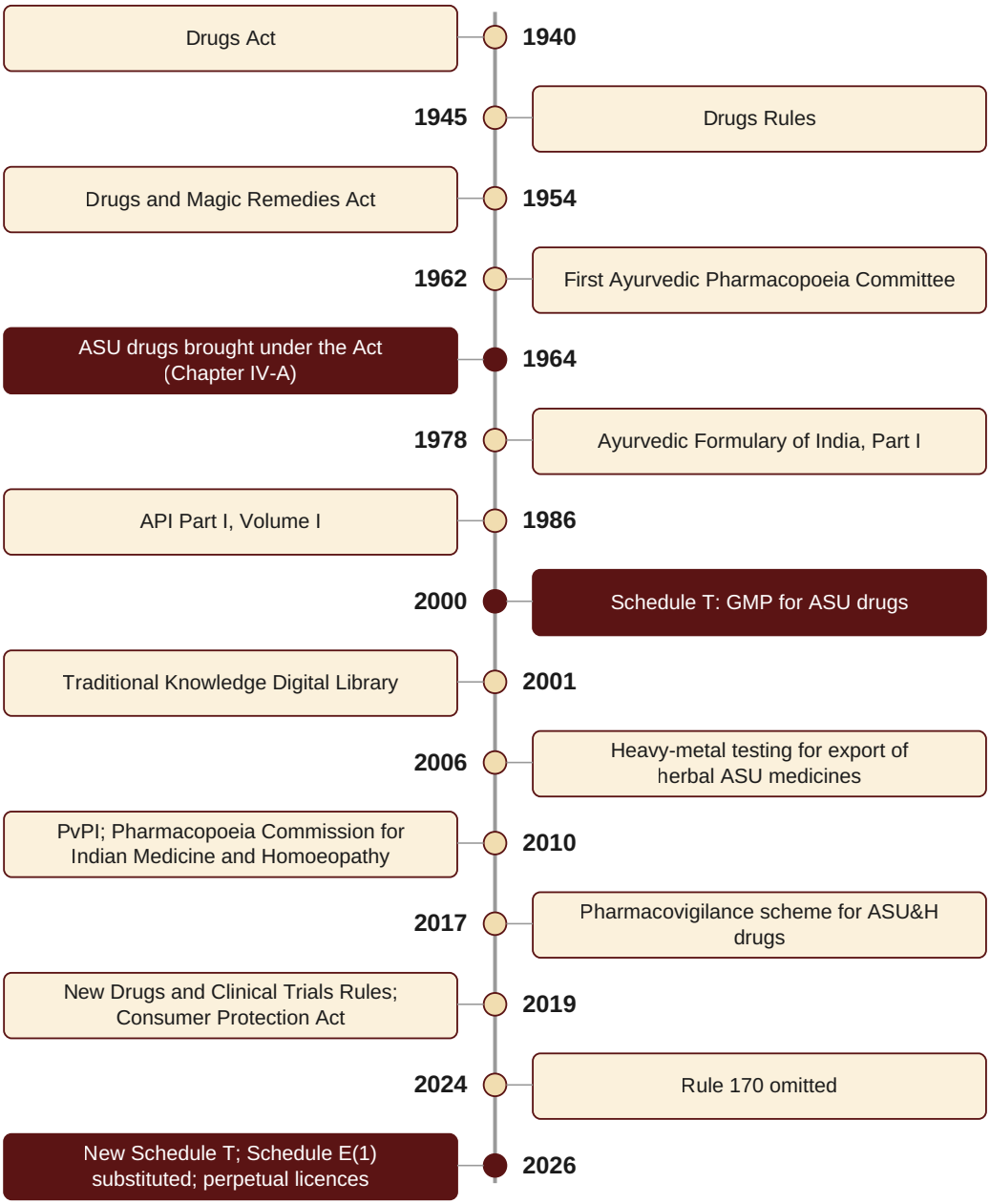


Figure AP4.1. Selected milestones in the regulation of Ayurvedic drugs in India, 1940–2026.

AP4.2 Chronology

Year	Event	Chapter
1883	Paris Convention for the Protection of Industrial Property	B7.3
1932	Pharmacy education in India begins at Banaras Hindu University	B1.2
1940	Drugs Act enacted (for modern drugs)	B8(i)
1945	Drugs Rules framed	B8(i)
1947	Nuremberg Code	A2.6

Year	Event	Chapter
1948	Pharmacy Act; Factories Act	B8(vi)
1954	Drugs and Magic Remedies (Objectionable Advertisements) Act; Prevention of Food Adulteration Act	B8(ii), B8(iii)
1955	First edition of the Indian Pharmacopoeia	B11.2
1959	The 3Rs of animal research proposed by Russell and Burch	A1d.7
1960	Prevention of Cruelty to Animals Act	A1d.7
1961	Thalidomide disaster recognised	A6.3
1962	First Ayurvedic Pharmacopoeia Committee, under Col. Sir R. N. Chopra	B10.3
1964	**Chapter IV-A** inserted in the Drugs and Cosmetics Act: ASU drugs brought under the Act. Declaration of Helsinki adopted	B8(i); A2.6
1968	WHO Programme for International Drug Monitoring begins	A6.6
1970	Patents Act; Patent Cooperation Treaty	B7.2, B7.3
1972	Ayurvedic Pharmacopoeia Committee reconstituted under Prof. A. N. Namjoshi	B10.3
1975	Pharmacopoeial Laboratory for Indian Medicine (PLIM), Ghaziabad	B10.3
1978	**Ayurvedic Formulary of India, Part I**	B10.4
1979	Belmont Report	A2.6
1985	Narcotic Drugs and Psychotropic Substances Act	B8(v)
1986	Consumer Protection Act. API Part I, Volume I (as tabulated by Yao and colleagues)	B8(vii); B10.4
1992	Convention on Biological Diversity	B7.3
1995	TRIPS Agreement in force	B7.3
1997	Turmeric wound-healing patent revoked in the United States	B7.5
1998	India joins the WHO drug monitoring programme; India joins the Paris Convention and PCT; Breeding of and Experiments on Animals (Control and Supervision) Rules	A6.6; B7.3; A1d.7
2000	**Schedule T** (GMP for ASU drugs) introduced. AFI Part II	B4.6; B10.4
2001	**Traditional Knowledge Digital Library** established (CSIR with the then Department of ISM&H)	B12.1
2002	Biological Diversity Act. WHO definition of pharmacovigilance. Classical LD50 test (OECD 401) deleted	B7.4; A6.1; A1d.4
2004	WHO guidelines on safety monitoring of herbal medicines	A6.6
2005	Patents (Amendment) Act: product patents for pharmaceuticals	B7.2
2006	Heavy-metal testing made mandatory for export of purely herbal ASU medicines (from 1 January). Food Safety and Standards Act	A4.8; B8(iv)

Year	Event	Chapter
2007	API Part II, Volume I (formulations)	B10.4
2009	AYUSH Mark (Standard and Premium) started by the Quality Council of India	B4.6.6
2010	Pharmacovigilance Programme of India (July). Pharmacopoeia Commission for Indian Medicine and Homoeopathy. Nagoya Protocol	A6.7; B10.3; B7.3
2011	PvPI National Coordination Centre moves to the Indian Pharmacopoeia Commission (15 April). PFA Act repealed (5 August)	A6.7; B8(iii)
2012	India’s first compulsory licence (sorafenib)	B7.2
2013	GCP guidelines for clinical trials in ASU medicine	A2.5
2014	Ministry of Ayush formed	B1.2
2016	Rule 161B (shelf life) substituted (12 August)	B3.4.3
2017	ICMR National Ethical Guidelines. Central Sector Scheme for Pharmacovigilance of ASU&H drugs (December)	A2.6; A6.8
2018	Ayush pharmacovigilance network functional (August). Rule 170 on ASU advertisements inserted	A6.8; B8(i)
2019	New Drugs and Clinical Trials Rules (19 March). Consumer Protection Act, 2019	A2.2; B8(vii)
2020	PCIM&H made a subordinate office of the Ministry of Ayush, with PLIM and HPL merged. OSH Code enacted	B10.3; B8(vi)
2021	CDC blood lead reference value lowered to 3.5 µg/dL	A4.4
2022	Ayurveda Aahara Regulations (FSSAI). AYUSHEXCIL launched (20 April). Wider access to TKDL approved	B8(iv); B9.1; B12.3
2023	CPCSEA renamed CCSEA (January). Ayush Visa (27 July). Biological Diversity Act amended	A1d.7; B9.5; B7.4
2024	Rule 170 omitted (1 July). WIPO Treaty on Intellectual Property, Genetic Resources and Associated Traditional Knowledge. Declaration of Helsinki revised (October). TACT2 trial reported	B8(i); B7.3; A2.6; A4.9
2025	ICH E6(R3) (January). Schedule TB for ASU nasal sprays (January). AyushSuraksha portal (30 May). Ph. Eur. online only (June). Supreme Court vacates stay on omission of Rule 170 (11 August). OSH Code in force (21 November)	A2.6; B8(i); A6.8; B11.5; B8(i); B8(vi)
2026	IP 2026 (10th edition). Jan Vishwas Act provisions on ASU drugs in force (1 July). **Drugs (Eleventh Amendment) Rules**: new Schedule T (compliance by 31 July 2029), perpetual licences, product code, QR code. **Drugs (Thirteenth Amendment) Rules**: Schedule E(1) substituted	B11.2; B8(i); B4.6.4; B8(i).8

Dates are those stated, with sources, in the chapters named. Where a chapter notes uncertainty about a date (for example, the year of the first API volume), the same caution applies here.

TEN DATES EVERY SCHOLAR SHOULD KNOW

- **1940** Drugs Act; **1945** Rules; **1964** ASU drugs brought under the Act.
- **1954** Drugs and Magic Remedies Act.
- **1962** First Ayurvedic Pharmacopoeia Committee; **1978** AFI Part I.
- **2000** Schedule T.
- **2001** TKDL.
- **2006** Heavy-metal testing for export of herbal ASU medicines.
- **2010** PvPI; PCIM&H.
- **2017** Ayush pharmacovigilance scheme.
- **2019** New Drugs and Clinical Trials Rules; Consumer Protection Act.
- **2026** New Schedule T and Schedule E(1).

PRINCIPAL SOURCES AND REFERENCES

Every reference number in this book, written **[R1]** to **[R53]**, is the number of the book in the list of **Reference Books** printed at the end of the Paper IV syllabus. A scholar can therefore trace every citation back to the syllabus. Where an entry in that list is incomplete or misprinted, the full, verified citation used in this book is given below, and each defect is noted. Other sources, such as research papers, Gazette notifications, answers given in Parliament, official websites and international guidelines, are cited in full in the reference box at the end of each chapter, with the date on which they were consulted (16 October 2026). They are not repeated here.

Principal sources used throughout this book

Classical Ayurvedic texts

- Sharma S, Shastri HD. *Rasatarangini*. 11th ed. Delhi: Motilal Banarasidas; 1979. *(Syllabus reference 7)*
- Rasa Vagbhata. *Rasa Ratna Samuccaya*. Kulkarni DA, editor. New Delhi: Meharchand Lachhmandas Publications; reprint 2006. *(Syllabus reference 3)*
- Sharangadhara. *Sharangadhara Samhita*, with the Dipika commentary of Adhamalla and the Gudartha Dipika of Kashirama. Shastri P, editor. 6th ed. Varanasi: Chaukhamba Orientalia; 2005. *(Syllabus reference 11)*
- *Charak Samhita New Edition*. Jamnagar: Charak Samhita Research, Training and Skill Development Centre; carakasamhitaonline.com (accessed 16 October 2026). Used to verify every verse number of the Charaka Samhita cited. *(Syllabus reference 26)*
- *Sushruta Samhita*: verse numbers as in the 1938 Nirnaya Sagar edition, checked against the Sushruta Project provisional edition of the Kalpasthana, saktumiva.org (accessed 16 October 2026). *(Syllabus reference 26)*

The editions of the first three texts are those used in the companion volume of this series, *Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana*.

Official compendia, laws and Government publications

- *The Ayurvedic Pharmacopoeia of India*, Part I (single drugs) and Part II (formulations). Government of India; now published by the Pharmacopoeia Commission for Indian Medicine and Homoeopathy, Ghaziabad. *(Syllabus reference 36)*
- *The Ayurvedic Formulary of India*, Parts I to III. Government of India. *(Syllabus reference 37)*
- Lohar D. *Protocol for Testing of Ayurvedic, Siddha and Unani Medicines*. Department of AYUSH, Ministry of Health and Family Welfare, Government of India; 2008. *(Syllabus reference 35)*
- *Indian Pharmacopoeia 2026* (10th edition). Ghaziabad: Indian Pharmacopoeia Commission; 2026. *(Syllabus reference 38)*
- *British Pharmacopoeia*. London: Medicines and Healthcare products Regulatory Agency (published annually). *(Syllabus reference 39)*
- *United States Pharmacopeia–National Formulary*. Rockville: United States Pharmacopeial Convention (online). *(Syllabus reference 40)*
- Lund W, editor. *The Pharmaceutical Codex: Principles and Practice of Pharmaceutics*. 12th ed. London: Pharmaceutical Press; 1994. *(Syllabus reference 41)*

- *The Drugs and Cosmetics Act, 1940 and the Drugs and Cosmetics Rules, 1945*, as amended up to the Drugs (Thirteenth Amendment) Rules, 2026. *(Syllabus reference 43)*
- *The Drugs and Magic Remedies (Objectionable Advertisements) Act, 1954*, as amended. India Code. *(Syllabus reference 44)*
- *The Prevention of Food Adulteration Act, 1954* (repealed with effect from 5 August 2011) and *The Food Safety and Standards Act, 2006*. *(Syllabus reference 45)*
- *The Narcotic Drugs and Psychotropic Substances Act, 1985*, as amended. *(Syllabus reference 46)*
- *The Factories Act, 1948* (repealed with effect from 21 November 2025; replaced by the Occupational Safety, Health and Working Conditions Code, 2020) and *The Pharmacy Act, 1948*. *(Syllabus reference 47)*
- *The Consumer Protection Act, 1986* (repealed) and *The Consumer Protection Act, 2019*. *(Syllabus reference 48)*

Modern texts

- Tripathi KD. *Essentials of Medical Pharmacology*. 9th ed. New Delhi: Jaypee Brothers Medical Publishers; 2024. *(Syllabus reference 29)*
- Brunton LL, Knollmann BC, editors. *Goodman & Gilman's The Pharmacological Basis of Therapeutics*. 14th ed. New York: McGraw Hill; 2023. *(Syllabus reference 32)*
- Brown MJ, Sharma P, Mir FA, Bennett PN. *Clinical Pharmacology*. 12th ed. Elsevier; 2018. *(Syllabus reference 30)*
- Vogel HG, editor. *Drug Discovery and Evaluation: Pharmacological Assays*. 3rd ed. Berlin: Springer; 2008. *(Syllabus reference 31)*
- Adejare A, editor. *Remington: The Science and Practice of Pharmacy*. 23rd ed. Elsevier; 2020. *(Syllabus reference 27)*
- Khar RK, Vyas SP, Ahmad FJ, Jain GK, editors. *Lachman/Lieberman's The Theory and Practice of Industrial Pharmacy*. 4th ed. New Delhi: CBS Publishers and Distributors; 2013. *(Syllabus reference 28)*

The syllabus list of reference books, with notes

The table reproduces the syllabus list as printed. The third column gives the correct title where the printed entry has an error, and the fourth shows where the book is cited in these notes. A text marked "recommended reading" is part of the course but has not been cited here for a specific statement; this is stated so that no citation is implied where none was used.

No.	As printed in the syllabus	Correction or note	Use in this book
1	Rasahridaya Tantra	—	B1 (history of pharmacy)
2	Rasarnava	—	B1 (history of pharmacy)
3	Rasaratna Samuccahaya	**Misprint**: Rasaratna Samuchchaya (Rasa Ratna Samuccaya)	B1; B4 (Rasashala)
4	Ayurved Prakasha	—	Recommended reading
5	Rasendrachudamani	—	Recommended reading
6	Rasendra Chintamani	—	Recommended reading
7	Rasatarangini	—	A4, A5, A6, B1

No.	As printed in the syllabus	Correction or note	Use in this book
8	Rasapraksha Sudhakar	**Misprint**: Rasaprakasha Sudhakara	Recommended reading
9	Rasamrita	—	Recommended reading
10	Rasa Chandanshu: CCRAS Publication	—	Recommended reading
11	Sharangadhara Samhita	—	A1(b), B1, B2, B3, B5, Appendix 2
12	Sharangadhara Darpan (BP Pandey)	—	Recommended reading
13	Bhavaprakasha	—	Recommended reading
14	Yoga Ratnakara	—	Recommended reading
15	Bhaishajya Ratnavali	—	Recommended reading
16	Siddha Bhaishajya Manimala	**Duplicate** of entry 20	Recommended reading
17	Ayurvediya Aushadhikarana – Puranik and Dhamanakar	—	Recommended reading
18	Bharat Bhaishajya Ratnakara	—	Recommended reading
19	Rasayoga Sagara	—	Recommended reading
20	Siddha Bhaishajya Manimala	**Duplicate** of entry 16	Cited as 16 if needed
21	Siddha Yoga Sangraha – Yadavaji Trikamji Acharya	—	Recommended reading
22	Sahasrayoga	—	Recommended reading
23	Vaidyaka Paribhasha Pradeepa	—	Recommended reading
24	Dravyaguna Vijnan Part 1 and 2 – Yadavji Trikamji	—	Recommended reading
25	Chakradatta – Ratnaprabha	—	Recommended reading
26	Relevant parts from Charaka Samhita, Sushruta Samhita, Kashyapa Samhita, Ashtanga Sangraha, Ashtanga Hridaya	Not a single book; the specific Sthana and verse are given at each citation	Throughout
27	Remington: Science and Practice of Pharmacy	Full title: *Remington: The Science and Practice of Pharmacy* (23rd ed., 2020)	B1–B6
28	Theory and Practice of Industrial Pharmacy – Leon Lachman et al	Current Indian edition: *Lachman/ Lieberman's The Theory and Practice of Industrial Pharmacy*, 4th ed., 2013	B2, B3, B4
29	Clinical Pharmacology, KD Tripathi	**Title error**: K. D. Tripathi's book is *Essentials of Medical Pharmacology*	Part A throughout
30	Clinical Pharmacology, Lawrence Benette	**Name error**: the book of D. R. Laurence and P. N. Bennett; current edition by Brown, Sharma, Mir and Bennett	A1(c), A2
31	Drug Discovery and Evaluation (Pharmacological assays) HG Vogel	—	A1(d)

No.	As printed in the syllabus	Correction or note	Use in this book
32	Pharmacological Basis of therapeutics – Goodman and Gilman	—	Part A throughout
33	Data Base of Medicinal Plants of CCRAS	Published title: *Database on Medicinal Plants Used in Ayurveda* (CCRAS, multi-volume)	B1
34	Quality and Standards of Medicinal Plants – ICMR publication	Published title: *Quality Standards of Indian Medicinal Plants* (ICMR, multi-volume)	B1
35	Quality Control of Ayurvedic Drugs – PLIM, Gaziabad	**Misprint**: Ghaziabad. Cited as Lohar D, *Protocol for Testing of ASU Medicines* (2008), prepared at PLIM	B10
36	Ayurvedic Pharmacopeia of India	—	A4, B3, B9, B10
37	Ayurvedic Formulary of India	—	A1(a), B6, B10, Appendices 1 and 2
38	Indian Pharmacopeia	—	B2, B3, B11
39	British Pharmacopeia	—	B2, B11
40	United States Pharmacopeia	—	B2, B11
41	Pharmacopeia Codex	**Title error**: *The Pharmaceutical Codex* (12th ed., 1994)	B11
42	Current Good Manufacturing Practices	Not a single publication; GMP is taught from Schedule T and WHO GMP	B4, B5
43	Drugs and Cosmetic Act 1940 and Rules 1945 with latest amendments	—	B3, B4, B5, B8(i), B9, A6
44	Drugs and Magic remedies (Objectionable advertisement) Act-1954	—	B5, B8(ii)
45	Prevention of Food Adulteration (PFA) act	Repealed in 2011; replaced by the FSS Act, 2006	B8(iii), B8(iv)
46	Laws pertaining to Narcotics	NDPS Act, 1985	B8(v)
47	Factory and Pharmacy Acts	Factories Act, 1948 repealed in 2025 (OSH Code, 2020); Pharmacy Act, 1948	B6, B8(vi)
48	Consumer Protection Act -1986	Repealed; replaced by the Consumer Protection Act, 2019	B8(vii)
49	Brief information on the peer reviewed journals, official websites and other official search engines along with their links (related with the subject)	**Not a citation**: a topic. It is answered in Chapter B1.4	B1
50	Rutleys Elements of Mineralogy	*Rutley's Elements of Mineralogy*	Recommended reading
51	Bhasma Vigyaniyam	—	Recommended reading
52	Kupipakva Vigyaniyam	—	Recommended reading

No.	As printed in the syllabus	Correction or note	Use in this book
53	Anupana Manjari	—	Recommended reading

The most important of these notes are also given under "About the references" in How to Use This Book.

GLOSSARY OF AYURVEDIC TERMS

This glossary covers the Sanskrit and Ayurvedic terms used in this book. Terms are given in plain Roman letters, without diacritical marks, in the same form as in the text. Where a term carries more than one sense, the sense used in this book is given.

A

Term	Meaning
Abhraka	Mica; a mineral used as Bhasma
Agada (Agada Tantra)	Antidote; the branch of Ayurveda dealing with poisons and their treatment
Agni	Digestive and metabolic fire
Agrahya	Unfit for use (of a raw drug)
Alpa Matra	Small dose
Amritikarana	A process after Marana to remove residual harmful qualities of a Bhasma
Anumana	Inference; a means of valid knowledge
Anupana	The vehicle or after-drink taken with a medicine
Anuraga	Devotion (of the attendant to the patient)
Apakva	Incompletely processed
Aptopadesha	The word of a reliable authority; a means of valid knowledge
Apunarbhava	Test that a Bhasma does not revert to metal when heated with Mitra Panchaka
Arishta	Fermented preparation made from a decoction
Arka	Distillate
Asava	Fermented preparation made from fresh juice or infusion without boiling
Ashuddha	Unpurified
Asthi Dhatu	Bone tissue
Atiyoga	Excessive effect (of a therapy)
Avaleha (Leha)	Confection; semi-solid preparation in a sugar or jaggery base
Ayoga	Inadequate effect (of a therapy)

B

Term	Meaning
Basti	Medicated enema
Bhaishajya Kalpana	The branch of Ayurveda dealing with the preparation of medicines
Bhasma	Fine powder obtained by incineration (Marana) of a metal or mineral
Bhasma Pariksha	Classical tests of quality of a Bhasma

Term	Meaning
Bhavana	Wet trituration of a drug with a liquid
Bheshaja Kala	Time of administration of medicine
Bharani	A large jar used for Sandhana

C

Term	Meaning
Chaturvimshati Upakrama	The twenty-four measures for treatment of poisoning (Charaka)
Chikitsa Chatushpada	The four pillars of treatment: physician, drug, attendant, patient
Churna	Powder

D

Term	Meaning
Dakshya	Skill (of the attendant)
Dashavidha Pariksha	Tenfold examination of the patient
Dhatu	Tissue of the body; also, a metal (in Rasashastra)
Dhatu Visha	Mineral poison
Dhatukshaya	Depletion of a tissue
Drava Dravya	Liquid medium used in a preparation
Dravadvaiguna	The rule of doubling the measure for liquids
Dushi Visha	Weak, residual poison that lingers in the body and vitiates the tissues

G

Term	Meaning
Gandhaka	Sulphur
Gara Visha	Artificial poison prepared from combinations, acting after a delay
Ghana	Solid extract obtained by evaporating a decoction
Ghrita	Medicated ghee
Godanti	Gypsum (calcium sulphate)
Grahya	Fit for use (of a raw drug)
Guggulu	Resin of *Commiphora wightii*; also a class of preparations with this base
Gunja	Seed of *Abrus precatorius*; a unit of weight (Ratti)
Gutika	Pill

H

Term	Meaning
Haratala	Orpiment (arsenic trisulphide)
Hima (Shita)	Cold infusion
Hingula	Cinnabar (red mercuric sulphide)
Hridayavarana	Protection of the heart (a measure in the treatment of poisoning)

J

Term	Meaning
Jangama	Of animal origin
Jasada (Yashada)	Zinc

K

Term	Meaning
Kajjali	Black, lustreless product of trituration of mercury with sulphur (mercuric sulphide)
Kala	Time; season
Kalka	Paste
Kalpana	Preparation; dosage form
Kapardika	Cowrie shell
Kharaliya Rasayana	Preparation made by trituration in a mortar, usually from Kajjali
Khalva (Kharala)	Mortar
Koshtha	Nature of the bowel
Kshara	Alkaline preparation obtained from plant ash
Kudava	A classical unit of measure (192 g)
Kupi	Glass bottle used in Kupipaka
Kupipakva Rasayana	Preparation made by heating in a glass bottle in a sand bath
Kwatha (Shrita)	Decoction

L

Term	Meaning
Lavana	Salt; a class of salt preparations
Lepa	External paste
Loha	Iron

M

Term	Meaning
Makaradhwaja	A Kupipakva preparation of mercury, sulphur and gold
Malahara	Ointment
Malla (Sankhiya)	Arsenic trioxide
Manahshila	Realgar (arsenic sulphide)
Mandura	Iron oxide (rust of iron); its Bhasma
Marana	Incineration of a metal or mineral to Bhasma
Masha	A classical unit of weight (1 g)
Matra	Dose
Mithya Yoga	Wrong use (of a therapy or medicine)
Mitra Panchaka	Five “friendly” substances (jaggery, Gunja, honey, ghee, borax) used in the Apunarbhava test
Mukta	Pearl

N

Term	Meaning
Naga	Lead
Nasya	Nasal administration of medicine
Niruttha	Test that a Bhasma does not combine with silver when heated with it
Nishchandratva	Absence of metallic lustre

P

Term	Meaning
Paka	Cooking; stage of processing (Mridu, Madhya, Khara)
Pala	A classical unit of weight (48 g)
Panchavidha Kashaya Kalpana	The five basic forms: Swarasa, Kalka, Kwatha, Hima, Phanta
Parada	Mercury
Paricharaka (Upasthata)	Attendant
Parpati	Flaky preparation made by melting Kajjali and pressing it
Parthiva	Of mineral (earth) origin
Pathya	Wholesome diet and conduct
Phanta	Hot infusion
Pishti	Fine powder made by trituration with liquids, without heat

Term	Meaning
Pottali	Bolus preparation cooked in a medium
Prabhava	Specific action not explained by other properties
Prakriti	Constitution
Prastha	A classical unit of measure (768 g)
Prativisha	Antidote poison used to counter a poison
Pratyaksha	Direct perception
Praval	Coral
Puta	Measured heating for Marana

R

Term	Meaning
Rajata	Silver
Rakta Dhatu	Blood tissue
Rasa Dravya	Mercury, metals and minerals used in Rasashastra
Rasa Sindura	A Kupipakva preparation of mercury and sulphur (red mercuric sulphide)
Rasashala	Pharmacy for Rasa preparations
Rasashastra	The branch of Ayurveda dealing with mercury, metals and minerals
Rasaushadhi (Rasayoga)	Medicine containing Rasa Dravya
Rasayana	Rejuvenative therapy
Ratti	A classical unit of weight (125 mg)
Rekhapurnatva	Test that a Bhasma fills the lines of the fingers

S

Term	Meaning
Samanya	The principle that like increases like
Sampat	Good quality (of a drug)
Samyak Yoga	Proper effect (of a therapy)
Sandhana	Fermentation; fermented preparations
Sankhiya	Arsenic trioxide
Saviryata Avadhi	Period of potency (shelf life)
Shankha	Conch shell
Shaucha	Cleanliness
Shodhana	Purification (of a drug); also purificatory therapy

Term	Meaning
Shuddha	Purified
Shukti	Pearl oyster shell
Sneha Kalpana	Preparation of medicated oils and ghee
Sneha Siddhi Lakshana	Signs that Sneha Paka is complete
Sthavara Visha	Poison of inanimate origin
Sukshma Churna	Fine powder
Swarasa	Expressed juice
Swarna	Gold

T

Term	Meaning
Taila	Medicated oil
Tamra	Copper
Tantra Yukti	Methods of interpreting a text
Tola (Karsha)	A classical unit of weight (12 g)

U

Term	Meaning
Upacharajnata	Knowledge of nursing and of preparing medicines
Upavisha	Sub-poisonous drugs

V

Term	Meaning
Vamana	Therapeutic emesis
Vanga	Tin
Varitaratva	Test that a Bhasma floats on water
Vastra Galita	Sieved through cloth
Vati	Tablet or pill
Vatsanabha	Aconite (*Aconitum ferox*)
Virechana	Therapeutic purgation
Viruddha Ahara	Incompatible food combinations
Visha	Poison
Vyapat	Complication (of a therapy)

Y

Term	Meaning
Yavakuta	Coarse powder for decoction
Yogavahi	Property of carrying and enhancing the action of other drugs
Yukti	Rational planning; reasoning from many factors

ABOUT THE AUTHOR



Dr Milind Pareshkumar Chatrabhuji is Associate Professor in the Postgraduate Department of Rasashastra and Bhaishajya Kalpana at Parul Institute of Ayurved and Research, Parul University, Vadodara.

He holds a BAMS and an MD in Ayurveda, with specialisation in Rasashastra and Bhaishajya Kalpana, and is presently a doctoral scholar in the same subject, working under the guidance of Dr Rohit Gokarn.

Alongside his academic work he has been in clinical practice for more than a decade at Suvarna Ayurveda Clinic, Vadodara.

His teaching and research interests lie in the pharmaceutical processing of metals and minerals, the standardisation of classical Ayurvedic preparations, and the application of modern analytical methods to the verification of classical pharmaceutical claims. He is the author of the *Textbook of Applied Basics of Rasashastra and Bhaishajya Kalpana* and its companion volumes. This book grew out of his teaching of the final-year postgraduate syllabus.

Correspondence and corrections. This is a first edition. The author would welcome corrections, comments and suggestions from teachers and scholars, so that later editions may be improved. Errors of fact, outdated regulatory references and omissions are of particular interest, and every such report will be acknowledged.

INDEX

Page numbers refer to the number printed at the foot of each page. Entries appearing on more than 40 pages have been omitted as too general to be useful.

A

ABC analysis 281, 290–291, 298–299, 301, 315, 409, 458
Abhraka 144, 174–175
Acrodynia 155–156, 172
Active surveillance 196, 211
Adverse drug reaction 4, 11, 65, 78–79, 85–87, 109, 191–193, 198, 209, 212, 214–215, 217, 219–221, 224, 417, 425, 468–469
Adverse event 78, 86, 114, 121, 129, 133, 191, 193, 200, 204, 206, 209–210, 215, 426, 465, 470
Affinity 31, 41, 48, 108, 163, 463, 467
Agada 84
Agni 45, 220, 258, 425
Agonist 30–34, 36, 41, 48–49, 93, 108–109, 225, 463
ALA dehydratase 151, 153, 168, 226, 466
Allergy 58, 65, 80, 82, 85–86, 95, 109, 160, 163, 198, 201, 274, 438, 467
Amritikarana 166, 181–187, 189, 210, 223, 227
Angle of repose 250–252, 260, 263, 478
Animal ethics 4, 11, 88, 99–100, 102–106, 109, 470, 472, 486
Antagonist 30–32, 34, 36, 48–49, 74–76, 84–85, 108, 200, 221, 225, 228, 463
Antidote 65, 76, 80, 83–86, 109, 140, 160, 201, 221
Anupana 45, 62, 73, 99, 167–168, 184, 201, 210, 237, 246, 256–258, 261, 274, 316–318, 320–321, 386, 424–425, 479
Aptopadesha 124, 240–241, 243
Apunarbhava 99, 167, 172, 185, 192, 258, 422–423, 479
Arishta 53, 73, 109, 246–247, 256, 267, 269–270, 272, 277, 279, 284, 286, 295, 362, 386, 412, 474
Arka 247–248, 256, 269–270, 272, 277, 386, 422
Arogyavardhini Vati 174–175, 178, 180, 182, 185, 187, 189, 227, 306
Arrhenius equation 254, 260, 262–263, 270, 478
Arsenic trioxide 157, 159, 168–169, 171, 175, 179, 185, 187, 223, 227, 345
Asava 53, 246–247, 256, 266–267, 269–270, 272, 277, 279, 284, 286, 295, 298, 362, 386, 412, 423, 474
Avaleha 15, 53, 247, 256, 258, 269–270, 272, 277, 284, 295, 320, 386, 423, 474, 479
Ayush pharmacovigilance network 208–210, 212–213
AyushSuraksha 196, 209–210, 212–214, 216, 351, 470, 486

B

Basophilic stippling 153–154, 168, 226, 466
Batch manufacturing record 210, 289, 295–296, 298–299, 301, 307, 409, 425, 430, 438, 440–441, 443–444, 447–448, 450, 474
Batch release 285, 303, 306–307, 309–310, 312, 430, 445, 447
Belmont Report 124–125, 130, 132–133, 226, 228, 465, 485
Bhasma Pariksha 85, 98–99, 106, 122, 166–169, 172, 184–185, 211, 258, 297, 418, 431, 445, 467, 474, 479
Bhavana 174, 178, 227, 257, 261, 273, 457, 479
Bheshaja Kala 65, 69–70, 87, 316, 320, 425
Bio-elementology 135, 138, 146–147, 222
Bioequivalence 4, 11, 20, 65, 70–73, 85–86, 109, 117, 120, 220–221, 455
Biological Diversity Act 327, 329–335, 339, 413, 477, 485–486
Biopharmaceutics Classification System 72, 253–254, 261, 263, 409, 412, 477–478
Biopiracy 331–332, 334–336, 379, 402, 409–410
Blinding 43, 68, 88, 97, 101, 105–106, 115–116, 119–121, 123, 130–131, 133, 221, 226, 228, 465, 471
Blister pack 267–270, 277, 279, 345, 412
Blood lead 153–154, 161, 166, 170–172, 181, 186, 227, 486
British Pharmacopoeia 235, 262, 393, 395–398, 410
Burton's line 154, 168, 172, 466

C

Cadmium 137, 149–152, 160, 162–163, 165, 168, 170–171, 226–228, 466–467
Calcium 15, 20, 30, 74–76, 84, 135–137, 139–140, 143–148, 151–154, 162–164, 166–167, 169, 175, 177–178, 181, 222, 226–227, 466–467
Calcium disodium EDTA 163–165, 169–170, 468
Causality assessment 86, 191, 197–200, 205–209, 212–215, 223–224, 426, 469
Chaturvimshati Upakrama 166
Chelation therapy 4, 9, 30, 71, 74, 84, 93, 149–151, 153–157, 159–171, 217–219, 222, 224, 466–468
Chikitsa Chatushpada 129, 234, 241–243, 320
Churna 53, 122, 246–247, 251–252, 256–257, 261, 263, 266, 268–270, 272, 276–279, 283–284, 295, 317, 320, 386, 422–423, 435–436, 438, 445, 447, 455, 457, 459, 474, 476, 479
Cinnabar 151, 155, 157, 168, 170–172, 227, 252, 345, 466
Clinical pharmacology 2, 4, 9, 14, 50, 87, 89, 111–113, 115–117, 119, 121, 123, 125, 127, 129–133, 219, 221
Clinical Trials Registry – India 114, 121, 130–131, 133, 226, 239, 464, 466
Community pharmacy 4, 231, 237, 313, 315, 317–323, 405, 407, 409, 411, 413
Compulsory licence 330, 334, 336, 413, 486
Consumer Protection Act 5, 226, 337, 339, 345, 351, 368–371, 410–411, 413, 476–477, 485–487
Copper 15, 135, 137, 139, 141, 143–144, 146, 151, 160, 163–166, 175, 181, 187, 226–227, 229, 345, 467
Copyright 326–327, 333, 336
Costing 282, 285–286, 301, 452, 459
Cretinism 141, 146, 148, 226

CYP enzymes 24, 39, 41, 47–48, 67, 69, 76, 85, 225, 462

D

Dashavidha Pariksha 65, 69, 85–87, 109, 211

Declaration of Helsinki 124–125, 130–133, 221, 226, 465, 485–486

Deferoxamine 84, 140, 160, 163–165, 169–170, 172, 227–228, 467

Dimercaprol 30, 84, 154, 156, 159, 163, 165–166, 168–170, 172, 222, 227–228, 437, 467–468

Dispensing 4, 14, 78, 231, 234, 237, 241, 283, 289, 313–317, 319–323, 366, 407, 409, 437, 439

Dissolution 20, 71–73, 93, 248–252, 254, 260, 262–263, 394, 412, 478

Distribution channel 304–305, 310, 312

DMSA 84, 154, 156, 159, 163, 165, 169–170, 227, 467

Dosage form design 4, 231, 245–249, 251, 253, 255–257, 259–263, 405, 407–409, 411, 477–478

Dose–response relationship 11–13, 16, 29, 32, 34–37, 40, 44–50, 91–92, 108, 135, 138, 147–148, 220, 225, 461, 463

Drug design 43, 245–246, 259, 261, 263, 406, 409

Drug interaction 4, 11, 24, 65, 73–74, 76, 85–86, 109, 192, 198, 201, 207, 217, 219–221, 224, 317

Drugs and Magic Remedies Act 4, 209, 226, 306, 311, 337–338, 345, 349–352, 372, 410–411, 476, 485, 487

Dushi Visha 84, 86, 109, 166, 168–169, 172, 223, 467

E

Economic order quantity 281, 291, 298–299, 301, 409, 412, 452, 458–459

Elemental composition 4, 9, 135–137, 139, 141, 143, 145–147, 218–219, 222, 224

Enzyme induction 24, 48, 75, 86, 108, 221, 462

Erethism 155–156, 168, 172, 466

Ethics committee 88, 99, 103, 111, 114, 121–122, 128–133, 221, 464–465, 472

Excipient 20, 71, 246, 248–250, 255, 260–261, 263, 393–395, 478

Experimental pharmacology 4, 11, 88, 90, 105, 133, 219–220

F

Factories Act 3, 292, 339, 364, 366–367, 372, 477, 485

Ferrochelatase 151, 153, 168, 226, 466

First-pass metabolism 17, 20, 46–48, 51–55, 58–59, 61–63, 68, 108–109, 221, 225, 248, 462

Flow properties 250–252, 260–261, 263, 409, 478

Fluorosis 142, 146–148, 222

Food Safety and Standards Act 3–4, 337, 339, 353–359, 372, 414, 476–477, 485–486

G

G-protein coupled receptor 32–33, 48–49, 221, 463

Gandhaka 15, 145, 291, 419–420

Gara Visha 84, 86, 166, 169, 467

Geographical indication 33, 326–327, 331–334, 336, 410

Ghrita 47, 53, 62, 246–247, 269–270, 272, 277, 279, 284, 315, 320, 386, 421

Good Clinical Practice 114, 120, 122–123, 125, 129–133, 222, 465, 486

Good pharmacy practice 313, 319, 321–323

Guggulu 77, 247, 257, 270, 272–273, 386, 458

H

Half-life 11–12, 26–28, 47–48, 72, 108, 150–153, 160, 221, 225, 451–453, 455, 462

Haratala 157, 175, 227, 287, 424

Haridra 15, 43, 46–47, 50, 73, 77, 85, 87, 225, 254, 325, 331–336, 400, 403, 410, 413, 463, 485

Hartwig scale 201, 212, 214–215, 426, 469

Hingula 155, 157, 175, 424

Hospital formulary 315, 321, 323

Hospital pharmacy 237, 283, 313–316, 320–321, 323, 409

I

ICH guidelines 88, 90, 104, 107, 120, 125, 129–130, 132, 191, 193, 204, 212–215, 236, 465, 471, 486

Identification of specimens 7, 418, 423

IMPPAT 239, 242

Indian Pharmacopoeia 5, 15, 196, 205–206, 214, 228, 236, 239, 262, 278, 393–394, 396–398, 410, 414, 469, 485–486

Indian Pharmacopoeia Commission 196, 205–206, 214, 228, 394, 396–397, 469, 486

Informed consent 111, 114, 121, 125–127, 129–131, 133, 221, 224, 226, 330, 425, 465

Intellectual property rights 4, 231, 325–327, 329–336, 339, 379, 399, 405–411, 413, 486

Inventory management 4, 281, 285, 289, 292, 298–299, 314–315, 409, 411, 433, 451

Iodine 22, 137, 141, 143–144, 146–148, 222, 224, 226, 229

Itai-itai disease 160, 168–170, 172, 223, 227, 466

J

Jasada Bhasma 135, 144, 146–147, 160, 174–177, 179–180, 182, 185, 187, 189, 227, 252

Jeevani 331, 334

K

Kajjali 145, 157, 161, 174–178, 185, 188–189, 227, 252, 287, 419–420, 423, 442

Kalka 39, 73, 247, 256, 260, 314–315, 421

Kharaliya Rasayana 174–175, 178, 185, 189, 247–248, 273, 291, 419

Kshara 247–248, 269–270, 272, 277, 287, 386

Kupipakva Rasayana 157, 174–177, 185, 189, 227–228, 247–248, 257, 270, 272–273, 277–279, 282, 284, 286–287, 293, 295, 297–298, 423–424, 473–474

Kwatha 53, 73, 93, 122, 237, 246–247, 256, 258, 260, 270, 272–273, 284, 315, 317, 320, 386, 421–422, 457

L

Lead 22, 145, 151–152, 154, 161, 163–164, 168–169, 171–172, 175, 181, 183, 188, 205, 222, 226–227, 271–273, 277–279, 287, 412, 467

Lipinski's rule of five 42, 50, 108, 253, 260

Lo-Coco trial 168, 171, 179, 185, 187, 228
Loan licence 237, 283, 296, 343, 346, 372, 476
Loha Bhasma 135, 144–147, 160, 174–175, 177

M

Makaradhwaja 174–177, 179–180, 182, 185–187, 189
Manahshila 157, 171, 175, 179–180, 185, 189, 227–228, 287, 345
Mandura 144–147, 160, 174–175, 177–179, 185, 187, 239–240, 272, 277
Marana 85, 161, 166–169, 174, 177, 184–185, 210–211, 257, 287, 297, 333, 422, 444, 457, 467, 474
Marketing 4, 14, 89, 113–114, 117, 125, 130–131, 193, 226, 231, 237, 241, 249, 252, 282, 285, 293, 298, 303–307, 309–312, 349, 407, 409, 464, 468
Materiovigilance 192, 206
MedDRA 203–204, 207–208, 212
Mees' lines 158, 168, 172, 227, 466
Methylmercury 150–152, 155–157, 163, 166, 168, 170, 172, 227–228, 466–467
Minamata disease 156, 168–170, 172, 222, 227, 466
Mitra Panchaka 422

N

Naranjo algorithm 199–200, 212, 214–215, 228, 426, 469
National Biodiversity Authority 329, 331–332, 334–335, 413
NDCT Rules 103–104, 107, 114, 124–125, 128–132, 196, 206, 214, 221, 226, 228, 464, 486–487
Neem patent 325, 331, 333–335, 400, 403
Niruttha 167, 172, 185, 258, 423
Nishchandratva 99, 167, 258, 419, 422, 479
Noyes–Whitney equation 250–251, 257, 260–263, 409, 412, 477–478

O

OECD guidelines 88, 90, 97–98, 105–107, 109, 220, 470–471, 485

P

Packaging 4, 114, 123, 202, 231, 248–250, 257–258, 260, 265–269, 271, 273–279, 295, 308, 315, 332, 393–394, 405–409, 478
Paka lakshana 258, 261, 263, 307, 316, 412, 418, 421, 479
Panchavidha Kashaya Kalpana 234, 247, 256, 260, 263, 479
Parpati 174–175, 185, 227, 247–248, 272–273, 277–278
Partition coefficient 40, 42–43, 250, 253, 260, 263, 412, 478
Patent 71, 122, 238, 325–336, 339–341, 343, 346–347, 372, 379, 399–404, 409–410, 413–414, 476–477, 485
Patents Act 326–328, 332–335, 339, 413, 477, 485
Pathya 123, 167–168, 184, 201, 237, 316–318, 320, 425
Penicillamine 84, 141, 163–166, 169–170, 227, 467
Periodic safety update report 193, 206, 215
Personnel management 4, 281, 285, 292, 298, 301, 409
Pharmacodynamics 3–4, 11–15, 29, 44–49, 51, 68, 71, 74–75, 85, 89–90, 104, 108, 112, 115, 217–218, 220, 461, 463, 471
Pharmacovigilance Programme of India 195–196, 198, 203, 205–206, 210, 212–215, 226, 228, 239, 469–470, 484, 486–487
Pharmacy Act 3, 5, 226, 236, 241, 320–322, 337, 339, 364, 366–367, 372, 410–411, 413, 476–477, 485
Phase 0 study 115, 130–131, 133, 221, 226, 466
Phase I study 23–24, 39, 47–49, 89, 115–116, 122, 124, 130–131, 133, 225–226, 462, 472
Phase II study 23–24, 39, 46–48, 89, 116–117, 124, 130, 133, 226, 462
Phase III study 89, 116–117, 124, 130, 133, 179, 187, 226
Phase IV study 89, 114, 117, 124, 130–131, 133, 221, 226, 466
Piperine 25, 45–46, 49–50, 73, 77, 85, 87, 221, 225, 254, 463
Pishti 53, 144, 174–175, 185, 189, 227, 247–248, 251, 270, 273, 277–278, 423
Placebo 43, 66, 68–69, 86, 115–120, 122, 124, 129–131, 133, 200, 221, 464–465
Polymorphism 71, 248, 250, 252, 254, 260–263, 409, 412, 478
Pre-formulation 4, 93, 231, 245–251, 253, 255–263, 405, 407–408, 411, 477–479
Preclinical evaluation 4, 11, 88–89, 98–99, 103–106, 109, 112–114, 121, 124, 130, 217, 463–465, 470, 472
Prescription 167, 186, 196, 207, 223, 275, 306, 313–321, 323, 345, 366, 409, 413–414, 456, 477
Prevention of Food Adulteration Act 4, 337–338, 353, 355–356, 407, 476, 485
Prodrug 23, 60, 67, 108, 259, 261, 263
Product life cycle 304–305, 310, 312, 409
Proportional reporting ratio 201–202, 212, 214–215, 469
Punarnavadi Mandura 174–175, 178–180, 185, 187, 239
Puta 174–175, 177, 282, 285–287, 293, 297, 418, 424, 444–445, 457, 473–474

Q

QSAR 40, 43, 259, 261, 263
Quality assurance 211, 237, 282, 285–286, 438, 449
Quality control 121, 161, 237, 282, 284–286, 288, 293, 295, 297, 306–307, 316, 383–384, 390, 424–425, 438, 468, 474–475, 480
Quarantine 284, 288, 295, 298, 307, 309–310, 312, 377, 413, 424, 431–432, 434, 437, 443, 447, 474

R

Randomisation 88, 97, 101, 105, 118–121, 123, 130–131, 221, 226, 228, 465, 471
Randomised controlled trial 111, 117–120, 124, 130–131, 179, 185–187, 221, 464, 466
Rasa Sindura 145, 157, 174–175, 177, 189, 227, 252
Rasashala 234, 297–299, 301, 365–366, 409, 442, 472, 474–475, 477
Rasatarangini 167, 170, 184, 187, 211, 213, 236, 242, 342
Rasayana 99, 106, 144–145, 174–176, 178, 185, 189, 227, 272, 279, 388, 419
Recall 118, 202, 285, 288, 294–296, 303, 307–312, 369, 409, 411, 413, 425, 430, 449–450, 473–474

Receptor 12, 14, 16, 22, 29–34, 36, 41, 47–49, 67–68, 74–76, 84, 91, 93–94, 105, 108, 220–221, 225, 228, 259, 461–463, 471
Rekhapurnatva 99, 167, 258, 419, 422, 431, 479
Reorder level 281, 292, 298, 301, 315, 434, 452, 458
Reverse pharmacology 99, 106, 123, 245, 259–263, 409
Rofecoxib 194
Routes of administration 4, 11–12, 51–52, 63–64, 108, 217–218, 220, 224, 461
Rule of three 117, 193–194, 211–213, 215, 468, 470

S

Safety in the pharmacy 274, 287, 384, 420–421, 423, 438, 480
Sandhana 247, 256, 258, 263, 269, 284–285, 293, 320, 362, 424, 473
Sankhiya 157, 175, 179, 287
Saviryata Avadhi 257–258, 261–263, 276, 278, 309, 311, 479
Schedule E(1) 47, 98, 122, 124, 273–279, 297, 338, 340–341, 343–347, 361–362, 369–370, 372, 408–412, 414, 474, 476–477, 486–487
Selenium 137–139, 142–143, 146–147, 222, 224, 226, 229
Serious adverse event 78, 86, 114, 129, 133, 197–198, 204, 206, 229, 308, 426
Sharangadhara Samhita 55, 58, 61, 64, 234–236, 239, 241–242, 260, 262, 276, 278, 311, 342, 457, 459
Shelf life 176, 236, 246, 250, 255, 258, 265–266, 270–274, 276–279, 282, 296, 298–299, 307, 338, 340–341, 344, 346–347, 407, 409, 411–412, 419, 424, 436, 476, 478–479, 486
Shodhana 2, 43–44, 47–48, 50, 85, 108, 144, 161, 166–169, 174, 177, 184–185, 207–208, 210–211, 257, 287, 297, 419, 422, 424, 435, 442, 444, 467, 470, 474
Signal detection 32–33, 48, 124, 193, 196–197, 201–202, 205–206, 208–209, 212–215, 220, 223, 250, 260, 468–469
Sneha Kalpana 47, 247, 256, 263, 283, 287, 315, 365, 421, 424
Solubility 19, 40, 46, 50, 59, 71, 151, 157, 170, 248–254, 256–258, 260–262, 478–479
Spontaneous reporting 194, 196–197, 211, 214, 228, 469
Stability testing 249, 255, 261, 263, 265, 271, 274, 278, 386, 409
Steady state 27–28, 47–48, 108, 225, 228, 454–455, 462
Store management 281, 283, 287
Structure–activity relationship 12, 40, 43, 48, 50, 220
Swarasa 247, 256, 260, 270, 314–315, 421, 479
Swarna Bhasma 15, 88, 99–101, 105–106, 144, 160, 174–175, 179–180, 188, 219–220, 291, 304, 470, 472

T

Taila 14, 47, 53, 60, 247, 269–270, 273, 277, 284, 315, 320, 386, 421, 423
Tamra Bhasma 15, 144, 160, 174–175, 181–183, 185–187, 189, 227, 271–273, 277–278, 412
Teratogenicity 42, 65, 67, 79, 81–82, 85–86, 109, 221, 469
Thalidomide 42, 81–82, 124–125, 194–195, 213–214, 235, 259, 468, 485
Therapeutic index 11–12, 36–37, 47–49, 72–73, 82, 108, 221, 224–225, 463
TKDL 5, 226, 231, 239, 242, 326–327, 329, 332, 334–335, 399–408, 410–412, 414, 485–487
Toxicity studies 94, 97, 105, 173, 182, 186–187, 189, 470–471
Trade mark 326–327, 333, 335–336, 413
Traditional knowledge 5, 231, 239, 325–335, 399–403, 406, 409, 411, 413–414, 485–486
TRIPS Agreement 325, 328, 330, 334–336, 410, 485
Turmeric patent 325, 331–336, 400, 403, 410, 413, 485

U

Unit dose 315, 320–321, 323, 409, 413
United States Pharmacopeia 235, 238, 241, 252, 262, 393–398, 410, 413
Uppsala Monitoring Centre 195, 198–199, 202–203, 205, 212–215, 227, 229, 426, 469

V

Validation 248–249, 285–286, 297, 301, 307, 437, 475
Varitaratva 99, 167, 258, 419, 422, 479
Vati 45, 53, 174–175, 182, 187, 189, 227, 246–248, 257, 266, 269–270, 272–273, 277, 279, 283–284, 295, 306, 386, 422–423, 474
Vatsanabha 40, 43–44, 47–48, 50, 82, 108, 207, 276, 287, 361–362, 434, 463
VED analysis 281, 290–291, 298–299, 301, 315, 409, 412, 458
VigiBase 196–197, 202–203, 205–206, 212–213, 215, 469
Viruddha Ahara 77, 87, 211
Volume of distribution 12, 21, 48, 67, 108, 225, 453, 462

W

WHO-UMC causality 198–199, 205, 212–215, 426, 469
Wilson's disease 141, 143–144, 146–148, 160, 164–165, 172, 222, 226, 467
World Flora Online 239, 241, 412, 436

X

X-ray diffraction 98–99, 106, 144, 167, 176, 183–184, 227, 250, 252, 256, 261, 423, 478

Y

Yellow Card scheme 195, 204, 212, 215
Yogavahi 45–49, 73, 108, 145, 176, 178, 185–186, 189, 221, 223, 227, 463
Yukti 124, 241, 260

Z

Zero-order kinetics 26, 48–49, 108, 225, 453
Zinc 15, 135–139, 141, 143–148, 150–154, 160, 163, 165–166, 175–179, 181, 185, 187, 217, 222, 226–227, 229, 252, 414, 466

One book for the whole of Paper IV.

Paper IV asks a scholar of Rasashastra and Bhaishajya Kalpana to think like a pharmacologist and to work like a pharmacy manager: from receptors and half-lives to heavy-metal toxicity and pharmacovigilance, from pre-formulation and packaging to GMP, patents and the law that governs Ayurvedic drugs.

This book follows that syllabus point by point. Each topic opens with the questions actually set in university papers from 2018 to 2026, so a scholar sees what is expected before the subject begins. The language is deliberately plain, and Sanskrit terms are given in ordinary Roman letters.

Classical concepts are set beside modern pharmacology, and the boundary between what is established and what is still proposed is marked plainly. Acts and rules are taught as they stand today, and every change since the syllabus was written is flagged.

WHAT THIS BOOK CONTAINS

- All 18 points of the Paper IV syllabus, Parts A and B, in the syllabus's own order
- All seventy questions of seven university papers, each placed under its topic
- 143 colour diagrams, flow charts and graphs
- Answer frames in every chapter, and nine long answers written out in full
- 120 multiple-choice questions with answers, and model question papers
- Specimen GMP documents, 25 solved numerical problems and a chronology of the law
- Ayurveda correlations with the evidence boundary marked
- Glossary, abbreviations and a full index

ABOUT THE AUTHOR



Dr Milind Pareshkumar Chatrabhuji

BAMS, MD (Ayurveda — Rasashastra and Bhaishajya Kalpana), PhD (Scholar). Associate Professor in the PG Department of Rasashastra and Bhaishajya Kalpana at Parul Institute of Ayurved and Research, Parul University, Vadodara, and in clinical practice for over a decade at Suvarna Ayurveda Clinic. His interests lie in the pharmaceutical processing of metals and minerals, the standardisation of classical preparations, and the verification of classical pharmaceutical claims by modern analytical methods.

Free to download. Free to print. Not for sale.
Licensed under Creative Commons Attribution-NonCommercial-NoDerivatives 4.0 International (CC BY-NC-ND 4.0).

ISBN 978-81-181614-2-4
9788118161424